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INDUSTRIAL STATISTICS-I: COURSE INTRODUCTION

In the core courses of this programme, namely, MST-001 to MST-005, you have studied the basic concepts of statistics and various applications of different statistical tools. You have also learnt about which statistical tool to apply in a given situation and how to interpret the results.

The course **Industrial Statistics-I** (MSTE-001) is the first elective course of the Industrial Statistics specialisation. The aim of this course is to enable you to apply statistical techniques and tools to data related to industries and business.

This course is divided into four blocks and each block comprises four units.

In **Block 1** entitled **Process Control**, we discuss the concepts of statistical quality control, process control and product control. We also discuss how we can control a **process** by using various control charts, such as the control charts for variables, namely, the $\bar{X}$ -chart, R-chart and S-chart and control charts for attributes, namely, the p-chart, np-chart, c-chart and u-chart.

In **Block 2** entitled **Product Control**, we discuss the techniques of **product control** and describe acceptance sampling plan and rectifying sampling plan. We also describe single and double sampling plans for attributes in this block.

Block 3 entitled **Decision and Game Theory** discusses some decision making criteria generally used by decision makers to arrive at good decisions in a given situation using available data. But the selection of a decision criterion depends on the environment in which a decision is taken. We discuss four types of environments, namely, decision making under certainty, uncertainty, risk and conflict.

The aim of **Block 4** entitled **Reliability** is to familiarise you with methods of evaluating the reliability of industrial systems. All devices used by human beings have an inherent possibility of failure. As an industrial statistician, you must understand the failure behaviour of a component or a system. But a better way of studying the failure of a component is to study the **reliability** of the component. In this block, we explain what is meant by the reliability of a component and how it is connected with failure. We study some simple and complex systems having components that cannot be repaired. You will also learn how systems with less reliable components can be made more reliable at design level.

BLOCK 1: PROCESS CONTROL

The quality of a product/service is the most important consideration for all of us while using it. The meaning of the term quality has evolved over time and now it means how well a product/service satisfies customer needs or demands. In the ever-growing market competition, industries and businesses cannot hope to survive and succeed until and unless they deliver products/services of high quality. Therefore, the main purpose of the manufacturer or businessman today is to achieve quality assurance and ensure that their products/services meet the desired specifications and requirements. To do so, a system of monitoring and maintaining quality has to be put in place and this requires the use of statistical tools and methods. The method of monitoring and maintaining the quality of products and services using statistical tools is known as **Statistical Quality Control**. It was developed in 1920s by Walter Shewhart, a researcher of Bell Telephone Laboratories. W. Edwards Deming expanded Shewhart's theory and demonstrated how it can be used successfully in industry.

Statistical quality control techniques are broadly classified into two categories:

- Statistical process control or simply Process control, and
- Product control.

In this block, we discuss **process control** and in the next block we shall discuss **product control**.

This block comprises four units.

In **Unit 1**, we introduce the concept of **statistical quality control (SQC)** and describe the elements and techniques of SQC, its advantages and disadvantages. We explain the basic concepts of **process control** and **product control**. We also discuss the causes of variation in a product and finally introduce the concept of **control chart**, which is the main technique of process control.

In **Unit 2**, we present the general technique of constructing a control chart. When we deal with a variable (measurable quality characteristic), it is necessary to control the mean (average) as well as the variability of the quality characteristic. So we discuss different **control charts for variables** for controlling the mean ($\bar{X}$ -chart) and variability (R-chart or S-chart) of the process.

For monitoring and controlling the quality characteristics that cannot be measured, we use **control charts for attributes**. The control charts for attributes are of two types: Control charts for defectives and control charts for defects. In **Unit 3**, we discuss the control charts for defectives and explain how to construct a control chart for fraction defective (p-chart) and control chart for number of defectives (np-chart).

Finally, in **Unit 4**, we discuss control charts for number of defects (c-chart) and control chart for number of defects per unit (u-chart). We also compare control charts for variables and attributes.

The following notations and symbols are used in this Block.

Notations and Symbols

Sec.	:	Section
Fig.	:	Figure
SQC	:	Statistical quality control
SPC	:	Statistical process control
CL	:	Centre line
UCL	:	Upper control limit
LCL	:	Lower control limit
USL & LSL	:	Upper and Lower specification limits
UNTL & LNTL	:	Upper and Lower natural tolerance limits
PC	:	Process capability
X	:	Random variable
$E(X)$ and $\text{Var}(X)$	:	Mean and variance of a random variable X, respectively
$SE(X)$	:	Standard error of a random variable X
$E(X) \pm 3SE(X)$	:	Control limits of a random variable X
$X_1, X_2, \dots, X_n$	:	Measurements on a random sample of size n
n	:	Sample size
$\bar{n}$	:	Average sample size
k	:	Number of samples
$\bar{X}$	:	Sample Mean
$\bar{\bar{X}}$	:	Grand mean
R	:	Sample range
$\bar{R}$	:	Mean of sample ranges
S	:	Sample standard deviation
$\bar{S}$	:	Mean of sample standard deviations
P	:	Process fraction (proportion) defective
p	:	Sample fraction (proportion) defective
$\bar{p}$	:	Average fraction (proportion) defective
c	:	Number of defects
$\bar{c}$	:	Average number of defects
u	:	Number of defects per unit/item
$\bar{u}$	:	Average number of defects per unit/item
d	:	Number of discarded samples which are out-of-control
P[.]	:	Probability
$\left. \begin{array}{l} A, A_2, A_3, B_3, \\ B_4, D_1, D_2, D_3, \\ D_4, d_1, d_2, c_4 \end{array} \right\}$	:	Constants/factors that depend on the size of the sample
μ	:	Process mean
σ	:	Process standard deviation
$N(\mu, \sigma^2)$	:	Normal distribution with mean μ and variance σ^2
λ	:	Average number of defects in the process

UNIT 1 INTRODUCTION TO STATISTICAL QUALITY CONTROL

Structure

- 1.1 Introduction
 - Objectives
- 1.2 Quality
 - Defining Quality
 - Dimensions of Quality
 - Quality Control
 - Historical Review of Quality Control
- 1.3 Statistical Quality Control (SQC)
 - Elements of SQC
 - Techniques of SQC
 - Statistical Process Control (SPC)
 - Product Control
- 1.4 Causes of Variation
 - Chance Causes
 - Assignable Causes
- 1.5 Control Charts
- 1.6 3σ Limits
- 1.7 Control Chart Patterns
 - Natural Patterns of Variation
 - Unnatural Patterns of Variation
- 1.8 Advantages and Limitations of SQC
- 1.9 Summary
- 1.10 Solutions/Answers

1.1 INTRODUCTION

In our day-to-day life, we use different products and services from the time we get up until we get back to bed. For example, we use different types of products such as toothpaste, soap, detergent, clothes, food, gas stove, vehicle, phone, computer, TV, electric bulb, etc. and we also use different types of services such as water supply, electricity, transportation, health care, education, etc.

All of us wish that these products and services should fulfil certain specifications when we use them. If the product/service fulfils the specifications needed for its proper use, we say that it is of **good quality**. If it does not fulfil the specifications, it is said to be of **poor quality**.

In the face of ever-growing market competition, the main objective of the manufacturer or producer is to achieve quality assurance in manufacturing and service organisations so that his/her product/service can meet the existing competition in the market. In order to achieve this objective, different statistical tools have been developed, which are useful for controlling the quality of products vis-a-vis certain specifications or standard. The technique of

controlling product quality against the specifications using statistical tools is known as **Statistical Quality Control (SQC)**.

In this unit, you will learn about the concepts and various aspects of SQC. In Sec. 1.2, we define the term **quality** and discuss dimensions of quality, quality control and historical review of quality control. In Sec. 1.3, we describe various aspects of SQC, e.g., the elements and techniques of SQC, statistical process control and product control. We also discuss the causes of variation, which may be due to chance or could be assigned to some factors in the production process in Sec. 1.4. In Sec. 1.5, we introduce the control chart – a tool used in statistical quality control to indicate whether a process is under control or out-of-control. We explain the concept of 3σ limits, different patterns of the control chart and advantages and limitations of SQC in Secs. 1.5 to 1.8. In the next unit, you will study the control charts for variables.

Objectives

After studying this unit, you should be able to:

- explain the concepts of quality, quality control and statistical quality control (SQC);
- describe the need for statistical quality control;
- distinguish between causes of variation (chance causes and assignable causes) in the production process;
- describe the techniques of SQC;
- define process control and product control;
- explain the concept of control chart, the principle underlying 3σ limits and various control chart patterns; and
- discuss advantages and limitations of SQC.

1.2 QUALITY

In the introduction of the unit, we have just given you the flavour of the notion of quality. We now define the term **quality**. The dictionary meaning of quality is **Degree of Excellence**. It means that both products and services should be excellent. This definition of the term **quality** is rather subjective as it depends on the perception of **excellence** and varies from situation to situation or from person to person. If 20 different people are asked to define quality, most probably there will be 20 different answers. However, **the best** will be widely used. If we ask what quality they want, mostly people say they want the best. This indicates that the general public opinion about the term quality is **the best**. However, if we observe the buying habits of people, we find that most people buy an item at a discount price in a sale or a mid priced item rather than the most expensive one. This indicates that while buying something, people want the best that they can afford. Further, apart from price, there are many other things to be considered for quality such as cost, size, performance, warranty, appearance, etc.

The concept of quality as goodness or the best or luxury is not adequate for professionals working in quality control because **the best** depends on an individual's perception. For example, a person may think that a car is a good vehicle for Delhi roads, but a comparatively poor vehicle for the forests of

Himachal Pradesh, Madhya Pradesh, etc. because the best vehicle for these places would be one with four-wheel drive. Similarly, a good walking shoe is good for walking, but unsuitable for running. Therefore, the **intended use** of the product or service needs to be included in the concept of quality. So we need to define the term **quality**.

1.2.1 Defining Quality

There are several ways of defining quality:

1. Initially, quality was defined as **conforming to specifications**. This means that any product should be manufactured according to given specifications. If a product fulfils its specifications, it is considered to be a **quality** product.

There are some problems with this definition. In some cases, it is found that even though a product conforms to all specifications, its utility is not up to the mark. For example, suppose a person wishes to purchase a touch screen mobile phone having clear sound. The shopkeeper shows him/her such a mobile set. But the person may feel that the size of the mobile phone is big. So he/she may not purchase it because it is not fit for his/her use.

Therefore, from the point of view of customers, such products are not useful. Thus, there is a need to redefine the term quality.

2. The definition of quality was modified to include **fitness for use**. This means that the manufactured product should be such that it conforms to its specifications and is fit for use. However, it was found that in some cases, a product conformed to all specifications and was fit for use but could not be sold as it did not appeal to the customer. Therefore, to sell any product it is necessary to incorporate the **customer's viewpoint**.
3. According to the third definition, quality was **customer satisfaction**. This means that a product, which brings satisfaction to the customer could be termed as a quality product. Then it was found that every customer would have plenty of demands in respect of each product that he/she wished to buy. However, customers seldom express all of their expectations. For example, suppose a person goes to a restaurant and orders a pizza. If the pizza given to him is not hot, that person would definitely be unhappy. But it is also true that the customer would never mention that he/she wanted a hot pizza! There are many situations where the customers are unhappy, even when all their expressed or stated needs are fulfilled. Hence, while defining quality, there is a need for considering unexpressed or unstated needs of customers.
4. The fourth definition given for quality was **delighting the customer**. Delight is one step ahead of satisfaction. When a product fulfils both the expressed and unexpressed needs of the customer, he/she is delighted. However, this definition was also improved upon.
5. The fifth definition given for quality was **enchancing the customers**. According to this definition, the manufacturer plays a dual role: firstly, he/she should know the need of the customers. Secondly, he/she should make customers aware of this fact and also make them feel that these are the products they want. For example, an electric bulb manufacturer needs to educate customers about LEDs and make them want to buy LEDs as these are low on consumption of electricity even though if these cost slightly more. Now-a-days, every manufacturer is expected to follow this definition.

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From the above discussion, we may conclude that in the manufacturing and services sector, the following aspects have to be incorporated in the definition of quality:

- conforming to specifications,
- fitness for use,
- customer satisfaction,
- delighting the customer, and
- enchanting the customers.

Having explained the concept of quality in industry, and defined it we now describe various dimensions of quality.

1.2.2 Dimensions of Quality

In the previous section, we have explained the concept of quality and defined it. In 1988, David A. Garvin summarised eight basic elements of quality, which are known as the **dimensions of quality**. We describe them briefly.

i) Performance

The first dimension of quality is **performance**. It refers to the primary operating characteristics of a product. Consumers judge the quality of any product based on its performance after comparing it with the competitor's products or the prevailing market standard. For example, a mobile phone can be judged by its clarity of sound, weight, size, functions, etc. Similarly, a motorcycle can be judged by its pick-up, fuel efficiency, etc.

ii) Features

Features constitute the second dimension of quality. These refer to additional characteristics available in products along with the primary operating characteristics. For example, complimentary drinks and snacks in a flight or hotel, Bluetooth and FM in mobile, etc. A customer's choice is also influenced by this dimension of quality.

iii) Reliability

Reliability refers to the probability of a product's failure within a specified time period. If a product fails frequently, we say that it is an unreliable product. For example, if a TV of a particular company requires frequent repair, we say that it is unreliable. There are many products such as laptops, TVs, automobiles, etc. in which customer's view about quality is influenced by the reliability of the product.

iv) Conformance (Agreement)

Conformance means meeting specifications. Customers obviously want that the product should meet its specifications. For example, when we purchase a motorcycle, we check whether the sitting space, weight, size, pick-up, fuel efficiency, etc. conform to the specifications mentioned by the company. This dimension also influences customer's choice.

v) Durability

Durability refers to the measure of product life. It can also be understood as the operational life of the product, i.e., how long the product can be used without replacement. The life of an electric bulb is an example of durability:

when its filament burns up, the bulb needs to be replaced because at this stage, repair is not possible. Durability also changes customers view.

vi) Serviceability

Serviceability is the sixth dimension of the quality. Consumers are concerned not only about a product break-down, but also about the time taken before the product is serviced restored. Serviceability is concerned with how readily a product can be serviced back into operational mode. For example, suppose a customer wishes to buy a washing machine. While deciding on the company, he/she may consider how long the company takes to service or repair it in case it breaks down.

vii) Aesthetics

The seventh dimension of quality is **aesthetics**, which means how a product looks, sounds, feels, etc. The aesthetic value of a product is purely subjective. For example, some people may find a particular car aesthetically appealing while others may not.

viii) Reputation

Reputation is related to the past performance of the company. In many cases, customers also check out the quality of products made by the company in the past. For instance, if a company launches a new car, it is usually assumed by the customer that the new model would be successful because the past performance of other cars of that company was good.

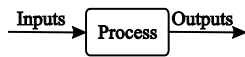
In order to design and manufacture products of high quality, it is necessary to incorporate all **eight dimensions of quality**.

You may like to pause here and check your understanding about the definition of quality and its dimensions by answering the following exercise.

E1) Choose the correct option from the following:

- i) Quality means
 - a) fitness for use
 - b) degree of excellence
 - c) conformance to requirement or specifications
 - d) all of the above
- ii) Primary operating characteristic of a product is known as
 - a) aesthetics
 - b) reliability
 - c) performance
 - d) features
- iii) Durability refers to a measure of
 - a) product life
 - b) specifications
 - c) past performance of the product
 - d) the probability of a product's failure within a specified time period

A process is a series of actions or operations that transforms input to output.



So far you have studied the definition of quality and its dimensions. We now introduce the concept of quality control.

1.2.3 Quality Control

We first understand what we mean by control.

The process/procedure/method that is applied to meet the specifications or standard is known as **control**. Process control works on feedback and comprises the following steps:

1. Choose the **parameter** to control, i.e., we first choose the characteristic that we intend to control such as length, height, weight, defects, etc.
2. Choose the **unit of measurement**, e.g., centimeter (cm), millimeter (mm), gram (g), etc.
3. Set the **standard** or goal for the parameter to control, e.g., 5 cm, 10 g, etc.
4. Select a sensing device, which can measure the parameter to control in terms of the unit of measure, e.g., scale, a weighing balance, etc.
5. Measure the actual performance.
6. Compare the actual performance with the standard.
7. Take necessary action when there is a difference between actual performance and the standard.

Thus, **quality control** can be defined as:

“The process by which we measure the quality characteristics of the product, compare them with the specifications or standard and take suitable actions whenever there is a difference between actual quality and the specifications or standard”.

Now-a-days, quality is controlled by using statistical tools. This technique is known as **Statistical Quality Control**, which we describe in the next section. However, before you study further, you may like to know how the concept of quality control evolved historically.

1.2.4 Historical Review of Quality Control

The notion of quality control is not new. We may say that it began when human beings lived in forests and ate raw plants and animals. In those times, they used natural materials and may have faced problems such as which plants were suitable for eating and which ones were poisonous for health. With the evolution of early technology, human beings started processing natural materials. They made products and examined themselves whether these worked properly for their purpose or not. For example, they produced rough hewn stone tools. They may have checked the point of the tool to see whether it was sharp enough for their purpose. In those times, human beings were both manufacturers and users. This situation changed with time.

As civilisations arose, jobs were divided and different people got specialised in production of different goods and services. For example, farmers cultivated crops, potters made pots; weavers wove the cloth required for everyone. This specialisation created a separation between the producer and the user. Thus, the producer/manufacturer and the user were no longer the same individual. However, the goods were not so complicated and the user had long familiarity

with the products through prior use. Therefore, the user could assure that the products were fit for use. The situation changed drastically with the advent of industrialisation and the notion of quality control underwent a sea change.

For the sake of interest, we describe below the important stages in the historical evolution of quality control:

1. Operator or craftsman quality control

From early civilisation until the industrial revolution, quality control was ensured/observed either by a single worker (operator or craftsman) or a very small number of workers who totally controlled the quality of their work. This situation continued until the nineteenth century.

2. Foreman or supervisor control

Large factories were established after the industrial revolution to meet the increasing demand of consumers. Therefore, the number of workers increased in the factories. This gave rise to the need of supervisors who guided, controlled and administered other workers. Generally, in those times, the supervisor was picked up from among the workers of the factory who had thorough knowledge of the work. Checking the quality of a job done by workers was also the responsibility of the supervisor functions. This situation continued until the First World War.

3. Inspector of quality control

The manufacturing system became more complex due to interchangeable components on a mass scale and it became essential to critically examine each component with the help of measuring instruments. Obviously, this job could not be performed by the supervisor because he/she had expertise only in one component of the product. Thus, a full time inspector came on the scene to inspect the finished goods and quality control was separated from production.

4. Statistical quality control

Initially, quality control inspectors inspected all products. However, this was costly and time consuming. In 1924, **Walter A. Shewhart**, a researcher of Bell Telephone Laboratories (a Research and Development Unit of the American Telephone and Telegraph Company) developed the technique of **statistical control charts** for the control of product variability. This is usually considered as the beginning of statistical quality control (SQC).

In the same decade, Harold F. Dodge and Harry G. Roming, researchers of Bell Telephone Laboratories, developed statistically based acceptance sampling as an alternative of 100% inspection.

The techniques of control chart and acceptance sampling were used in Bell system, but neither of these were widely adopted outside it. However, this changed during the Second World War. The ordinance department of the U.S. Army was facing the problem of how to get large quantities of arms and ammunitions from multiple suppliers at acceptable levels of quality. In 1942, a quality control section was established in the War Department and acceptance sampling came to be used widely.

After the Second World War, Japan was trying to revive from the devastations of world war. Japanese industries were almost destroyed and its leaders knew that rebuilding the industry was essential for the survival



Walter Andrew Shewhart
(1891-1967)

An American physicist,
engineer and statistician,
sometimes known as
the father of **statistical
quality control**.



William Edwards Deming
(1900-1993)

An American engineer,
statistician, professor,
author, lecturer and

of the nation. Japanese industrialists decided to engage an American statistician as their consultant. The man they chose was W. Edwards Deming, who had studied with Walter Shewhart. Deming developed theories on how statistics could be used to improve industrial quality. He gave a series of lectures in Japan on statistical quality improvement methods. Japan adopted these methods and within a few decades, became one of the most successful industrial nations on the earth.

So far, we have given you an overview of the concepts of quality and quality control. We now discuss statistical quality control.

1.3 STATISTICAL QUALITY CONTROL (SQC)

In the highly competitive market today, the main objective of manufacturers or producers is to achieve quality assurance in manufacturing and service organisations. In order to achieve this objective, different statistical tools have been developed, which are useful for controlling the quality of products vis-a-vis the specifications or standard. The technique of using statistical tools for controlling product quality vis-a-vis the specifications is known as **Statistical Quality Control (SQC)**.

Statistical quality control is defined as the technique of applying statistical methods based on the theory of probability and sampling to establish quality standard and to maintain it in the most economical manner.

Let us now outline the elements that constitute SQC.

1.3.1 Elements of SQC

The following are the main elements of SQC:

a) Sample Inspection

We know that 100% inspection needs huge expenditure of time, money, labour and resources. Further, if the nature of the product is such that it is completely destroyed during the process of inspection, e.g., a bulb, candle, ammunition, food, etc., 100% inspection is not practicable. Therefore, SQC is based on **sampling inspection**. In sampling inspection method, some items or units (called sample) are randomly selected from the process and then each and every unit of the sample is inspected.

b) Use of Statistical Methods

Some commonly used statistical tools such as random sampling, mean, range, standard deviation, mean deviation, standard error and concepts such as probability, binomial distribution, Poisson distribution, normal distribution, etc., are used in SQC. Since, quality control method involves extensive use of statistics, it is termed as **Statistical Quality Control**.

c) Fundamental Objective

The fundamental objective of SQC is to decide whether the unit produced is according to its specifications or not. If the unit produced is not according to its specifications and there is a variation in quality, it becomes necessary to trace the causes of variation and eliminate them if possible.

d) Decision Making

With the help of SQC, we **decide** whether the quality of the product or the process of manufacturing/producing goods is under control or not.

e) Specifications, Production and Inspection

SQC method helps in deciding about the specifications, production and inspection of an product.

We now describe the techniques of statistical quality control.

1.3.2 Techniques of Statistical Quality Control

The important techniques used for statistical quality control can be broadly classified into two categories:

- Statistical Process Control (SPC) or simply Process Control, and
- Product Control.

These techniques are further classified into different categories as shown in Fig.1.1.

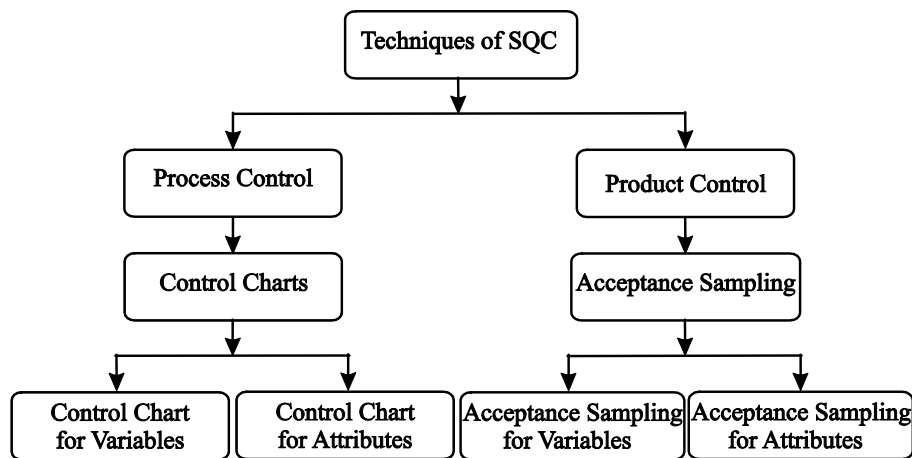


Fig. 1.1: Classification of SQC techniques.

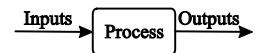
Let us discuss both categories of SQC techniques in some detail.

1.3.3 Statistical Process Control (SPC)

Statistical Process Control (SPC) or simply process control (PC) is the first part of SQC. For understanding SPC, first of all, we should understand the concept of **process** in quality control.

A process is a series of operations or actions that transforms input to output. It is said to be stable or repeatable if the resulting output product is of the given specifications or standard quality. But sometimes, due to certain causes such as poor quality of raw material, change in the machine settings, use of unskilled work force, improper machine, etc., the stable process is disturbed. In such situations, we require a tool or technique through which we can control the process. This technique is known as **statistical process control (SPC)**.

Statistical process control is a technique used for understanding and monitoring the process by collecting data on quality characteristics periodically from the process, analysing them and taking suitable actions whenever there is a difference between actual quality and the specifications or standard.



Statistical process control technique is widely used in almost all manufacturing processes for achieving process stability and making continuous improvements in product quality. Its major tools are:

1. Histogram
2. Check sheet
3. Pareto chart
4. Cause and effect diagram
5. Process flow diagram
6. Scatter diagram
7. Control chart

Of these seven tools we shall describe only the technique of “**control chart**” given by W.A. Shewhart in 1924 because it is the most preferred technique today. In fact, it is probably an outstanding technique for controlling and improving quality. We introduce the basics of control chart in Sec. 1.5. In Units 2, 3 and 4 of this course, we discuss various types of control charts.

1.3.4 Product Control

In many situations, the product is so complex that a manufacturer cannot make all components or parts of the product. Therefore, one or more component (s) of a product are purchased from outside agents or suppliers and the manufacturer does not have a direct control over the quality of such component (s). In such situations, the manufacturer faces the problem of controlling the quality of component (s) sourced from elsewhere. In addition, the manufacturer needs to **control the quality of the final product**. So he/she also faces the problem of ensuring that the final product meets its specifications and various lots of the product do not contain an excessively large number of defective items. Such types of problems come under the category of **product control**.

Product control means to control the products in such a way that these are free from defects and conform to their specifications.

Initially, product control was done by 100% inspection, which means that each and every unit produced or received from the outside suppliers was inspected. This type of inspection has the advantage of giving complete assurance that all defective units have been eliminated from the inspected lot. However, it is time consuming and costly. Also, if a product is destroyed under inspection, e.g., light bulb, crackers, ammunition, picture tube of the TV, etc., 100% inspection is not possible.

Sampling inspection or acceptance sampling was developed as an alternative of 100% inspection.

Acceptance sampling is a technique in which a small part or fraction of items/units is selected randomly from a lot and the selected items/units are inspected to decide whether the lot should be accepted or rejected on the basis of the information obtained from sample inspection. This is how product control is achieved through acceptance sampling.

Acceptance sampling is also of different types. We shall discuss it in detail in Unit 5.

You may like to pause now and check your understanding of various aspects of SQC discussed in this section. Answer the following exercises.

E2) Choose the correct option from the following:

- i) Statistical quality control (SQC) is a technique of
 - a) process control
 - b) product control
 - c) both (a) and (b)
- ii) The statistical techniques used in statistical quality control are
 - a) control charts
 - b) acceptance sampling plans
 - c) both (a) and (b)
- iii) Process control is achieved through
 - a) control charts
 - b) acceptance sampling plans
 - c) both (a) and (b)
- iv) Product control is achieved through
 - a) control charts
 - b) acceptance sampling plans
 - c) both (a) and (b)

You have learnt that SQC techniques are used for process control and product control so that the products of desired quality may be manufactured. We now explain the causes of variation in the process.

1.4 CAUSES OF VARIATION

Variation in manufactured products is inevitable; it is a fact of nature and industrial life. Even when a production process is well designed or carefully maintained, no two products are identical. The difference between any two products could be very large, moderate, very small or even undetectable depending on the sources of variation. For example, the weight of a particular model of automobile varies from unit to unit, the weight of packets of milk may differ very slightly from each other, the length of refills of ball pens, the diameter of cricket balls may also be different and so on. The existence of variation in products affects quality. So the aim of SQC is to trace the sources of such variation and try to eliminate them as far as possible. The causes of variation are broadly classified into two categories:

1. Chance causes, and
2. Assignable causes.

1.4.1 Chance Causes

Chance causes are also known as **random** or **natural** or **common causes**. Even in a well designed or carefully maintained production process, variability exists in the product due to some natural/random causes. Even if the process is operated under the same conditions, that is, the quality of raw materials used is same and there is no change in the machine settings, operators or the environment, there is a specific pattern of variability in the product. For

example, the diameter of ball bearings varies slightly, there is a slight variation in the weight of cricket balls, the fuel efficiency of a particular model of automobile is varies slightly and so on. Such variability is due to different common or chance causes, may which affect the process output in minor ways. Such causes are known as **chance causes of variation**. These may arise due to, inflexibility of aged machines, variability in purchased material, poor lighting, extent of worker training or other non obvious reasons. These may or may not be present at the same time, but when taken together produce random results. If the quality of the output varies too much due to chance causes, the process must be redesigned or modified to eliminate one or more of these causes. Since process redesigning or modification is the responsibility of the management, the elimination of common/chance causes of variation is usually the responsibility of the management and not that of workers. It may not be possible to eliminate all chance causes in a process. However, even **if variation due to chance causes is present in the production process, it is still said to be under statistical control.**

1.4.2 Assignable Causes

Another kind of variability may be present occasionally in the output of a process. The causes of such type of variability are **not** due to the process design, but take place because of changes in raw material, machine, operator, environment or any other component of the process. These causes are called **assignable causes** and are also known as **special** or **non-random** or **unnatural causes**. Accidental improper setting of the machine, a worker falling ill and still continuing to work, change of operators or shift, breakages, misreading of scales, batch of defective raw material, etc. are examples of assignable causes.

Since the effect of assignable causes is localised within a process, these may be eliminated by workers or their immediate supervisor. The variability due to assignable causes is generally larger than the variability due to chance causes and it usually represents an unacceptable level of process performance. A process that is operating in the presence of assignable causes is said to be an **out-of-control process**.

You can now check your understanding of the causes of variation by answering the following exercise.

-
- E3)** A company manufactures cricket balls. The statistical quality controller of the company finds that there is a variation in the weight of cricket balls. Answers the following:
- i) The variation in the weight of cricket balls may be due to
 - a) assignable causes
 - b) chance causes
 - c) both (a) and (b)
 - ii) If the variation in the weight is due to chance causes, it is
 - a) controllable
 - b) not controllable
 - c) both (a) and (b)
 - iii) If the variation in the weight is due to assignable causes, it is
 - a) controllable

- b) not controllable
- c) both (a) and (b)
- iv) The variation due to chance causes
 - a) is tolerable
 - b) does not affecting the quality of a product
 - c) is uncontrollable
 - d) all of the above
- v) The variation due to assignable causes
 - a) can be removed
 - b) cannot be removed
 - c) can be removed sometimes
 - d) can be removed most of the times

Having explained the causes of variations in the production process, we now introduce the technique of control chart. It is the main technique used for process control.

1.5 CONTROL CHARTS

You will agree that graphical representation is one of the most sensitive statistical instruments. So we can represent the quality characteristic of the output product such as weight, length, diameter, defects, etc. graphically to understand, describe or monitor process variation. The idea of representing quality characteristics graphically was first given by Walter A. Shewhart. He invented **control charts** for the industrial processes to distinguish acceptable (chance) variation from the assignable variation. He observed that with the help of control charts, the occurrence of assignable causes of variation could be detected quickly and corrective action could be taken to eliminate them.

A **control chart** is a two-dimensional graphical display of a quality characteristic that has been measured or computed in terms of mean or other statistic from samples and plotted against the sample number or time at which the sample is taken from the process.

The concept of control chart is based on the theory of sampling and probability. In a control chart, a sample statistic of a quality characteristic such as mean, range, proportion of defective units, etc. is taken along the Y-axis and the sample number or time is taken along the X-axis. A control chart consists of three horizontal lines, which are described below:

1. **Centre Line (CL)** – The centre line of a control chart represents the value which can have three different interpretations depending on the available data. First, it can be the **average value of the quality characteristic** or the **average of the plotted points**. Second, it can be a **standard or reference value**, based on representative prior data or an **aimed (targeted) value** based on specifications. Third, it can be the **population parameter** if that value is known. The **centre line is usually represented by a solid line**.

2. **Upper Control Line** – The upper control line represents the **upper value of the variation in the quality characteristic**. So this line is called **upper control limit (UCL)**. Usually, the **UCL is shown by a dotted line**.
3. **Lower Control Line** – The lower control line represents the **lower value of the variation in the quality characteristic**. So this line is called **lower control limit (LCL)**. Usually, the **LCL is shown by a dotted line**.

The UCL and LCL also have three interpretations depending on the available data same as the centre line. These limits are obtained using the concept of **3 σ (three sigma) limits**, which we shall describe in Sec. 1.6.

For construction of control charts, we select samples of few units/items at regular intervals from the process. Then we measure the quality characteristic, e.g., length of foil, diameter of ball bearing, weight of cricket ball, etc. for each unit of the selected samples or count the number of defects, number of defective units, etc. for each selected sample. After that, we calculate the statistic such as mean, standard deviation, range, proportion of defective unit, etc. for each sample. Then the values of the calculated statistic are plotted against the sample number or time. The sample points on the graph may be joined by a line. Joining the sample points by line segments is not compulsory but by doing so, we can easily understand their sequence (pattern) over time.

If all sample points lie on or in between the upper and lower control limits, the control chart indicates that the **process is under statistical control**. That is, only chance causes are present in the process. **No assignable cause is present** in the process. However, if one or more sample points lie outside the control limits, the control chart alarms (indicates) that the process is **not under statistical control**. Some **assignable causes are present** in the process. To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them.

However, a control chart cannot tell us what is wrong it. In fact, it is the responsibility of the supervisor or quality control manager to find out what has gone wrong.

Note: Sometimes the sample points may be inside the control limits but may behave in a systematic or non-random manner. This is also an indication that the process is out-of-control. For example, suppose 22 of the last 25 sample points lie below the centre line but above the lower control line and only 3 of these points lie above the centre line but below the upper control line. This pattern indicates that the process is **not under statistical control** because it is not random in appearance. We shall explore this aspect in more detail in Sec 1.7.

You may like try the following exercise before studying further.

E4) Choose the correct option from the following:

- i) Control chart is a
 - a) one-dimensional chart.
 - b) two-dimensional chart.
 - c) three-dimensional chart.
 - d) none of the above.
- ii) Control chart consists of
 - a) one control line.

- b) two control lines.
 - c) three control lines.
 - d) four control lines.
- iii) If one or more sample points lie outside the control limits, the control chart indicates that
- a) there is no assignable cause in the process and the process is under statistical control.
 - b) there is at least one assignable cause in the process and the process is under statistical control.
 - c) there is at least one assignable cause in the process and the process is out of statistical control.
 - d) there is no assignable cause in the process and the process is out of statistical control.
- iv) Control charts in statistical quality control are used for
- a) describing the pattern of variation.
 - b) checking whether the variability in the product is within the tolerance limits or not.
 - c) both (a) and (b).

In Sec. 1.5, you have learnt about the control chart, which contains a centre line (CL), lower control limit (LCL) and upper control limit (UCL). We now discuss how to obtain the centre line and control limits for a control chart. The UCL and LCL are also called 3σ limits.

1.6 3σ LIMITS

The quality characteristic can be described by a probability distribution or a frequency distribution. In most situations, a quality characteristic follows a **normal distribution** or can be approximated by a normal distribution. You have studied the normal distribution in Unit 14 of MST-003 entitled Probability Theory and you know the probability that a normally distributed random variable (X) lies between $\mu - 3\sigma$ and $\mu + 3\sigma$ is 0.9973 where μ and σ are the mean and the standard deviation of the random variable (X). Thus,

$$P[\mu - 3\sigma \leq X \leq \mu + 3\sigma] = 0.9973$$

So the probability that the random variable X lies outside the limits $\mu \pm 3\sigma$ is $1 - 0.9973 = 0.0027$, which is very small. It means that if we consider 100 samples, most probably 0.27 of these may fall outside the $\mu \pm 3\sigma$ limits. So if an observation falls outside the 3σ limits in 100 observations, it is logical to suspect that something might have gone wrong. Therefore, the control limits on a control chart are set up by using 3σ limits. The **UCL and LCL of a control chart are called 3σ limits** of the chart. The question is: How do we calculate 3σ limits?

Suppose M is a sample statistic (e.g., mean, range, proportion of defectives, etc.) that measures some quality characteristic of interest. Further suppose that

μ_M and σ_M are the mean and standard error (standard deviation) of the sample statistic M , respectively. Then the centre line and control limits for controlling the quality characteristic are given by:

$$\text{Centre line (CL)} = \mu_M \quad \dots (1)$$

$$\text{Upper control limit (UCL)} = \mu_M + 3\sigma_M \quad \dots (2)$$

$$\text{Lower control limit (LCL)} = \mu_M - 3\sigma_M \quad \dots (3)$$

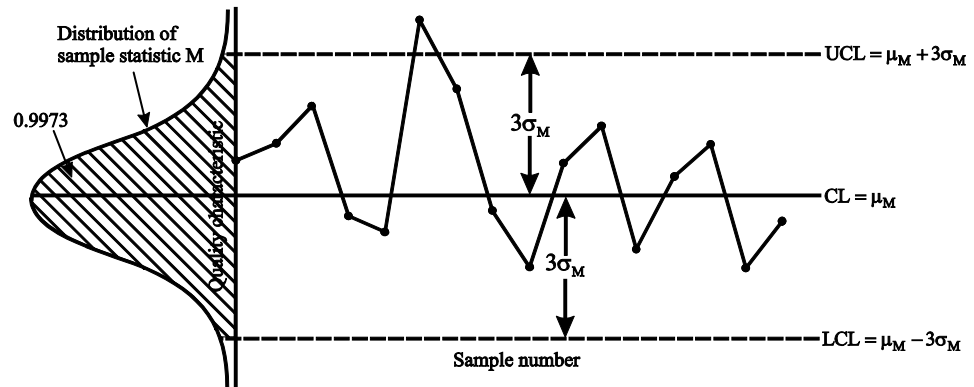


Fig. 1.2: 3σ limits.

The chart in Fig. 1.2 shows the centre line (CL), lower control limit (LCL) and upper control limit (UCL). The UCL and LCL are set at the distance $\pm 3\sigma_M$ from the centre line (μ_M). Note from Fig. 1.2 that the area covered between the UCL ($= \mu_M + 3\sigma_M$) and LCL ($= \mu_M - 3\sigma_M$) is 0.9973 (99.73%). So the probability that an observation falls outside these limits is 0.0027.

If the sample points fall between the control lines, the process is said to be **under statistical control**. But, if one or more points lie outside the control limits, control chart alarms (indicates) that the **process is not under statistical control**. Some assignable causes are present in the process. To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them and then continue the production process.

You may now like to check your understanding of 3σ limits by answering the following exercise.

E5) If μ and σ represent the mean and standard deviation of the process, the lower and upper three sigma control limits for a control chart are given by:

- a) $\mu - 3\sigma^2$ and $\mu + 3\sigma^2$
 - b) $\mu - 3\sigma$ and $\mu + 3\sigma$
 - c) $\mu^2 - 3\sigma^2$ and $\mu^2 + 3\sigma^2$
 - d) $\mu^2 - 3\sigma$ and $\mu^2 + 3\sigma$
-

In Sec. 1.6., you have learnt that the points on the control chart describe a pattern. If one or more sample points fall outside the control limits, the process

is said to be out-of-control and if all sample points are inside or on the control limits, the process is said to be under statistical control. But in many cases, the sample points may lie within the control limits and yet show an unnatural or specific pattern. This is also an indication of assignable causes. So to decide whether the process is under statistical control or not, it is also important to analyse the **pattern of the sample points**. This is what we discuss in Sec. 1.7.

1.7 CONTROL CHART PATTERNS

The **patterns of the control chart** are broadly classified into two categories:

1. Natural patterns of variation, and
2. Unnatural patterns of variation.

We first discuss the natural patterns of variation.

1.7.1 Natural Patterns of Variation

In Unit 1 of MST-002 entitled Analysis of Quantitative Data, you have studied that the central tendency is the characteristic of the distribution. Most observations tend to concentrate near the centre (mean) of the distribution and very few points lie near the tails. Since normal distribution is symmetrical about its mean (μ), i.e., the centre line is at μ ($CL = \mu$), we expect that half the points will lie above the centre line and half below it. We also know that for the normal distribution

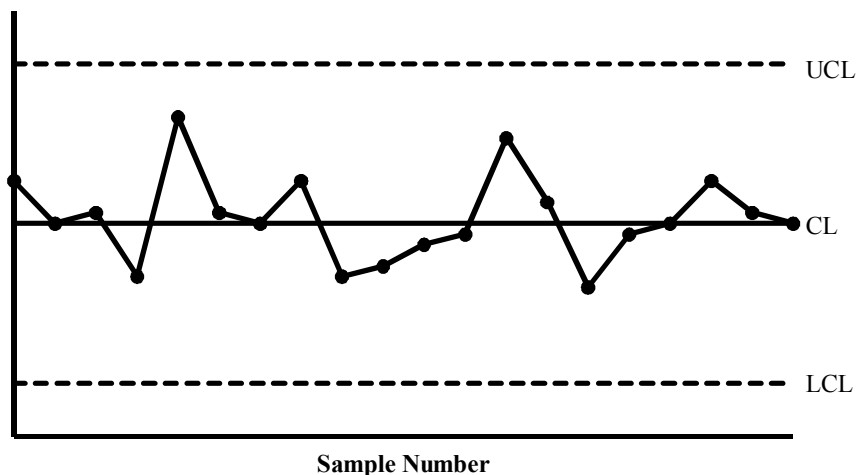
$$P[\mu - 3\sigma \leq X \leq \mu + 3\sigma] = 0.9973$$

This means that 99.73% observations lie between the 3σ limits. So of a total of 100 observations, 99.73 observations will lie inside the 3σ limits and only 0.27 observations may lie outside the 3σ limits.

We may conclude that a control chart having a natural pattern of variation has the following three characteristics:

- i) Most points lie near the centre line of the chart.
- ii) Very few points lie near the control limits.
- iii) None of the points fall outside the control limits.

A typical control chart with 3σ limits is shown in Fig.1.3.



1.7.2 Unnatural Patterns of Variation

The Western Electric Company engineers have identified 15 unnatural typical patterns in control charts. In this section, we discuss the most important types of unnatural patterns:

1. Extreme Variation

If one or more samples point are significantly different from the other points and lie outside the control limits of the control chart as shown in Fig.1.4, we say that there is an extreme variation in the chart. Some assignable causes are present in the process and corrective action is necessary to bring the process under control.

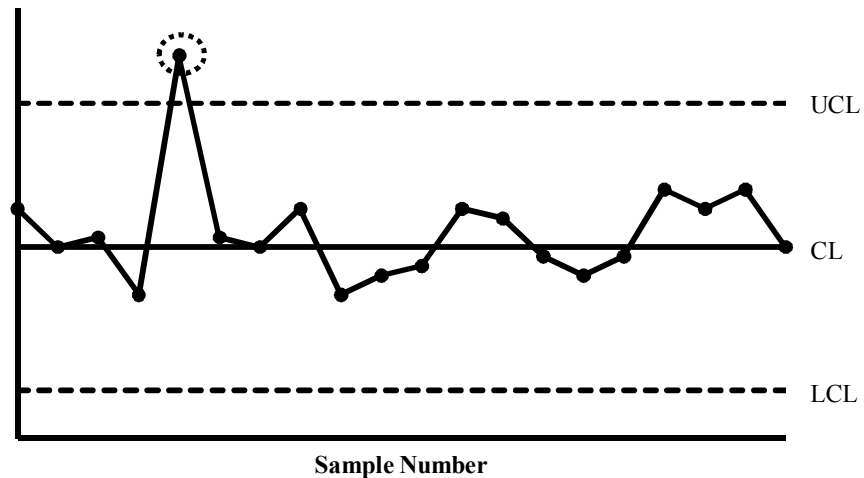


Fig. 1.4: Extreme variation.

Causes of extreme variation are:

- i) Error in measurement, recording and calculations,
- ii) Wrong setting and defective machine tools or erroneous use,
- iii) Power failures for short time,
- iv) Use of a new tool, failure of the component at the time of test, etc.

A sample point falling outside the control limits is a clear indication of the presence of assignable causes. There are other situations where in the pattern of sample points on the chart indicates the presence of assignable causes, although all points may lie within the control limits. Such situations are discussed below.

2. Trend

If consecutive points on the control chart tend to move upward or downward as shown in Fig. 1.5, it can be assumed that the process indicates a trend. If proper care or corrective action is not taken, the process may go out-of-control.

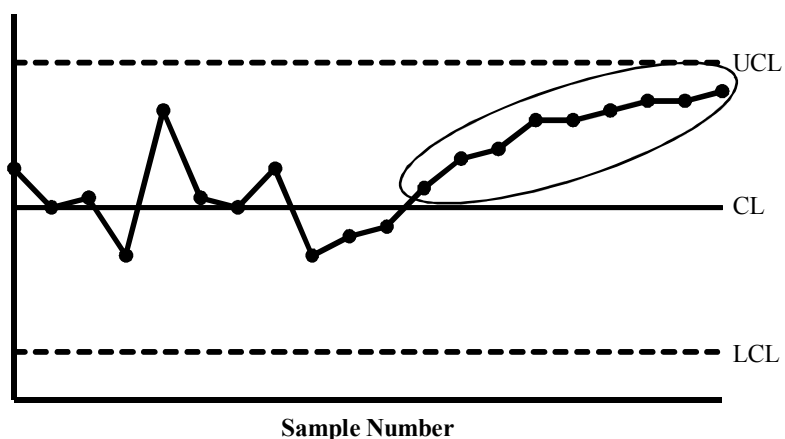


Fig. 1.5: Trend pattern.

Causes of trend pattern are:

- i) Tool or die wear,
- ii) Gradual change in temperature or humidity,
- iii) Gradual wearing of operating machine parts,
- iv) Gradual deterioration of equipment, etc.

3. Cycles

When consecutive points exhibit a cyclic pattern (as shown in Fig. 1.6), it is also an indication that assignable causes are present in the process which affect it periodically.

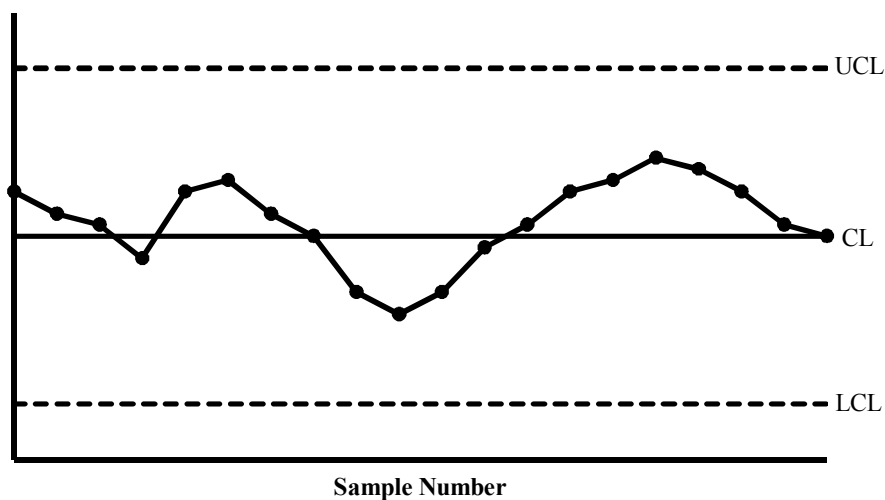


Fig. 1.6: Cyclic pattern.

Causes of cyclic pattern are:

- i) Rotation of operators,
- ii) Periodic changes in temperature and humidity,
- iii) Periodicity in the mechanical or chemical properties of the material,
- iv) Seasonal variation of incoming component, etc.

4. Shifts

When a series of consecutive points falls above or below the centre line of the chart, it can be assumed that a shift in the process has taken place. This indicates the presence of some assignable causes. Such a pattern is shown in Fig. 1.7.

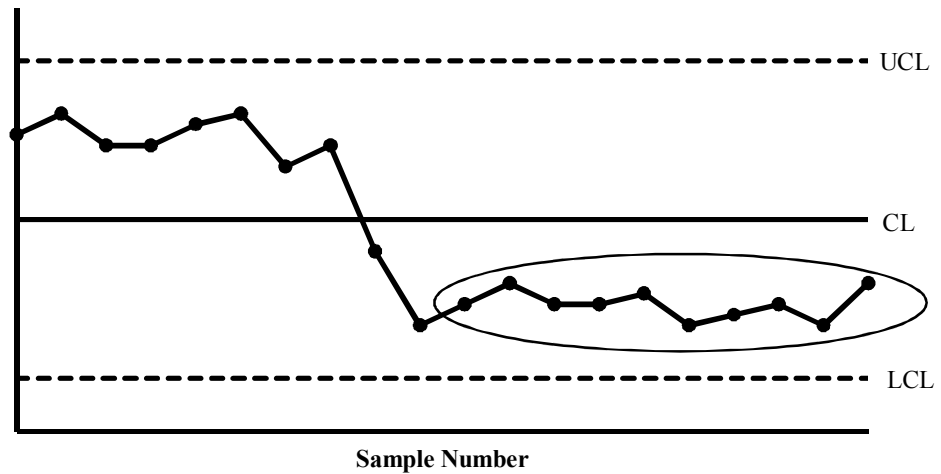


Fig. 1.7: Shift pattern.

Generally, it is assumed that when 7 consecutive points lie above or below the centre line, a shift has occurred.

Causes of shift pattern are:

- i) Change in material,
- ii) Change in machine setting,
- iii) Change in operator, inspector, inspection equipment, etc.

5. Erratic Fluctuations

When the sample points of control chart tend to fall near or slightly outside the control limits with relatively few points near the centre line as shown in Fig. 1.8, it can be assumed that erratic fluctuation has taken place. This indicates the presence of some assignable causes. The causes for erratic fluctuations are slightly difficult to identify. These may be due to different causes acting at different times in the process.

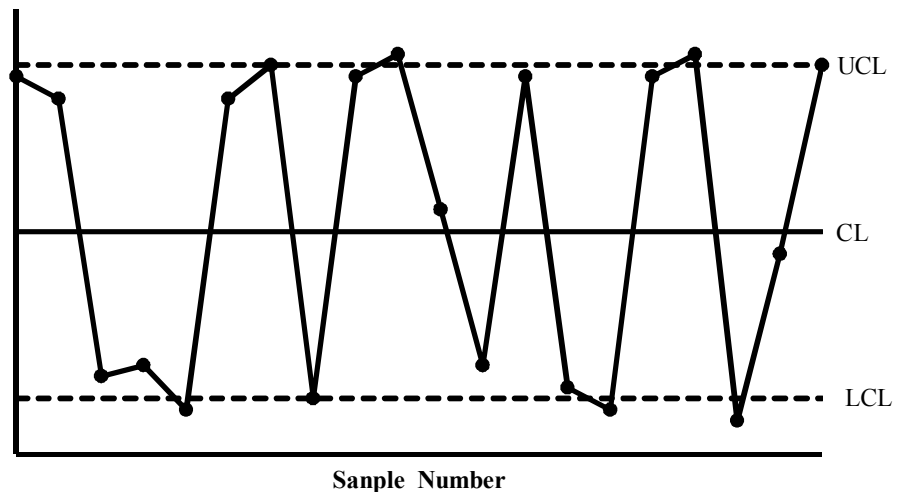


Fig. 1.8: Erratic fluctuations.

Causes of erratic fluctuations are:

- i) Different types of materials being processed,
- ii) Change in operator, machine, inspection, equipment,
- iii) Frequent adjustment of machine, etc.

You may now like to review what you have studied about control charts patterns.

E6) Choose the correct option from the following:

- i) If the points on the chart have continuous movements upward/downward, the pattern is called
 - a) freak pattern b) shift pattern
 - c) trend pattern d) cyclic pattern
 - ii) If all points on the control chart are within the control limits and the pattern of the points show trend, the process is said to be
 - a) under control b) out-of-control
 - c) both (a) and (b)
 - iii) To check whether the process is under control or out-of-control, we see
 - a) the pattern of the sample points on the chart
 - b) the position of the sample points on the control chart
 - c) both (a) and (b)
-

1.8 ADVANTAGES AND LIMITATIONS OF SQC

When a lot of items/units is manufactured, the manufacturer has two methods to check the quality of the lot: firstly, he/she could check each and every item and decide about the quality of the product, i.e., 100% inspection. Secondly, he/she could use the statistical quality control methods, i.e., inspect a small number of items and decide about the quality of the entire lot of the produced product. SQC has many advantages over 100% inspections, which are listed below:

A lot is the collection of units or items.

1. Ease of Application

An excellent feature of statistical quality control is that it is easy to apply. While developing the statistical methods for quality control, skilled and intelligent persons are required. However, even those persons who have not had extensive specialised training can apply statistical methods easily.

2. Reduction in Costs

The cost of inspection is reduced. In SQC as only a part or fraction of a lot is taken and inspected.

3. Greater Efficiency

Inspection of every item is bound to reduce the efficiency of a quality control inspectors because of dullness. Inspectors are more alert while using SQC as only a part is inspected.

4. Early Detection of Faulty Units

SQC consists of continuous checking of the quality of the product. When a sample point falls outside the control limits it gives the signal that the process is not under statistical control. If some assignable causes are present in the process, necessary corrective action can be taken. Therefore, SQC ensures an early detection of faults and results in minimum wastage of items.

5. Helpful in Specification

Using SQC, we can find out whether the produced item is under control or not, that is, whether the item meets the specifications within the tolerance limits or not. If the variation is beyond the tolerance limits, SQC gives a danger signal and necessary corrective action can be taken. So, as long as statistical control continues, specifications can be accurately predicted for the future, which cannot be guaranteed by 100% inspection.

6. Ensures Overall Coordination

SQC methods ensure coordination between managers managing specifications, production and inspection. It provides a basis to resolve the differences arising among the various interests in an organisation.

7. Determination of the Effect of Change in the Process

With the help of control charts, we can easily detect whether or not a change in the production process results in a significant change in the quality.

8. Equilibrium in Consumer's and Producer's Risk

Methods such as quality control and acceptance sampling help in maintaining equilibrium between the consumer's risk and producer's risk.

9. Wider Applications

It is not only useful in the examination of items produced in small numbers, but also when bulk production has to be done.

10. Unique Method

Statistical quality control is helpful for those items that get destroyed on being examined for a given quality characteristic, for example, the intensity of match sticks, average life of compact fluorescent lamp (CFL), strength of glass, etc., In such cases, 100% inspection will spoil the entire lot and create a huge loss.

However, SQC has some **limitations**, which are described below:

1. When a sample of the items drawn from the lot is not a true representative of the entire lot, does not have the same characteristics as the lot from which it is drawn. Then a good lot may be rejected and a bad one may be accepted. This is the **main limitation** of SQC.
2. SQC cannot be used mechanically for any production process without studying the process and without adequate knowledge about it.
3. SQC applied on a production process provides only the information that the process is under control or out-of-control. It cannot take any action for improvement.

We end this unit by giving a summary of its contents.

1.9 SUMMARY

1. **Quality** means:

- conforming to specifications,
 - fitness for use,
 - customer satisfaction,
 - delighting the customer, and
 - enchanting the customers.
2. Quality has eight **dimensions**:
 - i) Performance ii) Features iii) Reliability iv) Conformance
 - v) Durability vi) Serviceability vii) Aesthetics
 - viii) Reputation
 3. Quality control is defined as the process by which we measure the quality characteristics of the product, compare them with the specifications or standard and take suitable actions whenever there is a difference between actual quality and the specifications or standard.
 4. The technique of controlling product quality vis-a-vis the specifications using statistical tools is known as **Statistical Quality Control (SQC)**.
 5. Statistical quality control can be broadly classified into two categories:
 - Statistical Process Control (SPC) or simply Process Control, and
 - Product Control or Acceptance Sampling.
 6. **Statistical process control** is a technique used for understanding and monitoring the process by collecting the data on quality characteristic periodically from the process, analysing them and take suitable actions whenever there is a difference between actual quality and the specifications.
 7. **Product control** means to control the products in such a way that these are free from defects and conform to their specifications.
 8. The causes of variation are broadly classified into two categories:
 - Chance or random or natural or common causes, and
 - Assignable or non-random or unnatural or special causes.
 9. The control chart is a two-dimensional graphical display of a quality characteristic that has been measured or computed in terms of means or other statistics from a sample and plotted against the sample number or time at which the sample is taken from the process.
 10. A control chart consists of three horizontal lines:

Centre Line (CL) – The centre line of a control chart represents the value which can have three different interpretations depending on the available data. First, it can be the **average value of the quality characteristic** or the **average of the plotted points**. Second, it can be a **standard or reference value**, based on representative prior data or an **aimed (targeted) value** based on specifications. Third, it can be the **population parameter** if that value is known. The **centre line is usually represented by a solid line**.

Upper Control Line – The upper control line represents the **upper value of the variation in the quality characteristic**. So this line is called **upper control limit (UCL)**. Usually, the **UCL is shown by a dotted line**.

Lower Control Line – The lower control line represents the **lower value of the variation in the quality characteristic**. So this line is called **lower control limit (LCL)**. Usually, the **LCL is shown by a dotted line**.

11. If all sample points **lie on or in between the upper and lower control limits**, the control chart indicates that the process is **under statistical control**. However, if **one or more points lie outside the control limits**, the control chart alarms (indicates) that the process is **not under statistical control**. Some assignable causes are present in the process.
12. There are two types of patterns of sample points on the control chart:
 - Natural patterns of variation, and
 - Unnatural patterns of variation

1.10 SOLUTIONS/ANSWERS

- E1)**
- i) Option (d) is the correct option because we know that quality has to incorporate the following:
 - conforming to specifications,
 - fitness for use,
 - customer satisfaction,
 - delighting the customer, and
 - enchanting the customers.
 - ii) Option (c) is the correct option because we know that the **performance** is the primary operating characteristics of a product. **Features** are the additional characteristics available in product along with the primary operating characteristics. **Reliability** refers to the probability of a product's failure within a specified time period. **Aesthetics** means how a product looks, sounds, feels, etc.
 - iii) Option (a) is the correct option because we know that **durability** means a measure of product life. **Conformance** means meeting the specifications or standard. **Reputation** is related to the past performance of the company. **Reliability** refers to the probability of a product's failure within a specified time period.
- E2)**
- i) Option (c) is the correct option because we know that statistical quality control is a technique of both process control and product control.
 - ii) Option (c) is the correct option because we know that statistical quality control is a technique of both process control and product control. The control charts are used for process control and acceptance sampling plans are used for product control.
 - iii) Option (a) is the correct option because we know that process control is achieved through control charts, whereas product control is achieved through acceptance sampling plans.

- iv) Option (b) is the correct option because we know that product control is achieved through acceptance sampling plans, whereas process control is achieved through control charts.
- E3)**
- i) Option (c) is the correct option because we know that the variation in quality characteristic may be due to assignable causes and chance causes.
 - ii) Option (b) is the correct option because we know that the variation due to chance causes is not controllable whereas the variation due to assignable causes is controllable.
 - iii) Option (a) is the correct option because we know that the variation due to assignable causes is controllable whereas the variation due to chance causes is not controllable.
 - iv) Option (d) is the correct option because we know that chance causes affect the process output in minor ways.
 - v) Option (a) is the correct option because we know that the variation due to assignable causes can be removed always.
- E4)**
- i) Option (b) is the correct option because we know that the control chart is a two-dimensional graphical display of a quality characteristic.
 - ii) Option (c) is the correct option because we know that the control chart consists of the centre line, upper control line, and lower control line.
 - iii) Option (c) is the correct option because we know that if one or more points lie outside the control limits, the process is not under statistical control. Some assignable causes are present in the process.
 - iv) Option (c) is the correct option because we know that control charts in statistical quality control are used for describing the pattern of variation and checking whether the variability in the product is within the tolerance limits or not.
- E5)** Option (b) is the correct option because we know that the lower and upper three sigma control limits for a control chart are $\mu \pm 3\sigma$ where μ and σ are the mean and the standard deviation of a random variable.
- E6)**
- i) Option (c) is the correct option because we know that if some consecutive points on the control chart tend to move upward or downward, it is called a **trend** pattern. If one sample point is significantly different from the other points and lies outside the control limits of the control chart, we say that there is an **extreme variation (freak)** in the chart. If a series of consecutive points falls above or below the centre line of the chart, it can be assumed that a **shift** in the process has taken place. If points on the chart have peaks and troughs which repeat themselves, we say that there is a **cyclic** pattern. These patterns are also an indication of the assignable causes.
 - ii) Option (b) is the correct option because we know that if the sample points show trend, it is also an indication of assignable causes and the process is out-of-control.

Process Control

- iii) Option (c) is the correct option because for checking whether the process is under control or out-of-control, we see the pattern of the sample points on the chart as well as the position of the sample points on the control chart.

UNIT 2 CONTROL CHARTS FOR VARIABLES

Structure

- 2.1 Introduction
 - Objectives
- 2.2 Control Chart Technique
- 2.3 Control Charts for Variables
- 2.4 Control Chart for Mean ($\bar{X}$ -Chart)
- 2.5 Range Chart (R-Chart)
- 2.6 Standard Deviation Chart (S-Chart)
- 2.7 Process Capability Analysis
- 2.8 Summary
- 2.9 Solutions/Answers

2.1 INTRODUCTION

In Unit 1, you have learnt that the **control chart** is the most important tool for process control. With the help of control chart, we quickly detect the occurrence of assignable causes of variation and can take corrective action to eliminate them.

Several control charts may be designed for different situations. Each chart has its own field of applications and its own advantages and disadvantages. However, all control charts have some characteristics in common and are interpreted in almost the same manner. Generally, quality characteristics are of two types (variables and attributes). So control charts are broadly classified into two categories:

1. Control charts for variables (Control charts for measurable characteristics)
2. Control charts for attributes (Control charts for non-measurable characteristics)

In this unit, you study the construction and interpretation of control chart for variables. We introduce the general technique of constructing a control chart in Sec. 2.2. We also discuss control charts for variables, i.e., control chart for mean and control chart for variability in Secs. 2.3 to 2.6. Finally, we introduce the process capability analysis, which is used to check whether the process is able to meet the specifications or not in Sec. 2.7. The control charts for attributes are taken up in Units 3 and 4.

Objectives

After studying this unit, you should be able to:

- explain different types of control charts for variables;
- decide which control chart for variables to use in a given situation;
- describe the procedure for constructing a control chart;
- construct and interpret the control chart for process mean ($\bar{X}$ -chart);

- construct and interpret the control chart for process variability (R-chart and S-chart);
- define the concepts of specification limits, natural tolerance limits and process capability.

2.2 CONTROL CHART TECHNIQUE

There are two types of control charts: one for measurable characteristics known as **control charts for variables** and second for non-measurable characteristics known as **control charts for attributes**. The technique for drawing a control chart is the same for all types.

The main steps are as follows:

1. Select the Quality Characteristic

In any manufacturing plant (small or big) of a product, there exist a large number of quality characteristics in the product. A single product/item/unit usually has several quality characteristics such as weight, length, width, strength, thickness, etc. It is, therefore, impossible to construct a control chart for each characteristic because **it is drawn for controlling a single quality characteristic**. So it is necessary to make a judicious selection of quality characteristic.

For example, a cricket ball has many quality characteristics such as weight, diameter, surface, elasticity, etc. A manufacturer of cricket ball would like to find out which of these quality characteristics, should be given priority for quality control. While selecting a quality characteristic, generally, we give higher priority to the one that causes more non-conforming (defective) items and increases cost. For this, we use **Pareto diagram**. It is a bar diagram of the quality characteristics of a product and was used for the first time by Joseph Moses Juran for quality control. For drawing a Pareto diagram, the frequency of non-conformities (defects) of each quality characteristic such as weight, diameter, surface, etc. is noted. In Pareto diagram, quality characteristics are taken along the X-axis and the percentage of defects occurring in the quality is taken along the Y-axis. For example, consider the data for quality characteristics for 200 cricket balls given in Table 1.

Table 1: Quality Characteristics Data for a Sample of 200 Cricket Balls

S. No.	Quality Characteristic	Frequency of Defects	Percentage of Defects
1	Weight	90	45
2	Diameter	70	35
3	Rough Surface	20	10
4	Elasticity	10	5
5	Other	10	5
Total		200	100

Joseph Moses Juran (December 24, 1904-February 28, 2008) was a Romanian-born American engineer and management consultant. He is principally remembered for quality and quality management.

The Pareto diagram drawn from the above data is as follows:

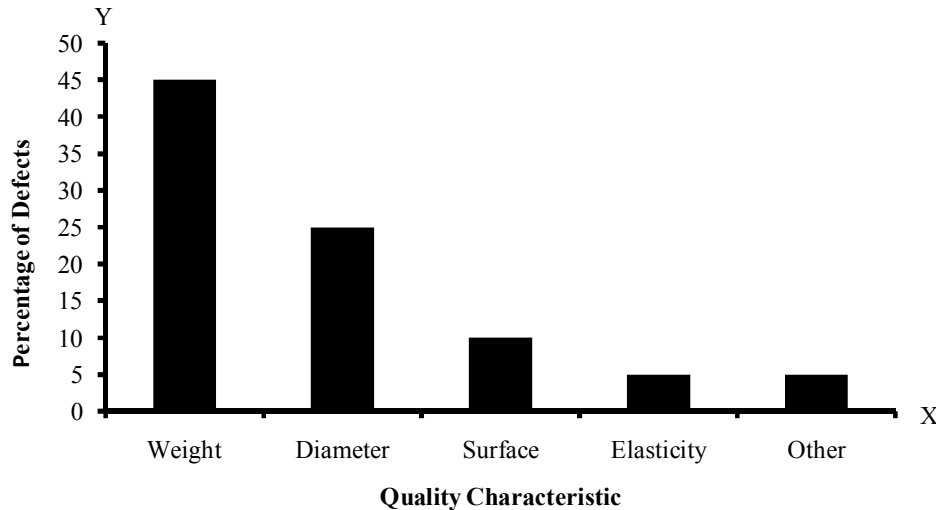


Fig. 2.1: Pareto diagram for data of Table 1.

From the Pareto diagram shown in Fig. 2.1, we observe that the weight and diameter are serious problems accounting for most of the non-conforming (defective) cricket balls produced. Rough surface occurs **less frequently**, elasticity and other problems are relatively **rare**. So for this problem, we should select quality characteristics, weight and diameter of the balls for quality control.

2. Select the Type of Control Chart

The choice of control chart depends upon the measurement, quality characteristic and the cost involved. For selection of control chart, we first select the category of the chart to be used as variables or attributes. If the characteristic to be controlled is **measurable** such as weight, length, diameter, etc., we use the **control chart for variables**. In the example of cricket balls, we have selected weight and diameter for control and both are measurable. So we will use control charts for variables for both quality characteristics.

If a characteristic to be controlled is **non-measurable**, e.g., colour, surface roughness, etc., we use the **control chart for attributes**. If many characteristics need control, it is too costly to use control charts for variables because these charts are used for controlling only one characteristic at a time. Therefore, each characteristic has its variables chart. In such situations, a single chart for attributes can be used in place of all the charts for variables.

After selecting the category of the chart to be used (variables or attributes), we select the actual chart to be used. For this, we need to know which chart can be used for a given situation and also the purpose of the control chart. We shall explain this aspect while describes each control chart.

3. Selection of Rational Subgroups

We know that control charts for quality characteristics are drawn with the help of sample data. Now, the question may arise “how are the data collected for analysis and controlling processes”? For this Walter A. Shewhart developed and introduced the concept of **rational subgroups** for control charts. He suggested that **the differences between rational**

Control Charts for Variables

For drawing the bar chart, please recall Unit 14 of MST-001 entitled Foundation in Mathematics and Statistics.

Process Control

Recall the procedure of drawing a sample from a population (process), which has been explained in Unit 1 of MST-005 (Statistical Techniques).

For statistic and standard error please recall Unit 1 of MST-004.

The procedure of drawing a sample is called sampling.

subgroups are an indication of process changes (assignable causes) while differences within rational subgroup are an indication of inherent variability, i.e., chance causes. So the subgroups are selected in such a way that we are able to differentiate between assignable causes and chance causes. If assignable causes are present in the process, they will show up as differences between the subgroups rather than differences within a subgroup.

A rational subgroup is a small set of items/units that are produced under similar conditions within a relatively short time, i.e., the variation within the subgroup is only due to chance causes.

Generally, in industries, the word '**sample**' is used for a subgroup. So we use words, **sample and subgroup**, in this course.

4. Size of Subgroup (Sample Size)

To provide maximum homogeneity within a subgroup/sample, the size of the sample should be as small as possible. However, a sample size of four or five units/items is quite common in the industry. It is also valid on statistical grounds. As we know from sampling distribution theory, the distribution of the sample mean $\bar{X}$ is nearly normal for samples of four or more, even though the samples are taken from a non-normal population. This fact is helpful in interpretation of control chart limits. When a sample of size 5 is used there is ease in the computation of the average.

When we have to make the control chart more sensitive, samples of size 10 or 20 are used. This is because the standard error of a statistic is inversely proportion to sample size, i.e., as the sample size increases, the standard error decreases. Therefore, 3σ limits (upper and lower control limits) will lie closer to the centre line. However, if the items produced are destroyed under inspection or are expensive, a small sample of size 2 or 3 is used.

5. Frequency of Subgroups (Number of Samples)

There is no hard and fast rule for the number of subgroups/samples, but it should be such that it is enough to detect trend, pattern of level shifts and process changes. To decide the number of samples, we need to keep in mind that the cost of taking samples must be balanced with the data obtained.

There are two ways of taking samples:

- i) Taking larger samples at less frequent intervals, or
- ii) Taking smaller samples at more frequent intervals.

It is best to take samples more frequently in the beginning of the process. As the processes are brought into control, frequency of sampling can be reduced. For example, in a firm, the analyst may take a sample every hour at the beginning of the process. If the process remains under statistical control for two or three days, the frequency of sampling may be reduced to every 2 or 3 hour. If the process continuously remains under statistical control for a few weeks, the frequency may be further reduced. However, if difficulties are encountered in keeping the process under statistical control, samples can be taken more frequently for inspection.

6. Design the Forms for Data Collection

After taking the decision about the quality characteristic, subgroup/sample size and number of subgroups, we design the **data recording form**. The form for data recording should be designed in accordance with the control chart to be used. A typical data recording form for the $\bar{X}$ -chart is shown below:

Name of the Chart:				Plot No.:		Pin:			
Part Name:			Operation:			SPEC:			
Operator:		Machine:		Gauge:		Unit of Measurement:			
Subgroup Number	Date	Time	Measurements				Mean	SD	Comment
			X ₁	X ₂	X ₃	X ₄			
1	17/5	10:00	52	52	50	51	51.25	0.83	
2		12:00	50	53	52	53	52.00	1.22	
3		02:00	64	60	63	61	62.00	1.58	New temporary operator
4		04:00	51	52	50	53	51.50	1.12	
5	18/5	10:00	50	52	51	50	50.75	0.83	
6		12:00	50	50	52	51	50.75	0.83	
7		02:00	52	51	53	50	51.50	1.12	
8		04:00	52	53	52	55	53.00	1.22	
9	19/5	10:00	42	41	40	43	41.50	1.12	Bad material
10		12:00	52	52	54	52	52.50	0.87	
11		02:00	52	53	54	49	52.00	1.87	
12		04:00	52	51	54	51	52.00	1.22	
13	20/5	10:00	51	51	52	52	51.50	0.50	
14		12:00	50	50	51	52	50.75	0.83	
15		02:00	50	51	53	51	51.25	1.09	
16		04:00	52	50	49	53	51.00	1.58	
17	21/5	10:00	52	51	54	51	52.00	1.22	
18		12:00	51	51	50	52	51.00	0.71	
19		02:00	52	50	53	45	50.00	3.08	
20		04:00	50	49	50	48	49.25	0.83	
Total							1027.50	23.68	

Generally, a data recording form is divided into two parts. The top part of the form contains information about part name, operation machine, gauge used, unit of measurement and specifications. The remaining segment of the form contains information about subgroup/sample number, the date and time when the sample was selected and raw values of the observations. A column for comment is used to incorporate remark about the process.

7. Determination of Trial Centre Line and Control Limits

You have learnt in Unit 1 that the quality characteristic can be described by a probability distribution. In most cases, it follows a normal distribution or can be approximated by a normal distribution. You have learnt in Sec. 14.2

of Unit 14 of MST-003 (Probability Theory) that the probability of a normally distributed random variable (X) that lies between $\mu - 3\sigma$ and $\mu + 3\sigma$ is 0.9973 where μ and σ are the mean and standard deviation of random variable X . That is,

$$P[\mu - 3\sigma \leq X \leq \mu + 3\sigma] = 0.9973$$

The probability that the random variable X lies outside the limits $\mu \pm 3\sigma$ is $1 - 0.9973 = 0.0027$, that is, it is very small. It means that if we consider 100 samples, then most probably 0.27 items fall outside $\mu \pm 3\sigma$ limits. Therefore, if an observation falls outside 3σ limits, it is logical to suspect that something might have gone wrong. For this reason, the control limits are set up using 3σ limits.

Suppose M is a sample statistic (e.g., mean, range, proportion of defectives, etc.) that measures some quality characteristic of interest. Further suppose that μ_M and σ_M are the mean and standard error (standard deviation) of the sample statistic M , respectively. Then the centre line and control limits for controlling the quality characteristic are given by:

$$\text{Centre line (CL)} = E(M) = \mu_M$$

$$\text{Upper control limit (UCL)} = \mu_M + 3\sigma_M$$

$$\text{Lower control limit (LCL)} = \mu_M - 3\sigma_M$$

8. Constructing a Control Chart

The centre line and control limits are different for different charts. We shall explain the method of determining these limits while describing each type of chart.

After obtaining centre line and control limits, we construct the control chart. In a control chart, the statistic (e.g., mean, range, number of defects, etc.) is taken along the Y-axis and the sample number or time is taken along the X-axis. We represent the centre line by a solid line and the control limits by dotted lines. We plot the value of statistic for each sample against the sample number and consecutive sample points are joined by line segments.

9. Drawing Preliminary Conclusions from the Control Chart

After constructing the control chart, we draw preliminary conclusions from it. We check whether the plotted sample points lie on or in between the upper and lower control limits or some of them lie outside these control limits. If one or more points lie outside the control limits, we indicate each such point by drawing a **dotted circle** around it. The control limits on this chart are called **trial control limits**.

Recall Sec. 1.5 of Unit 1. If all sample points lie on or in between the upper and lower control limits and there is no unnatural patterns of variation, the chart indicates that the process is under statistical control.

Even if a process is under statistical control, small variations may exist due to machine performances, operator performance, material characteristics, etc. Such small variations are considered to be a part of a stable process. It means that only chance causes are present in the process and no assignable causes are present.

When a control chart indicates that the process is under statistical control, periodic samples are taken from the process to determine whether it remains under statistical control continuously. When we assure that the process has been under statistical control after the analysis of the preliminary data, the trial control limits are used to control future production. Then $\bar{X}$, σ , R , etc. can be considered as representative of the process and their values are taken as standard values. However, when more data accumulates, the limits may be revised from time to time or whenever necessary. To ensure that the process remains under statistical control, periodic samples are taken once a week, once a month or once every 25, 50 or 100 items.

Most processes are not under statistical control at the time of the initial analysis, i.e., some points lie outside the control limits or there are long runs or unusual patterns of variation. Then the control chart indicates that the process is not under statistical control. Some assignable causes are present in the process. We investigate the reasons of assignable causes and take corrective action to eliminate them from the process. Suppose these are due to raw materials supplied by new vendor, we eliminate the cause by choosing a new vendor or maintaining the quality of raw material at vendor's point. Rectification is made by deleting the out-of-control points and calculating the revised centre line and control limits for the chart. These revised limits are known as the **revised control limits**. This procedure is continued till the process is being under statistical control.

So far you have studied the general technique for constructing a control chart. You have learnt how a control chart is constructed and how to determine whether a process is under statistical control or out of control on the basis of the control chart. We now discuss the control charts for variables.

2.3 CONTROL CHARTS FOR VARIABLES

In any production process, it is not possible to ensure that all items produced are alike in respect of certain characteristics. Some variations are always present in the items. These variations may be due to raw material quality, unskilled work force, machines, faulty equipment, etc. The existence of variation in products affects the quality of the product. These variations may be due to chance causes or assignable causes. So the aim of Statistical Process Control is to trace the sources of such variation and as far as possible try to eliminate them. If the quality characteristic in which we have to control the variation is measurable, we use control chart for variables.

In quality control, the term **variable** means the quality characteristic which can be measured, e.g., diameter of ball bearings, length of refills, weight of cricket balls, etc. The control charts based on measurements of quality characteristics are called **control charts for variables**.

When we deal with a measurable quality characteristic, it is necessary to control the central tendency (average) as well as the dispersion (variability) of the quality characteristic or the process. For central tendency, we usually apply mean and for variability we calculate range or standard deviation. Therefore, there are different control charts for variables for controlling the mean and variability of the process. The most frequently used control charts for variables are:

Process Control

To monitor and control variability process, we use R-chart or S-chart. But R-chart is most widely used in quality control as compared to S-chart because in quality control, we generally use samples of sizes less than 10. We know that for $n \leq 10$, the range and standard deviation reflect almost the same information about the variability. However, the range is easier to calculate and interpret in comparison to the standard deviation.

1. Control chart for mean
2. Control chart for variability

We generally use control chart for mean ($\bar{X}$ -chart, read as X-bar chart) to control the mean quality level or process mean. We discuss this chart with examples in Secs. 2.4.

Experience reveals that process variability also affects product quality. When variation in the production process is high, that is, the produced items have a wider range of values then the quality of the product is assumed to be poor. So to improve the quality of the product we must reduce the variability.

The following charts are used to control process variability:

1. Range chart (R-chart), and
2. Standard deviation chart (S-chart).

We discuss each of these charts with examples in Secs. 2.5 and 2.6.

You may now like to check your understanding of different types of control charts by answering the following exercise.

E1) Choose the correct option from the following:

- i) The control chart for variables is/are the
 - a) $\bar{X}$ -chart
 - b) R-chart
 - c) S-chart
 - d) all of these
 - ii) The control chart for process mean is the
 - a) $\bar{X}$ -chart
 - b) R-chart
 - c) S-chart
 - d) p-chart
 - iii) The control chart for process variability is the
 - a) $\bar{X}$ -chart
 - b) R-chart
 - c) c-chart
 - d) p-chart
-

We now discuss the control chart for the mean.

2.4 CONTROL CHART FOR MEAN ($\bar{X}$ -CHART)

If we want to control the process mean, in any production process, we use the **control chart for mean**, i.e., the $\bar{X}$ -chart. With the help of the $\bar{X}$ -chart, we monitor the variation in the mean of the samples that have been drawn from time to time from the process. We plot the sample means instead of individual measurements on the control chart because sample means are calculated from n individuals and give additional information. The $\bar{X}$ -chart is more sensitive than the individual chart for detecting the changes in the process mean.

We now explain various steps for constructing the $\bar{X}$ -chart.

Steps Involved in the Construction of the $\bar{X}$ -chart

The main steps for the construction of the $\bar{X}$ -chart are as follows:

Step 1: We select the measurable quality characteristic for which the $\bar{X}$ -chart has to be constructed. We have discussed how to select the quality characteristic in Sec.2.2.

Step 2: Then we decide the subgroup/sample size as explained in Sec. 2.2. Generally, four or five items are selected in a sample and twenty to twenty five samples are collected for the $\bar{X}$ -chart.

Step 3: After deciding the size of the sample and the number of samples, we select sample units/items randomly from the process so that each unit/item has an equal chance of being selected.

Step 4: We measure the quality characteristic decided in **Step 1** for each selected item/unit in each sample. Errors in measurement may be due to many reasons. These may be due to:

- i) use of faulty machine,
- ii) different methods of taking measurements,
- iii) inept use of instruments by unskilled or inexperienced work force, etc.

Therefore, care should be taken that measurements are made by experts, using standard instruments.

Step 5: Calculate the sample mean for each sample.

Suppose we measure the quality characteristic, say, X for each unit/item of the sample. Also suppose $X_1, X_2, \dots, X_n$ are the measurements for the units of a sample of size n . Then sample mean is given by

$$\bar{X} = \frac{1}{n}(X_1 + X_2 + \dots + X_n) = \frac{1}{n} \sum_{i=1}^n X_i \quad \dots (1)$$

where X_i represents the measurement of the quality characteristic of the i^{th} unit.

Suppose there are k samples each of size n . Then we calculate the sample mean for each sample by using equation (1). Suppose the means for $1^{\text{st}}, 2^{\text{nd}}, \dots, k^{\text{th}}$ samples are represented by $\bar{X}_1, \bar{X}_2, \dots, \bar{X}_k$, respectively.

Step 6: Set the control limits

To find out whether the process is under statistical control or out-of-control, we set up the control limits. The 3σ control limits for the $\bar{X}$ -chart are given by

$$\text{Centre line} = E(\bar{X})$$

$$\text{Lower control limit (UCL)} = E(\bar{X}) - 3SE(\bar{X})$$

$$\text{Upper control limit (UCL)} = E(\bar{X}) + 3SE(\bar{X})$$

where $E(\bar{X})$ and $SE(\bar{X})$ are the mean and the standard error of the sampling distribution of $\bar{X}$, respectively.

For getting the control limits of the $\bar{X}$ -chart we need the sampling distribution of the sample mean. From Sec. 2.2 of Unit 2 of MST-004, we know that if $X_1, X_2, \dots, X_n$ is a sample of size n drawn from a normal population (distribution) with mean μ and variance σ^2 , the

Control Charts for Variables

Recall the procedure of drawing a sample from a population (process) which has been explained in Unit 1 of MST-005 (Statistical Techniques).

sample mean $\bar{X}$ is normally distributed with mean μ and variance σ^2/n . It means that, if

$$X_i \sim N(\mu, \sigma^2) \text{ then } \bar{X} \sim N(\mu, \sigma^2/n),$$

$$E(\bar{X}) = \mu \text{ and } \text{Var}(\bar{X}) = \frac{\sigma^2}{n} \quad \dots (2)$$

We also know that

$$\text{SE}(X) = \sqrt{\text{Var}(X)}$$

$$\therefore \text{SE}(\bar{X}) = \sqrt{\text{Var}(\bar{X})} = \sqrt{\frac{\sigma^2}{n}} = \frac{\sigma}{\sqrt{n}} \quad \dots (3)$$

Therefore, the centre line and control limits for the $\bar{X}$ -chart are given as follows:

$$\text{Centre line (CL)} = E(\bar{X}) = \mu \quad \dots (4a)$$

$$\text{Lower control limit (LCL)} = E(\bar{X}) - 3\text{SE}(\bar{X}) = \mu - 3\sigma/\sqrt{n}$$

$$\text{or} \quad \text{LCL} = \mu - A\sigma \quad \dots (4b)$$

$$\text{Upper control limit (UCL)} = E(\bar{X}) + 3\text{SE}(\bar{X}) = \mu + 3\sigma/\sqrt{n}$$

$$\text{or} \quad \text{UCL} = \mu + A\sigma \quad \dots (4c)$$

where $A = 3/\sqrt{n}$ is a constant and depends on the size of the sample. It has been tabulated for various sample sizes in Table I given at the end of this block.

he values of μ and σ are not known. Therefore, these are estimated from the samples, which are taken when the process is thought to be under statistical control. We use their **best possible estimates**. In Unit 5 of MST-004, we have seen that the best estimators of the population mean (μ) and population standard deviation (σ) of normal population are the sample mean and sample standard deviation, respectively. So in this case, the best estimator of the process mean (μ) is the grand mean ($\bar{\bar{X}}$), that is, the mean of all sample means given by

$$\bar{\bar{X}} = \frac{1}{k}(\bar{X}_1 + \bar{X}_2, \dots, \bar{X}_k) = \frac{1}{k} \sum_{i=1}^k \bar{X}_i \quad \dots (5)$$

To calculate the grand mean, we first calculate the mean of each sample from equation (1). Then we calculate the grand mean from equation (5).

Now, we need the best estimate of the unknown standard deviation σ . It can either be estimated by sample range (R) or sample standard deviation (S). So we have two cases:

Case I: When standard deviation is estimated by the sample range R.

In this case, the standard deviation σ is estimated as follows:

$$\hat{\sigma} = \bar{R}/d_2 \quad \dots (6)$$

where $\bar{R}$ represents the mean of all sample ranges given by

$$\bar{R} = \frac{1}{k}(R_1 + R_2 \dots, R_k) = \frac{1}{k} \sum_{i=1}^k R_i \quad \dots (7)$$

Here R_i is the sample range and is given by

$$R_i = X_{\max} - X_{\min} \quad \dots (8)$$

where $X_{\max}$ and $X_{\min}$ represent the maximum (or largest) and minimum (or smallest) measurements of the quality characteristic of a sample, respectively. Also d_2 is a constant and depends on the size of the sample. It has been tabulated for various sample sizes in Table I given at the end of this block.

The control limits for the $\bar{X}$ -chart when μ is estimated by $(\bar{\bar{X}})$ and σ is estimated by $\bar{R}/d_2$ are given by:

$$\text{Centre line (CL)} = \hat{\mu} = \bar{\bar{X}} \quad \dots (9a)$$

$$\text{Lower control limit (LCL)} = \hat{\mu} - \frac{3}{\sqrt{n}} \hat{\sigma} = \bar{\bar{X}} - \frac{3}{\sqrt{n}} \frac{\bar{R}}{d_2}$$

$$\text{or} \quad \text{LCL} = \bar{\bar{X}} - A_2 \bar{R} \quad \dots (9b)$$

$$\text{Upper control limit (UCL)} = \hat{\mu} + \frac{3}{\sqrt{n}} \hat{\sigma} = \bar{\bar{X}} + \frac{3}{\sqrt{n}} \frac{\bar{R}}{d_2} = \bar{\bar{X}} + A_2 \bar{R}$$

$$\text{or} \quad \text{UCL} = \bar{\bar{X}} + A_2 \bar{R} \quad \dots (9c)$$

where $A_2 = \frac{3}{d_2 \sqrt{n}}$ is a constant and depends on the size of the sample. It has been also tabulated for various sample sizes in Table I given at the end of this block.

Case II: When standard deviation is estimated by the sample standard deviation.

In this case, the standard deviation σ is estimated as follows:

$$\hat{\sigma} = \bar{S} / c_4 \quad \dots (10)$$

where $\bar{S}$ represents the mean of all sample standard deviations given by

$$\bar{S} = \frac{1}{k}(S_1 + S_2 \dots, S_k) = \frac{1}{k} \sum_{i=1}^k S_i \quad \dots (11)$$

Here S_i is the sample standard deviation given by

$$S_i = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_i)^2} \quad \dots (12)$$

Therefore, the control limits for the $\bar{X}$ -chart when μ is estimated by $(\bar{\bar{X}})$ and σ is estimated by $\bar{S}$ are given by

$$\text{Centre line (CL)} = \hat{\mu} = \bar{\bar{X}} \quad \dots (13a)$$

$$\text{Lower control limit (LCL)} = \hat{\mu} - \frac{3}{\sqrt{n}} \hat{\sigma} = \bar{\bar{X}} - \frac{3}{\sqrt{n}} \frac{\bar{S}}{c_4}$$

$$\text{or} \quad \text{LCL} = \bar{\bar{X}} - A_3 \bar{S} \quad \dots (13b)$$

$$\text{Upper control limit (UCL)} = \hat{\mu} + \frac{3}{\sqrt{n}} \hat{\sigma} = \bar{\bar{X}} + \frac{3}{\sqrt{n}} \frac{\bar{S}}{c_4}$$

$$\text{or} \quad \text{UCL} = \bar{\bar{X}} + A_3 \bar{S} \quad \dots (13c)$$

where $A_3 = \frac{3}{c_4 \sqrt{n}}$ is a constant and depends on the size of the sample. It has been also tabulated for various sample sizes in Table I given at the end of this block.

Note: Generally, for the $\bar{X}$ -chart, we select samples of four or five units/items. So for this chart, we use the sample range to estimate the standard deviation of the process instead of the sample standard deviation because the range is easier to calculate as compared to the standard deviation. Moreover, it reflects almost the same information about variability as the standard deviation.

Step 7: Construct the $\bar{X}$ -chart

After setting the centre line and control limits, we construct the $\bar{X}$ -chart as explained in Step 8 of Sec.2.2. It means that we take the sample number on the X-axis (horizontal scale) and the sample mean ($\bar{X}$) on the Y-axis (vertical scale). We plot the value of sample mean for each sample against the sample number and then join the consecutive sample points by line segments.

Step 8: Interpret the result

If all sample points on the plot lie on or in between upper and lower control limits, **the process is under statistical control**. In this situation, only chance causes are present in the process. If one or more points lie outside the control limits the control chart alarms (indicates) that **the process is not under statistical control**. Some assignable causes are present in the process.

To bring the process under statistical control, we investigate the assignable causes and take corrective action to eliminate them. Once the assignable causes are eliminated from the process, we delete the out-of-control points (samples) and calculate the revised centre line and control limits for the $\bar{X}$ -chart by using the remaining samples. These limits are known as **revised control limits**. To calculate the revised limits for the $\bar{X}$ -chart, we first calculate new $\bar{\bar{X}}$ and new $\bar{R}$ as follows:

$$\bar{\bar{X}}_{\text{new}} = \frac{\sum_{i=1}^k \bar{X}_i - \sum_{j=1}^d \bar{X}_j}{k - d} \quad \dots (14)$$

and

$$\bar{R}_{\text{new}} = \frac{\sum_{i=1}^k R_i - \sum_{j=1}^d R_j}{k - d} \quad \dots (15)$$

where d – the number of discarded samples,

$\sum_{j=1}^d \bar{X}_j$ – the sum of means of discarded samples, and

$\sum_{j=1}^d R_j$ – the sum of ranges of discarded samples.

After finding the $\bar{\bar{X}}_{\text{new}}$ and $\bar{R}_{\text{new}}$, we reconstruct the centre line and control limits, replacing $\bar{\bar{X}}$ by $\bar{\bar{X}}_{\text{new}}$ and $\bar{R}$ by $\bar{R}_{\text{new}}$ in equations (9a to 9c) as follows:

$$\text{Centre line (CL)} = \bar{\bar{X}}_{\text{new}} \quad \dots (16a)$$

$$\text{Lower control limit (UCL)} = \bar{\bar{X}}_{\text{new}} - A_2 \bar{R}_{\text{new}} \quad \dots (16b)$$

$$\text{Upper control limit (UCL)} = \bar{\bar{X}}_{\text{new}} + A_2 \bar{R}_{\text{new}} \quad \dots (16c)$$

If we use the sample standard deviation to estimate the process standard deviation, then the revised control line and control limits are given as follows:

$$\text{Centre line (CL)} = \bar{\bar{X}}_{\text{new}} \quad \dots (17a)$$

$$\text{Lower control limit (UCL)} = \bar{\bar{X}}_{\text{new}} - A_3 \bar{S}_{\text{new}} \quad \dots (17b)$$

$$\text{Upper control limit (UCL)} = \bar{\bar{X}}_{\text{new}} + A_3 \bar{S}_{\text{new}} \quad \dots (17c)$$

where

$$\bar{S}_{\text{new}} = \frac{\sum_{i=1}^k S_i - \sum_{j=1}^d S_j}{k - d} \quad \text{and} \quad \dots (18)$$

$\sum_{j=1}^d S_j$ – the sum of standard deviations of discarded samples.

Note: Generally, we control the process mean and the process variability at the same time. So we use both $\bar{X}$ and R-charts or $\bar{X}$ and S-charts together.

So far, we have discussed various steps involved in the construction of the $\bar{X}$ -chart. Let us now take up some examples to illustrate this method.

Process Control

Example 1: A statistical quality controller uses the $\bar{X}$ -chart for monitoring a quality characteristic of a product. If the process mean (μ) and process standard deviation (σ) are 100 and 5, respectively, find the centre line and control limits for the $\bar{X}$ -chart. It is given that $A = 1.5$ for $n = 4$.

Solution: Here we are given that

$$\mu = 100, \sigma = 5 \text{ and } A = 1.5$$

Therefore, we can calculate the centre line and control limits for the $\bar{X}$ -chart using equations (4a to 4c) as follows:

$$\text{Centre line (CL)} = \mu = 100$$

$$\text{Upper control limit (UCL)} = \mu - A\sigma = 100 - 1.5 \times 5 = 100 - 7.5 = 92.5$$

$$\text{Lower control limit (LCL)} = \mu + A\sigma = 100 + 1.5 \times 5 = 100 + 7.5 = 107.5$$

Example 2: An automatic machine (shown in Fig. 2.2) is used to fill and seal 20 ml tube of medicine. The $\bar{X}$ -control chart has been used to monitor this process. The process is sampled in samples of four and the values of $\bar{X}$ and R are computed for each sample. After 25 samples, $\sum \bar{X} = 525$ and $\sum R = 90$.

Estimate the process mean and standard deviation. Compute the control limits for the $\bar{X}$ -chart. It is given that $d_2 = 2.088$ and $A_2 = 0.729$ for $n = 4$.

Solution: It is given that

$$k = 25, n = 4, \sum \bar{X} = 525, \sum R = 90, d_2 = 2.088 \text{ and } A_2 = 0.729.$$

The process mean is estimated by the grand sample mean, which is calculated as follows:

$$\hat{\mu} = \bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{25} \times 525 = 21$$

The process standard deviation is estimated by sample range, which is calculated as follows:

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{1}{d_2} \frac{1}{k} \sum R = \frac{1}{2.088} \times \frac{1}{25} \times 90 = 1.724$$

Since the process average (μ) and process variability (σ) are unknown in this case, we use equations (9a to 9c) to calculate the centre line and the control limits for the $\bar{X}$ -chart. These are given by

$$\text{CL} = \bar{\bar{X}} = 21$$

$$\text{UCL} = \bar{\bar{X}} + A_2 \bar{R} = 21 + 0.729 \times 3.6 = 23.6244$$

$$\text{LCL} = \bar{\bar{X}} - A_2 \bar{R} = 21 - 0.729 \times 3.6 = 18.3756$$

Example 3: A new process of producing ball bearings is started. For monitoring the outside diameter of the ball bearings, the quality controller takes the sample of five ball bearings at 10.00 AM, 12.00 PM, 2.00 PM, 4.00 PM and 6.00 PM and measures the outside diameter (in mm) of each selected ball bearing (Fig. 2.3). The results of the test over a 4-day production period are as follows:



Fig. 2.2

The CL, UCL and LCL for the $\bar{X}$ -chart when μ and σ are unknown are

$$\text{CL} = \bar{\bar{X}}$$

$$\text{UCL} = \bar{\bar{X}} + A_2 \bar{R}$$

$$\text{LCL} = \bar{\bar{X}} - A_2 \bar{R}$$



Fig. 2.3

Control Charts for Variables

Day	Sample Number	Time	Observations					$\bar{X}$	R
			X ₁	X ₂	X ₃	X ₄	X ₅		
Monday	1	10.00AM	52	52	50	51	51	51.2	2
	2	12.00PM	50	53	52	53	51	51.8	3
	3	2.00 PM	54	51	50	52	53	52.0	4
	4	4.00PM	56	55	53	55	53	54.4	3
	5	6.00 PM	51	52	50	53	53	51.8	3
Tuesday	6	10.00AM	50	52	51	50	51	50.8	2
	7	12.00PM	50	50	52	51	53	51.2	3
	8	2.00 PM	52	51	53	50	50	51.2	3
	9	4.00PM	52	53	52	55	53	53.0	3
	10	6.00 PM	51	51	50	51	52	51.0	2
Wednesday	11	10.00AM	52	52	54	52	52	52.4	2
	12	12.00PM	49	48	50	50	51	49.6	3
	13	2.00 PM	52	53	54	49	52	52.0	5
	14	4.00PM	52	51	54	51	54	52.4	3
	15	6.00 PM	51	51	52	52	51	51.4	1
Thursday	16	10.00AM	50	50	51	52	51	50.8	2
	17	12.00PM	50	51	53	51	53	51.6	3
	18	2.00 PM	52	50	49	53	50	50.8	4
	19	4.00PM	52	51	54	51	51	51.8	3
	20	6.00 PM	51	51	50	52	52	51.2	2
Total								1032.4	56

Calculate and draw the centre line and control limits of the $\bar{X}$ -chart. Draw the conclusion about the process, assuming assignable causes for any out-of-control point. If the process is out-of-control, calculate the revised centre line and control limits.

Solution: Since the process average (μ) and process variability (σ) are unknown, we use equations (9a to 9c) for determining the centre line and control limits for the $\bar{X}$ -chart.

From Table I given at the end of this block, we have $A_2 = 0.577$ for $n = 5$.

To calculate the centre line and control limits, we first calculate the values of $\bar{\bar{X}}$ and $\bar{R}$ using equations (5) and (7) as follows:

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{20} \times 1032.4 = 51.62$$

$$\bar{R} = \frac{1}{k} \sum R = \frac{1}{20} \times 56 = 2.80$$

Putting the values of $\bar{\bar{X}}$, $\bar{R}$ and A_2 in equations (9a to 9c), we have

$$CL = \bar{\bar{X}} = 51.62$$

Process Control

The CL, UCL and LCL for the $\bar{X}$ -chart when μ and σ are unknown are

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 51.62 + 0.577 \times 2.80 = 53.236$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 51.62 - 0.577 \times 2.80 = 50.004$$

We now construct the $\bar{X}$ -chart by taking the sample number on the X-axis and the average diameter of the ball bearing ($\bar{X}$) on the Y-axis as shown in Fig. 2.4.

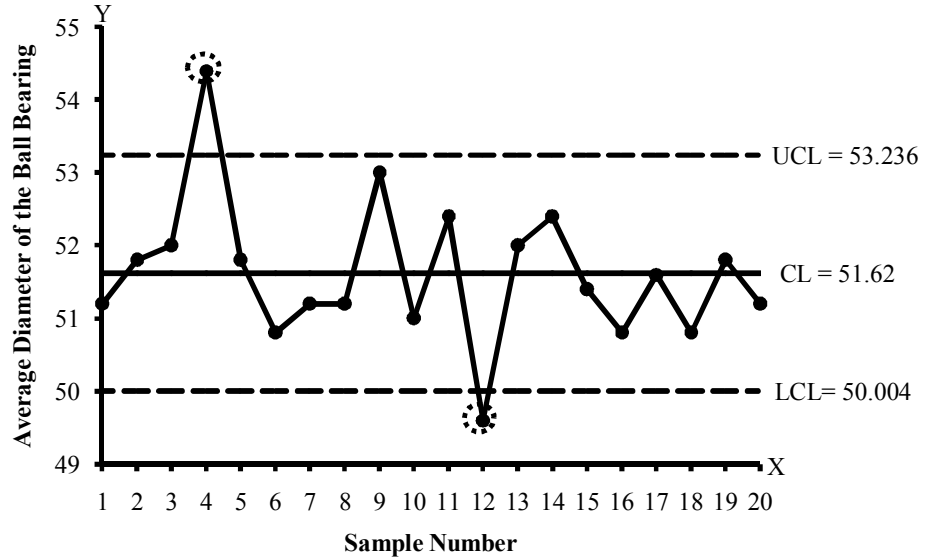


Fig. 2.4: The $\bar{X}$ -chart for the average diameter of the ball bearings.

Interpretation of the result

From Fig. 2.4, we observe that the points corresponding to samples 4 and 12 lie outside the control limits. Therefore, the process is out-of-control and some assignable causes are present in the process. To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them.

After eliminating the assignable causes from the process, we delete the out-of-control points (4 and 12 samples) and calculate the revised centre line and control limits for the $\bar{X}$ -chart using the remaining samples. For revised limits of the $\bar{X}$ -chart, we first calculate the new $\bar{\bar{X}}$ and new $\bar{R}$ by using equations (14) and (15).

In our case, $k = 20, d = 2, \sum_{j=1}^d \bar{X}_j = 54.4 + 49.6 = 104$ and $\sum_{j=1}^d R_j = 3 + 3 = 6$.

$$\therefore \bar{\bar{X}}_{\text{new}} = \frac{\sum_{i=1}^k \bar{X}_i - \sum_{j=1}^d \bar{X}_j}{k - d} = \frac{1032.4 - 104}{20 - 2} = 51.578 \text{ and}$$

$$\bar{R}_{\text{new}} = \frac{\sum_{i=1}^k R_i - \sum_{j=1}^d R_j}{k - d} = \frac{56 - 6}{20 - 2} = 2.778$$

We calculate the revised centre line and control limits of the $\bar{X}$ -chart using equations (16a to 16c) as follows:

$$CL = \bar{\bar{X}}_{\text{new}} = 51.578$$

Formulas for calculating the new $\bar{\bar{X}}$ and new $\bar{R}$ are

$$\bar{\bar{X}}_{\text{new}} = \frac{\sum_{i=1}^k \bar{X}_i - \sum_{j=1}^d \bar{X}_j}{k - d} \text{ and}$$

$$\bar{R}_{\text{new}} = \frac{\sum_{i=1}^k R_i - \sum_{j=1}^d R_j}{k - d}$$

where d – the number of discarded samples,

$\sum_{j=1}^d \bar{X}_j$ – the sum of means

of discarded sample, and

$\sum_{j=1}^d R_j$ – the sum of ranges

of discarded samples.

$$LCL = \bar{\bar{X}}_{\text{new}} - A_2 \bar{R}_{\text{new}} = 51.578 - 0.577 \times 2.778 = 49.975$$

$$UCL = \bar{\bar{X}}_{\text{new}} + A_2 \bar{R}_{\text{new}} = 51.578 + 0.577 \times 2.778 = 53.181$$

You should now check your understanding about the $\bar{X}$ -chart by solving the following exercises.

E2) Choose the correct option from the following:

- i) The control limits for the $\bar{X}$ -chart are
 - a) $\bar{\bar{X}} \pm A_2 \bar{R}$ b) $\bar{\bar{X}} \pm 3\bar{R}$
 - c) $\bar{\bar{X}} \pm B_2 \bar{R}$ d) $\bar{\bar{X}} \pm A\bar{R}$
- ii) If one point of the $\bar{X}$ -chart lies outside the control limits, the process is
 - a) in control b) out-of-control c) both (a) and (b)
- iii) The $\bar{X}$ -chart is used for
 - a) variables b) attributes c) defects d) defectives

E3) Samples of size 5 are taken from a manufacturing process at regular intervals. A normally distributed quality characteristic is measured and $\bar{X}$ and R are calculated for each sample. After 20 samples, we have $\bar{\bar{X}} = 6.40$ and $\bar{R} = 0.0877$. Compute the centre line and control limits for the $\bar{X}$ -chart.

E4) A factory was producing electrical bulbs. To test whether the process was under statistical control or not with respect to the average life of the bulbs, 16 samples of 3 bulbs were drawn at regular intervals and the life of each sampled bulb was measured. The measurements are given below:

Sample Number	Life of Bulb (in hour) (X)			Total	$\bar{X}$
1	1152	650	1090	2893	964.00
2	688	540	1074	2304	767.33
3	876	1342	1187	3408	1135.00
4	725	1070	847	2646	880.67
5	900	846	1254	3005	1000.00
6	1080	790	871	2747	913.67
7	952	1087	1126	3172	1055.00
8	741	786	578	2113	701.67
9	1005	1244	1140	3398	1129.67
10	1230	1005	654	2899	963.00
11	1200	870	859	2940	976.33
12	745	964	1050	2771	919.67
13	926	1172	1200	3311	1099.33
14	781	876	981	2652	879.33
15	1120	1350	750	3235	1073.33
16	880	975	768	2639	874.33

Test whether the process is under statistical control or not by drawing the $\bar{X}$ -chart when the average life and the standard deviation of the life of the bulbs are

- i) known to be 1000 hour and 215 hour, respectively.
- ii) unknown.

It is given that $A = 1.732$ and $A_2 = 1.023$ for $n = 3$.

In Sec.2.4, you have studied the $\bar{X}$ -chart and learnt when it is used and how it is constructed. We now discuss the charts which are used for controlling the process variability. We first take up the R-chart.

2.5 RANGE CHART (R-CHART)

You have learnt in Sec. 2.3 that we use R-chart or S-chart to control the process variability and the R-chart is widely used in quality control.

The underlying logic and basic form of an R-chart are similar to the $\bar{X}$ -chart.

The primary difference between the $\bar{X}$ -chart and the R-chart is that instead of plotting the sample means and monitoring their variation, we plot sample range and monitor the variation in sample ranges.

To get the control limits for the R-chart, we need to know the sampling distribution of the range. However, the derivation of the sampling distribution of the range is beyond the scope of this course. So we state the result without proof. If $X_1, X_2, \dots, X_n$ is a random sample of size n taken from a normal population with mean μ and variance σ^2 , the mean and variance of sampling distribution of range are given as follows:

$$E(R) = d_2 \sigma \quad \dots (19)$$

and

$$\text{Var}(R) = d_3^2 \sigma^2 \quad \dots (20)$$

where d_2 and d_3 are constants and depend on the size of the sample. These constants have been tabulated for various sample sizes in Table I given at the end of this block.

We also know that $SE(X) = \sqrt{\text{Var}(X)}$

$$\text{Therefore, } SE(R) = \sqrt{\text{Var}(R)} = \sqrt{d_3^2 \sigma^2} = d_3 \sigma \quad \dots (21)$$

Although the sampling distribution of range is not a normal distribution, it is common in practice to use the $\pm 3\sigma$ limits. These limits assure that the probability of an observation outside these limits is very small. Therefore, we define CL, LCL and UCL for the R-chart as follows:

$$\text{Centre line (CL)} = E(R) = d_2 \sigma \quad \dots (22a)$$

$$\text{Lower control limit (LCL)} = E(R) - 3 SE(R) = d_2 \sigma - 3d_3 \sigma$$

$$\text{or} \quad LCL = (d_2 - 3d_3) \sigma = D_1 \sigma \quad \dots (22b)$$

$$\text{Upper control limit (UCL)} = E(R) + 3 SE(R) = d_2 \sigma + 3d_3 \sigma$$

$$\text{or} \quad UCL = (d_2 + 3d_3) \sigma = D_2 \sigma \quad \dots (22c)$$

where $D_1 = d_2 - 3d_3$ and $D_2 = d_2 + 3d_3$ are constants and depend on the size of the samples. These constants have been tabulated for various sample sizes in Table I given at the end of this block.

In practice, the value of σ is not known. Therefore, it is estimated from the samples, which are taken when the process is thought to be under statistical control. Since the R-chart is used for small sample size (<10), σ is estimated by sample range (R). If $X_1, X_2, \dots, X_n$ is a sample of size n , we estimate σ from equation (6) as:

$$\hat{\sigma} = \bar{R}/d_2$$

So the centre line and control limits for the R-chart when σ is estimated by $\bar{R}/d_2$ are given as:

$$\text{Centre line (CL)} = E(R) = d_2 \hat{\sigma} = d_2 \frac{\bar{R}}{d_2} = \bar{R} \quad \dots (23a)$$

$$\text{Lower control limit (LCL)} = E(R) - 3SE(R) = d_2 \hat{\sigma} - 3d_3 \hat{\sigma} = d_2 \frac{\bar{R}}{d_2} - 3d_3 \frac{\bar{R}}{d_2}$$

$$\text{or} \quad LCL = \bar{R} - \frac{3d_3}{d_2} \bar{R} = \left(1 - \frac{3d_3}{d_2}\right) \bar{R} = D_3 \bar{R} \quad \dots (23b)$$

$$\text{Upper control limit (UCL)} = E(R) + 3SE(R) = d_2 \hat{\sigma} + 3d_3 \hat{\sigma} = d_2 \frac{\bar{R}}{d_2} + 3d_3 \frac{\bar{R}}{d_2}$$

$$\text{or} \quad UCL = \bar{R} + \frac{3d_3}{d_2} \bar{R} = \left(1 + \frac{3d_3}{d_2}\right) \bar{R} = D_4 \bar{R} \quad \dots (23c)$$

where $D_3 = \left(1 - \frac{3d_3}{d_2}\right)$ and $D_4 = \left(1 + \frac{3d_3}{d_2}\right)$ are constants and depend on the size of the sample. These constants have been tabulated for various sample sizes in Table I given at the end of this block.

After setting the centre line and the control limits, we construct the R-chart by taking the sample number on the X-axis (horizontal scale) and the sample range (R) on the Y-axis (vertical scale). We represent the centre line by solid line and UCL and LCL by dotted lines. We plot the value of sample range for each sample against the sample number. The consecutive sample points are joined by line segments.

Interpretation of the result

If all sample points lie on or between the upper and lower control limits, the R-chart indicates that the process is under statistical control. Otherwise, it is out-of-control. To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them. For this, we delete the out-of-control points and calculate the **revised centre line** and **control limits** as we did for the control chart of process mean. We

$$\text{Centre Line (CL)} = \bar{R}_{\text{new}} \quad \dots (24a)$$

$$\text{Lower control limit (LCL)} = D_3 \bar{R}_{\text{new}} \quad \dots (24b)$$

$$\text{Upper control limit (UCL)} = D_4 \bar{R}_{\text{new}} \quad \dots (24c)$$

define the revised limits for the R-chart as follows:

The following points should be kept in mind while using the $\bar{X}$ and R-charts:

- In order to determine whether a process is under statistical control, we control both process mean and process variability. For that, we use the $\bar{X}$ and R-charts together. We first control the process variability. So we analyse the **R-chart before the $\bar{X}$ -chart**.
- If the R-chart indicates that the process variability is out-of-control, we first control the process variability by investigating the assignable causes and taking corrective action to eliminate them. Only then we analyse the $\bar{X}$ -chart because when the R-chart is brought under statistical control, many assignable causes for the $\bar{X}$ -chart are eliminated automatically.
- If the R-chart indicates that the process variability is under statistical control, but the $\bar{X}$ -chart indicates that the process mean is out-of-control, we continue to use the R-chart, but revise the $\bar{X}$ -chart. The centre line and control limits of the $\bar{X}$ -chart are revised by eliminating these points. As a result, the position of the control limits in the revised $\bar{X}$ -chart shift. But the distance of control limits remains the same as in out-of-control case because we do not calculate $\bar{R}$ again.
- If both $\bar{X}$ and R-charts indicate that the process variability and process mean are under statistical control, $\bar{\bar{X}}$ and $\bar{\bar{R}}$ can be considered as representative of the process and their values are taken as standard values.

The following examples will help you understand how to construct and interpret the $\bar{X}$ and R-charts together.

Example 4: A statistical quality controller uses the $\bar{X}$ and R-charts together for monitoring the quality characteristic of a product. Samples of size 5 are taken from the manufacturing process at regular intervals. A normally distributed quality characteristic is measured and the $\bar{X}$ and R are calculated for each sample. After 20 samples have been analysed, we have $\bar{\bar{X}} = 6.40$ and $\bar{\bar{R}} = 0.0877$. Compute the centre line and control limits for the $\bar{X}$ and R-charts.

Solution: Here we are given that

$$n = 5, k = 20, \bar{\bar{X}} = 6.40, \bar{\bar{R}} = 0.0877$$

Note that the process mean (μ) and process variability (σ) are unknown in this case. Therefore, we use equations (9a to 9c) and (23a to 23c) for calculating the centre and control limits for the $\bar{X}$ -chart and the R-chart, respectively.

From Table I, we have

$$A_2 = 0.577, D_3 = 0, D_4 = 2.114 \text{ for } n = 5$$

Substituting the values of $\bar{\bar{X}}$, $\bar{R}$ and A_2 in equations (9a to 9c), we get the centre line and control limits of the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 6.40$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 6.40 + 0.577 \times 0.0877 = 6.451$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 6.40 - 0.577 \times 0.0877 = 6.349$$

Similarly, substituting the values of $\bar{R}$, D_3 and D_4 in equations (23a to 23c), we get the centre line and control limits of the R-chart as follows:

$$CL = \bar{R} = 0.0877$$

$$UCL = D_4 \bar{R} = 2.114 \times 0.0877 = 0.185$$

$$LCL = D_3 \bar{R} = 0 \times 0.0877 = 0$$

Example 5: A milk company uses automatic machines to fill 500 ml milk packets. A quality control inspector inspected four packets for each sample at given time intervals and measured the weight of each filled packet. The data for 20 samples are shown in the following table:

Sample Number	Weight of Filled Milk Packet (in ml)				$\bar{X}$	R
1	500	520	500	500	506.67	20
2	500	490	520	530	503.33	40
3	490	550	570	540	536.67	80
4	510	520	500	520	510.00	20
5	510	480	490	490	493.33	30
6	520	500	520	500	513.33	20
7	520	510	530	510	520.00	20
8	530	490	520	500	513.33	40
9	510	490	500	510	500.00	20
10	520	520	490	520	510.00	30
11	520	500	510	500	510.00	20
12	480	500	520	510	500.00	40
13	530	510	520	510	520.00	20
14	500	510	510	500	506.67	10
15	490	520	500	510	503.33	30
16	520	500	530	500	516.67	30
17	520	560	490	510	523.33	70
18	500	490	500	510	496.67	20
19	520	500	530	500	516.67	30
20	500	490	500	490	496.67	10
Total					10196.67	600

Control Charts for Variables

The CL, UCL and LCL for the $\bar{X}$ -chart when μ and σ are unknown are

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

The CL, UCL and LCL for the R-chart when σ is unknown are

$$CL = \bar{R}$$

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

Process Control

Using the $\bar{X}$ and R-charts together, draw the conclusion about the process by assuming assignable causes for any out-of-control points. If the process is out-of-control, calculate the revised centre line and control limits to bring the process under statistical control.

Solution: Since the $\bar{X}$ and R-charts are to be used together, we draw and analyse the R-chart first.

Here, the variability of the process (σ) is unknown. Therefore, we use equations (23a to 23c) for calculating the centre and control limits for the R-chart

From Table I, we have

$$A_2 = 0.729, D_3 = 0 \text{ and } D_4 = 2.282 \text{ for } n = 4$$

We first calculate the value of $\bar{R}$ for the centre line and the control limits from the given data using equation (7) as follows:

$$\bar{R} = \frac{1}{k} \sum R = \frac{1}{20} \times 600 = 30$$

Substituting the values of $\bar{R}$, D_3 and D_4 in equations (23a to 23c) we get

$$CL = \bar{R} = 30$$

$$UCL = D_4 \bar{R} = 2.282 \times 30 = 68.46$$

$$LCL = D_3 \bar{R} = 0 \times 30 = 0$$

We now construct the R-chart by taking the sample number on the X-axis and the sample range (R) of the milk packets on the Y-axis as shown in Fig. 2.5.

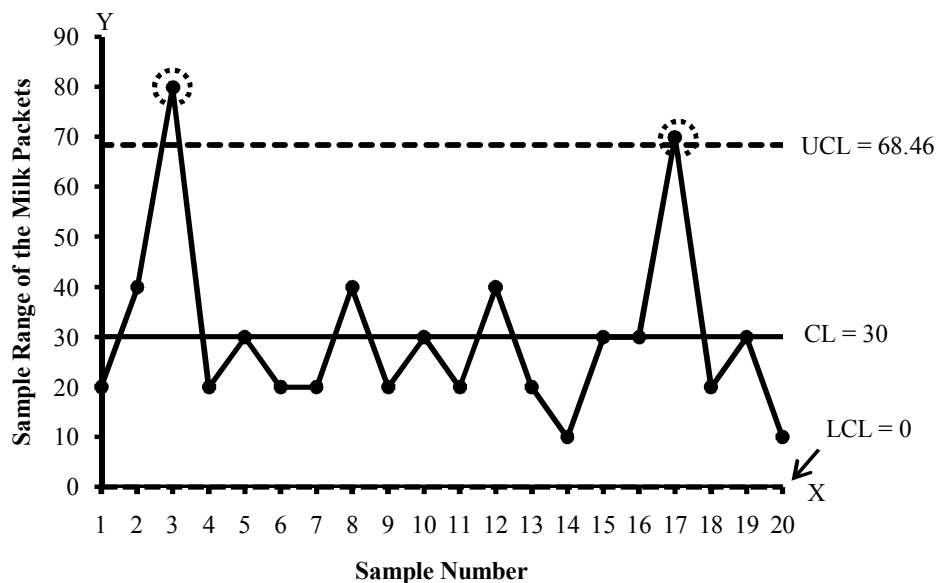


Fig. 2.5: The R-chart for milk packets.

Interpretation of the result

From Fig. 2.5, we observe that the points corresponding to samples 3 and 17 lie outside the control limits. Therefore, the process variability is out-of-control and some assignable causes are present in the process.

The CL, UCL and LCL for the R-chart when σ is unknown are

$$CL = \bar{R}$$

$$UCL = D_4 \bar{R}$$

$$LCL = D_3 \bar{R}$$

For calculating the revised control limits, we first delete the out-of-control points and calculate the $\bar{R}_{new}$ for the remaining samples.

In this example, $k = 20, d = 2, \sum_{j=1}^d R_j = 80 + 70 = 150$

$$\therefore \bar{R}_{new} = \frac{\sum_{i=1}^k R - \sum_{j=1}^d R_j}{k - d} = \frac{600 - 150}{20 - 2} = \frac{450}{18} = 25$$

We calculate the revised centre line and control limits of the R-chart using equations (24a to 24c) as follows:

$$CL = \bar{R}_{new} = 25$$

$$UCL = D_4 \bar{R}_{new} = 2.282 \times 25 = 57.05$$

$$LCL = D_3 \bar{R}_{new} = 0 \times 25 = 0$$

Fig. 2.6 shows the revised R-chart.

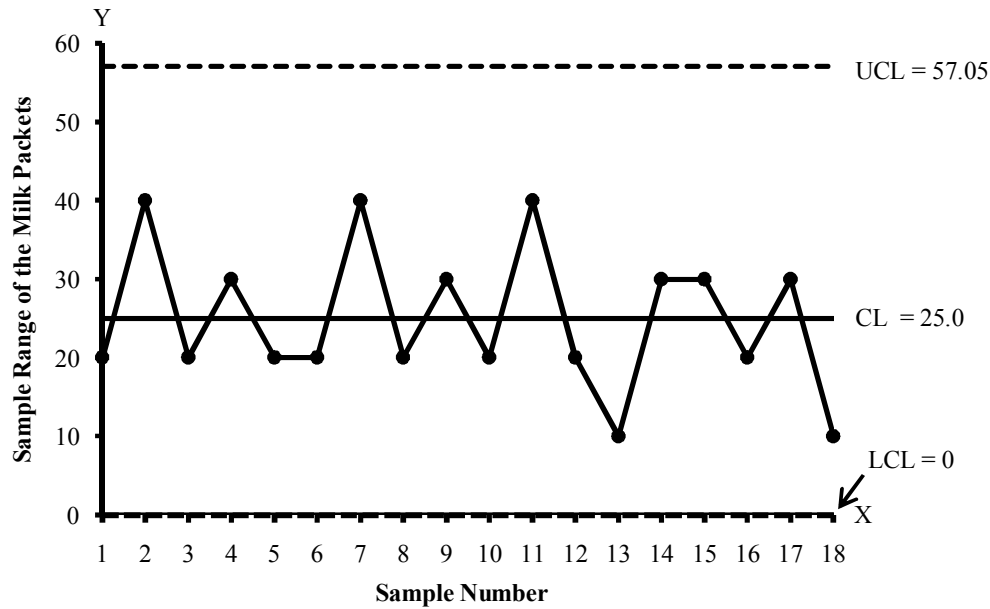


Fig. 2.6: The revised R-chart.

Interpretation of the result

The revised R-chart (shown in Fig. 2.6) indicates that all points lie within the control limits. Hence, we say that the process variability is under statistical control.

Note: If one or more points lie outside the revised control limits, we calculate the revised control limits for the R-chart again. This is continued until the process is brought under statistical control.

After controlling the process variability, we study the process mean. For this we calculate the control limits of the $\bar{X}$ -chart for the remaining samples. We first calculate the new $\bar{\bar{X}}$ from equation (14).

In this example, $k = 20, d = 2, \sum_{j=1}^d \bar{X}_j = 536.67 + 523.33 = 1060$

Control Charts for Variables

Formula for calculating the new $\bar{R}$ is

$$\bar{R}_{new} = \frac{\sum_{i=1}^k R_i - \sum_{j=1}^d R_j}{k - d}$$

where d – the number of discarded samples, and

$\sum_{j=1}^d R_j$ – sum of ranges of discarded samples.

Process Control

The CL, UCL and LCU for the $\bar{X}$ -chart when out-of-control samples are deleted given by

$$CL = \bar{\bar{X}}_{\text{new}}$$

$$UCL = \bar{\bar{X}}_{\text{new}} + A_2 \bar{R}_{\text{new}}$$

$$LCL = \bar{\bar{X}}_{\text{new}} - A_2 \bar{R}_{\text{new}}$$

$$\therefore \bar{\bar{X}}_{\text{new}} = \frac{\sum_{i=1}^k \bar{X} - \sum_{j=1}^d \bar{X}_j}{k-d} = \frac{10196.67 - 1060}{20-2} = \frac{9136.67}{18} = 507.593$$

We now calculate the revised centre line and control limits of the $\bar{X}$ -chart using equations (16a to 16c) as follows:

$$CL = \bar{\bar{X}}_{\text{new}} = 507.593$$

$$UCL = \bar{\bar{X}}_{\text{new}} + A_2 \bar{R}_{\text{new}} = 507.593 + 0.729 \times 25 = 525.818$$

$$LCL = \bar{\bar{X}}_{\text{new}} - A_2 \bar{R}_{\text{new}} = 507.593 - 0.729 \times 25 = 489.368$$

We can construct the $\bar{X}$ -chart by taking sample the number on the X-axis and the average weight of the milk packets on the Y-axis as shown in Fig. 2.7.

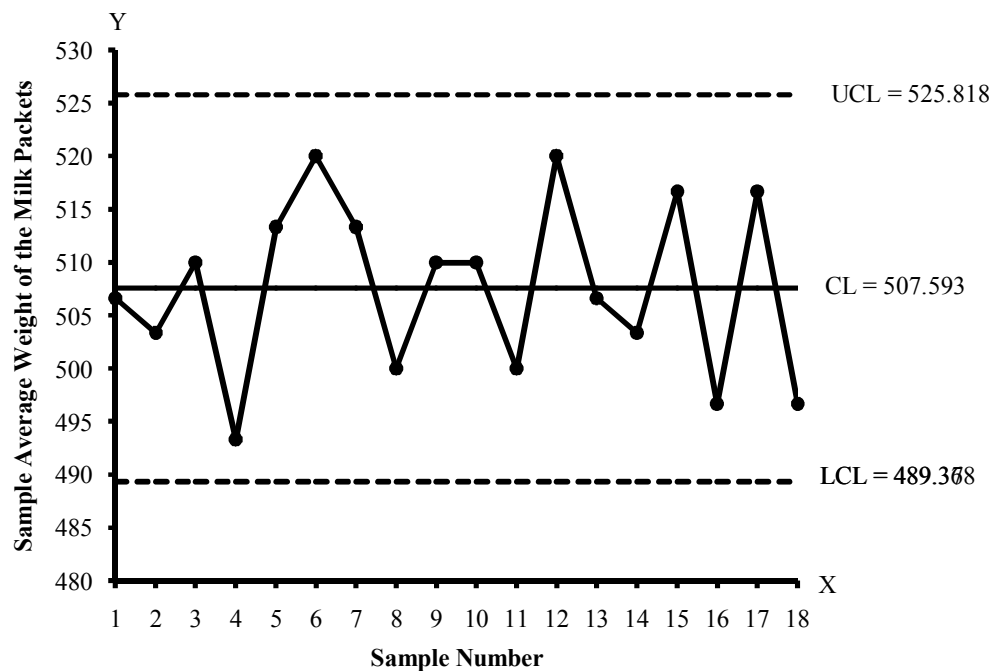


Fig. 2.7: The $\bar{X}$ -chart.

Interpretation of the result

Since no point lies outside the control limits of the $\bar{X}$ -chart (shown in Fig. 2.7), it indicates that the process mean is under statistical control.

You may now like to construct the $\bar{X}$ -chart and the R-chart for practice.

E5) Choose the correct option from the following:

- The upper control limit for the R-chart is
 - $D_4 \bar{R}$
 - $D_3 \bar{R}$
 - $\bar{R}$
 - $\bar{\bar{X}} + D_4 \bar{R}$
- Which control chart is used for controlling the process variability?
 - $\bar{\bar{X}}$ -chart
 - R-chart
 - p-chart
 - c-chart

- E6)** A high-voltage power station supplies a nominal output voltage of 250 V. A random sample of five outputs is taken each day and the mean and range calculated for all samples are tabulated below:

Control Charts for Variables

Day	1	2	3	4	5	6	7	8
$\bar{X}$	260	240	250	260	290	240	260	250
R	20	20	60	30	20	60	50	30
Day	9	10	11	12	13	14	15	16
$\bar{X}$	240	300	240	260	250	270	250	240
R	50	40	50	20	20	60	22	40

Test whether the process is under statistical control or not by drawing the $\bar{X}$ and R-charts.

In this section, you have learnt about the R-chart and how it is constructed. We now discuss the S-chart which is also used for controlling process variability.

2.6 STANDARD DEVIATION CHART (S-CHART)

In Secs. 2.4 and 2.5, you have studied the control charts for process mean ($\bar{X}$ -chart) and process variability (R-chart). Although the R-chart is easy to calculate and explain, for large sample size (usually greater than 10) it does not give a correct picture of the process variability. In Unit 2 of MST-002 entitled Descriptive Statistics, you have studied that the range depends only on maximum and minimum values of the data whereas the standard deviation depends on all observations. As the number of observations increases, the range of the data fluctuates more than the standard deviation. So, when the number of observations increases (≥ 10), we use standard deviation chart (S-chart) instead of the range chart (R-chart).

To get the control limits for the S-chart, we need to know the sampling distribution of the standard deviation (S). Since the derivation of the sampling distribution of S is beyond the scope of this course, we state the result without proof. If $X_1, X_2, \dots, X_n$ is a random sample of size n drawn from a normal distribution with mean μ and variance σ^2 ,

$$E(S) = c_4 \sigma \quad \dots (25)$$

and

$$\text{Var}(S) = (1 - c_4^2) \sigma^2 \quad \dots (26)$$

where c_4 is a constant and depends on the size of the sample. It has been tabulated for various sample sizes in Table I given at the end of this block.

We also know that $SE(X) = \sqrt{\text{Var}(X)}$

$$\therefore SE(S) = \sqrt{\text{Var}(S)} = \sqrt{(1 - c_4^2) \sigma^2} = \sigma \sqrt{1 - c_4^2} \quad \dots (27)$$

Although the sampling distribution of the standard deviation is not normal, it is common to use the $\pm 3\sigma$ limits. These limits assure that the probability of an observation outside these limits is very small. Therefore, we can calculate the CL, LCL and UCL for the S-chart as follows:

$$\text{Centre line (CL)} = E(S) = c_4 \sigma \quad \dots (28a)$$

$$\text{Lower control limit (LCL)} = E(S) - 3SE(S) = c_4 \sigma - 3\sigma \sqrt{1 - c_4^2}$$

$$\text{or} \quad \text{LCL} = \left(c_4 - 3\sqrt{1 - c_4^2} \right) \sigma = B_5 \sigma \quad \dots (28b)$$

$$\text{Upper control limit (UCL)} = E(S) + 3SE(S) = c_4 \sigma + 3\sigma \sqrt{1 - c_4^2}$$

$$\text{or} \quad \text{UCL} = \left(c_4 + 3\sqrt{1 - c_4^2} \right) \sigma = B_6 \sigma \quad \dots (28c)$$

where $B_5 = c_4 - 3\sqrt{1 - c_4^2}$ and $B_6 = c_4 + 3\sqrt{1 - c_4^2}$ are constants and depend on the size of the sample. These constants have been tabulated for various sample sizes in Table I given at the end of this block.

In practice, the value of σ is not known. Therefore, it is estimated from the samples which are taken when the process is thought to be under statistical control. Since the S-chart is used for large sample size (≥ 10), σ is estimated by the sample standard deviation (S) as we have described in Case II of Sec. 2.4.

Hence, the control limits for the S-chart when σ is estimated by the average sample standard deviation are given by

$$\text{CL} = E(S) = c_4 \hat{\sigma} = c_4 \frac{\bar{S}}{c_4} = \bar{S} \quad \dots (29a)$$

$$\text{LCL} = E(S) - 3SE(S) = c_4 \hat{\sigma} - 3\hat{\sigma} \sqrt{1 - c_4^2}$$

$$\text{or} \quad \text{LCL} = c_4 \frac{\bar{S}}{c_4} - 3 \frac{\bar{S}}{c_4} \sqrt{1 - c_4^2} = \left(1 - \frac{3}{c_4} \sqrt{1 - c_4^2} \right) \bar{S} = B_3 \bar{S} \quad \dots (29b)$$

$$\text{UCL} = E(S) + 3SE(S) = c_4 \hat{\sigma} + 3\hat{\sigma} \sqrt{1 - c_4^2}$$

$$\text{or} \quad \text{UCL} = c_4 \frac{\bar{S}}{c_4} + 3 \frac{\bar{S}}{c_4} \sqrt{1 - c_4^2} = \left(1 + \frac{3}{c_4} \sqrt{1 - c_4^2} \right) \bar{S} = B_4 \bar{S} \quad \dots (29c)$$

where $B_3 = 1 - \frac{3}{c_4} \sqrt{1 - c_4^2}$ and $B_4 = 1 + \frac{3}{c_4} \sqrt{1 - c_4^2}$ are constants and depend on the size of the sample. These constants have been tabulated for various sample sizes in Table I given at the end of this block.

The construction of the S-chart is similar to the R-chart. We take the sample number on the X-axis (horizontal scale) and the sample standard deviation on the Y-axis (vertical scale). We represent the centre line by solid line and UCL and LCL by dotted lines. We also plot the value of sample standard deviation

for each sample against the sample number. The consecutive sample points are joined by line segments.

Interpretation of the result

If all sample points lie on or in between the upper and lower control limits, the S-chart indicates that the process is under statistical control. Otherwise the process is out-of-control. To bring the process under statistical control it is necessary to investigate the assignable causes and take corrective action to eliminate them. For this, we delete the out-of-control points (samples) and calculate the revised centre line and control limits as was done in the $\bar{X}$ -chart. These limits are known as the **revised control limits**. For revised limits for the S-chart, we first calculate $\bar{S}_{\text{new}}$ by using the following formula:

$$\bar{S}_{\text{new}} = \frac{\sum_{i=1}^k S - \sum_{j=1}^d S_j}{k - d} \quad \dots (30)$$

where

d – the number of discarded samples, and

$\sum_{j=1}^d S_j$ – the sum of standard deviations of discarded samples.

We reconstruct the centre line and control limits of the S-chart by replacing $\bar{S}$ by $\bar{S}_{\text{new}}$ in equations (29a to 29c) as follows:

$$\text{Centre Line (CL)} = \bar{S}_{\text{new}} \quad \dots (30a)$$

$$\text{Lower control limit (LCL)} = B_3 \bar{S}_{\text{new}} \quad \dots (30b)$$

$$\text{Upper control limit (UCL)} = B_4 \bar{S}_{\text{new}} \quad \dots (30c)$$

Let us illustrate the construction and interpretation of the S-chart with the help of an example.

Example 6: The mean and standard deviation of 15 samples of size 12 for a production process are given in the following table:

Sample No	Mean	Standard Deviation	Sample No	Mean	Standard Deviation
1	9.5	1.2	9	8.75	2.1
2	7.5	1.5	10	9.75	3.5
3	10.0	2.6	11	7.25	2.6
4	8.25	3.2	12	10.25	1.1
5	7.25	1.9	13	9.25	2.2
6	9.0	2.6	14	10.0	1.6
7	10.5	5.4	15	10.5	3.2
8	9.25	2.4			

Determine the control limits for the $\bar{X}$ and S-charts. Comment on whether the process variability and process mean are under statistical control or not. Use the quality control factors $A_3 = 0.886$, $B_3 = 0.354$, $B_4 = 1.646$ for $n = 12$.

Process Control

The CL, UCL and LCL for the $\bar{X}$ -chart when σ is estimated by S are

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_3 \bar{S}$$

$$LCL = \bar{\bar{X}} - A_3 \bar{S}$$

The CL, UCL and LCL for the S-chart when σ is estimated by S are

$$CL = \bar{S}$$

$$UCL = B_4 \bar{S}$$

$$LCL = B_3 \bar{S}$$

Solution: Here, we want to control the process variability and process mean. Since the sample size is greater than 10, we use S and $\bar{X}$ -charts.

Since sample size $n > 10$, we use equations (13a to 13c) and (29a to 29c) for calculating the centre and control limits for the $\bar{X}$ -chart and S-chart, respectively.

It is given that

$$n = 12, k = 15, A_3 = 0.886, B_3 = 0.354 \text{ and } B_4 = 1.646$$

We first calculate the values of $\bar{\bar{X}}$ and $\bar{S}$ for centre line and control limits using equations (5) and (11) as follows:

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{15} (9.5 + 7.5 + \dots + 10.5) = \frac{1}{15} \times 137 = 9.133$$

$$\bar{S} = \frac{1}{k} \sum S = \frac{1}{15} (1.2 + 1.5 + \dots + 3.2) = \frac{1}{15} \times 37.1 = 2.473$$

To check whether the process is under statistical control or not, we first draw and analyse the S-chart.

Substituting the values of $\bar{S}$, B_3 and B_4 in equations (29a to 29c), we get the centre line and control limits of the S-chart as follows:

$$CL = \bar{S} = 2.473$$

$$UCL = B_4 \bar{S} = 1.646 \times 2.473 = 4.071$$

$$LCL = B_3 \bar{S} = 0.354 \times 2.473 = 0.875$$

We construct the S-chart by taking the sample number on the X-axis and the sample standard deviation (S) on the Y-axis as shown in Fig. 2.8.

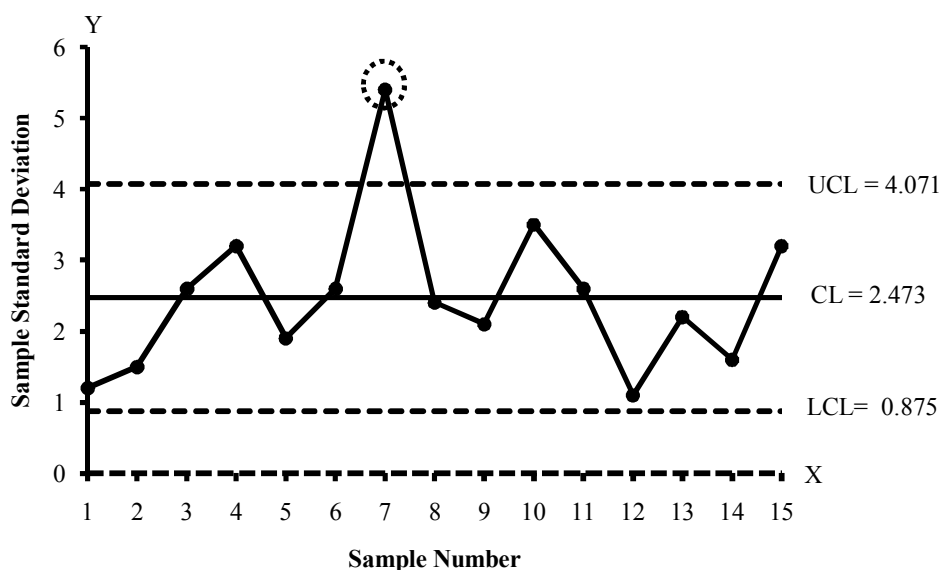


Fig. 2.8: The S-chart for Example 6.

Interpretation of the result

The S-chart shown in Fig. 2.8 indicates that the point corresponding to sample 7 lies outside the upper control limits. Therefore, the process variability is

out-of-control and some assignable causes are present in the process. To bring the process under statistical control it is necessary to investigate the assignable causes and take corrective action to eliminate them.

We calculate the centre line and control limits of the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 9.133$$

$$UCL = \bar{\bar{X}} + A_3 \bar{S} = 9.133 + 0.886 \times 2.473 = 11.324$$

$$LCL = \bar{\bar{X}} - A_3 \bar{S} = 9.133 - 0.886 \times 2.473 = 6.942$$

We construct the $\bar{X}$ -chart by taking the sample number on the X-axis and the sample average ($\bar{X}$) on the Y-axis as shown in Fig. 2.9.

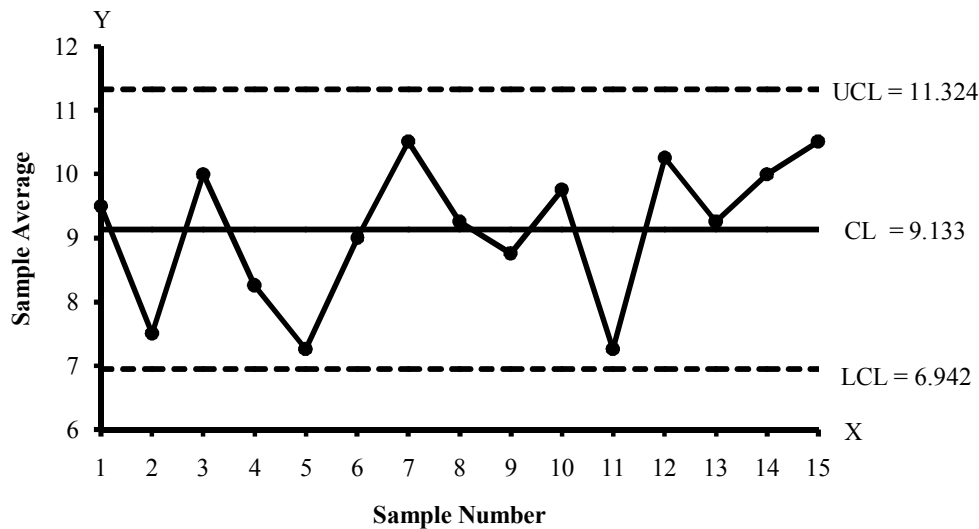


Fig. 2.9: The $\bar{X}$ -chart for Example 6.

Interpretation of the result

The $\bar{X}$ -chart shown in Fig. 2.9 indicates that all the sample points lie within the control limits. So the process mean is under statistical control.

Hence, with the help of S and $\bar{X}$ -chart we conclude that the process variability is not under statistical control, whereas the process average is under statistical control.

You may like to check your understanding about the S-chart by answering the following exercise.

-
- E7)** In a factory, quality is controlled using mean ($\bar{X}$) and standard deviation (S) charts. Twenty samples are chosen, each having 10 units for which $\sum \bar{X} = 645$ and $\sum S = 6$. Determine the 3σ limits for the $\bar{X}$ and S-charts. It is given that $A_3 = 0.97$, $B_3 = 0.284$ and $B_4 = 1.716$ for $n = 10$.
-

So far we have discussed one aspect of statistical process control, that is, to ensure that the process is under statistical control with the help of control chart. If a process is under statistical control, it merely indicates that there are no assignable causes of variations and that it is stable. Once a process is under statistical control, the next step is to check whether the process is able to meet

the **specifications** or not. For this, we use process capability analysis, which we discuss in the next section.

2.7 PROCESS CAPABILITY ANALYSIS

Variation in manufactured products is inevitable and it is a fact of nature and industrial life. Even if the production process is well designed or carefully maintained, no two objects are identical. The difference in two items/units may range from very large to very small or it may even be undetectable depending on the sources of variation.

Sometimes, it is possible that the process is under statistical control, but the customer is not satisfied with the product. Such a situation may arise when the product does not conform to its specifications. Then we need to carry out the process capability analysis to determine whether a product conforms to its specifications.

To conduct the process capability study, it is important to distinguish between the specification limits of a product, the control limits of the process and the natural tolerance limits of the product. We now explain these limits.

Specification Limits

The limits specified by the consumer for the value of a variable characteristic of the product or set by the management or the manufacturing engineers at the design and development stages of the product are called **specification limits**. Thus, the specification limits are determined externally. The upper value of the specification limits is called upper specification limit (USL) and the lower value of the specification limits is called lower specification limit (LSL). If a product fails to meet the specifications, it is called a **defective product**. This means that any product manufactured outside these limits would have to be scrapped and reworked. Therefore, the process should be set in a way that these limits are met (if possible) by all products.

For example, the weight of a cricket ball may be set by the customer or by the manufacturing engineers as 160.0 ± 3.0 gram. Then the weight of a cricket ball produced by the manufacturer can vary from 157.0 ($= 160.0 - 3.0$) gram to 163.0 ($= 160.0 + 3.0$) gram. Thus, the USL is 163.0 gram and LSL is 157.0 gram. If any ball produced by the manufacturer has weight greater than 163.0 gram or less than 157.0 gram, it may have to be scrapped.

Control Limits

In Sec. 2.2, you have learnt about the control limits for a process. You know that the control limits are usually determined by samples collected from a process. If a sample point lies outside the control limits, the process is said to be **out-of-control**. However, even in this case the individual product may not necessarily be defective because in the control charts, we plot a statistic such as the sample mean, range instead of individual measurement of the quality characteristic. It means that a single product that falls outside the control limits will neither cause the process to be out-of-control nor defective.

Natural Tolerance Limits

The natural tolerance limits (NTLs) are the limits usually determined by individual measurements of the characteristic of the product manufactured. These limits represent the natural variability of products in the process and are set at the 3σ limits from the mean, i.e., $\mu \pm 3\sigma$ where μ and σ are the process

mean and standard deviation, respectively, for individual measurements of the characteristic of the products produced. Thus, $\mu + 3\sigma$ is known as the **upper natural tolerance limit (UNTL)** and $\mu - 3\sigma$ is known as the **lower natural tolerance limit (LNTL)**. If μ and σ are unknown, their values are estimated by the grand sample mean and range or standard deviation, respectively, as we have described in Sec.2.4 for the $\bar{X}$ -chart.

It is not that there is no relationship between the specification limits, control limits and natural tolerance limits just because the specification limits are specified by the customer and the control limits and natural tolerance limits are driven by the variability of the process.

We now distinguish between the specification limits of a product, the control limits of the manufacturing process and the natural tolerance limits of the product with the help of an example.

Example 7: A cricket ball manufacturing company wants to control the weight of the ball. Twenty-five samples, each of size 4 were collected. The sum of averages and sum of sample ranges were found to be 4010 gram and 72 gram, respectively. If the customer specified the weight of the ball as 160 ± 3.0 gram, find the estimates of process mean and standard deviation. Also find the specification limits, control limits and natural tolerance limits. It is given that, $d_2 = 2.088$ and $A_2 = 0.729$ for $n = 4$.

Solution: It is given that

$$k = 25, n = 4, \sum \bar{X} = 4010, \sum R = 72, d_2 = 2.059 \text{ and } A_2 = 0.729$$

The process mean is estimated by the grand sample mean which is calculated as follows:

$$\hat{\mu} = \bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{25} \times 4010 = 160.4$$

The process standard deviation is estimated by the sample range which is calculated as follows:

$$\hat{\sigma} = \frac{\bar{R}}{d_2} = \frac{1}{d_2} \frac{1}{k} \sum R = \frac{1}{2.059} \times \frac{1}{25} \times 72 = 1.399$$

It is given that the customer specified the weight of the ball as:

160.0 ± 3.0 gram. Therefore,

The targeted or aimed value (μ_0) = 160.0

Upper specification limit (USL) = $160.0 + 3.0 = 163.0$

Lower specification limit (USL) = $160.0 - 3.0 = 157.0$

Since the manufacturer has to control the mean weight of the balls, we use $\bar{X}$ -chart. Here μ and σ are unknown, therefore, we can calculate the centre line and control limits for the $\bar{X}$ -chart using equations (9a to 9c) as follows:

Centre line (CL) = $\bar{\bar{X}} = 160.4$

Upper control limit (UCL) = $\bar{\bar{X}} + A_2 \bar{R} = 160.4 + 0.729 \times 2.88 = 162.50$

Lower control limit (LCL) = $\bar{\bar{X}} - A_2 \bar{R} = 160.4 - 0.729 \times 2.88 = 158.30$

The natural tolerance limits can be obtained as follows:

$$\text{Process mean } (\mu) = \hat{\mu} = \bar{\bar{X}} = 160.4$$

$$\text{Upper natural tolerance limit (UNTL)} = \hat{\mu} + 3\hat{\sigma} = 160.4 + 3 \times 1.399 = 164.597$$

$$\text{Lower natural tolerance limit (LNTL)} = \hat{\mu} - 3\hat{\sigma} = 160.4 - 3 \times 1.399 = 156.203$$

So far we have explained the specification limits, control limits and natural tolerance limits and basic difference among them. We now discuss process capability.

Process Capability

Process capability is a measure of the ability of a process to meet specifications. It tells us how good an individual product is. Even though the process may seem to be under statistical control, it is necessary to see whether the process is capable of manufacturing products within the specific limits or not. This can be done by process capability analysis.

Process capability may be defined as the spread of a quality characteristic measurement (i.e., natural variability) when the process is under statistical control (stable).

If the standard deviation of the stable process is σ , the process capability is defined as follows:

$$\text{Process capability} = 6\sigma$$

This indicates that products can be manufactured within $\pm 3\sigma$ limits. Generally, standard deviation of a process is not known and we estimate σ by range or standard deviation as follows:

$$\hat{\sigma} = \frac{\bar{R}}{d_2} \quad \text{or} \quad \hat{\sigma} = \frac{\bar{S}}{c_4}$$

To test whether the process is capable of manufacturing the product within the specific limits, we compare process capability with tolerance. Tolerance is defined as the difference between specification limits, i.e.,

$$\text{Tolerance} = \text{USL} - \text{LSL}$$

When we compare process capability with tolerance, the following situations may arise:

Case I: $6\sigma < \text{USL} - \text{LSL}$

In this case, the process is capable of producing products of desired specifications, even though the process is under statistical control or out-of-control.

Case II: $6\sigma = \text{USL} - \text{LSL}$

In this case, the process is capable of meeting specifications only if no shift in the process mean or a change in process variability takes place.

Case III: $6\sigma > \text{USL} - \text{LSL}$

In this situation, the products do not meet the desired specifications, even though the process is under statistical control.

The relation of process capability with the specification limits can be understood from the following example of your daily life.

Consider a car and a garage (Fig. 2.10). The garage defines the specification limits and the car defines the output of the process. If the car is only a little bit smaller than the garage, we can park it right in the middle of the garage (centre of the specification). If the car is wider than the garage, it does not matter if we have it centred; it will not fit. If the car is a lot smaller than the garage, it does not matter if we park it exactly in the middle; it will fit and we have plenty of room on either side.

If we have a process in control with little variation, we should be able to (park the car easily within the garage) manufacture a product that meets its specifications as well as customer requirements.

Now, we take up an example based on process capability.

Example 8: A manufacturing process produces a certain type of bolt of mean diameter 2 inch with a standard deviation of 0.05 inch. The lower and upper specification limits of the process are 1.90 and 2.05 inch. Calculate the process capability. Does it appear that the manufacturing process is capable of meeting the specification requirements?

Solution: We know the process capability (PC) of any process is defined as

$PC = 6\sigma$ where σ is the standard deviation of the process.

It is given that

$$\mu = 2, \sigma = 0.05, USL = 2.05 \text{ and } LSL = 1.90$$

$$\therefore PC = 6\sigma = 6 \times 0.05 = 0.30$$

Now we have to test whether the process is capable of meeting the required specifications. We first calculate tolerance and then compare PC with the tolerance:

$$\begin{aligned} \text{Tolerance} &= USL - LSL \\ &= 2.05 - 1.90 = 0.15 \end{aligned}$$

Since process capability > tolerance, the products do not meet the desired specifications.

Now, you can check your understanding of process capability by answering the following exercise.

-
- E8)** An automatic machine is used to fill and seal 20 ml tube of medicine. The process is sampled in samples of four and the values of $\bar{X}$ and R are computed for each sample. After 25 samples, $\sum \bar{X} = 525$ and $\sum R = 30$. If the specification limits are 20 ± 1.5 , what are the conclusions regarding the ability of the process to produce medicine tubes conforming to specifications? It is given that $d_2 = 2.059$ for $n = 4$.
-

We end this unit by giving a summary of what we have covered in it.

2.8 SUMMARY

1. **Pareto diagram** is a bar diagram, used for selection of quality characteristic for control in statistical quality control.
2. If the characteristic to be controlled is **measureable** such as weight, length, diameter, etc., **control chart for variables** can be used.



Fig. 2.10

3. If a characteristic **cannot be measured**, e.g., colour, surface roughness, etc., **control chart for attributes** can be used.
4. A **rational subgroup** is a small set of items that are produced under similar conditions within a relatively short time, such that the variation within the group is only due to chance causes.
5. If all sample points lie on or in between the upper and lower control limits and there is no unnatural pattern of variation, the control chart indicates that the process is under statistical control.
6. If some points lie outside the control limits or there is a long run or unnatural patterns of variation, the control chart indicates that the process is not under statistical control.
7. If we wish to control the **process mean**, we use **control chart for mean** ($\bar{X}$ -chart). If we wish to control **process variability**, we use the **R-chart and S-chart**.

8. The centre line and control limits of the $\bar{X}$ -chart are given as

$$CL = \bar{\bar{X}}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$\text{where } \bar{\bar{X}} = \frac{1}{k} \sum_{i=1}^k \bar{X}_i, \quad \bar{R} = \frac{1}{k} \sum_{i=1}^k R_i \text{ and } A_2 \text{ is a constant.}$$

9. The centre line and control limits of the R-chart are given as

$$CL = \bar{R}$$

$$LCL = D_3 \bar{R}$$

$$UCL = D_4 \bar{R}$$

where D_3 and D_4 are constants.

10. The centre line and control limits of the S-chart are given as

$$CL = \bar{S}$$

$$LCL = B_3 \bar{S}$$

$$UCL = B_4 \bar{S}$$

$$\text{where } \bar{S} = \frac{1}{k} \sum_{i=1}^k S_i, \quad S_i = \sqrt{\frac{1}{n-1} \sum_{j=1}^n (X_{ij} - \bar{X}_i)^2} \text{ and } B_3 \text{ and } B_4 \text{ are constants.}$$

11. The **specification limits** are the limits specified by the consumer or set by the management or the manufacturing engineers.
12. The **natural tolerance limits** (NTLs) are the limits usually determined by individual measurements of the characteristic of the product manufactured. These limits represent the natural variability of products in the process and are set at the 3σ limits.
13. **Process capability** is a measure of the ability of a process to meet specifications.

2.9 SOLUTION /ANSWERS

Control Charts for Variables

- E1)** i) Option (d) is the correct option because we know that control charts for variables are the $\bar{X}$ -chart, R-chart and S-chart.
- ii) Option (a) is the correct option because we use the $\bar{X}$ -chart for controlling the process mean, whereas the R-chart and the S-chart are used for controlling the process variability. We use the p-chart for controlling fraction defective of the process. You will study the p-chart in Unit 3.
- iii) Option (b) is the correct option because the R-chart and the S-chart are used for controlling process variability. Whereas the $\bar{X}$ -chart is used for process mean and the p-chart for fraction defective.
- E2)** i) Option (a) is the correct option because we know that the control limits for the $\bar{X}$ -chart are $\bar{\bar{X}} \pm A_2 \bar{R}$.
- ii) Option (b) is the correct option because we know that if one or more points lie outside the control limits, process is said to be out-of-control.
- iii) Option (a) is the correct option because we know that the $\bar{X}$, R and S-charts are control charts for variables, whereas the p, np, c and u-charts are control charts for attributes.

E3) It is given that

$$n = 5, k = 20, \bar{\bar{X}} = 6.40, \bar{R} = 0.0877$$

Since the process average (μ) and the process variability (σ) are unknown, in this case, we use equations (9a to 9c) to calculate the centre line and control limits of the $\bar{X}$ -chart.

From Table I given at the end of this block, we have $A_2 = 0.577$ for $n = 5$.

Therefore, we can calculate the centre line and control limits for the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 6.40$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 6.40 + 0.577 \times 0.0877 = 6.451$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 6.40 - 0.577 \times 0.0877 = 6.349$$

E4) i) It is given that

$$\mu = 1000, \sigma = 215, k = 16 \text{ and } n = 3$$

To draw the $\bar{X}$ -chart, we first calculate the centre line and control limits when the average life and the standard deviation of the life of bulbs are known as follows:

$$\text{Centre line (CL)} = \mu = 1000$$

$$\text{Upper control limit (UCL)} = \mu + A\sigma = 1000 + 1.732 \times 215 = 1372.380$$

The CL, UCL and LCL for the $\bar{X}$ -chart when μ and σ are unknown are

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R}$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R}$$

$$\text{Lower control limit (LCL)} = \mu - A\sigma = 1000 - 1.732 \times 215 = 627.620$$

To test whether the process under statistical control or not, we plot the sample mean ($\bar{X}$) against the sample number by taking the sample number on the X-axis and the sample average life of the bulb ($\bar{X}$) on the Y-axis as shown in Fig. 2.11.

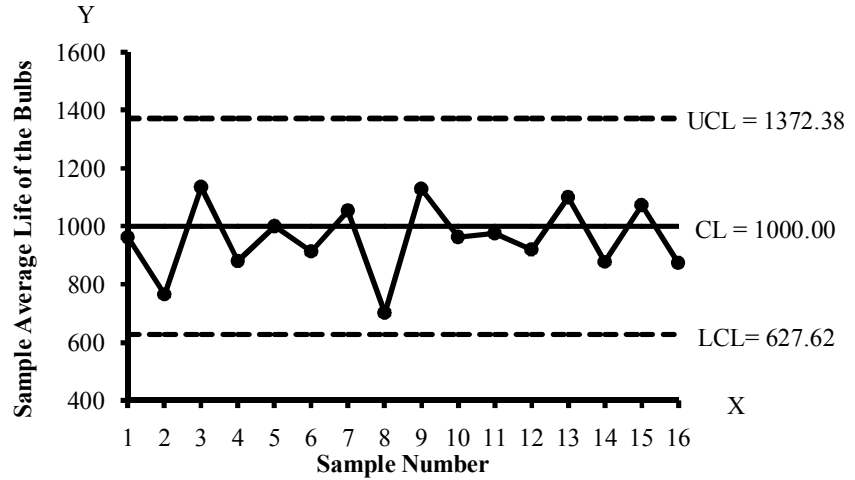


Fig. 2.11: The $\bar{X}$ -chart for average life of the bulbs.

Interpretation of the result

From Fig. 2.11, we observe that no point lies outside the control limits of the $\bar{X}$ -chart and also there is no specific pattern of the sample points on the chart. So this indicates that the process is under statistical control with respect to the average life of the bulbs.

- ii) In this case, the average life and the standard deviation of the life of the bulbs are not known. So we use equations (9a to 9c) to calculate the centre line and control limits of the $\bar{X}$ -chart.

For calculating the centre line and control limits of the $\bar{X}$ -chart, we first calculate the grand mean ($\bar{\bar{X}}$), range (R) and mean of range ($\bar{R}$) by the formulas:

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X}, R = \text{Max}(X) - \text{Min}(X) \text{ and } \bar{R} = \frac{1}{k} \sum R$$

Sample Number	Life of Bulb (in hour) (X)			Total	$\bar{X}$	R
1	1152	650	1090	2893	964.00	502
2	688	540	1074	2304	767.33	534
3	876	1342	1187	3408	1135.00	466
4	725	1070	847	2646	880.67	345
5	900	846	1254	3005	1000.00	408
6	1080	790	871	2747	913.67	290
7	952	1087	1126	3172	1055.00	174
8	741	786	578	2113	701.67	208
9	1005	1244	1140	3398	1129.67	239

10	1230	1005	654	2899	963.00	576
11	1200	870	859	2940	976.33	341
12	745	964	1050	2771	919.67	305
13	926	1172	1200	3311	1099.33	274
14	781	876	981	2652	879.33	200
15	1120	1350	750	3235	1073.33	600
16	880	975	768	2639	874.33	207
Total					15332.33	5669

From the above table, we have

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{16} \times 15332.33 = 958.271$$

$$\bar{R} = \frac{1}{k} \sum R = \frac{1}{16} \times 5669 = 354.313$$

We can calculate the centre line and control limits of the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 958.271$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 958.271 + 1.023 \times 354.313 = 1320.733$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 958.271 - 1.023 \times 354.313 = 595.809$$

We draw the $\bar{X}$ -chart as in case (i) (see Fig. 2.12).

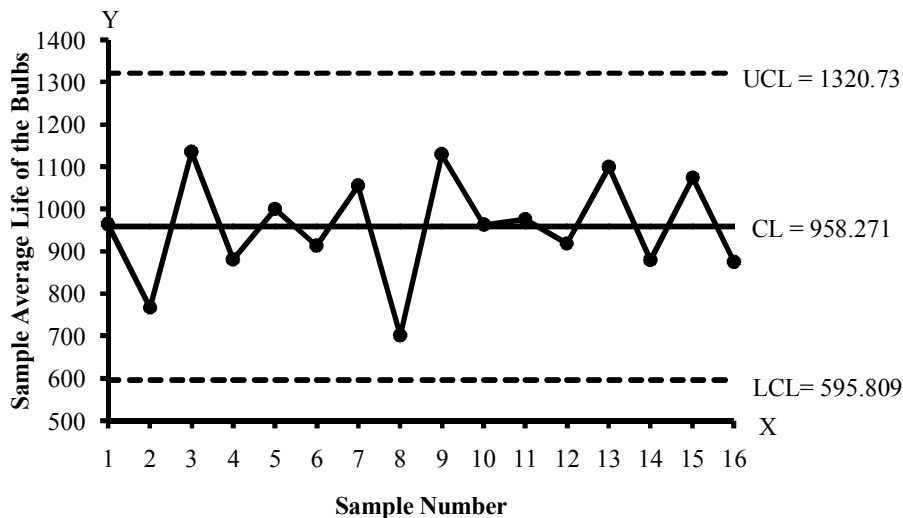


Fig. 2.12: The $\bar{X}$ -chart for average life of the bulbs.

Interpretation of the result

From Fig. 2.12, we observe that no point lies outside the control limits of the $\bar{X}$ -chart and there is no specific pattern of the sample points on the chart. So this indicates that the process mean is under statistical control with respect to the average life of the bulbs.

- E5) i) Option (a) is the correct option because the UCL and LCL of the R-chart are $D_4 \bar{R}$ and $D_3 \bar{R}$.

- ii) Option (b) is the correct option because we know that for controlling the process variability, we use either the R-chart or the S-chart. The $\bar{X}$ -chart is used for the process mean, the p-chart is used for the fraction defective and the c-chart is used for the number of defects.

E6) Here we are given that

$$n = 5 \text{ and } k = 20$$

Note that the process mean (μ) and the process variability (σ) are unknown in this case. Therefore, we use equations (9a to 9c) and (23a to 23c) for calculating the centre and control limits for the $\bar{X}$ -chart and R-chart, respectively.

From Table I, we have

$$A_2 = 0.577, D_3 = 0, D_4 = 2.114 \text{ for } n = 5$$

We first calculate $\bar{\bar{X}}$ and $\bar{R}$ using equations (5) and (7) as follows:

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{16} (260 + 240 + \dots + 240) = \frac{1}{16} \times 4100 = 256.25$$

$$\bar{R} = \frac{1}{k} \sum R = \frac{1}{16} (20 + 20 + \dots + 40) = \frac{1}{16} \times 592 = 37$$

To check whether the process is under statistical control or not, we first draw and analyse the R-chart.

Substituting the values of $\bar{R}$, D_3 and D_4 in equations (23a to 23c), we get the centre line and control limits of the R-chart as follows:

$$CL = \bar{R} = 37$$

$$UCL = D_4 \bar{R} = 2.117 \times 37 = 78.218$$

$$LCL = D_3 \bar{R} = 0 \times 37 = 0$$

We now construct the R-chart by taking the sample number on the X-axis and the sample voltage range (R) on the Y-axis as shown in Fig. 2.13.

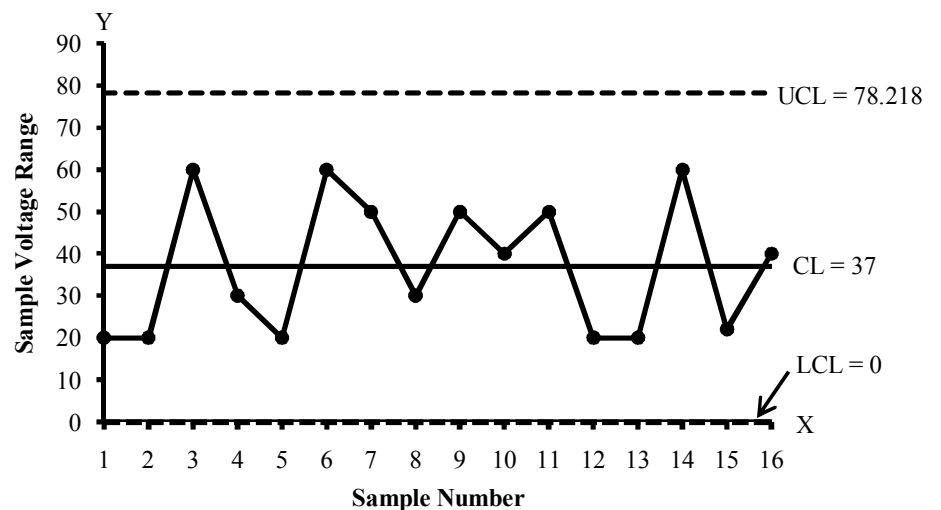


Fig. 2.13: The R-chart for process variability.

Interpretation of the result

The R-chart (shown in Fig. 2.13) indicates that all the points lie within the control limits and there is no specific pattern of the sample points on the chart. So the process variability is under statistical control.

We now check whether the process mean is under statistical control or not using the $\bar{X}$ -chart.

Substituting the values of $\bar{\bar{X}}$, $\bar{R}$ and A_2 in equations (9a to 9c), we get the centre line and control limits of the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 256.25$$

$$UCL = \bar{\bar{X}} + A_2 \bar{R} = 256.25 + 0.577 \times 37 = 277.599$$

$$LCL = \bar{\bar{X}} - A_2 \bar{R} = 256.25 - 0.577 \times 37 = 234.901$$

We construct the $\bar{X}$ -chart by taking the sample number on the X-axis and the sample average voltage ($\bar{X}$) on the Y-axis as shown in Fig. 2.14.

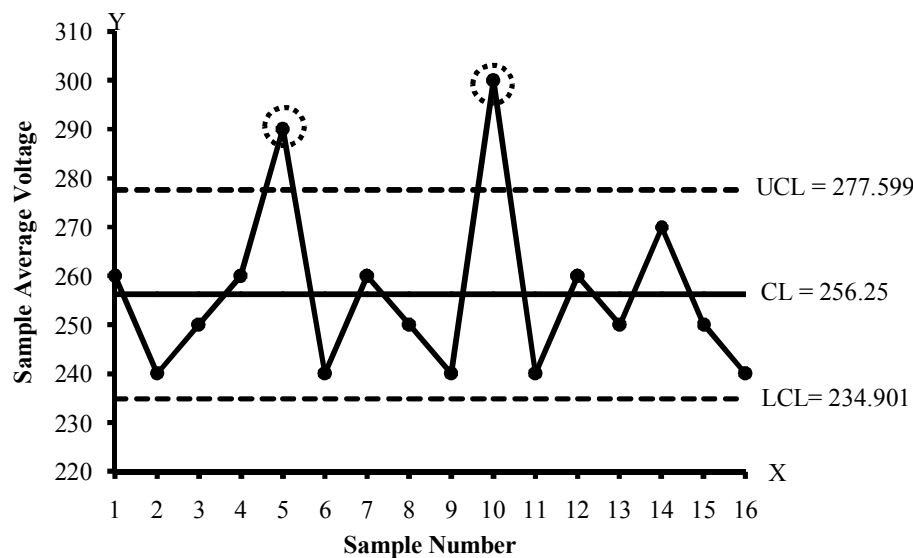


Fig. 2.14: The $\bar{X}$ -chart for process mean.

Interpretation of the result

The $\bar{X}$ -chart (shown in Fig. 2.14) indicates that the points corresponding to samples 5 and 10 lie outside the control limits. Therefore, the process average is out-of-control and some assignable causes are present in the process.

Hence, we conclude that the process variability is under statistical control, whereas the process average is out-of-control.

E7) It is given that

$$k = 20, n = 10, \sum \bar{X} = 645, \sum S = 6$$

Since the sum of sample standard deviations is given and the sample size is 10 so in this case we use equations (13a to 13c) and (29a to 29c) to

Process Control

The CL, UCL and LCL for the $\bar{X}$ -chart when σ is estimated by S are

$$CL = \bar{\bar{X}}$$

$$UCL = \bar{\bar{X}} + A_3 \bar{S}$$

$$LCL = \bar{\bar{X}} - A_3 \bar{S}$$

The CL, UCL and LCL for the S -chart when σ is estimated by S are

$$CL = \bar{S}$$

$$UCL = B_4 \bar{S}$$

$$LCL = B_3 \bar{S}$$

calculate the centre line and control limits of the $\bar{X}$ -chart and S -chart, respectively.

We first calculate $\bar{\bar{X}}$ and $\bar{S}$ using equations (5) and (11) as follows:

$$\bar{\bar{X}} = \frac{1}{k} \sum \bar{X} = \frac{1}{20} \times 645 = 32.25 \text{ and } \bar{S} = \frac{1}{k} \sum S = \frac{1}{20} \times 6 = 0.30$$

Substituting the values of $\bar{\bar{X}}$, A_3 and $\bar{S}$ in equations (13a to 13c) we get the centre line and control limits of the $\bar{X}$ -chart as follows:

$$CL = \bar{\bar{X}} = 32.25$$

$$UCL = \bar{\bar{X}} + A_3 \bar{S} = 32.25 + 0.975 \times 0.30 = 32.543$$

$$LCL = \bar{\bar{X}} - A_3 \bar{S} = 32.25 - 0.975 \times 0.30 = 31.958$$

Similarly, substituting the values of $\bar{S}$, B_3 and B_4 in equations (29a to 29c) we get the centre line and control limits of the S -chart as follows:

$$CL = \bar{S} = 0.30$$

$$UCL = B_4 \bar{S} = 1.716 \times 0.30 = 0.515$$

$$LCL = B_3 \bar{S} = 0.284 \times 0.30 = 0.085$$

E8) Here we want to check the ability of the process. For this, we have to compare the process capability (PC) with tolerance.

We know the process capability (PC) and tolerance (T) of any process are defined as

$$PC = 6\hat{\sigma} = 6 \frac{\bar{R}}{d_2}$$

and

$$T = USL - LSL$$

It is given that

$$n = 4, k = 25, \sum R = 30, d_2 = 2.059$$

Therefore,

$$USL = 20 + 1.5 = 21.5 \text{ and } LSL = 20 - 1.5 = 18.5$$

$$\bar{R} = \frac{1}{k} \sum R = \frac{1}{25} \times 30 = 1.2$$

Therefore, we can calculate the process capability and tolerance as follows:

$$PC = 6 \frac{\bar{R}}{d_2} = 6 \times \frac{1.2}{2.059} = 3.50$$

$$\begin{aligned} T &= USL - LSL \\ &= 21.5 - 18.5 = 3.0 \end{aligned}$$

Since process capability is greater than tolerance, the tubes of medicine do not meet the desired specifications.

Control Charts for Variables

UNIT 3 CONTROL CHARTS FOR ATTRIBUTES

Structure

- 3.1 Introduction
 - Objectives
- 3.2 Control Charts for Attributes
- 3.3 Control Chart for Fraction Defective (p-Chart)
- 3.4 p-Chart for Variable Sample Size
- 3.5 Control Chart for Number of Defectives (np-Chart)
- 3.6 Summary
- 3.7 Solutions/Answers

3.1 INTRODUCTION

In Unit 2, you have learnt about control charts for variables, which are used to control the measurable quality characteristics. You have also learnt how to construct control charts for mean ($\bar{X}$ -chart) and control chart for variability (R-chart and S-chart). However, there are many situations in which measurement is not possible, e.g., the number of failures in a production run, number of defects in a bolt of cloth, surface roughness of a cricket ball, etc. In such cases, we cannot use the control chart for variables. Moreover, control charts for variables can be used only for **one** measurable characteristic at a time. This may be problematic if a manufacturing company or a factory makes a product or a part having more than one measurable quality characteristic, say, 10 quality characteristics and the quality controller would like to control all the characteristics. Then for each quality characteristic, he/she would require a separate control chart for variables. However, it will be impracticable and uneconomical to have 10 such charts. In such situations, we use control charts for attributes, which provide overall information about quality.

You will learn about various aspects of control charts for attributes in this unit. In Sec. 3.2, we explain the meaning of the term **attributes** in industry and introduce different types of control charts for attributes. In Secs. 3.3 to 3.5, we discuss how to construct the control charts for attributes and interpret the result obtained from these charts for constant and variable sample size, respectively. In the next unit, we discuss the control charts for defects.

Objectives

After studying this unit, you should be able to:

- distinguish between the control charts for variables and attributes;
- explain the need for control charts for attributes;
- describe different types of control charts for attributes;
- decide on which control charts for attributes to use in a given situation;
- construct and interpret the control chart for fraction defective (p-chart);
- construct and interpret the p-chart for variable sample size; and
- construct and interpret the control chart for number of defectives (np-chart).

3.2 CONTROL CHARTS FOR ATTRIBUTES

We first define the term **attributes** as used in quality control terminology.

The term **attributes** in quality control refers to those quality characteristics, which classify the items/units into one of the two classes: conforming or non-conforming, defective or non-defective, good or bad.

There are two types of attributes:

- i) Where numerical measurements of the quality characteristics are not possible, for example, colour, scratches, damages, missing parts, etc.
- ii) Where numerical measurements of the quality characteristics are possible and items are classified as defective or non-defective on the basis of the inspection. For example, the diameter of a cricket ball can be measured by the micrometer but sometimes it may be more convenient to classify the balls as defective and non-defective using a Go-No-Go gauge (read the margin remark).

In inspection of attributes, actual measurements are not done, but the number of defective items (defectives) or number of defects in the item is counted. The size of defect and its location is not important. Items are inspected and either accepted or rejected.

There are different types of control charts for attributes for different situations. We classify the control charts for attributes into two groups as follows:

- Control charts for defectives, and
- Control charts for defects.

Control charts for defectives are mainly of two types as given below:

1. Control chart for fraction defective (p-chart), and
2. Control chart for number of defectives (np-chart).

Both control charts for defectives are based on the binomial distribution.

Control charts for defects are also of two types as given below:

1. Control chart for number of defects (c-chart), and
2. Control chart for number of defects per unit (u-chart).

Control charts for defects are based on Poisson distribution and these charts will be described in the next unit. In this unit, we focus on control **charts for defectives** and discuss the p-chart first.

3.3 CONTROL CHARTS FOR FRACTION DEFECTIVE (p-CHART)

The most widely used control chart for attributes is the **fraction (proportion) defective chart**, that is, the p-chart. The p-chart may be applied to quality characteristics, which cannot be measured or impracticable and uneconomical to measure it. These items/units are classified as defective or non-defective on the basis of certain criteria (defects). You have learnt that the control charts for variables are used to control only one quality characteristic at a time. If the quality controller would like to control two characteristics, he/she must use two

A Go-No-Go gauge is an inspection tool which is used to check an item or a unit or a piece against its allowed tolerances. The name **Go-No-Go** derives from its use. It means that we check the item and if the item is acceptable (fulfil the specifications), we say **Go** and if unacceptable, we say **No-Go**.

control charts for variables. But a single p-chart may be applied to more than one quality characteristic, in fact, to as many quality characteristics as we want.

Before describing the control chart for fraction defective (p-chart), we explain the meaning of **fraction defective**.

Fraction defective (p) is defined as the ratio of the number of defective terms/units/articles found in any inspection to the total number of items/units/articles inspected. Symbolically, we write

$\text{Fraction defective (p)} = \frac{\text{Number of defective items}}{\text{Total number of items inspected}} \quad \dots (1)$

For example, if 500 cricket balls are inspected and 20 balls are found defective, the fraction defective of the balls is given as:

$$\begin{aligned} \text{Fraction defective (p)} &= \frac{\text{Number of defective balls}}{\text{Total number of balls inspected}} \\ &= \frac{20}{500} = 0.04 \end{aligned}$$

Fraction defective is always less than or equal to 1 and expressed as a decimal or fraction.

We now explain how to construct the p-chart. The procedure of constructing a p-chart is similar to that of the control charts for variables. The main steps of constructing a p-chart are as follows:

Steps Involved in Construction of a p-chart

- Step 1:** We define non-conformity (defectiveness) properly because it depends on the product, its use, consumer needs, etc. For example, a scratch mark in a machine might not be considered non-conformity whereas the same scratch on a mobile phone, optical lens, laptop, etc. would be considered non-conformity.
- Step 2:** We determine the subgroup (sample) size and the number of subgroups (samples).

The selection of sample size for control charts for attributes is very important. The sample size for control charts for attributes should be large enough to allow non-conformities (non-conforming/defective items) to be observed in the samples. The sample size for control charts for attributes is a function of the proportion of non-conforming (defective) items. For example, if a process has 2% defective items, a sample of 20 items is not sufficient because the average number of defective items per sample is 0.40. It means that there is only 40% chance that defective items could be observed in the sample.

Therefore, misleading inference could be made. So for defective items, we would require many more samples. In such a situation, sample size of 100 should be sufficient. Thus, if a process is of very good quality, the p-chart requires large sample size to determine lack of control. If the process is of poor quality, small sample size is sufficient.

Generally, we decide the sample size as per the table given below:

Percentage of	Sample Size
---------------	-------------

Defective	
0 to 10%	20
0 to 6%	100
1.06 to 2.94%	1000
1.81 to 2.19%	50000

Note that determining the sample size for a p-chart, we require some preliminary observation to obtain a rough idea about the proportion of defective items and their average number.

The number of samples depends upon the production rate and cost of sampling in addition to other factors. Generally, 25 samples are gathered to collect sufficient data.

- Step 3:** After deciding the size of the sample and the number of samples, we select sample units from the item produced randomly, so that each item has an equal chance of being selected.
- Step 4:** We collect the data for a p-chart in the same way as for the control chart for variables. However, instead of recording measurements of the items, we record the number of items inspected and the number of defective items.
- Step 5:** We calculate the fraction defective for each sample from equation (1).

If there are k samples of the same size n and $d_1, d_2, \dots, d_k$ are the numbers of defectives in the 1st, 2nd, ..., k^{th} sample, respectively, then for each sample, we calculate the fraction defective as follows:

$$p_1 = \frac{d_1}{n}, p_2 = \frac{d_2}{n}, \dots, p_k = \frac{d_k}{n} \quad \dots (2)$$

- Step 6:** We set the 3σ control limits to find out whether the process is under control or out-of-control with respect to the fraction defective. The 3σ control limits for the p-chart are given by:

$$\text{Centre line} = E(p) \quad \dots (3a)$$

$$\text{Upper control limit (UCL)} = E(p) + 3SE(p) \quad \dots (3b)$$

$$\text{Lower control limit (LCL)} = E(p) - 3SE(p) \quad \dots (3c)$$

For calculating the control limits of the p-chart, we need the sampling distribution of the fraction defectives.

You have learnt in Unit 2 of MST-004 that the mean and variance of the sampling distribution of the fraction (proportion) defective is given by

$$E(p) = P \text{ and } \text{Var}(p) = \frac{PQ}{n} \quad \dots (4a)$$

Here P is the probability or fraction or proportion of defective in the process and $Q = 1 - P$.

We also know the standard error of a random variable X is

$$SE(X) = \sqrt{\text{var}(X)}$$

Therefore, the standard error of sample fraction (proportion) defective is given by

$$SE(p) = \sqrt{\text{Var}(p)} = \sqrt{\frac{PQ}{n}} \quad \dots (4b)$$

You have learnt that the sampling distribution of the fraction defectives is not a normal distribution. However, if the sample size is sufficiently large, such that $nP > 5$ and $nQ > 5$, you know from the **centre limit theorem**, that the sampling distribution of the fraction defectives is approximately normally distributed with mean P and variance PQ/n . Therefore, the centre line and control limits for the p-chart are given as follows:

Control Charts for Attributes

You have studied the **Central Limit Theorem** in Unit1 of MST-004.

$$CL = E(p) = P \quad \dots (5a)$$

$$UCL = E(p) + 3SE(p) = P + 3\sqrt{\frac{P(1-P)}{n}} \quad \dots (5b)$$

$$LCL = E(p) - 3SE(p) = P - 3\sqrt{\frac{P(1-P)}{n}} \quad \dots (5c)$$

In practice, fraction defective (P) of the process is not known. So equations (5a to 5c) are not used to construct the control chart. It is necessary to estimate the fraction defective of the process. We estimate it using the sample fractions. The best estimator of the process fraction defective is the **average fraction defective** of the samples. Suppose, we draw k samples of the same size and for each sample, we calculate the fraction defective using equation (2). If $p_1, p_2, \dots, p_k$ are the fraction defectives of the 1st, 2nd, ..., k^{th} sample, respectively, we calculate the **average fraction defective** as follows:

$$\bar{p} = \frac{1}{k}(p_1 + p_2, \dots, p_k) = \frac{1}{k} \sum_{i=1}^k p_i \quad \dots (6)$$

Generally, this formula is used when we have sample fraction defectives. If the number of defective items for each sample is given, it is convenient to calculate the average fraction defective as follows:

$$\bar{p} = \frac{\text{Sum of defective items in all samples}}{\text{Total number of items inspected}} = \frac{1}{nk} \sum_{i=1}^k d_i \quad \dots (7)$$

In this case, the centre line and control limits of the p-chart are given as:

$$CL = E(p) = \hat{P} = \bar{p} \quad \dots (8a)$$

$$UCL = E(p) + 3SE(p) = \hat{P} + 3\sqrt{\frac{\hat{P}(1-\hat{P})}{n}} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \dots (8b)$$

$$LCL = E(p) - 3SE(p) = \hat{P} - 3\sqrt{\frac{\hat{P}(1-\hat{P})}{n}} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} \quad \dots (8c)$$

Step 7: Having set the centre line and control limits, we construct the p-chart by taking the sample number on the X-axis and the sample fraction defective (p) on the Y-axis. We draw the control line as a solid line, and the UCL and LCL as dotted lines. Then we plot the value of the sample fraction defective against the sample number. The consecutive sample points are joined by line segments.

Step 8: We interpret the result. If all sample points lie on or in between the upper and lower control limits, the control chart indicates that the process is under control. That is, only chance causes are present in the process. If one or more points lie outside the upper or lower control limits, the control chart alarms (indicates) that the process is not under statistical control and some assignable causes are present in the process.

To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them. Once the assignable causes are eliminated, we delete the out-of-control points (samples) and calculate the **revised centre line** and **control limits** for the p-chart by using the remaining samples. These limits are known as the **revised control limits**. To calculate the revised limits of the p-chart, we first calculate new $\bar{p}$ as follows:

$$\bar{p}_{\text{new}} = \frac{\sum_{i=1}^k p_i - \sum_{j=1}^d p_j}{k-d} \quad \text{or} \quad \bar{p}_{\text{new}} = \frac{\sum_{i=1}^k d_i - \sum_{j=1}^d d_j}{n(k-d)} \quad \dots (9)$$

where d – the number of discarded samples,

$\sum_{j=1}^d p_j$ – the sum of fraction defectives in the discarded samples, and

$\sum_{j=1}^d d_j$ – the sum of number of defects in the discarded samples.

After finding $\bar{p}_{\text{new}}$, we reconstruct the centre line and control limits of the p-chart by replacing $\bar{p}$ by $\bar{p}_{\text{new}}$ in equations (8a to 8c) as follows:

$$CL = \bar{p}_{\text{new}} \quad \dots (10a)$$

$$UCL = \bar{p}_{\text{new}} + 3\sqrt{\frac{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}{n}} \quad \dots (10b)$$

$$LCL = \bar{p}_{\text{new}} - 3\sqrt{\frac{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}{n}} \quad \dots (10c)$$

Note 1: If the value of lower control limit is negative, we change the lower control limit to zero because a negative fraction defective is impossible.

Note 2: If the p-chart continuously indicates an increase in the value of the average fraction (proportion) defective, the management should investigate the reasons behind this increase rather than constantly revising the centre line and

control limits. The possible reasons behind the increase in the defective items may be poor incoming quality from vendors, tightening of specification limits, etc.

So far, we have discussed various steps involved in the construction of the p-chart. Let us now take up some examples to illustrate this method.

Example 1: In a company manufacturing cricket ball, the quality controller inspects the balls and classifies them as defective or non-defective on the basis of certain defects. The company manager wants to maintain the process so that an average of not more than 5 percent of the output is defective. Suggest a suitable control chart for this purpose. If the company can work with a sample of size 500, calculate the centre line and control limits for this chart.

Solution: Here the balls are classified as defective or non-defective. So we use the control chart for attributes. The quality controller has to maintain the process so that an average of not more than 5% of the output is defective. It means that the process proportion defective is given. So we use the p-chart.

We are given that $P = 5\% = 0.05$ and $n = 500$.

Therefore, we can calculate the centre line and control limits from equations (5a to 5c) as follows:

$$CL = P = 0.05$$

$$\begin{aligned} UCL &= P + 3\sqrt{\frac{P(1-P)}{n}} = 0.05 + 3\sqrt{\frac{0.05 \times (1-0.05)}{500}} \\ &= 0.05 + 3\sqrt{0.000095} = 0.0792 \end{aligned}$$

$$\begin{aligned} LCL &= P - 3\sqrt{\frac{P(1-P)}{n}} = 0.05 - 3\sqrt{\frac{0.05 \times (1-0.05)}{500}} \\ &= 0.05 - 3\sqrt{0.000095} = 0.0208 \end{aligned}$$

Example 2: A factory manufacturing small bolts. To check the quality of the bolts, the manufacturer selected 20 samples of same size 100 from the manufacturing process time to time. He/she visually inspected each selected bolt for certain defects. After the inspection, he/she obtained the following data:

Sample Number	1	2	3	4	5	6	7	8	9	10
Proportion Defective	0.10	0.04	0.08	0.15	0.08	0	0.01	0.05	0.05	0.08
Sample Number	11	12	13	14	15	16	17	18	19	20
Proportion Defective	0.10	0	0.06	0.05	0.03	0.20	0.05	0.07	0.01	0.08

Estimate the proportion defective of the process. Does the process appear to be under control with respect to the proportion of defective bolts?

Solution: We know that the proportion defective of the process can be estimated by average proportion defective of the samples. We can calculate it using equation (6) as follows:

$$\hat{P} = \bar{p} = \frac{1}{k} \sum p = \frac{1}{20} \times 1.29 = 0.065$$

Process Control

Here the manufacturer visually inspected each selected bolt and classified it as defective or non-defective. So we use the control chart for attributes. Since the proportion defective for each sample is given, we can use the p-chart.

Here the process proportion defective is not known. So we use equations (8a to 8c) to obtain the centre line and control limits for the p-chart as follows:

$$CL = \bar{p} = 0.065$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.065 + 3\sqrt{\frac{0.065 \times (1-0.065)}{100}}$$

$$= 0.065 + 3 \times 0.025 = 0.140$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.065 - 3\sqrt{\frac{0.065 \times (1-0.065)}{100}}$$

$$= 0.065 - 3 \times 0.025 = -0.01 \approx 0$$

We now construct the p-chart. We draw the centre line as a solid line and control limits as dotted lines on the graph and plot the points by taking the sample number on the X-axis and the proportion defective (p) on the Y-axis as shown in Fig. 3.1.

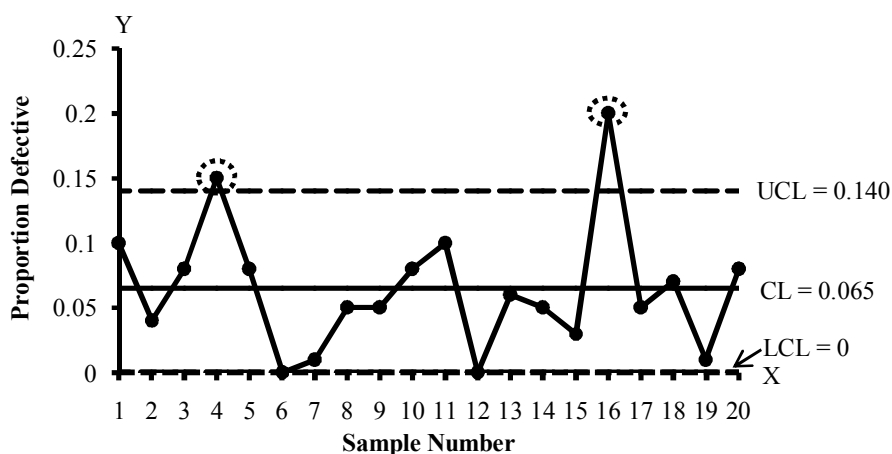


Fig. 3.1: The p-chart for proportion of defective bolts.

Interpretation of the result

From the p-chart shown in Fig. 3.1, we observe that the points corresponding to sample numbers 4 and 16 lie outside the control limits. So we can say that the process is not under statistical control with respect to the proportion of defective bolts.

Example 3: The following data are found during the inspection of the first 15 samples of size 100 each from a lot of two-wheelers manufactured by an automobile company:

Sample Number	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Number of Defectives	3	4	6	2	12	5	3	6	3	5	4	15	5	2	3

Draw the chart for fraction defective (p) and comment on the state of control. If the process is out-of-control, calculate the revised centre line and control limits by assuming assignable causes for any out-of-control point.

Solution: To draw the p-chart, we need to calculate the centre line and control limits. Here, the process fraction defective is not known. So in this case we use equations (8a to 8c).

To calculate the control limits, we first calculate the fraction defective for each sample and then $\bar{p}$.

Sample Number	Sample Size (n)	Number of Defectives (d)	Proportion Defective (p = d/n)
1	100	3	0.03
2	100	4	0.04
3	100	6	0.06
4	100	2	0.02
5	100	12	0.12
6	100	5	0.05
7	100	3	0.03
8	100	6	0.06
9	100	3	0.03
10	100	5	0.05
11	100	4	0.04
12	100	15	0.15
13	100	5	0.05
14	100	2	0.02
15	100	3	0.03
Total		78	

$$\text{Proportion defective } p = \frac{\text{Number of defectives}}{\text{Sample size}}$$

From the above table, we have

$$\bar{p} = \frac{1}{nk} \sum d = \frac{1}{15 \times 100} \times 78 = 0.052$$

Therefore, we can calculate the centre line and control limits as follows:

$$CL = \bar{p} = 0.052$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.052 + 3\sqrt{\frac{0.052 \times (1-0.052)}{100}}$$

$$= 0.052 + 3 \times 0.022 = 0.118$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}} = 0.052 - 3\sqrt{\frac{0.052 \times (1-0.052)}{100}}$$

$$= 0.052 - 3 \times 0.022 = -0.014 \approx 0$$

We now draw the p-chart by taking the sample number on the X-axis and the proportion defective (p) on the Y-axis as shown in Fig. 3.2.

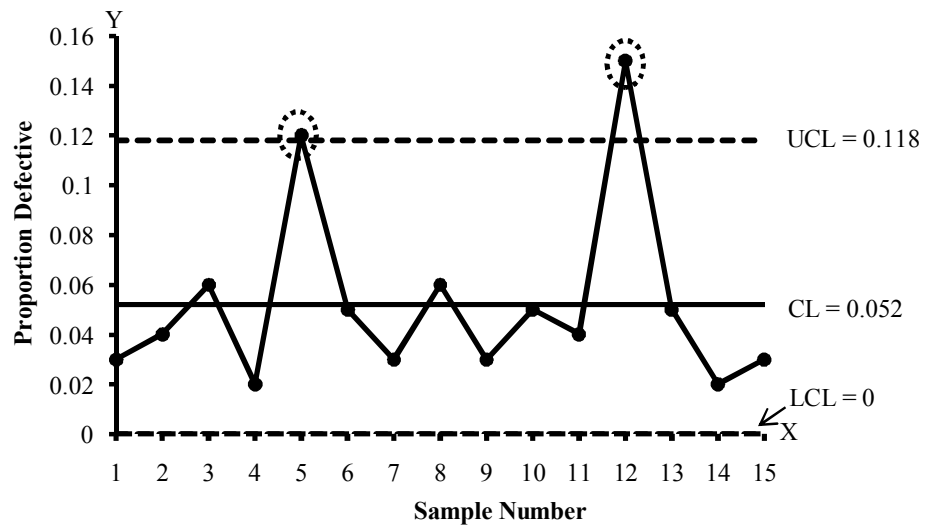


Fig. 3.2: The p-chart for fraction defective of two-wheelers.

Interpretation of the result

From the p-chart shown in Fig. 3.2., we observe that the points corresponding to sample numbers 5 and 12 lie outside the upper control limits. Therefore, the process is out-of-control. It means that some assignable causes are present in the process.

For calculating the revised limits for the p-chart, we first delete the out-of-control points (5 and 12) and calculate the new $\bar{p}$ using remaining samples.

In our case $n = 100$, $k = 15$, $d = 2$, $\sum_{j=1}^2 d_j = 12 + 15 = 27$

$$\therefore \bar{p}_{\text{new}} = \frac{\sum_{i=1}^k d_i - \sum_{j=1}^d d_j}{n(k-d)} = \frac{78-27}{100(20-2)} = \frac{51}{1800} = 0.028$$

After finding the $\bar{p}_{\text{new}}$, we reconstruct the centre line and control limits of the chart using equations (10a to 10c) as follows:

$$CL = \bar{p}_{\text{new}} = 0.028$$

$$\begin{aligned} UCL &= \bar{p}_{\text{new}} + 3\sqrt{\frac{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}{n}} = 0.028 + 3\sqrt{\frac{0.028 \times (1-0.028)}{100}} \\ &= 0.028 + 3 \times 0.016 = 0.0716 \end{aligned}$$

$$\begin{aligned} LCL &= \bar{p}_{\text{new}} - 3\sqrt{\frac{\bar{p}_{\text{new}}(1-\bar{p}_{\text{new}})}{n}} = 0.028 - 3\sqrt{\frac{0.028 \times (1-0.028)}{100}} \\ &= 0.028 - 3 \times 0.016 = -0.020 \approx 0 \end{aligned}$$

You may like to pause here and check your understanding of the p-chart by answering the following exercises.

E1) Choose the correct option from the following:

- i) The control chart used for attributes is the
 - a) p-chart b) np-chart c) c-chart d) all a), b) and c)

- ii) The control chart for proportion defective is the
 a) $\bar{X}$ -chart b) p-chart c) np-chart d) c-chart
- iii) The upper control limit for the p-chart is
 a) $\bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$ b) $\bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n}}$
 c) $\bar{p} - 3\sqrt{\bar{p}(1-\bar{p})}$ d) $\bar{p} + 3\sqrt{\bar{p}(1-\bar{p})}$
- iv) If the lower control limit has a negative value in a p-chart, it is treated as
 a) zero b) unity c) negative only d) positive only

- E2)** List the different types of control charts for attributes.
- E3)** Write the control limits of the p-chart when the fraction defective of the process is not known.
- E4)** A daily sample of 30 shirts was taken over a period of 15 days in order to monitor the manufacturing process of the shirts. Each shirt was inspected and classified as defective or non-defective. If a total of 22 defective shirts were found in 15 days, what should be the upper and lower control limits of the proportion of defective shirts?
- E5)** A mobile manufacturer inspects 30 mobiles at the end of the day of production and notes the number of defective mobiles. This procedure is continued up to 12 days and 2, 1, 3, 0, 2, 1, 0, 5, 2, 0, 3, 1 defective mobile are found. Is the production process under control with respect to the proportion defective?

3.4 p-CHART FOR VARIABLE SAMPLE SIZE

In Sec. 3.3, we have discussed the p-chart for constant sample size, that is, we have inspected the same number of items in each sample and the number of defective items has been counted. Sometimes, situations arise wherein we cannot take the same number of items in each sample. Generally, this happens when the p-chart is used for 100% inspection of output that varies from day to day. The variation in the output per day can be due to breakdown of machines, different production requirements, etc. When the sample size is not uniform (same or constant), we use the p-chart for variable sample size.

The procedure of constructing the p-chart for variable sample size is similar to the p-chart with uniform (constant) sample size except for the control limits.

Since the control limits are function of sample size (n), these will vary with the sample size. So for the p-chart for variable sample size, the control limits are calculated separately for each sample. The control limits in such a situation are known as **variable control limits**.

Generally, two approaches are used for constructing variable control limits for the p-chart.

First Approach

According to the first approach, we calculate the control limits for each sample. Let n_i be the size of the i^{th} sample. The centre line and control limits of the i^{th} sample for the p-chart when the standard value of fraction defective P is known are given as follows:

$$CL = P \quad \dots (11a)$$

$$UCL = P + 3\sqrt{\frac{P(1-P)}{n_i}} \quad \dots (11b)$$

$$LCL = P - 3\sqrt{\frac{P(1-P)}{n_i}} \quad \dots (11c)$$

When P is unknown, it is estimated by the sample average fraction (proportion) defective, i.e., $\bar{p}$ as we have discussed in Sec. 3.3. That means, we first calculate the fraction defective for each sample using equation (1).

If $d_1, d_2, \dots, d_k$ represent the number of defectives in 1st, 2nd, ..., k^{th} samples of size $n_1, n_2, \dots, n_k$, respectively, then

$$p_1 = \frac{d_1}{n_1}, p_2 = \frac{d_2}{n_2}, \dots, p_k = \frac{d_k}{n_k} \quad \dots (12a)$$

and we calculate $\bar{p}$ as follows:

$$\bar{p} = \frac{d_1 + d_2 + \dots + d_k}{n_1 + n_2 + \dots + n_k} = \frac{\sum_{i=1}^k d_i}{\sum_{i=1}^k n_i} \quad \text{or} \quad \bar{p} = \frac{1}{k} \sum_{i=1}^k p_i \quad \dots (12b)$$

The centre line remains constant for each sample and is given by

$$CL = \bar{p} \quad \dots (13a)$$

The control limits change due to variable sample size. The control limits for the i^{th} sample are given by

$$UCL_i = \hat{P} + 3\sqrt{\frac{\hat{P}(1-\hat{P})}{n_i}} = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}} \quad \dots (13b)$$

$$LCL_i = \hat{P} - 3\sqrt{\frac{\hat{P}(1-\hat{P})}{n_i}} = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_i}} \quad \dots (13c)$$

Second Approach

According to the second approach, the control limits are calculated using the average sample size. This approach is used only when there is no large variation in the sample sizes and it is expected that the future sample sizes will not differ significantly from the average sample size. Using this approach, we get constant control limits just as we get in the case of constant sample size.

We calculate the average sample size as follows:

$$\bar{n} = \frac{n_1 + n_2 + \dots + n_k}{k} = \frac{1}{k} \sum_{i=1}^k n_i \quad \dots (14)$$

The centre line and control limits when P is known are given as follows:

$$CL = P \quad \dots (15a)$$

$$UCL = P + 3\sqrt{\frac{P(1-P)}{\bar{n}}} \quad \dots (15b)$$

$$LCL = P - 3\sqrt{\frac{P(1-P)}{\bar{n}}} \quad \dots (15c)$$

However, when P is not known, the centre line and control limits are given as

$$CL = \bar{p} \quad \dots (16a)$$

$$UCL = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} \quad \dots (16b)$$

$$LCL = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} \quad \dots (16c)$$

follows:

Note: The p-chart for variable sample size should only be used when samples of constant size are not possible, unavailable or unrealistic.

Let us explain how to construct this chart.

Example 4: In the production of tyres of an automobile, the output of a given size was inspected every day prior to the tyres being given to finished goods stores. The number of defective tyres found every day inspection is summarised in the following table:

Number of Sample	Number of Tyres Inspected	Number of Defective Tyres	Number of Sample	Number of Tyres Inspected	Number of Defective Tyres
1	650	70	11	670	71
2	510	74	12	660	75
3	600	58	13	600	77
4	590	61	14	550	78
5	630	65	15	540	64
6	650	115	16	610	90
7	700	82	17	670	96
8	740	55	18	660	110
9	580	80	19	650	78
10	600	90	20	590	60

Draw the appropriate control chart and comment on the state of the process.

Solution: Since the number of defectives is given and sample size varies, we use the p-chart for variable sample size. We shall solve this problem using both approaches described in this section.

First Approach

According to the first approach, the control limits of the p-chart are calculated for each sample.

Since the average fraction defective of the process is not given, we use equations (13a to 13c) to calculate the centre line and control limits of the i^{th} sample.

For this, we first calculate the proportion of defective tyres (p) for each sample and $\bar{p}$ using equations (12a to 12b).

Number of Sample	Number of Tyres Inspected (n)	Number of Defective Tyres (d)	Proportion of Defective Tyres (p = d/n)	CL	UCL	LCL
1	650	70	0.108	0.124	0.163	0.085
2	510	74	0.145	0.124	0.168	0.080
3	600	58	0.097	0.124	0.164	0.084
4	590	61	0.103	0.124	0.165	0.083
5	630	65	0.103	0.124	0.163	0.085
6	650	115	0.177	0.124	0.163	0.085
7	700	82	0.117	0.124	0.161	0.087
8	740	55	0.074	0.124	0.160	0.088
9	580	80	0.138	0.124	0.165	0.083
10	600	90	0.150	0.124	0.164	0.084
11	670	71	0.106	0.124	0.162	0.086
12	660	75	0.114	0.124	0.162	0.086
13	600	77	0.128	0.124	0.164	0.084
14	550	78	0.142	0.124	0.166	0.082
15	540	64	0.119	0.124	0.167	0.081
16	610	90	0.148	0.124	0.164	0.084
17	670	96	0.143	0.124	0.162	0.086
18	660	110	0.167	0.124	0.162	0.086
19	650	78	0.120	0.124	0.163	0.085
20	590	60	0.102	0.124	0.165	0.083
Total	12450	1549				

From the above table, we have

$$\bar{p} = \frac{\sum_{i=1}^k d_i}{\sum_{i=1}^k n_i} = \frac{1549}{12450} = 0.124$$

The centre line and control limits (shown in the above table) are calculated using equations (13a to 13c) as follows:

$$CL = \bar{p} = 0.124$$

$$UCL_1 = \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_1}} = 0.124 + 3\sqrt{\frac{0.124 \times (1-0.124)}{650}}$$

$$= 0.124 + 3 \times 0.013 = 0.163, \text{ and so on.}$$

$$LCL_1 = \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{n_1}} = 0.124 - 3\sqrt{\frac{0.124 \times (1-0.124)}{650}}$$

$$= 0.124 - 3 \times 0.013 = 0.085, \text{ and so on.}$$

We draw the p-chart by taking the sample number on the X-axis and the proportion defective (p) of tyres on the Y-axis as shown in Fig. 3.3.

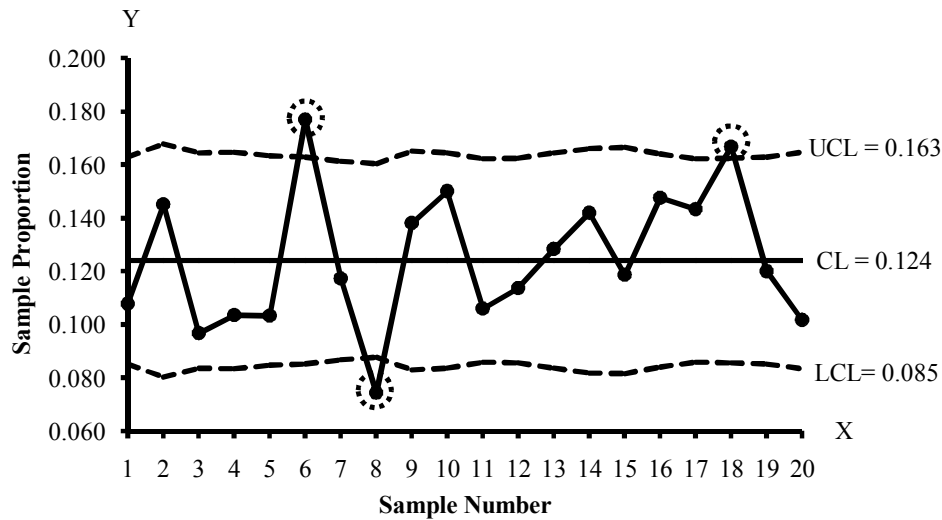


Fig. 3.3: The p-chart for proportion of defective tyres.

Interpretation of the result

From the p-chart shown in Fig. 3.3, we observe that the points corresponding to sample numbers 6, 8 and 18 lie outside the control limits. Therefore, we can say that the process is out-of-control with respect to the proportion of defective tyres and some assignable causes are present in it.

Let us now solve this example by using the second approach.

Second Approach

According to this approach, the control limits are calculated using the average sample size and we get constant control limits just as we get in the case of constant sample size. We can calculate the average sample size using equation (14) as follows:

$$\bar{n} = \frac{1}{k} \sum_{i=1}^k n_i = \frac{1}{20} \times 12450 = 622.5$$

Using equations (16a to 16c), we calculate the centre line and control limits as follows:

$$CL = \bar{p} = 0.124$$

$$\begin{aligned} UCL &= \bar{p} + 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.124 + 3\sqrt{\frac{0.124 \times (1-0.124)}{622.5}} \\ &= 0.124 + 3 \times 0.013 = 0.163 \end{aligned}$$

$$\begin{aligned} LCL &= \bar{p} - 3\sqrt{\frac{\bar{p}(1-\bar{p})}{\bar{n}}} = 0.124 - 3\sqrt{\frac{0.124 \times (1-0.124)}{622.5}} \\ &= 0.124 - 3 \times 0.013 = 0.085 \end{aligned}$$

Then, we plot the p-chart (see Fig. 3.4).

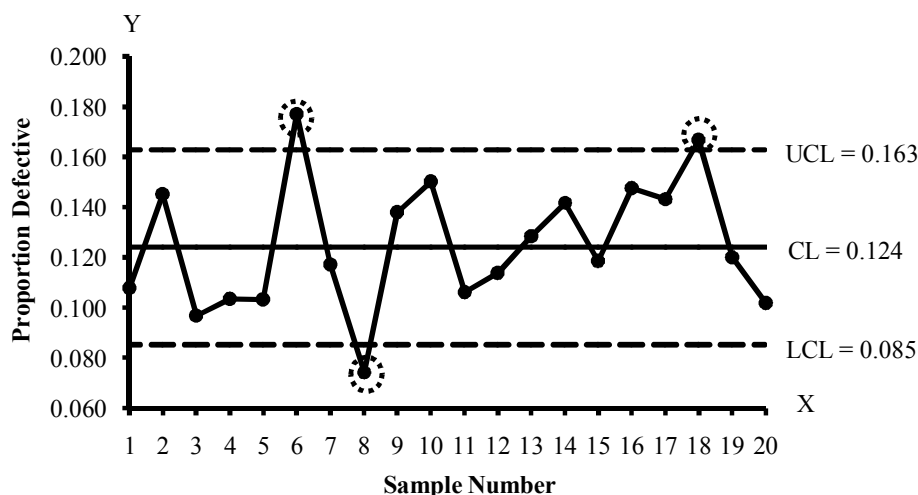


Fig. 3.4: The p-chart for proportion of defective tyres.

Interpretation of the result

From Fig. 3.4, we note that the points corresponding to sample numbers 6, 8 and 18 lie outside the control limits. Therefore, the process is out-of-control.

Hence, both approaches give the same conclusion about the process.

Now, you can try the following exercise and draw the p-chart for variable sample size.

- E6)** To monitor the manufacturing process of laptops, a quality control engineer randomly selects a number of laptops from the production line, each day over a period of 20 days. The laptops are inspected for certain defects and the number of defective laptops found each day is recorded in the following table:

Day	Number of Laptops Inspected	Number of Defective Laptops	Day	Number of Laptops Inspected	Number of Defective Laptops
1	50	4	11	55	6
2	52	5	12	60	1
3	57	6	13	55	5
4	50	10	14	55	8
5	50	4	15	52	2
6	48	3	16	48	2
7	51	4	17	50	7
8	54	7	18	56	9
9	52	8	19	52	2
10	50	4	20	53	4

Construct the p-chart for variable sample size with the help of both approaches and comment on the process.

3.5 CONTROL CHART FOR NUMBER OF DEFECTIVES (np-CHART)

In Sec. 3.4, you have learnt that for the p-chart, the fraction (proportion) defective for each sample is calculated and plotted against the sample number. Alternatively, instead of calculating the proportion defective, we can **count** the number of defective items in the samples and plot them against the sample number. The chart thus obtained is known as the np-chart.

The np-chart is an adaptation of the basic p-chart. The purpose of this chart is to provide a better understanding to the operator who may understand actual number of defective items easily rather than the proportion defective. However, there is one drawback of the np-chart: It is used only for equal sample size because the number of defectives cannot be compared for variable sample size. For example, suppose we get one defective item in a sample of size 10 and one defective item in a sample of size 100. In both samples, we get one defective item, but the first sample shows a much higher rate of problems, that is 10%, in comparison to the second.

np-chart is also known as d-chart.

The name np is due to the reason that the proportion defective p was obtained by dividing the actual number of defectives (d) by the sample size (n), i.e., $p = d/n$. So $d = np$. Therefore, the actual number of defectives may be represented by np.

The underlying logic and basic form of the np-chart are similar to the p-chart. The primary difference between the np-chart and p-chart is that instead of plotting the sample fraction defectives and monitoring their variation, we plot the number of defectives per sample and monitor them. Therefore, in this section, we describe the centre line and control limits of the np-chart.

For obtaining the control limits of the np-chart, we need the sampling distribution of the number of defectives (d). Recall the sampling distribution of the number of defectives described in Unit 2 of MST-004. Suppose the items or units of a process (population) are classified into two mutually exclusive groups as defective (success) and non-defective (failure) and a random sample of size n is drawn from it. Then the number of defectives follows a binomial distribution with mean nP and variance nPQ where P is the fraction or proportion of defectives in the process (population) and $Q = 1 - P$:

$$E(d) = nP \text{ and } \text{Var}(d) = nP(1 - P) \quad \dots (17)$$

We also know that the standard error of a random variable (X) is

$$SE(X) = \sqrt{\text{var}(X)}$$

Therefore, the standard error of the number of defectives is given by

$$SE(d) = \sqrt{\text{Var}(d)} = \sqrt{nP(1 - P)} \quad \dots (18)$$

The sampling distribution of number of defectives is not a normal distribution. However, if the sample size is sufficiently large, you know from the **centre limit theorem** that the sampling distribution of number of defectives is approximately normally distributed with mean nP and variance nPQ. Therefore, the centre line and control limits for the np-chart are given as follows:

$$\text{Centre line (CL)} = E(d) = nP \quad \dots (19a)$$

$$\text{Upper control limit (UCL)} = E(d) + 3SE(d) = nP + 3\sqrt{nP(1 - P)} \quad \dots (19b)$$

$$\text{Lower control limit (LCL)} = E(d) - 3SE(d) = nP - 3\sqrt{nP(1 - P)} \quad \dots (19c)$$

In practice, the fraction (proportion) defective (P) of the process is not known. Therefore, it is necessary to estimate it by the average sample fraction defective as we have discussed in Sec.3.3.

If we draw k samples of same size n and $d_1, d_2, \dots, d_k$ are the numbers of defective items in 1st, 2nd, ..., kth sample, respectively, we can calculate the average fraction defective of the samples as follows:

$$\bar{p} = \frac{\text{Sum of defective items}}{\text{Total number of items inspected}} = \frac{1}{nk} \sum_{i=1}^k d_i \quad \dots (20)$$

Hence, in this case, the centre line and control limits of the np-chart are obtained by replacing P by $\bar{p}$ in equations (19a to 19c) as follows:

$$CL = E(d) = n\hat{P} = n\bar{p} \quad \dots (21a)$$

$$UCL = E(d) + 3SE(d) = n\hat{P} + 3\sqrt{n\hat{P}(1-\hat{P})} = n\bar{p} + 3\sqrt{n\bar{p}(1-\bar{p})} \quad \dots (21b)$$

$$LCL = E(d) - 3SE(d) = n\hat{P} - 3\sqrt{n\hat{P}(1-\hat{P})} = n\bar{p} - 3\sqrt{n\bar{p}(1-\bar{p})} \quad \dots (21c)$$

Having set the centre line and control limits, we construct the np-chart by taking the sample number on the X-axis and the number of defective items (d) in the sample on the Y-axis. We draw the control line as a solid line and UCL and LCL as dotted lines. Then we plot the number of defective items for each sample against the sample number. The consecutive sample points are joined by line segments.

Interpretation of the result

If all sample points lie on or in between upper and lower control limits, the control chart indicates that the process is under statistical control, that is, only chance causes are present in the process. If one or more points lie outside the upper or lower control limits, the control chart alarms (indicates) that the process is not under statistical control and some assignable causes are present in the process.

To bring the process under statistical control it is necessary to investigate the assignable causes and take corrective action to eliminate them. Once the assignable causes are eliminated, we delete the out-of-control points and calculate the revised centre line and control limits for the np-chart by using the remaining samples. These limits are known as the **revised control limits**. To obtain the revised limits for the np-chart, we first calculate the new $\bar{p}$ as follows:

$$\bar{p}_{\text{new}} = \frac{\sum_{i=1}^k d_i - \sum_{j=1}^d d_j}{n(k-d)} \quad \dots (22)$$

where d – the number of discarded samples, and

$\sum_{j=1}^d d_j$ – the sum of number of defectives in the discarded samples.

We then reconstruct the centre line and control limits of the chart by replacing $\bar{p}$ by $\bar{p}_{new}$ in equations (21 to 21c) as follows:

$$CL = n\bar{p}_{new} \quad \dots (23a)$$

$$UCL = n\bar{p}_{new} + 3\sqrt{n\bar{p}_{new}(1-\bar{p}_{new})} \quad \dots (23b)$$

$$LCL = n\bar{p}_{new} - 3\sqrt{n\bar{p}_{new}(1-\bar{p}_{new})} \quad \dots (23c)$$

Let us explain how to draw the np-chart with the help of examples.

Example 5: To monitor the manufacturing process of laptops, a quality control engineer randomly selects 50 laptops from the production line, each day over a period of 20 days. The laptops are inspected for certain defects and the number of defective laptops found each day is recorded in the following table:

Day	Number of Laptops Inspected	Number of Defective Laptops	Day	Number of Laptops Inspected	Number of Defective Laptops
1	50	4	11	50	6
2	50	8	12	50	1
3	50	6	13	50	5
4	50	10	14	50	3
5	50	4	15	50	2
6	50	3	16	50	3
7	50	4	17	50	7
8	50	7	18	50	9
9	50	8	19	50	2
10	50	4	20	50	4

Construct the appropriate control chart and state whether the process is in control.

Solution: In this case, the quality control engineer visually inspects each selected laptop to determine whether it is defective or not. So we use the control chart for attributes. Since the number of defectives is given, we can use p or np-chart. But it is convenient to use the np-chart.

We do not know the process fraction defective. Therefore, we use equations (21a to 21c) to calculate the centre and control limits for the np-chart.

We are given that

the number of laptops inspected in each sample (n) = 50 and

the number of samples (k) = 20.

The total number of defectives is $\sum_{i=1}^k d_i = 4 + 8 + \dots + 2 + 4 = 100$.

We calculate $\bar{p}$ from equation (20) as follows:

$$\bar{p} = \frac{1}{nk} \sum_{i=1}^k d_i = \frac{1}{50 \times 20} \times 100 = 0.10$$

UNIT 4 CONTROL CHARTS FOR DEFECTS

Structure

- 4.1 Introduction
 - Objectives
- 4.2 Control Charts for Number of Defects (c-Chart)
- 4.3 Control Charts for Number of Defects per Unit (u-Chart)
- 4.4 Comparison between Control Charts for Variables and Attributes
- 4.5 Summary
- 4.6 Solutions/Answers

4.1 INTRODUCTION

In Unit 2, you have studied the $\bar{X}$, R and S control charts used for monitoring and controlling measurable quality characteristics such as diameter, length, width, weight, etc. In Unit 3, you have studied the p and np control charts applied for monitoring and controlling quality characteristics which are not measurable and the items or units are classified as defective or non-defective. These types of charts are broadly useful in statistical quality control.

Sometimes, an item may contain many defects, but it may not be classified as defective. For example, suppose a company manufactures motorcycles. Suppose there are one or two minor scratches on the body of the motorcycle, which do not affect its functioning. So the motorcycle cannot be classified as defective. However, this is a defect. But if there are too many scratches, the motorcycle should be classified as defective because the scratches would be noticeable and affect its sale. So there are so many situations in industries in which we require to control the number of defects rather than the number of defective items. The c-chart and u-chart are used to control the number of defects.

In Secs. 4.2 and 4.3, we discuss the c-chart which is used to control the number of defects and its applications. We describe the u-chart in Sec 4.4. It is used to monitor and control the number of defects per unit when the sample size varies. In Sec 4.5, we compare the control charts for variables and attributes.

Objectives

After studying this unit, you should be able to:

- distinguish between defective and defect;
- describe the need of control charts for defects;
- select the appropriate control chart for defects;
- construct and interpret the control chart for number of defects (c-chart); and
- construct and interpret the control chart for number of defects per unit (u-chart).

4.2 CONTROL CHARTS FOR NUMBER OF DEFECTS (c-CHART)

You have learnt about the p-chart and np-chart in Unit 3. These control charts are used to control the fraction (proportion) defective and the number of defectives, respectively, on the basis of criterion: whether the item is defective or non-defective. But in many situations, it is advantageous economically to know the **number of defects** in an item instead of classifying it as defective or non-defective. For example, a glass bottle may have one small air bubble and the another glass bottle may have many small air bubbles. Even though both bottle sets are defective, the second one is far more serious as it has many defects. So it is also important to control the number of defects within the item. The c-chart is used for monitoring and controlling the number of defects in an item/unit. Traditionally, the number of defects was denoted by c . Therefore, the control chart for number of defects is known as the c-chart. It is used when the item/unit is largely assembled, e.g., TV, computer, mobile, laptop, aircraft, etc. In such cases, there are many opportunities for a defect to occur and the probability of the occurrence of a defect is very small. For example, in the case of commercial airplane there is a large number of rivets, but a small chance of any rivet having defects.

The sample/subgroup size for the c-chart is a single item/unit/article or constant size and we count the number of defects within the item/unit/article.

The underlying logic and procedure of constructing the c-chart are similar to the np-chart. The primary difference between the c-chart and the np-chart is that, instead of collecting data relating to the number of defective items, we collect data about the number of defects within the item and plot the number of defects against the sample number.

For obtaining the centre line and control limits of the c-chart, we require the sampling distribution of the number of defects. In this case, when there are many opportunities for a defect to occur and the probability of the occurrence of a defect is very small, the Poisson distribution is applied (recall Unit 10 of MST-003 entitled Poisson Distribution). So the number of defects follows the Poisson distribution. If λ is the average number of defects in the process, the mean and variance of the Poisson distribution are given as follows:

$$E(c) = \lambda \text{ and } \text{Var}(c) = \lambda \quad \dots (1)$$

We know the standard error of a random variable X is

$$SE(X) = \sqrt{\text{Var}(X)}$$

$$\therefore SE(c) = \sqrt{\text{Var}(c)} = \sqrt{\lambda} \quad \dots (2)$$

In Unit 10 of MST-003, you have also learnt that the Poisson distribution is not symmetrical. Therefore, the upper and lower 3σ limits do not correspond to equal probabilities of a point on the control chart. But if the sample size is sufficiently large, then by the centre limit theorem, the sampling distribution of the number of defects is approximately normally distributed with mean λ and variance λ . Therefore, the centre line and control limits for the c-chart can be obtained as follows:

$$\text{Centre line (CL)} = E(c) = \lambda \quad \dots (3a)$$

$$\text{Upper control limit (UCL)} = E(c) + 3SE(c) = \lambda + 3\sqrt{\lambda} \quad \dots (3b)$$

$$\text{Lower control limit (LCL)} = E(c) - 3SE(c) = \lambda - 3\sqrt{\lambda} \quad \dots (3c)$$

Generally, λ is not known. So it is estimated from the sample information. Suppose we draw k samples of constant size or size 1 and $c_1, c_2, \dots, c_k$ are the numbers of defects in the 1st, 2nd, ..., k^{th} sample, respectively. The average number of defects in the process is estimated by the average number of defects in the sample which is calculated by the formula given below:

$$\bar{c} = \frac{\text{Sum of defects}}{\text{Total number of samples inspected}} = \frac{1}{k} \sum_{i=1}^k c_i \quad \dots (4)$$

In this case, centre line and control limits of the c -chart are obtained by replacing λ by $\bar{c}$ in equations (3a to 3c) as follows:

$$\text{Centre line (CL)} = \hat{\lambda} = \bar{c} \quad \dots (5a)$$

$$\text{Upper control limit (UCL)} = \hat{\lambda} + 3\sqrt{\hat{\lambda}} = \bar{c} + 3\sqrt{\bar{c}} \quad \dots (5b)$$

$$\text{Lower control limit (UCL)} = \hat{\lambda} - 3\sqrt{\hat{\lambda}} = \bar{c} - 3\sqrt{\bar{c}} \quad \dots (5c)$$

We construct the c -chart by taking the sample number on the X-axis and the number of defects in the sample on the Y-axis. We draw the centre line as a solid line and control limits as dotted lines on the chart. We plot the number of defects in the sample against the sample number. The consecutive sample points are joined by line segments.

Interpretation of the result

If all sample points lie on or between the upper and lower control limits, the control chart indicates that the process is under statistical control. Only chance causes are present in the process and no assignable cause is present. However, if one or more sample points lie outside the upper or lower control limit, the control chart alarms (indicates) that the process is not under statistical control and some assignable causes are present in the process.

To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them. Once the assignable causes are eliminated, we delete the out-of-control points (samples) and calculate the revised centre line and control limits for the c -chart by using the remaining samples. These limits are known as the **revised control limits**. For the revised limits of the c -chart, we first calculate new $\bar{c}$ as follows:

$$\bar{c}_{\text{new}} = \frac{\sum_{i=1}^k c_i - \sum_{j=1}^d c_j}{k - d} \quad \dots (6)$$

where d – the number of discarded samples, and

$\sum_{j=1}^d c_j$ – the sum of the number of defects within the discarded samples.

After finding the $\bar{c}_{\text{new}}$, we reconstruct the centre line and control limits for the c-chart by replacing $\bar{c}$ by $\bar{c}_{\text{new}}$ in equations (5a to 5c) as follows:

$$\text{Centre line (CL)} = \bar{c}_{\text{new}} \quad \dots (7a)$$

$$\text{Upper control limit (UCL)} = \bar{c}_{\text{new}} + 3\sqrt{\bar{c}_{\text{new}}} \quad \dots (7b)$$

$$\text{Lower control limit (UCL)} = \bar{c}_{\text{new}} - 3\sqrt{\bar{c}_{\text{new}}} \quad \dots (7c)$$

Note: If the value of lower control limit is negative, we change the lower control limit to zero because a negative number of defects is not possible.

Let us take up some examples to illustrate how to construct the c-chart for practical situations.

Example 1: A control chart is to be formed for a process in which laptops are produced. The inspection unit is one laptop and control chart for the number of defects is to be used. Preliminary data are recorded and 45 defects are found in 30 laptops. Obtain the control limits for the chart.

Solution: Since we need to control the number of defects and the inspection unit is one laptop, we use the c-chart for number of defects.

The average number of defects in the process is not given. So, we use equations (5a to 5c) to calculate the centre line and control limits of the c-chart.

It is given that

the total number of defects in laptops = 45, and

the total number of laptops inspected (k) = 30.

$$\therefore \bar{c} = \frac{1}{k} \sum_{i=1}^k c_i = \frac{1}{30} \times 45 = 1.5$$

Hence, using equations (5a to 5c), we calculate the centre line and control limits of the c-chart as follows:

$$\text{CL} = \bar{c} = 1.5$$

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 1.5 + 3 \times \sqrt{1.5} = 1.5 + 3 \times 1.225 = 5.175$$

$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} = 1.5 - 3 \times \sqrt{1.5} = 1.5 - 3 \times 1.225 = -2.175 \approx 0$$

Example 2: The number of scratch marks on a particular piece of furniture is recorded. The data for 20 samples are given below:

Sample Number	1	2	3	4	5	6	7	8	9	10
Scratch Mark	6	3	14	7	2	5	12	4	7	3
Sample Number	11	12	13	14	15	16	17	18	19	20
Scratch Mark	2	7	6	8	4	10	5	4	13	9

Draw the appropriate control chart and write the comments about the state of the process when:

- i) the management sets a goal of 5 scratch marks on an average per piece.
- ii) the management does not set the average number of marks per piece.

Solution: Here the number of defects on a particular piece of furniture is given. So we use the c-chart.

- i) The management sets a goal of 5 scratch marks on an average per piece. It means that the average number of defects in the process is known. Therefore, we use equations (3a to 3c) to calculate the centre line and control limits of the c-chart.

It is given that

the average number of defects per sample (λ) = 5, and

the number of items in the sample (n) = 1.

Therefore, we calculate the centre line and control limits of the c-chart using equations (3a to 3c) as follows:

$$CL = \lambda = 5$$

$$UCL = \lambda + 3\sqrt{\lambda} = 5 + 3 \times \sqrt{5} = 5 + 3 \times 2.236 = 11.708$$

$$LCL = \lambda - 3\sqrt{\lambda} = 5 - 3 \times \sqrt{5} = 5 - 3 \times 2.236 = -1.708 \approx 0$$

For constructing the c-chart, we plot the points by taking the sample number on the X-axis and the number of defects (c) on the Y-axis. We draw the centre line as a solid line and control limits as dotted lines on the graph as shown in Fig. 4.1.

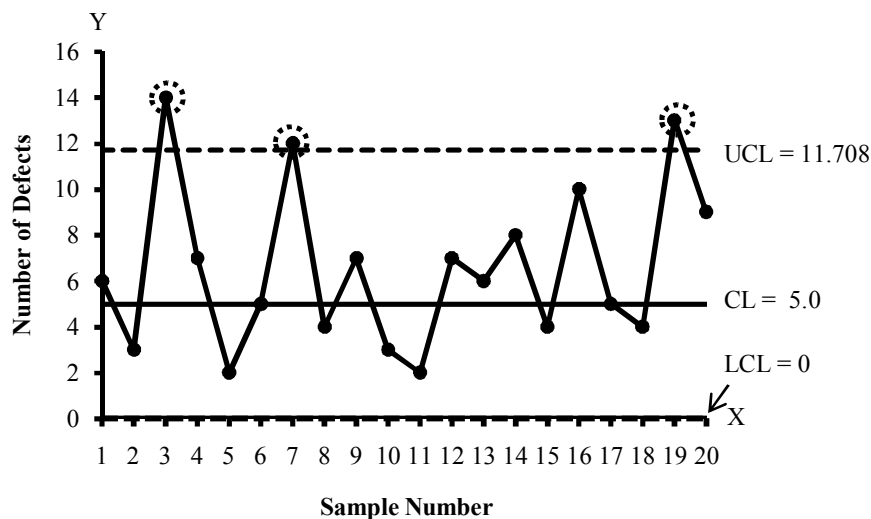


Fig. 4.1: The c-chart for number of defectes on an average per piece of the furniture, which is 5.

Interpretation of the result

From Fig. 4.1, we observe that the points corresponding to the sample numbers 3, 7 and 19 lie outside the upper control limit. Therefore, the process is not under statistical control with respect to the average number of defects per piece, which is 5.

- ii) Here, the average number of defects in the process is not given. So we use equations (5a to 5c) to calculate the centre line and control limits of the c-chart.

Here total number of defects = $6 + 3 + \dots + 13 + 9 = 131$.

$$\therefore \bar{c} = \frac{1}{k} \sum_{i=1}^k c_i = \frac{1}{20} \times 131 = 6.55$$

Therefore, using equations (5a to 5c), we calculate the centre line and control limits of the c-chart as follows:

$$CL = \bar{c} = 6.55$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 6.550 + 3 \times \sqrt{6.55} = 6.550 + 3 \times 2.559 = 14.227$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 6.550 - 3 \times \sqrt{6.55} = 6.550 - 3 \times 2.559 = -1.127 \approx 0$$

We construct the c-chart by taking the sample number on the X-axis and the number of defects (c) on the Y-axis as shown in Fig. 4.2.

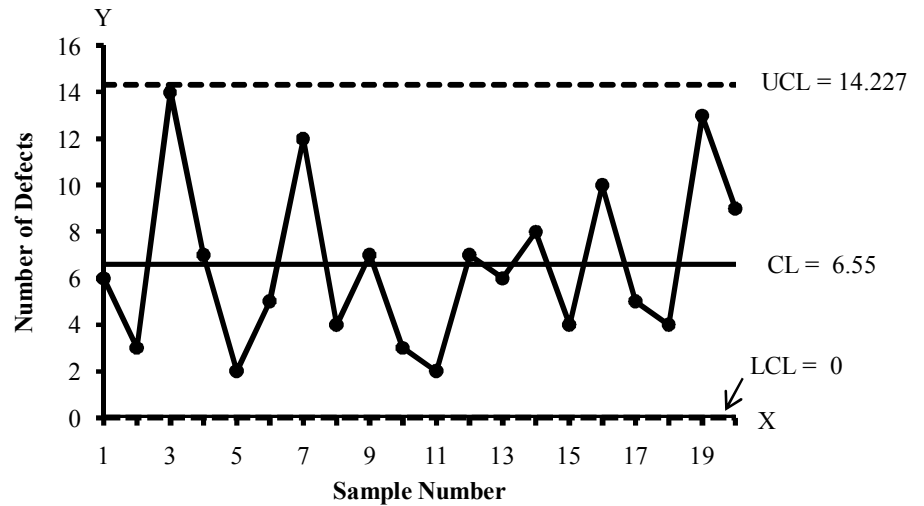


Fig. 4.2: The c-chart for number of defects on an average per piece of the furniture, which is 6.55.

Interpretation of the result

From the c-chart (shown in Fig. 4.2), we observe that no point lies outside the control limits and there is no specific pattern of the sample points on the chart. So it indicates that the process is under control with respect to the average number of defects per piece, which is 6.55.

Example 3: As part of an overall quality improvement programme, a textile manufacturer decides to monitor the number of defects found in each inspected bolt (large bundle) of cloth. The data from 20 inspections are recorded in the table given below:

Bolt of Cloth	1	2	3	4	5	6	7	8	9	10
Number of Defects	10	19	5	9	2	8	7	13	3	2
Bolt of Cloth	11	12	13	14	15	16	17	18	19	20
Number of Defects	22	4	6	9	7	2	5	12	4	2

- i) Which control chart should be used in this case? Calculate the control limits for this chart.
- ii) Do these data from a controlled process? If not, calculate the revised control limits.

Solution:

- i) Here the manufacturer monitors the number of defects found in each inspected bolt of cloth and only one bolt is inspected in each sample. So we can use the c-chart. The average number of defects in the bolt of cloth is not given in this case. So we use equations (5a to 5c) to calculate the centre line and control limits of the c-chart.

It is given that $k = 20$ and $n = 1$.

The total number of defects in bolts = $10 + 19 + \dots + 2 = 151$

$$\therefore \bar{c} = \frac{1}{k} \sum_{i=1}^k c_i = \frac{1}{20} \times 151 = 7.550$$

Hence, using equations (5a to 5c), we calculate the centre line and control limits of the c-chart as follows:

$$CL = \bar{c} = 7.550$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 7.550 + 3 \times \sqrt{7.550} = 7.550 + 3 \times 2.748 = 15.794$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 7.550 - 3 \times \sqrt{7.550} = 7.550 - 3 \times 2.748 = -0.694 \approx 0$$

We construct the c-chart by taking the sample number on the X-axis and the number of defects (c) on the Y-axis as shown in Fig. 4.3.

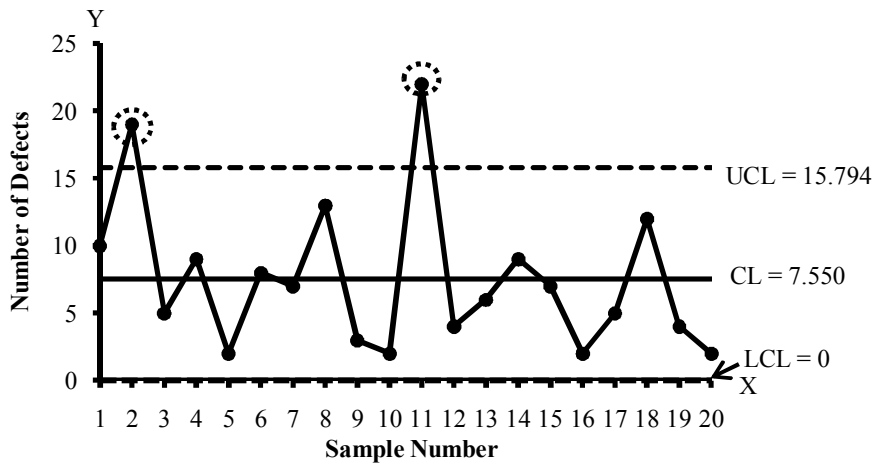


Fig. 4.3: The c-chart for number of defects in the bolt of cloth.

Interpretation of the result

From Fig. 4.3, we observe that the points corresponding to the sample numbers (bolt) 2 and 11 lie outside the upper control limit. Therefore, the process is not under statistical control with respect to the number of defects. To bring the process under statistical control, it is necessary to investigate the assignable causes and take corrective action to eliminate them.

- ii) For calculating the revised control limits for the c-chart, we first delete the out-of-control points (2 and 11) and then calculate the new $\bar{c}$ by using remaining samples.

In our case, $k = 20, d = 2, \sum_{j=1}^d c_j = 19 + 22 = 41$

$$\bar{c}_{\text{new}} = \frac{\sum_{i=1}^k c_i - \sum_{j=1}^d c_j}{k - d} = \frac{151 - 41}{20 - 2} = 6.111$$

After finding the $\bar{p}_{\text{new}}$, we calculate the revised centre line and control limits of the chart using equations (7a to 7c) as follows:

$$CL = \bar{c}_{\text{new}} = 6.111$$

$$UCL = \bar{c}_{\text{new}} + 3\sqrt{\bar{c}_{\text{new}}} = 6.111 + 3 \times \sqrt{6.111} = 6.111 + 3 \times 2.472 = 13.527$$

$$LCL = \bar{c}_{\text{new}} - 3\sqrt{\bar{c}_{\text{new}}} = 6.111 - 3 \times \sqrt{6.111} = 6.111 - 3 \times 2.472 = -1.305 \approx 0$$

So far you have studied the c-chart and learnt when it is used and how it is constructed. We now discuss the application of the c-chart.

Applications of the c-chart

Although the applications of the c-chart are limited compared to the $\bar{X}$, R, p and np-charts, yet a number of practical situations exist where we apply the c-chart. Some area where the c-chart is applied are listed below:

1. c is the number of defects observed in a television, computer, laptop, mobile, etc.
2. c is the number of defects observed in a woollen carpet, cloth, paper, etc.
3. c is the number of air bubbles in a glass, bottle, paper weight, etc.
4. c is the number of defects observed in aircraft engines.
5. c is the number of breakdowns at weak spots in insulation in a given length of insulated wire.
6. c is the number of imperfections observed in a bolt (large bundle) of cloth.
7. c is the number of surface defects observed in a galvanised sheet, furniture, automobile, etc.
8. The c-chart is also used in
 - a) chemical laboratories,
 - b) accident statistics both in highway accidents and industrial accidents.

You may like to check your understanding of the c-chart by answering the following exercises.

E1) Choose the correct option from the following:

- i) The control chart for number of defects is called the
 - a) $\bar{X}$ -chart b) p-chart c) np-chart d) c-chart
- ii) The control limits for the c-chart are

$$\text{a) } \lambda \pm 3\sqrt{\frac{\lambda}{n}} \quad \text{b) } \lambda \pm 3\sqrt{\lambda}$$

c) $\lambda \pm \sqrt{\frac{\lambda}{n}}$ d) $n\lambda \pm 3\sqrt{n\lambda}$

iii) The control chart for number of defects per unit is called the

- a) u-chart b) p-chart c) np-chart d) c-chart

E2) Write the control limits of the c-chart when the average number of defects of the process is not known.

E3) Twenty five samples of packets of match boxes containing 10 match boxes each, were selected at regular intervals. The match boxes in each sample were inspected and the number of defects with regard to the number of broken sticks, paste on the sticks, labeling, etc. were noted in the following table:

Sample Number	1	2	3	4	5	6	7	8	9	10	11	12	13
Number of Defects	7	3	0	5	8	2	4	5	6	3	8	2	4
Sample Number	14	15	16	17	18	19	20	21	22	23	24	25	
Number of Defects	9	6	7	10	3	8	4	2	10	4	1	3	

Comment on the state of the process.

E4) A quality control technician has recorded the number of defects per 100 square meter on white papers. The results of 25 inspections are shown in the following table:

Sample Number	Number of Defects	Sample Number	Number of Defects	Sample Number	Number of Defects
1	6	10	6	18	7
2	8	11	10	19	6
3	12	12	9	20	10
4	5	13	6	21	5
5	10	14	8	22	8
6	18	15	20	23	10
7	5	16	5	24	8
8	8	17	9	25	7
9	8				

Construct a control chart for the number of defects. Revise the control limits, assuming special causes for the out-of-control.

4.3 CONTROL CHARTS FOR NUMBER OF DEFECTS PER UNIT (u-CHART)

In Sec. 4.2, you have learnt that the c-chart is used for monitoring and controlling the number of defects. But it is applied only when the sample (subgroup) size is one or constant, i.e., we have inspected the same number of units or items in each sample. To overcome this drawback, we use the u-chart.

When the sample size varies due to some reason, such as machinery, raw material, workers, etc., the u-chart is constructed to monitor the number of defects per unit. The u-chart can also be used when the **sample size is constant**.

The procedure for drawing the u-chart for variable sample size is similar to the c-chart with constant size. The primary difference between the c-chart and u-chart is that instead of plotting the number of defects per sample, we plot the number of defects per item/unit and monitor them.

Since the control limits are function of the sample size (n), these will vary with the sample size. So we should calculate the control limits separately for each sample. In such cases, the control limits are known as **variable control limits**.

Generally, two approaches are used for constructing variable control limits for the u-chart.

First Approach

In this approach, we calculate the control limits for each sample. Suppose $c_1, c_2, \dots, c_k$ represent the number of defects in 1st, 2nd, ..., kth samples of size $n_1, n_2, \dots, n_k$, respectively, we first calculate the number of defects per item for each sample as follows:

$$u_1 = \frac{c_1}{n_1}, u_2 = \frac{c_2}{n_2}, \dots, u_k = \frac{c_k}{n_k} \quad \dots (8)$$

The average number of defects per item is given by

$$\bar{u} = \frac{1}{k} \sum_{i=1}^k u_i \quad \text{or} \quad \bar{u} = \frac{\sum_{i=1}^k c_i}{\sum_{i=1}^k n_i} \quad \dots (9)$$

The centre line remains constant for each sample because it is not a function of sample size and is given by

$$CL = \bar{u} \quad \dots (10a)$$

The control limits change due to variable sample size. The control limits for the ith sample are given by

$$UCL_i = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_i}} \quad \dots (10b)$$

$$LCL_i = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_i}} \quad \dots (10c)$$

Second Approach

In this approach, we calculate the control limits by using the average sample size. This approach is used only when there is no large variation in the sample sizes and it is expected that the future sample sizes will not differ significantly from the average sample size. Using this approach, we get constant control

limits just as we get in the case of constant sample size. We calculate the average sample size as follows:

$$\bar{n} = \frac{n_1 + n_2 + \dots + n_k}{k} = \frac{1}{k} \sum_{i=1}^k n_i \quad \dots (11)$$

$$CL = \bar{u} \quad \dots (12a)$$

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{\bar{n}}} \quad \dots (12b)$$

$$LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{\bar{n}}} \quad \dots (12c)$$

Therefore, the centre line and control limits of the u-chart are as follows:

The construction and interpretation of the u-chart are similar to that of the c-chart.

We now consider an example to illustrate the use of the u-chart.

Example 4: Twenty samples of carpets are inspected for varying sample size and the number of defects in each sample is noted in the following table:

Sample Number	Number of Carpet Inspected	Number of Defects	Sample Number	Number of Carpet Inspected	Number of Defects
1	25	12	11	20	9
2	20	5	12	20	12
3	25	7	13	15	14
4	15	7	14	25	6
5	25	10	15	20	7
6	15	4	16	25	12
7	20	6	17	15	5
8	15	2	18	25	6
9	15	4	19	15	8
10	25	10	20	25	4

Construct a suitable control chart for the number of defects per carpet.

Solution: Since the number of defects is given for varying sample size, we use the u-chart.

We solve this example using both approaches.

First Approach

We use equations (10a to 10c) to calculate the centre line and control limits of the i^{th} sample.

To calculate the control limits and draw the u-chart, we first calculate the number of defects per carpet using equation (8) and then $\bar{u}$.

Sample Number	Number of Carpet Inspected (n)	Number of Defects (c)	$u = c/n$	CL	UCL	LCL
1	25	12	0.480	0.370	0.736	0.004
2	20	5	0.250	0.370	0.778	- 0.038 □ 0
3	25	7	0.280	0.370	0.736	0.004
4	15	7	0.467	0.370	0.841	- 0.101 □ 0
5	25	10	0.400	0.370	0.736	0.004
6	15	4	0.267	0.370	0.841	- 0.101 □ 0
7	20	6	0.300	0.370	0.778	- 0.038 □ 0
8	15	2	0.133	0.370	0.841	- 0.101 □ 0
9	15	4	0.267	0.370	0.841	- 0.101 □ 0
10	25	10	0.400	0.370	0.736	0.004
11	20	9	0.450	0.370	0.778	- 0.038 □ 0
12	20	12	0.600	0.370	0.778	- 0.038 □ 0
13	15	14	0.933	0.370	0.841	- 0.101 □ 0
14	25	6	0.240	0.370	0.736	0.004
15	20	7	0.350	0.370	0.778	- 0.038 □ 0
16	25	12	0.480	0.370	0.736	0.004
17	15	5	0.333	0.370	0.841	- 0.101 □ 0
18	25	6	0.240	0.370	0.736	0.004
19	15	8	0.533	0.370	0.841	- 0.101 □ 0
20	25	4	0.160	0.370	0.736	0.004
Total	405	150				

From the above table and equation (9), we have

$$\bar{u} = \frac{\sum_{i=1}^k c_i}{\sum_{i=1}^k n_i} = \frac{150}{405} = 0.370$$

Therefore, using equations (10a to 10c), we calculate the centre line and control limits which are shown in the table as follows:

$$CL = \bar{u} = 0.370$$

$$UCL_1 = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_1}} = 0.370 + 3\sqrt{\frac{0.370}{25}} \\ = 0.370 + 3 \times 0.122 = 0.736, \text{ and so on}$$

$$LCL_1 = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_1}} = 0.370 - 3\sqrt{\frac{0.370}{25}} \\ = 0.370 - 3 \times 0.122 = 0.004, \text{ and so on}$$

We construct the u-chart by taking the sample number on the X-axis and the number of defects per carpet (u) on the Y-axis. We draw the centre line as a solid

line and control limits as dotted lines on the graph and plot the points as shown in Fig. 4.4.

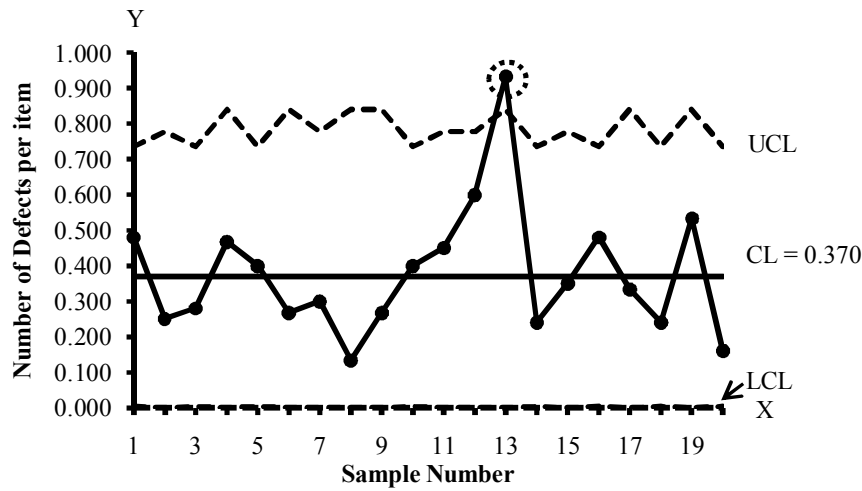


Fig. 4.4: The u-chart for number of defects per carpet.

Interpretation of the result

From the u-chart shown in Fig. 4.4, we observe that the point corresponding to sample 13 lies outside the upper control limit. Therefore, the process is not under statistical control with respect to the number of defects per carpet.

Second Approach

According to this approach, the control limits are calculated using the average sample size and we get constant control limits. We first calculate the average sample size using equation (11) as follows:

$$\bar{n} = \frac{1}{k} \sum_{i=1}^k n_i = \frac{1}{20} \times 405 = 20.25$$

Using equations (12a to 12c), we calculate the centre line and control limits as follows:

$$CL = \bar{u} = 0.370$$

$$UCL = \bar{u} + 3\sqrt{\frac{\bar{u}}{\bar{n}}} = 0.370 + 3\sqrt{\frac{0.370}{20.25}} = 0.370 + 3 \times 0.135 = 0.775$$

$$LCL = \bar{u} - 3\sqrt{\frac{\bar{u}}{\bar{n}}} = 0.370 - 3\sqrt{\frac{0.370}{20.25}} = 0.370 - 3 \times 0.135 = -0.035 \approx 0$$

We draw the u-chart in the same way as in the first approach (see Fig. 4.5).

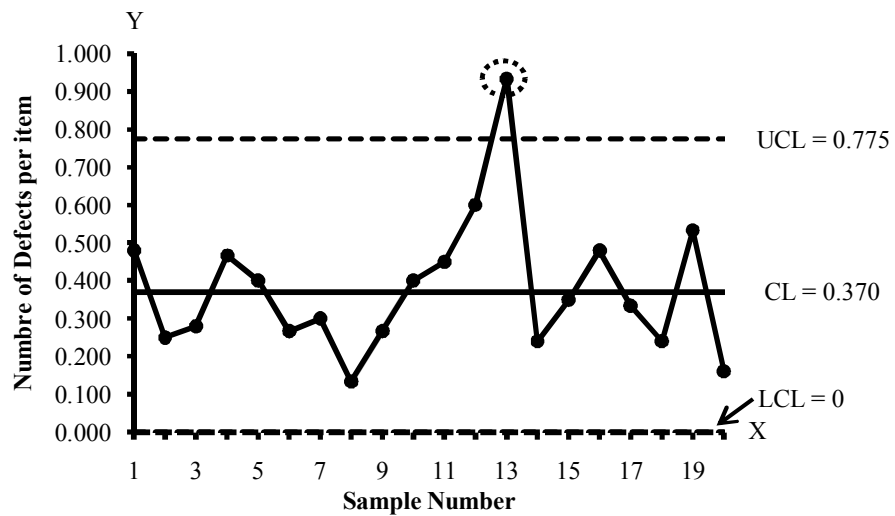


Fig. 4.5: The u-chart for number of defects per carpet.

Interpretation of the result

From the u-chart shown in Fig. 4.5, we observe that the point corresponding to sample 13 lies outside the upper control limit. Therefore, the process is not under statistical control with respect to the number of defects per carpet.

You can check your understanding of the u-chart by answering the following exercises.

- E6) A quality control technician notes the number of defects per 100 square meter on white paper, but the amount of paper inspected for each sample varies. The results of 25 inspections are shown in the following table:

Sample Number	Amount of the Paper Inspected (in square meter)	Number of Defects	Sample Number	Amount of the Paper Inspected (in square meter)	Number of Defects
1	300	7	14	250	8
2	200	8	15	300	6
3	250	5	16	250	5
4	150	5	17	150	9
5	250	10	18	200	7
6	100	4	19	150	6
7	200	5	20	300	10
8	150	8	21	200	5
9	150	8	22	250	8
10	250	6	23	200	5
11	200	5	24	100	8
12	250	9	25	150	5
13	100	6			

Construct a control chart for the number of defects per 100 square meter.

4.4 COMPARISON BETWEEN CONTROL CHARTS FOR VARIABLES AND ATTRIBUTES

Now that you have studied the control charts for variables and attributes, you know which type of control chart to use in a given situation. In the table below, we present a comparison of the control charts for variables and attributes.

S. No.	Control Charts for Variables	Control Charts for Attributes
1	These charts are used for measurable characteristics.	These charts are used for non-measurable characteristics and the situations where the control charts for variables are not applied due to time and cost factors.
2	The types of charts are the $\bar{X}$ -chart, R-chart and S-chart.	The types of charts are the p-chart, np-chart, c-chart and u-chart.
3	A separate control chart is needed for each quality characteristic.	A single chart is enough for a number of quality characteristics.
4	Small samples serve the purpose very well.	Large samples are required for correct conclusions.
5	The cost of inspection of units is large.	The cost of inspection of units is small since no measurements are made.
6	These charts provide the better quality control.	These charts are not very effective in quality control.
7	These charts are not easily understood.	These charts are easily understood.

With this we end the unit and summarise our discussion.

4.5 SUMMARY

1. The c-chart is used when we want to control the number of defects and sample size is one unit or constant.
2. The u-chart is used when we want to control the number of defects per unit and sample size varies.

3. The centre line and control limits of the c-chart are given as

$$\text{Centre line (CL)} = \bar{c}$$

$$\text{Upper control limit (UCL)} = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{Lower control limit (UCL)} = \bar{c} - 3\sqrt{\bar{c}}$$

$$\text{where } \bar{c} = \frac{\text{Sum of defects}}{\text{Total number of samples inspected}} = \frac{1}{k} \sum_{i=1}^k c_i.$$

4. The centre line and control limits of the u-chart are given as

$$\text{Centre line (CL)} = \bar{u}$$

$$\text{Upper control limit (UCL)} = \bar{u} + 3\sqrt{\frac{\bar{u}}{\bar{n}}}$$

$$\text{Lower control limit (UCL)} = \bar{u} - 3\sqrt{\frac{\bar{u}}{\bar{n}}}$$

$$\text{where } \bar{u} = \frac{1}{k} \sum_{i=1}^n u_i \text{ or } \bar{u} = \frac{\sum_{i=1}^n c_i}{\sum_{i=1}^n n_i} \text{ and } \bar{n} = \frac{n_1 + n_2 + \dots + n_k}{k} = \frac{1}{k} \sum_{i=1}^k n_i$$

4.6 SOLUTIONS / ANSWERS

- E1)** i) Option (d) is the correct option because we know that the c-chart and u-chart are the control charts for number of defects.
- ii) Option (b) is the correct option because we know that the control limits for the c-chart are $= \lambda \pm 3\sqrt{\lambda}$.
- iii) Option (a) is the correct option because we know that the u-chart is the control chart for number of defects per unit whereas the c-chart is the control chart for number of defects.

- E2)** The control limits of the c-chart when the average number of defects of the process is not known are

$$\text{Centre line (CL)} = \bar{c}$$

$$\text{Upper control limit (UCL)} = \bar{c} + 3\sqrt{\bar{c}}$$

$$\text{Lower control limit (UCL)} = \bar{c} - 3\sqrt{\bar{c}}$$

$$\text{where } \bar{c} = \frac{\text{Sum of defects}}{\text{Total number of samples inspected}} = \frac{1}{k} \sum_{i=1}^k c_i.$$

- E3)** Since the sample size is constant, we can use the c-chart. Also, the average number of defects in the process is not given. So we use equations (5a to 5c) to calculate the centre line and control limits of the c-chart.

It is given that $k = 25$ and $n = 10$.

The total number of defects in all match boxes $= 7 + 1 + \dots + 3 = 124$

$$\therefore \bar{c} = \frac{1}{k} \sum_{i=1}^k c_i = \frac{1}{25} \times 124 = 4.960$$

Hence, using equations (5a to 5c), we calculate the centre line and control limits of the c-chart as follows:

$$\text{CL} = \bar{c} = 4.960$$

$$\text{UCL} = \bar{c} + 3\sqrt{\bar{c}} = 4.960 + 3 \times \sqrt{4.960} = 4.960 + 3 \times 2.227 = 11.641$$

$$\text{LCL} = \bar{c} - 3\sqrt{\bar{c}} = 4.960 - 3 \times \sqrt{4.960} = 4.960 - 3 \times 2.227 = -1.721 \square 0$$

We construct the c-chart by taking the sample number on the X-axis and the number of defects (c) per sample on the Y-axis as shown in Fig. 4.6.

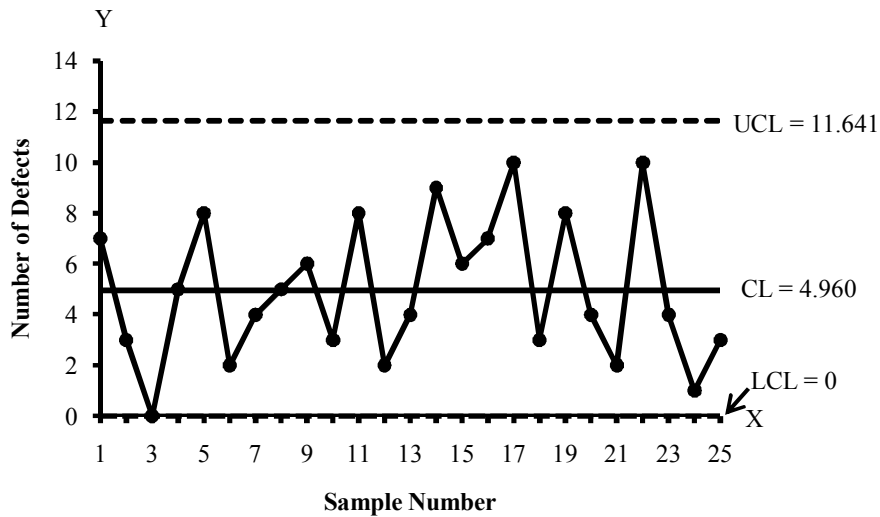


Fig. 4.6: The c-chart for the number of defects in packets of match box.

Interpretation of the result

From the c-chart shown in Fig. 4.6, we observe that no point lies outside the control limits and there is no specific pattern of the sample points on the chart. It indicates that the process is under statistical control.

- E4) Since the number of defects on the white paper per 100 square meter is given. It means that the sample size is constant. So we use the c-chart.

Since the average number of defects of the process is not given, So we use equations (5a to 5c) to calculate the centre line and control limits of the c-chart.

It is given that $k = 25$ and $n = 1$.

The total number of defects = $6 + 8 + \dots + 7 = 214$

$$\therefore \bar{c} = \frac{1}{k} \sum_{i=1}^k c_i = \frac{1}{25} \times 214 = 8.560$$

Hence, using equations (5a to 5c), we calculate the centre line and control limits of the c-chart as follows:

$$CL = \bar{c} = 8.560$$

$$UCL = \bar{c} + 3\sqrt{\bar{c}} = 8.560 + 3 \times \sqrt{8.560} = 8.560 + 3 \times 2.926 = 17.338$$

$$LCL = \bar{c} - 3\sqrt{\bar{c}} = 8.560 - 3 \times \sqrt{8.560} = 8.560 - 3 \times 2.926 = -0.218 \approx 0$$

We construct the c-chart by taking the sample number on the X-axis and the number of defects (c) on the white paper on the Y-axis as shown in Fig. 4.7.

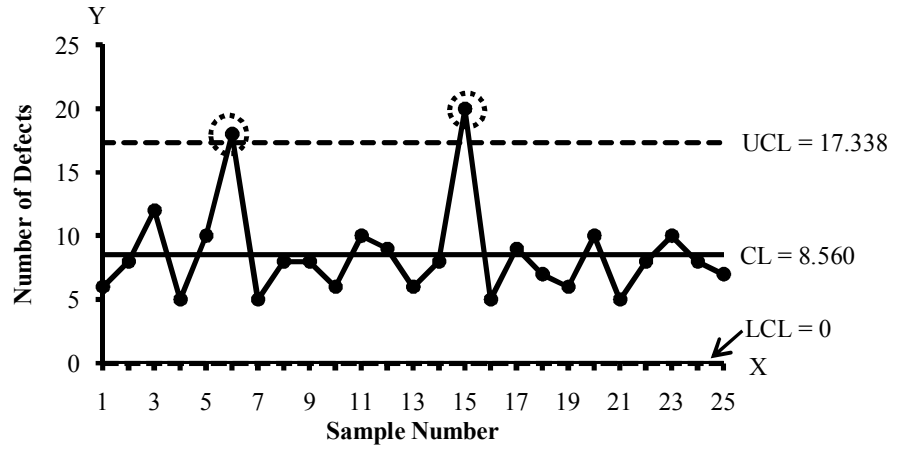


Fig. 4.7: The c-chart for the number of defects on the white paper.

Interpretation of the result

From Fig. 4.7, we observe that the points corresponding to the sample numbers 6 and 15 lie outside the upper control limit. Therefore, the process is not under statistical control with respect to the number of defects. To bring the process under statistical control it is necessary to investigate the assignable causes and take corrective action to eliminate them.

For calculating the revised limits of the c-chart, we first delete the out-of-control points (6 and 15) and calculate the new $\bar{c}$ using remaining samples.

In our case, $k = 25, d = 2, \sum_{j=1}^d c_j = 18 + 20 = 38$

$$\therefore \bar{c}_{\text{new}} = \frac{\sum_{i=1}^k c_i - \sum_{j=1}^d c_j}{k - d} = \frac{214 - 38}{25 - 2} = \frac{176}{23} = 7.652$$

After finding the $\bar{p}_{\text{new}}$, we calculate the revised centre line and control limits using equations (7a to 7c) as follows:

$$CL = \bar{c}_{\text{new}} = 7.652$$

$$UCL = \bar{c}_{\text{new}} + 3\sqrt{\bar{c}_{\text{new}}} = 7.652 + 3 \times \sqrt{7.652} = 7.652 + 3 \times 2.766 = 15.950$$

$$\begin{aligned} LCL &= \bar{c}_{\text{new}} - 3\sqrt{\bar{c}_{\text{new}}} = 7.652 - 3 \times \sqrt{7.652} \\ &= 7.652 - 3 \times 2.766 = -0.646 \approx 0 \end{aligned}$$

- E5)** Since the number of defects is given, we may use either c-chart or u-chart. Since the sample size varies, we use the u-chart.

For calculating the centre line and control limits of the u-chart, we first calculate the defects per unit (u) and then $\bar{u}$.

Sample Number	Amount of the Carpet Inspected	Number of Unit Inspected (n)	Number of Defects (c)	$\bar{u} = c/n$	CL	UCL	LCL
1	300	3.0	7	2.333	3.301	6.486	0.168
2	200	2.0	8	4.000	3.301	7.196	-0.542 □ 0
3	250	2.5	5	2.000	3.301	6.788	-0.134 □ 0
4	150	1.5	5	3.333	3.301	7.795	-1.141 □ 0
5	250	2.5	10	4.000	3.301	6.788	-0.134 □ 0
6	100	1.0	4	4.000	3.301	8.799	-2.145 □ 0
7	200	2.0	5	2.500	3.301	7.196	-0.542 □ 0
8	150	1.5	8	5.333	3.301	7.795	-1.141 □ 0
9	150	1.5	8	5.333	3.301	7.795	-1.141 □ 0
10	250	2.5	6	2.400	3.301	6.788	-0.134 □ 0
11	200	2.0	5	2.500	3.301	7.196	-0.542 □ 0
12	250	2.5	9	3.600	3.301	6.788	-0.134 □ 0
13	100	1.0	6	6.000	3.301	8.799	-2.145 □ 0
14	250	2.5	8	3.200	3.301	6.788	-0.134 □ 0
15	300	3.0	6	2.000	3.301	6.486	0.168
16	250	2.5	5	2.000	3.301	6.788	-0.134 □ 0
17	150	1.5	9	6.000	3.301	7.795	-1.141 □ 0
18	200	2.0	7	3.500	3.301	7.196	-0.542 □ 0
19	150	1.5	6	4.000	3.301	7.795	-1.141 □ 0
20	300	3.0	10	3.333	3.301	6.486	0.168
21	200	2.0	5	2.500	3.301	7.196	-0.542 □ 0
22	250	2.5	8	3.200	3.301	6.788	-0.134 □ 0
23	200	2.0	5	2.500	3.301	7.196	-0.542 □ 0
24	100	1.0	8	8.000	3.301	8.799	-2.145 □ 0
25	150	1.5	5	3.333	3.301	7.795	-1.141 □ 0
Total	5050	50.5	168				

From the above table and equation (9), we have

$$\bar{u} = \frac{\sum_{i=1}^n c_i}{\sum_{i=1}^n n_i} = \frac{168}{50.5} = 3.327$$

Therefore, using equation (10a to 10c), we calculate the centre line and control limits which are shown in the table as follows:

$$CL = \bar{u} = 3.327$$

$$UCL_1 = \bar{u} + 3\sqrt{\frac{\bar{u}}{n_1}} = 3.327 + 3\sqrt{\frac{3.327}{3}} \\ = 3.327 + 3 \times 1.053 = 6.486, \text{ and so on.}$$

$$LCL_1 = \bar{u} - 3\sqrt{\frac{\bar{u}}{n_1}} = 3.327 - 3\sqrt{\frac{3.327}{3}}$$

$$= 3.327 - 3 \times 1.053 = 0.168, \text{ and so on.}$$

We construct the u-chart by taking the sample number on the X-axis and the number of defects per 100 square meter paper (u) on the Y-axis as shown in Fig. 4.8.

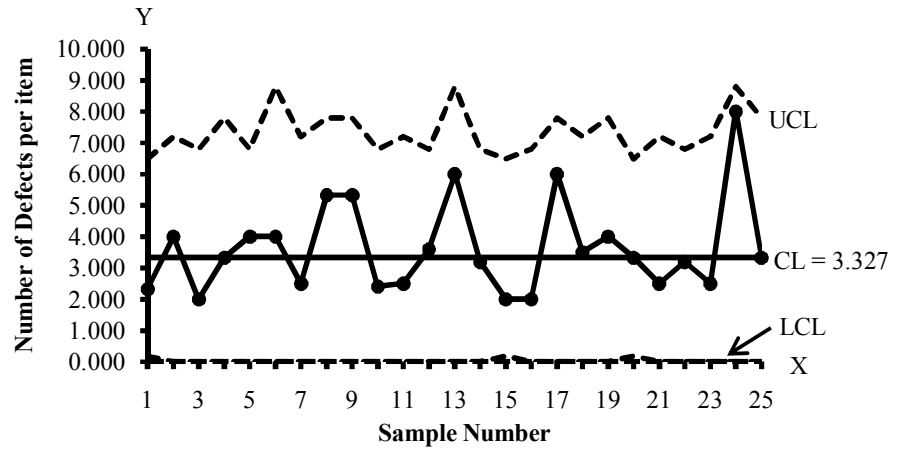


Fig. 4.8: The u-chart for the number of defects per 100 square meter paper.

Interpretation of the result

From Fig. 4.8, we observe that no point lies outside the control limits of the u-chart and there is no specific pattern of the sample points on the chart. So it indicates that the process is under statistical control with respect to the number of defects per 100 square meter paper.

APPENDIX

Table-I: Constants/Factors for Variable Control Charts

Sample Size (n)	c ₄	d ₂	d ₃	A	A ₂	A ₃	B ₃	B ₄	D ₁	D ₂	D ₃	D ₄
2	0.798	1.128	0.853	2.121	1.880	2.659	0	3.267	0	3.686	0	3.267
3	0.886	1.693	0.888	1.732	1.023	1.954	0	2.568	0	4.358	0	2.574
4	0.921	2.059	0.880	1.500	0.729	1.628	0	2.266	0	4.698	0	2.282
5	0.940	2.326	0.864	1.342	0.577	1.427	0	2.089	0	4.918	0	2.114
6	0.952	2.534	0.848	1.225	0.483	1.287	0.030	1.970	0	5.078	0	2.004
7	0.959	2.704	0.833	1.134	0.419	1.182	0.118	1.882	0.204	5.204	0.076	1.924
8	0.965	2.847	0.820	1.061	0.373	1.099	0.185	1.815	0.388	5.306	0.136	1.864
9	0.969	2.970	0.808	1.000	0.337	1.032	0.239	1.761	0.547	5.393	0.184	1.816
10	0.973	3.078	0.797	0.949	0.308	0.975	0.284	1.716	0.687	5.469	0.223	1.777
11	0.975	3.173	0.787	0.905	0.285	0.927	0.321	1.679	0.811	5.535	0.256	1.744
12	0.978	3.258	0.778	0.866	0.266	0.886	0.354	1.646	0.922	5.594	0.283	1.717
13	0.979	3.336	0.770	0.832	0.249	0.850	0.382	1.618	1.025	5.647	0.307	1.693
14	0.981	3.407	0.763	0.802	0.235	0.817	0.406	1.594	1.118	5.696	0.328	1.672
15	0.982	3.472	0.756	0.775	0.223	0.789	0.428	1.572	1.203	5.741	0.347	1.653
16	0.984	3.532	0.750	0.750	0.212	0.763	0.448	1.552	1.282	5.782	0.363	1.637
17	0.985	3.588	0.744	0.728	0.203	0.739	0.466	1.534	1.356	5.820	0.378	1.622
18	0.985	3.640	0.739	0.707	0.194	0.718	0.482	1.518	1.424	5.856	0.391	1.608
19	0.986	3.689	0.734	0.688	0.187	0.698	0.497	1.503	1.487	5.891	0.403	1.597
20	0.987	3.735	0.729	0.671	0.180	0.680	0.510	1.490	1.549	5.921	0.415	1.585

Block

2

PRODUCT CONTROL

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BLOCK 2: PRODUCT CONTROL

In Block 1, you have studied that statistical quality control techniques can be broadly classified into two categories:

- Statistical process control (SPC), and
- Product control.

You have learnt about statistical process control in Block 1. Statistical process control is a technique used for understanding and monitoring a process by collecting the data on quality characteristic periodically from the process, analysing them and taking suitable actions whenever there is a difference between actual quality and the specification or standard. Generally, process control can be achieved through **control charts**.

In this Block, we discuss **product control**. Product control refers to control the products in such a way that they are free from defects and conform to given specifications. Generally, product control can be achieved through **acceptance sampling**, which was given by Harold F. Dodge and Harry G. Roming, researchers of Bell Telephone Laboratories in 1920.

In process control, we control the methods used at the time of producing a unit, whereas in product control, we inspect the quality of the units which are produced or received from suppliers or vendors.

This block comprises four units.

In **Unit 5**, we introduce the **acceptance sampling plans for attributes**. We explain what an acceptance sampling plan is and how to implement it in industry. We also discuss the basic terminology related to it, such as lot, probability of accepting a lot, acceptance quality level (AQL), lot tolerance percent defective (LTPD), producer's risk and consumer's risk.

In **Unit 6**, we discuss the **rectifying sampling plans for attributes** and their implementation in industry. We introduce average outgoing quality (AOQ), operating characteristic (OC) curve, average sample number (ASN) and average total inspection (ATI).

In **Units 7 and 8**, we discuss the **single and double sampling plans for attributes**, respectively. We describe various features of both sampling plans, such as operating characteristic (OC) curve, average outgoing quality (AOQ), producer's risk, consumer's risk, average sample number (ASN), average total inspection (ATI) and how to design these plans.

The following notations and symbols are used in the block.

Notations and Symbols

Sec.	:	Section
Fig.	:	Figure
AQL	:	Acceptance quality level
LTPD	:	Lot tolerance percent defective
AOQ	:	Average outgoing quality
OC	:	Operation characteristic
ASN	:	Average sample number
ATI	:	Average total inspection
N	:	Lot size
n	:	Sample size
n_1	:	Size of first sample
n_2	:	Size of second sample
p	:	Proportion defective (proportion of defective units or lot quality)
p_1	:	Acceptance quality level
p_2	:	Lot tolerance percent defective
d	:	Number of defective units in a sample
d_1	:	Number of defective units in first sample
d_2	:	Number of defective units in second sample
c	:	Acceptance number
P[.]	:	Probability
$P_a(p)$ or P_a	:	Probability of accepting a lot of quality p
$P_p(p)$ or P_p	:	Producer's risk
$P_c(p)$ or P_c	:	Consumer's risk
α	:	Probability of rejecting a lot of AQL quality
β	:	Probability of accepting a lot of LTPD quality

UNIT 5 ACCEPTANCE SAMPLING PLANS

Structure

- 5.1 Introduction
 - Objectives
- 5.2 Inspection
- 5.3 Acceptance Sampling Plan
 - Advantages and Limitations of Acceptance Sampling
 - Types of Acceptance Sampling Plans
- 5.4 Implementation of Acceptance Sampling Plan for Attributes
- 5.5 Terms Used in Acceptance Sampling Plans
- 5.6 Producer's Risk and Consumer's Risk
- 5.7 Summary
- 5.8 Solutions/Answers

5.1 INTRODUCTION

In many situations, the product is so complex that all components/parts of the product are not made by a manufacturer. In such cases, one or more components of the product are purchased from an outside agent or supplier and the manufacturer does not have direct control over the quality of such components. Since the final product is produced by the manufacturer, he/she faces problems such as:

- How to control the quality of components received from others?
- How to ensure that the lots produced do not contain an excessively large proportion of defective products?
- Do the products meet the desired specifications?

Such problems belong to the category of **product control**.

Product control refers to control the products in such a way that these are free from defects and conform to specifications.

You have learnt in Unit 1 that product control can be done by 100% inspection, i.e., each and every unit produced or received from the outside suppliers is inspected. This type of inspection has the advantage of ensuring that all defective units are eliminated. However, it is very time consuming and costly. Also, if the unit is destroyed under investigation, e.g., light bulb, cracker, ammunition, etc., 100% inspection is not practicable.

In 1920, Harold F. Dodge and Harry G. Roming developed statistical methods for product control known as **acceptance sampling** or **sampling inspection** as an alternative of 100% inspection. Now-a-days, product control is achieved through acceptance sampling.

In this unit, we introduce the acceptance sampling plans and the related terminology. In Sec. 5.2, we explain the meaning of the term **inspection** as used in industry. We discuss the concept of acceptance sampling plan, its advantages and limitations and the types of acceptance sampling plans in Sec. 5.3. In Sec 5.4, we discuss the implementation of the acceptance sampling plan. In Secs. 5.5 and 5.6, we explain the basic terminology related to acceptance

sampling plan, such as lot, probability of accepting a lot, acceptance quality level (AQL), lot tolerance percent defective (LTPD), producer's risk and consumer's risk. In the next unit, we shall discuss the rectifying sampling plans.

Objectives

After studying this unit, you should be able to:

- distinguish between 100% inspection and sampling inspection;
- define acceptance sampling plan and discuss the procedure of its implementation;
- compute the probability of accepting or rejecting a lot;
- define acceptance quality level (AQL) and lot tolerance percent defective (LTPD) of the lot; and
- compute the producer's risk and consumer's risk for an acceptance sampling plan.

5.2 INSPECTION

You have learnt briefly about 100% inspection and sampling inspection in Sec. 1.3.1 of Unit 1. In this section, we discuss these concepts in some more detail.

When a manufacturer produces a product or buys some parts of the product from outside agents or suppliers, he/she would like to ensure that the final product is as per specifications. For this purpose, he/she inspects the lot at every strategic point. The method of checking, measuring or testing one or more quality characteristics of the product or the parts and determining whether it satisfies the required specifications or not, is called **inspection**.

Inspection is of two types:

i) Inspection of Variables

In this type of inspection, the quality characteristic (s) of an item/unit is (are) measured and compared with the required specifications. For example, the desired specification for the diameter of a ball bearing is 50 mm, length of the refill of a ball pen is 14 cm, weight of a cricket ball is 162 gram, and so on.

ii) Inspection of Attributes

In inspection of attributes, actual measurements are not taken. Instead, the item/units are categorised as defective or non-defective on the basis of Go-No-Go gauges. It means that if a unit fulfils all required quality characteristics, it is categorised as non-defective and if not, it is categorised as defective. In such type of inspection, the number of defective units (or defects) is counted.

Methods of Inspection

There are two methods of inspection:

i) 100% Inspection

In this method of inspection, each and every item/unit of any given lot is inspected. A decision regarding the quality of the entire lot is taken on the basis of **all** inspected units of the lot. This procedure needs huge expenditure of time,

A Go-No-Go gauge is an inspection tool which is used to check an item or a unit or a piece against its allowed tolerances. The name **Go-No-Go** derives from its use. It means that we check the item and if the item is acceptable (fulfil the specifications), we say **Go** and if unacceptable, we say **No-Go**.

money, labour and resources. Also, if the product is such that it is completely destroyed under the process of inspection (e.g., a cracker, ammunition, etc.) then in such cases, 100% inspection is neither practicable nor economical.

As an alternative, we use sampling inspection.

ii) Sampling Inspection

In sampling inspection method, some items/units (called sample) are randomly selected from a lot in such a way that the selected sample is a true representative of the entire lot. Then each and every unit of the selected sample is inspected. A decision regarding the quality of the entire lot is taken on the basis of the information obtained from the sampled units.

The question now is: How do we take any decision about the quality of a lot on the basis of sampling inspection? For this, we need to learn about the **acceptance sampling plan**.

5.3 ACCEPTANCE SAMPLING PLAN

Let us first define acceptance sampling.

A sampling inspection in which a decision about acceptance or rejection of a lot is based on one or more samples that have been inspected is known as **acceptance sampling**.

In other words:

Acceptance sampling is a technique in which a small part or a fraction of the units/items is selected randomly from a lot and the selected units are inspected to decide whether the lot should be accepted or rejected on the basis of the information provided by the sample inspection.

For example, suppose a manufacturer of cricket balls supplies them in lots of 500. A buyer wants to inspect 20 balls from each lot before accepting the lot and takes a decision about acceptance or rejection of the lot on the basis of the information provided by this sample. This is an example of acceptance sampling because the decision about the lot is taken on the basis of a sample.

Consider another example. Suppose a mobile phone company packs the mobile phones it produces in lots of 100 units. To check the quality of the lots, the quality inspector of the company draws a random sample of size 10 from each lot. He/she takes a decision about acceptance or rejection of the lot on the basis of the information provided by this sample.

When a lot is rejected on the basis of the acceptance sampling, it does not mean that all units of the lot are of no use or all units are defective. It means that all units of the lot have been carefully examined and the defective units have been either repaired or removed. Thus, the remaining units in the lot conform to a particular quality level. The rejected units may be sold at a lower price or the defects in the units may be removed so that it can be retained. For instance, the clearance sales of, e.g., ready-made clothes or shoes, etc. are carried out to sell those items which are not up to the specifications or have been rejected. This is a loss to the producer or the manufacturer, which can be prevented if he/she could maintain the desired level of quality. Therefore, manufacturers should be very careful about the quality of goods, which they supply to the customers.

We can now answer the question: **What is an acceptance sampling plan?**

An acceptance sampling plan is a specific plan that clearly states the rules for sampling and the associated criteria for acceptance or rejection of a lot.

Acceptance sampling plans can be used for the inspection of:

1. Manufactured units/items,
2. Components,
3. Raw materials,
4. Operations,
5. Materials in process,
6. Supplies in storage,
7. Maintenance operations,
8. Data or records, and
9. Administrative procedures.

Acceptance sampling has many advantages and some limitations, which we now describe.

5.3.1 Advantages and Limitations of Acceptance Sampling

The main **advantages of acceptance sampling** are as follows:

- i) It is less expensive in terms of money, time and labour in comparison to 100% inspection.
- ii) For items, which cannot be used after single inspection, such as crackers, bulbs, tube lights, food, etc., 100% inspection is not practicable. Sampling inspection is the only way for inspecting such items.
- iii) In acceptance sampling, a sample of a small number of items or units is inspected and hence smaller inspection staff is required.
- iv) In many cases, acceptance sampling provides better outgoing quality. In general, it is seen and agreed that **good 100% inspection** removes only **85% to 90%** of the defective items, whereas **very good 100% inspection** removes only **99%** of the defective items. However, due to human error, it usually does not reach the 100% mark. In other words, we can say that 100% inspection is not always reliable because it involves too much routine work for the persons inspecting each and every item. Due to this, defective items may also be labelled as satisfactory and may also be accepted at times when these persons are distracted. Hence, an appropriate sampling plan is preferable.
- v) Due to quick inspection through the acceptance sampling, the scheduling and delivery times are saved.

Acceptance sampling has some **limitations** which are given below:

- i) Since, in acceptance sampling, the entire lot is accepted or rejected on the basis of conclusions drawn from one or more samples, there is always some risk of making wrong inference about the quality of the lot. These risks are termed the producer's risk and consumer's risk. You will learn more about these in Sec. 5.6.
- ii) The success of acceptance sampling depends on the randomness of the sample, quality characteristics to be tested, lot size, acceptance criteria,

etc. Therefore, it is a specialised job requiring careful planning and execution and every one cannot undertake it.

So far you have learnt about the acceptance sampling plan and its advantages and limitations. It is also important for you to know about different types of acceptance sampling plans used in industries.

5.3.2 Types of Acceptance Sampling Plans

There are several types of acceptance sampling plans based on different approaches that can influence the decision about a lot. These are categorised as follows:

1. Acceptance sampling plans for attributes, and
2. Acceptance sampling plans for variables.

1. Acceptance Sampling Plans for Attributes

Acceptance sampling plans in which **actual measurements** of the quality characteristics are **not made**, but the units/items are categorised as **defective or non-defective** on the basis of Go-No-Go gauges are called **acceptance sampling plans for attributes**. These plans are of two types:

A. Lot-By-Lot Acceptance Sampling Plans for Attributes

Lot-by-lot acceptance sampling plans for attributes are the most commonly used sampling plans and, therefore, are simply called acceptance sampling plans for attributes. Such plans are used whenever the units to be inspected can be conveniently grouped into **batches** or **lots**.

In lot-by-lot acceptance sampling plan for attributes, generally, we use the following plans:

i) Single sampling plans

In the single sampling plan, the decision about the acceptance or rejection of a lot is based on a single sample that has been inspected. This is the simplest type of sampling plan.

ii) Double sampling plans

In the double sampling plan, the decision about the acceptance or rejection of a lot requires the evidence of two samples drawn from the lot. If the lot quality is good (or bad), the lot is accepted (or rejected) on the basis of the first sample. If the first sample shows an intermediate quality, the decision about the lot is taken on the evidence of the first and second sample combined. This is more complicated than the single sampling plan.

iii) Multiple sampling plans

A multiple sampling plan is an extension of the double sampling plan. This sampling plan may require more than two samples to reach a decision about the acceptance or rejection of a lot.

iv) Sequential sampling plans

In the sequential sampling plan, the sample size is not pre-fixed as in the case of single, double and multiple sampling plans. The units/items are drawn from the lot, one at a time and inspected. The decision about accepting or rejecting of the lot or continuing with the inspection by taking one more unit from the lot, is made on the basis of information available up to that stage.

The **multiple** and **sequential sampling plans** are beyond the scope of this course. However, in Units 7 and 8, we shall discuss the single sampling plans and double sampling plans in detail.

B. Continuous Production Acceptance Sampling Plans for Attributes

Continuous production acceptance sampling plans for attributes are used when the units/items to be inspected cannot be grouped into lots or batches. Many manufacturing operations do not create lots because in these operations, the units are produced in a continuous process on a conveyor belt or other straight-line systems.

Continuous production acceptance sampling plans for attributes are beyond the scope of this course.

2. Acceptance Sampling Plans for Variables

The acceptance sampling plans in which **actual measurements** of the quality characteristics are taken are called **acceptance sampling plans for variables**.

The acceptance sampling plans for variables are beyond the scope of this course. So we now focus on the acceptance sampling plans for attributes and describe the general procedure for implementing it.

5.4 IMPLEMENTATION OF ACCEPTANCE SAMPLING PLAN FOR ATTRIBUTES

Suppose that lots of the same size, say, N , are received from the supplier or the final assembly line and submitted for inspection one at a time. The procedure for implementing the acceptance sampling plan to arrive at a decision about the lot is described in the following steps:

- Step 1:** We draw a random sample of size n from the lot received from the supplier or the final assembly.
- Step 2:** We inspect each and every unit of the sample and classify it as defective or non-defective on the basis of certain criteria. At the end of the inspection, we count the number of defective units found in the sample.
- Step 3:** We compare the number of defective units found in the sample with the acceptance criteria.
- Step 4:** If the acceptance criteria are satisfied, we accept the entire lot. Otherwise, we reject the entire lot.

The steps described above are shown in Fig. 5.1.

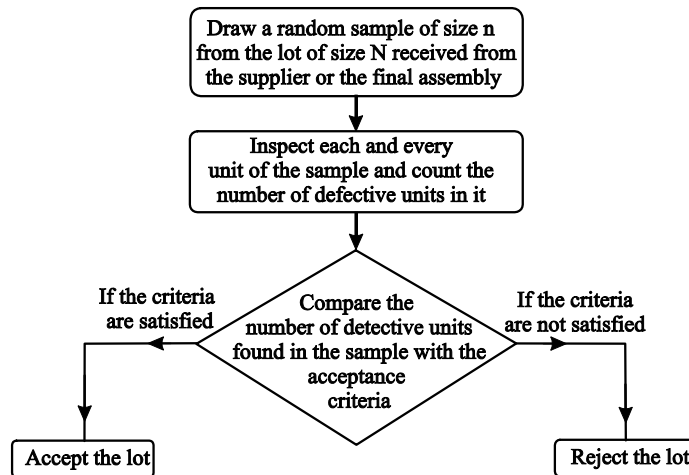


Fig. 5.1: Procedure for implementation of the acceptance sampling plan.

Let us explain these steps further with the help of an example.

Suppose a buyer of cricket balls wants to inspect a sample of 20 balls from each lot to check their quality. The acceptance criterion is that if the sample contains at most one defective ball, the lot would be accepted. Otherwise, it would be rejected. The buyer draws 20 balls from each lot and inspects each and every ball of the sample and classifies each ball as defective or non-defective on the basis of certain defects. At the end of the inspection, he/she counts the number of defective balls found in the sample and compares the number of defective balls with the acceptance criterion. If the number of defective balls in the sample is more than 1, he/she rejects the lot. If the number of defective balls is zero or 1, he/she accepts the lot.

We shall discuss different types of sampling inspection plans in detail in the remaining units. But, before studying these plans, you should be aware about the related terminology. This is what we discuss in Sec. 5.5.

5.5 TERMS USED IN ACCEPTANCE SAMPLING PLANS

In this section, we define some terms used in acceptance sampling plan.

1. Lot

A lot is the collection of units or items from which a sample is taken and inspected to determine its acceptability.

It is found that **lot formation** influences the effectiveness of the acceptance sampling plan. Therefore, formation of the lot is important for the success of an acceptance sampling plan. Some guidelines for the lot formation are as follows:

- i) Units in the lot should be homogeneous. It means that the lot should consist of units produced by the same machine, same operators, using the same raw materials and approximately during the same time period.
- ii) Units in the lots should be packed so that the shipping costs and handling risks are minimum. This also makes the selection of units in the sample easy.

The decision about the acceptance or rejection of all units of the lot is taken on the basis of the results of inspection of the sample. Therefore, the sample should be such that it is a true representative of the lot.

2. Sentencing

The act of accepting or rejecting the entire lot is called **sentencing** the lot.

3. Lot Size

The number of units in a lot is called **lot size**. It is denoted by N.

4. Sample Size

The number of units inspected to sentence a lot is called **sample size**. It is denoted by n.

5. Lot Quality

The proportion of defective units in a lot is called **lot quality** or **proportion defective**. It is denoted by p and defined as

$$p = \frac{\text{Number of defective units in a lot}}{\text{lot size}} \quad \dots (1)$$

From the definition of p given by equation (1), you should note that for the same lot size, as p increases, the lot quality decreases.

6. Acceptance Number

When a deal is finalised between a seller and a buyer, they decide on the maximum number of allowable defective units in a sample. This number is called **acceptance number** and is denoted by c. The rule for acceptance or rejection of a lot is that if the number of defective units observed in the sample is less than or equal to the acceptance number c, the lot will be accepted. Otherwise, it will be rejected.

Let us explain the above terminology with the help of an example.

Example 1: Suppose a cricket ball manufacturing company supplies lots of 500 balls. To check the quality of the lots, a buyer draws a random sample of size 20 balls from each lot and accepts the lot if the inspected sample contains at most one defective ball. Otherwise, he/she rejects the lot. If the lot consists of 10 defective balls, find N, n, p and c.

Solution: Since the lot contains 500 balls, the **lot size** N = 500.

Since the buyer draws a sample of size 20 balls from each lot to take the decision about the lot, the **sample size** n = 20.

The buyer accepts the lot if the inspected sample contains at most one defective ball. So the **acceptance number** c = 1.

The lot consists of 10 defective balls. So from equation (1), the **lot quality** is given as

$$p = \frac{\text{Number of defective units in a lot}}{\text{lot size}} = \frac{10}{500} = 0.02$$

Now that you have learnt about terms such as lot size, lot quality, sample size, etc., you may like to know: **How can we be confident that a lot of good quality will be accepted and a lot of bad quality will be rejected?** We can answer such questions by calculating the probability of accepting a lot of specified quality. This probability will be 1 if all units in the lot are accepted

and 0, if all units in the lot are rejected. The probability of accepting a lot is denoted by $P_a(p)$ or in short as P_a .

Acceptance Sampling Plans

7. Probability of accepting a lot (P_a)

Suppose that the lots of the same size, say, N , are received from the supplier or the final assembly line and submitted for inspection one at a time. The lot is accepted if the number of defective units observed in the sample is less than acceptance number (c). Otherwise, it is rejected. Suppose a random sample of size n is drawn from each lot for inspection.

If X represents the number of defective units in the sample, the lot is accepted if $X \leq c$. It means that we accept the lot if $X = 0$ or 1 or 2, ... or c . Therefore, the probability of accepting the lot is given by

$$P_a(p) = P_a = P[X \leq c] = P[X = 0 \text{ or } 1 \text{ or } 2 \text{ or } 3, \dots, \text{ or } c] \quad \dots (2)$$

Since $X = 0, 1, \dots, c$ are mutually exclusive events, from the addition theorem of probability, we have

$$P_a(p) = P[X \leq c] = P[X = 0] + P[X = 1] + \dots + P[X = c]$$

$$\text{or} \quad P_a(p) = \sum_{x=0}^c P[X = x] \quad \dots (3)$$

We can easily calculate this probability if we know the probability distribution of X . Generally, in quality control, a random sample is drawn without replacement. So the number of defective units (X) in the sample follows a hypergeometric distribution. However, we know that when the lot size (N) is large compared to the sample size, i.e., $N \geq 10n$, the hypergeometric distribution approximates a binomial distribution with parameters n and p where p is the lot quality. It is easier to calculate the probabilities with the help of the binomial distribution rather than with the hypergeometric distribution.

Therefore, the probability of getting exactly x defective units in a sample of size n using the binomial distribution is given by

$$P[X = x] = {}^nC_x p^x (1-p)^{n-x} \quad \dots (4)$$

Hence, from equation (3), we have

$$P_a(p) = P[X \leq c] = \sum_{x=0}^c P[X = x] = \sum_{x=0}^c {}^nC_x p^x (1-p)^{n-x} \quad \dots (5)$$

However, for rapid calculation, we can use **Table I** entitled **Cumulative Binomial Probability Distribution** given at the end of this block.

Let us now compute $P_a(p)$ for Example 1.

In this example, we have

$$N = 500, n = 20, c = 1, p = 0.02$$

Suppose X represents the number of defective balls in the sample. The buyer accepts the lot if the number of defective balls (X) in the sample is at most one, i.e., $c = 1$. Therefore, the probability of accepting the lot of quality $p = 0.02$ is given by

$$P_a(p) = P[X \leq c] = P[X \leq 1] = P[X = 0] + P[X = 1]$$

You have studied in Unit 3 of MST-003 that if A and B are mutually exclusive events then
 $P[A \text{ or } B] = P[A] + P[B]$

The notation nC_x can also be repersened as $\binom{n}{x}$.

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$$= \sum_{x=0}^1 P[X = x]$$

Since the lot size (N) is large compared to the sample size (n), i.e., $N \geq 10n$, X approximately follows the binomial distribution with parameters n and p where p is the lot quality. Therefore, the probability of accepting the lot of quality p is given by

$$P_a(p) = \sum_{x=0}^1 P[X = x] = \sum_{x=0}^1 {}^nC_x p^x (1-p)^{n-x}$$

For rapid calculation, we can use Table I for obtaining this probability.

From Table I, for $n = 20$, $x = c = 1$ and $p = 0.02$, we have

$$\sum_{x=0}^1 {}^nC_x p^x (1-p)^{n-x} = 0.9401$$

Therefore, the probability of accepting the lot of quality $p = 0.02$ is given by

$$P_a(p) = \sum_{x=0}^1 {}^nC_x p^x (1-p)^{n-x} = 0.9401 \quad \dots (6)$$

It means that if there are several lots of the same quality $p = 0.02$, about 94.01% of these will be accepted and about 5.99% will be rejected.

We can also calculate this probability manually using the scientific calculator for $n = 20$, $p = 0.02$ as follows:

$$\begin{aligned} P[X = x] &= {}^nC_x p^x (1-p)^{n-x} \\ P_a(p) &= P[X \leq 1] = P[X = 0] + P[X = 1] \\ &= {}^{20}C_0 (0.02)^0 (1-0.02)^{20-0} + {}^{20}C_1 (0.02)^1 (1-0.02)^{20-1} \\ &= (0.98)^{20} + 20 \times 0.02 \times (0.98)^{19} \\ &= 0.6676 + 0.2725 = 0.9401 \end{aligned}$$

Now, we can see the effect of the lot quality on the probability of accepting the lot.

Suppose the lot in Example 1 contains 20 defective balls instead of 10. Then the lot quality is:

$$p = \frac{\text{Number of defective units in the lot}}{\text{lot size}} = \frac{20}{500} = 0.04$$

We can calculate the probability of accepting the lot of quality $p = 0.04$ with the help of Table I as discussed above.

From Table I, for $n = 20$, $x = c = 1$ and $p = 0.04$, we have

$$P_a(p) = P[X \leq 1] = 0.8103$$

In this case, about 81.03% lots will be accepted and about 18.97% lots will be rejected.

If we compare this probability with the probability obtained for $p = 0.02$, we observe that as the lot quality decreases (from $p = 0.02$ to $p = 0.04$), the

probability of accepting the lot also decreases (from $P_a = 0.9401$ to $P_a = 0.8103$).

We can also see the effect of the acceptance number on the probability of accepting the lot.

Suppose the buyer accepts the lot if the inspected sample contains at most two defective balls, that is, $c = 2$. Then we can obtain the probability of accepting the lot using Table I.

From Table I, for $n = 20$, $x = c = 2$ and $p = 0.02$, we have

$$P_a(p) = P[X \leq c] = P[X \leq 2] = 0.9929$$

It means that if there are several lots of the same quality $p = 0.02$, then for the acceptance number $c = 2$, about 99.29% lots will be accepted and about 0.71% lots will be rejected.

If we compare this probability with the probability obtained for $p = 0.02$ and $c = 1$, we observe that when the acceptance number increases (from $c = 1$ to $c = 2$), the probability of accepting the lot also increases (from $P_a = 0.9401$ to $P_a = 0.9929$).

It is time for you to pause and solve some exercises for practice.

-
- E1)** A shopkeeper purchases pens from a pen company in cartons (lots) that usually contain one thousand pens. To check the quality of the pens, the shopkeeper selects 25 pens at random from each carton and visually inspects each selected pen for certain defects. The shopkeeper accepts the lot if the inspected sample contains at most two defective pens. Otherwise, he/she rejects the lot. If there are 40 defective pens in each carton, find N , n , p , c and P_a .
- E2)** For a sampling plan with $n = 5$ and $c = 0$, find the probability of accepting a lot that has 2% defective units by assuming that the number of defective units in a sample follows a binomial distribution.
-

8. Acceptance Quality Level (AQL)

The seller and the buyer generally know that the supply of completely defect-free lots is not possible. So they usually negotiate and arrive at an agreement that the buyer will accept all lots which have at most a definite quality level or definite percentage of defective units. This definite quality level is known as **acceptance quality level (AQL)**. It is denoted by p_1 . Hence, AQL can be defined as follows:

Acceptance quality level is the quality level decided in a negotiation of the seller and the buyer: If the proportion of defective units in a lot is less than or equal to AQL, the buyer will have to definitely accept the lot. Otherwise, the buyer may accept or reject the lot.

Suppose 100% inspection of all lots is carried out and the lot quality (p) for each lot is computed. Then all lots with lot quality $p \leq \text{AQL}$ will be accepted and all lots with $p > \text{AQL}$ may either be accepted or rejected.

In Example 1, if the manufacturing company and the buyer decide that the buyer will accept all lots which have at most 2% defective units, then acceptance quality level (AQL) of this deal is 2%.

Even though the seller and the buyer decide that the buyer would accept all lots of AQL quality, this may not happen in practical situations. In Example 1, the probability of accepting the lot of the lot quality $p = \text{AQL} = 0.02$ was

$$P_a(p) = P[X \leq 1] = 0.9401$$

It means that if there are several lots of the same quality $p = 0.02 = \text{AQL}$, about 94.01% of these would be accepted and about 5.99% would be rejected. This is obviously a risk of the manufacturer because it was agreed upon by both that all lots of quality 0.02 will be accepted whereas the buyer is rejecting 5.99% of them. This risk is known as producer's risk which is explained in detail in Sec.5.6.

9. Lot Tolerance Percent Defective (LTPD)

In order to reduce the producer's risk, the buyer agrees to tolerate a lot quality worse than acceptance quality level (AQL) up to a certain limit but not beyond it. This limiting value is known as the **lot tolerance percent or proportion defective (LTPD)**. It is also known as **rejectable quality level (RQL)**, **unsatisfactory quality level** or **limiting quality level (LQL)**.

LTPD is also decided at the time when the acceptance quality level is decided in the negotiation between the producer and the buyer. They make an agreement that the buyer will definitely reject the lot of quality equal to or greater than LTPD. Therefore, LTPD can be defined as follows:

Lot tolerance percent or proportion defective is the quality level decided in negotiation between the producer and the buyer: If the proportion of defective units in a lot is equal to or greater than this level, the buyer will definitely reject the lot. Otherwise, the buyer may accept or reject the lot.

The lot tolerance percent defective (LTPD) of a sampling plan is the level of quality at which the lot is routinely rejected by the sampling plan. LTPD is greater than AQL.

In Example 1, if the manufacturing company and the buyer decide that the buyer will accept all lots which have at most 2% defective units and reject all lots which have 5% or more defective units, then for this plan the AQL is 2% and the LTPD is 5%.

This level represents the dividing line between good and bad lots.

Therefore, lot tolerance percent defective is the quality level of a lot that the buyer considers bad and he/she would like to reject all lots that have this level of quality. LTPD is denoted by p_2 .

In acceptance sampling, the acceptance or rejection of the entire lot depends on the conclusions drawn from the sample. Thus, there is always a chance of making a wrong decision. It means that a lot of good quality may be rejected and a lot of poor quality may be accepted. This leads to two kinds of risks:

- i) Producer's risk, and
- ii) Consumer's risk.

We now discuss these risks in some details.

5.6 PRODUCER'S RISK AND CONSUMER'S RISK

In quality control, a producer may be defined as follows:

A **person** or a **firm** or an **organisation** that **produces/manufactures** goods or provides services for use or consumption of another person or firm or organisation is known as a **producer**.

In Example 1, the manufacturing company that produces the cricket balls, is the producer. Let us explain what is meant by producer's risk.

Producer's Risk

It may happen in practice that a sampling inspection plan leads to the rejection of a lot of satisfactory or good quality. This means that there is a possibility of rejecting a lot having a quality level less than or equal to the acceptance quality level (AQL) due to sampling inspection. If a lot of good quality is rejected, the producer suffers loss. Therefore, the producer always faces the risk of a good lot being rejected. Such a risk is known as **producer's risk** and is defined as follows:

The probability of rejecting a lot of acceptance quality level (AQL) is known as the producer's risk. It is denoted by $P_p(p)$ or in short as P_p and given by

$$P_p(p) = P_p = P[\text{rejecting a lot of acceptance quality level}] = \alpha$$

This can be written as

$$\begin{aligned} P_p(p) &= 1 - P[\text{accepting a lot of acceptance quality level}] \\ &= 1 - P_a(p - p_1) \end{aligned} \quad \dots (7)$$

The producer's risk is equivalent to the type I error in hypothesis testing discussed in Unit 8 of MST-004 entitled Statistical Inference. Therefore, it is also denoted by α .

For computing the producer's risk, we have to compute the probability of accepting a lot of quality $p = \text{AQL}$. Then we use equation (7) to compute the producer's risk.

Let us explain how to compute the producer's risk with the help of an example.

Example 2: Suppose in Example 1, the manufacturing company and the buyer agree that $\text{AQL} = 0.02$. Find the producer's risk for this plan.

Solution: We know that the producer's risk is

$$P_p(p) = 1 - P[\text{accepting a lot of acceptance quality level}] = 1 - P_a(p)$$

For computing the producer's risk, we first compute the probability of accepting a lot of quality $(p) = \text{AQL} = 0.02$. We have already calculated this probability in Example 1. So here we use the result:

$$P_a(p) = 0.9401$$

Therefore, the producer's risk for this plan is

$$P_p(p) = 1 - P_a(p) = 1 - 0.9401 = 0.0599$$

It means that if there are several lots of the same quality $p = 0.02$ as AQL, out of these lots, about 5.99% will be rejected. This is obviously a risk of the manufacturer (producer) because it was agreed upon by both that all lots of quality $p = \text{AQL} = 0.02$ will be accepted whereas the buyer is rejecting 5.99% of the lots.

Let us now explain the consumer's risk.

Consumer's Risk

A **person** or a **firm** or an **organisation** that **purchases** goods for its own need or consumption or for use in the production of other goods (not for resale to another person or firm or organization directly or indirectly) is known as a **consumer**.

In Example 1, since the buyer purchases the cricket balls from the manufacturer, he/she is the consumer.

Just as the producer has a risk of a lot of good quality being rejected, a consumer also has a risk of buying a lot of unsatisfactory quality. Such a risk is known as the **consumer's risk**. If p_2 is the maximum proportion of defective (LTPD) in the lot, which the consumer is ready to tolerate, the consumer's risk may be defined as follows:

The probability of accepting a lot of unsatisfactory quality, i.e., LTPD is known as consumer's risk. It is denoted by $P_c(p)$ or in short as P_c .

Thus,

$$P_c(p) = P[\text{accepting a lot of quality} = \text{LTPD}] = \beta \quad \dots (8)$$

Note that $P_c(p)$ is the same as $P_a(p)$ for $p = \text{LTPD}$.

The consumer's risk is denoted by β because it is equivalent to the type II error described in Unit 8 of MST-004.

Let us explain how to compute the consumer's risk with the help of an example.

Example 3: Suppose in Example 1, the manufacturing company and the buyer agree that $\text{LTPD} = 0.05$. Find the consumer's risk for this plan.

Solution: We know that the consumer's risk is given by

$$P_c(p) = P[\text{accepting a lot of quality} = \text{LTPD}] = P_a(p = 0.05)$$

It is given that

$$N = 500, n = 20, c = 1 \text{ and } \text{LTPD} = 0.05$$

We first calculate the probability of accepting the lot of quality $p = 0.05$ as follows:

$$P_a(p) = P[X \leq c] = P[X \leq 1]$$

We can calculate this probability as we have discussed in Example 1.

From Table I, for $n = 20$, $x = c = 1$ and $p = \text{LTPD} = 0.05$, we have

$$P_a(p) = P[X \leq 1] = 0.7358$$

Therefore the consumer's risk is given by

$$P_c(p) = P_a(p = 0.05) = 0.7358$$

It means that if there are several lots of the same quality $p = 0.05$, out of these about 73.58% of them will be accepted by the buyer even though this quality is unsatisfactory. This is obviously the buyer's risk.

We take up an example to further explain the terms AQL, LTPD, producer's risk and consumer's risk.

Example 4: A mobile manufacturing company has decided to purchase the mobile batteries from a battery manufacturing company. Both manufacturing companies have decided that the batteries are to be supplied in lots of 1000 batteries each. The lot will be accepted up to quality level $p = 0.05$ and rejected at more than quality level $p = 0.20$. Acceptance sampling plan is based on a sample of size 25 drawn from each lot and the lot is accepted if inspected sample contains at most one defective battery. Otherwise, the lot is rejected. Identify which company is the producer and which one is the consumer in the plan. Calculate the producer's risk and the consumer's risk.

Solution: Here the mobile manufacturing company purchases the mobile batteries from a battery manufacturing company. So it is a consumer. The battery manufacturing company supplies batteries to the mobile manufacturing company. So it is a producer.

It is given that

$$N = 1000, AQL = 0.05, LTPD = 0.20, n = 25 \text{ and } c = 1$$

We know that the producer's risk is defined as

$$P_p = P[\text{rejecting a lot of acceptance quality level}]$$

$$= 1 - P[\text{accepting a lot of quality} = AQL = 0.05] = 1 - P_a(p = 0.05) \dots (i)$$

To calculate the producer's risk, we have to calculate the probability of accepting a lot of quality $p = 0.05$.

Suppose X denotes the number of defective mobiles in the sample. The mobile company accepts the lot if the number of defective mobiles (X) in the sample is at most one. Therefore, the probability of accepting the lot of quality p is given by

$$P_a(p) = P[X \leq c] = P[X \leq 1]$$

Since the lot size (N) is large compared to the sample size (n), i.e., $N \geq 10n$, we can use the binomial distribution and can use Table I for obtaining the probability $P[X \leq 1]$.

From Table I, for $n = 25$, $x = c = 1$ and $p = 0.05$, we have

$$P[X \leq 1] = 0.6424$$

Therefore, the probability of accepting the lot of quality $p = 0.05$ is given by

$$P_a(p = 0.05) = 0.6424$$

From (i), we calculate the producer's risk as follows:

$$P_p = 1 - P_a(p = 0.05) = 1 - 0.6424 = 0.3576$$

Similarly, we know that the consumer's risk is defined as

$$P_c = P[\text{accepting a lot of quality } p = LTPD = 0.20] = P_a(p = 0.20)$$

From Table I, for $n = 25$, $x = c = 1$ and $p = 0.2$, we have

$$P_c = P_a(p = 0.20) = 0.0274$$

You may like to solve some exercises for practice.

-
- E3)** Identify the consumer and producer in the exercise E1. If the shopkeeper and the pen company have decided that $AQL = 0.03$ and $LTPD = 0.10$, calculate the producer's risk and the consumer's risk.
- E4)** If in the exercise E2, the acceptance quality level (AQL) and the Lot tolerance percent defective (LTPD) are 1% and 5%, respectively, calculate the producer's risk and the consumer's risk for this plan.
-

We end this unit by giving a summary of what we have covered in it.

5.7 SUMMARY

1. The technique of controlling the quality of products in such a way that these are free from defects and conform to their specifications, is called **product control**.
2. The method of checking, measuring, testing one or more quality characteristics of a product (unit) to determine whether it satisfies the required specifications or not is called **inspection**.
3. If quality characteristic of a unit is measured on a continuous scale, the inspection is called **inspection by variables**.
4. If the actual measurements of the quality characteristic of a unit are not taken, but the unit is categorised as defective or non-defective on the basis of Go-No-Go gauges, the inspection is called **inspection by attributes**.
5. A **lot** is the collection of units or items from which a sample is taken and inspected to determine its acceptability.
6. If each and every unit of a lot is inspected, the inspection is known as **100% inspection**.
7. If some items or units are randomly selected from a lot in such a way that a selected sample is a true representative of the entire lot and each and every unit of the selected sample is inspected, the inspection is known as **sample inspection**.
8. **Acceptance sampling** is a technique in which a small part or a fraction of units is selected randomly from a lot and the selected units are inspected to decide whether the lot should be accepted or rejected on the basis of the information supplied by the sample inspection.
9. The proportion of defective units in a lot is called **lot quality** or **proportion defective**, which is denoted by p and defined as follows:
$$p = \frac{\text{Number of defective units in a lot}}{\text{lot size}}$$
10. The **acceptance quality level (AQL)** is the quality level decided mutually by the manufacturer and the buyer. If the proportion of defective units in a lot is less than or equal to AQL, the buyer has to accept the lot. Otherwise, the buyer may accept or reject the lot.
11. The **lot tolerance percent or proportion defective (LTPD)** is the quality level decided mutually by the manufacturer and the buyer. If the proportion of defects in a lot is greater than or equal to this level, the buyer will definitely reject the lot. Otherwise, the buyer may accept or reject the lot.

12. The probability of rejecting a lot of acceptance quality level (AQL) is known as producer's risk $P_p(p)$.
13. The probability of accepting a lot of unsatisfactory quality (LTPD) is known as consumer's risk $P_c(p)$.

5.8 SOLUTIONS/ANSWERS

E1) The carton contains 1000 pens and so the lot size $N = 1000$.

Since the shopkeeper draws a sample of size 25 pens, the sample size $n = 25$.

The shopkeeper accepts the lot if the inspected sample contains at most two defective pens. Otherwise, he/she rejects the lot.

$\therefore$ The acceptance number $c = 2$

The lot contains 40 defective pens

$$\therefore p = \frac{\text{Number of defective pens in the lot}}{\text{lot size}} = \frac{40}{1000} = 0.04$$

If X represents the number of defective pens in the sample, then the shopkeeper accepts the lot if $X \leq c = 2$. Therefore, the probability of accepting the lot is given by

$$P_a = P[X \leq 2] = P[X = 0] + P[X = 1] + P[X = 2]$$

Since the lot size is large as compared to the sample size ($N \geq 10n$), X approximately follows the binomial distribution with parameters n and p . Therefore, we can obtain this probability by using Table I.

From Table I, for $n = 25$, $x = c = 2$, and $p = 0.04$, we have

$$P_a = P[X \leq c = 2] = 0.9235$$

E2) It is given that

$$n = 5, c = 0 \text{ and } p = 2\% = 0.02$$

Suppose X represents the number of defective units in the sample, the probability of accepting the lot is given by

$$P_a = P[X \leq c] = P[X \leq 0]$$

Since X follows the binomial distribution, we can use Table I to obtain this probability.

From Table I, for $n = 5$, $x = c = 0$, and $p = 0.02$, we have

$$P[X \leq 0] = 0.9039$$

Hence, the probability of accepting the lot is given by

$$P_a = P[X \leq 0] = 0.9039$$

E3) The shopkeeper purchases pens from a pen company and, therefore, he/she is a consumer. The pen company is a producer since it sells pens.

We have

$$n = 25, c = 2, \text{ AQL} = 0.03 \text{ and LTPD} = 0.10$$

Product Control

The producer's risk is given by

$$\begin{aligned}P_p &= \alpha = P[\text{rejecting a lot of acceptance quality level}] \\&= 1 - P[\text{accepting a lot of AQL} = 0.03] \\&= 1 - P_a(p = 0.03)\end{aligned}$$

From Table I, for $n = 25$, $x = c = 2$ and $p = \text{AQL} = 0.03$, we have

$$\begin{aligned}P_a(p = 0.03) &= 0.9620 \\ \therefore P_p &= 1 - P_a(p = 0.03) = 1 - 0.9620 = 0.0380\end{aligned}$$

The consumer's risk is given by

$$\begin{aligned}P_c &= P[\text{accepting a lot of LTPD quality}] \\&= P_a(p = 0.10)\end{aligned}$$

From Table I, for $n = 25$, $x = c = 2$ and $p = \text{LTPD} = 0.10$, we have

$$\begin{aligned}P_a(p = 0.10) &= 0.5371 \\ \therefore P_c &= P_a(p = 0.10) = 0.5371\end{aligned}$$

E4) It is given that

$$n = 5, c = 0, \text{AQL} = 1\% = 0.01 \text{ and } \text{LTPD} = 5\% = 0.05$$

The producer's risk is

$$\begin{aligned}P_p &= \alpha = P[\text{rejecting a lot of acceptance quality level}] \\&= 1 - P_a(p = 0.01)\end{aligned}$$

From Table I, for $n = 5$, $x = c = 0$ and $p = \text{AQL} = 0.01$, we have

$$\begin{aligned}P_a(p = 0.01) &= 0.9510 \\ \therefore P_p &= 1 - P_a(p = 0.01) = 1 - 0.9510 = 0.0490\end{aligned}$$

The consumer's risk is

$$\begin{aligned}P_c &= P[\text{accepting a lot of LTPD quality}] \\&= P_a(p = 0.05)\end{aligned}$$

From Table I, for $n = 5$, $x = c = 0$ and $p = \text{LTPD} = 0.05$, we have

$$\begin{aligned}P_a(p = 0.05) &= 0.7738 \\ \therefore P_c &= P_a(p = 0.05) = 0.7738\end{aligned}$$

UNIT 6 RECTIFYING SAMPLING PLANS

Structure

- 6.1 Introduction
 - Objectives
- 6.2 Rectifying Sampling Plan
 - Implementation of Rectifying Sampling Plan for Attributes
- 6.3 Average Outgoing Quality (AOQ)
- 6.4 Operating Characteristic (OC) Curve
- 6.5 Average Sample Number (ASN)
- 6.6 Average Total Inspection (ATI)
- 6.7 Summary
- 6.8 Solutions/Answers

6.1 INTRODUCTION

In Unit 5, you have learnt about the acceptance sampling plans and how these are implemented in industry. You have also learnt about different terms related to it such as acceptance quality level (AQL), lot tolerance percent defective (LTPD), producer's risk and consumer's risk. In an acceptance sampling plan, if the consumer rejects the lot on the basis of the information provided by the inspected sample and the producer is not able to sell the rejected lot, he/she suffers loss. If this continues, the producer might face a huge loss and not be able to continue with production. He/she may even have to shut it down. To avoid such situations, the concept of the rectifying sampling inspection plan was introduced.

In this unit, you will learn about the rectifying sampling plans for attributes. In Sec. 6.2, we discuss what is the rectifying sampling plan and how it implement in industry. In Secs. 6.3 to 6.6, we introduce some more parameters such as average outgoing quality (AOQ), operating characteristic (OC) curve, average sample number (ASN) and average total inspection (ATI). In the next unit, you will study single sampling plans.

Objectives

After studying this unit, you should be able to:

- explain why rectifying sampling plans are needed;
- describe a rectifying sampling plan;
- define average outgoing quality (AOQ) and the average outgoing quality limit (AOQL);
- describe the operating characteristic (OC) curve; and
- define average sample number (ASN) and average total inspection (ATI).

6.2 RECTIFYING SAMPLING PLAN

In an acceptance sampling plan, the consumer accepts the lot if the acceptance criteria are satisfied. Otherwise, he/she rejects the lot. This increases the

producer's risk. Rectifying sampling plans are used to reduce the producer's risk. In such a plan, the entire lot is not rejected. Instead, each and every unit/item of the lot is inspected. It means that 100% inspection of the rejected lot is carried out and the defective units found in the lot are replaced by non-defective units. This procedure is known as **rectifying** or **screening** or **detailing** the rejected lots.

The sampling plan in which 100% inspection is carried out for rejected lots is called the **rectifying sampling inspection plan**. It is also called the **rectifying sampling plan** or **rectifying inspection plan**. Let us explain the procedure for implementing this plan.

6.2.1 Implementation of Rectifying Sampling Plan for Attributes

Suppose that lots of the same size (N) are received from the supplier or the final assembly line and submitted for inspection one at a time. The procedure for implementing the rectifying sampling plan to arrive at a decision about the lot is described in the following steps:

- Step 1:** We draw a random sample of size n from the lot received from the supplier or the final assembly.
- Step 2:** We inspect each and every unit of the sample and classify it as defective or non-defective on the basis of certain criteria. At the end of the inspection, we count the number of defective units found in the sample.
- Step 3:** We compare the number of defective units found in the sample with the acceptance criteria.
- Step 4:** If acceptance criteria are satisfied, we accept the entire lot by replacing all defective units **in the sample** by non-defective units. If the criteria are not satisfied, we accept the lot by inspecting **the entire lot** and replacing all defective units in it by non-defective units.

The steps described above are shown in Fig. 6.1.

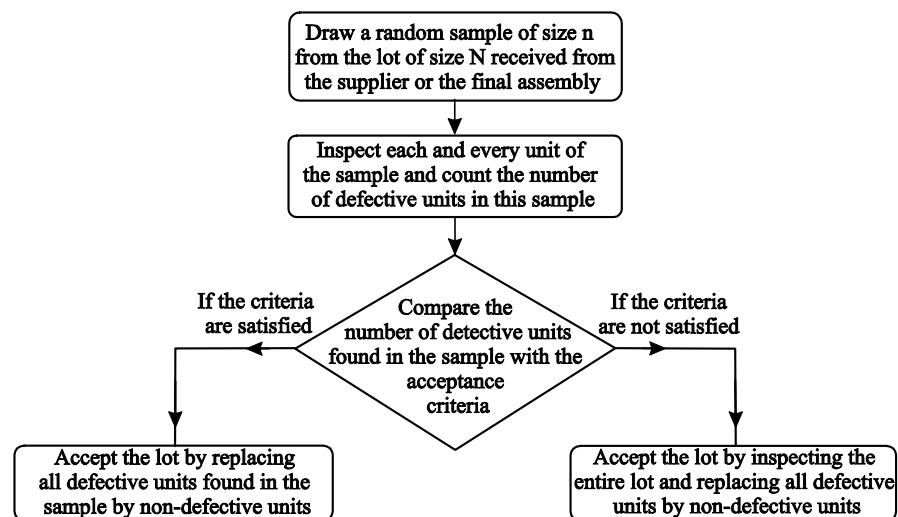


Fig. 6.1: Procedure for implementation of the rectifying sampling plan.

Let us explain these steps further with the help of an example.

Example 1: Suppose a cricket ball manufacturing company supplies lots of 200 balls. To check the quality of the lots, the buyer and the manufacturing

company decide that a sample of size 20 will be drawn from each lot. The lot would be accepted if the inspected sample contains at most one defective ball. Otherwise, the lot would be rejected. If they use the rectifying sampling plan, explain the procedure for implementing it.

Solution: For implementing the rectifying sampling plan, the buyer draws 20 balls from each lot, inspects each and every ball of the sample and classifies it as defective or non-defective on the basis of certain criteria. At the end of the inspection, he/she counts the number of defective balls found in the sample and compares the number of defective balls with the acceptance criteria. If the number of defective balls in the sample is less than or equal to the acceptance number, he/she accepts the lot by replacing all defective balls in the sample by non-defective balls. If the number of defective balls is greater than the acceptance number, he/she carries out 100% inspection for this lot instead of rejecting the lot. The defective balls found in the lot are replaced by non-defective balls and the lot is accepted.

You may like to explain the procedure for implementing a rectifying sampling plan yourself. Try the following exercise.

E1) A hospital receives disposable injection syringes in lots of 2000. A single sampling plan with $n = 25$ and $c = 2$ is being used for accepting the lots. Explain the procedure for implementing the rectifying sampling plan in this case.

So far you have learnt about the rectifying sampling plan. In Secs. 6.3 to 6.6, we explain various terms associated with the rectifying sampling plans.

6.3 AVERAGE OUTGOING QUALITY (AOQ)

In a sampling inspection plan, the items or units produced/manufactured by the producer are formed in lots. The average quality level of the lots is set by the producer and the consumer through negotiation and the producer sends the lots to the consumer for inspection. The quality of the lots before the inspection is known as **incoming quality** and the quality of the lots which have been accepted after the inspection is known as **outgoing quality**. In an acceptance sampling plan, the lots are either accepted or rejected. So the outgoing quality is the same as the incoming quality. However, in a rectifying sampling inspection plan, the rejected lots are rectified or screened. So the outgoing quality will differ from the incoming quality. Therefore, the concept of **average outgoing quality (AOQ)** is particularly useful for the evaluation of a rectifying sampling plan.

You have learnt in Sec.6.2 that the consumer takes a sample from each lot to check its quality. He/she inspects each item or unit of the sample for defects. If the number of defective units in the sample is less than the acceptance number, the lot is accepted by replacing all defective units found in the sample by non-defective units. If the number of defective units is greater than the acceptance number, each and every unit of the lot is inspected. It means that 100% inspection is carried out for each rejected lot and all defective units found in the lot are replaced by non-defective units. Therefore, these lots are accepted after 100% inspection with zero percent defective. As a result, the accepted stores will consist of lots of varying quality level, ranging from quality levels

lower than acceptance quality level to lots with zero defective. Therefore, we need to define the concept **average outgoing quality**.

When all lots are considered together, their average quality level may be considerably different from the incoming quality. The average outgoing quality (AOQ) is defined as follows:

The expected quality of the lots after the application of sampling inspection is called the **average outgoing quality**. It is calculated as follows:

$$AOQ = \frac{\text{Number of defective units in the lot after inspection}}{\text{Total number of units in the lot}} \quad \dots(1)$$

But we do not know the number of defective units in the entire lot after the inspection of samples. So we have an alternative way of calculating AOQ. If N is the size of each lot, n is the sample size inspected in each lot, p is the incoming quality of the lots and P_a is the probability of accepting the lot of incoming quality p , AOQ for single sampling plan is given by

$$AOQ = \frac{p(N - n)P_a}{N} \quad \dots(2)$$

We shall derive this formula in Unit 7. The concept of AOQ is shown in Fig. 6.2.

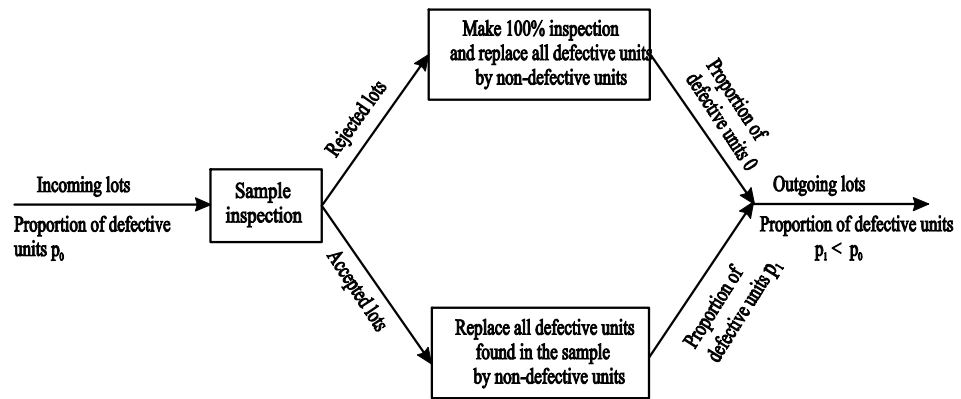


Fig. 6.2: Concept of average outgoing quality.

For the acceptance sampling plan in which rectification is not done, the AOQ is the same as the incoming quality.

If we draw a graph of AOQ versus the incoming quality, the curve so obtained is known as the **AOQ curve**. A typical AOQ curve is shown in Fig. 6.3.

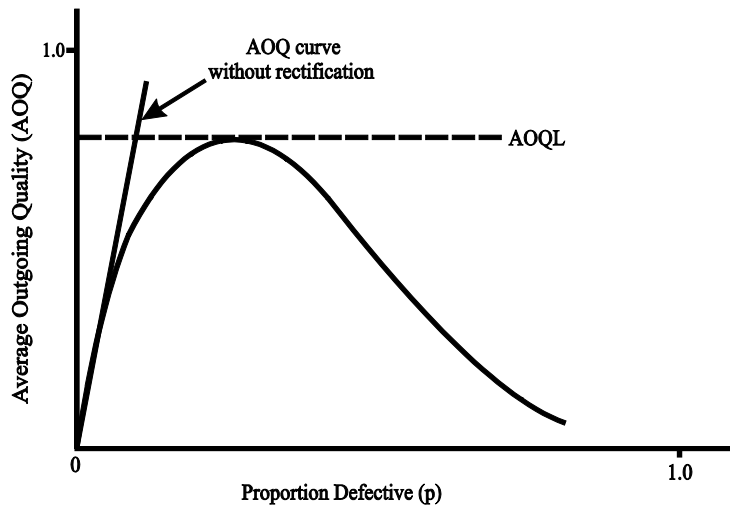


Fig. 6.3: The AOQ curve.

If we analyse the AOQ curve (shown in Fig. 6.3), we observe that for $p = 0$, the incoming lots have no defective units and hence there are no defective units in the outgoing lots. The consumer accepts all lots. So in this case, $AOQ = 0$. When the incoming quality is good, a large proportion of the lots will be accepted by the rectifying sampling plan and only a smaller fraction will be screened. Hence, the outgoing quality will be good. However, when the incoming quality is not good, a large proportion of the lots will be screened. In such cases, the outgoing quality will also be good because defective units or items will either be replaced or rectified. Between these extremes, the AOQ increases up to a maximum and then decreases. The maximum value of AOQ represents the worst possible average for the outgoing quality. It is known as the **average outgoing quality limit (AOQL)**. The AOQL tells us that no matter how poor the incoming quality is, on an average, the outgoing quality will never be poorer than AOQL.

Let us take up an example based on AOQ.

Example 2: Suppose a cricket ball manufacturing company formed lots of 500 balls. To check the quality of the lots, the buyer draws 20 balls from each lot and accepts the lot if the sample contains at most one defective ball. Otherwise, he/she rejects it. Suppose the quality of the submitted lots is 0.03. Calculate the AOQ for this sampling plan. If rejected lots are screened and all defective balls are replaced by non-defective ones, calculate the AOQ again.

Solution: It is given that

$$N = 500, n = 20, c = 1 \text{ and } p = 0.03$$

We know that in an acceptance sampling plan, the lots are either accepted or rejected and the average outgoing quality is the same as the incoming quality. Therefore, in the first case, when the lot is either accepted or rejected, the AOQ is the same as the quality of the submitted lot, i.e., $AOQ = 0.03$.

In the second case, when the rejected lots are screened and all defective balls are replaced by non-defective ones, we calculate the AOQ using equation (2).

For calculating the AOQ in this case, we have to calculate the probability of accepting a lot (P_a). Recall the method of finding it described in Sec. 5.5 of Unit 5.

If X represents the number of defective balls in the sample, the buyer accepts the lot if $X \leq c$. Here $c = 1$. Therefore, the probability of accepting the lot is given by

$$P_a = P[X \leq c = 1] = \sum_{x=0}^1 P[X = x]$$

Since $N \geq 10n$, we can use the binomial distribution and Table I given at the end of this block to obtain P_a .

From Table I, for $n = 20$, $x = c = 1$ and $p = 0.03$, we have

$$P_a = P[X \leq 1] = \sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x} = 0.8802$$

On putting the values of N , n , p and P_a in equation (2), we get

$$AOQ = \frac{p(N-n)P_a}{N} = \frac{0.03 \times (500-20) \times 0.8802}{500} = 0.025$$

You may like to calculate the AOQ in the following exercises.

-
- E2)** Suppose, in Example 2, the quality of the submitted lots is 0.01, calculate AOQ for this sampling plan.
- E3)** Suppose, in E1, the quality of the submitted lot is 0.05, calculate AOQ for this sampling plan.
-

We now describe another feature of the sampling plan known as the operating characteristic curve.

6.4 OPERATING CHARACTERISTIC (OC) CURVE

You have learnt that in acceptance sampling, the lot is either rejected or accepted on the basis of conclusions drawn from the sample. Sometimes, a good lot may be rejected and a bad lot may be accepted because we infer the quality of all units in the lot on the basis of a sample (small part of the lot).

So we require an acceptance sampling plan which ensures that good lots are always accepted and the bad lots are always rejected. It means that the plan should perfectly discriminate between good and bad lots. Therefore, we introduce a tool called the **operating characteristic curve**, which measures the discriminating ability of a plan. The operating characteristic (OC) curve of an acceptance sampling plan reflects the ability of the plan to distinguish between good and bad lots.

The OC curve for a sampling plan is a graph of the probability of accepting the lot versus the proportion defective (proportion of defective units) in the lot. It is plotted by taking the proportion defective (p) along the X-axis and the probability of accepting the lot along the Y-axis.

When the acceptance sampling plan is used, there is a conflict of interest between the producer and the consumer. The producer wants that all acceptable lots to be accepted and the consumer wants that all unsatisfactory lots to be rejected. A sampling plan which perfectly discriminates between good and bad lots is called an **ideal sampling plan**. The OC curve for an ideal sampling plan is called an **ideal OC curve** and is shown in Fig. 6.4.

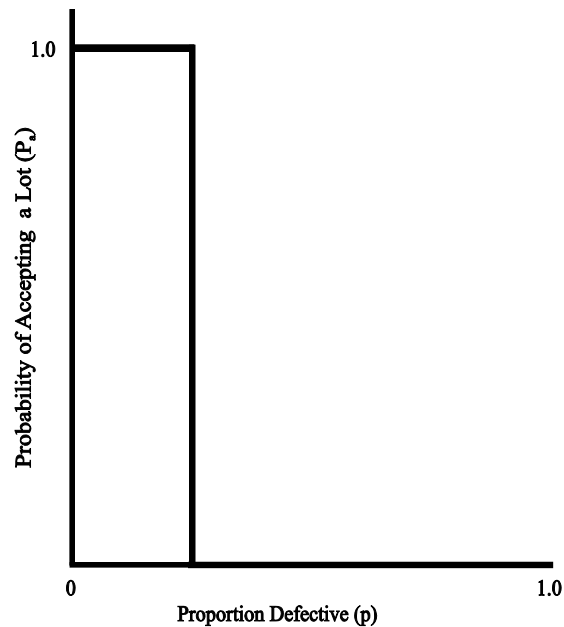


Fig. 6.4: Ideal OC curve.

The ideal OC curve runs horizontally at a probability of acceptance $P_a = 1.0$ until a level of lot quality that is considered **bad** is reached. At that point, the curve drops vertically to a probability of acceptance $P_a = 0$. Then the curve runs horizontally again for all lot proportion defectives greater than the undesirable level. If such a sampling plan could be employed, all lots of **bad** quality would be rejected and all lots of **good** quality would be accepted. But, the ideal OC curve can almost never be obtained in practice. In theory, it could be realised by 100% inspection, if the inspections were error free. But for 100% inspection, the cost is high and the time of inspection is much more. So we use sampling plans. The ideal OC curve shape can be approached by increasing sample size. A typical OC curve is shown in Fig. 6.5.

The OC curve shows the acceptance probability of a lot of certain quality. It represents the relationship between the probability of acceptance and the quality of the lot. If the lot has no defective units, it is certain to be accepted and if it has all defective units then it is equally certain to be rejected. Hence, the OC curve has to start with $P_a = 1$ when $p = 0$. As the proportion defective (p) of the lot increases, the probability P_a of lot acceptance decreases until it reaches 0. Thus, $P_a = 0$ when $p = 1$.

The OC curve for a particular combination of n and c shows how well the plan discriminates between good and bad lots. In Fig. 6.5, we have shown an OC curve for $n = 100$ and $c = 2$. In this plan, if 0, 1, or 2 defective units are found in the sample of size $n = 100$, the lot would be considered acceptable and if more than 2 defective units are found, it would be rejected. Suppose, the actual lot quality is 0.01, i.e., there are 1 percent defective units in the lot, the OC curve shows that the probability of accepting the lot is approximately 0.99. It means that this sampling plan accepts the lot about 99% of the time and rejects the lot about 1% of the time. However, if the actual quality of a lot is 0.05, i.e., there are 5 percent defective units, the OC curve shows that the probability of accepting the lot is approximately 0.19.

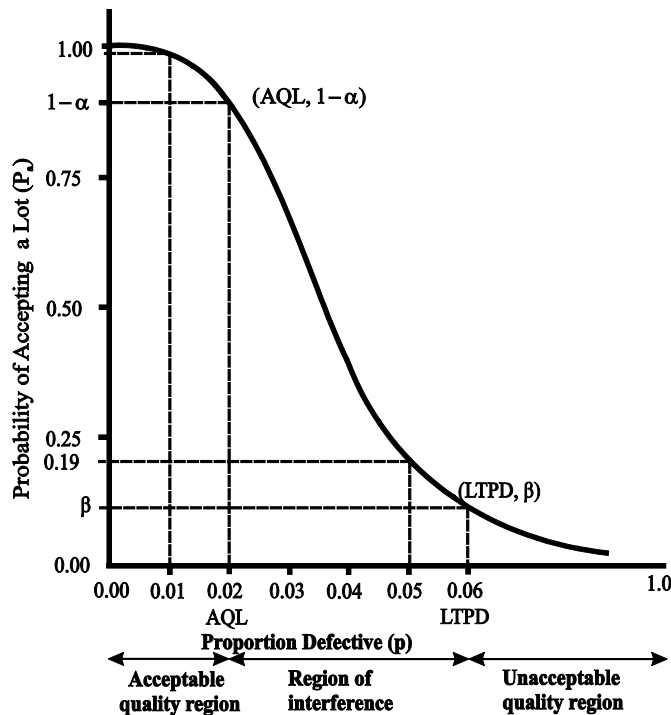


Fig. 6.5: the OC curve.

Hence, if the actual quality of a lot is good, the plan provides high probability of acceptance and if its quality is poor, the probability of acceptance of the lot is low. Thus, the OC curve shows how well a given plan discriminates between good and bad quality.

The OC curve of a sampling plan passes through two points agreed upon by the producer and the consumer. These points are $(AQL, 1 - \alpha)$ and $(LTPD, \beta)$ where the average quality level (AQL) and the lot tolerance percent defective (LTPD) are decided by the producer and the consumer after negotiation with each other. You know from Unit 5 that α and β are the producer's risk and the consumer's risk, respectively. Since the OC curve represents the probability of accepting a lot, instead of taking α (probability of rejecting a lot of quality AQL) we take $1 - \alpha$ (probability of accepting a lot of quality AQL).

In practice, the performance of acceptance sampling plan for discriminating good and bad lots depends mainly on the sample size (n) taken from the lot and the acceptance number (c) that can be permitted in the sample. Since n and c are different in different sampling plans, the OC curve will be different for different sampling plans. Let us see how the sample size (n) and the acceptance sampling number (c) affect the OC curve.

Effect of increasing the sample size n on OC curve

Study Fig. 6.6. It shows the effect of increasing sample size on OC curve. Here we have three OC curves for three sampling plans with $n = 100, c = 1$; $n = 200, c = 2$ and $n = 300, c = 3$. From Fig. 6.6, we note that even though each plan uses the same percent defective which can be allowed for an acceptance lot ($1/100, 2/200, 3/300 = 0.01$, i.e., 1 percent), the OC curve becomes steeper and lies closer to the origin as the sample size increases.

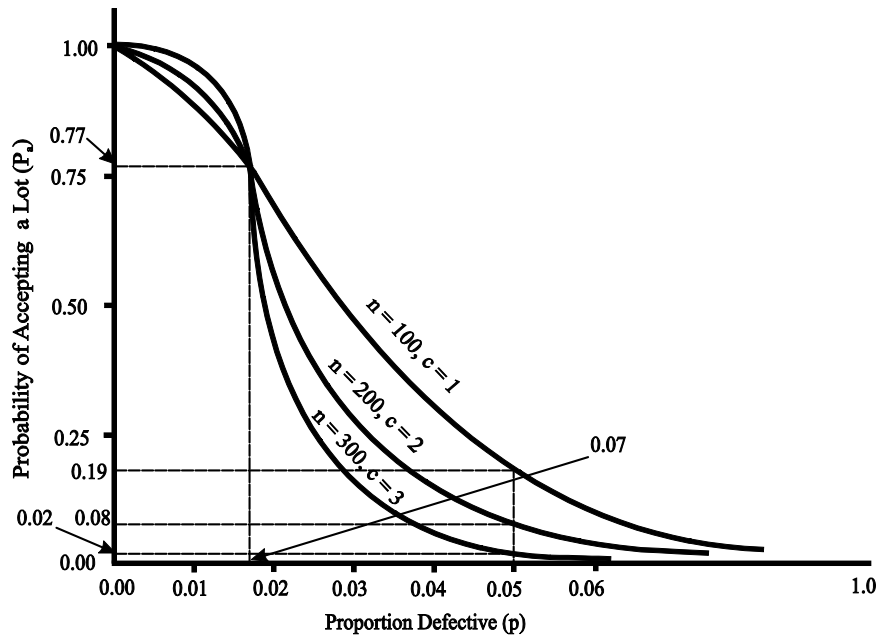


Fig. 6.6: The OC curves for $n = 100, c = 1$; $n = 200, c = 2$ and $n = 300, c = 3$.

If we compare the discrimination power of the three plans, we see that all three plans would accept lots with about 0.017, i.e., 1.7 percent defective units, about 77 percent of the time. However, if the actual quality falls to 5 percent defective units, the plan $n = 100, c = 1$ accepts the lot about 19 percent of the time, $n = 200, c = 2$ about 8 percent of the time and $n = 300, c = 3$ about 2 percent of the times. Therefore, the plans with larger sample sizes have definitely more discrimination power. It means that the plans with larger sample sizes are more effective in protecting the consumer against the acceptance of relatively bad lots and also give protection to the producer against rejection of relatively good lots.

Effect of decreasing the acceptance number (c) on OC curve

In Fig. 6.6, you have seen that the OC curve changes as the sample size changes. Generally, changing the acceptance number does not dramatically change the slope of the OC curve. As the acceptance number is decreased, the OC curve shifts to the left. For the same n , plans with smaller values of c provide discrimination at lower levels of the lot proportion defective than the plans with larger values of c . You can see this in Fig. 6.7, which shows the effect of acceptance number on the OC curve. It has three OC curves for $n = 200$ and $c = 1, 2$ and 3 .

The greater the slope of the OC curve, greater the discriminatory power.

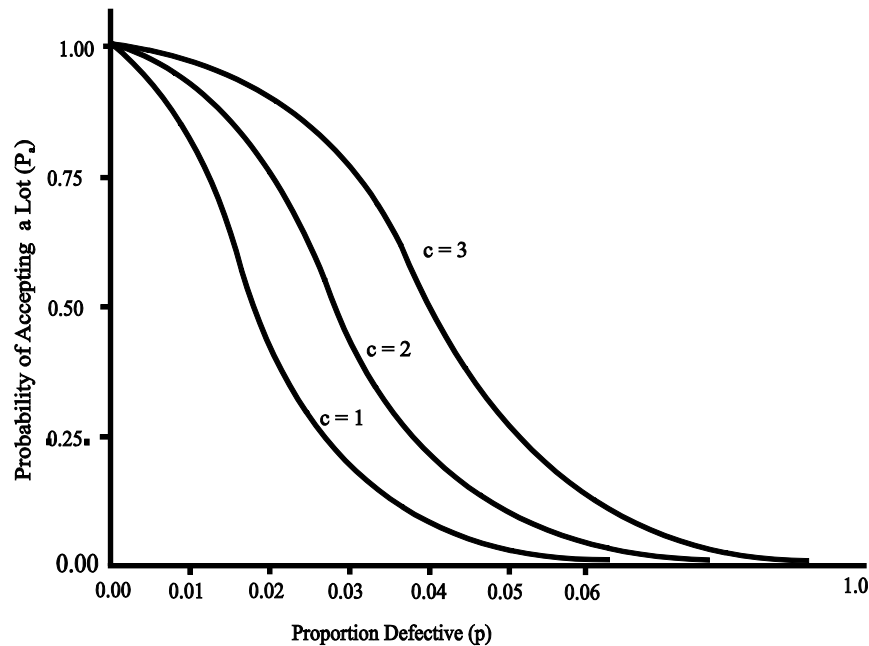


Fig. 6.7

Note that in Fig. 6.7, the effect of decreasing c is that the OC curve becomes steeper. Thus, when we decrease the acceptance number (c), the sampling plan becomes **tighter or stricter**. It means that the plan with smaller acceptance number has more discrimination power.

Now that you know what the OC curve of a sampling plan is and why it is used, let us summarize its properties.

Properties of the Operating Characteristic Curve

The OC curve possesses the following properties:

- i) The OC curve of an acceptance sampling plan shows the ability of the plan to distinguish between good and bad lots.
- ii) The OC curve with larger sample size is steeper. It means that the plan with larger sample size has more discrimination power. The plan with larger sample size protects the consumer against the acceptance of relatively bad lots and also gives protection to the producer against rejection of relatively good lots.
- iii) The OC curve for a fixed sample size and smaller acceptance sample number is steeper. It means that the plan with a smaller acceptance number has more discrimination power.
- iv) The OC curves with fixed sample sizes give constant quality protection, whereas the OC curves with relative sample sizes give different quality protections.

We now discuss another useful concept of sampling plans called the **average sample number (ASN)**.

6.5 AVERAGE SAMPLE NUMBER (ASN)

In acceptance sampling plan, the decision of acceptance or rejection of a lot is based on the information provided by the sample (s) drawn from the lot. So the average sample number can be defined as follows:

The **average sample number (ASN)** is defined as the average (expected) number of sample units per lot, which is required to arrive at a decision about the acceptance or rejection of the lot under the **acceptance sampling plan**.

The curve drawn between the ASN and the lot quality (p) is known as the **ASN curve**.

In a single sampling plan, we take the decision of acceptance or rejection of the lot on the basis of only a single sample of size n . Hence, the ASN in a single sampling plan is simply the sample size n , which means that ASN is constant. Therefore, the ASN curve for a single sampling plan is a straight line as shown in Fig. 6.8.

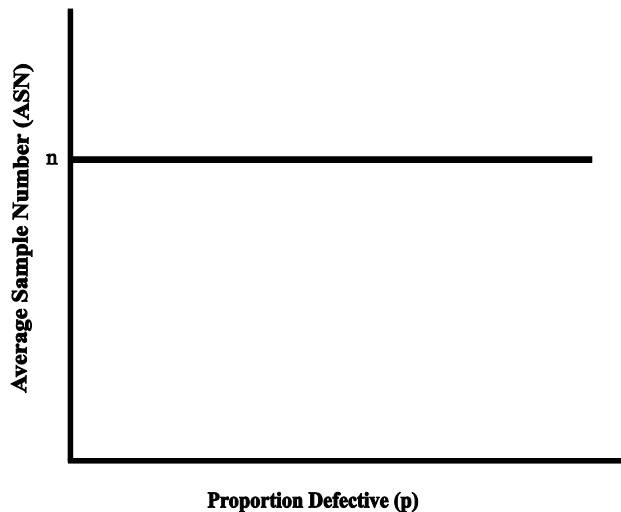


Fig. 6.8: The ASN curve for a single sampling plan.

ASN for a double sampling plan varies with the quality of the lot and is a function of the actual proportion defective (p) in the lot. We shall discuss the method of finding ASN and ASN curve for a double sampling plan in Unit 8.

The concept of average sample number is considered for the acceptance sampling plan. The average total inspection (ATI) plays the same role in a rectifying sampling plan.

6.6 AVERAGE TOTAL INSPECTION (ATI)

The average total inspection (ATI) for a rectifying sampling plan is defined as follows:

The average number of units inspected per lot under the rectifying sampling plan is called the average total inspection (ATI).

In other words,

The **average total inspection (ATI)** is the number of units inspected per lot to take the decision for acceptance or rejection of the lot under rectifying sampling plan calling for 100% inspection of the rejected lots.

If the lot quality is p , the average total inspection (ATI) for a single sampling plan is given by

$$ATI = n + (1 - P_a)(N - n) \quad \dots(3)$$

We shall derive this formula in Unit 7.

Therefore,

$$ATI = ASN + (\text{average number of units inspected in the rejected lots})$$

Thus, if the lot is accepted on the basis of the rectifying sampling plan, then

$$ATI = ASN$$

Otherwise,

$$ATI > ASN$$

If a lot contains no defective unit, it will obviously be accepted by the sampling plan and only n items will be inspected. Therefore, in this case ATI will be equal to the sample size n . If all units of the lot are defective, the lot will be rejected and 100% inspection of the lot will be called. Therefore, in this case the ATI will be equal to the lot size N . If the lot quality lies between 0 and 1, i.e., $0 < p < 1$, the ATI will lie between the sample size n and the lot size N . This means that ATI is a function of the lot quality p .

The curve drawn between ATI and the lot quality (p) is known as **ATI curve**. A typical ATI curve for a single sample plan is shown in Fig. 6.9 for $N = 2000$, $n = 30$ and $c = 2$.

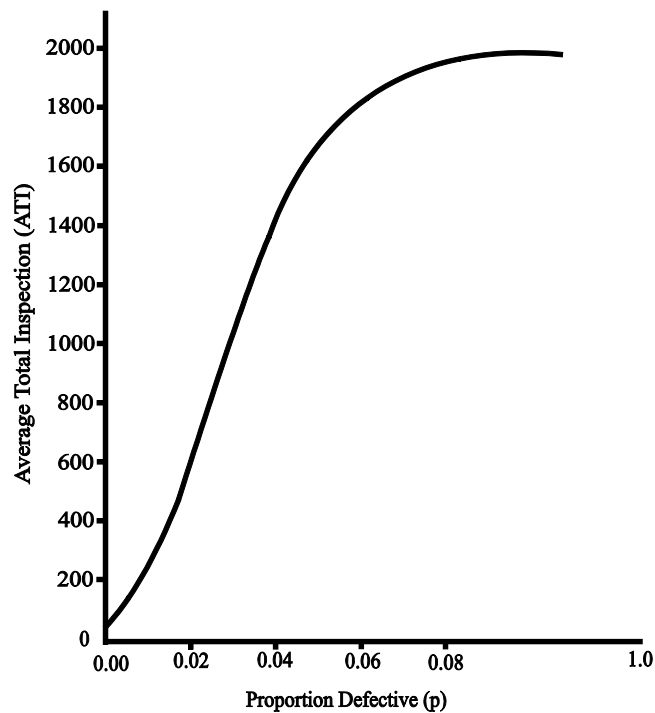


Fig. 6.9: The ATI curve.

We shall discuss the methods of obtaining ATI and ATI curves for single and double sampling plans in Units 7 and 8.

Let us now compute ASN and ATI for a real life situation.

Example 3: A hospital receives disposable injection syringes in lots of 2000. A single sampling plan with $n = 25$ and $c = 2$ is being used for inspection. Calculate the average sample number (ASN). Suppose the incoming lots

contain 5% defective units. Calculate the average total inspection (ATI) for this plan, if the rejected lots are 100% inspected and all defective syringes are replaced by non-defective syringes.

Solution: It is given that

$$N = 2000, n = 25, c = 2, p = 0.05$$

In the first case, the decision of acceptance or rejection of the lot is taken only on a single sample of size $n = 25$. Therefore, the ASN for this plan is simply the sample size, i.e., $ASN = n = 25$.

We calculate ATI using equation (3).

We first calculate the probability of accepting a lot of quality $p = 0.05$.

If X represents the number of defective injection syringes in the sample, the hospital accepts the lot if $X \leq c = 2$. Therefore, the probability of accepting the lot is given by

$$P_a = P[X \leq 2] = \sum_{x=0}^2 P[X = x]$$

Since $N \geq 10n$, we can obtain this probability by using Table I.

From Table I, for $n = 25$, $x = c = 2$ and $p = 0.05$, we have

$$P_a = P[X \leq 2] = \sum_{x=0}^2 {}^n C_x p^x (1-p)^{n-x} = 0.8729$$

On putting the values of N , n and P_a in equation (3), we get

$$\begin{aligned} ATI &= n + (1 - P_a)(N - n) \\ &= 25 + (1 - 0.8729)(2000 - 25) \\ &= 25 + 251.02 = 276.02 \approx 277 \end{aligned}$$

It is time for you to do the following exercises based on ASN and ATI.

E4) Explain the following terms:

- i) Average sample number (ASN), and
- ii) Average total inspection (ATI).

E5) A computer manufacturer purchases computer chips from a company in lots of 400. Twenty computer chips are sampled from each lot at random and inspected for defects. The computer manufacturer accepts the lot if inspected sample contains at most one defective chip. Otherwise, he rejects it. Suppose the incoming lots contain 3% defective chips. Calculate ASN for this plan. If rejected lots are screened and all defective computer chips are replaced by non-defective ones, calculate AOQ for this plan.

You have studied four curves, namely, AOQ, OC, ASN and ATI in Secs. 6.3 to 6.6. Their main purpose is to assess the efficiency and compare the costs of sampling plans. The AOQ and OC curves serve to assess protection of the consumer and the producer in terms of their risks given by the sampling plan and also give an idea of the efficiency of the sampling plan. The ASN and ATI curves give an indication of the cost of inspection. With the help of these curves, we arrive at a balance between the costs of inspection and the degree of

protection. If inspection (sampling) costs are more than the costs of wrong decisions (without the aid of such inspection), then that particular inspection plan cannot be justified on economic considerations.

We now end this unit by giving a summary of what we have covered in it.

6.7 SUMMARY

1. An acceptance sampling plan in which rejected lots are 100% inspected is called a **rectifying sampling plan**.
2. The quality of the lots before inspection is known as **incoming quality** and the quality of the lots which have been accepted after the inspection is known as **outgoing quality**.
3. The expected quality of the lots after the application of sampling inspection is called **average outgoing quality (AOQ)** and is given by

$$AOQ = \frac{\text{Number of defective units in the lot after inspection}}{\text{Total number of units in the lot}}$$

4. The maximum value of AOQ represents the worst possible average for the outgoing quality and is known as the **average outgoing quality limit (AOQL)**.
5. The **OC curve** for a sampling plan is a graph of the probability of accepting the lot versus the proportion or fraction defective in the lot. It shows the ability of the plan to distinguish between good and bad lots.
6. A sampling plan which perfectly discriminates between good and bad lots is called an **ideal sampling plan** and the OC curve for this sampling plan is called an **ideal OC curve**.
7. The **precision** with which a sampling plan differentiates between good and bad lots increases with the **size of the sample**: The **greater the slope** of the OC curve, the **greater is the discrimination power** of the plan. The plan with a **smaller acceptance number** has **more discrimination power**.
8. The **average sample number (ASN)** is defined as the average (expected) number of sample units per lot, which is required to arrive at a decision about the acceptance or rejection of the lot under acceptance sampling plan. The curve drawn between ASN and lot quality (p) is known as the **ASN curve**.
9. The average number of units inspected per lot under the rectifying sampling plan is called the **average total inspection**. The curve drawn between ATI and lot quality (p) is known as the **ATI curve**.

6.8 SOLUTIONS/ANSWERS

- E1)** To check the quality of the lot, the quality inspector of the hospital draws 25 syringes randomly from each lot of disposable injection syringes and inspects each and every syringe of the sample. He/she classifies each syringe as defective or non-defective on the basis of certain defects. At the end of the inspection, he/she counts the number of defective syringes found in the sample and compares the number of defective syringes found in the sample with the acceptance number. If the number of defective syringes in the sample is greater than $c = 2$,

say, 3, instead of rejecting the lot, he/she calls for 100% inspection of this lot and replaces all defective syringes found in the lot by non-defective syringes. Then he/she accepts the lot. If the number of defective syringes is less than or equal to $c = 2$, say, 1, he/she accepts the lot by replacing all defective syringes in the sample by non-defective syringes.

E2) It is given that

$$N = 500, n = 20, c = 1 \text{ and } p = 0.01$$

We calculate AOQ for the single sampling plan using equation (2). We first calculate P_a . Since $N \geq 10n$, we can obtain this probability by using Table I.

From Table I, for $n = 20$, $x = c = 1$ and $p = 0.01$, we have

$$P_a = P[X \leq 1] = \sum_{x=0}^1 {}^nC_x p^x (1-p)^{n-x} = 0.9831$$

On putting the values of N , n , p and P_a in equation (2), we get

$$\begin{aligned} \text{AOQ} &= \frac{p(N-n)P_a}{N} = \frac{0.01 \times (500-20) \times 0.9831}{500} \\ &= \frac{4.7189}{500} = 0.009 \end{aligned}$$

E3) It is given that

$$N = 2000, n = 25, c = 2 \text{ and } p = 0.05$$

We calculate AOQ for the single sampling plan using equation (2).

Since here $N \geq 10n$, the probability of accepting the lot can be obtained as in E2.

$$\therefore P_a = P[X \leq 2] = \sum_{x=0}^2 {}^nC_x p^x (1-p)^{n-x} = 0.8729$$

On putting the values of N , n , p and P_a in equation (2), we get

$$\begin{aligned} \text{AOQ} &= \frac{p(N-n)P_a}{N} = \frac{0.05 \times (2000-25) \times 0.8729}{2000} \\ &= \frac{86.1989}{2000} = 0.043 \end{aligned}$$

E4) Refer to Sec. 6.5 for (i) and Sec. 6.6 for (ii).

E5) In this sampling plan, the decision of acceptance or rejection of the lot is taken only using a single sample of size $n = 20$. Therefore, the ASN for this plan is simply the sample size $n = 20$.

It is given that

$$N = 400, n = 20, c = 1, p = 3\% = 0.03$$

We calculate ATI for the single sampling plan using equation (3).

Since $N \geq 10n$, we can obtain the probability of accepting a lot of quality $p = 3\% = 0.03$ using Table I.

Process Control

From Table I, for $n = 20$, $x = c = 1$ and $p = 0.03$, we have

$$P_a(p) = P[X \leq 1] = \sum_{x=0}^1 {}^nC_x p^x (1-p)^{n-x} = 0.8802$$

On putting the values of N , n and P_a in equation (3) we get

$$\begin{aligned}ATI &= n + (1 - P_a)(N - n) \\&= 20 + (1 - 0.8802)(400 - 20) \\&= 20 + 45.524 = 65.524 \approx 66\end{aligned}$$

UNIT 7 SINGLE SAMPLING PLANS

Structure

- 7.1 Introduction
 - Objectives
- 7.2 Single Sampling Plan
- 7.3 Operating Characteristics (OC) Curve
- 7.4 Producer's Risk and Consumer's Risk
- 7.5 Average Outgoing Quality (AOQ)
- 7.6 Average Sample Number (ASN) and Average Total Inspection (ATI)
- 7.7 Design of Single Sampling Plans
 - Stipulated Producer's Risk
 - Stipulated Consumer's Risk
 - Stipulated Producer's Risk and Consumer's Risk
 - Larson Binomial Nomograph
- 7.8 Summary
- 7.9 Solutions/Answers

7.1 INTRODUCTION

In Units 5 and 6, you have learnt the various features of a sampling inspection plan such as AQL, LTPD, producer's risk, consumer's risk, OC curve, ASN, ATI, etc. The same features characterise different types of sampling plans. In Unit 5, you have also learnt that the main types of acceptance sampling plans for attributes are:

- i) Single sampling plan,
- ii) Double sampling plan,
- iii) Multiple sampling plan, and
- iv) Sequential sampling plan

In this unit, we focus on the **single sampling plans for attributes**. In Sec. 7.2, we explain the single sampling plan and its implementation. We describe various features of the single sampling plan such as the operating characteristic (OC) curve, producer's risk, consumer's risk, average sample number (ASN) and average total inspection (ATI) in Secs. 7.3 to 7.6. Finally, we describe the design of the single sampling plans in Sec. 7.7. In the next unit, we shall discuss the double sampling plans for attributes.

Objectives

After studying this unit, you should be able to:

- describe a single sampling plan;
- compute the probability of accepting or rejecting a lot of given incoming quality in a single sampling plan;
- construct the operating characteristics (OC) curve of a single sampling plan;
- compute the consumer's risk and producer's risk in a single sampling plan;

- compute the average sample number (ASN) and the average total inspection (ATI) for a single sampling plan; and
- design single sampling plans.

7.2 SINGLE SAMPLING PLAN

A sampling plan in which a decision about the acceptance or rejection of a lot is based on a single sample that has been inspected is known as a **single sampling plan**. For example, suppose a buyer purchases cricket balls in lots of 500 from a company manufacturing cricket balls. To check the quality of the lots, the buyer draws a random sample of size 20 from each lot and takes a decision about accepting or rejecting of the lot on the basis of the information provided by this sample. Since the buyer takes the decision about the lot on the basis of a single sample, this sampling plan is a single sampling plan.

A single sampling plan requires the specification of two quantities which are known as **parameters** of the single sampling plan. These parameters are

n – size of the sample, and

c – acceptance number for the sample.

Let us suppose that the lots are of the same size (N) and are submitted for inspection one at a time. The procedure for implementing the single sampling plan to arrive at a decision about the lot is described in the following steps:

Step 1: We draw a random sample of size n from the lot received from the supplier or the final assembly.

Step 2: We inspect each and every unit of the sample and classify it as defective or non-defective. At the end of the inspection, we count the number of defective units found in the sample. Suppose the number of defective units found in the sample is d .

Step 3: We compare the number of defective units (d) found in the sample with the stated acceptance number (c).

Step 4: We take the decision of acceptance or rejection of the lot on the basis of the sample as follows:

Under acceptance sampling plan

If the number of defective units (d) in the sample is less than or equal to the stated acceptance number (c), i.e., if $d \leq c$, we accept the lot and if $d > c$, we reject the lot.

Under rectifying sampling plan

If $d \leq c$, we accept the lot and replace all defective units found in the sample by non-defective units and if $d > c$, we accept the lot after inspecting the entire lot and replacing all defective units in the lot by non-defective units.

The steps described above are shown in Fig. 7.1.

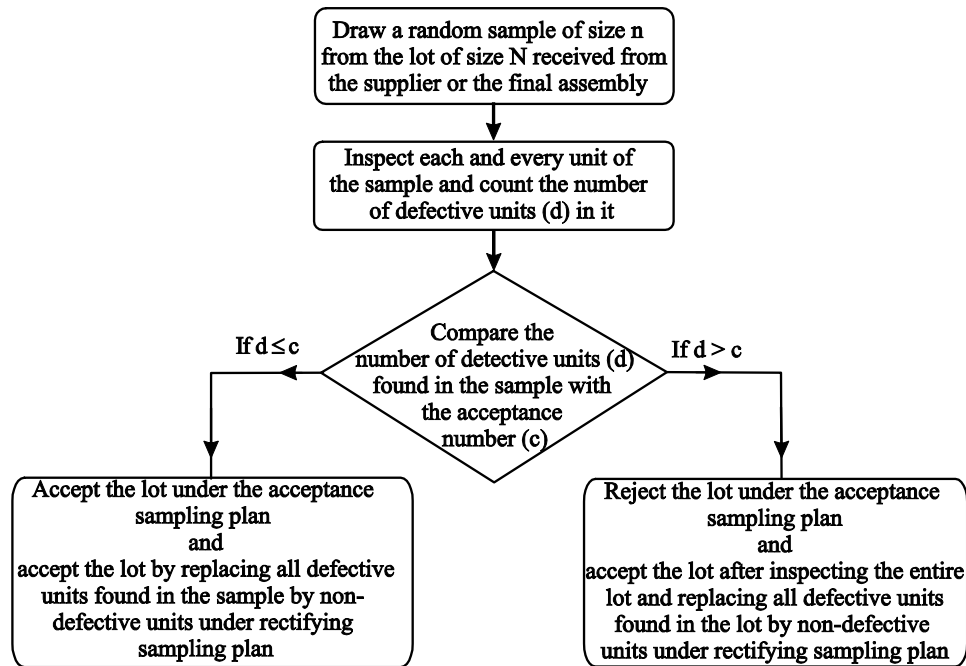


Fig. 7.1: Procedure for implementing a single sampling plan.

Let us explain these steps further with the help of an example.

Example 1: Suppose a mobile phone company produces mobile phones in lots of 100 phones. To check the quality of the lots, the quality inspector of the company uses a single sampling plan with $n = 15$ and $c = 1$. Explain the procedure for implementing it.

Solution: For implementing the single sampling plan, the quality inspector of the company randomly draws a sample of 15 mobile phones from each lot and classifies each mobile of the sample as defective or non-defective. At the end of the inspection, he/she counts the number of defective mobiles (d) found in the sample and compares it with the acceptance number (c). If $d \leq c (= 1)$, he/she accepts the lot and if $d > c (= 1)$, he/she rejects the lot under the acceptance sampling plan. Under rectifying sampling plan, if $d \leq c (= 1)$, he/she accepts the lot by replacing all defective mobiles found in the sample by non-defective mobiles and if $d > c$, he/she accepts the lot by inspecting the entire lot and replacing all defective mobiles in the lot by non-defective mobiles.

You may like to explain the procedure for implementing a single sampling plan yourself. Try the following exercise.

-
- E1)** A manufacturer of silicon chip produces lots of 1000 chips for shipment. A buyer uses a single sampling plan with $n = 50$ and $c = 2$ to test for bad outgoing lots. Explain the procedure for implementing it under acceptance sampling plan.
-

So far you have learnt about the single sampling plan and how it is implemented. We describe various features of the single sampling plan in Secs.7.3 to 7.6.

7.3 OPERATING CHARACTERISTIC (OC) CURVE

Product Control

You have studied in Unit 6 that the operating characteristics (OC) curve is an important aspect of an acceptance sampling plan. This curve displays the discriminatory power of the sampling plan. That is, it shows the probability that a lot submitted with a certain fraction defective will be either accepted or rejected. In this section, we discuss how to construct the OC curve for a single sampling plan.

You have learnt that for constructing an OC curve, we require the probabilities of accepting a lot corresponding to different quality levels. Therefore, we first compute the probability of accepting a lot of incoming quality p for a single sampling plan.

You have studied in Sec. 7.2 that in a single sample plan, we accept the lot if the number of defective units (d) in the sample is less than or equal to the acceptance number (c). It means that if X represents the number of defective units in the sample, we accept the lot if $X \leq c$, i.e., $X = 0$ or 1 or 2 , ..., or c . Therefore, the probability of accepting the lot of incoming quality p is given by

$$\begin{aligned} P_a(p) &= P[X \leq c] = P[X = 0 \text{ or } 1, \dots, \text{ or } c] \\ &= P[X = 0] + P[X = 1] + \dots + P[X = c] \quad \left(\because X = 0, 1, 2, \dots, c \text{ are } \right. \\ &\quad \left. \text{mutually exclusive} \right) \\ &= \sum_{x=0}^c P[X = x] \end{aligned} \quad \dots (1)$$

We can calculate this probability if we know the distribution of X . Generally, in quality control, a random sample is drawn from a lot of finite size without replacement. So in such situations, the number of defective units (X) in the sample follows a hypergeometric distribution. In a lot of size N and incoming quality p , the number of defective units is Np and the number of non-defective units is $N - Np$. Therefore, the probability of getting exactly x defective units in a sample of size n from this lot is given by

$$P[X = x] = \frac{{}^{Np}C_x {}^{N-Np}C_{n-x}}{{}^NC_n}; \quad x = 0, 1, \dots, \min(Np, n) \quad \dots (2)$$

Thus, we can obtain the probability of accepting a lot of quality p in a single sampling plan by putting the value of $P[X = x]$ in equation (1) as follows:

$$P_a(p) = \sum_{x=0}^c P[X = x] = \sum_{x=0}^c \frac{{}^{Np}C_x {}^{N-Np}C_{n-x}}{{}^NC_n} \quad \dots (3)$$

We know from industrial experience that n is usually small for any economically worthwhile production process. Therefore, when sample size n is small compared to lot size (N), i.e., when $N \geq 10n$, we know that the hypergeometric distribution is approximated by the binomial distribution with parameters n and p where p is the lot quality. It is far easier to calculate the probabilities with the help of the binomial distribution in comparison with the hypergeometric distribution.

Therefore, we can take

$$P[X = x] = {}^nC_x p^x (1-p)^{n-x}$$

Thus, the probability of accepting a lot of quality p using binomial approximation is given by

You have studied in Unit 3 of MST-003 that if A and B are mutually exclusive events then

$$P[A \text{ or } B] = P[A] + P[B]$$

The notation ${}^{Np}C_x$ can also be repersened as $\binom{Np}{x}$.

$$P_a(p) = \sum_{x=0}^c P[X=x] = \sum_{x=0}^c {}^n C_x p^x (1-p)^{n-x} \quad \dots (4)$$

However, for rapid calculation, we can use Table I entitled **Cumulative Binomial Probability Distribution** which is given at the end of this block. We can also approximate the binomial distribution to another distribution. When p is small and n is large such that np is finite, we know that the binomial distribution approaches the Poisson distribution with parameter $\lambda = np$. Therefore, the probability of accepting a lot of quality p using the Poisson approximation is given by

$$P_a(p) = \sum_{x=0}^c \frac{e^{-\lambda} \lambda^x}{x!} \quad \dots (5)$$

For Poisson distribution

$$P[X=x] = \frac{e^{-\lambda} \lambda^x}{x!}$$

We can use Table II entitled **Cumulative Poisson Probability Distribution** given at the end of this block for calculating this probability.

Note: Table I and Table II do not list a tabulated value for each value of p and λ , respectively. In such cases, we interpolate it as we have discussed in Unit 4 of MST-004 or use a scientific calculator to calculate the probability of accepting a lot.

We now illustrate how to compute the probability of accepting a lot in different situations with the help of an example.

Example 2: A manufacturer of silicon chip produces chips in lots of 1000. A single sampling plan is used to test for bad outgoing lots. If the quality of incoming lot is 0.01, calculate the probability of accepting the lot in the following cases:

- i) $n = 12$ and $c = 1$, and
- ii) $n = 60$ and $c = 2$.

Solution:

- i) It is given that

$$N = 1000, n = 12, c = 1 \text{ and } p = 0.01$$

If X represents the number of defective chips in the sample, the lot is accepted if $X \leq c$, i.e., $X \leq 1$. Therefore, the probability of accepting the lot is given by

$$\begin{aligned} P_a(p) &= P[X \leq c] = P[X \leq 1] = P[X=0] + P[X=1] \\ &= \sum_{x=0}^1 P[X=x] = \sum_{x=0}^1 \frac{{}^{Np} C_x {}^{N-Np} C_{n-x}}{{}^N C_x} \end{aligned}$$

Since $N \geq 10n$, we can use the binomial distribution (with parameters n and p where p is the lot quality) as the approximation of the hypergeometric distribution. Therefore, the probability of accepting the lot of quality p is given by

$$P_a(p) = P[X \leq 1] = \sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x}$$

For rapid calculation, we can use Table I for obtaining this probability.

From Table I, for $n = 12$, $x = c = 1$ and $p = 0.01$, we have

$$\sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x} = 0.9938$$

Therefore, the probability of accepting the lot of quality $p = 0.01$ is given by

$$P_a(p) = P[X \leq 1] = \sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x} = 0.9938$$

ii) It is given that

$$N = 1000, n = 60, c = 2 \text{ and } p = 0.01$$

Since p is small, n is large and $np = 0.65$ is finite, we can use the Poisson distribution (with parameters $\lambda = np$) as the approximation of the hypergeometric distribution. Therefore, the probability of accepting the lot of quality p is given by

$$P_a(p) = P[X \leq 2] = \sum_{x=0}^2 \frac{e^{-\lambda} \lambda^x}{x!}$$

where $\lambda = np = 60 \times 0.01 = 0.60$.

For rapid calculation, we can use Table II for obtaining this probability.

From Table II, for $\lambda = 0.60$ and $x = c = 2$, we have

$$\sum_{x=0}^2 \frac{e^{-\lambda} \lambda^x}{x!} = 0.9769$$

Hence, the probability of accepting a lot of quality $p = 0.01$ is given by

$$P_a(p) = P[X \leq 2] = \sum_{x=0}^2 \frac{e^{-\lambda} \lambda^x}{x!} = 0.9769$$

You can calculate the probability of acceptance of a lot for different lot qualities in the same way.

We now take up the construction of the OC curve for a single sampling plan.

As you know, the OC curve is constructed by taking the quality level (proportion defective) on the X-axis and the probability of accepting the lot on the Y-axis. So for construction of the OC curve, we first consider different quality levels such as $p = 0.01, 0.02, 0.03 \dots$ and then calculate the corresponding probability of accepting a lot as discussed in Example 2.

Let us consider an example to demonstrate the construction of the OC curve.

Example 3: Suppose a consumer receives lots of 500 candles from a new supplier. To check the quality of the lot, the consumer draws one sample of size 20 and accepts the lot if the inspected sample contains at most one defective candle. Otherwise, he/she rejects the lot. Construct the OC curve for this plan.

Solution: It is given that

$$N = 500, n = 20, c = 1$$

For constructing the OC curve, we have to calculate the probabilities of accepting the lot corresponding to different quality levels.

If X represents the number of defective candles in the sample, the consumer accepts the lot if $X \leq c$, i.e., $X \leq 1$. Therefore, the probability of accepting the lot is given by

$$P_a(p) = P[X \leq c] = P[X \leq 1] = \sum_{x=0}^1 P[X = x]$$

Since $N \geq 10n$, we can use binomial distribution. Therefore,

$$P_a(p) = P[X \leq 1] = \sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x}$$

We use Table I for calculating the probabilities of accepting the lot corresponding to different quality levels such as $p = 0.01, 0.02, 0.03, \dots$ as we have discussed in Example 2. These probabilities are shown in the table given below:

Incoming Lot Quality	Probability of Accepting the Lot
0	1
0.01	0.9831
0.02	0.9401
0.04	0.8103
0.06	0.6605
0.08	0.5169
0.10	0.3917
0.12	0.2891
0.14	0.2084
0.16	0.1471
0.18	0.1018
0.20	0.0692

We now construct the OC curve by taking the quality level (proportion defective) on the X-axis and the probability of accepting the lot on the Y-axis as shown in Fig. 7.2.

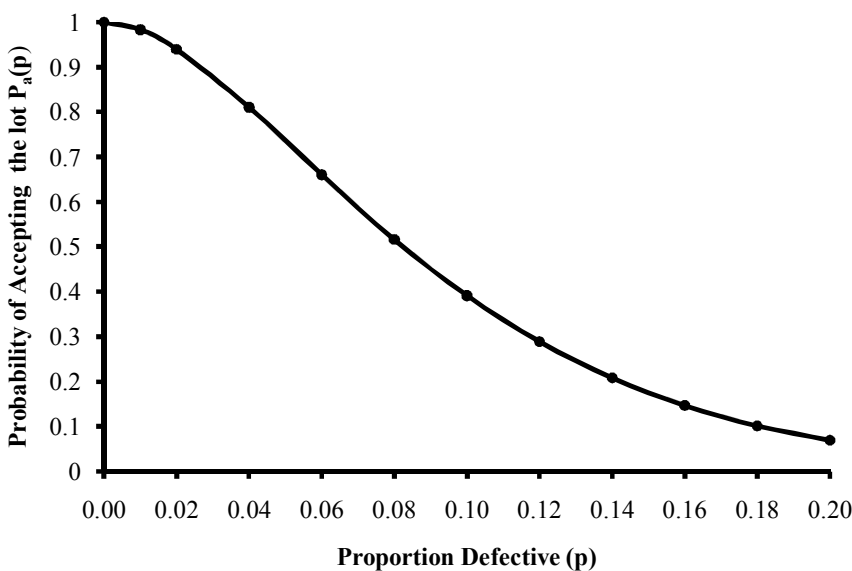


Fig. 7.2: The OC curve for the single sampling plan.

You may now like to construct the OC curve for the following exercise.

E2) A hospital receives disposable injection syringes in lots of 2000. A single sampling plan with $n = 25$ and $c = 2$ is being used for inspection by the quality inspector of the hospital. Construct the OC curve for this plan.

You have learnt in Unit 5 that the acceptance or rejection of the entire lot depends on the conclusions drawn from the sample. Thus, there is always a chance of making a wrong decision. It means that a lot of good quality may be rejected and a lot of poor quality may be accepted. This leads to two kinds of risks:

1. Producer's risk, and
2. Consumer's risk.

We now discuss these risks for the single sampling plan.

7.4 PRODUCER'S RISK AND CONSUMER'S RISK

In Unit 5, we have defined the producer's risk as follows:

The probability of rejecting a lot of acceptance quality level (AQL) p_1 is known as the producer's risk.

Therefore, the producer's risk for a single sampling plan is given by

$$\begin{aligned} P_p &= P[\text{rejecting a lot of acceptance quality level } p_1] \\ &= 1 - P[\text{accepting a lot of acceptance quality level } p_1] \\ &= 1 - P_a(p_1) \end{aligned} \quad \dots (6)$$

We can compute $P_a(p_1)$ from equation (3) by replacing the quality level p with p_1 as follows:

$$P_a(p_1) = \sum_{x=0}^c P[X = x] = \sum_{x=0}^c \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n-x}}{{}^NC_x}$$

Therefore, from equation (6), the producer's risk is given by

$$P_p = 1 - P_a(p_1) = 1 - \sum_{x=0}^c \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n-x}}{{}^NC_x} \quad \dots (7)$$

For rapid calculation of the producer's risk for a single sampling plan, we can also use approximations as we have discussed in Sec. 7.4. Therefore, if we use the approximation of the hypergeometric distribution to the binomial distribution with parameters n and p_1 , the producer's risk is given by

$$P_p = 1 - P_a(p_1) = 1 - \sum_{x=0}^c {}^nC_x p_1^x (1-p_1)^{n-x} \quad \dots (8)$$

We now explain the **Consumer's Risk** for a single sampling plan:

By definition, the probability of accepting a lot of unsatisfactory quality (LTPD) p_2 is known as the consumer's risk.

Therefore, the consumer's risk for a single sampling plan is given by

$$P_c = P[\text{accepting a lot of quality } p_2]$$

We can compute the consumer's risk for the single sampling plan from equation (4) by replacing the quality level p with p_2 as follows:

$$P_c = P_a(p_2) = \sum_{x=0}^c P[X=x] = \sum_{x=0}^c \frac{{}^{Np_2}C_x {}^{N-Np_2}C_{n-x}}{{}^NC_x} \quad \dots (9)$$

If we approximate the hypergeometric distribution to the binomial distribution with parameters n and p_2 , the consumer's risk is given by

$$P_c = P_a(p_2) = \sum_{x=0}^c {}^nC_x p_2^x (1-p_2)^{n-x} \quad \dots (10)$$

Let us take up an example from real life to explain these concepts.

Example 4: Suppose a tyre supplier ships tyres in lots of size 400 to the buyer. A single sampling plan with $n = 15$ and $c = 0$ is being used for the lot inspection. The supplier and the buyer's quality control inspector decide that $AQL = 0.01$ and $LTPD = 0.10$. Compute the producer's risk and consumer's risk for this single sampling plan.

Solution: It is given that

$$N = 400, n = 15, c = 0, AQL(p_1) = 0.01 \text{ and } LTPD(p_2) = 0.10$$

Since $N \geq 10n$, we can use the binomial distribution. Therefore, we can use equation (8) to calculate the producer's risk for the single sampling plan.

We first calculate the probability of accepting a lot of quality $p = p_1 = 0.01$ using Table I.

From Table I, for $n = 15$, $x = c = 0$ and $p = p_1 = 0.01$, we have

$$P_a(p_1) = P[X \leq 0] = \sum_{x=0}^0 {}^nC_x p_1^x (1-p_1)^{n-x} = 0.8601$$

Therefore, from equation (8), we calculate the producer's risk as follows:

$$P_p = 1 - P_a(p_1) = 1 - 0.8601 = 0.1399$$

It means that if there are several lots of the same quality $p = 0.01$, about 13.99% of these will be rejected. This is obviously a risk for the supplier because it was agreed upon by both that lots of quality 0.01 will be accepted whereas the quality inspector is rejecting 13.99% of those.

Similarly, we can calculate the consumer's risk using equation (10).

We first calculate the probability of accepting a lot of quality $p = p_2 = 0.10$ using Table I.

From Table I, for $n = 15$, $x = c = 0$ and $p = 0.10$, we have

$$P_a(p_2) = P[X \leq 0] = \sum_{x=0}^0 {}^nC_x p_2^x (1-p_2)^{n-x} = 0.2059$$

Therefore, from equation (10), the consumer's risk is given by

$$P_c = P_a(p_2) = 0.2059$$

It means that if there are several lots of the same quality $p = 0.10$, about 20.59% of these will be accepted by the quality inspector even though this quality is unsatisfactory. This is obviously the buyer's risk.

For practice you can also compute the producer's risk and consumer's risk in the following exercise.

E3) Suppose in E2, the acceptance quality level (AQL) and lot tolerance percent defective (LTPD) are 0.04 and 0.10, respectively. Calculate the producer's risk and consumer's risk for this plan.

7.5 AVERAGE OUTGOING QUALITY (AOQ)

You have studied in Unit 6 that the concept of average outgoing quality is particularly useful in the rectifying sampling plan where the rejected lots are inspected 100% and all defective units are replaced by non-defective units. The AOQ is defined as follows:

The expected quality of the lots after the applications of sampling inspection is called the **average outgoing quality (AOQ)**. It is calculated from the formula given below:

$$AOQ = \frac{\text{Number of defective units in the lot after the inspection}}{\text{Total number of units in the lot}} \dots (11)$$

So in a single sample plan, we can obtain the formula for average outgoing quality by considering the following situations:

- i) If the lot of size N is accepted on the basis of a sample of size n , $(N - n)$ units remain un-inspected. If the incoming quality of the lot is p , we expect that $p(N - n)$ defective units are left in the lot after the inspection. However, the probability that the lot will be accepted by the sampling plan is P_a . Therefore, the expected number of defective units per lot in the outgoing stage is $p(N - n) P_a$.
- ii) If the lot is rejected, all units of the lot go for 100% inspection and all defective units found in the lot are replaced by non-defective units. So there is no defective unit at the outgoing stage. The probability that the lot will be rejected is $(1 - P_a)$. Therefore, the expected number of defective units per lot at the outgoing stage is $0 \times (1 - P_a) = 0$.

Thus, the expected number of defective units per lot after sampling inspection is $p(N - n)P_a + 0 = p(N - n)P_a$.

Hence, the average proportion defective in the outgoing stage or average outgoing quality (AOQ) is given by

$$AOQ = \frac{\text{Number of defective units in the lot after the inspection}}{\text{Total number of units in the lot}}$$

$$\text{or } AOQ = \frac{p(N - n)P_a}{N} \dots (12)$$

If the sample size n is very small in proportion to the lot size N , i.e., $n/N \ll 0$, equation (12) for AOQ becomes

$$AOQ = p \left(1 - \frac{n}{N} \right) P_a \approx pP_a \dots (13)$$

Let us now discuss the construction of the AOQ curve for a single sampling plan.

As you know, the AOQ curve is constructed by taking the quality level (proportion defective) on the X-axis and the AOQ on the Y-axis. So for constructing the AOQ curve, we first consider different quality levels such as $p = 0.01, 0.02, 0.03 \dots$ and then calculate the corresponding AOQ.

Let us consider some examples for calculating AOQ and constructing the AOQ curve.

Example 5: Suppose in Example 3 the submitted lot quality is $p = 0.02$. The rejected lots are screened and all defective candles are replaced by the non-defective candles. Calculate the average outgoing quality (AOQ) for this plan.

Solution: The submitted lot quality is $p = 0.02$ and we have to calculate the AOQ for this single sampling plan.

It is given that

$$N = 500, n = 20, c = 1 \text{ and } p = 0.02$$

From equation (12), the AOQ for the single sampling plan is

$$AOQ = \frac{p(N - n)P_a}{N}$$

where P_a is the probability of accepting the lot of quality p . Therefore, for calculating AOQ, we have to calculate P_a . We have already calculated this probability in Example 3. So we directly use the result:

$$P_a = 0.9401$$

On putting the values of $N = 500, n = 20, p = 0.02$ and $P_a = 0.9401$ in equation (12), we get

$$AOQ = \frac{p(N - n)P_a}{N} = \frac{0.02 \times (500 - 20) \times 0.9401}{500} = 0.018$$

In the same way, you can calculate AOQ for different submitted lot qualities.

Example 6: Suppose in Example 4, the rejected lots are screened and all defective tyres are replaced by non-defective tyres. Construct the AQO curve for this plan.

Solution: It is given that

$$N = 400, n = 15, c = 0$$

For constructing the AOQ curve, we first calculate the probabilities of accepting the lot corresponding to different quality levels using Table I. Then we calculate the AOQ for each quality level using equation (12).

The probabilities of accepting the lot and the AOQs corresponding to different quality levels are given in the following table:

Incoming Lot Quality	Probability of Accepting the Lot	AOQ
0	1	0
0.01	0.8601	0.0083
0.02	0.7386	0.0142

0.04	0.5421	0.0209
0.06	0.3953	0.0228
0.08	0.2863	0.0220
0.10	0.2059	0.0198
0.12	0.1470	0.0170
0.14	0.1041	0.0099
0.16	0.0731	0.0079
0.18	0.0510	0.0061
0.20	0.0352	0.0000

We now construct the AOQ curve by taking the quality level (proportion defective) on the X-axis and the corresponding AOQ values on the Y-axis as shown in Fig. 7.3.

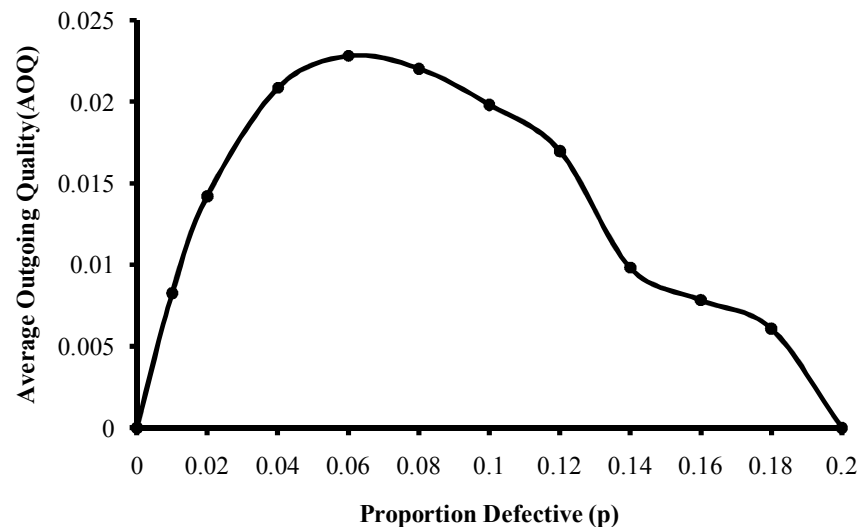


Fig. 7.3: The AOQ curve for Example 5.

You may now like to calculate the AOQ and construct the AOQ curve. Try the following exercises.

-
- E4)** Assuming that the lot size is large relative to the sample size, calculate the approximate average outgoing quality (AOQ) for the single sampling plan with $n = 10$ and $c = 0$ containing 20% defective units.
- E5)** A computer manufacturer purchases computer chips from a company in lots of 200. Twelve computer chips are sampled at random and inspected for defects. The computer manufacturer accepts the lot if the inspected sample contains at most one defective chip. Otherwise, he/she rejects the lot. If the rejected lots are screened and all defective computer chips are replaced by non-defective chips. Construct the AOQ curve for this plan.
-

You have learnt about the OC curve, producer's risk, consumer's risk and AOQ for a single sampling plan. We now discuss ASN and ATI of the plan.

7.6 AVERAGE SAMPLE NUMBER (ASN) AND AVERAGE TOTAL INSPECTION (ATI)

Two other features that are also useful for any sampling plan are the average sample number (ASN) and the average total inspection (ATI). We now discuss these for a single sampling plan in some detail.

Average Sample Number (ASN)

You have studied in Unit 6 that the average sample number is the expected number of sample units per lot, which is required to arrive at a decision about the acceptance or rejection of the lot under the acceptance sampling plan.

In acceptance single sampling plan, the decision about the acceptance or rejection of a lot is taken on the basis of a single sample that has been inspected. Therefore, the ASN in a single sampling plan is simply the sample size n . It means that ASN is constant in a single sampling plan.

Therefore,

$$\text{ASN} = n \quad \dots (14)$$

The curve drawn between the ASN and the lot quality (p) is known as the **ASN curve**.

The ASN curve for a single sampling plan is a straight line as shown in Fig. 7.4.

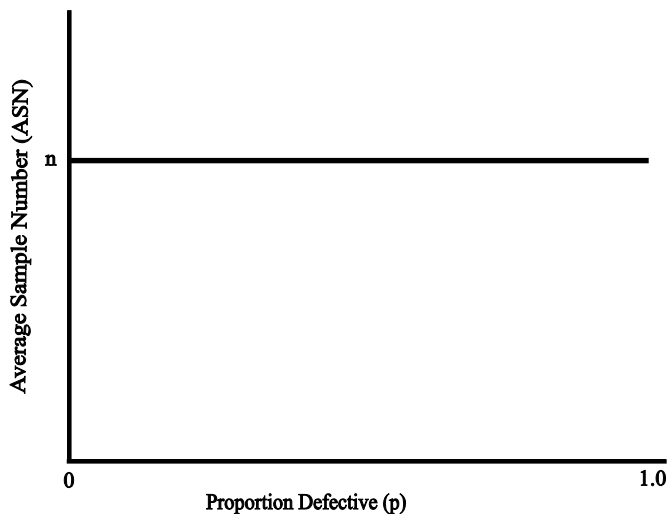


Fig. 7.4: The ASN curve for a single sampling plan.

Let us consider Example 4. In this example, the quality control inspector takes the decision of acceptance or rejection of the lot on the basis of the single sample of size $n = 15$. So the ASN for this single sampling plan is 15.

Average Total Inspection (ATI)

You know that the concept of average total inspection (ATI) is considered under rectifying sampling plan in which rejected lots under go 100% inspection. It is defined as follows:

The average number of units inspected per lot under the rectifying sampling plan is called the **average total inspection (ATI)**.

So in a rectifying single sampling plan, the number of units to be inspected will depend on two situations given below:

- i) If the lot of size N is accepted on the basis of a sample of size n , the number of units inspected is n and the probability of the accepting the lot is P_a .
- ii) If the lot is rejected on the basis of a sample, we inspect the entire lot of size N and the probability of rejecting the lot is $(1 - P_a)$.

Therefore, we can compute the ATI for a single sampling plan as follows:

$$\text{ATI} = \text{Average number of units inspected per lot}$$

$$= \sum (\text{inspected number of units} \times \text{probability of taking decision})$$

$$ATI = n \times P_a + N \times (1 - P_a)$$

This can also be written as

$$ATI = n + (1 - P_a)(N - n) \quad \dots (15)$$

The curve drawn between ATI and lot quality (p) is known as the **ATI curve**.

Let us take up an example to illustrate this concept.

Example 7: Calculate the ASN for the plan given in Example 4. If the rejected lots are screened and all defective tyres are replaced by non-defective tyres, construct the ATI curve for this plan.

Solution: It is given that

$$N = 400, n = 15, c = 0$$

In the first case, the decision of acceptance or rejection of the lot is taken only on a single sample of size $n = 15$. Therefore, the ASN for this plan is simply the sample size, i.e., $ASN = n = 15$.

For construction of the ATI curve, we first calculate the probabilities of accepting the lot corresponding to different quality levels using Table I. Then we calculate the ATI for each quality level using equation (15).

We have already calculated these probabilities in Example 6 and we use those results. Substituting the values of N , n , p and P_a in equation (15) we can calculate ATI.

The probabilities of accepting the lot and the ATIs corresponding to different quality levels are given in the following table:

Incoming Lot Quality	Probability of Accepting the Lot	ATI
0	1	15.00
0.01	0.8601	68.86
0.02	0.7386	115.64
0.04	0.5421	191.29
0.06	0.3953	247.81
0.08	0.2863	289.77
0.10	0.2059	320.73
0.12	0.1470	343.41
0.14	0.1041	359.92
0.16	0.0731	371.86
0.18	0.0510	380.37
0.20	0.0352	386.45

We now construct the ATI curve by taking the quality level (proportion defective) on the X-axis and the corresponding ATI values on the Y-axis as shown in Fig. 7.5.

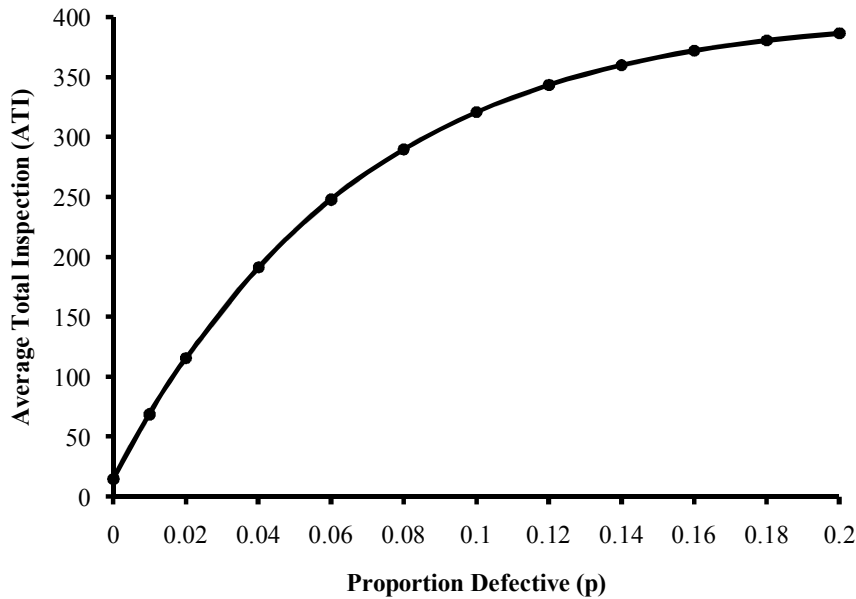


Fig. 7.5: The ATI curve for Example 6.

You can try the following exercise based on ASN and ATI for practice.

E6) Calculate the ASN and construct the ATI curve for the plan given in E5.

So far you have learnt the various features of the single sampling plan. We now discuss how to design a single sampling plan.

7.7 DESIGN OF SINGLE SAMPLING PLANS

The design of a single sampling plan implies the determination of the parameters of the plan, i.e., the sample size n and the acceptance number c . These numbers have to be decided in advance before applying the single sampling plan technique. There are several approaches for determining the parameters n and c for the single sampling. Sometimes, more than one plan will satisfy the criteria, but the best plan is the one with the largest sample size. It provides adequate protection to both producer and consumer. We now discuss some of the approaches.

7.7.1 Stipulated Producer's Risk

In this approach, the producer's risk α and its corresponding acceptance quality level (AQL) p_1 are specified. We would like to design a single sampling plan in such a way that the lots of quality level p_1 are accepted $100(1 - \alpha)\%$ of the time.

According to this approach, we first select an acceptance number (c) and then find the value of np_1 corresponding to c and α with the help of Table III given at the end of this block. Then the value of n is obtained by dividing np_1 by p_1 ($=$ AQL) as follows:

$$n = \frac{np_1}{p_1} \quad \dots (16)$$

If the computed value of n is a fraction, it is rounded off to the next integer.

For different values of the acceptance number (c), we may get different values of sample size (n) (as you will see in Example 8). This means that for the same

producer's risk and acceptance quality level, we have different single sampling plans. If we draw the OC curve for each sampling plan, we will see that each plan has a different consumer's risk. So out of these, we choose the sampling plan which gives the best protection to the consumer against acceptance of poor quality lots.

We now describe this procedure with the help of an example.

Example 8: Suppose a tyre supplier ships tyres in lots of size 400 to the buyer. The supplier and quality control inspector of the buyer decide the acceptance quality level (AQL) to be 2%. Design a sampling plan which ensures that the lots of quality 2% will be rejected 5% of the time.

Solution: The supplier and the quality control inspector desire the sampling plan for which AQL (p_1) = 2% and the producer's risk (α) = 5%.

To find the desired sampling plan, first of all, we choose the acceptance number c as $c = 1$. Then we look up the value of np_1 corresponding to $c = 1$ and $\alpha = 0.05$ from Table III. We have

$$np_1 = 0.355$$

Therefore, we can obtain the sample size as follows:

$$n = \frac{np_1}{p_1} = \frac{0.355}{0.02} = 17.75 \approx 18$$

Hence, the required single sampling plan is

$$n = 18 \text{ and } c = 1$$

Similarly, for $c = 2$ and $\alpha = 0.05$, the value of np_1 is 0.818.

Therefore, the sample size is

$$n = \frac{np_1}{p_1} = \frac{0.818}{0.02} = 40.9 \approx 41$$

Hence, the required single sampling plan is

$$n = 41 \text{ and } c = 2$$

Similarly, for $c = 5$ and $\alpha = 0.05$, the value of np_1 is 2.613.

Therefore,

$$n = \frac{np_1}{p_1} = \frac{2.613}{0.02} = 130.65 \approx 131$$

Hence, the required single sampling plan is

$$n = 131 \text{ and } c = 5$$

The OC curves for the three sampling plans (as discussed in Sec. 7.4) are shown in Fig. 7.6.

Incoming Lot Quality	Probability of Accepting the Lot		
	$n = 18 \text{ and } c = 1$	$n = 41 \text{ and } c = 2$	$n = 131 \text{ and } c = 5$
0	1.0000	1.0000	1.0000
0.01	0.9862	0.9920	0.9978
0.02	0.9505	0.9514	0.9513
0.04	0.8393	0.7750	0.5731
0.06	0.7055	0.5505	0.1959

0.08	0.5719	0.3526	0.0445
0.10	0.4503	0.2086	0.0075
0.12	0.3460	0.1156	0.0010
0.14	0.2602	0.0607	0.0001
0.16	0.1920	0.0303	0.0000
0.18	0.1391	0.0145	0.0000
0.20	0.0991	0.0066	0.0000

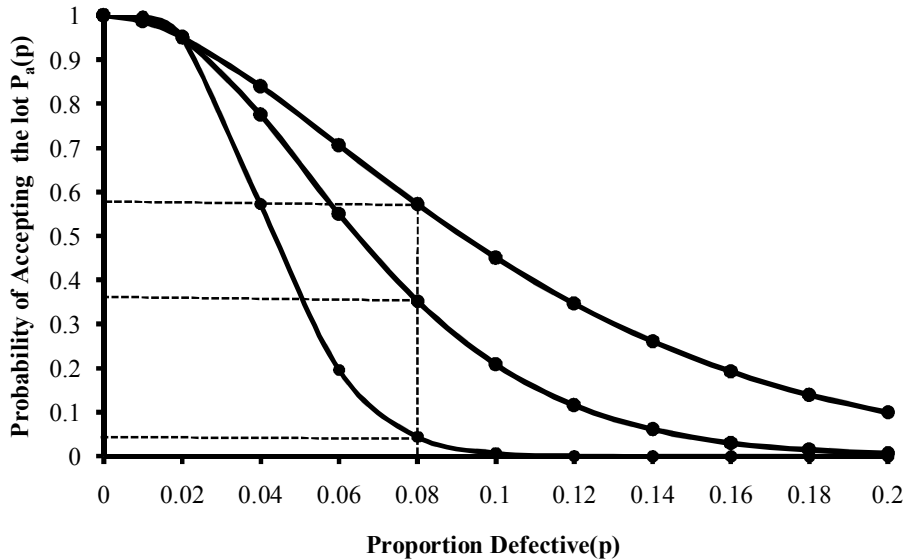


Fig. 7.6: The OC curves for Example 8.

From Fig. 7.6, we conclude that out of the three sampling plans, the sampling plan with $n = 131$, $c = 5$ provides the best protection to the consumer because it has the lowest probability of accepting poor quality lots. However, this sampling plan has the largest sample size ($n = 131$), which increases the inspection cost.

7.7.2 Stipulated Consumer's Risk

According to this approach, the consumer risk β and its corresponding lot tolerance percent defective (LTPD) p_2 are specified. We desire to determine a single sampling plan in such a way that we will be accepted lots of quality level p_2 $100\beta\%$ of the time.

To design the plan using this approach, we first select an acceptance number (c) and then find the value of np_2 corresponding to c and β with the help of Table III. Then the value of n can be obtained by dividing np_2 by $p_2 = \text{LTPD}$ as follows:

$$n = \frac{np_2}{p_2} \quad \dots (17)$$

For different values of c , a number of sampling plans may satisfy this criterion. If we draw the OC curve for each sampling plan, we will see that each plan has a different producer's risk. So out of these, we choose the sampling plan which gives the best protection to the producer against the rejection of good quality lots.

We now describe this procedure with the help of an example.

Example 9: Suppose, in Example 8, the supplier and the quality control inspector decide the lot tolerance percent defective (LTPD) to be 5%. Determine the single sampling plans using $c = 1, 2$ and 8 which ensure that the lots of quality 5% will be accepted 10% of the time.

Solution: It is given that

$$\text{LTPD } (p_2) = 5\% \text{ and the consumer's risk } (\beta) = 10\%$$

To find the desired sampling plan, we first look up the value of np_2 corresponding to $c = 1$ and $\beta = 0.10$ from Table III. We have

$$np_2 = 3.890$$

Therefore, the sample size is

$$n = \frac{np_2}{p_2} = \frac{3.890}{0.10} = 38.90 \approx 39$$

Hence, the required single sampling plan is

$$n = 39, c = 1$$

Similarly, for $c = 2$ and $\beta = 0.10$, the value of np_2 is 5.322.

Therefore, the sample size is

$$n = \frac{np_2}{p_2} = \frac{5.322}{0.10} = 53.22 \approx 54$$

Hence, the required single sampling plan is

$$n = 54, c = 2$$

Similarly, for $c = 8$ and $\beta = 0.10$ the value of np_2 is 12.995.

Therefore,

$$n = \frac{np_2}{p_2} = \frac{12.955}{0.10} = 129.55 \approx 130$$

Hence, the required single sampling plan is

$$n = 130, c = 8$$

7.7.3 Stipulated Producer's Risk and Consumer's Risk

According to this approach, the producer's risk with its corresponding acceptance quality level (AQL) and consumer's risk with its corresponding lot tolerance percent defective (LTPD) are specified. We have to design single sampling plans which satisfy both producer's and consumer's risks, such that the lots of AQL are to be rejected no more than $100\alpha\%$ of the time and lots of LTPD are to be accepted no more than $100\beta\%$ of the time.

Here the criteria are more stringent than the previous approaches and we may not have much flexibility in choosing the acceptance number and the associated sampling plan. So it may be difficult to design a sampling plan that will satisfy both producer's and consumer's risks.

In this approach, we first find the operating ratio R as follows:

$$R = \frac{p_2}{p_1} \quad \dots (18)$$

The values of R corresponding to various acceptance number (c) and α and β are also listed in Table III. We choose a value of R from Table III which is exactly equal to the desired value of R corresponding to the desired α and β . Generally, the tabulated value of R is not equal to the desired value of R. So in such situations, we take the tabulated values of R between which the desired value of R lies. Then we look up the corresponding values of acceptance number (c) in Table III and find the value of sample size (n). The following two approaches are used to find n:

1. Satisfy Producer's Risk Stipulation Exactly and come Close to Consumer's Risk

According to this approach, we find the values of n as we have discussed in Sec. 7.7.1. It means that we first find the value of np_1 corresponding to c and α with the help of Table III. Then we find n from equation (16). Thus,

$$n = \frac{np_1}{p_1}$$

The values of n are obtained for both values of c. In this way, we get two sampling plans which satisfy producer's risk exactly.

Out of these, we choose the sampling plan which is close to satisfying the consumer's risk. For that we find the value of p_2 for each plan. We find the values of np_2 corresponding to the desired β and each c from Table III. Then we find the value of p_2 for each plan as follows:

$$p_2 = \frac{np_2}{n} \quad \dots (19)$$

We choose the sampling plan for which the calculated p_2 is closer to the desired p_2 .

Another criterion for choosing the sampling plan is that we select the sampling plan which has the smallest sample size in order to minimize inspection costs. Alternatively, we can choose the sampling plan which has the largest sample size in order to get maximum information.

2. Satisfy Consumer's Risk Stipulation Exactly and come Close to Producer's Risk

According to this approach, we find the values of n as we have discussed in Sec. 7.7.2. It means that we first find the value of np_2 corresponding to c and β with the help of Table III. Then we find n from equation (17). Thus,

$$n = \frac{np_2}{p_2}$$

The values of n are obtained for both values of c. In this way, we get two sampling plans which satisfy consumer's risk exactly.

Out of these, we choose the sampling plan which is close to satisfying the producer's risk. For that we find the value of p_1 for each plan. We find the values of np_1 corresponding to the desired α and each c from Table III. Then we find the value of p_1 for each plan as follows:

$$p_1 = \frac{np_1}{n} \quad \dots (20)$$

We choose the sampling plan for which the calculated p_1 is closer to the desired p_1 .

Another criterion for choosing the sampling plan is that we select the sampling plan which has the smallest sample size in order to minimize inspection costs. Alternatively, we can choose the sampling plan which has the largest sample size in order to get maximum information.

We now describe this procedure with the help of an example.

Example 10: Suppose, in Example 8, the supplier and the quality control inspector decide the acceptance quality level (AQL) to be 2% and the lot tolerance percent defective (LTPD) to be 8%. Design a sampling plan which ensures that lots of quality 2% will be rejected 5% of the time and lots of quality 8% will be accepted 5% of the time.

Solution: It is given that

$$\text{AQL} = p_1 = 2\% = 0.02 \text{ and } \alpha = 5\% = 0.05$$

$$\text{LTPD} = p_2 = 8\% = 0.08 \text{ and } \beta = 5\% = 0.05$$

To design the desired sampling plan, we first calculate the operating ratio (R) from equation (18) as follows:

$$R = \frac{p_2}{p_1} = \frac{0.08}{0.02} = 4.0$$

From Table III, we see that the desired value of $R = 4.0$ lies between 4.023 and 3.604 for $\alpha = 0.05$ and $\beta = 0.05$. From Table III, the corresponding acceptance numbers (c) are 5 and 6. We have to find the value of sample size (n).

We first find the plans which satisfy the desired producer's risk exactly.

For this, we find the value of np_1 corresponding to $c = 5$ and $\alpha = 0.05$ with the help of Table III.

From Table III, for $c = 5$ and $\alpha = 0.05$, we have $np_1 = 2.613$. Therefore, from equation (16), we have

$$n = \frac{np_1}{p_1} = \frac{2.613}{0.02} = 130.65 \approx 131$$

Similarly, for $c = 6$ and $\alpha = 0.05$, the value of np_1 is 3.286. Therefore,

$$n = \frac{np_1}{p_1} = \frac{3.286}{0.02} = 164.3 \approx 165$$

So both the plans with $n = 131$, $c = 5$ and $n = 165$, $c = 6$ satisfy the producer's risk exactly. Out of these plans, we have to find the plan which is closer to satisfying the desired consumer's risk. For this we find np_2 corresponding to c and β .

From Table III, for $c = 5$, $\beta = 0.05$, we have $np_2 = 10.513$.

Therefore, from equation (19), we have

$$p_2 = \frac{np_2}{n} = \frac{10.513}{131} = 0.08$$

Similarly, for $c = 6$, $\beta = 0.05$, we have $np_2 = 11.842$.

Therefore,

$$p_2 = \frac{np_2}{n} = \frac{11.842}{165} = 0.07$$

Since the value of $p_2 = 0.08$ corresponding to the plan $n = 131$, $c = 5$ is equal to the desired value 0.08, the plan with $n = 131$, $c = 5$ is the best single sampling plan.

We now find the plans which satisfy the desired consumer's risk exactly.

From Table III, for $c = 5$ and $\beta = 0.05$, we have $np_2 = 10.513$. Therefore, from equation (17), we have

$$n = \frac{np_2}{p_2} = \frac{10.513}{0.08} = 131.4 \approx 132$$

Similarly, for $c = 6$ and $\beta = 0.05$, we have $np_2 = 11.842$. Therefore,

$$n = \frac{np_2}{p_2} = \frac{11.842}{0.08} = 148.02 \approx 149$$

So both the plans with $n = 132$, $c = 5$ and $n = 149$, $c = 6$ are satisfied the consumer's risk exactly. Out of these plans, we have to find the plan which is closer to the desired producer's risk. For this we find np_1 corresponding to c and α .

From Table III, for $c = 5$, $\alpha = 0.05$, we have $np_1 = 2.613$

Therefore,

$$p_1 = \frac{np_1}{n} = \frac{2.613}{132} = 0.0198$$

Similarly, for $c = 6$, $\alpha = 0.05$, we have $np_1 = 3.286$.

Therefore,

$$p_1 = \frac{np_1}{n} = \frac{3.286}{149} = 0.022$$

Since the value of $p_1 = 0.0198$ corresponding to the plan $n = 132$, $c = 5$ is approximate equal to the desired value 0.02, the plan $n = 132$, $c = 5$ is the best single sampling plan.

7.7.4 Larson Binomial Nomograph

There is also a graphical method for designing the single sampling plans when the producer's risk with corresponding acceptance quality level (AQL) and consumer's risk with corresponding lot tolerance percent defective (LTPD) are specified.

This method is based on Larson binomial nomograph and is used when we take the binomial approximate to the hypergeometric distribution, i.e., when

$N \geq 10n$. The Larson binomial nomograph (shown in Fig. 7.7) is a graph of the cumulative binomial distribution. It has two scales. On the left scale,

proportion defective (p) is shown as **probability of occurrences on a single trial (p)**. This scale is known as **p-scale**. On the right scale, the **probability of acceptance (P_a)** is shown as **probability of less than or equal to c occurrences in n trials (p)**. This scale is known as **P_a -scale**.

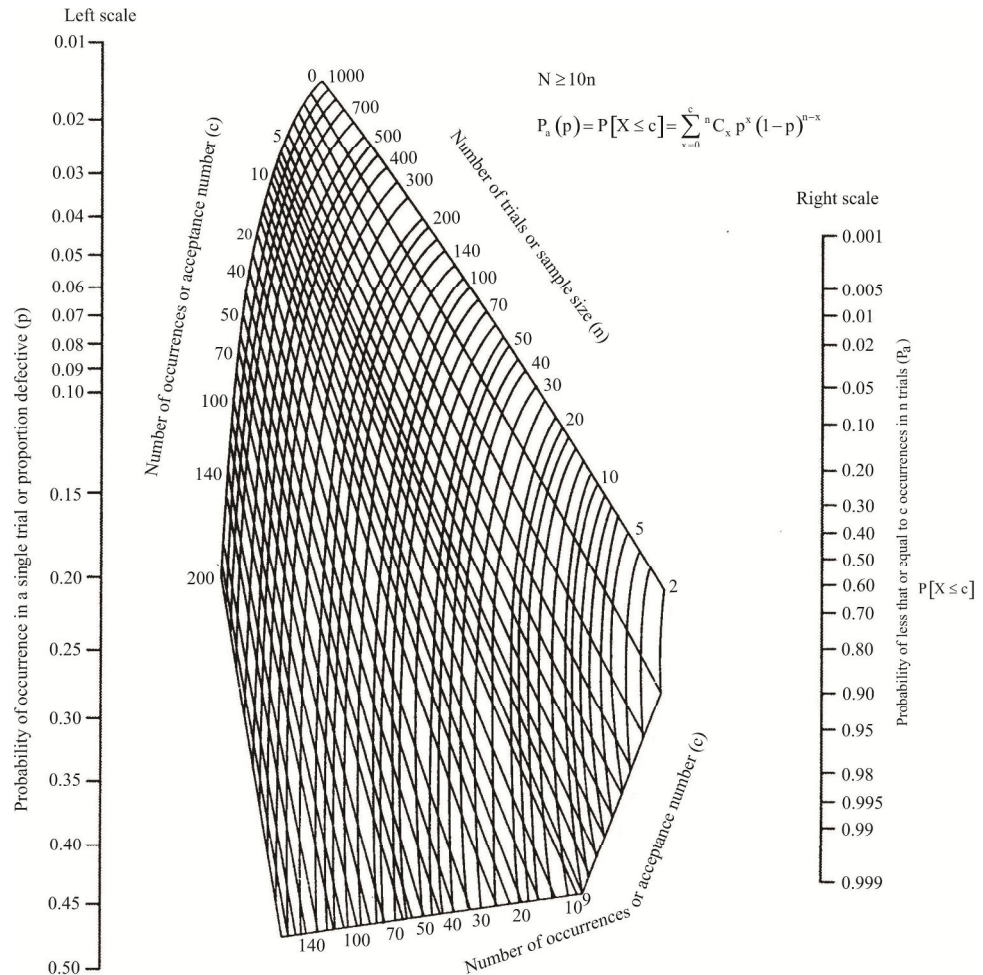


Fig. 7.7: Larson binomial nomograph.

The procedure of designing a single sampling plan by using the nomograph is quite simple. We plot the values of AQL (p_1) and LTPD (p_2) on the left scale and the corresponding values of $(1 - \alpha)$ and β on the right scale. Then we join p_1 with $(1 - \alpha)$ and p_2 with β by straight lines. We read the values of sample size (n) and acceptance number (c) at the intersection of the lines on the grid.

To demonstrate this method, we consider Example 10.

It is given that

$$\text{AQL} = p_1 = 2\% = 0.02 \text{ and } \alpha = 5\% = 0.05$$

$$\text{LTPD} = p_2 = 8\% = 0.08 \text{ and } \beta = 5\% = 0.05$$

For designing a single sampling plan by using Larson nomograph, we first plot $p_1 = 0.02$ and $p_2 = 0.08$ on the p -scale on the nomograph. Then we plot $1 - \alpha (= 0.95)$ and $\beta = 0.05$ on the P_a -scale on the nomograph. After plotting these points, we draw a straight line joining $p_1 (= 0.02)$ and $1 - \alpha (= 0.95)$ and another straight line joining $p_2 (= 0.08)$ and $\beta (= 0.05)$ as shown in Fig. 7.8. At the intersection of the two lines, we read the values of n and c .

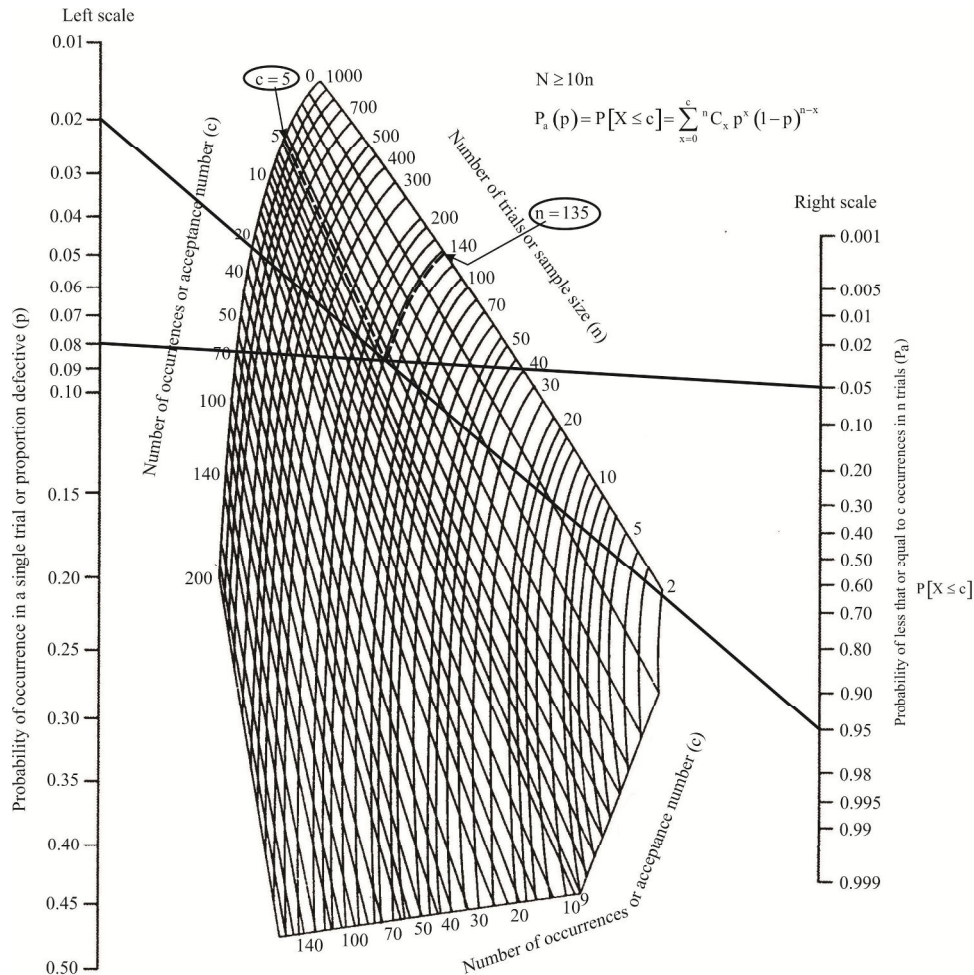


Fig. 7.8

From Fig. 7.8 we have, $n = 135$ and $c = 5$.

You can now check your understanding of how to design a single sampling plan by answering the following exercises.

-
- E7)** A ball bearing supplier and an automobile company have decided to check the quality of ball bearings in lots of size 1000 with acceptance quality level (AQL) as 1%. Design the single sampling plans using $c = 2, 4$ and 6 which ensure that lots of quality 1% will be rejected 1% of the time.
- E8)** A consumer receives lots of 5000 candles from a new supplier. To check the quality of lots, the consumer and supplier want to use the single sampling plan which satisfies a consumer's risk of 5% for lots of quality 5%. Determine sampling plans for the specified consumer's risk and LTPD for acceptance number $c = 3$ and 6 .
-

We end this unit by giving a summary of what we have covered in it.

7.8 SUMMARY

1. The main acceptance sampling plans for attributes are:
 - i) Single sampling plan,
 - ii) Double sampling plan,

- iii) Multiple sampling plan, and
 - iv) Sequential Sampling Plan.
2. A sampling plan in which a decision about the acceptance or rejection of a lot is based on a single sample that has been inspected is known as a **single sampling plan**. There are two parameters of a single sampling plan:
 n – size of the sample, and
 c – acceptance number for the sample.
 3. In a single sampling plan, if number of defective units (d) in the sample is less than or equal to the stated acceptance number (c), i.e., if $d \leq c$, we accept the lot and if $d > c$, we reject the lot under acceptance sampling plan.
 4. In a single sampling plan, if $d \leq c$, we accept the lot and replace all defective units found in the sample by non-defective units and if $d > c$, we accept the lot by inspecting the entire lot and replacing all defective units in the lot by non-defective units under rectifying sampling plan.
 5. The probability of accepting a lot of quality p for a single sampling plan is given by

$$P_a(p) = P[X \leq c] = \sum_{x=0}^c {}^n C_x p^x (1-p)^{n-x}$$

6. The produce's risk and consumer's risk for a single sampling plan are given by

$$P_p = 1 - P_a(p_1) = 1 - \sum_{x=0}^c {}^n C_x p_1^x (1-p_1)^{n-x} \text{ and}$$

$$P_c = P_a(p_2) = \sum_{x=0}^c {}^n C_x p_2^x (1-p_2)^{n-x}$$

7. The AOQ for a single sampling plan is

$$AOQ = \frac{p(N-n)P_a}{N}$$

8. The ASN and ATI for a single sampling plan are

$$ASN = n \text{ and } ATI = n + (1 - P_a)(N - n)$$

9. Designing a single sampling plan implies determining the sample size (n) and acceptance number (c).

7.9 SOLUTIONS/ANSWERS

- E1) To check the quality of the lots, the buyer randomly draws 50 silicon chips from each lot. After that he/she inspects each and every chip drawn from the lot for certain defects and classifies each chip of the sample as defective or non-defective. At the end of the inspection, he/she counts the number of defective chips (d) found in the sample and then compares the number of defective chips (d) with the acceptance number (c). If $d \leq c = 2$, he/she accepts the lot and if $d > c = 2$, he/she rejects the lot on the basis of the inspected sample. It means that if the

buyer finds 0 or 1 or 2 defective chips in the sample, he/she accepts the lot. Otherwise, he/she rejects the lot.

E2) It is given that

$$N = 2000, n = 25, c = 2$$

For constructing the OC curve, we have to calculate the probabilities of accepting the lot corresponding to different quality levels.

If X represents the number of defective syringes in the sample, the quality inspector accepts the lot if $X \leq c = 2$. Therefore, the probability of accepting the lot is given by

$$P_a(p) = P[X \leq c] = P[X \leq 2] = \sum_{x=0}^2 P[X = x]$$

Since $N \geq 10n$, we use the binomial distribution. We can use Table I for calculating the probabilities of accepting the lot corresponding to different quality levels such as $p = 0.01, 0.02, 0.03 \dots$. These probabilities are shown in the table given below:

Incoming Lot Quality	Probability of Accepting the Lot
0	1
0.01	0.9980
0.02	0.9868
0.04	0.9235
0.06	0.8129
0.08	0.6768
0.10	0.5371
0.12	0.4088
0.14	0.3000
0.16	0.2130
0.18	0.1467
0.20	0.0982

We construct the OC curve by taking the quality level (proportion defective) on the X-axis and the probability of accepting the lot on the Y-axis as shown in Fig. 7.9.

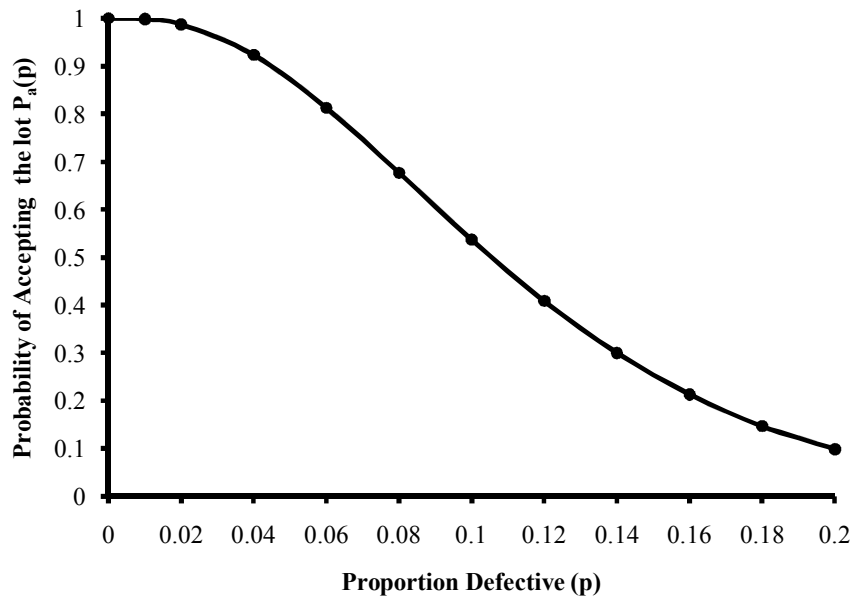


Fig. 7.9: The OC curve for E2.

E3) It is given that

$$N = 2000, n = 25, c = 2, AQL(p_1) = 0.04 \text{ and } LTPD(p_2) = 0.10$$

Since $N \geq 10n$, we use the binomial distribution. Therefore, we use equation (8) to calculate the producer's risk for the single sampling plan.

We first calculate the probability of accepting the lot of quality $p = p_1 = AQL = 0.04$ using Table I.

From Table I, for $n = 25$, $x = c = 2$ and $p = p_1 = 0.04$, we have

$$P_a(p_1) = P[X \leq 2] = \sum_{x=0}^2 {}^n C_x p_1^x (1-p_1)^{n-x} = 0.9235$$

Therefore, from equation (8), we have

$$P_p = 1 - P_a(p_1) = 1 - 0.9235 = 0.0765$$

It means that if there are several lots of the same quality $p = 0.04$, about 7.65% of these will be rejected. This is obviously a risk for the supplier because it was agreed upon by both that lots of quality 0.04 will be accepted whereas the quality inspector is rejecting 7.65% of them.

Similarly, we can calculate the consumer's risk using equation (10).

We first calculate the probability of accepting the lot of quality $p = p_2 = LTPD = 0.10$ using Table I.

From Table I, for $n = 25$, $x = c = 2$ and $p = p_2 = 0.10$, we have

$$P_a(p_2) = P[X \leq 2] = \sum_{x=0}^2 {}^n C_x p_2^x (1-p_2)^{n-x} = 0.5371$$

Therefore, from equation (10), we have the consumer's risk

$$P_c = P_a(p_2) = 0.5371$$

It means that if there are several lots of the same quality $p = 0.10$, about 53.71% out of these will be accepted by the quality inspector even though this quality is unsatisfactory. This is obviously a risk for the quality inspector.

E4) It is given that

$$n = 10, c = 0, p = 20\% = 0.20$$

Since the lot size is large relative to the sample size, we can calculate the average outgoing quality (AOQ) for the single sampling plan using equation (13).

For calculating AOQ, we have to calculate the probability of accepting the lot corresponding to $p = 0.20$.

If X represents the number of defective units in the sample, the lot is accepted if $X \leq c = 0$. Therefore, the probability of accepting the lot is given by

$$P_a(p) = P_a = P[X \leq 0]$$

Since the lot size is large relative to the sample size, we can calculate this probability by using Table I.

From Table I, for $n = 10$, $x = c = 0$ and $p = 0.20$, we have

$$P_a = P[X \leq 0] = \sum_{x=0}^0 {}^n C_x p^x (1-p)^{n-x} = 0.1074$$

On putting the values of p and P_a in equation (13), we get

$$AOQ = pP_a = 0.20 \times 0.1074 = 0.0215 = 2.15\%$$

E5) It is given that

$$N = 200, n = 12, c = 1$$

Since $N \geq 10n$, we can use the binomial distribution. Therefore, we use equation (12) to calculate the AOQ.

For constructing the AOQ curve, we have to calculate the probabilities of accepting the lot corresponding to different quality levels.

If X represents the number of defective chips in the sample, the manufacturer accepts the lot if $X \leq c = 1$. Therefore, the probability of accepting the lot is given by

$$P_a(p) = P[X \leq 1] = \sum_{x=0}^1 P[X = x]$$

We can use Table I for calculating the probabilities of accepting the lot corresponding to different quality levels such as $p = 0.01, 0.02, 0.03 \dots$. Then we calculate AOQ for each quality level by using equation (12). The probabilities of accepting the lot and AOQs corresponding to different quality levels are given in the following table:

Incoming Lot Quality	Probability of Accepting the Lot	AOQ
0	1.0000	0
0.01	0.9938	0.0093
0.02	0.9769	0.0184

Product Control

0.04	0.9191	0.0346
0.06	0.8405	0.0474
0.08	0.7513	0.0565
0.10	0.6590	0.0619
0.12	0.5686	0.0641
0.14	0.4834	0.0636
0.16	0.4055	0.0610
0.18	0.3359	0.0568
0.20	0.2749	0.0517

We construct the AOQ curve by taking the quality level (proportion defective) on the X-axis and the corresponding AOQ values on the Y-axis as shown in Fig.7.10.

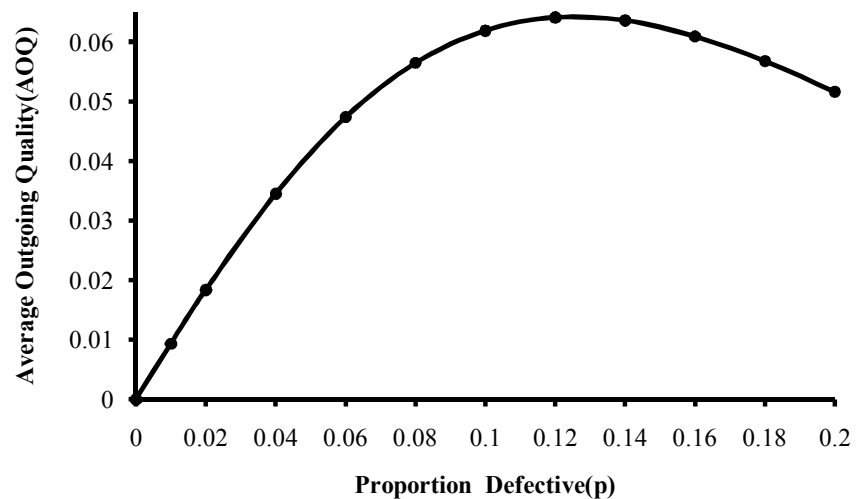


Fig. 7.10: The AOQ curve for E5.

E6) It is given that

$$N = 200, n = 12, c = 1$$

In the first case, the decision of acceptance or rejection of the lot is taken only on a single sample of size $n = 12$. Therefore, the ASN for this plan is simply the sample size, i.e., $ASN = n = 12$.

For construction of the ATI curve, we first calculate the probabilities of accepting the lot corresponding to different quality levels using Table I. Then we calculate the ATI for each quality level using equation (15).

We have already calculated these probabilities in E5 and we those results. Substituting the values of N , n , p and P_a in equation (15) we can calculate ATI.

The probabilities of accepting the lot and the ATIs corresponding to different quality levels are given in the following table:

Incoming Lot Quality	Probability of Accepting the Lot	ATI
0	1.0000	12.00
0.01	0.9938	13.17
0.02	0.9769	16.34
0.04	0.9191	27.21
0.06	0.8405	41.99

0.08	0.7513	58.76
0.10	0.6590	76.11
0.12	0.5686	93.10
0.14	0.4834	109.12
0.16	0.4055	123.77
0.18	0.3359	136.85
0.20	0.2749	148.32

We now construct the ATI curve by taking the quality level (proportion defective) on the X-axis and the corresponding ATI values on the Y-axis as shown in Fig.7.11.

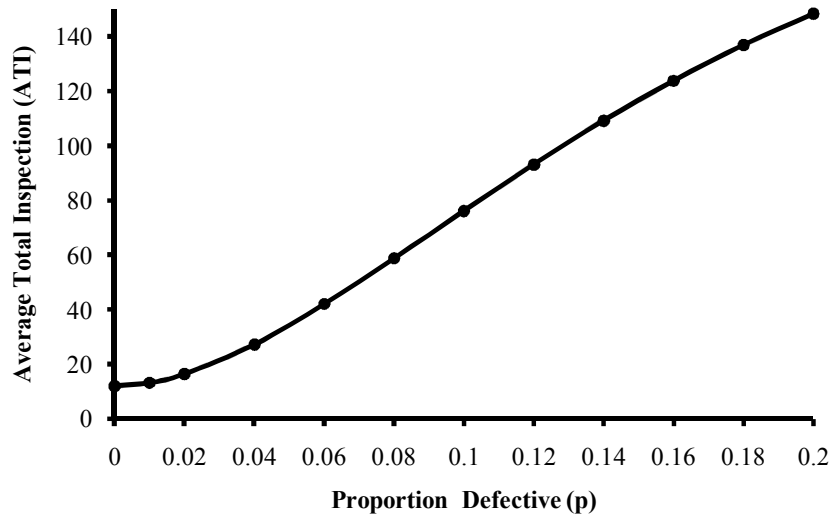


Fig. 7.11: The ATI curve for E6.

E7) We have

$$AQL = 1\% = 0.01 \text{ and } \alpha = 1\% = 0.01$$

To design the desired sampling plan, we first look up the value of np_1 corresponding to $c = 2$ and $\alpha = 0.01$ from Table III. We have

$$np_1 = 0.436$$

Therefore, the sample size is

$$n = \frac{np_1}{p_1} = \frac{0.436}{0.01} = 43.6 \approx 44$$

Hence, the required single sampling plan is

$$n = 44 \text{ and } c = 2$$

Similarly, for $c = 4$ and $\alpha = 0.01$, the value of np_1 is 1.279.

Therefore, the sample size is

$$n = \frac{np_1}{p_1} = \frac{1.279}{0.01} = 127.9 \approx 128$$

Hence, the required single sampling plan is

$$n = 128 \text{ and } c = 4$$

Similarly, for $c = 6$ and $\alpha = 0.01$, the value of np_1 is 2.330.

Therefore,

$$n = \frac{np_1}{p_1} = \frac{2.330}{0.01} = 233$$

Hence, the required single sampling plan is

$$n = 233 \text{ and } c = 6$$

E8) We have

$$\text{LTPD} = p_2 = 5\% = 0.05 \text{ and } \beta = 5\% = 0.05$$

To design the desired sampling plan, we look up the value of np_2 corresponding to $c = 3$ and $\beta = 0.05$ from Table III. We have

$$np_2 = 7.754$$

Therefore, the sample size is

$$n = \frac{np_2}{p_2} = \frac{7.754}{0.05} = 155.08 \approx 156$$

Hence, the required single sampling plan is

$$n = 156 \text{ and } c = 3$$

Similarly, for $c = 6$ and $\beta = 0.05$, the value of np_2 is 11.842.

Therefore, the sample size is

$$n = \frac{np_2}{p_2} = \frac{11.842}{0.05} = 236.84 \approx 237$$

Hence, the required single sampling plan is

$$n = 237 \text{ and } c = 6$$

UNIT 8 DOUBLE SAMPLING PLANS

Structure

- 8.1 Introduction
Objectives
- 8.2 Double Sampling Plan
Implementation of Double Sampling Plan
Advantages of Double Sampling Plan
Difference between Single and Double Sampling Plans
- 8.3 Operating Characteristic (OC) Curve
- 8.4 Producer's Risk and Consumer's Risk
- 8.5 Average Outgoing Quality (AOQ)
- 8.6 Average Sample Number (ASN)
- 8.7 Average Total Inspection (ATI)
- 8.8 Design of Double Sampling Plans
- 8.9 Summary
- 8.10 Solutions/Answers

8.1 INTRODUCTION

In Unit 7, you have learnt about the single sampling plan and the procedure for implementing it. In a single sampling plan, we take the decision of accepting or rejecting a lot on the basis of a single sample.

Sometimes, situations arise when it is not possible to decide whether to accept or reject the lot on the basis of a single sample. In such situations, we use a sampling plan known as the **double sampling plan**. In this plan, the decision of acceptance or rejection of a lot is taken on the basis of **two samples**. A lot may be accepted immediately if the first sample is good or may be rejected if it is bad. If the first sample is neither good nor bad, the decision is based on the evidence of the first and second sample combined.

In this unit, we explain the concept of the double sampling plan and the procedure for implementing it (Sec. 8.2). In Secs. 8.3 to 8.7, we discuss various features of the double sampling plan viz. the OC curve, producer's risk and consumer's risk, AOQ, ASN and ATI. In the last section of the unit (Sec. 8.8), we explain how to design the double sampling plan. In the next unit, we shall introduce decision theory.

Objectives

After studying this unit, you should be able to:

- describe a double sampling plan and explain how to implement it;
- differentiate between single and double sampling plans;
- compute the probability of accepting or rejecting a lot of incoming quality in a double sampling plan;
- construct the operating characteristic (OC) curve for a double sampling plan;

- compute the producer's risk and consumer's risk for a double sampling plan;
- compute the average sample number (ASN) and the average total inspection (ATI) for a double sampling plan; and
- design double sampling plans.

8.2 DOUBLE SAMPLING PLAN

A sampling plan in which a decision about the acceptance or rejection of a lot is based on two samples that have been inspected is known as a **double sampling plan**.

The double sampling plan is used when a clear decision about acceptance or rejection of a lot cannot be taken on the basis of a single sample. In double sampling plan, generally, the decision of acceptance or rejection of a lot is taken on the basis of two samples. If the first sample is bad, the lot may be rejected on the first sample and a second sample need not be drawn. If the first sample is good, the lot may be accepted on the first sample and a second sample is not needed. But if the first sample is neither good nor bad and there is a doubt about its results, we take a second sample and the decision of acceptance or rejection of a lot is taken on the basis of the evidence obtained from both the first and the second samples.

For example, suppose a buyer purchases cricket balls in lots of 500 from a company. To check the quality of the lots, the buyer and the company decide that the buyer will draw two samples of sizes 10 (first sample) and 20 (second sample) and the acceptance numbers for the plan are 1 and 3. The buyer takes two samples and makes the decision of acceptance or rejection of the lot on the basis of two samples. Since the decision of acceptance or rejection of the lot is taken on the basis of two samples, this is a double sampling plan.

A double sampling plan requires the specification of four quantities which are known as its **parameters**. These parameters are

n_1 – size of the first sample,

c_1 – acceptance number for the first sample,

n_2 – size of the second sample, and

c_2 – acceptance numbers for both samples combined.

Therefore, the parameters of the double sampling plan in the above example are

the size of the first sample (n_1) = 10,

the acceptance number for the first sample (c_1), = 1

the size of the second sample (n_2) = 20, and

the acceptance numbers for both the samples combined (c_2) = 3.

So far you have learnt the definition of the double sampling plan and why it is used. We now describe the procedure for implementing it and its advantages over the single sampling plan.

8.2.1 Implementation of Double Sampling Plan

Suppose, lots of the same size, say N , are received from the supplier or the final assembly line and submitted for inspection one at a time. The procedure

for implementing the double sampling plan to arrive at a decision about the lot is described in the following steps:

- Step 1:** We draw a random sample (first sample) of size n_1 from the lot received from the supplier or the final assembly.
- Step 2:** We inspect each and every unit of the sample and classify it as defective or non-defective. At the end of the inspection, we count the number of defective units found in the sample. Suppose the number of defective units found in the first sample is d_1 .
- Step 3:** We compare the number of defective units (d_1) found in the first sample with the stated acceptance numbers c_1 and c_2 .
- Step 4:** We take the decision on the basis of the first sample as follows:

Under acceptance sampling plan

If the number of defective units (d_1) in the first sample is less than or equal to the stated acceptance number (c_1) for the first sample, i.e., if $d_1 \leq c_1$, we accept the lot and if $d_1 > c_2$, we reject the lot. But if $c_1 < d_1 \leq c_2$, the first (single) sample is failed.

Under rectifying sampling plan

If $d_1 \leq c_1$, we accept the lot and replace all defective units found in the sample by non-defective units. If $d_1 > c_2$, we accept the lot after inspecting the entire lot and replacing all defective units in the lot by non-defective units. But if $c_1 < d_1 \leq c_2$, the first (single) sample is failed.

- Step 5:** If $c_1 < d_1 \leq c_2$, we draw a second random sample of size n_2 from the lot.
- Step 6:** We inspect each and every unit of the second sample and count the number of defective units found in it. Suppose the number of defective units found in the second sample is d_2 .
- Step 7:** We combine the number of defective units (d_1 and d_2) found in both samples and consider $d_1 + d_2$ for taking the decision about the lot on the basis of the second sample as follows:

Under acceptance sampling plan

If $d_1 + d_2 \leq c_2$, we accept the lot and if $d_1 + d_2 > c_2$, we reject the lot.

Under rectifying sampling plan

If $d_1 + d_2 \leq c_2$, we accept the lot and replace all defective units found in the second sample by non-defective units. If $d_1 + d_2 > c_2$, we accept the lot after inspecting the entire lot and replacing all defective units in the lot by non-defective units.

The steps described above are shown in Fig. 8.1 for the acceptance sampling plan.

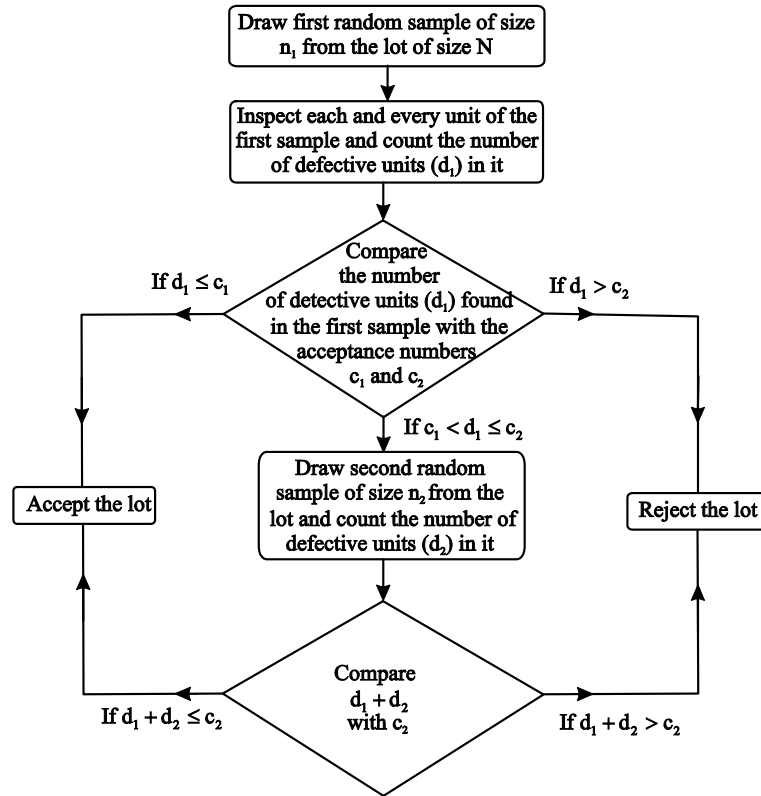


Fig. 8.1: Procedure for implementation of an acceptance double sampling plan.

Let us explain these steps further with the help of an example.

Example 1: Suppose a mobile phone company produces mobile phones in lots of 400 phones each. To check the quality of the lots, the quality inspector of the company uses a double sampling plan with $n_1 = 15$, $c_1 = 1$, $n_2 = 30$, $c_2 = 3$. Explain the procedure for implementing it under acceptance sampling plan.

Solution: For implementing the double sampling plan, the quality inspector of the company randomly draws first sample of 15 mobiles from the lot and classifies each mobile of the first sample as defective or non-defective. At the end of the inspection, he/she counts the number of defective mobiles (d_1) found in the first sample and compares d_1 with the acceptance numbers c_1 and c_2 . If $d_1 \leq c_1 = 1$, he/she accepts the lot and if $d_1 > c_2 = 3$, he/she rejects the lot. If $c_1 < d_1 \leq c_2$, it means that if the number of defective mobiles is 2 or 3, he/she draws the second sample from the lot. He/she counts the number of defective mobiles (d_2) found in the second sample and compares the total number of defective mobiles ($d_1 + d_2$) in both samples with the acceptance number c_2 . If $d_1 + d_2 \leq c_2 = 3$, he/she accepts the lot and if $d_1 + d_2 > c_2 = 3$, he/she rejects the lot.

8.2.2 Advantages of Double Sampling Plan

A double sampling plan has the following two main advantages over a single sampling plan:

- The principal advantage of the double sampling plan over the single sample plan is that for the same degree of protection (i.e., the same probability of accepting a lot of a given quality), the double sampling plan may have a smaller average sample number (ASN) than that corresponding to the single sampling plan. The underlying reason is that the size (n_1) of the first sample in the double sampling plan is always smaller than the sample size

(n) of an equivalent single sampling plan. Thus, if a decision is taken on the basis of the first sample, ASN will be lower for the double sampling plan or if a decision is taken after the second sample, the ASN will be reduced.

- ii) The double sampling plan has the psychological advantage of giving a lot a second chance. From the viewpoint of the producer/manufacture, it is unfair to reject a lot on the basis of a single sample. The double sampling plan permits the decision to be made on the basis of two samples.

However, double sampling plans are costlier to administer in comparison with the single sampling plans.

Try the following exercise to explain the procedure for implementing a double sampling plan.

E1) A manufacturer of silicon chip produces lots of 1000 chips for shipment. A buyer uses a double sampling plan with $n_1 = 5$, $c_1 = 0$, $n_2 = 20$, $c_2 = 2$ to test the quality of the lots. Explain the procedure for implementing it under acceptance sampling plan.

8.2.3 Difference between Single and Double Sampling Plans

The differences between single and double sampling plans are given below:

1. Single sampling plans are easier to design and operate in comparison with the double sampling plans.
2. The advantage of double sampling plans over single sampling plans seems to be psychological. It looks unreasonable to reject a lot on the basis of one sample and it is more convincing to say that the lot was rejected on the basis of inspection of two samples.
3. Under the double sampling plan, the good lot will generally be accepted and the bad lot will usually be rejected on the basis of the first sample. Thus, in all cases, where a decision to accept or reject is taken on the basis of the first sample, there is a considerable saving in the amount of inspection than required by a comparable single sampling plan. Moreover, when a second sample is taken, it may be possible to reject the lot without completely inspecting the entire second sample (see Sec.8.6).
4. In the double sampling plan, on an average, 25% to 33 % less number of items/units need to be inspected as compared to the single sampling plan.
5. The operating characteristic (OC) curve for a double sampling plan is steeper than a single sampling plan, i.e., the discriminatory power of the double sampling plan is higher than that of the single sampling plan.

So far you have learnt what a double sampling plan is and how it is implemented in industry. You have also learnt the differences between the double and single sampling plans. In Secs. 8.3 to 8.7, we describe various features of the double sampling plan.

8.3 OPERATING CHARACTERISTIC (OC) CURVE

You have learnt in Unit 6 that the operating characteristic (OC) curve displays the discriminatory power of the sampling plan. That is, it shows the probability that a lot submitted with a certain fraction defective will either be accepted or rejected.

Product Control

You have learnt in Units 6 and 7 that for constructing an OC curve, we require the probabilities of accepting a lot corresponding to different quality levels. Therefore, we now describe how to compute the probability of accepting a lot in a double sampling plan.

You have learnt in Sec. 8.2.1 that in a double sampling plan, the decision of acceptance or rejection of the lot is taken on the basis of two samples. The lot is accepted on the first sample if the number of defective units (d_1) in the first sample is less than the acceptance number c_1 . The lot is accepted on the second sample if the number of defective units ($d_1 + d_2$) in both samples is greater than c_1 and less than or equal to the acceptance number c_2 . Therefore, if $P_{a1}(p)$ and $P_{a2}(p)$ denote the probabilities of accepting a lot on the first sample and the second sample, respectively, the probability of accepting a lot of quality level p is given by:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) \quad \dots (1)$$

If X and Y denote the observed number of defective units in the first and second samples, respectively, we accept the lot on the first sample if $X \leq c_1$, i.e., if $X = 0$ or 1 or $2, \dots$, or c_1 .

Therefore,

$$\begin{aligned} P_{a1}(p) &= P[X \leq c_1] = P[X = 0 \text{ or } 1 \text{ or } 2, \dots, \text{ or } c_1] \\ &= P[X = 0] + P[X = 1] + \dots + P[X = c_1] \quad \left(\because X = 0, 1, 2, \dots, c_1 \text{ are } \right. \\ &\quad \left. \text{mutually exclusive} \right) \\ &= \sum_{x=0}^{c_1} P[X = x] \quad \dots (2) \end{aligned}$$

We can calculate this probability if we know the distribution of X . You have learnt in Unit 5 that generally, in quality control, a random sample is drawn from a lot of finite size without replacement. So in such situations, the number of defective units (X) in the sample follows a hypergeometric distribution.

In a lot of size N and incoming quality p , the number of defective units is Np and non-defective units is $N - Np$. Therefore, we can compute the probability of getting exactly x defective units in a sample of size n_1 using the hypergeometric distribution as follows:

$$P[X = x] = \frac{{}^{Np}C_x {}^{N-Np}C_{n_1-x}}{{}^NC_{n_1}}; \quad x = 0, 1, \dots, \min(Np, n_1) \quad \dots (3)$$

Thus, we can obtain the probability of accepting a lot of quality p on the first sample by putting the value of $P[X = x]$ in equation (2) as follows:

$$P_{a1}(p) = \sum_{x=0}^{c_1} P[X = x] = \sum_{x=0}^{c_1} \frac{{}^{Np}C_x {}^{N-Np}C_{n_1-x}}{{}^NC_{n_1}} \quad \dots (4)$$

We accept the lot of quality p on the second sample if $c_1 < X + Y \leq c_2$. It means that we accept the lot if

$$X = c_1 + 1 \text{ and } Y \leq c_2 - X = c_2 - c_1 - 1$$

$$\text{or } X = c_1 + 2 \text{ and } Y \leq c_2 - c_1 - 2$$

You have studied in Unit 3 of MST-003 that if A and B are mutually exclusive events then

$$P[A \text{ or } B] = P[A] + P[B]$$

∴

or $X = c_2$ and $Y \leq c_2 - c_1 = 0$.

Therefore, from the addition theorem of probability, we can obtain the probability of accepting a lot of quality p on the second sample as follows:

$$\begin{aligned} P_{a2}(p) &= P[X = c_1 + 1]P[Y \leq c_2 - c_1 - 1] + P[X = c_1 + 2]P[Y \leq c_2 - c_1 - 2] \\ &\quad + \dots + P[X = c_2]P[Y \leq 0] \\ &= \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} P[X = x]P[Y = y / X = x] \end{aligned} \quad \dots (5)$$

where $P[Y = y / X = x]$ is the conditional probability of observing y defective units in the second sample under the condition that x defective units have already appeared in the first sample.

Since it is given that x defective units have already appeared in the first sample, the remaining defective units and non-defective units in the lot on the second sample are $Np - x$ and $(N - n_1) - (Np - x)$, respectively. Thus, we can compute the probability $P[Y = y / X = x]$ by using the hypergeometric distribution as follows:

$$P(Y = y / X = x) = \frac{{}^{Np-x}C_y {}^{N-n_1-(Np-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \quad \dots (6)$$

On putting the values of $P[X = x]$ and $P[Y = y / X = x]$ from equations (4) and (6) in equation (5), we get

$$P_{a2}(p) = \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{{}^{Np}C_x {}^{N-Np}C_{n_1-x}}{{}^N C_{n_1}} \frac{{}^{Np-x}C_y {}^{N-n_1-(Np-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \quad \dots (7)$$

Therefore, the required probability of accepting a lot of quality p in a double sampling plan is given by

$$\begin{aligned} P_a(p) &= \sum_{x=0}^{c_1} \frac{{}^{Np}C_x {}^{N-Np}C_{n_1-x}}{{}^N C_{n_1}} \\ &\quad + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{{}^{Np}C_x {}^{N-Np}C_{n_1-x}}{{}^N C_{n_1}} \frac{{}^{Np-x}C_y {}^{N-n_1-(Np-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \end{aligned} \quad \dots (8)$$

However, we can approximate the hypergeometric distribution to the binomial distribution with parameters n and p as we have explained in Unit 7. It means that if the sample sizes n_1 and n_2 are small compared to the lot size (N), the probability of accepting a lot of quality p (using the binomial approximation) is given by

$$\begin{aligned} P_a(p) &= \sum_{x=0}^{c_1} {}^{n_1}C_x p^x (1-p)^{n_1-x} \\ &\quad + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} {}^{n_1}C_x p^x (1-p)^{n_1-x} {}^{n_2}C_y p^y (1-p)^{n_2-y} \end{aligned} \quad \dots (9)$$

However, for rapid calculation, we can use Table I entitled **Cumulative Binomial Probability Distribution** given at the end of this block for calculating this probability.

When p is small and n is large such that np is finite, we know that the binomial distribution approaches the Poisson distribution with parameter $\lambda = np$. Therefore, the probability of accepting a lot of quality p using the Poisson approximation is given by

$$P_a(p) = \sum_{x=0}^{c_1} \frac{e^{-\lambda_1} \lambda_1^x}{x!} + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{e^{-\lambda_1} \lambda_1^x}{x!} \frac{e^{-\lambda_2} \lambda_2^y}{y!} \quad \dots (10)$$

where $\lambda_1 = n_1 p$ and $\lambda_2 = n_2 p$.

We can use Table II entitled **Cumulative Poisson Probability Distribution** given at the end of this block for calculating this probability.

We now illustrate how to compute the probability of accepting a lot in a double sampling plan.

Example 2: Suppose, in Example 1, the incoming quality of the lot is 0.05. What is the probability of accepting the lot on the first sample? What is the probability of final acceptance?

Solution: It is given that

$$N = 400, n_1 = 15, c_1 = 1, n_2 = 30, c_2 = 3 \text{ and } p = 0.05$$

If $P_{a1}(p)$ and $P_{a2}(p)$ denote the probabilities of accepting the lot on the first sample and the second sample, respectively, we can calculate the probability of accepting the lot of quality level (p) in a double sampling plan as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) \quad \dots (i)$$

If X represents the number of defective mobiles in the first sample of size 15, the lot is accepted on the first sample if $X \leq c_1 = 1$. Therefore,

$$P_{a1}(p) = P[X \leq c_1] = P[X \leq 1] \quad \dots (ii)$$

Since $N \geq 10n_1$, we can use the binomial distribution and calculate $P_{a1}(p)$ using Table I.

From Table I, for $n = n_1 = 15, x = c_1 = 1$ and $p = 0.05$, we have

$$P_{a1}(p) = P[X \leq 1] = \sum_{x=0}^1 {}^n C_x p^x (1-p)^{n-x} = 0.8290 \quad \dots (iii)$$

Hence, the probability of accepting the lot on the first sample is 0.8290. It means that about 82.9% of the lot of quality $p = 0.05$ will be accepted on the first sample.

If Y represents the number of defective mobiles in the second sample, the lot is accepted on the second sample if $c_1 < X + Y \leq c_2$, i.e., $1 < X + Y \leq 3$. It means that we shall accept the lot if

$$X = c_1 + 1 = 1 + 1 = 2 \text{ and } Y \leq c_2 - X = 3 - 2 = 1$$

$$\text{or } X = c_1 + 2 = 1 + 2 = 3 \text{ and } Y \leq c_2 - X = 3 - 3 = 0.$$

Therefore,

$$P_{a2}(p) = P[X = 2]P[Y \leq 1] + P[X = 3]P[Y \leq 0] \quad \dots (iv)$$

The probabilities $P[X = 2]$ and $P[X = 1]$ can also be obtained using Table I, which contains the cumulative probabilities of the binomial distribution, using the following expression:

$$P[X = x] = P[X \leq x] - P[X \leq x - 1] \quad \dots (v)$$

From Table I, for $n = n_1 = 15$ and $p = 0.05$, we have

$$P[X = 2] = P[X \leq 2] - P[X \leq 1] = 0.9638 - 0.8290 = 0.1348$$

$$P[X = 1] = P[X \leq 1] - P[X \leq 0] = 0.8290 - 0.4633 = 0.3657$$

From Table I, for $n = n_2 = 30$ and $p = 0.05$, we have

$$P[Y \leq 1] = 0.5535 \text{ and } P[Y \leq 0] = 0.2146$$

On putting these values in equation (iv), we get

$$\begin{aligned} P_{a2}(p) &= P[X = 2]P[Y \leq 1] + P[X = 3]P[Y \leq 0] \\ &= 0.1348 \times 0.5535 + 0.3657 \times 0.2146 = 0.1531 \end{aligned}$$

Hence, the probability of accepting the lot on the second sample is 0.1531. It means that about 15.31% of the lot of quality $p = 0.05$ will be accepted on the second sample.

From equation (i), we get the probability of accepting the lot in the double sampling plan as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) = 0.8290 + 0.1531 = 0.9821$$

Thus, overall, 98.21% of the lots will be accepted by this sampling plan.

In a similar way, we can calculate the probability of accepting a lot for different lot qualities.

The construction of the OC curve for a double sampling plan is beyond the scope of this course. If you are interested in constructing the OC curve, you may consider different quality levels such as $p = 0.01, 0.02, 0.03 \dots$ and then calculate the corresponding probabilities of accepting the lot as explained in Example 2. The OC curve may then be constructed by taking the quality level (proportion defective) on the X-axis and the probability of accepting the lot on the Y-axis as explained for the single sampling plan in Unit 7.

You may now like to calculate the probability of accepting a lot for practice. Try the following exercise.

E2) Suppose that in E1, the incoming quality of the lot is 0.03. Calculate the probabilities of accepting the lot on the first sample and on the second sample. What is the probability of accepting the lot?

8.4 PRODUCER'S RISK AND CONSUMER'S RISK

In Units 5 and 7, we have defined the **producer's risk** as follows:

The probability of rejecting a lot of acceptance quality level (p_1) is known as the producer's risk.

Thus, the producer's risk for a double sampling plan is given by

$$\begin{aligned} P_p &= P[\text{rejecting a lot of acceptance quality level } p_1] \\ &= 1 - P[\text{accepting a lot of acceptance quality level } p_1] \\ &= 1 - P_a(p_1) \end{aligned} \quad \dots (11)$$

We can compute $P_a(p_1)$ from equation (8) by replacing the quality level p with p_1 as follows:

$$\begin{aligned} P_a(p_1) &= \sum_{x=0}^{c_1} \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n_1-x}}{{}^NC_{n_1}} \\ &\quad + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n_1-x}}{{}^NC_{n_1}} \cdot \frac{{}^{Np_1-x}C_y {}^{N-n_1-(Np_1-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \end{aligned} \quad \dots (12)$$

Therefore, from equation (11), the producer's risk for a double sampling plan is given by

$$\begin{aligned} P_p &= 1 - \sum_{x=0}^{c_1} \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n_1-x}}{{}^NC_{n_1}} \\ &\quad - \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{{}^{Np_1}C_x {}^{N-Np_1}C_{n_1-x}}{{}^NC_{n_1}} \cdot \frac{{}^{Np_1-x}C_y {}^{N-n_1-(Np_1-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \end{aligned} \quad \dots (13)$$

For rapid calculation of the producer's risk for a double sampling plan, we can also use the approximations as explained in Sec. 8.3. Therefore, if we use the approximation of the hypergeometric distribution to the binomial distribution, the producer's risk is given by

$$\begin{aligned} P_p &= 1 - \sum_{x=0}^{c_1} {}^{n_1}C_x p_1^x (1-p_1)^{n_1-x} \\ &\quad - \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} {}^{n_1}C_x p_1^x (1-p_1)^{n_1-x} {}^{n_2}C_y p_1^y (1-p_1)^{n_2-y} \end{aligned} \quad \dots (14)$$

We now explain the **Consumer's Risk** for a double sampling plan:

By definition, the probability of accepting a lot of unsatisfactory quality (LTPD) p_2 is known as the consumer's risk.

Therefore, the consumer's risk for a double sampling plan is given by

$$P_c = P[\text{accepting a lot of quality (LTPD) } p_2] = P_a(p_2)$$

We can compute the consumer's risk for a double sampling plan from equation (8) by replacing the quality level p with the quality LTPD (p_2) as follows:

$$\begin{aligned} P_c = P_a(p_2) &= \sum_{x=0}^{c_1} \frac{{}^{Np_2}C_x {}^{N-Np_2}C_{n_1-x}}{{}^NC_{n_1}} \\ &\quad + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} \frac{{}^{Np_2}C_x {}^{N-Np_2}C_{n_1-x}}{{}^NC_{n_1}} \cdot \frac{{}^{Np_2-x}C_y {}^{N-n_2-(Np_2-x)}C_{n_2-y}}{{}^{N-n_1}C_{n_2}} \end{aligned} \quad \dots (15)$$

If we approximate the hypergeometric distribution to the binomial distribution, the consumer's risk is given by

$$P_c = P_a(p_2) = \sum_{x=0}^{c_1} {}^{n_1}C_x p_2^x (1-p_2)^{n_1-x} + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} {}^{n_1}C_x p_2^x (1-p_2)^{n_1-x} {}^{n_2}C_y p_2^y (1-p_2)^{n_2-y} \dots (16)$$

Let us consider an example from a real life situation to explain these concepts.

Example 3: Suppose a shirt manufacturing company supplies shirts in lots of size 500 to the buyer. A double sampling plan with $n_1 = 10$, $c_1 = 0$, $n_2 = 25$, $c_2 = 1$ is being used for the lot inspection. The company and the buyer's quality control inspector decide that $AQL = 0.04$ and $LTPD = 0.10$. Compute the producer's risk and consumer's risk for this sampling plan.

Solution: It is given that

$$N = 500, n_1 = 10, c_1 = 0, n_2 = 25, c_2 = 1,$$

$$AQL(p_1) = 0.04 \text{ and } LTPD(p_2) = 0.10$$

We know that the producer's risk for a double sampling plan is given by

$$\begin{aligned} P_p &= P[\text{rejecting a lot of acceptance quality level } p_1] \\ &= 1 - P[\text{accepting a lot of acceptance quality level } p_1] \\ &= 1 - P_a(p_1) \end{aligned} \dots (i)$$

Therefore, for calculating the producer's risk, we first calculate the probability of accepting the lot of quality $p = p_1 = AQL = 0.04$ as we have explained in Sec. 8.3.

In a double sampling plan, the lot may be accepted either on the first sample or on the second sample. So if $P_{a1}(p)$ and $P_{a2}(p)$ denote the probabilities of accepting the lot on the first sample and on the second sample, respectively, we can calculate the probability of accepting the lot of quality level p , as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) \dots (ii)$$

If X represents the number of defective shirts in the first sample, the lot will be accepted on the first sample if $X \leq c_1 = 0$. Therefore,

$$P_{a1}(p) = P[X \leq c] = P[X \leq 0] \dots (iii)$$

Since $N \geq 10n$, we can use the binomial distribution and calculate the probability using Table I.

From Table I, for $n = n_1 = 10$, $x = c_1 = 0$ and $p = p_1 = 0.04$, we have

$$P_{a1}(p_1) = P[X \leq 0] = \sum_{x=0}^0 {}^{n_1}C_x p_1^x (1-p_1)^{n_1-x} = 0.6648 \dots (iv)$$

If Y represents the number of defective shirts in the second sample, the lot will be accepted on the second sample if $c_1 < X + Y \leq c_2$. It means that we shall accept the lot if $X = c_1 + 1 = 0 + 1 = 1$ and $Y \leq c_2 - X = 1 - 1 = 0$.

Therefore,

$$P_{a2}(p_1) = P[X=1]P[Y \leq 0] \quad \dots (v)$$

We also know that $P[X=x] = P[X \leq x] - P[X \leq x-1]$

From Table I, for $n = n_1 = 10$ and $p = p_1 = 0.04$, we have

$$P[X=1] = P[X \leq 1] - P[X \leq 0] = 0.9418 - 0.6648 = 0.2770$$

From Table I, for $n = n_2 = 25$, $x = y = 0$ and $p = p_1 = 0.04$, we have

$$P[Y \leq 0] = 0.3604$$

On putting these values in equation (v), we get

$$\begin{aligned} P_{a2}(p_1) &= P[X=1]P[Y \leq 0] \\ &= 0.2770 \times 0.3604 = 0.0998 \end{aligned}$$

Hence, from equation (ii), we get the probability of accepting the lot in this double sampling plan as follows:

$$P_a(p_1) = P_{a1}(p_1) + P_{a2}(p_1) = 0.6648 + 0.0998 = 0.7646$$

Therefore, we calculate the producer's risk for this plan using equation (i) as follows:

$$P_p = 1 - P_a(p_1) = 1 - 0.7646 = 0.2354$$

It means that if there are several lots of the same quality $p = 0.04$, about 23.54% out of these will be rejected. This is obviously a risk for the manufacturing company because it was agreed upon by both that lots of quality 0.04 will be accepted whereas the quality inspector is rejecting 23.54% of them.

Similarly, we can calculate the consumer's risk as follows:

We know the consumer's risk for a double sampling plan is given by

$$P_c = P[\text{accepting a lot of quality (LTPD)} p_2] = P_a(p_2) \quad \dots (vi)$$

We first calculate the probability of accepting the lot of quality $p = p_2 = \text{LTPD} = 0.10$ using equations (ii), (iii) and (v).

From Table I, for $n = n_1 = 10$, $x = c_1 = 0$ and $p = p_2 = 0.10$, we have

$$P_{a1}(p_2) = P[X \leq 0] = \sum_{x=0}^0 {}^{n_1}C_x p_2^x (1-p_2)^{n-x} = 0.3487 \quad \dots (vii)$$

From Table I, for $n = n_1 = 10$ and $p = p_2 = 0.10$, we have

$$P[X=1] = P[X \leq 1] - P[X \leq 0] = 0.7361 - 0.3487 = 0.3874$$

From Table I, for $n = n_2 = 25$, $x = y = 0$ and $p = p_2 = 0.10$, we have

$$P[Y \leq 0] = 0.0718$$

On putting these values in equation (v), we get

$$\begin{aligned} P_{a2}(p_2) &= P[X=1]P[Y \leq 0] \\ &= 0.3874 \times 0.0718 = 0.0278 \quad \dots (viii) \end{aligned}$$

On putting the values of $P_{a1}(p_2)$ and $P_{a2}(p_2)$ in equation (ii), we get

$$P_a(p_2) = 0.3487 + 0.0278 = 0.3765$$

Hence, from equation (vi), we get the consumer's risk for this plan as follows:

$$P_c = P_a(p_2) = 0.3765$$

It means that if there are several lots of the same quality $p = 0.10$, about 37.65% out of these will be accepted by the quality inspector even though this quality is unsatisfactory. This is obviously the buyer's risk.

For practice, you can also compute the producer's risk and consumer's risk in the following exercise.

E3) Suppose that in E2, the acceptance quality level (AQL) and lot tolerance percent defective (LTPD) are 0.05 and 0.14, respectively. Calculate the producer's risk and consumer's risk for this plan.

8.5 AVERAGE OUTGOING QUALITY (AOQ)

In Unit 6, you have learnt that the average outgoing quality (AOQ) is defined as follows:

The expected quality of the lots after the application of sampling inspection is called the **average outgoing quality (AOQ)**. It is given by:

$$AOQ = \frac{\text{Number of defective units in the lot after the inspection}}{\text{Total number of units in the lot}} \dots (17)$$

You know that the concept of AOQ is particularly useful in the rectifying sampling plan wherein the rejected lots are inspected 100% and all defective units are replaced by non-defective units. So in a double sample plan, we can obtain the formula for average outgoing quality by considering the following situations:

- i) If the lot of size N is accepted on the first sample of size n_1 , $(N - n_1)$ units remain un-inspected. If the incoming quality of the lot is p , we expect that $p(N - n_1)$ defective units are left in the lot after the inspection on the first sample. The probability that the lot will be accepted on the first sample is P_{a1} . Therefore, the expected number of defective units per lot in the outgoing stage is $p(N - n_1)P_{a1}$.
- ii) If the lot is rejected on the first sample, all units of the lot go for 100% inspection and all defective units found in the lot are replaced by non-defective units. So there is no defective unit in the outgoing stage. However, the probability that the lot will be rejected on the first sample is $(1 - P_{a1})$. Therefore, the expected number of defective units per lot at the outgoing stage is $0 \times (1 - P_{a1}) = 0$.
- iii) If the lot is accepted on the second sample of size n_2 , $(N - n_1 - n_2)$ units remain un-inspected. If the incoming quality of the lot is p , we expect that $p(N - n_1 - n_2)$ defective units are left in the lot after the inspection on the second sample. The probability that the lot will be accepted on the second

sample is P_{a2} . Therefore, the expected number of defective units per lot in the outgoing stage is $p(N - n_1 - n_2)P_{a2}$.

- iv) If the lot is rejected on the second sample, all units of the lot go for 100% inspection and all defective units found in the lot are replaced by non-defective units. So there is no defective unit in the outgoing stage. However, the probability that the lot will be rejected on the second sample is $(1 - P_{a2})$. Therefore, the expected number of defective units per lot at the outgoing stage is $0 \times (1 - P_{a2}) = 0$.

Thus, the expected number of defective units per lot after sampling inspection is $p(N - n_1)P_{a1} + 0 + p(N - n_1 - n_2)P_{a2} + 0 = p[(N - n_1)(P_{a1} + P_{a2}) - n_2P_{a2}]$.

Hence, from equation (17), the AOQ for a double sampling plan is given by

$$\begin{aligned} \text{AOQ} &= \frac{\text{Number of defective units in the lot after the inspection}}{\text{Total number of units in the lot}} \\ &= \frac{p[(N - n_1)(P_{a1} + P_{a2}) - n_2P_{a2}]}{N} \end{aligned}$$

$$\text{or} \quad \text{AOQ} = p \left[\left(1 - \frac{n_1}{N} \right) (P_{a1} + P_{a2}) - \frac{n_2}{N} P_{a2} \right] \quad \dots (18)$$

If the sample sizes n_1 and n_2 are very small in proportion to the lot size N , i.e., $n_1 / N \approx 0$ and $n_2 / N \approx 0$, equation (18) for AOQ becomes

$$\text{AOQ} = p(P_{a1} + P_{a2}) \quad \dots (19)$$

Let us consider an example to illustrate how to calculate the AOQ for a double sampling plan.

Example 4: Suppose that in Example 2, the rejected lots are screened and all defective mobiles are replaced by non-defective ones. Calculate the average outgoing quality (AOQ) for this plan.

Solution: It is given that

$$N = 400, n_1 = 15, c_1 = 1, n_2 = 30, c_2 = 3 \text{ and } p = 0.05$$

Since $N > 10n_1$ and $N > 10n_2$, we can calculate the average outgoing quality (AOQ) for the double sampling plan using equation (18).

For calculating AOQ, we have to calculate the probabilities of accepting the lot on the first sample and on the second sample corresponding to $p = 0.05$.

We have already calculated these probabilities in Example 2. So we directly use the results:

$$P_{a1} = 0.8290 \text{ and } P_{a2} = 0.1531$$

Substituting the values of N , n_1 , n_2 , P_{a1} , P_{a2} and p in equation (18), we get

$$\begin{aligned} \text{AOQ} &= p \left[\left(1 - \frac{n_1}{N} \right) (P_{a1} + P_{a2}) - \frac{n_2}{N} P_{a2} \right] \\ &= 0.05 \left[\left(1 - \frac{15}{400} \right) (0.8290 + 0.1531) - \frac{30}{400} \times 0.1531 \right] \end{aligned}$$

$$= 0.05[0.9625 \times 0.9821 - 0.0115] = 0.0467$$

In the same way, you can calculate the AOQ for different lot qualities.

The construction of the AOQ curve for a double sampling plan is beyond the scope of this course. If you are interested in constructing the AOQ curve, you may consider different quality levels such as $p = 0.01, 0.02, 0.03 \dots$ and then calculate the corresponding probabilities of accepting the lot. The AOQ can then be constructed as explained in Units 6 and 7.

You may now like to calculate the average outgoing quality (AOQ) in the following exercise.

E4) Suppose that in E2, the rejected lots are screened and all defective silicon chips are replaced by non-defective ones. Calculate the average outgoing quality (AOQ) for this plan.

So far you have learnt about the OC curve, producer's risk, consumer's risk and AOQ for double sampling plan. We now explain ASN for a double sampling plan.

8.6 AVERAGE SAMPLE NUMBER (ASN)

You have learnt in Unit 6 that the average sample number (ASN) is the expected number of sample units per lot which is required to arrive at a decision about the acceptance or rejection of the lot under the acceptance sampling plan.

In acceptance double sampling plan, the number of units inspected to arrive at a decision of acceptance or rejection of the lot depends upon whether the decision is taken only on the first sample or on the second sample as well. Therefore, we have two situations:

- i) If the decision is taken on the first sample of size n_1 , the number of units inspected is n_1 . However, the probability of taking decision of acceptance or rejection of the lot on the first sample is P_1 .
- ii) If the decision is taken on the second sample of size n_2 , the number of units inspected is $(n_1 + n_2)$. The probability that a second sample is necessary is $(1 - P_1)$.

Therefore, the ASN for a double sampling plan can be obtained as follows:

$$\begin{aligned} \text{ASN} &= \text{Expected number of units inspected per lot} \\ &= \sum (\text{inspected number of units} \times \text{probability of taking decision}) \\ \text{ASN} &= n_1 \times P_1 + (n_1 + n_2) \times (1 - P_1) = n_1 P_1 + (n_1 + n_2)(1 - P_1) \quad \dots (20) \end{aligned}$$

where P_1 is the probability of making a decision about acceptance or rejection of the lot on the first sample and can be calculated as follows:

$$\begin{aligned} P_1 &= P[\text{lot is accepted on the first sample}] \\ &\quad + P[\text{lot is rejected on the first sample}] \quad \dots (21) \end{aligned}$$

Let us take up an example to illustrate this concept.

Example 5: Calculate the average sample number (ASN) for Example 2.

Solution: It is given that

Product Control

$$N = 400, n_1 = 15, c_1 = 1, n_2 = 30, c_2 = 3 \text{ and } p = 0.05$$

For computing the ASN, we can use equation (20):

$$ASN = n_1 P_1 + (n_1 + n_2)(1 - P_1)$$

where P_1 is the probability of making a decision about acceptance or rejection of the lot on the first sample. We calculate it using equation (21).

So we have to find the probability P_1 first. In this double sampling plan, the lot is accepted on the first sample if $X \leq c_1 = 1$, i.e., if $X \leq 1$. Therefore,

$$P[\text{lot is accepted on the first sample}] = P[X \leq c_1] = P[X \leq 1] = P_{a1}(p)$$

We have already calculated this probability in Example 2. So we directly use the results. Therefore,

$$P_{a1}(p) = P[X \leq c_1] = P[X \leq 1] = 0.8290 \quad \dots (i)$$

The lot will be rejected on the first sample if $X > c_2$, i.e., $X > 3$. Therefore,

$$P[\text{lot is rejected on the first sample}] = P[X > 3] = 1 - P[X \leq 3] \quad \dots (ii)$$

From Table I, for $n = n_1 = 15$, $x = 3$ and $p = 0.05$, we have

$$P[X \leq 3] = \sum_{x=0}^3 {}^{n_1}C_x p^x (1-p)^{n_1-x} = 0.9945$$

Therefore, from equation (ii), we have

$$\begin{aligned} P[\text{lot is rejected on the first sample}] &= 1 - P[X \leq 3] \\ &= 1 - 0.9945 = 0.0055 \quad \dots (iii) \end{aligned}$$

Thus, on putting the values of equations (ii) and (iii) in equation (21), we get

$$P_1 = 0.8290 + 0.0055 = 0.8345$$

Hence, on putting the values of n_1 , n_2 , and P_1 in equation (20), we get the ASN for the plan as follows:

$$\begin{aligned} ASN &= n_1 P_1 + (n_1 + n_2)(1 - P_1) \\ &= 15 \times 0.8345 + (15 + 30)(1 - 0.8345) = 19.965 \approx 20 \end{aligned}$$

In Sec. 8.2.2, you have learnt the advantages of the double sampling plan over the single sampling plan: The average sample number (ASN) for a double sampling plan is expected to be less than that for an equivalent single sampling plan (i.e., the same probability of accepting a lot of a given quality). If we plot the ASN curves for equivalent double and single sampling plans for different values of quality level p , we obtain the curves shown in Fig. 8.2.

We know that
 $P[X > A] = 1 - P[X \leq A]$

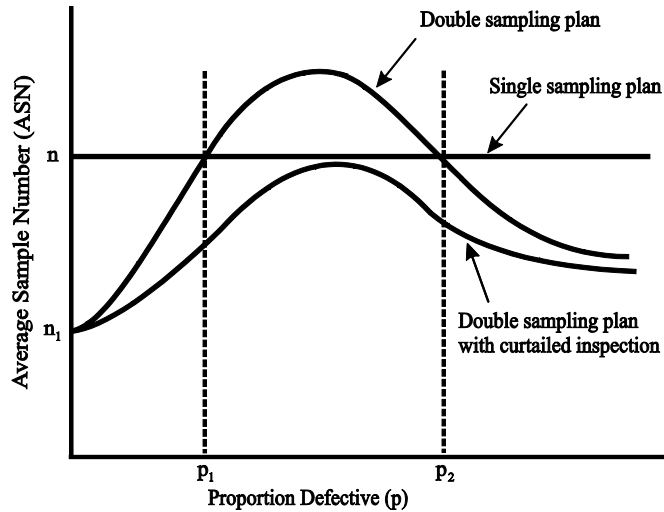


Fig. 8.2: The ASN curves for single and double sampling plans.

If we compare the ASN curves (shown in Fig. 8.2) for a double sampling plan with an equivalent single sampling plan, we observe that when the lots are of very good quality, they are usually accepted on the first sample. However, if the lots are of very bad quality, they are usually rejected on the first sample. In such cases, the ASN for the double sampling plan is smaller than the ASN for the equivalent single sampling plan because the size of the first sample in the double sampling plan is generally smaller than the size of the sample in the single sampling plan. However, if the lots are of intermediate quality (neither good nor bad), the second sample is usually required for making a decision about the acceptance or rejection of the lot in double sampling plan. In this case, the ASN for the double sampling plan is greater than the ASN for the single sampling plan.

In practice, inspection of the second sample is usually terminated and the lot is rejected as soon as the number of observed defective units in the combined samples (first and second) exceeds the acceptance number for the second sample. This termination is known as **curtailment**.

Thus, if we use the curtailed inspection for the second sample, the ASN for the double sampling plan is lower than ASN for the single sampling plan for the intermediate quality.

You can now calculate the ASN for the following exercise.

E5) Calculate the average sample number (ASN) for E2.

8.7 AVERAGE TOTAL INSPECTION (ATI)

Another important feature of the rectifying sampling plan is the average total inspection (ATI). Under rectifying sampling plan, the rejected lots are 100% inspected. The ATI is defined as follows:

The expected number of units inspected per lot under the rectifying sampling plan is called the average total inspection (ATI).

So in the rectifying double sampling plan, the number of units to be inspected will depend on the three situations given below:

- i) If the lot of size N is accepted on the first sample of size n_1 , the number of units inspected is n_1 and the probability of accepting the lot on the first sample is $P_{a1}(p)$.
- ii) If the lot is accepted on the second sample of size n_2 , the number of units inspected is $(n_1 + n_2)$ and the probability of accepting the lot on the second sample is $P_{a2}(p)$.
- iii) If the lot is rejected on the first or second sample, the entire lot of size (N) is inspected and the probability of rejecting the lot is $1 - P_a(p)$.

Therefore, we can compute the ATI for a double sampling plan as follows:

$$\begin{aligned}
 \text{ATI} &= \text{Expected number of units inspected per lot} \\
 &= \sum (\text{inspected number of units} \times \text{probability of taking decision}) \\
 \text{ATI} &= n_1 P_{a1}(p) + (n_1 + n_2) P_{a2}(p) + N[1 - P_a(p)] \quad \dots (22)
 \end{aligned}$$

Let us take up an example.

Example 6: Suppose that in Example 2, the rejected lots are screened and all defective mobiles are replaced by non-defective ones. Calculate the average total inspection (ATI) for this plan.

Solution: It is given that

$$N = 400, n_1 = 15, c_1 = 1, n_2 = 30, c_2 = 3 \text{ and } p = 0.05$$

We can calculate the ATI for the double sampling plan using equation (22).

For calculating ATI, we have to calculate the probabilities of accepting the lot on the first sample and the second sample corresponding to $p = 0.05$.

We have already calculated these probabilities in Example 2. So we directly use the results as follows:

$$P_{a1}(p) = 0.8290, P_{a2}(p) = 0.1531 \text{ and } P_a(p) = 0.9821$$

Substituting the values of N, n_1, n_2, P_{a1} and P_{a2} in equation (22), we get

$$\begin{aligned}
 \text{ATI} &= n_1 P_{a1}(p) + (n_1 + n_2) P_{a2}(p) + N[1 - P_a(p)] \\
 &= 15 \times 0.8290 + (15 + 30) \times 0.1531 + 400 \times (1 - 0.9821) \\
 &= 12.4350 + 6.8895 + 7.16 = 26.4845 \approx 27
 \end{aligned}$$

In the same way, you can calculate the ATI for different submitted lot qualities.

E6) Calculate the average total inspection (ATI) for E3.

We now explain how to design double sampling plans.

8.8 DESIGN OF DOUBLE SAMPLING PLANS

The design of a double sampling plan implies the determination of the parameters of the plan, i.e., the first sample size n_1 , the second sample size n_2 and the acceptance numbers c_1 and c_2 . These numbers have to be decided in advance before applying the double sampling plan technique. There are several approaches for determining the parameters n_1, n_2, c_1 and c_2 . Here we discuss an approach for designing the double sampling plan in which the producer's risk

with its corresponding acceptance quality level (AQL) and the consumer's risk with its corresponding lot tolerance percent defective (LTPD) are specified.

We have to design a double sampling plan, which satisfies both the producer's and the consumer's risk, such that the lots of AQL are to be rejected no more than $100\alpha\%$ of the time and lots of LTPD are to be accepted no more than $100\beta\%$ of the time.

For designing the plan in such situations, we use a pair of tables known as Grubbs Tables (Tables IV and V given at the end of this block). Tables IV and V are based on a relationship imposed on the parameters n_1 and n_2 , that $n_1 = n_2$ and $n_1 = 2n_2$, respectively. Both tables are based on a producer's risk (α) of 0.05 and a consumer's risk (β) of 0.10. Table IV is used when $n_1 = n_2$ and Table V is used when $n_1 = 2n_2$. In this approach, we first find the operating ratio R as follows:

$$R = \frac{p_2}{p_1} \quad \dots (23)$$

The values of R for double sampling plan are also listed in Table IV or V. We choose a value of R which is exactly equal to its desired value. Generally, the tabulated value of R is not equal to the desired value of R. In such situations, we take the value closest to the calculated value of R. Then we look up the corresponding values of c_1 , c_2 and n_1p from the appropriate table. There are two columns for n_1p , one for $\alpha = 0.05$ and another for $\beta = 0.10$.

The following two approaches are used for deciding the value of n_1 :

1. Satisfy Producer's Risk Stipulation Exactly and come close to Consumer's Risk

According to this approach, we look up the value of n_1p for $\alpha = 0.05$. Then we obtain the value of n_1 by dividing n_1p by $p = p_1 = \text{AQL}$ as follows:

$$n_1 = \frac{n_1p}{p_1} \quad \dots (24)$$

2. Satisfy Consumer's Risk Stipulation Exactly and come close to Producer's Risk

According to this approach, we look up the value of n_1p for $\beta = 0.10$. Then we obtain the value of n_1 by dividing n_1p by $p = p_2 = \text{LTPD}$ as follows:

$$n_1 = \frac{n_1p}{p_2} \quad \dots (25)$$

If the computed value of n_1 is a fraction, it is rounded off to the next integer.

We then obtain the value of n_2 by taking $n_1 = n_2$ or $n_1 = 2n_2$ as the case may be.

Let us consider an example to illustrate how to design a specific double sampling plan.

Example 7: Suppose a tyre supplier ships tyres in lots of size 500 to the buyer. The supplier and the quality control inspector of the buyer decide the acceptance quality level (AQL) to be 2% and the lot tolerance percent defective (LTPD) to be 8%. Design a double sampling plan which ensures that lots of quality 2% will be rejected about 5% of the time and lots of quality 8% will be accepted about 10% of the time. The sample sizes are equal.

Solution: It is given that

Product Control

$$AQL = p_1 = 2\% = 0.02 \text{ and } \alpha = 5\% = 0.05$$

$$LTPD = p_2 = 8\% = 0.08 \text{ and } \beta = 10\% = 0.10$$

$$n_1 = n_2$$

To design the desired sampling plan, we first calculate the operating ratio (R) from equation (23) as follows:

$$R = \frac{p_2}{p_1} = \frac{0.08}{0.02} = 4.0$$

Since sample sizes are equal, we use Table IV to look up the value of R. We see from Table IV that the desired value of $R = 4.0$ lies closest to 3.88. Then we look up the corresponding value of c_1 and c_2 . From Table IV, we have $c_1 = 2$ and $c_2 = 5$. We have to find the value of the first sample size (n_1).

We first find the plan which satisfies the desired producer's risk exactly.

For this we look up the value of $n_1 p$ corresponding to $c_1 = 2$, $c_2 = 5$ and $\alpha = 0.05$ in Table IV.

From Table IV, we have $n_1 p = 1.43$. Therefore, from equation (24), we have

$$n_1 = \frac{n_1 p}{p_1} = \frac{1.43}{0.02} = 71.5 \approx 72$$

Hence, the required double sampling plan is

$$n_1 = 72, n_2 = 72, c_1 = 2 \text{ and } c_2 = 5$$

We now find the plan which satisfies the desired consumer's risk exactly.

From Table IV, we have $n_1 p = 5.55$ corresponding to $c_1 = 2$, $c_2 = 5$ and $\beta = 0.10$. Therefore, from equation (25), we have

$$n_1 = \frac{n_1 p}{p_2} = \frac{5.55}{0.10} = 55.5 \approx 56$$

Hence, the required double sampling plan is

$$n_1 = 56, n_2 = 56, c_1 = 2 \text{ and } c_2 = 5$$

You may like to design a sampling plan yourself. Try the following exercise.

-
- E7)** A computer manufacturer purchases a computer part from a supplier in lots of 2000. The supplier and the quality control inspector of the company decide the acceptance quality level (AQL) to be 1.5% and the lot tolerance percent defective (LTPD) to be 10%. Assuming that the lot of the second sample is twice that of the first sample, design a double sampling plan which ensures that lots of quality 1.5% will be rejected about 5% of the time and lots of quality 10% will be accepted about 10% of the time.
-

We now end this unit by giving a summary of what we have covered in it.

8.9 SUMMARY

1. A sampling plan in which a decision about the acceptance or rejection of a lot is based on the inspection of two samples is known as a **double sampling plan**. There are four parameters of a double sampling plan:

- n_1 – size of the first sample,
 c_1 – acceptance number for the first sample,
 n_2 – size of the second sample, and
 c_2 – acceptance numbers for both samples combined.

2. The probability of accepting a lot of quality p under a double sampling plan is given by

$$P_a(p) = \sum_{x=0}^{c_1} {}^{n_1}C_x p^x (1-p)^{n_1-x} + \sum_{x=c_1+1}^{c_2} \sum_{y=0}^{c_2-x} {}^{n_1}C_x p^x (1-p)^{n_1-x} {}^{n_2}C_y p^y (1-p)^{n_2-y}$$

3. The AOQ for a double sampling plan is

$$AOQ = p \left[\left(1 - \frac{n_1}{N} \right) (P_{a1} + P_{a2}) - \frac{n_2}{N} P_{a2} \right]$$

4. The ASN for a double sampling plan is

$$ASN = n_1 P_1 + (n_1 + n_2)(1 - P_1)$$

where P_1 is the probability of making a decision about the acceptance or rejection of the lot on the first sample and is given by

$$P_1 = P[\text{lot is accepted on the first sample}] + P[\text{lot is rejected on the first sample}]$$

5. The ATI for a double sampling plan is

$$ATI = n_1 P_{a1} + (n_1 + n_2) P_{a2} + N [1 - P_a]$$

8.10 SOLUTIONS/ANSWERS

- E1)** For implementing the double sampling plan, the buyer randomly draws the first sample of 5 chips from the lot and classifies each chip of the first sample as defective or non-defective. At the end of the inspection, he/she counts the number of defective chips (d_1) found in the first sample and compares the number of defective chips (d_1) with the acceptance numbers c_1 and c_2 . If $d_1 \leq c_1 = 0$, he/she accepts the lot and if $d_1 > c_2 = 2$, he/she rejects the lot. If $c_1 < d_1 \leq c_2$, it means that if the number of defective chips is 1, he/she draws the second sample of size 20 from the lot. He/she counts the number of defective chips (d_2) found in the second sample and compares the total number of defective chips ($d_1 + d_2$) in both samples with the acceptance numbers c_2 . If $d_1 + d_2 \leq c_2 = 2$, he/she accepts the lot and if $d_1 + d_2 > c_2 = 2$, he/she rejects the lot.

- E2)** It is given that

$$N = 1000, n_1 = 5, c_1 = 0, n_2 = 20, c_2 = 2 \text{ and } p = 0.03$$

If $P_{a1}(p)$ and $P_{a2}(p)$ denote the probabilities of accepting the lot on the first sample and the second sample, respectively, we can calculate the probability of accepting the lot of quality level p in a double sampling plan as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) \quad \dots (i)$$

Product Control

If X represents the number of defective chips in the first sample of size n_1 , the lot is accepted on the first sample if $X \leq c_1 = 0$. Therefore,

$$P_{a1}(p) = P[X \leq c_1] = P[X \leq 0] \quad \dots (ii)$$

Since $N \geq 10n_1$, we can use the binomial distribution and calculate $P_{a1}(p)$ using Table I.

From Table I, for $n = n_1 = 5$, $x = c_1 = 0$ and $p = p_1 = 0.03$, we have

$$P_{a1}(p) = P[X \leq 0] = \sum_{x=0}^0 {}^{n_1}C_x p^x (1-p)^{n_1-x} = 0.8587 \quad \dots (iii)$$

Hence, the probability of accepting the lot on the first sample is 0.8587. It means that about 85.87% of the lot of quality $p = 0.03$ will be accepted on the first sample.

If Y represents the number of defective chips in the second sample, the lot is accepted on the second sample if $c_1 < X + Y \leq c_2$. It means that we shall accept the lot if

$$X = c_1 + 1 = 0 + 1 = 1 \text{ and } Y \leq c_2 - X = 2 - 1 = 1 \text{ or}$$

$$X = c_1 + 2 = 0 + 2 = 2 \text{ and } Y \leq c_2 - X = 2 - 2 = 0.$$

Therefore,

$$P_{a2}(p) = P[X = 1]P[Y \leq 1] + P[X = 2]P[Y \leq 0] \quad \dots (iv)$$

$$\text{We also know that } P[X = x] = P[X \leq x] - P[X \leq x - 1]$$

From Table I, for $n = n_1 = 5$ and $p = 0.03$, we have

$$P[X = 1] = P[X \leq 1] - P[X \leq 0] = 0.9915 - 0.8587 = 0.1328$$

$$P[X = 2] = P[X \leq 2] - P[X \leq 1] = 0.9997 - 0.9915 = 0.0082$$

From Table I, for $n = n_2 = 20$ and $p = 0.03$, we have

$$P[Y \leq 1] = 0.8802 \text{ and } P[Y \leq 0] = 0.5438$$

On putting these values in equation (iv), we get

$$\begin{aligned} P_{a2}(p) &= P[X = 1]P[Y \leq 1] + P[X = 2]P[Y \leq 0] \\ &= 0.1328 \times 0.8802 + 0.008 \times 0.5438 = 0.1213 \end{aligned}$$

Hence, the probability of accepting the lot on the second sample is 0.1213. It means that about 12.13% of the lot of quality $p = 0.03$ will be accepted on the second sample.

Thus, from equation (i), we get the probability of accepting a lot in a double sampling plan as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) = 0.8587 + 0.1213 = 0.9800$$

Thus, overall, 98% of the lots will be accepted by this sampling plan.

E3) It is given that

$$N = 1000, n_1 = 5, c_1 = 0, n_2 = 20, c_2 = 2,$$

$$AQL = p_1 = 0.05 \text{ and } LTPD = p_2 = 0.14$$

Double Sampling Plans

For calculating the producer's risk, we can use equation (11):

$$\begin{aligned} P_p &= P[\text{rejecting a lot of acceptance quality level } p_1] \\ &= 1 - P_a(p_1) \end{aligned}$$

Therefore, for calculating the producer's risk, we first calculate the probability of accepting the lot of quality $p = p_1 = AQL = 0.05$.

In a double sampling plan, the lot may be accepted either on the first sample or on the second sample. So if $P_{a1}(p)$ and $P_{a2}(p)$ denote the probabilities of accepting the lot on the first sample and the second sample, respectively, we can calculate the probability of acceptance of the lot of quality level p as follows:

$$P_a(p) = P_{a1}(p) + P_{a2}(p) \quad \dots (i)$$

If X represents the number of defective chips in the first sample, the lot will be accepted on the first sample if $X \leq c_1$, i.e., $X \leq 0$. Therefore,

$$P_{a1}(p) = P[X \leq c] = P[X \leq 0] \quad \dots (ii)$$

Since $N \geq 10n_1$, we can use the binomial distribution and calculate the probability using Table I.

From Table I, for $n = n_1 = 5$, $x = c_1 = 0$ and $p = p_1 = 0.05$, we have

$$P_{a1}(p_1) = P[X \leq 0] = \sum_{x=0}^0 {}^{n_1}C_x p_1^x (1-p_1)^{n_1-x} = 0.7738 \quad \dots (iii)$$

If Y represents the number of defective chips in the second sample, the lot will be accepted on the second sample if $c_1 < X + Y \leq c_2$. It means that we shall accept the lot if

$$X = c_1 + 1 = 0 + 1 = 1 \text{ and } Y \leq c_2 - X = 2 - 1 = 1$$

$$\text{or } X = c_1 + 2 = 0 + 2 = 2 \text{ and } Y \leq c_2 - X = 2 - 2 = 0.$$

Therefore,

$$P_{a2}(p_2) = P[X = 1]P[Y \leq 1] + P[X = 2]P[Y \leq 0] \quad \dots (iv)$$

We also know that $P[X = x] = P[X \leq x] - P[X \leq x - 1]$

From Table I, for $n = n_1 = 5$ and $p = p_1 = 0.05$, we have

$$P[X = 1] = P[X \leq 1] - P[X \leq 0] = 0.9774 - 0.7738 = 0.2036$$

$$P[X = 2] = P[X \leq 2] - P[X \leq 1] = 0.9988 - 0.9774 = 0.0214$$

From Table I, for $n = n_2 = 20$ and $p = p_1 = 0.05$, we have

$$P[Y \leq 0] = 0.3585 \text{ and } P[Y \leq 1] = 0.7358$$

On putting these values in equation (iv), we get

$$P_{a2}(p_2) = P[X = 1]P[Y \leq 1] + P[X = 2]P[Y \leq 0]$$

$$= 0.2036 \times 0.7358 + 0.0214 \times 0.3585 = 0.1575$$

Hence, from equation (i), we get the probability of accepting the lot of quality $p = p_1 = 0.05$ in this double sampling plan as follows:

$$P_a(p_1) = P_{a1}(p_1) + P_{a2}(p_1) = 0.7738 + 0.1575 = 0.9313$$

Therefore, we calculate the producer's risk for this plan using equation (11) as follows:

$$P_p = 1 - P_a(p_1) = 1 - 0.9313 = 0.0687$$

Similarly, we can calculate the consumer's risk as follows:

We can calculate the consumer's risk for the double sampling plan using equation (12) which is given by

$$P_c(p) = P[\text{accepting a lot of quality (LTPD)} p_2] = P_a(p_2) \quad \dots (v)$$

We first calculate the probability of accepting the lot of quality $p = p_1 = \text{LTPD} = 0.14$ using equations (i), (ii) and (iv).

From Table I, for $n = n_1 = 5$, $x = c_1 = 0$ and $p = p_2 = 0.14$, we have

$$P_{a1}(p_2) = P[X \leq 0] = \sum_{x=0}^0 {}^{n_1}C_x p_2^x (1-p_2)^{n_1-x} = 0.4704 \quad \dots (vi)$$

From Table I, for $n = n_1 = 5$ and $p = p_2 = 0.14$, we have

$$P[X = 1] = P[X \leq 1] - P[X \leq 0] = 0.8533 - 0.4704 = 0.3829$$

$$P[X = 2] = P[X \leq 2] - P[X \leq 1] = 0.9780 - 0.8533 = 0.1247$$

From Table I, for $n = n_2 = 20$ and $p = p_2 = 0.14$, we have

$$P[Y \leq 0] = 0.0490 \text{ and } P[Y \leq 1] = 0.2084$$

On putting these values in equation (iv), we get

$$\begin{aligned} P_{a2}(p_2) &= P[X = 1]P[Y \leq 1] + P[X = 2]P[Y \leq 0] \\ &= 0.3829 \times 0.2084 + 0.1247 \times 0.0490 = 0.0859 \quad \dots (vii) \end{aligned}$$

On putting the values of $P_{a1}(p_2)$ and $P_{a2}(p_2)$ in equation (i), we get

$$P_a(p_2) = 0.4704 + 0.0859 = 0.5563$$

Hence, from equation (v), we get, the consumer's risk for this plan as follows:

$$P_c = P_a(p_2) = 0.5563$$

E4) It is given that

$$N = 1000, n_1 = 5, c_1 = 0, n_2 = 20, c_2 = 2 \text{ and } p = 0.03$$

It is noted that the sample sizes $n_1 = 5$ and $n_2 = 20$ are very small in proportion to the lot size $N = 1000$, i.e., $n_1 / N \approx 0$ and $n_2 / N \approx 0$. So we can calculate the average outgoing quality (AOQ) for the double sampling plan using equation (19).

For calculating AOQ, we have to calculate the probabilities of accepting the lot on the first sample and the second sample corresponding to $p = 0.03$.

We have already calculated these probabilities in E2. So we directly use the results:

$$P_{a1}(p) = 0.8587 \text{ and } P_{a2}(p) = 0.1213$$

Substituting the values of P_{a1} , P_{a2} and p in equation (19), we get

$$AOQ = p(P_{a1} + P_{a2}) = 0.03 \times (0.8587 + 0.1213) = 0.0294$$

E5) It is given that

$$N = 1000, n_1 = 5, c_1 = 0, n_2 = 20, c_2 = 2 \text{ and } p = 0.03$$

For computing ASN for the double sampling plan, we can use equation (20):

$$ASN = n_1 P_1 + (n_1 + n_2)(1 - P_1)$$

where P_1 is the probability of making a decision about the acceptance or rejection of the lot on the first sample and can be calculated as follows:

$$P_1 = P[\text{lot is accepted on the first sample}] + P[\text{lot is rejected on the first sample}] \quad \dots (i)$$

So we have to find the probability P_1 first. In the double sampling plan, the lot is accepted on the first sample if $X \leq c_1$, i.e., if $X \leq 0$. Therefore,

$$P[\text{lot is accepted on the first sample}] = P[X \leq c_1] = P[X \leq 0] = P_{a1}$$

We have already calculated this probability in E2. So we directly use the results. Therefore,

$$P_{a1}(p) = P[X \leq 0] = 0.8587 \quad \dots (ii)$$

The lot will be rejected on the first sample if $X > c_2$, i.e., if $X > 2$.

Therefore,

$$P[\text{lot is rejected on the first sample}] = P[X > 2] = 1 - P[X \leq 2] \quad \dots (iii)$$

From Table I, for $n = n_1 = 5$, $x = c_2 = 2$ and $p = 0.03$, we have

$$P[X \leq 2] = \sum_{x=0}^2 {}^{n_1}C_x p^x (1-p)^{n_1-x} = 0.9997$$

Therefore, from equation (iii), we have

$$\begin{aligned} P[\text{lot is rejected on the first sample}] &= 1 - P[X \leq 2] \\ &= 1 - 0.9997 = 0.0003 \quad \dots (iv) \end{aligned}$$

Thus, on putting the values from equations (ii) and (iv) in equation (i), we get

$$P_1 = 0.8587 + 0.0003 = 0.8590$$

Hence, on putting the values of n_1 , n_2 , and P_1 in equation (20), we get the ASN for the plan as follows:

$$\begin{aligned} \text{ASN} &= n_1 P_1 + (n_1 + n_2)(1 - P_1) \\ &= 5 \times 0.8590 + (5 + 20)(1 - 0.8590) = 7.82 \approx 8 \end{aligned}$$

E6) It is given that

$$N = 1000, n_1 = 5, c_1 = 0, n_2 = 20, c_2 = 2 \text{ and } p = 0.03$$

We can calculate the ATI for the double sampling plan using equation (22).

For calculating ATI, we have to calculate the probabilities of accepting the lot on the first sample and the second sample corresponding to $p = 0.03$.

We have already calculated these probabilities in E2. So we directly use the results:

$$P_{a1}(p) = 0.8587, P_{a2}(p) = 0.1213 \text{ and } P_a(p) = 0.9800$$

Substituting the values of N , n_1 , n_2 , P_{a1} and P_{a2} in equation (22), we get

$$\begin{aligned} \text{ATI} &= n_1 P_{a1}(p) + (n_1 + n_2) P_{a2}(p) + N[1 - P_a(p)] \\ &= 5 \times 0.8587 + (5 + 20) \times 0.1213 + 1000 \times (1 - 0.9800) \\ &= 4.2935 + 3.0325 + 20 = 27.3260 \approx 28 \end{aligned}$$

E7) It is given that

$$\text{AQL} = p_1 = 1.5\% = 0.015 \text{ and } \alpha = 5\% = 0.05$$

$$\text{LTPD} = p_2 = 10\% = 0.10 \text{ and } \beta = 10\% = 0.10$$

$$n_2 = 2n_1$$

To design the desired sampling plan, we first calculate the operating ratio (R) from equation (23) as follows:

$$R = \frac{p_2}{p_1} = \frac{0.10}{0.015} = 6.67$$

Since $n_2 = 2n_1$, we use Table V to look up the value of R. We see from Table V that the desired value of $R = 6.67$ lies closest to 6.48. Then we look up the corresponding values of c_1 and c_2 . From Table V, we have $c_1 = 1$ and $c_2 = 3$. We have to find the value of the first sample size (n_1).

We first find the plan which satisfies the desired producer's risk exactly.

For this we look up the value of $n_1 p$ corresponding $c_1 = 1$, $c_2 = 3$ and $\alpha = 0.05$ in Table V.

From Table V, we have $n_1 p = 0.60$. Therefore, from equation (24), we have

$$n_1 = \frac{n_1 p}{p_1} = \frac{0.60}{0.015} = 40$$

Hence, the required double sampling plan is

$$n_1 = 40, n_2 = 80, c_1 = 1 \text{ and } c_2 = 3$$

We now find the plan which satisfies the desired consumer's risk exactly.

From Table V, we have $n_1 p = 3.89$ corresponding to $c_1 = 1$, $c_2 = 3$ and $\beta = 0.10$. Therefore, from equation (25), we have

$$n_1 = \frac{n_1 p}{p_2} = \frac{3.89}{0.10} = 38.9 \approx 39$$

Hence, the required double sampling plan is

$$n_1 = 39, n_2 = 78, c_1 = 1 \text{ and } c_2 = 3$$

n	x	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09	0.10	0.11	0.12	0.14	0.16	0.18	0.20
1	0	0.9900	0.9800	0.9700	0.9600	0.9500	0.9400	0.9300	0.9200	0.9100	0.9000	0.8900	0.8800	0.8600	0.8400	0.8200	0.8000
	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	0	0.9801	0.9604	0.9409	0.9216	0.9025	0.8836	0.8649	0.8464	0.8281	0.8100	0.7921	0.7744	0.7396	0.7056	0.6724	0.6400
	1	0.9999	0.9996	0.9991	0.9984	0.9975	0.9964	0.9951	0.9936	0.9919	0.9900	0.9879	0.9856	0.9804	0.9744	0.9676	0.9600
	2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	0	0.9703	0.9412	0.9127	0.8847	0.8574	0.8306	0.8044	0.7787	0.7536	0.7290	0.7050	0.6815	0.6361	0.5927	0.5514	0.5120
	1	0.9997	0.9988	0.9974	0.9953	0.9928	0.9896	0.9860	0.9818	0.9772	0.9720	0.9664	0.9603	0.9467	0.9314	0.9145	0.8960
	2	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9995	0.9993	0.9990	0.9987	0.9983	0.9973	0.9959	0.9942	0.9920
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
4	0	0.9606	0.9224	0.8853	0.8493	0.8145	0.7807	0.7481	0.7164	0.6857	0.6561	0.6274	0.5997	0.5470	0.4979	0.4521	0.4096
	1	0.9994	0.9977	0.9948	0.9909	0.9860	0.9801	0.9733	0.9656	0.9570	0.9477	0.9376	0.9268	0.9032	0.8772	0.8491	0.8192
	2	1.0000	1.0000	0.9999	0.9998	0.9995	0.9992	0.9987	0.9981	0.9973	0.9963	0.9951	0.9937	0.9902	0.9856	0.9798	0.9728
	3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9998	0.9996	0.9993	0.9990	0.9984
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
5	0	0.9510	0.9039	0.8587	0.8154	0.7738	0.7339	0.6957	0.6591	0.6240	0.5905	0.5584	0.5277	0.4704	0.4182	0.3707	0.3277
	1	0.9990	0.9962	0.9915	0.9852	0.9774	0.9681	0.9575	0.9456	0.9326	0.9185	0.9035	0.8875	0.8533	0.8165	0.7776	0.7373
	2	1.0000	0.9999	0.9997	0.9994	0.9988	0.9980	0.9969	0.9955	0.9937	0.9914	0.9888	0.9857	0.9780	0.9682	0.9563	0.9421
	3	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9995	0.9993	0.9991	0.9983	0.9971	0.9955	0.9933
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
6	0	0.9415	0.8858	0.8330	0.7828	0.7351	0										

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7	0	0.9321	0.8681	0.8080	0.7514	0.6983	0.6485	0.6017	0.5578	0.5168	0.4783	0.4423	0.4087	0.3479	0.2951	0.2493	0.2097
	1	0.9980	0.9921	0.9829	0.9706	0.9556	0.9382	0.9187	0.8974	0.8745	0.8503	0.8250	0.7988	0.7444	0.6885	0.6323	0.5767
	2	1.0000	0.9997	0.9991	0.9980	0.9962	0.9937	0.9903	0.9860	0.9807	0.9743	0.9669	0.9584	0.9380	0.9134	0.8846	0.8520
	3	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9993	0.9988	0.9982	0.9973	0.9961	0.9946	0.9906	0.9847	0.9769	0.9667
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9996	0.9991	0.9983	0.9971	0.9953
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
8	0	0.9227	0.8508	0.7837	0.7214	0.6634	0.6096	0.5596	0.5132	0.4703	0.4305	0.3937	0.3596	0.2992	0.2479	0.2044	0.1678
	1	0.9973	0.9897	0.9777	0.9619	0.9428	0.9208	0.8965	0.8702	0.8423	0.8131	0.7829	0.7520	0.6889	0.6256	0.5634	0.5033
	2	0.9999	0.9996	0.9987	0.9969	0.9942	0.9904	0.9853	0.9789	0.9711	0.9619	0.9513	0.9392	0.9109	0.8774	0.8392	0.7969
	3	1.0000	1.0000	0.9999	0.9998	0.9996	0.9993	0.9987	0.9978	0.9966	0.9950	0.9929	0.9903	0.9832	0.9733	0.9603	0.9437
	4	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9996	0.9993	0.9990	0.9979	0.9962	0.9935	0.9896
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9993	0.9988
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
9	0	0.9135	0.8337	0.7602	0.6925	0.6302	0.5730	0.5204	0.4722	0.4279	0.3874	0.3504	0.3165	0.2573	0.2082	0.1676	0.1342
	1	0.9966	0.9869	0.9718	0.9522	0.9288	0.9022	0.8729	0.8417	0.8088	0.7748	0.7401	0.7049	0.6343	0.5652	0.4988	0.4362
	2	0.9999	0.9994	0.9980	0.9955	0.9916	0.9862	0.9791	0.9702	0.9595	0.9470	0.9328	0.9167	0.8798	0.8371	0.7895	0.7382
	3	1.0000	1.0000	0.9999	0.9997	0.9994	0.9987	0.9977	0.9963	0.9943	0.9917	0.9883	0.9842	0.9731	0.9580	0.9385	0.9144
	4	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9995	0.9991	0.9986	0.9979	0.9959	0.9925	0.9875	0.9804
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9996	0.9991	0.9983	0.9969
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
10	0	0.9044	0.8171	0.7374	0.6648	0.5987	0.5386	0.4840	0.4344	0.3894	0.3487	0.3118	0.2785	0.2213	0.1749	0.1374	0.1074
	1	0.9957	0.9838	0.9655	0.9418	0.9139	0.8824	0.8483	0.8121	0.7746	0.7361	0.6972	0.6583	0.5816	0.5080	0.4392	0.3758
	2	0.9999	0.9991	0.9972	0.9938	0.9885	0.9812	0.9717	0.9599	0.9460	0.9298	0.9116	0.8913	0.8455	0.7936	0.7372	0.6778
	3	1.0000	1.0000	0.9999	0.9996	0.9990	0.9980	0.9964	0.9942	0.9912	0.9872	0.9822	0.9761	0.9600	0.9386	0.9117	0.8791
	4	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994	0.9990	0.9984	0.9975	0.9963	0.9927	0.9870	0.9787	0.9672

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	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9996	0.9990	0.9980	0.9963	0.9936
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
11	0	0.8953	0.8007	0.7153	0.6382	0.5688	0.5063	0.4501	0.3996	0.3544	0.3138	0.2775	0.2451	0.1903	0.1469	0.1127	0.0859
	1	0.9948	0.9805	0.9587	0.9308	0.8981	0.8618	0.8228	0.7819	0.7399	0.6974	0.6548	0.6127	0.5311	0.4547	0.3849	0.3221
	2	0.9998	0.9988	0.9963	0.9917	0.9848	0.9752	0.9630	0.9481	0.9305	0.9104	0.8880	0.8634	0.8085	0.7479	0.6836	0.6174
	3	1.0000	1.0000	0.9998	0.9993	0.9984	0.9970	0.9947	0.9915	0.9871	0.9815	0.9744	0.9659	0.9440	0.9154	0.8803	0.8389
	4	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9995	0.9990	0.9983	0.9972	0.9958	0.9939	0.9881	0.9793	0.9666	0.9496
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9995	0.9992	0.9982	0.9963	0.9932	0.9883
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9990	0.9980
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
12	0	0.8864	0.7847	0.6938	0.6127	0.5404	0.4759	0.4186	0.3677	0.3225	0.2824	0.2470	0.2157	0.1637	0.1234	0.0924	0.0687
	1	0.9938	0.9769	0.9514	0.9191	0.8816	0.8405	0.7967	0.7513	0.7052	0.6590	0.6133	0.5686	0.4834	0.4055	0.3359	0.2749
	2	0.9998	0.9985	0.9952	0.9893	0.9804	0.9684	0.9532	0.9348	0.9134	0.8891	0.8623	0.8333	0.7697	0.7010	0.6298	0.5583
	3	1.0000	0.9999	0.9997	0.9990	0.9978	0.9957	0.9925	0.9880	0.9820	0.9744	0.9649	0.9536	0.9250	0.8886	0.8448	0.7946
	4	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991	0.9984	0.9973	0.9957	0.9935	0.9905	0.9819	0.9690	0.9511	0.9274
	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9995	0.9991	0.9986	0.9967	0.9935	0.9884	0.9806
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9996	0.9990	0.9979	0.9961
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
13	0	0.8775	0.7690	0.6730	0.5882	0.5133	0.4474	0.3893	0.3383	0.2935	0.2542	0.2198	0.1898	0.1408	0.1037	0.0758	0.0550

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	1	0.9928	0.9730	0.9436	0.9068	0.8646	0.8186	0.7702	0.7206	0.6707	0.6213	0.5730	0.5262	0.4386	0.3604	0.2920	0.2336
	2	0.9997	0.9980	0.9938	0.9865	0.9755	0.9608	0.9422	0.9201	0.8946	0.8661	0.8349	0.8015	0.7296	0.6537	0.5769	0.5017
	3	1.0000	0.9999	0.9995	0.9986	0.9969	0.9940	0.9897	0.9837	0.9758	0.9658	0.9536	0.9391	0.9033	0.8586	0.8061	0.7473
	4	1.0000	1.0000	1.0000	0.9999	0.9997	0.9993	0.9987	0.9976	0.9959	0.9935	0.9903	0.9861	0.9740	0.9562	0.9319	0.9009
	5	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9995	0.9991	0.9985	0.9976	0.9947	0.9896	0.9817	0.9700
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9992	0.9981	0.9962	0.9930
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9988
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
14	0	0.8687	0.7536	0.6528	0.5647	0.4877	0.4205	0.3620	0.3112	0.2670	0.2288	0.1956	0.1670	0.1211	0.0871	0.0621	0.0440
	1	0.9916	0.9690	0.9355	0.8941	0.8470	0.7963	0.7436	0.6900	0.6368	0.5846	0.5342	0.4859	0.3969	0.3193	0.2531	0.1979
	2	0.9997	0.9975	0.9923	0.9833	0.9699	0.9522	0.9302	0.9042	0.8745	0.8416	0.8061	0.7685	0.6889	0.6068	0.5256	0.4481
	3	1.0000	0.9999	0.9994	0.9981	0.9958	0.9920	0.9864	0.9786	0.9685	0.9559	0.9406	0.9226	0.8790	0.8258	0.7649	0.6982
	4	1.0000	1.0000	1.0000	0.9998	0.9996	0.9990	0.9980	0.9965	0.9941	0.9908	0.9863	0.9804	0.9641	0.9406	0.9093	0.8702
	5	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9992	0.9985	0.9976	0.9962	0.9918	0.9843	0.9727	0.9561
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994	0.9985	0.9968	0.9936	0.9884
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9988	0.9976
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
15	0	0.8601	0.7386	0.6333	0.5421	0.4633	0.3953	0.3367	0.2863	0.2430	0.2059	0.1741	0.1470	0.1041	0.0731	0.0510	0.0352
	1	0.9904	0.9647	0.9270	0.8809	0.8290	0.7738	0.7168	0.6597	0.6035	0.5490	0.4969	0.4476	0.3583	0.2821	0.2187	0.1671
	2	0.9996	0.9970	0.9906	0.9797	0.9638	0.9429	0.9171	0.8870	0.8531	0.8159	0.7762	0.7346	0.6480	0.5608	0.4766	0.3980
	3	1.0000	0.9998	0.9992	0.9976	0.9945	0.9896	0.9825	0.9727	0.9601	0.9444	0.9258	0.9041	0.8524	0.7908	0.7218	0.6482

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	4	1.0000	1.0000	0.9999	0.9998	0.9994	0.9986	0.9972	0.9950	0.9918	0.9873	0.9813	0.9735	0.9522	0.9222	0.8833	0.8358
	5	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9993	0.9987	0.9978	0.9963	0.9943	0.9879	0.9773	0.9613	0.9389
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994	0.9990	0.9976	0.9948	0.9898	0.9819
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9996	0.9990	0.9979	0.9958
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
16	0	0.8515	0.7238	0.6143	0.5204	0.4401	0.3716	0.3131	0.2634	0.2211	0.1853	0.1550	0.1293	0.0895	0.0614	0.0418	0.0281
	1	0.9891	0.9601	0.9182	0.8673	0.8108	0.7511	0.6902	0.6299	0.5711	0.5147	0.4614	0.4115	0.3227	0.2487	0.1885	0.1407
	2	0.9995	0.9963	0.9887	0.9758	0.9571	0.9327	0.9031	0.8689	0.8306	0.7892	0.7455	0.7001	0.6074	0.5162	0.4302	0.3518
	3	1.0000	0.9998	0.9989	0.9968	0.9930	0.9868	0.9779	0.9658	0.9504	0.9316	0.9093	0.8838	0.8237	0.7540	0.6777	0.5981
	4	1.0000	1.0000	0.9999	0.9997	0.9991	0.9981	0.9962	0.9932	0.9889	0.9830	0.9752	0.9652	0.9382	0.9012	0.8542	0.7982
	5	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9990	0.9981	0.9967	0.9947	0.9918	0.9829	0.9685	0.9473	0.9183
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9995	0.9991	0.9985	0.9962	0.9920	0.9847	0.9733
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9993	0.9984	0.9964	0.9930
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9993	0.9985
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
17	0	0.8429	0.7093	0.5958	0.4996	0.4181	0.3493	0.2912	0.2423	0.2012	0.1668	0.1379	0.1138	0.0770	0.0516	0.0343	0.0225
	1	0.9877	0.9554	0.9091	0.8535	0.7922	0.7283	0.6638	0.6005	0.5396	0.4818	0.4277	0.3777	0.2901	0.2187	0.1621	0.1182
	2	0.9994	0.9956	0.9866	0.9714	0.9497	0.9218	0.8882	0.8497	0.8073	0.7618	0.7142	0.6655	0.5676	0.4734	0.3867	0.3096

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	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
19	0	0.8262	0.6812	0.5606	0.4604	0.3774	0.3086	0.2519	0.2051	0.1666	0.1351	0.1092	0.0881	0.0569	0.0364	0.0230	0.0144
	1	0.9847	0.9454	0.8900	0.8249	0.7547	0.6829	0.6121	0.5440	0.4798	0.4203	0.3658	0.3165	0.2331	0.1682	0.1191	0.0829
	2	0.9991	0.9939	0.9817	0.9616	0.9335	0.8979	0.8561	0.8092	0.7585	0.7054	0.6512	0.5968	0.4911	0.3941	0.3090	0.2369
	3	1.0000	0.9995	0.9978	0.9939	0.9868	0.9757	0.9602	0.9398	0.9147	0.8850	0.8510	0.8133	0.7292	0.6380	0.5451	0.4551
	4	1.0000	1.0000	0.9998	0.9993	0.9980	0.9956	0.9915	0.9853	0.9765	0.9648	0.9498	0.9315	0.8842	0.8238	0.7524	0.6733
	5	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9986	0.9971	0.9949	0.9914	0.9865	0.9798	0.9599	0.9300	0.8890	0.8369
	6	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991	0.9983	0.9970	0.9952	0.9887	0.9772	0.9589	0.9324
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9995	0.9991	0.9974	0.9939	0.9874	0.9767
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9986	0.9968	0.9933
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9984
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
20	0	0.8179	0.6676	0.5438	0.4420	0.3585	0.2901	0.2342	0.1887	0.1516	0.1216	0.0972	0.0776	0.0490	0.0306	0.0189	0.0115
	1	0.9831	0.9401	0.8802	0.8103	0.7358	0.6605	0.5869	0.5169	0.4516	0.3917	0.3376	0.2891	0.2084	0.1471	0.1018	0.0692
	2	0.9990	0.9929	0.9790	0.9561	0.9245	0.8850	0.8390	0.7879	0.7334	0.6769	0.6198	0.5631	0.4550	0.3580	0.2748	0.2061
	3	1.0000	0.9994	0.9973	0.9926	0.9841	0.9710	0.9529	0.9294	0.9007	0.8670	0.8290	0.7873	0.6959	0.5990	0.5026	0.4114
	4	1.0000	1.0000	0.9997	0.9990	0.9974	0.9944	0.9893	0.9817	0.9710	0.9568	0.9390	0.9173	0.8625	0.7941	0.7151	0.6296
	5	1.0000	1.0000	1.0000	0.9999	0.9997	0.9991	0.9981	0.9962	0.9932	0.9887	0.9825	0.9740	0.9493	0.9130	0.8644	0.8042
	6	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994	0.9987	0.9976	0.9959	0.9933	0.9847	0.9696	0.9463	0.9133
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9992	0.9986	0.9962	0.9912	0.9823	0.9679
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9992	0.9979	0.9951	0.9900
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9989	0.9974

APPENDIX

[illegible]

APPENDIX

	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
22	0	0.8016	0.6412	0.5117	0.4073	0.3235	0.2563	0.2026	0.1597	0.1256	0.0985	0.0770	0.0601	0.0362	0.0216	0.0127	0.0074
	1	0.9798	0.9290	0.8598	0.7808	0.6982	0.6163	0.5381	0.4652	0.3988	0.3392	0.2864	0.2403	0.1659	0.1120	0.0740	0.0480
	2	0.9987	0.9907	0.9728	0.9441	0.9052	0.8576	0.8032	0.7442	0.6826	0.6200	0.5582	0.4983	0.3877	0.2929	0.2154	0.1545
	3	0.9999	0.9991	0.9962	0.9895	0.9778	0.9602	0.9362	0.9059	0.8696	0.8281	0.7821	0.7328	0.6283	0.5226	0.4224	0.3320
	4	1.0000	0.9999	0.9996	0.9985	0.9960	0.9913	0.9838	0.9727	0.9575	0.9379	0.9136	0.8847	0.8144	0.7305	0.6381	0.5429
	5	1.0000	1.0000	1.0000	0.9998	0.9994	0.9985	0.9967	0.9936	0.9888	0.9818	0.9721	0.9593	0.9235	0.8730	0.8086	0.7326
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9988	0.9976	0.9956	0.9926	0.9881	0.9738	0.9499	0.9147	0.8670
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991	0.9984	0.9971	0.9925	0.9834	0.9679	0.9439
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9994	0.9982	0.9954	0.9898	0.9799
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9989	0.9972	0.9939
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9984
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
23	0	0.7936	0.6283	0.4963	0.3911	0.3074	0.2410	0.1884	0.1469	0.1143	0.0886	0.0685	0.0529	0.0312	0.0181	0.0104	0.0059
	1	0.9780	0.9233	0.8493	0.7658	0.6794	0.5947	0.5146	0.4408	0.3742	0.3151	0.2634	0.2186	0.1478	0.0976	0.0630	0.0398
	2	0.9985	0.9895	0.9695	0.9376	0.8948	0.8431	0.7846	0.7219	0.6570	0.5920	0.5283	0.4673	0.3566	0.2640	0.1900	0.1332
	3	0.9999	0.9990	0.9955	0.9877	0.9742	0.9541	0.9269	0.8930	0.8528	0.8073	0.7575	0.7047	0.5946	0.4859	0.3851	0.2965
	4	1.0000	0.9999	0.9995	0.9981	0.9951	0.9895	0.9805	0.9674	0.9496	0.9269	0.8991	0.8665	0.7884	0.6972	0.5993	0.5007
	5	1.0000	1.0000	1.0000	0.9998	0.9992	0.9981	0.9958	0.9920	0.9860	0.9774	0.9656	0.9504	0.9082	0.8502	0.7779	0.6947
	6	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9993	0.9984	0.9968	0.9942	0.9903	0.9847	0.9667	0.9376	0.8956	0.8402
	7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994	0.9988	0.9977	0.9960	0.9899	0.9780	0.9583	0.9285

	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9991	0.9974	0.9934	0.9858	0.9727
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9983	0.9959	0.9911
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9990	0.9975
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
24	0	0.7857	0.6158	0.4814	0.3754	0.2920	0.2265	0.1752	0.1352	0.1040	0.0798	0.0610	0.0465	0.0268	0.0152	0.0085	0.0047
	1	0.9761	0.9174	0.8388	0.7508	0.6608	0.5735	0.4918	0.4173	0.3508	0.2925	0.2420	0.1987	0.1315	0.0849	0.0535	0.0331
	2	0.9983	0.9882	0.9659	0.9307	0.8841	0.8282	0.7657	0.6994	0.6316	0.5643	0.4992	0.4375	0.3274	0.2374	0.1671	0.1145
	3	0.9999	0.9988	0.9947	0.9857	0.9702	0.9474	0.9170	0.8793	0.8352	0.7857	0.7323	0.6762	0.5613	0.4504	0.3500	0.2639
	4	1.0000	0.9999	0.9994	0.9977	0.9940	0.9873	0.9767	0.9614	0.9409	0.9149	0.8835	0.8471	0.7612	0.6634	0.5607	0.4599
	5	1.0000	1.0000	0.9999	0.9997	0.9990	0.9975	0.9947</									

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APPENDIX

	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
26	0	0.7700	0.5914	0.4530	0.3460	0.2635	0.2001	0.1516	0.1144	0.0861	0.0646	0.0483	0.0360	0.0198	0.0107	0.0057	0.0030
	1	0.9723	0.9052	0.8172	0.7208	0.6241	0.5323	0.4481	0.3731	0.3076	0.2513	0.2036	0.1637	0.1037	0.0640	0.0385	0.0227
	2	0.9978	0.9852	0.9580	0.9160	0.8614	0.7973	0.7272	0.6543	0.5813	0.5105	0.4435	0.3814	0.2743	0.1907	0.1285	0.0841
	3	0.9999	0.9983	0.9929	0.9811	0.9613	0.9326	0.8952	0.8499	0.7979	0.7409	0.6807	0.6189	0.4966	0.3838	0.2864	0.2068
	4	1.0000	0.9999	0.9991	0.9967	0.9915	0.9823	0.9679	0.9477	0.9211	0.8882	0.8493	0.8051	0.7046	0.5952	0.4858	0.3833
	5	1.0000	1.0000	0.9999	0.9995	0.9985	0.9962	0.9920	0.9851	0.9747	0.9601	0.9410	0.9168	0.8536	0.7725	0.6783	0.5775
	6	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9983	0.9965	0.9932	0.9881	0.9806	0.9701	0.9385	0.8906	0.8263	0.7474
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9993	0.9985	0.9970	0.9946	0.9909	0.9780	0.9550	0.9190	0.8687
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9994	0.9987	0.9976	0.9932	0.9840	0.9674	0.9408
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9995	0.9982	0.9951	0.9886	0.9768
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9987	0.9966	0.9921
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9991	0.9977
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
27	0	0.7623	0.5796	0.4394	0.3321	0.2503	0.1881	0.1409	0.1053	0.0784	0.0581	0.0430	0.0317	0.0170	0.0090	0.0047	0.0024
	1	0.9703	0.8989	0.8063	0.7058	0.6061	0.5123	0.4274	0.3524	0.2876	0.2326	0.1865	0.1484	0.0919	0.0555	0.0326	0.0187

APPENDIX

	2	0.9976	0.9836	0.9538	0.9082	0.8495	0.7814	0.7076	0.6318	0.5567	0.4846	0.4171	0.3553	0.2504	0.1704	0.1123	0.0718
	3	0.9999	0.9981	0.9918	0.9785	0.9563	0.9245	0.8834	0.8342	0.7784	0.7179	0.6546	0.5904	0.4654	0.3529	0.2580	0.1823
	4	1.0000	0.9998	0.9989	0.9960	0.9900	0.9793	0.9628	0.9398	0.9100	0.8734	0.8307	0.7827	0.6755	0.5614	0.4499	0.3480
	5	1.0000	1.0000	0.9999	0.9994	0.9981	0.9954	0.9903	0.9821	0.9699	0.9529	0.9309	0.9034	0.8327	0.7441	0.6437	0.5387
	6	1.0000	1.0000	1.0000	0.9999	0.9997	0.9991	0.9979	0.9956	0.9916	0.9853	0.9762	0.9637	0.9266	0.8717	0.7996	0.7134
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9991	0.9980	0.9961	0.9931	0.9884	0.9725	0.9447	0.9023	0.8444
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9991	0.9983	0.9968	0.9911	0.9794	0.9587	0.9263
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9993	0.9975	0.9933	0.9848	0.9696
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994	0.9981	0.9951	0.9890
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9995	0.9986	0.9965
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9990
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	27	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
28	0	0.7547	0.5680	0.4262	0.3189	0.2378	0.1768	0.1311	0.0968	0.0713	0.0523	0.0383	0.0279	0.0147	0.0076	0.0039	0.0019
	1	0.9682	0.8925	0.7953	0.6909	0.5883	0.4929	0.4073	0.3326	0.2688	0.2152	0.1707	0.1344	0.0814	0.0480	0.0276	0.0155
	2	0.9973	0.9820	0.9494	0.9001	0.8373	0.7652	0.6880	0.6094	0.5325	0.4594	0.3917	0.3305	0.2282	0.1520	0.0979	0.0612
	3	0.9998	0.9978	0.9907	0.9757	0.9509	0.9159	0.8711	0.8180	0.7585	0.6946	0.6285	0.5622	0.4353	0.3237	0.2318	0.1602
	4	1.0000	0.9998	0.9987	0.9953	0.9883	0.9760	0.9573	0.9314	0.8982	0.8579	0.8114	0.7597	0.6461	0.5280	0.4153	0.3149
	5	1.0000	1.0000	0.9998	0.9993	0.9977	0.9944	0.9884	0.9787	0.9645	0.9450	0.9199	0.8889	0.8107	0.7149	0.6088	0.5005

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	6	1.0000	1.0000	1.0000	0.9999	0.9996	0.9989	0.9974	0.9945	0.9896	0.9821	0.9713	0.9565	0.9135	0.8513	0.7716	0.6784
	7	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9995	0.9988	0.9974	0.9950	0.9912	0.9855	0.9660	0.9330	0.8839	0.8182
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9988	0.9977	0.9958	0.9885	0.9738	0.9486	0.9100
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9990	0.9966	0.9911	0.9801	0.9609
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9991	0.9974	0.9933	0.9851
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9993	0.9980	0.9950
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9998	0.9995	0.9985
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	27	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	28	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
29	0	0.7472	0.5566	0.4134	0.3061	0.2259	0.1662	0.1219	0.0891	0.0649	0.0471	0.0341	0.0245	0.0126	0.0064	0.0032	0.0015
	1	0.9660	0.8860	0.7842	0.6760	0.5708	0.4739	0.3880	0.3138	0.2510	0.1989	0.1562	0.1216	0.0721	0.0416	0.0233	0.0128
	2	0.9970	0.9802	0.9447	0.8917	0.8249	0.7489	0.6684	0.5873	0.5087	0.4350	0.3674	0.3069	0.2077	0.1354	0.0853	0.0520
	3	0.9998	0.9975	0.9894	0.9726	0.9452	0.9069	0.8583	0.8013	0.7381	0.6710	0.6024	0.5344	0.4064	0.2962	0.2077	0.1404
	4	1.0000	0.9997	0.9984	0.9946	0.9864	0.9724	0.9512	0.9223	0.8856	0.8416	0.7912	0.7360	0.6166	0.4953	0.3823	0.2839
	5	1.0000	1.0000	0.9998	0.9991	0.9973	0.9933	0.9862	0.9749	0.9585	0.9363	0.9079	0.8734	0.7877	0.6850	0.5740	0.4634
	6	1.0000	1.0000	1.0000	0.9999	0.9995	0.9986	0.9967	0.9932	0.9873	0.9784	0.9656	0.9484	0.8991	0.8295	0.7423	0.6429
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9993	0.9984	0.9967	0.9938	0.9890	0.9820	0.9587	0.9199	0.8636	0.7903
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9993	0.9984	0.9970	0.9946	0.9854	0.9673	0.9369	0.8916

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	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9993	0.9986	0.9955	0.9883	0.9744	0.9507
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9997	0.9988	0.9964	0.9909	0.9803
	11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9990	0.9972	0.9931
	12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9992	0.9978
	13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9994
	14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
	15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	25	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	26	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	27	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	28	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	29	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
30	0	0.7397	0.5455	0.4010	0.2939	0.2146	0.1563	0.1134	0.0820	0.0591	0.0424	0.0303	0.0216	0.0108	0.0054	0.0026	0.0012
	1	0.9639	0.8795	0.7731	0.6612	0.5535	0.4555	0.3694	0.2958	0.2343	0.1837	0.1427	0.1100	0.0638	0.0359	0.0197	0.0105
	2	0.9967	0.9783	0.9399	0.8831	0.8122	0.7324	0.6487	0.5654	0.4855	0.4114	0.3442	0.2847	0.1887	0.1204	0.0741	0.0442
	3	0.9998	0.9971	0.9881	0.9694	0.9392	0.8974	0.8450	0.7842	0.7175	0.6474	0.5766	0.5071	0.3785	0.2705	0.1856	0.1227
	4	1.0000	0.9997	0.9982	0.9937	0.9844	0.9685	0.9447	0.9126	0.8723	0.8245	0.7705	0.7118	0.5871	0.4635	0.3509	0.2552
	5	1.0000	1.0000	0.9998	0.9989	0.9967	0.9921	0.9838	0.9707	0.9519	0.9268	0.8951	0.8569	0.7637	0.6547	0.5395	0.4275
	6	1.0000	1.0000	1.0000	0.9999	0.9994	0.9983	0.9960	0.9918	0.9848	0.9742	0.9593	0.9394	0.8835	0.8064	0.7120	0.6070
	7	1.0000	1.0000	1.0000	1.0000	0.9999	0.9997	0.9992	0.9980	0.9959	0.9922	0.9865	0.9779	0.9503	0.9055	0.8418	0.7608
	8	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9996	0.9990	0.9980	0.9961	0.9931	0.9816	0.9597	0.9237	0.8713
	9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9990	0.9981	0.9941	0.9850	0.9677	0.9389
	10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9995	0.9983	0.9951	0.9880	0.9744

APPENDIX

[illegible]

APPENDIX

Table II: Cumulative Poisson Probability Distribution $P[X \leq c] = \sum_{x=0}^c \frac{e^{-\lambda} \lambda^x}{x!}$

x	$\lambda = 0$	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0	1.0000	0.9900	0.9802	0.9704	0.9608	0.9512	0.9418	0.9324	0.9231	0.9139
1	1.0000	1.0000	0.9998	0.9996	0.9992	0.9988	0.9983	0.9977	0.9970	0.9962
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999
3	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 0.1$	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9	1.0
0	0.9048	0.8187	0.7408	0.6703	0.6065	0.5488	0.4966	0.4493	0.4066	0.3679
1	0.9953	0.9825	0.9631	0.9384	0.9098	0.8781	0.8442	0.8088	0.7725	0.7358
2	0.9998	0.9989	0.9964	0.9921	0.9856	0.9769	0.9659	0.9526	0.9371	0.9197
3	1.0000	0.9999	0.9997	0.9992	0.9982	0.9966	0.9942	0.9909	0.9865	0.9810
4	1.0000	1.0000	1.0000	0.9999	0.9998	0.9996	0.9992	0.9986	0.9977	0.9963
5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9998	0.9997	0.9994
6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
7	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 1.1$	1.2	1.3	1.4	1.5	1.6	1.7	1.8	1.9	2.0
1	0.3329	0.3012	0.2725	0.2466	0.2231	0.2019	0.1827	0.1653	0.1496	0.1353
2	0.6990	0.6626	0.6268	0.5918	0.5578	0.5249	0.4932	0.4628	0.4337	0.4060
3	0.9004	0.8795	0.8571	0.8335	0.8088	0.7834	0.7572	0.7306	0.7037	0.6767
4	0.9743	0.9662	0.9569	0.9463	0.9344	0.9212	0.9068	0.8913	0.8747	0.8571
5	0.9946	0.9923	0.9893	0.9857	0.9814	0.9763	0.9704	0.9636	0.9559	0.9473
6	0.9990	0.9985	0.9978	0.9968	0.9955	0.9940	0.9920	0.9896	0.9868	0.9834
7	0.9999	0.9997	0.9996	0.9994	0.9991	0.9987	0.9981	0.9974	0.9966	0.9955
8	1.0000	1.0000	0.9999	0.9999	0.9998	0.9997	0.9996	0.9994	0.9992	0.9989
9	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9998	0.9998
10	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 2.1$	2.2	2.3	2.4	2.5	2.6	2.7	2.8	2.9	3.0
0	0.1225	0.1108	0.1003	0.0907	0.0821	0.0743	0.0672	0.0608	0.0550	0.0498
1	0.3796	0.3546	0.3309	0.3084	0.2873	0.2674	0.2487	0.2311	0.2146	0.1991
2	0.6496	0.6227	0.5960	0.5697	0.5438	0.5184	0.4936	0.4695	0.4460	0.4232
3	0.8386	0.8194	0.7993	0.7787	0.7576	0.7360	0.7141	0.6919	0.6696	0.6472
4	0.9379	0.9275	0.9162	0.9041	0.8912	0.8774	0.8629	0.8477	0.8318	0.8153
5	0.9796	0.9751	0.9700	0.9643	0.9580	0.9510	0.9433	0.9349	0.9258	0.9161
6	0.9941	0.9925	0.9906	0.9884	0.9858	0.9828	0.9794	0.9756	0.9713	0.9665
7	0.9985	0.9980	0.9974	0.9967	0.9958	0.9947	0.9934	0.9919	0.9901	0.9881
8	0.9997	0.9995	0.9994	0.9991	0.9989	0.9985	0.9981	0.9976	0.9969	0.9962
9	0.9999	0.9999	0.9999	0.9998	0.9997	0.9996	0.9995	0.9993	0.9991	0.9989
10	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9998	0.9998	0.9997
11	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999
12	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 3.1$	3.2	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4.0
0	0.0450	0.0408	0.0369	0.0334	0.0302	0.0273	0.0247	0.0224	0.0202	0.0183
1	0.1847	0.1712	0.1586	0.1468	0.1359	0.1257	0.1162	0.1074	0.0992	0.0916
2	0.4012	0.3799	0.3594	0.3397	0.3208	0.3027	0.2854	0.2689	0.2531	0.2381

APPENDIX

3	0.6248	0.6025	0.5803	0.5584	0.5366	0.5152	0.4942	0.4735	0.4532	0.4335
4	0.7982	0.7806	0.7626	0.7442	0.7254	0.7064	0.6872	0.6678	0.6484	0.6288
5	0.9057	0.8946	0.8829	0.8705	0.8576	0.8441	0.8301	0.8156	0.8006	0.7851
6	0.9612	0.9554	0.9490	0.9421	0.9347	0.9267	0.9182	0.9091	0.8995	0.8893
7	0.9858	0.9832	0.9802	0.9769	0.9733	0.9692	0.9648	0.9599	0.9546	0.9489
8	0.9953	0.9943	0.9931	0.9917	0.9901	0.9883	0.9863	0.9840	0.9815	0.9786
9	0.9986	0.9982	0.9978	0.9973	0.9967	0.9960	0.9952	0.9942	0.9931	0.9919
10	0.9996	0.9995	0.9994	0.9992	0.9990	0.9987	0.9984	0.9981	0.9977	0.9972
11	0.9999	0.9999	0.9998	0.9998	0.9997	0.9996	0.9995	0.9994	0.9993	0.9991
12	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998	0.9997
13	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999
14	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 4.1$	4.2	4.3	4.4	4.5	4.6	4.7	4.8	4.9	5.0
0	0.0166	0.0150	0.0136	0.0123	0.0111	0.0101	0.0091	0.0082	0.0074	0.0067
1	0.0845	0.0780	0.0719	0.0663	0.0611	0.0563	0.0518	0.0477	0.0439	0.0404
2	0.2238	0.2102	0.1974	0.1851	0.1736	0.1626	0.1523	0.1425	0.1333	0.1247
3	0.4142	0.3954	0.3772	0.3594	0.3423	0.3257	0.3097	0.2942	0.2793	0.2650
4	0.6093	0.5898	0.5704	0.5512	0.5321	0.5132	0.4946	0.4763	0.4582	0.4405
5	0.7693	0.7531	0.7367	0.7199	0.7029	0.6858	0.6684	0.6510	0.6335	0.6160
6	0.8786	0.8675	0.8558	0.8436	0.8311	0.8180	0.8046	0.7908	0.7767	0.7622
7	0.9427	0.9361	0.9290	0.9214	0.9134	0.9049	0.8960	0.8867	0.8769	0.8666
8	0.9755	0.9721	0.9683	0.9642	0.9597	0.9549	0.9497	0.9442	0.9382	0.9319
9	0.9905	0.9889	0.9871	0.9851	0.9829	0.9805	0.9778	0.9749	0.9717	0.9682
10	0.9966	0.9959	0.9952	0.9943	0.9933	0.9922	0.9910	0.9896	0.9880	0.9863
11	0.9989	0.9986	0.9983	0.9980	0.9976	0.9971	0.9966	0.9960	0.9953	0.9945
12	0.9997	0.9996	0.9995	0.9993	0.9992	0.9990	0.9988	0.9986	0.9983	0.9980
13	0.9999	0.9999	0.9998	0.9998	0.9997	0.9997	0.9996	0.9995	0.9994	0.9993
14	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998
15	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999
16	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 5.1$	5.2	5.3	5.4	5.5	5.6	5.7	5.8	5.9	6.0
0	0.0061	0.0055	0.0050	0.0045	0.0041	0.0037	0.0033	0.0030	0.0027	0.0025
1	0.0372	0.0342	0.0314	0.0289	0.0266	0.0244	0.0224	0.0206	0.0189	0.0174
2	0.1165	0.1088	0.1016	0.0948	0.0884	0.0824	0.0768	0.0715	0.0666	0.0620
3	0.2513	0.2381	0.2254	0.2133	0.2017	0.1906	0.1800	0.1700	0.1604	0.1512
4	0.4231	0.4061	0.3895	0.3733	0.3575	0.3422	0.3272	0.3127	0.2987	0.2851
5	0.5984	0.5809	0.5635	0.5461	0.5289	0.5119	0.4950	0.4783	0.4619	0.4457
6	0.7474	0.7324	0.7171	0.7017	0.6860	0.6703	0.6544	0.6384	0.6224	0.6063
7	0.8560	0.8449	0.8335	0.8217	0.8095	0.7970	0.7841	0.7710	0.7576	0.7440
8	0.9252	0.9181	0.9106	0.9027	0.8944	0.8857	0.8766	0.8672	0.8574	0.8472
9	0.9644	0.9603	0.9559	0.9512	0.9462	0.9409	0.9352	0.9292	0.9228	0.9161
10	0.9844	0.9823	0.9800	0.9775	0.9747	0.9718	0.9686	0.9651	0.9614	0.9574
11	0.9937	0.9927	0.9916	0.9904	0.9890	0.9875	0.9859	0.9841	0.9821	0.9799
12	0.9976	0.9972	0.9967	0.9962	0.9955	0.9949	0.9941	0.9932	0.9922	0.9912
13	0.9992	0.9990	0.9988	0.9986	0.9983	0.9980	0.9977	0.9973	0.9969	0.9964
14	0.9997	0.9997	0.9996	0.9995	0.9994	0.9993	0.9991	0.9990	0.9988	0.9986

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15	0.9999	0.9999	0.9999	0.9998	0.9998	0.9998	0.9997	0.9996	0.9996	0.9995
16	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9998
17	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999
18	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 6.1$	6.2	6.3	6.4	6.5	6.6	6.7	6.8	6.9	7.0
0	0.0022	0.0020	0.0018	0.0017	0.0015	0.0014	0.0012	0.0011	0.0010	0.0009
1	0.0159	0.0146	0.0134	0.0123	0.0113	0.0103	0.0095	0.0087	0.0080	0.0073
2	0.0577	0.0536	0.0498	0.0463	0.0430	0.0400	0.0371	0.0344	0.0320	0.0296
3	0.1425	0.1342	0.1264	0.1189	0.1118	0.1052	0.0988	0.0928	0.0871	0.0818
4	0.2719	0.2592	0.2469	0.2351	0.2237	0.2127	0.2022	0.1920	0.1823	0.1730
5	0.4298	0.4141	0.3988	0.3837	0.3690	0.3547	0.3406	0.3270	0.3137	0.3007
6	0.5902	0.5742	0.5582	0.5423	0.5265	0.5108	0.4953	0.4799	0.4647	0.4497
7	0.7301	0.7160	0.7017	0.6873	0.6728	0.6581	0.6433	0.6285	0.6136	0.5987
8	0.8367	0.8259	0.8148	0.8033	0.7916	0.7796	0.7673	0.7548	0.7420	0.7291
9	0.9090	0.9016	0.8939	0.8858	0.8774	0.8686	0.8596	0.8502	0.8405	0.8305
10	0.9531	0.9486	0.9437	0.9386	0.9332	0.9274	0.9214	0.9151	0.9084	0.9015
11	0.9776	0.9750	0.9723	0.9693	0.9661	0.9627	0.9591	0.9552	0.9510	0.9467
12	0.9900	0.9887	0.9873	0.9857	0.9840	0.9821	0.9801	0.9779	0.9755	0.9730
13	0.9958	0.9952	0.9945	0.9937	0.9929	0.9920	0.9909	0.9898	0.9885	0.9872
14	0.9984	0.9981	0.9978	0.9974	0.9970	0.9966	0.9961	0.9956	0.9950	0.9943
15	0.9994	0.9993	0.9992	0.9990	0.9988	0.9986	0.9984	0.9982	0.9979	0.9976
16	0.9998	0.9997	0.9997	0.9996	0.9996	0.9995	0.9994	0.9993	0.9992	0.9990
17	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998	0.9998	0.9997	0.9997	0.9996
18	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
19	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 7.1$	7.2	7.3	7.4	7.5	7.6	7.7	7.8	7.9	8.0
0	0.0008	0.0007	0.0007	0.0006	0.0006	0.0005	0.0005	0.0004	0.0004	0.0003
1	0.0067	0.0061	0.0056	0.0051	0.0047	0.0043	0.0039	0.0036	0.0033	0.0030
2	0.0275	0.0255	0.0236	0.0219	0.0203	0.0188	0.0174	0.0161	0.0149	0.0138
3	0.0767	0.0719	0.0674	0.0632	0.0591	0.0554	0.0518	0.0485	0.0453	0.0424
4	0.1641	0.1555	0.1473	0.1395	0.1321	0.1249	0.1181	0.1117	0.1055	0.0996
5	0.2881	0.2759	0.2640	0.2526	0.2414	0.2307	0.2203	0.2103	0.2006	0.1912
6	0.4349	0.4204	0.4060	0.3920	0.3782	0.3646	0.3514	0.3384	0.3257	0.3134
7	0.5838	0.5689	0.5541	0.5393	0.5246	0.5100	0.4956	0.4812	0.4670	0.4530
8	0.7160	0.7027	0.6892	0.6757	0.6620	0.6482	0.6343	0.6204	0.6065	0.5925
9	0.8202	0.8096	0.7988	0.7877	0.7764	0.7649	0.7531	0.7411	0.7290	0.7166
10	0.8942	0.8867	0.8788	0.8707	0.8622	0.8535	0.8445	0.8352	0.8257	0.8159
11	0.9420	0.9371	0.9319	0.9265	0.9208	0.9148	0.9085	0.9020	0.8952	0.8881
12	0.9703	0.9673	0.9642	0.9609	0.9573	0.9536	0.9496	0.9454	0.9409	0.9362
13	0.9857	0.9841	0.9824	0.9805	0.9784	0.9762	0.9739	0.9714	0.9687	0.9658
14	0.9935	0.9927	0.9918	0.9908	0.9897	0.9886	0.9873	0.9859	0.9844	0.9827
15	0.9972	0.9969	0.9964	0.9959	0.9954	0.9948	0.9941	0.9934	0.9926	0.9918
16	0.9989	0.9987	0.9985	0.9983	0.9980	0.9978	0.9974	0.9971	0.9967	0.9963
17	0.9996	0.9995	0.9994	0.9993	0.9992	0.9991	0.9989	0.9988	0.9986	0.9984
18	0.9998	0.9998	0.9998	0.9997	0.9997	0.9996	0.9996	0.9995	0.9994	0.9993
19	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998	0.9998	0.9997

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20	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999
21	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 8.1$	8.2	8.3	8.4	8.5	8.6	8.7	8.8	8.9	9.0
0	0.0003	0.0003	0.0002	0.0002	0.0002	0.0002	0.0002	0.0002	0.0001	0.0001
1	0.0028	0.0025	0.0023	0.0021	0.0019	0.0018	0.0016	0.0015	0.0014	0.0012
2	0.0127	0.0118	0.0109	0.0100	0.0093	0.0086	0.0079	0.0073	0.0068	0.0062
3	0.0396	0.0370	0.0346	0.0323	0.0301	0.0281	0.0262	0.0244	0.0228	0.0212
4	0.0940	0.0887	0.0837	0.0789	0.0744	0.0701	0.0660	0.0621	0.0584	0.0550
5	0.1822	0.1736	0.1653	0.1573	0.1496	0.1422	0.1352	0.1284	0.1219	0.1157
6	0.3013	0.2896	0.2781	0.2670	0.2562	0.2457	0.2355	0.2256	0.2160	0.2068
7	0.4391	0.4254	0.4119	0.3987	0.3856	0.3728	0.3602	0.3478	0.3357	0.3239
8	0.5786	0.5647	0.5507	0.5369	0.5231	0.5094	0.4958	0.4823	0.4689	0.4557
9	0.7041	0.6915	0.6788	0.6659	0.6530	0.6400	0.6269	0.6137	0.6006	0.5874
10	0.8058	0.7955	0.7850	0.7743	0.7634	0.7522	0.7409	0.7294	0.7178	0.7060
11	0.8807	0.8731	0.8652	0.8571	0.8487	0.8400	0.8311	0.8220	0.8126	0.8030
12	0.9313	0.9261	0.9207	0.9150	0.9091	0.9029	0.8965	0.8898	0.8829	0.8758
13	0.9628	0.9595	0.9561	0.9524	0.9486	0.9445	0.9403	0.9358	0.9311	0.9261
14	0.9810	0.9791	0.9771	0.9749	0.9726	0.9701	0.9675	0.9647	0.9617	0.9585
15	0.9908	0.9898	0.9887	0.9875	0.9862	0.9848	0.9832	0.9816	0.9798	0.9780
16	0.9958	0.9953	0.9947	0.9941	0.9934	0.9926	0.9918	0.9909	0.9899	0.9889
17	0.9982	0.9979	0.9977	0.9973	0.9970	0.9966	0.9962	0.9957	0.9952	0.9947
18	0.9992	0.9991	0.9990	0.9989	0.9987	0.9985	0.9983	0.9981	0.9978	0.9976
19	0.9997	0.9997	0.9996	0.9995	0.9995	0.9994	0.9993	0.9992	0.9991	0.9989
20	0.9999	0.9999	0.9998	0.9998	0.9998	0.9998	0.9997	0.9997	0.9996	0.9996
21	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998
22	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999
23	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 9.1$	9.2	9.3	9.4	9.5	9.6	9.7	9.8	9.9	10.0
0	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0001	0.0000
1	0.0011	0.0010	0.0009	0.0009	0.0008	0.0007	0.0007	0.0006	0.0005	0.0005
2	0.0058	0.0053	0.0049	0.0045	0.0042	0.0038	0.0035	0.0033	0.0030	0.0028
3	0.0198	0.0184	0.0172	0.0160	0.0149	0.0138	0.0129	0.0120	0.0111	0.0103
4	0.0517	0.0486	0.0456	0.0429	0.0403	0.0378	0.0355	0.0333	0.0312	0.0293
5	0.1098	0.1041	0.0986	0.0935	0.0885	0.0838	0.0793	0.0750	0.0710	0.0671
6	0.1978	0.1892	0.1808	0.1727	0.1649	0.1574	0.1502	0.1433	0.1366	0.1301
7	0.3123	0.3010	0.2900	0.2792	0.2687	0.2584	0.2485	0.2388	0.2294	0.2202
8	0.4426	0.4296	0.4168	0.4042	0.3918	0.3796	0.3676	0.3558	0.3442	0.3328
9	0.5742	0.5611	0.5479	0.5349	0.5218	0.5089	0.4960	0.4832	0.4705	0.4579
10	0.6941	0.6820	0.6699	0.6576	0.6453	0.6329	0.6205	0.6080	0.5955	0.5830
11	0.7932	0.7832	0.7730	0.7626	0.7520	0.7412	0.7303	0.7193	0.7081	0.6968
12	0.8684	0.8607	0.8529	0.8448	0.8364	0.8279	0.8191	0.8101	0.8009	0.7916
13	0.9210	0.9156	0.9100	0.9042	0.8981	0.8919	0.8853	0.8786	0.8716	0.8645
14	0.9552	0.9517	0.9480	0.9441	0.9400	0.9357	0.9312	0.9265	0.9216	0.9165
15	0.9760	0.9738	0.9715	0.9691	0.9665	0.9638	0.9609	0.9579	0.9546	0.9513
16	0.9878	0.9865	0.9852	0.9838	0.9823	0.9806	0.9789	0.9770	0.9751	0.9730
17	0.9941	0.9934	0.9927	0.9919	0.9911	0.9902	0.9892	0.9881	0.9870	0.9857

18	0.9973	0.9969	0.9966	0.9962	0.9957	0.9952	0.9947	0.9941	0.9935	0.9928
19	0.9988	0.9986	0.9985	0.9983	0.9980	0.9978	0.9975	0.9972	0.9969	0.9965
20	0.9995	0.9994	0.9993	0.9992	0.9991	0.9990	0.9989	0.9987	0.9986	0.9984
21	0.9998	0.9998	0.9997	0.9997	0.9996	0.9996	0.9995	0.9995	0.9994	0.9993
22	0.9999	0.9999	0.9999	0.9999	0.9999	0.9998	0.9998	0.9998	0.9997	0.9997
23	1.0000	1.0000	1.0000	1.0000	0.9999	0.9999	0.9999	0.9999	0.9999	0.9999
24	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
x	$\lambda = 11$	12	13	14	15	16	17	18	19	20
0	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
1	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
2	0.0012	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
3	0.0049	0.0023	0.0011	0.0005	0.0002	0.0001	0.0000	0.0000	0.0000	0.0000
4	0.0151	0.0076	0.0037	0.0018	0.0009	0.0004	0.0002	0.0001	0.0000	0.0000
5	0.0375	0.0203	0.0107	0.0055	0.0028	0.0014	0.0007	0.0003	0.0002	0.0001
6	0.0786	0.0458	0.0259	0.0142	0.0076	0.0040	0.0021	0.0010	0.0005	0.0003
7	0.1432	0.0895	0.0540	0.0316	0.0180	0.0100	0.0054	0.0029	0.0015	0.0008
8	0.2320	0.1550	0.0998	0.0621	0.0374	0.0220	0.0126	0.0071	0.0039	0.0021
9	0.3405	0.2424	0.1658	0.1094	0.0699	0.0433	0.0261	0.0154	0.0089	0.0050
10	0.4599	0.3472	0.2517	0.1757	0.1185	0.0774	0.0491	0.0304	0.0183	0.0108
11	0.5793	0.4616	0.3532	0.2600	0.1848	0.1270	0.0847	0.0549	0.0347	0.0214
12	0.6887	0.5760	0.4631	0.3585	0.2676	0.1931	0.1350	0.0917	0.0606	0.0390
13	0.7813	0.6815	0.5730	0.4644	0.3632	0.2745	0.2009	0.1426	0.0984	0.0661
14	0.8540	0.7720	0.6751	0.5704	0.4657	0.3675	0.2808	0.2081	0.1497	0.1049
15	0.9074	0.8444	0.7636	0.6694	0.5681	0.4667	0.3715	0.2867	0.2148	0.1565
16	0.9441	0.8987	0.8355	0.7559	0.6641	0.5660	0.4677	0.3751	0.2920	0.2211
17	0.9678	0.9370	0.8905	0.8272	0.7489	0.6593	0.5640	0.4686	0.3784	0.2970
18	0.9823	0.9626	0.9302	0.8826	0.8195	0.7423	0.6550	0.5622	0.4695	0.3814
19	0.9907	0.9787	0.9573	0.9235	0.8752	0.8122	0.7363	0.6509	0.5606	0.4703
20	0.9953	0.9884	0.9750	0.9521	0.9170	0.8682	0.8055	0.7307	0.6472	0.5591
21	0.9977	0.9939	0.9859	0.9712	0.9469	0.9108	0.8615	0.7991	0.7255	0.6437
22	0.9990	0.9970	0.9924	0.9833	0.9673	0.9418	0.9047	0.8551	0.7931	0.7206
23	0.9995	0.9985	0.9960	0.9907	0.9805	0.9633	0.9367	0.8989	0.8490	0.7875
24	0.9998	0.9993	0.9980	0.9950	0.9888	0.9777	0.9594	0.9317	0.8933	0.8432
25	0.9999	0.9997	0.9990	0.9974	0.9938	0.9869	0.9748	0.9554	0.9269	0.8878
26	1.0000	0.9								

APPENDIX

[illegible]

APPENDIX

Table III: Value for Constructing Single Sampling Plan

c	np ₁			np ₂			R = p ₂ /p ₁			R = p ₂ /p ₁		
	$\alpha = 0.01$	$\alpha = 0.05$	$\alpha = 0.10$	$\beta = 0.01$	$\beta = 0.05$	$\beta = 0.10$	$\alpha = 0.01$ $\beta = 0.01$	$\alpha = 0.01$ $\beta = 0.05$	$\alpha = 0.01$ $\beta = 0.10$	$\alpha = 0.05$ $\beta = 0.01$	$\alpha = 0.05$ $\beta = 0.05$	$\alpha = 0.05$ $\beta = 0.10$
0	0.010	0.051	0.105	4.605	2.996	2.303	458.210	298.073	229.105	89.781	58.404	44.890
1	0.149	0.355	0.532	6.638	4.744	3.890	44.686	31.933	26.184	18.681	13.349	10.946
2	0.436	0.818	1.102	8.406	6.296	5.322	19.278	14.439	12.206	10.280	7.699	6.509
3	0.823	1.366	1.745	10.045	7.754	6.681	12.202	9.418	8.115	7.352	5.675	4.890
4	1.279	1.970	2.433	11.605	9.154	7.994	9.072	7.156	6.249	5.890	4.646	4.057
5	1.785	2.613	3.152	13.108	10.513	9.275	7.343	5.889	5.195	5.017	4.023	3.549
6	2.330	3.286	3.895	14.571	11.842	10.532	6.253	5.082	4.520	4.435	3.604	3.206
7	2.906	3.981	4.656	16.000	13.148	11.771	5.506	4.524	4.050	4.019	3.303	2.957
8	3.507	4.695	5.432	17.403	14.434	12.995	4.962	4.115	3.705	3.707	3.074	2.768
9	4.130	5.426	6.221	18.783	15.705	14.206	4.548	3.803	3.440	3.462	2.895	2.618
10	4.771	6.169	7.021	20.145	16.962	15.407	4.222	3.555	3.229	3.265	2.750	2.497
11	5.428	6.924	7.829	21.490	18.208	16.598	3.959	3.354	3.058	3.104	2.630	2.397
12	6.099	7.690	8.646	22.821	19.442	17.782	3.742	3.188	2.915	2.968	2.528	2.312
13	6.782	8.464	9.470	24.139	20.668	18.958	3.559	3.047	2.795	2.852	2.442	2.240
14	7.477	9.246	10.300	25.446	21.886	20.128	3.403	2.927	2.692	2.752	2.367	2.177
15	8.181	10.035	11.135	26.743	23.098	21.292	3.269	2.823	2.603	2.665	2.302	2.122

APPENDIX

**Table IV: Grubbs Table: Value for Constructing Double Sampling Plan
where $n_1 = n_2$ ($\alpha = 0.05, \beta = 0.10$)**

Plan Number	$R = p_2/p_1$	Acceptance Numbers		Approximate Value of n, p for	
		c_1	c_2	$\alpha = 0.05$	$\beta = 0.10$
1	11.9	0	1	0.21	2.5
2	7.54	1	2	0.52	3.92
3	6.79	0	2	0.43	2.96
4	5.39	1	3	0.76	4.11
5	4.65	2	4	1.16	5.39
6	4.25	1	4	1.04	4.42
7	3.88	2	5	1.43	5.55
8	3.63	3	6	1.87	6.78
9	3.38	2	6	1.72	5.82
10	3.21	3	7	2.15	6.91
11	3.09	4	8	2.62	8.1
12	2.85	4	9	2.9	8.26
13	2.6	5	11	3.68	9.56
14	2.44	5	12	4	9.77
15	2.32	5	13	4.35	10.08

APPENDIX

**Table V: Grubbs Table: Value for Constructing Double Sampling Plan
where $n_2 = 2n_1$ ($\alpha = 0.05, \beta = 0.10$)**

Plan Number	$R = p_2/p_1$	Acceptance Numbers		Approximate Value of n_1p for	
		c_1	c_2	$\alpha = 0.05$	$\beta = 0.10$
1	14.50	0	1	0.16	2.32
2	8.07	0	2	0.30	2.42
3	6.48	1	3	0.60	3.89
4	5.39	0	3	0.49	2.64
5	5.09	1	4	0.77	3.92
6	4.31	0	4	0.68	2.93
7	4.19	1	5	0.96	4.02
8	3.60	1	6	1.16	4.17
9	3.26	2	8	1.68	5.47
10	2.96	3	10	2.27	6.72
11	2.77	3	11	2.46	6.82
12	2.62	4	13	3.07	8.05
13	2.46	4	14	3.29	8.11
14	2.21	3	15	3.41	7.55
15	1.97	4	20	4.75	9.35

Block

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BLOCK 3 DECISION AND GAME THEORIES

In Blocks 1 and 2, you have studied applications of statistical tools in process and product control in different types of manufacturing industries.

For any individual business/organisation/company/industry, one of the key aspects of achieving its goals is that it should make the right decisions at the right time. In fact, decision making is both a science and an art. In Block 3, we introduce decision theory and discuss certain criteria of decision making. However, the selection of a criteria depends on the environment of decision making and in this block we discuss decision making under the following environments:

- Decision making under certainty,
- Decision making under uncertainty,
- Decision making under risk, and
- Decision making under conflict.

In **Unit 9** entitled **Introduction to Decision Theory** we discuss the criteria of decision making under the environments of certainty and uncertainty.

Unit 10 entitled **Decision Making Process** discusses the criteria of decision making under the environment of risk.

The environment of conflict is generally present in games. In game theory, games may broadly be classified as:

- Games with saddle point, and
- Games without saddle point.

In this block, we restrict our discussion to two-person zero-sum games.

Unit 11 entitled **Two-Person Zero-Sum Games with Saddle Point** deals with the basic concepts in game theory and solutions of games with saddle point.

In **Unit 12**, entitled **Two-Person Zero-Sum Games without Saddle Point**, we discuss some methods of solving two-person zero-sum games without saddle point.

Notations and Symbols

Sec.	:	Section
Secs.	:	Sections
Fig.	:	Figure
A_i	:	i^{th} course of action
N_i	:	i^{th} state of nature
x_{ij}	:	Payoff value corresponding to i^{th} state of nature and j^{th} course of action
v	:	Value of the game

UNIT 9 INTRODUCTION TO DECISION THEORY

Structure

- 9.1 Introduction
 - Objectives
- 9.2 Decision Making: Is it a Science or an Art?
- 9.3 Types of Decisions
- 9.4 Components of Decision Making
 - Courses of Action (Alternatives)
 - States of Nature (Events)
 - Payoff Values and Payoff Matrix
 - Opportunity Loss Table or Regret Table
- 9.5 Types of Environment
 - Decision Making under Certainty
 - Decision Making under Uncertainty
- 9.6 Summary
- 9.7 Solutions/Answers

9.1 INTRODUCTION

Decision making is an integral part of our lives. Some of these decisions are related to routine matters. For example, a vegetable seller has to decide which vegetables to purchase so that he can make maximum profit, a student has to decide which trouser and shirt to wear while going to a farewell party, a home maker has to decide whether to serve tea, coffee or cold drinks to her guests, and so on! Similarly, a manager of an organisation also has to take decisions to fulfil a given set of objectives such as increase in profit and turnover, reduction in cost, advertisement, etc. Moreover, decisions may affect an individual or a group of people or an organisation or the entire society. For example, decisions taken by business managers affect their consumers, shareholders and the employees of the organisation. In brief, we can say that all of us are involved in decision making either in our capacity as decision makers or suffer the consequences of decisions made by other people!

Some decisions are very important and require a great deal of consideration and thought. For instance, we would like to decide whether to invest in a particular share or mutual funds only after giving it careful thought. In the same way, a company requires extensive planning while taking a decision about whether to launch a new product in a market or not. In such cases, decisions have to be based on some criteria and standard tools and we can use techniques which increase the chances that our decisions are correct. Therefore, in this unit we present an introduction to decision theory.

Now-a-days, business related decisions are increasingly being taken on the basis of analysis and interpretation of data using statistical concepts and techniques. Especially, business managers and professionals need suitable data to help them in making decisions in different situations. You will understand this better when you go through the situations discussed in examples and

exercises in this unit as well as in the next unit. The use of statistical concepts and techniques in decision making enables us to:

- solve problems in a systematic manner;
- add substance to decision making; and
- reduce the guess factor and obtain reliable results.

In effect, the decision maker needs to know and use decision support systems based on statistical models. Units 9 and 10 are devoted to the discussion of some methods which will benefit the people who have to make decisions in different situations. However, before we discuss these methods, we would like to answer the question “Is decision making a science or an art?” This is what we do in Sec. 9.2. In Sec. 9.3, we describe various types of decisions. We introduce the components of decision making in Sec. 9.4. A knowledge of the decision making environment helps the decision maker in arriving at good decisions. In Sec. 9.5 we discuss two types of environments in which the decision maker may have to take decisions, namely, decision making under certainty and decision making under uncertainty. We also discuss certain criteria that may be used for decision making under an environment of uncertainty which provide solutions of the problems in certain situations.

In the next unit, we shall discuss the environment of decision making under risk and decision tree analysis.

Objectives

After studying this unit, you should be able to:

- answer the question: “Is decision making a science or an art?”;
- explain the concepts of structured and unstructured decisions;
- define the components of decision making;
- explain different types of environment in which a decision maker may have to make decisions;
- state the criteria which are applicable in the environment of “Decision Making under Certainty”; and
- solve numerical problems using different criteria in the environment of “Decision Making under Uncertainty”.

9.2 DECISION MAKING: IS IT A SCIENCE OR AN ART?

The first and foremost question we would like to answer is:

Is decision making a science or an art?

This question arises because in decision making we make inferences about the **unknown**, and unknown phenomena may be predicted using two approaches: one based on objective application of appropriate scientific knowledge and the other using a subjective approach. Many a times, it is felt that the choice of a particular decision is related to the subjectivity of the decision maker. It may be argued that experienced people make right decisions without the aid of scientific tools and techniques and that decision making is largely a matter of intuition based on experience. For example, experienced fishermen can forecast the weather conditions by simply looking at the sky without applying any tools or techniques. Their decisions are completely based on their experience. Decision making in situations like this may be termed as an **art** because

decisions are taken on the basis of the experience of an individual, rather than applying scientific tools or techniques.

On the other hand, there are many aspects of the decision making process, especially in industries and business, which lend themselves to scientific analysis and treatment. In such situations, a statistician breaks a decision making problem into small components and makes an effort to detect the pattern, which is relevant to a particular decision making process. He/she tries to analyse whether the pattern conforms to any set of rules or can be evaluated against a set of standards. These standards are set on the basis of knowledge about past events and experiences. There are no concrete rules and policies, which would enable a statistician to foresee events before they happen. The task of a decision maker in an industry or business becomes easier when he/she uses standards and tools. Statisticians try to develop standards and tools of measurement to solve a particular problem. Decision making that involves the application of scientific tools and techniques to the situation at hand rather than being entirely based on the intuition or experiences of individuals is known as a **science**.

From the above discussion, we may infer that decision making in the field of industry or business need not be exclusively subjective in character. There are many areas where it is possible to apply statistical tools and techniques and thereby make decision making far easier. In general, we can say that, as such, there are no specific criteria in decision making suitable for all situations. Hence, decision making may be treated as a **science** or an **art** depending upon the situations and objectives of the problem to be solved.

9.3 TYPES OF DECISIONS

Since we are taking decisions all the time as individuals, workers, managers or owners of companies, there exists a huge variety of decisions all around us. As you may know, for any scientific study, we first try to classify the subject of investigation. So we first need to classify decisions and select the types of decisions that we shall study in decision theory.

Classification of decisions is not unique. Decisions are sometimes categorised as:

- Personal Decisions, and
- Organisational Decisions,

and also as:

- Routine Decisions, and
- Basic Decisions.

In **decision theory**, decisions are generally classified as follows:

- Structured Decisions, and
- Unstructured Decisions.

Decisions related to routine situations or problems or preplanned decisions belong to the category of **structured decisions**. On the other hand, decisions related to problems arising accidentally or unplanned decisions belong to the category of **unstructured decisions**.

Let us further explain both these types of decisions.

Structured Decisions

Structured decisions are decisions which can be programmed (planned) in advance and are made under established situations. Structured or programmed decisions are the ones which an organisation or its employees take repeatedly on the basis of past data and advance planning.

For instance, the decisions of an organisation for reducing the cost of a product fall under the category of structured decisions because these can be planned in advance. Structured decisions are repetitive in nature and involve definite procedures and processes for handling problems. Managers make structured decisions for routine tasks all the time. For example, they routinely give additional facilities/benefits on the purchase of products to promote sales. A manager of a fast food outlet may incentivise sales by offering a free drink to his/her customers with their orders. Such promotional activities can always be preplanned or programmed in advance to fulfil the needs of the customers and maximise profit.

Structured decisions are deterministic in nature and the outcome of the decision can be determined with certainty if a specified sequence of activities is performed. Therefore, it is possible to make use of decision procedures or decision rules or algorithm to make structured decisions.

Unstructured Decisions

Sometimes, organisations face problems that have not arisen before and hence, there are no fixed processes for solving them. Decisions taken to solve such unforeseen problems are called **unstructured decisions**. These types of decisions are taken for the problems which arise occasionally and are unique in nature. Since such problems appear occasionally, there is no pre-defined method or predetermined procedure for solving them and a great deal of creativity may be required for decision making. Such decisions are quite complex and the risk factor involved in making unstructured decisions is quite high. Unstructured decisions are non-repetitive and vital. In fact, higher level managers face such situations quite often. For instance, they may have to handle the grievances of employees related to wages in an organisation or face a sudden labour strike in a factory. There are no predetermined procedures to resolve such problems and unstructured decisions need to be taken. We can say that unstructured decisions are taken in situations that are uncertain, unclear and not preplanned. An important feature of unstructured decisions is that they cannot be delegated.

So far you have learnt about the types of decisions we shall be dealing with in decision theory. We now discuss the components of decision making.

9.4 COMPONENTS OF DECISION MAKING

Decision theory teaches ways of assessing and managing a real life situation generally faced by decision makers working in a small or big organisation at the executive level or any other level. To apply techniques of decision theory to any real-life situation, we first need to define the following components:

- Courses of Action (Alternatives),
- States of Nature (Events), and
- Payoff Values and Payoff Matrix.

In order to apply some of the decision criteria, we also need to define the **opportunity loss matrix**.

We now explain each one of these components.

9.4.1 Courses of Action (Alternatives)

The problem of decision making arises in a situation when we have more than one option. If there is only one course of action with no alternative then we have no choices and decision making is simple: We just follow that course of action! For example, if a student has a single pair of shoes, he/she wears just that pair and does not have to decide which pair of shoes to wear. So the problem of decision making arises in situations where only one course of action has to be taken out of two or more available courses of action or alternatives. For example, if a person is given the option of drinking tea or coffee, he will have to decide on only one of them! A manager in a fast food outlet may like to decide on whether to give the offer of 'buy one, get one free' or offer a free drink with an order or give a coupon for future redemption to increase sales. So, all **possible options** available in a given situation of decision making, which are **under the control** of the decision maker are known as **courses of action** or **alternatives**.

If A_1 and A_2 denote the courses of action in a given situation with two options, the **action space** is denoted by A and given by

$$A = \{A_1, A_2\}$$

In general, if n options are available to the decision maker in a given situation, which are under his/her control, the action space is given by

$$A = \{A_1, A_2, \dots, A_n\} \text{ where } A_i, (i = 1, 2, \dots, n) \text{ are } n \text{ courses of action.}$$

A situation with three courses of action available to the decision maker is shown in Fig. 9.1.

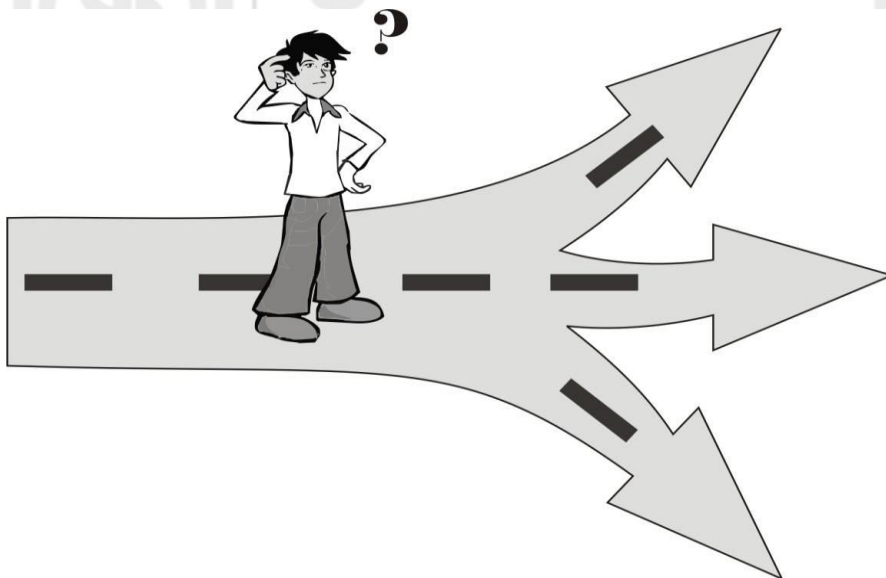


Fig. 9.1: A typical situation for three courses of action.

Let us explain the concept of courses of action (alternatives) in a given situation with the help of an example.

Example 1: Suppose a fruit seller buys apples at the rate of Rs 50 per kg and sells them at the rate of Rs 60 per kg. Assume that an apple not sold during the day is treated as rotten and thrown away. The daily sales of apples in the past has never been less than 48 kg. It has also not been more than 50 kg. State the courses of action in this situation and determine the action space.

Solution: On the basis of the given information, it is clear that the fruit seller should not buy less than 48 kg of apples. He should also not buy more than 50 kg of apples. Thus, he/she has three options or courses of action (alternatives): Buying 48 kg, 49 kg or 50 kg of apples and these options are under his/her control. If we denote these three courses of action by A_1 , A_2 , A_3 , respectively, we have

$A_1 = 48$, $A_2 = 49$, $A_3 = 50$ and the action space is given by $A = \{48, 49, 50\}$.

9.4.2 States of Nature (Events)

You have learnt that courses of action are alternatives presently available to the decision maker, which are under his/her control. But a situation may have an outcome, which may not be under the control of the decision maker. **States of nature or events** are related to future outcomes, conditions or situations, which are **not under the control** of the decision maker. For example, consider Example 1. In this example, the sale of apples is the future outcome, which is not under control of the fruit seller. Hence, the sales of apples (in kg), correspond to the states of nature for Example 1.

In Example 1, the sales of apples may be 48 kg, 49 kg or 50 kg. If we denote these possible states of nature or events by N_1 , N_2 , N_3 , respectively, we have

$N_1 = 48$, $N_2 = 49$, $N_3 = 50$

Note that the fruit seller cannot predict which of the three events N_1 , N_2 or N_3 would take place, that is, these behave like random events.

A decision maker can, at most, associate probabilities with the outcomes of the events based on the experience of past data. But he/she may face two situations:

- 1) Situations where **data are not available to associate probabilities**, and
- 2) Situations where **data are available to associate probabilities**.

Decision making in the first situation is known as **decision making under uncertainty**. In the second situation, it is known as **decision making under risk**.

Let us further explain both these categories of decision making.

Decision making under uncertainty: This refers to decision making in situations where the probabilities associated with the occurrence of different states of nature are **not known** to the decision maker. Thus, decision making in situations where the decision maker does not have any prior data that could provide the probabilities of occurrence of different states of nature is known as **decision making under uncertainty**. For example, decision making for launching a new product lies under this category because the probabilities of success or failure of the product are not known in advance. Since the product is being launched for the first time, no past data about its performance in the market are available and the probabilities of its success or failure are unknown.

Decision making under risk: Decision making in situations where the probabilities associated with the occurrence of different states of nature are **known** from the past experience or past records falls under the category of **decision making under risk**. If the probabilities of the occurrence of the states of nature are given, the best decision to be selected is the one that gives the highest expected output among the available alternatives.

9.4.3 Payoff Values and Payoff Matrix

The possible outcomes from the combination of each course of action and each state of nature represented in numerical form (or in rupees) are termed as **payoff values**. In other words, a value of an output when represented in terms of gain or loss is called a **payoff value**. Payoff values may be presented in the form of a table or a matrix as shown in Tables 9.1 and 9.2. The resulting table or matrix is called the **payoff table** or **payoff matrix**. Payoff values are also known as **conditional profits** if they represent profits and **conditional losses** if they represent losses. The word conditional is added because of the states of nature, which are not under the control of the decision maker.

Let us consider Example 1 again to explain these concepts. The courses of action (apples bought) are $A_1 = 48$, $A_2 = 49$, $A_3 = 50$, and the states of nature (future sales of apples) are $N_1 = 48$, $N_2 = 49$, $N_3 = 50$. Also, the cost price is Rs 50 per kg and the sale price is Rs 60 per kg.

Therefore, the profit on sold apples = Rs $(60 - 50) = \text{Rs } 10$ per kg.

Loss on unsold apples = Rs 50 per kg.

Therefore, the payoff values for this example are given by:

$$\begin{aligned}\text{Payoff value} &= (\text{Profit per kg}) \times (\text{Number of kg of apples sold}) \\ &\quad - (\text{Loss per kg}) \times (\text{Number of kg of apples unsold}) \\ &= 10 \times (\text{Number of kg of apples sold}) \\ &\quad - 50 \times (\text{Number of kg of apples unsold})\end{aligned}$$

Now, if x_{ij} , ($i, j = 1, 2, 3$) denotes the payoff value corresponding to the state of nature N_i and course of action A_j , then we have:

$$\begin{aligned}x_{11} &= \text{Payoff value corresponding to state of nature } N_1 \text{ and course of action } A_1 \\ &= \text{Rs } (10 \times 48 - 50 \times 0) = \text{Rs } 480\end{aligned} \quad \left[\begin{array}{l} \because N_1 = 48 \text{ means that demand is of} \\ 48 \text{ kg and } A_1 = 48 \text{ means he/she has} \\ \text{already bought 48 kg. So there is no} \\ \text{loss of unsold apples.} \end{array} \right]$$

$$\begin{aligned}x_{12} &= \text{Payoff value corresponding to state of nature } N_1 \text{ and course of action } A_2 \\ &= \text{Rs } (10 \times 48 - 50 \times 1) = \text{Rs } 430\end{aligned} \quad \left[\begin{array}{l} \because N_1 = 48 \text{ means that demand is of} \\ 48 \text{ kg and } A_2 = 49 \text{ means he/she has} \\ \text{already bought 49 kg. So he/she is} \\ \text{facing a loss for 1 kg.} \end{array} \right]$$

$$\begin{aligned}x_{13} &= \text{Payoff value corresponding to state of nature } N_1 \text{ and course of action } A_3 \\ &= \text{Rs } (10 \times 48 - 50 \times 2) = \text{Rs } 380\end{aligned} \quad \left[\begin{array}{l} \because N_1 = 48 \text{ means that demand is of} \\ 48 \text{ kg and } A_3 = 50 \text{ means that he/she} \\ \text{has already bought 50 kg. So he/she} \\ \text{is facing a loss for 2 kg.} \end{array} \right]$$

$$\begin{aligned}x_{21} &= \text{Payoff value corresponding to state of nature } N_2 \text{ and course of action } A_1 \\ &= \text{Rs } (10 \times 48 - 50 \times 0) = \text{Rs } 480\end{aligned} \quad \left[\begin{array}{l} \because N_2 = 49 \text{ means that demand is of} \\ 49 \text{ kg and } A_1 = 48 \text{ means that he/she} \\ \text{has already bought 48 kg. So there is} \\ \text{no loss of unsold apples.} \end{array} \right]$$

Similarly, we can calculate the payoff values x_{22} , x_{23} , x_{31} , x_{32} and x_{33} for the different states of nature and different courses of action and present them in a tabular form as shown in Table 9.1.

Table 9.1: Payoff Table for the Fruit Seller

States of Nature (Sale)	Courses of Action (Purchase of Apples)		
	$A_1 (=48)$	$A_2 (=49)$	$A_3 (=50)$
$N_1 (=48)$	$x_{11} = 480$	$x_{12} = 430$	$x_{13} = 380$
$N_2 (=49)$	$x_{21} = 480$	$x_{22} = 10 \times 49 - 50 \times 0 = 490$	$x_{23} = 10 \times 49 - 50 \times 1 = 440$
$N_3 (=50)$	$x_{31} = 10 \times 48 - 50 \times 0 = 480$	$x_{32} = 10 \times 49 - 50 \times 0 = 490$	$x_{33} = 10 \times 50 - 50 \times 0 = 500$

In general, if in a given situation there are m states of nature $N_1, N_2, \dots, N_m$ and n courses of action $A_1, A_2, \dots, A_n$, the format of a **payoff table** (also called the **payoff matrix**) is as shown in Table 9.2. Here x_{ij} , ($i = 1, 2, \dots, m$; $j = 1, 2, \dots, n$) denotes the value of payoff corresponding to the i^{th} state of nature N_i and j^{th} course of action A_j .

Table 9.2: General Format of a Payoff Table

States of Nature (Events)	Courses of Action (Alternatives)						
	A_1	A_2	A_3	...	A_j	...	A_n
N_1	x_{11}	x_{12}	x_{13}	...	x_{1j}	...	x_{1n}
N_2	x_{21}	x_{22}	x_{23}	...	x_{2j}	...	x_{2n}
N_3	x_{31}	x_{32}	x_{33}	...	x_{3j}	...	x_{3n}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
N_i	x_{i1}	x_{i2}	x_{i3}	...	x_{ij}	...	x_{in}
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
N_m	x_{m1}	x_{m2}	x_{m3}	...	x_{mj}	...	x_{mn}

Let us take another example to help you understand how to calculate payoff values.

Example 2: A shopkeeper gets the following offer from a company:

- The company will provide 4 TV sets for Rs 40000 to the shopkeeper, where the retail price of each TV is Rs 15000.
- The shopkeeper cannot return the unsold TV set, if any, to the company.
- The shopkeeper has only two options: he can either accept the offer or reject it.

Assume that the unsold TV is a direct loss to the shopkeeper and prepare the payoff table for the shopkeeper.

Solution: First of all, we have to define the courses of action and the states of nature. Here the shopkeeper has two courses of action: he can either accept the offer or reject it. These are under his/her control. Let A_1 and A_2 denote these courses of action, respectively. Now, as far as the states of nature are concerned, there are five possibilities: he is able to sell none, one, two, three or all four TV sets. These are future outcomes and therefore, not under the control of the shopkeeper. So these form the states of nature. Let

N_0, N_1, N_2, N_3, N_4 denote these five states of nature, respectively. If he accepts the offer and cannot sell any TV (corresponding to the state of nature N_0), he faces a loss of Rs 40000. We can represent it as a gain of INR -40000 . Similarly, if he accepts the offer and only one TV is sold (corresponding to the state of nature N_1), he faces a loss of INR 25000 ($= 40000 - 15000$) or a gain of

–25000 and so on. The complete payoff table where each output is gain is shown in Table 9.3.

Table 9.3: Payoff Table for the Shopkeeper

States of Nature	Courses of Action	
	Accept the Offer (A_1)	Reject the Offer (A_2)
N_0	–40000	0
N_1	–25000	0
N_2	–10000	0
N_3	5000	0
N_4	20000	0

From Table 9.3, note that to earn the maximum profit, the shopkeeper should accept the offer and should be able to sell all four TV sets. In this way, he will earn the maximum profit of Rs 20000.

So far, we have defined three components of decision making: courses of action, states of nature and payoff values. We have also explained how to construct a payoff table. We have to define these components in all decision making criteria to solve decision making problems. But some criteria of solving decision making problems such as the Salvage Criterion (discussed in this unit) and Expected Opportunity Loss (EOL) criterion (discussed in the next unit) use one more component of decision making known as **opportunity loss values** or **regret values**. We now discuss how to calculate opportunity loss values or regret values and construct the opportunity loss table or regret table.

9.4.4 Opportunity Loss Table or Regret Table

To obtain the opportunity loss table or regret table, we first need to form the payoff table. Then we have to examine whether the payoff values represent profits or losses. If the payoff values represent profits, the opportunity loss table is obtained as follows:

Step 1: We select the maximum payoff values for each state of nature.

Step 2: We subtract the payoff values of different courses of action for each state of nature from the maximum payoff value selected for that state of nature.

The opportunity loss table corresponding to the payoff Table 9.2 assuming that the payoff values represent profits is shown in Table 9.4, where

M_i = Maximum payoff value corresponding to the i^{th} state of nature, $i = 1, 2, \dots, m$.

Table 9.4: Opportunity Loss or Regret Table when Payoff Values x_{ij} Represent Profits or Gains

States of Nature	Courses of Action (Alternatives)						
	A_1	A_2	A_3	...	A_j	...	A_n
N_1	$M_1 - x_{11}$	$M_1 - x_{12}$	$M_1 - x_{13}$	...	$M_1 - x_{1j}$	...	$M_1 - x_{1n}$
N_2	$M_2 - x_{21}$	$M_2 - x_{22}$	$M_2 - x_{23}$	...	$M_2 - x_{2j}$	...	$M_2 - x_{2n}$
N_3	$M_3 - x_{31}$	$M_3 - x_{32}$	$M_3 - x_{33}$	...	$M_3 - x_{3j}$	...	$M_3 - x_{3n}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$		$\ddots$	$\vdots$
N_i	$M_i - x_{i1}$	$M_i - x_{i2}$	$M_i - x_{i3}$	...	$M_i - x_{ij}$	...	$M_i - x_{in}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$		$\ddots$	$\vdots$
N_m	$M_m - x_{m1}$	$M_m - x_{m2}$	$M_m - x_{m3}$	...	$M_m - x_{mj}$	...	$M_m - x_{mn}$

But if payoff values of the payoff table represent losses, then the opportunity loss table is obtained as follows:

Step 1: Select minimum payoff value for each state of nature.

Step 2: For each state of nature, subtract the minimum payoff value from all payoff values corresponding to different courses of action.

The opportunity loss table corresponding to payoff Table 9.2 assuming that the payoff values represent losses is shown in Table 9.5, where

m_i = min payoff value corresponding to the i^{th} state of nature, $i = 1, 2, 3, \dots, m$.

Table 9.5: Opportunity Loss or Regret Table when Payoff Values x_{ij} Represent Losses or Costs

States of Nature (Events)	Courses of Action (Alternatives)						
	A_1	A_2	A_3	$\dots$	A_j	$\dots$	A_n
N_1	$x_{11} - m_1$	$x_{12} - m_1$	$x_{13} - m_1$	$\dots$	$x_{1j} - m_1$	$\dots$	$x_{1n} - m_1$
N_2	$x_{21} - m_2$	$x_{22} - m_2$	$x_{23} - m_2$	$\dots$	$x_{2j} - m_2$	$\dots$	$x_{2n} - m_2$
N_3	$x_{31} - m_3$	$x_{32} - m_3$	$x_{33} - m_3$	$\dots$	$x_{3j} - m_3$	$\dots$	$x_{3n} - m_3$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
N_i	$x_{i1} - m_i$	$x_{i2} - m_i$	$x_{i3} - m_i$	$\dots$	$x_{ij} - m_i$	$\dots$	$x_{in} - m_i$
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\ddots$	$\vdots$
N_m	$x_{m1} - m_m$	$x_{m2} - m_m$	$x_{m3} - m_m$	$\dots$	$x_{mj} - m_m$	$\dots$	$x_{mn} - m_m$

Let us consider an example to explain the calculations involved in obtaining an opportunity loss table.

Example 3: Obtain the opportunity loss table for the data given in Example 1.

Solution: We have already obtained the payoff table for the data of Example 1 in Sec. 9.4.3 and the same is shown in Table 9.1.

Since payoff values in the payoff table shown in Table 9.1 represent profit of the fruit seller, the opportunity loss table is obtained by following the two steps given below.

Step 1: We select the maximum payoff value (Max PV) for each state of nature given in Table 9.1:

$$\begin{aligned}\therefore \text{Max PV for the state of nature } N_1 &= \max(N_1) \\ &= \max\{480, 430, 380\} \\ &= 480\end{aligned}$$

Similarly,

$$\begin{aligned}\max(N_2) &= \max\{480, 490, 440\} = 490 \\ \max(N_3) &= \max\{480, 490, 500\} = 500\end{aligned}$$

Step 2: We subtract all payoff values corresponding to different courses of action for each state of nature from the maximum payoff value for that state of nature. That is, for the state of nature N_1 , we subtract all payoff values from 480. Similarly, we subtract all payoff values from 490 and 500, respectively, for the states of nature N_2 and N_3 as shown in Table 9.6:

Table 9.6: Opportunity Loss Table for Example 3

States of Nature	Courses of Action		
	A ₁ (48)	A ₂ (49)	A ₃ (50)
N ₁ (48)	$480 - 480 = 0$	$480 - 430 = 50$	$480 - 380 = 100$
N ₂ (49)	$490 - 480 = 10$	$490 - 490 = 0$	$490 - 440 = 50$
N ₃ (50)	$500 - 480 = 20$	$500 - 490 = 10$	$500 - 500 = 0$

You may now like to try the following exercises based on the components of decision making.

E1) A bangle seller gets the following offer:

- The cost of a pack of 10 dozen bangles is Rs 150.
- The bangle seller can either accept the offer or reject it.
- Each pack of 10 dozen bangles may have 40%, 30%, 20% or 10% defective bangles. Defective bangles cannot be returned by the bangle seller. Thus, defective bangles in a pack mean a direct loss to the bangle seller.

The bangle seller sells them at the rate of Rs 20 per dozen. Assume that if he/she accept the offer, all the bangles are sold out. On the basis of this information,

- (i) Identify the courses of action,
- (ii) Identify states of nature, and
- (iii) Obtain the payoff table.

E2) Obtain the opportunity loss table for the data of E1.

The challenge for the decision maker is to select one of the possible courses of action in a given situation. The ultimate goal of this unit and the next unit is to enable you to face this challenge. An important aspect of decision making is to know the environment in which a decision is to be made. This is what we discuss in Sec. 9.5.

9.5 TYPES OF ENVIRONMENT

For a decision maker to arrive at a good decision in a given situation, he/she has to consider all the available data, courses of action or alternatives and states of nature. He/she should have a knowledge of the decision making environment, and also know how to use the relevant quantitative approach for decision making. In this section, we discuss different types of environment under which a decision maker may have to make a decision. This discussion is important because the selection of decision criteria depends on the environment under which a decision is to be taken. The following types of environment are categorised on the basis of the degree of certainty:

- Decision Making under Certainty,
- Decision Making under Uncertainty, and
- Decision Making under Risk.

We discuss the first two types of environment in this section and the third one in detail in Unit 10. We shall discuss one more type of environment, namely, “Decision Making under Conflict” in Units 11 and 12 on Game Theory since this type of environment is present in games between two or more players.

9.5.1 Decision Making under Certainty

In certain problems, the information available to the decision maker is almost complete so that he/she knows all the facts about the states of nature and also knows which state of nature would occur. He/she also knows the consequences. In such a situation, decision making is easy. The decision maker has to choose that alternative, which will give him/her the maximum payoff in terms of utility or output under the state of nature. In the payoff table, there will be only one column indicating the state of nature and he/she will pick up the alternative which optimises his/her payoff. For instance, the decision of purchasing Kisan Vikas Patra (KVP) or National Saving Certificate (NSC) ensures the future benefit the stakeholder will get at the time of maturity. In such investments, a stakeholder knows the amount of maturity in advance. Decision making under certainty is simple and easy for decision makers.

At first sight, it appears to be a simple problem, to select an alternative which optimises the payoff when there is a single state of nature. However, when the courses of action are large, the number of rows would be very large. For example, if there are 25 jobs to be assigned to 25 machines with different outputs and different costs of production, the total number of alternatives/rows would be in millions. In such complex situations, the decision making method becomes less efficient. Then the tools of operations research, such as linear/non-linear programming, etc. need to be used to arrive at decisions. You will study linear programming problems in detail in Block 1 of the course MSTE-002.

9.5.2 Decision Making under Uncertainty

We have already defined decision making under uncertainty in Sec. 9.4.2. Recall that under this scenario, the decision maker has no prior information about the probabilities of different states of nature. Undoubtedly, he/she can collect information, if possible, about the probabilities of the states of nature (e.g., by conducting market surveys).

In business situations, there are many problems of this nature and the choice of a course of action depends on the personality or judgment of the decision maker. The responses of decision makers may be termed as optimistic, pessimistic or those of least regret, etc. Consider the example of a half-filled glass of water. An optimist would say that the glass is half-filled **while** a pessimist would say that the glass is half-empty. And the Manager would say that the glass is twice as large as it needs to be!

Some criteria used for decision making under such situations are as follows:

- Optimistic (Maximax/Minimin) Criterion
- Pessimistic (Maximin) Criterion
- Hurwicz (Coefficient of optimism or realism) Criterion
- Regret (Salvage) Criterion
- Laplace (Equal Probability or Rationality) Criterion

We now discuss each one of these criteria with examples.

i) **Optimistic (Maximax/Minimin) Criterion**

Under this criterion, the decision maker first selects the maximum payoff value in case of profit or gain and the minimum payoff value in case of loss or cost from among the payoff values for each course of action. Then he/she selects the maximum in case of gains (or minimum in case of losses) from among the

selected payoff values. Since the approach of the decision maker is completely positive, it is called the optimistic criterion. Thus, if payoff values represent profits or gains, the decision maker selects the maximum from among maximum payoff values for each course of action. Hence, it is known as the **maximax criterion**. Similarly, if payoff values represent losses or costs, the decision maker selects the minimum from among the minimum payoff values for each course of action. Hence, it is known as a **minimin criterion**. For solving the problems under optimistic criterion, we follow the steps given below:

Step 1: We select the maximum payoff value in case of profit or gain or the minimum payoff value in case of loss or cost for each course of action.

Step 2: We select the maximum from among the payoff values selected in Step 1 in case of profit or gain or the minimum from among the payoff values selected in Step 1 in case of loss or cost.

Let us explain the procedure with the help of an example.

Example 4: The profit of a milk booth of a particular brand under small, medium and large orders subject to low, moderate and high demands are shown in Table 9.7.

Table 9.7: Profits of the Owner of the Milk Booth

Order for Milk	Demand of Milk at the Booth		
	Low	Moderate	High
Small	1200	1200	1200
Medium	1000	1500	1500
Large	800	1300	1800

Using the optimistic criterion, take a decision on which order is more beneficial for the owner.

Solution: The quantity of the milk ordered is under the control of the owner of the booth. So the courses of action are small (A_1), medium (A_2) and large (A_3). But the demand for any day is not under the control of the owner of the booth. So the states of nature are low (N_1), moderate (N_2) and high (N_3). This information can be expressed in the payoff table (Table 9.8) as follows:

Table 9.8: Payoff Table for the Owner of the Booth

States of Nature	Courses of Action		
	Small (A_1)	Medium (A_2)	Large (A_3)
Low (N_1)	1200	1000	800
Moderate (N_2)	1200	1500	1300
High (N_3)	1200	1500	1800

We now follow the two steps under the optimistic criterion:

Step 1: We look at the payoff values and assess whether these values represent profits or losses. In this case, the values represent profits. So we select the maximum payoff value for each course of action. These are shown in Table 9.9 against the stub 'Step 1'.

Step 2: We select the maximum among the payoff values selected in Step 1. Thus, $\max \{1200, 1500, 1800\} = 1800$, and it corresponds to the course of action A_3 as shown in the Table 9.9 against the stub 'Step 2'. Hence, the decision maker should suggest to the owner that he/she should go with the large order.

Recall from Unit 13 of MST-001 that **stubs** are row headings and explain what information the rows present. Stubs are placed at the extreme left of the table.

Table 9.9: Payoff Table for the Owner of the Booth and Calculations of Step 1 and Step 2

States of Nature	Courses of Action		
	Small (A_1)	Medium (A_2)	Large (A_3)
Low (N_1)	1200	1000	800
Moderate (N_2)	1200	1500	1300
High (N_3)	1200	1500	1800
Step 1	1200	1500	1800
Step 2	$\max\{1200, 1500, 1800\} = 1800$		

ii) Pessimistic (Maximin) Criterion

Under this criterion, the decision maker first selects the minimum payoff value for each course of action. Then he/she selects the maximum among these selected payoff values. Thus, the approach of the decision maker is pessimistic and so this criterion is known as pessimistic (maximin, that is, maximum among minima) criterion. It is also known as Wald's criterion. For solving problems, we adopt the following two steps:

Step 1: We select the minimum payoff value for each course of action.

Step 2: We select the maximum from among the minimum payoff values selected in Step 1. The course of action corresponding to this maximum payoff value is selected under this criterion.

Let us apply this criterion to the data of Example 4 given in Table 9.10.

Step 1: We first select the minimum payoff value for each course of action as shown against the stub 'Step 1' in Table 9.10.

Step 2: We select the maximum from among the minimum payoff values selected in Step 1. Thus, we select $\max\{1200, 1000, 800\} = 1200$, which corresponds to the course of action A_1 . It is also shown in Table 9.10 against the stub 'Step 2'. Hence, the owner of the booth should go with the small order under this criterion.

Table 9.10: Payoff Table for the Owner of the Booth and Calculation of Step 1 and Step 2

States of nature	Courses of action		
	Small (A_1)	Medium (A_2)	Large (A_3)
Low (N_1)	1200	1000	800
Moderate (N_2)	1200	1500	1300
High (N_3)	1200	1500	1800
Step 1	1200	1000	800
Step 2	$\max\{1200, 1000, 800\} = 1200$		

iii) Hurwicz Criterion

In the optimistic criterion, the approach of the decision maker was completely positive while in the pessimistic criterion, it was pessimistic. But Hurwicz criterion suggests that a decision maker should neither be too optimistic nor too pessimistic. For this purpose, Hurwicz introduced the idea of the **coefficient of optimism**. It is generally denoted by α . The value of α varies from 0 to 1 (i.e., $0 \leq \alpha \leq 1$), and measures the degree of optimism of the decision maker. The two extreme values $\alpha = 1$ and $\alpha = 0$ indicate completely optimistic and pessimistic approaches, respectively. Thus, the optimistic criterion and pessimistic criterion are particular cases of Hurwicz criterion. Also, $(1 - \alpha)$ is known as the **coefficient of pessimism**.

The steps used to solve problems using the Hurwicz criterion are given below:

Step 1: We select the value of α as per the degree of optimism of the decision maker and consequently $1 - \alpha$, the coefficient of pessimism is known.

Step 2: For each course of action, we identify the maximum as well as minimum payoff values and obtain the value of H, where

$$H = \alpha \times (\text{Max PV corresponding to a course of action})$$

$$+ (1 - \alpha) \times (\text{Min PV corresponding to the same course of action})$$

Step 3: We select the course of action which corresponds to the maximum value of H obtained in Step 2. The course of action thus obtained will be the optimum course of action for the decision maker under Hurwicz criterion for the selected value of α .

Let us consider the data of Example 4 again to explain the calculation involved in this criterion for, say, $\alpha = 0.6$:

We have already defined the courses of action and states of nature for Example 4. So, we go directly to Step 1.

Step 1: We have selected $\alpha = 0.6$ and, therefore, $1 - \alpha = 0.4$.

Step 2: Calculations of this step are shown in Table 9.11 against the stub 'Step 2'.

Table 9.11: Payoff Table for the Owner of the Booth and Calculation of Step 1 and Step 2

States of Nature		Courses of Action		
		Small (A_1)	Medium (A_2)	Large (A_3)
Low (N_1)		1200	1000	800
Moderate (N_2)		1200	1500	1300
High (N_3)		1200	1500	1800
Step 2	Max PV for each Course of action	1200	1500	1800
	Min PV for each Course of action	1200	1000	800
	Value of H for each Course of action	$0.6 \times 1200 + 0.4 \times 1200 = 1200$	$0.6 \times 1500 + 0.4 \times 1000 = 1300$	$0.6 \times 1800 + 0.4 \times 800 = 1400$
Step 3		$\max\{1200, 1300, 1400\} = 1400$		

Step 3: In this step we select the maximum among the values of H obtained in Step 2.

$\therefore \max\{1200, 1300, 1400\} = 1400$, which corresponds to the course of action A_3 . Hence, the large order (A_3) is the optimum course of action under this criterion.

iv) Regret (Salvage) Criterion

In this criterion, first of all we convert the payoff table into an **opportunity loss table** about which you have learnt in Sec. 9.4.4. Under this criterion, the goal of the decision maker is to minimise the opportunity loss which he/she may face by not selecting the best opportunity or by adopting a wrong course of action. For example, suppose two years ago you had surplus money and were planning to buy a flat from the investment point of view. Suppose at that time, flats were available within your budget at three locations, say X, Y and Z at the same price. You had to choose only one location and you chose location X. Now, after two years, the rates of the flats at the three locations X, Y, Z have increased by 30%, 20% and 42%, respectively. From the point of view of making a profit, you would regret your decision, because had you selected location Z, you would have earned $(42 - 30)\% = 12\%$ more compared to what

you earned for location X. This is an example of opportunity loss. You had an opportunity two years ago, which you missed because of making a wrong selection. We would like to minimise the opportunity loss due to the possibility of wrong decision. The regret criterion suggests **the course of action, which minimises our opportunity loss**. The following three steps are involved in using the regret criterion for solving problems:

- Step 1:** We form the opportunity loss matrix by calculating the amount of regret corresponding to each course of action and each state of nature. You have learnt these calculations in Sec. 9.4.4.
- Step 2:** We identify the maximum regret amount for each course of action from the opportunity loss table obtained in Step 1.
- Step 3:** We select the minimum opportunity loss value from among the regret amounts selected in Step 2. The optimum course of action corresponds to the minimum amount selected from among the regret amounts obtained in Step 2.

Let us consider the data of Example 4 to explain the calculations involved in this criterion.

We have already defined the courses of action and states of nature for Example 4. We have also obtained the payoff table for the data of Example 4. So, we go directly to Step 1.

- Step 1:** Since the payoff values represent profits, to obtain the opportunity loss table, we first calculate the maximum payoff value (Max PV) for each state of nature as shown below:

$$\text{Max PV for state of nature } N_1 = \max \{1200, 1000, 800\} = 1200$$

$$\text{Max PV for state of nature } N_2 = \max \{1200, 1500, 1300\} = 1500$$

$$\text{Max PV for state of nature } N_3 = \max \{1200, 1500, 1800\} = 1800$$

We then subtract all payoff values corresponding to different courses of action for the state of nature N_1 from 1200. We repeat the calculation for the states of nature N_2 and N_3 as shown in Table 9.12.

Table 9.12: Opportunity Loss Table

Status of Nature	Courses of Action		
	Small (A_1)	Medium (A_2)	Large (A_3)
Low (N_1)	$1200 - 1200 = 0$	$1200 - 1000 = 200$	$1200 - 800 = 400$
Moderate (N_2)	$1500 - 1200 = 300$	$1500 - 1500 = 0$	$1500 - 1300 = 200$
High (N_3)	$1800 - 1200 = 600$	$1800 - 1500 = 300$	$1800 - 1800 = 0$
Step 2	600	300	400
Step 3	$\min\{600, 300, 400\} = 300$		

- Step 2:** Maximum amount of regret for the course of action $A_1 = \max \{0, 300, 600\} = 600$. Similarly, the maximum amount of regret for the courses of action A_2 and A_3 are 300 and 400, respectively, as shown in Table 9.12 against the stub 'Step 2'.

- Step 3:** We select the minimum regret amount from among the amounts obtained in Step 2.

$$\therefore \min \{600, 300, 400\} = 300.$$

This corresponds to the course of action A_2 . Hence, A_2 is the optimum course of action under this criterion. It is also shown in Table 9.12 against the stub 'Step 3'.

v) Laplace Criterion

This criterion is a method to solve the problems that involve “decision making under uncertainty”. You have learnt in Sec. 9.4.2 that in the situations of decision making under uncertainty, we do not have any information about the probabilities of occurrence of different states of nature. But this criterion works by giving equal probabilities to each state of nature. The point to be noted here is that we apply this criterion in a given situation only if we do not have sufficient information about the inequality of the probabilities of different states of nature. This is because we are assuming equal probabilities for different states of nature for each course of action.

The following steps are involved in the calculations for this criterion:

Step 1: We assign an equal probability to each state of nature. That is, if there are m states of nature, then we assign the probability $1/m$ to each state of nature.

Step 2: We calculate the expected payoff value for each course of action. For example, if $x_{1j}, x_{2j}, \dots, x_{mj}$ denote the payoff values for the j^{th} course of action corresponding to m states of nature, then the **expected payoff value** (EPV) of the j^{th} course of action is given by:

$$\text{EPV for } j^{\text{th}} \text{ course of action} = \frac{1}{m} (x_{1j} + x_{2j} + \dots + x_{mj})$$

Step 3: We select the maximum expected payoff value from among the values calculated in Step 2. The optimum course of action corresponds to this maximum expected payoff value.

If, in Example 4, we do not have information that the three states of demands (i.e., low, moderate and high) have different probabilities, we can apply this criterion. The calculations involved in this criterion are explained below:

Step 1: Here we have three states of nature and we know that under this criterion each state of nature is assigned equal probability. So each state of nature will be assigned the probability $1/3$.

Step 2: The expected payoff value (EPV) for each course of action is given by:

$$\text{EPV for course of action } A_1 = \frac{1}{3} (1200 + 1200 + 1200) = \frac{3600}{3} = 1200$$

$$\text{EPV for course of action } A_2 = \frac{1}{3} (1000 + 1500 + 1500) = \frac{4000}{3} \approx 1333.33$$

$$\text{EPV for course of action } A_3 = \frac{1}{3} (800 + 1300 + 1800) = \frac{3900}{3} = 1300$$

Step 3: We select the maximum EPV from among the values obtained in Step 2. $\max\{1200, 1333.33, 1300\} = 1333.33$, which corresponds to the course of action A_2 . Hence, A_2 is the optimum course of action under this criterion.

Let us consider another example.

Example 5: Consider the following payoff table

States of Nature	Courses of Action			
	A_1	A_2	A_3	A_4
N_1	400	900	900	1000
N_2	200	400	700	-300
N_3	600	200	500	700

Identify the optimum course of action under:

- (i) Optimistic criterion (assume that payoff values represent profits),
- (ii) Pessimistic criterion,
- (iii) Hurwicz criterion (for $\alpha = 0.8$),
- (iv) Regret criterion (assume that payoff values represent losses), and
- (v) Laplace criterion (assume that we do not have sufficient information that the states of nature N_1, N_2, N_3 have different probabilities).

Solution:

- (i) The payoff values represent profits. So under the optimistic criterion we follow the two steps given below:

Step 1: We select a maximum payoff value (Max PV) for each course of action:

$$\text{Max PV for course of action } A_1 = \max \{400, 200, 600\} = 600$$

The maximum payoff values for the courses of action A_2, A_3, A_4 are 900, 900, 1000, respectively.

Step 2: We select the maximum from among the payoff values selected in Step 1. Thus, $\max \{600, 900, 900, 1000\} = 1000$, which corresponds to the course of action A_4 . It is shown in Table 9.13 against the stub 'Step 2'. Hence A_4 is the optimum course of action.

Table 9.13: Calculation for Optimistic Criterion

States of Nature	Courses of Action			
	A_1	A_2	A_3	A_4
N_1	400	900	900	1000
N_2	200	400	700	-300
N_3	600	200	500	700
Step 1	600	900	900	1000
Step 2	$\max \{600, 900, 900, 1000\} = 1000$			

- (ii) To identify an optimum course of action under the pessimistic criterion, we follow the two steps given below:

Step 1: We select the minimum payoff value for each course of action:

$$\text{Min PV for the course of action } A_1 = \min \{400, 200, 600\} = 200.$$

Similarly, the minimum payoff values for the courses of action A_2, A_3, A_4 are 200, 500, -300, respectively. These values are shown in Table 9.14 against the stub 'Step 1'.

Step 2: We select the maximum from among the payoff values selected in Step 1. Thus, $\max \{200, 200, 500, -300\} = 500$, which corresponds to the course of action A_3 . It is shown in Table 9.14 against the stub 'Step 2'. Hence, A_3 is the optimum course of action under the pessimistic criterion.

Table 9.14: Calculation for the Pessimistic Criterion

States of Nature	Courses of Action			
	A ₁	A ₂	A ₃	A ₄
N ₁	400	900	900	1000
N ₂	200	400	700	-300
N ₃	600	200	500	700
Step 1	200	200	500	-300
Step 2	max{200, 200, 500, -300} = 500			

(iii) To identify an optimum course of action under the Hurwicz criterion for $\alpha = 0.8$, we follow the three steps given below:

Step 1: Here we are given that $\alpha = 0.8$ and so $1 - \alpha = 0.2$.

Step 2: We identify the maximum as well as the minimum payoff values for each course of action and obtain H, where

$$\begin{aligned}
 H &= \alpha \times (\text{Max PV corresponding to a course of action}) \\
 &\quad + (1 - \alpha) \times (\text{Min PV corresponding to the same course of action}) \\
 &= 0.8 \times (\text{Max PV corresponding to a course of action}) \\
 &\quad + 0.2 \times (\text{Min PV corresponding to the same course of action})
 \end{aligned}$$

Calculations of this step for each course of action are shown in Table 9.15 against 'Step 2'.

Table 9.15: Calculation for Hurwicz Criterion

States of Nature		Courses of Action			
		A ₁	A ₂	A ₃	A ₄
N ₁		400	900	900	1000
N ₂		200	400	700	-300
N ₃		600	200	500	700
Step 2	Max PV	600	900	900	1000
	Min PV	200	200	500	-300
	Value of H	$0.8 \times 600 + 0.2 \times 200 = 520$	$0.8 \times 900 + 0.2 \times 200 = 760$	$0.8 \times 900 + 0.2 \times 500 = 820$	$0.8 \times 1000 + 0.2 \times (-300) = 740$
Step 3		max{520, 760, 820, 740} = 820			

Step 3: We select the maximum value of H from among the values obtained in Step 2.

$\therefore \text{Max } \{520, 760, 820, 740\} = 820$, which corresponds to the course of action A₃. It is also shown against the stub 'Step 3' in Table 9.15. Hence, A₃ is the optimum course of action under the Hurwicz criterion.

(iv) It is given that payoff values represent losses. So, under the regret criterion, we follow the three steps given below:

Step 1: We first obtain the opportunity loss matrix. Since it is given that the payoff values represent losses, to obtain opportunity loss table, we calculate the minimum payoff value (Min PV) for each state of nature as shown below:

For the state of nature N₁, it is $\min \{400, 900, 900, 1000\} = 400$,

For the state of nature N_2 , it is $\min \{200, 400, 700, -300\} = -300$, and

For the state of nature N_3 , it is $\min \{600, 200, 500, 700\} = 200$.

We now subtract the minimum payoff value (400) from all payoff values corresponding to different courses of action for the state of nature N_1 . We do the same calculations for the states of nature

N_2 and N_3 as shown in Table 9.16.

Step 2: We identify the maximum regret amount for each course of action from the opportunity loss table obtained in Step 1. This is shown in Table 9.16 against the stub 'Step 2'.

Step 3: We select the minimum regret amount from among the regret amounts obtained in Step 2. Thus,

$\min \{500, 700, 1000, 600\} = 500$, which corresponds to the course of action A_1 . It is also shown against the stub 'Step 3' in Table 9.16. Hence, A_1 is the optimum course of action under the regret criterion.

Table 9.16: Calculation for the Regret Criterion

States of Nature	Courses of Action			
	A_1	A_2	A_3	A_4
N_1	$400 - 400 = 0$	$900 - 40 = 500$	$900 - 40 = 500$	$1000 - 40 = 600$
N_2	$200 - (-300) = 500$	$400 - (-300) = 700$	$700 - (-300) = 1000$	$-300 - (-300) = 0$
N_3	$600 - 20 = 400$	$200 - 200 = 0$	$500 - 20 = 300$	$700 - 200 = 500$
Step 2	500	700	1000	600
Step 3	$\min \{500, 700, 1000, 600\} = 500$			

(v) It is given that there is no sufficient information that the states of nature N_1 , N_2 and N_3 have different probabilities. So, we can opt for the Laplace criterion. The calculations involved in this criterion are explained in the following three steps:

Step 1: We assign equal probability to each state of nature. Here, we have three states of nature. So each state of nature will be assigned the probability $1/3$.

Step 2: We calculate the expected payoff value (EPV) for each course of action as follows:

$$\text{EPV for the course of action } A_1 = \frac{400 + 200 + 600}{3} = \frac{1200}{3} = 400$$

$$\text{EPV for the course of action } A_2 = \frac{900 + 400 + 200}{3} = \frac{1500}{3} = 500$$

$$\text{EPV for the course of action } A_3 = \frac{900 + 700 + 500}{3} = \frac{2100}{3} = 700$$

$$\text{EPV for the course of action } A_4 = \frac{1000 + (-300) + 700}{3} = \frac{1400}{3} = 466.67$$

Step 3: We select the maximum EPV from among the values obtained in Step 2. Thus, $\text{Max} \{400, 500, 700, 466.67\} = 700$, which corresponds to the course of action A_3 . Hence, A_3 is the optimum course of action under the Laplace criterion.

You may now like to identify the optimum course of action for the following exercise.

E3) In the following payoff table, the payoff values represent losses:

Table 9.17: Payoff Table for E3 where Payoff Values Represent Losses

States of Nature	Courses of Action		
	A ₁	A ₂	A ₃
N ₁	1500	1800	2500
N ₂	2400	1600	2000
N ₃	1200	2000	3000
N ₄	800	1000	400

On the basis of this information, identify the optimum course of action under each of the following criteria:

- Optimistic Criterion,
- Pessimistic Criterion,
- Hurwicz Criterion (for $\alpha = 0.7$),
- Regret Criterion, and
- Laplace criterion (assuming that we have no information that the states of nature N₁, N₂, N₃, N₄ have different probabilities)

Let us summarise the main points that we have discussed in this unit:

9.6 SUMMARY

- Decision making in situations where decisions are taken on the basis of the experience of an individual, rather than by applying scientific tools or techniques is known as an **art**.
- Decision making in situations where decisions are taken by applying scientific tools or techniques rather than on the basis of the experiences of an individual is known as a **science**.
- Structured decisions are made under established situations. **Structured or programmed decisions** are the ones where the organisation has already faced such decisions in the past and the employees are used to solving such problems.
- Decisions, that are taken to solve problems which have not been faced before in the organisation and cannot be solved using any known procedures are called **unstructured decisions**.
- To apply techniques of decision theory in any real life situation, we first need to break down the given situation into the following components:
 - Courses of Action (Alternatives):** Possible options available in a given situation of decision making which are **under the control** of the decision maker are known as **courses of action** or **alternatives**.
 - States of Nature (Events):** States of nature or events are related to **future outcomes**, conditions or situations which are **not under the control** of the decision maker.
 - Payoff Values and Payoff Matrix:** Total possible outcomes which result from combination of each course of action and each state of nature in terms of money or some other value of interest are known as **payoff values**. We can represent payoff values in the form of a **payoff**

table or payoff matrix. Payoff values are also known as **conditional profits** if they represent profits and **conditional losses** if they represent losses (because of the states of nature which are not under the control of the decision maker).

6) The selection of a decision criterion depends on the environment in which decision is to be taken. The following types of environment are categorised on the basis of the degree of certainty:

- Decision Making under Certainty,
- Decision Making under Uncertainty, and
- Decision Making under Risk.

7) Several criteria are used for decision making under uncertainty. Some of these are:

- i) Optimistic (Maximax/Minimin) Criterion
- ii) Pessimistic (Maximin) Criterion
- iii) Regret (Salvage) Criterion
- iv) Hurwicz (Coefficient of optimism or realism) Criterion
- v) Laplace (Equal Probability or Rationality) Criterion

9.7 SOLUTIONS/ANSWERS

E1) (i) The bangle seller has two options: he/she can either accept the offer or reject it. Also accepting or rejecting the offer is under his/her control, and so the courses of action, A_1 and A_2 , are

A_1 : accept the offer.

A_2 : reject the offer.

(ii) It is assumed that if he/she accepts the offer, there is no issue of demand as all the bangles are sold out. So, demand does not form states of nature. But the number of defective bangles is a future event and may be 40%, 30%, 20% or 10%. Since the number of defective bangles are not under the control of the bangle seller, these form the states of nature. If we denote these states of nature by N_1, N_2, N_3, N_4 , respectively, then

N_1 : represents the state of 40% defective bangles

N_2 : represents the state of 30% defective bangles

N_3 : represents the state of 20% defective bangles

N_4 : represents the state of 10% defective bangles

iii) In the first two parts we have identified the courses of action and the states of nature. The payoff values are nothing but the outcomes in terms of money of all possible combinations of courses of action and states of nature. If he/she rejects the offer (i.e., he/she adopts the course of action A_2), then there will be no profit and no loss and so payoff values corresponding to all states of nature will be zero. But if he/she accepts the offer, then his/her profit will vary according to the states of nature as explained below:

Case I: For the state of nature N_1 ,

The total number of bangles in one packing = 10 dozen

Number of defective bangles = 40% of 10 dozen = 4 dozen

Bangles which can be sold out = $10 - 4 = 6$ dozen

Cost price of bangles = Rs $\frac{150}{10} = \text{Rs } 15$ per dozen

Selling price of bangles = Rs 20 per dozen

Profit per dozen on good bangles = Rs $(20 - 15) = \text{Rs } 5$

$\therefore$ Payoff value = $5 \times (\text{Number of good bangles in dozen})$

$$\begin{aligned} & -15 \times (\text{Number of defective bangles in dozen}) \\ & = 5 \times 6 - 15 \times 4 = 30 - 60 = -\text{Rs } 30 \end{aligned}$$

Case II: For the state of nature N_2 ,

The number of defective bangles = 30% of 10 dozen

= 3 dozen

$\therefore$ Payoff value = $5 \times (10 - 3) - 15 \times 3 = 35 - 45 = -\text{Rs } 10$

Case III: For the state of nature N_3

The number of defective bangles = 20% of 10 dozen

= 2 dozen

$\therefore$ Payoff value = $5 \times (10 - 2) - 15 \times 2 = 40 - 30 = \text{Rs } 10$

Case IV: For the state of nature N_4 ,

The number of defective bangles = 10% of 10 dozen

= 1 dozen

$\therefore$ Payoff value = $5 \times (10 - 1) - 15 \times 1 = 45 - 15 = \text{Rs } 30$

Thus, the payoff table for the bangle seller is given as follows:

Table 9.18: Payoff Table for Bangle Seller

States of Nature	Courses of Action	
	A_1 (Accept the Offer)	A_2 (Reject the Offer)
N_1 (40%)	-30	0
N_2 (30%)	-10	0
N_3 (20%)	10	0
N_4 (10%)	30	0

- E 2)** We have already obtained the payoff table for the data of E1. Also, the payoff values in the payoff table represent profits of the bangle seller. So, the opportunity loss table is obtained by using the same two steps as explained in Example 3 and is given below.

Table 9.19: Opportunity Loss Table

States of Nature	Courses of Action	
	A_1 (Accept the Offer)	A_1 (Reject the Offer)
N_1 (40%)	$0 - (-30) = 30$	$0 - 0 = 0$
N_2 (30%)	$0 - (-10) = 10$	$0 - 0 = 0$
N_3 (20%)	$10 - 10 = 0$	$10 - 0 = 10$
N_4 (10%)	$30 - 30 = 0$	$30 - 0 = 30$

- E3) i)** It is given that the payoff values represent losses. So under the optimistic criterion, we proceed as follows:

Step 1: We select a minimum payoff value for each course of action.

$\therefore$ For the course of action A_1 , it is $\min \{1500, 2400, 1200, 800\} = 800$

Similarly, the minimum payoff values for the courses of action A_2, A_3 are 1000, 400, respectively.

Step 2: We select the minimum from among the payoff values selected in Step 1. Thus, $\min \{800, 1000, 400\} = 400$, which corresponds to the course of action A_3 . It is also shown in Table 9.20 against the stub 'Step 2'. Hence, A_3 is the optimum course of action.

Table 9.20: Calculation for Optimistic Criterion

States of Nature	Courses of Action		
	A_1	A_2	A_3
N_1	1500	1800	2500
N_2	2400	1600	2000
N_3	1200	2000	3000
N_4	800	1000	400
Step 1	800	1000	400
Step 2	$\min \{800, 1000, 400\} = 400$		

- ii) To identify an optimum course of action under the pessimistic criterion, we follow the two steps given below:

Step 1: We select the maximum payoff value for each course of action.

$\therefore$ for the course of action A_1 , it is $\max \{1500, 2400, 1200, 800\} = 2400$

Similarly, the maximum payoff values for the courses of action A_2, A_3 are 2000, 3000, respectively, and are shown in Table 9.21 against the stub 'Step 1'.

Step 2: We select the minimum from among the payoff values selected in Step 1. $\min \{2400, 2000, 3000\} = 2000$, which corresponds to the course of action A_2 and is shown in Table 9.21 against the stub 'Step 2'. Hence, A_2 is the optimum course of action under the pessimistic criterion.

Table 9.21: Calculation for Pessimistic Criterion

States of Nature	Courses of Action		
	A_1	A_2	A_3
N_1	1500	1800	2500

N_2	2400	1600	2000
N_3	1200	2000	3000
N_4	800	1000	400
Step 1	2400	2000	3000
Step 2	$\min\{2400, 2000, 3000\} = 2000$		

iii) To identify an optimum course of action under the Hurwicz criterion for $\alpha = 0.7$, we follow the three steps given below:

Step 1: We select the value of α as per the degree of optimism of the decision maker. It is given that $\alpha = 0.7$ and so the coefficient of pessimism is $1 - \alpha = 0.3$.

Step 2: We identify the maximum as well as the minimum payoff values for each course of action and obtain H, where

$$\begin{aligned}
 H &= \alpha \times (\text{Max PV corresponding to a course of action}) \\
 &\quad + (1 - \alpha) \times (\text{Min PV corresponding to the same course of action}) \\
 &= 0.7 \times (\text{Max PV corresponding to a course of action}) \\
 &\quad + 0.3 \times (\text{Min PV corresponding to the same course of action})
 \end{aligned}$$

Calculations of H for each course of action are shown against the stub 'Step 2' in Table 9.22.

Table 9.22: Calculation for Hurwicz Criterion

States of Nature		Courses of Action		
		A_1	A_2	A_3
N_1		1500	1800	2500
N_2		2400	1600	2000
N_3		1200	2000	3000
N_4		800	1000	400
Step 2	Max PV	2400	2000	3000
	Min PV	800	1000	400
	Value of H	0.7×2400 $+ 0.3 \times 800$ $= 1920$	0.7×2000 $+ 0.3 \times 1000$ $= 1700$	0.7×3000 $+ 0.3 \times 400$ $= 2180$
Step 3		$\max\{1920, 1700, 2180\} = 2180$		

Step 3: We select the maximum value of H from among the values obtained in Step 2.

Therefore, $\max\{1920, 1700, 2180\} = 2180$, which corresponds to the course of action A_3 and is also shown against the stub 'Step 3' in Table 9.22. Hence, A_3 is the optimum course of action under Hurwicz criterion.

iv) It is given that the payoff values represent losses. So, under the regret criterion we follow the three steps given below:

Step 1: In this step, we first obtain the opportunity loss matrix. Since it is given that payoff values represent losses, to obtain opportunity loss table, we calculate the minimum payoff value (Min PV) for each state of nature as follows:

Min PV for state of nature $N_1 = \{1500, 1800, 2500\} = 1500$,
 Min PV for state of nature $N_2 = \{2400, 1600, 2000\} = 1600$,
 Min PV for state of nature $N_3 = \{1200, 2000, 3000\} = 1200$, and
 Min PV for state of nature $N_4 = \{800, 1000, 400\} = 400$.

Now, we subtract the minimum payoff value (1500) from all payoff values corresponding to different courses of action for the state of nature N_1 . Similarly, the values are calculated for the states of nature N_2, N_3 and N_4 and shown in Table 9.23.

Step 2: We identify the maximum regret amount for each course of action from the opportunity loss table obtained in Step 1. This is shown in Table 9.23 against the stub 'Step 2'.

Step 3: We select the minimum regret amount from among the regret amounts obtained in Step 2.

Min $\{800, 800, 1800\} = 800$, which corresponds to the courses of action A_1 and A_2 and is also shown against the stub 'Step 3' in Table 9.23. Hence, both A_1 and A_2 are the optimum courses of action under the regret criterion.

Table 9.23: Calculation for Regret Criterion

States of Nature	Courses of Action		
	A_1	A_2	A_3
N_1	$1500 - 1500 = 0$	$1800 - 1500 = 300$	$2500 - 1500 = 1000$
N_2	$2400 - 1600 = 800$	$1600 - 1600 = 0$	$2000 - 1600 = 400$
N_3	$1200 - 1200 = 0$	$2000 - 1200 = 800$	$3000 - 1200 = 1800$
N_4	$800 - 400 = 400$	$1000 - 400 = 600$	$400 - 400 = 0$
Step 2	800	800	1800
Step 3	$\min\{800, 800, 1800\} = 800$		

v) We do not have the information that the states of nature N_1, N_2, N_3 and N_4 have different probabilities. So we can use the Laplace criterion. The calculations are explained in the following three steps:

Step 1: We assign equal probability to each state of nature. Here we have four states of nature. So each state of nature is assigned the probability $1/4$.

Step 2: We calculate the expected payoff value (EPV) for each course of action as follows:

$$\text{EPV for course of action } (A_1) = \frac{1500 + 2400 + 1200 + 800}{4} = 1475$$

$$\text{EPV for course of action } (A_2) = \frac{1800 + 1600 + 2000 + 1000}{4} = 1600$$

$$\text{EPV for course of action } (A_3) = \frac{2500 + 2000 + 3000 + 400}{4} = 1975$$

Step 3: We select the minimum EPV from among the values obtained in Step 2:

$\min\{1475, 1600, 1975\} = 1475$, which corresponds to the course of action A_1 . Hence, A_1 is the optimum course of action under the Laplace criterion.

UNIT 10 DECISION MAKING PROCESS

Structure

- 10.1 Introduction
 - Objectives
- 10.2 Decision Making Under Risk
 - Expected Monetary Value (EMV) Criterion
 - Expected Opportunity Loss (EOL) Criterion
 - Expected Profit with Perfect Information (EPPI) and Expected Value of Perfect Information (EVPI)
- 10.3 Decision Tree Analysis
- 10.4 Summary
- 10.5 Solutions/Answers

10.1 INTRODUCTION

In Unit 9, we have introduced the components of decision making. You have learnt about courses of action, states of nature, payoff values, payoff matrix and opportunity loss table. We have also introduced various types of decision making environments, namely:

- Decision making under certainty;
- Decision making under uncertainty; and
- Decision making under risk.

The first two environments have been discussed in Unit 9 and we discuss the third environment in Sec. 10.2 of this unit. Note that the situations discussed in Unit 9 and Sec. 10.2 are the ones wherein the decision maker has to make only one decision. However, in many real life situations, the decision maker has to make a sequence of decisions. This means that he/she has to select an optimum course of action more than once, because the decision taken by the decision maker at a given stage generally leads to the next stage. Such multistage decision making problems are solved using decision tree analysis and roll-back technique and we discuss them in Sec. 10.3.

In the next unit, we shall introduce game theory and solve two-person-zero-sum games with saddle point.

Objectives

After studying this unit, you should be able to:

- select an optimum course of action by applying the expected monetary value (EMV) and expected opportunity loss (EOL) criteria;
- draw decision tree diagrams for multistage decision making problems; and
- solve multistage decision making problems by applying the roll-back technique.

10.2 DECISION MAKING UNDER RISK

In the environment, 'decision making under certainty' discussed in Unit 9, we had only one state of nature and so there was no question of decision making for different states of nature. In the environment of 'decision making under uncertainty' there were more than one states of nature but we had no data

regarding the occurrence of the states of nature in terms of probabilities. For the environment of uncertainty, we have discussed five criteria to select an optimum course of action.

We now discuss decision making under risk. In this environment, we have additional information about the occurrence of the states of nature: we have the past data containing information about the occurrence of different states of nature. We know either:

- 1) The directly available probabilities of occurrence of different states of nature
or
- 2) The frequency data for different states of nature which can be converted into probabilities using the relative frequency approach of probability (we have explained relative frequency approach of probability in Unit 2 of MST-003)
or
- 3) Subjective probabilities on the basis of experience of individuals (we have explained subjective probabilities in Unit 2 of MST-003).

In all cases, we have with us the probabilities of occurrence of different states of nature in the environment of decision making under risk. The following criteria are used to select an optimum course of action in this environment:

- (i) Expected Monetary Value (EMV) Criterion,
- (ii) Expected Opportunity Loss (EOL) Criterion.

Let us now explain these criteria.

10.2.1 Expected Monetary Value (EMV) Criterion

In this criterion, we first form the payoff table or payoff matrix if it is not already given. Then for each course of action we find the expected value by multiplying the payoff value for each course of action one at a time with the probabilities of the corresponding state of nature. The resulting values are called the **expected monetary values** (EMVs). Next we select the maximum of the EMVs in the case of profit or gain, and the minimum of the EMVs in the case of loss or cost. In the case of profit or gain the course of action corresponding to the maximum expected monetary value is the optimum course of action according to this criterion. And in the case of loss or cost, the course of action corresponding to the minimum expected monetary value is the optimum course of action according to this criterion.

We follow the steps given below for the calculations of this criterion:

Step 1: If the payoff table or payoff matrix is already given, Step 1 is not needed. Otherwise, we first define the courses of action and states of nature and then obtain the payoff table or payoff matrix for the given situation. We also add one more column to the table indicating the probabilities of different states of nature.

Step 2: To obtain the expected monetary value (EMV) for each course of action, we multiply the payoff value of each course of action with the probability of the corresponding state of nature and then add the results. For example, let $x_{1j}, x_{2j}, \dots, x_{mj}$ be the payoff values for the j^{th} course of action corresponding to m states of nature $N_1, N_2, \dots, N_m$ and let $p_1, p_2, \dots, p_m$ be the corresponding probabilities of these m states of

nature, respectively. Then the expected monetary value (EMV) for the j^{th} course of action is given as:

$$\text{EMV for the } j^{\text{th}} \text{ course of action} = p_1x_{1j} + p_2x_{2j} + \dots + p_mx_{mj} = \sum_{i=1}^m p_i x_{ij} \dots (1)$$

Step 3: We select the maximum expected monetary value from among the expected monetary values obtained in Step 2 if payoff values represent profit or gain. We select the minimum EMV if the values represent loss or cost.

Step 4: Under this criterion, the course of action corresponding to the maximum (or minimum) EMV selected in Step 3 will be the optimum course of action.

Let us consider an example to explain the steps involved in this criterion.

Example 1: A vendor buys newspapers at the rate of Rs 3 per newspaper and sells at the rate of Rs 4 per newspaper. Assume that a newspaper which is not sold on the same day goes to scrap and pays him Rs 0.50 as regret value. The information for the past 200 days about the sale of the newspapers is shown in Table 10.1.

Table 10.1: Frequency Table representing Demand of Newspapers over past 200 Days

Number of Newspapers Demanded	200	204	206	208	Total
Number of Days	40	100	40	20	200

On the basis of this information, how many newspapers should be bought by the vendor so that his profit is maximum?

Solution:

Step 1: On the basis of the given information, it is clear that the vendor should buy either 200, 204, 206 or 208 newspapers per day. Since the number of newspapers he will buy is under his control, purchases of newspapers form the courses of action. If we denote these courses of action by A_1, A_2, A_3, A_4 , respectively, then we have

$$A_1 = 200, A_2 = 204, A_3 = 206, A_4 = 208$$

But the future demand of newspapers on any day is not under his control. So the demands of newspapers form the states of nature. If we denote these states of nature by N_1, N_2, N_3, N_4 , respectively, then we have

$$N_1 = 200, N_2 = 204, N_3 = 206, N_4 = 208$$

Now, the frequencies corresponding to these demands can be used to calculate probabilities (using the relative frequency approach of probability discussed in Unit 2 of MST-003) and for the states of nature N_1, N_2, N_3, N_4 , these are given as:

$$\frac{40}{200} = 0.2, \frac{100}{200} = 0.5, \frac{40}{200} = 0.2, \frac{20}{200} = 0.1, \text{ respectively.}$$

Let us now calculate the payoff values.

The cost of a newspaper = Rs 3

The selling price of a newspaper = Rs 4

Profit gained by the vendor on selling one newspaper = Rs (4 – 3)

$$= \text{Rs } 1$$

Loss to the vendor on an unsold newspaper = Rs $(3 - 0.5) = \text{Rs } 2.5$

$$\begin{aligned} \therefore \text{conditional profit} &= (\text{profit on a sold newspaper}) \times (\text{Number of newspapers sold}) \\ &\quad - (\text{Loss on an unsold newspaper}) \times (\text{Number of newspapers unsold}) \\ &= 1 \times (\text{Number of newspapers sold}) \\ &\quad - (2.5) \times (\text{Number of newspapers unsold}) \end{aligned}$$

For example, for $A_1 = 200$, $N_1 = 200$,

$$\text{conditional profit} = 1 \times 200 - 2.5 \times 0 = 200$$

For $A_1 = 200$, $N_2 = 204$,

$$\text{conditional profit} = 1 \times 200 - 2.5 \times 0 = 200$$

Similarly, other calculations can be done as shown in Table 10.2 given below:

Table 10.2: Payoff Table for the Vendor

States of Nature	Courses of Action				
	Prob.	200 (A_1)	204 (A_2)	206 (A_3)	208 (A_4)
200 (N_1)	0.2	200	$1 \times 200 - 2.5 \times 4 = 190$	$1 \times 200 - 2.5 \times 6 = 185$	$1 \times 200 - 2.5 \times 8 = 180$
204 (N_2)	0.5	200	$1 \times 204 - 2.5 \times 0 = 204$	$1 \times 204 - 2.5 \times 2 = 199$	$1 \times 204 - 2.5 \times 4 = 194$
206 (N_3)	0.2	200	204	206	$1 \times 206 - 2.5 \times 2 = 201$
208 (N_4)	0.1	200	204	206	208

Step 2: In this step, we have to calculate the expected monetary value for each course of action. Expected monetary values (EMVs) for different courses of action are given by:

$$\begin{aligned} \text{EMV for } A_1 &= 0.2 \times 200 + 0.5 \times 200 + 0.2 \times 200 + 0.1 \times 200 \\ &= (0.2 + 0.5 + 0.2 + 0.1) \times 200 = 1 \times 200 = 200 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_2 &= 0.2 \times 190 + 0.5 \times 204 + 0.2 \times 204 + 0.1 \times 204 \\ &= 38 + 102 + 40.8 + 20.4 = 201.2 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_3 &= 0.2 \times 185 + 0.5 \times 199 + 0.2 \times 206 + 0.1 \times 206 \\ &= 37 + 99.5 + 41.2 + 20.6 = 198.3 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_4 &= 0.2 \times 180 + 0.5 \times 194 + 0.2 \times 201 + 0.1 \times 208 \\ &= 36 + 97 + 40.2 + 20.8 = 194 \end{aligned}$$

Step 3: $\max \{200, 201.2, 198.3, 194\} = 201.2$

Step 4: Max EMV corresponds to the course of action A_2 . Hence, under this criterion, A_2 is the optimum course of action.

10.2.2 Expected Opportunity Loss (EOL) Criterion

We have already explained how to obtain the opportunity loss table in Sec. 9.4.4 of Unit 9. This criterion suggests the course of action which minimizes our expected opportunity loss. The steps involved in the procedure of this criterion are the same as in the expected monetary value (EMV) criterion except that

instead of dealing with payoff values, here we deal with opportunity loss values. We follow the steps explained below:

Step 1: If the payoff table or payoff matrix is already given, then Step 1 is not needed. Otherwise, we first define the courses of action, states of nature and then obtain the payoff table. We also add one more column indicating the probabilities of different states of nature.

Step 2: We obtain the opportunity loss values or regret values or conditional opportunity loss values for **each** state of nature by subtracting

- all payoff values corresponding to each state of nature from their respective maximum payoff values in case of profit or gain.
- or
- the minimum payoff value corresponding to each state of nature from all other payoff values of the states of nature in case of cost or loss.

The calculation has been explained in Tables 9.4 and 9.5, respectively, in Unit 9.

Step 3: Next, we obtain the expected opportunity loss values for each course of action by finding the sum of the products of the opportunity loss values of the course of action with the probabilities of the corresponding states of nature as explained in Example 2 given below.

Step 4: Finally, we select the minimum from among the expected opportunity loss values calculated in Step 3. The course of action corresponding to the minimum expected opportunity loss value will be the optimum course of action.

Example 2: A company has decided to buy an equipment for its powerhouse station located in a remote area. But this equipment contains an expensive part, which is subject to random failure. The failure data of the same part on the basis of the experience of other users is given in Table 10.3.

Table 10.3: Probability Distribution of the Random Variable, Number of Failures

Number of Failures	0	1	2	3 and more
Probability of Failure	0.70	0.20	0.10	0.00

If the company purchases spares of this part at the time of purchasing the equipment, it costs Rs 6000 per unit. If it is ordered after the failure of the part during its operation, the total cost including the cost of down time of the equipment is Rs 30000. Assume that there is no scrap value of the part. On the basis of this information, what should the suggestion of a decision maker to the company be in each of the following cases?

- (i) What is the optimal number of spares the company should buy at the time of purchasing the equipment using the EMV criterion?
- (ii) What is the optimal course of action using the EOL criterion?

Solution: We first define the states of nature and courses of action. Then we shall obtain the payoff matrix. The purchases of spare parts of the equipment, (when the equipment is purchased or after the failure of the parts) are under the control of the company and so form the courses of action. If we denote these courses of action by A_1 , A_2 and A_3 , then we can write:

A_1 : No spare part was purchased at the time of purchasing the equipment

A_2 : One spare part was purchased at the time of purchasing the equipment

Down time cost includes all those costs and losses that occur during the entire period in which the system remains in non-operating mode.

A_3 : Two spare parts were purchased at the time of purchasing the equipment

But the number of spares required by the company depends on the number of failures, which is a random event and not under the control of the company. So these numbers correspond to the states of nature. Since the probability of 3 or more failures is zero from Table 10.3, there are only three states of nature

N_1, N_2, N_3 as given below:

N_1 : No failure occurs

N_2 : One failure occurs

N_3 : Two failures occur

Let x_{ij} denotes the payoff value corresponding to the i^{th} state of nature N_i and j^{th} course of action A_j . We first carry out the calculations shown in Table 10.4.

Table 10.4: Calculation(s) of Total Cost for Different Combinations of Courses of Action and States of Nature

States of Nature	Courses of Action	Cost when Spare(s) is/are Purchased at the Time when Equipment is Purchased	Cost when Company Orders after Occurrence of Failure	Total Cost
0	0	0	0	0
0	1	6000	0	6000
0	2	12000	0	12000
1	0	0	30000	30000
1	1	6000	0	6000
1	2	12000	0	12000
2	0	0	60000	60000
2	1	6000	30000	36000
2	2	12000	0	12000

Then we obtain the payoff table (Table 10.5) using the 9 payoff values given in the last column of Table 10.4.

Table 10.5: Payoff Table for the Data of Example 2

States of Nature	Probability	Courses of Action		
		A_1	A_2	A_3
N_1	0.70	0	6000	12000
N_2	0.20	30000	6000	12000
N_3	0.10	60000	36000	12000

(i) We now find the optimal number of spares that should be purchased by the company using the EMV criterion.

Step 1: We have already obtained the payoff table, which also includes a column indicating probabilities of different states of nature.

Step 2: Next, we calculate the expected monetary value (EMV) for each course of action.

$$\begin{aligned}\text{EMV for } A_1 &= 0.70 \times 0 + 0.20 \times 30000 + 0.10 \times 60000 \\ &= 0 + 6000 + 6000 = 12000\end{aligned}$$

$$\begin{aligned}\text{EMV for } A_2 &= 0.70 \times 6000 + 0.20 \times 6000 + 0.10 \times 36000 \\ &= 4200 + 1200 + 3600 = 9000\end{aligned}$$

$$\begin{aligned}\text{EMV for } A_3 &= 0.70 \times 12000 + 0.20 \times 12000 + 0.10 \times 12000 \\ &= 8400 + 2400 + 1200 = 12000\end{aligned}$$

Step 3: Here payoff values represent a cost to the company. So, in this step, instead of the maximum EMV we select the minimum EMV from among the values obtained in Step 2.

$$\therefore \min \{12000, 9000, 12000\} = 9000$$

Step 4: Minimum EMV as obtained in Step 3 corresponds to the course of action A_2 . Hence, the suggestion of the decision maker to the company under EMV criterion is that the company should buy one spare part of the equipment to minimise the cost.

(ii) Now, we use an EOL criterion to find the optimum course of action.

Step 1: We have already obtained the payoff table, which includes a column indicating the probabilities of different states of nature.

Step 2: In this step, we obtain the opportunity loss values (or regret values or conditional opportunity loss values) for each state of nature. In this case, the payoff values represent costs. So, the opportunity loss values are obtained by subtracting the minimum payoff value corresponding to each state of nature from all other payoff values of the states of nature as shown in Table 10.6.

Table 10.6: Opportunity Loss Table for the Company

States of Nature	Prob.	Courses of Action		
		A_1	A_2	A_3
N_1	0.70	$0 - 0 = 0$	$6000 - 0 = 6000$	$12000 - 0 = 12000$
N_2	0.20	$30000 - 6000 = 24000$	$6000 - 6000 = 0$	$12000 - 6000 = 6000$
N_3	0.10	$60000 - 12000 = 48000$	$36000 - 12000 = 24000$	$12000 - 12000 = 0$

Step 3: We now obtain the expected opportunity loss (EOL) value for each course of action as follows:

$$\begin{aligned} \text{EOL for } A_1 &= 0.70 \times 0 + 0.20 \times 24000 + 0.10 \times 48000 \\ &= 0 + 4800 + 4800 = 9600 \end{aligned}$$

$$\begin{aligned} \text{EOL for } A_2 &= 0.70 \times 6000 + 0.20 \times 0 + 0.10 \times 24000 \\ &= 4200 + 0 + 2400 = 6600 \end{aligned}$$

$$\begin{aligned} \text{EOL for } A_3 &= 0.70 \times 12000 + 0.20 \times 6000 + 0.10 \times 0 \\ &= 8400 + 1200 + 0 = 9600 \end{aligned}$$

Step 4: We select the minimum value from among the EOL values obtained in Step 3:

$$\min \{9600, 6600, 9600\} = 6600$$

This corresponds to the course of action A_2 . Hence, the suggestion of the decision maker to the company under EOL criterion is that the company should buy one spare part of the equipment to minimise the cost.

Now, before giving you some exercises to solve for practice, we define two more terms commonly used in decision making, namely, EPPI and EVPI.

10.2.3 Expected Profit with Perfect Information (EPPI) and Expected Value of Perfect Information (EVPI)

We first define EPPI.

Expected Profit with Perfect Information (EPPI)

Expected profit with perfect information (EPPI) is the expected profit that we could make if we had perfect information about the occurrence of a particular state of nature. That is, it represents the expected profit under the environment of certainty and is defined as:

$$EPPI = \sum_i P(N_i) \times \left(\begin{array}{l} \text{payoff value of optimum course} \\ \text{of action for the state of nature } N_i \end{array} \right) \quad \dots (2)$$

Thus, EPPI is the maximum attainable value of EMV under perfect information about the occurrence of the states of nature.

We explain how to calculate EPPI in Example 3, after defining the expected value of perfect information (EVPI).

Expected Value of Perfect Information (EVPI)

The expected value of perfect information (EVPI) represents the excess of the amount earned under the environment of certainty over the expected monetary value (EMV) under the environment of risk. In other words, it represents the value of the maximum amount to be paid to get perfect information. If EMV* represents the expected monetary value under the environment of risk (i.e., probabilities are associated with states of nature) then EVPI is defined as

$$EVPI = EPPI - EMV^* \quad \dots (3)$$

The following example will help you understand the calculations of both EPPI and EVPI.

Example 3: Using the information given in Example 2, obtain the expected profit with perfect information (EPPI) and the expected value of perfect information (EVPI).

Solution: First of all, we have to calculate the expected profit with perfect information (EPPI). By definition,

$$EPPI = \sum_i P(N_i) \times \left(\begin{array}{l} \text{payoff value of optimum course} \\ \text{of action for the state of nature } N_i \end{array} \right)$$

The calculations are shown in Table 10.7:

Table 10.7: Calculations of Expected Profit with Perfect Information (EPPI)

States of Nature	Prob.	Optimum Course of Action	Payoff Value for Optimum Course of Action	Weighted Opportunity Loss
(1)	(2)	(3)	(4)	(2) × (4)
N_1	0.70	A_1	0	$0.70 \times 0 = 0$
N_2	0.20	A_2	6000	$0.20 \times 6000 = 1200$
N_3	0.10	A_3	12000	$0.10 \times 12000 = 1200$

$$\therefore EPPI = -(0 + 1200 + 1200) = -2400$$

[The negative sign is taken because the payoff values represent costs.]

We now calculate the expected monetary value (EMV*) under the environment of risk.

In the solution of part (i) of Example 2, we have already obtained the expected monetary value (EMV) under the environment of risk as Rs 9000. But in Example 2, payoff values represent costs. So if we want to convert the EMV into profit, we have to put a negative sign with it.

Therefore, if EMV^* represents the expected monetary value in the environment of risk, we have

$$EMV^* = -Rs\ 9000$$

Now, we know that the expected value of perfect information (EVPI) is given by

$$EVPI = EPPI - EMV^*$$

$$\therefore EVPI = -2400 - (-9000) = Rs\ 6600$$

You may like to try the following exercises on decision making.

- E1)** A flower seller buys flowers at the rate of Rs 8 per dozen and sells them at the rate of Rs 15 per dozen. He has the following data on the demand for past 100 days:

Table 10.8: Frequency Distribution of Demand of Flowers for Past 100 Days

Demand (in Dozens)	50	51	52	53	Total
Number of Days	10	40	40	10	100

The seller has to give orders to the supplier on the previous day for the next day's sale. Also, the seller donates all the unsold flowers to an orphanage. On the basis of this information, as a decision maker, what will you, suggest to the flower seller about the order to the supplier on the previous day for meeting the next day's demand? Assume that he purchases and sells flowers only in dozens.

- E2)** Identify the optimum course of action under the expected opportunity loss (EOL) criterion for the data of E1.
- E3)** Using the data of E1, obtain the expected profit with perfect information (EPPI) and the expected value of perfect information (EVPI).

10.3 DECISION TREE ANALYSIS

So far we have studied decision criteria under different types of environment to solve decision making problems. But in all those criteria, we were identifying an optimum course of action among the available courses of action at a given point in time. That is, we were dealing with the types of situations in which the decision maker has to take the decision at a single stage. But there may exist situations wherein the decision maker has to make a sequence of decisions. That is, he/she has to select an optimum course of action more than once, because generally one decision taken by the decision maker leads to the next. **Decision tree analysis** is the most useful technique for solving such complex problems. You are familiar with the basic shape of a tree shown in Fig. 10.1. It looks like a sequence of branches and sub branches.



Fig. 10.1: Shape of a tree.

If different branches and sub branches of a tree are associated with courses of action, the states of nature, probabilities of the states of nature and payoff values of a given decision making problem, then the tree so obtained is known as a **decision tree**. But, generally we keep the direction of moving the decision trees from left to right, unlike the natural tree which moves in the vertical direction. A typical representation of the decision tree is shown in Fig. 10.2. Study Fig. 10.2 carefully for a better understanding of the decision tree and the terms given below:

Decision Point: A point in the pictorial representation of a decision tree having courses of action as immediate sub branches is known as a **decision point**. The symbol '□' (square) is used to represent a decision point as shown in Fig. 10.2. In other words, a point from which the branches representing the courses of action come out is known as a decision point.

Chance Point: A point in the pictorial presentation of decision tree having states of nature as immediate sub branches is known as a **chance point, chance node or event point**. The symbol '○' (circle) is used to represent an event point as shown in Fig. 10.2. In other words, a point from which the branches representing the states of nature come out is known as the **chance point or chance node**.

Root Node: We have already mentioned that, generally, we keep the direction of moving the decision tree from left to right and so the leftmost node is known as the **root node**. Root node is represented by the same symbol '□' (square) as the one used for the decision point.

Decision and Chance Branches: We know that any branch of a decision tree either represents a course of action or a state of nature. So the branches which represent the courses of action are known as **decision branches** and the branches which represent the states of nature are known as **chance branches**. Also, probabilities corresponding to different states of nature are written along the corresponding chance branches.

End Points: The points in the pictorial representation of a decision tree which are neither followed by any decision branch nor by any chance branch are known as its **end points**.

Terminal Branches: Any branch which ends at an **end point** is known as the **terminal branch**.

You should learn all the terms explained above with the help of Fig. 10. 2, before studying further.

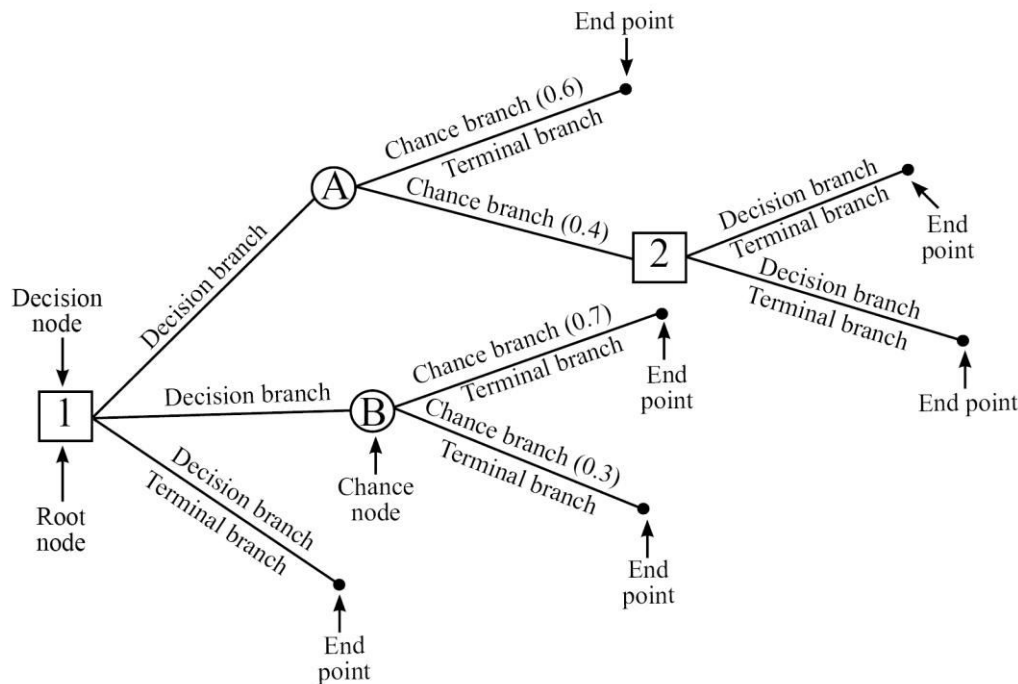


Fig. 10.2: Pictorial representation of a typical decision tree.

To solve multistage decision making problems using decision tree analysis, we apply the roll-back technique, which is explained below:

In the situations where the decision maker has to make a sequence of decisions in multiple stages, the ultimate consequence of the decision taken at the first stage depends on the output of all subsequent decisions that will be taken in future as a result of this decision. Also, the output of the last result is of primary concern for the decision maker. That is why, in decision tree analysis, we start evaluating the output of the decisions from the last decision and move in the backward direction until we reach the root node. Since we work in the backward direction, this technique is known as the **roll-back technique**.

Let us consider some examples to explain the application of the roll-back technique in numerical problems.

Example 4: Mr. Singh had to decide whether or not to drill a tubewell at his farm. In his village, only 40% of the tubewells were successful at 60 feet of depth. Some farmers who did not get water at 60 feet drilled up to 150 feet, but only 30% struck water at 150 feet. The cost of drilling is Rs 300 per foot. Mr. Singh estimated that he would have to pay Rs 20000 for the next 5 years, if he continued to buy water from his neighbour instead of drilling the tubewell, which would have a life of 5 years. Also, if he struck water, the total cost of drawing water for 5 years from his own tubewell would be Rs 3000. If this problem is given to a decision maker, what should his/her suggestion be to Mr. Singh? Assume that all amounts are calculated in terms of the present value.

Solution: So far you have learnt that for solving a decision making problem, we first have to state/identify the courses of action and states of nature. Recall the discussion about the courses of action and states of nature from Sec. 9.4 of Unit 9. The courses of action are under the control of the decision maker. But the states of nature are future outcomes and beyond the control of the decision maker. For this situation, you may like to think for a while as to which of the following situations is under the control of the decision maker and which one is beyond his/her control?

- Deciding whether to drill the tubewell or not.

- Status of water struck at a given depth.

We hope that you have got the correct answer. Deciding whether to drill the tubewell or not is under the control of the decision maker and so forms the courses of action. Thus, the courses of action and states of nature in this situation are as under:

Courses of Action

A₁: Not to drill any tubewell and continue buying water from the neighbour.

A₂: To drill the tubewell up to a depth of 60 feet.

A₃: To drill the tubewell up to a depth of 150 feet, if water is not struck water at 60 feet.

A₄: Not to drill the tubewell up to 150 feet, if water is not struck at 60 feet.

States of Nature

N₁: Water is struck at 60 feet.

N₂: Water is not struck at 60 feet.

N₃: Water is struck at 150 feet.

N₄: Water is not struck at 150 feet.

Further, it is given that

the cost of drilling = Rs 300 per foot

Therefore, the cost of drilling up to 60 feet = Rs 60 × 300 = Rs 18000, and

the cost of drilling further 90 feet (= 150 – 60) = Rs 90 × 300 = Rs 27000

If Mr. Singh does not drill the tubewell, he will have to pay Rs 20000 to his neighbour, as water charges for the next 5 years. But if water is struck, then the cost of watering from his own tubewell for the next five years is Rs 3000.

Now, all these amounts represent costs. So to convert them into profit we have to put a negative sign before each amount. Further, it is a multistage decision making problem. So we have to solve it using the roll-back technique as explained below:

Roll-back Technique: We know that to solve the problem using roll-back technique, we have to draw the decision tree for the problem at hand. As explained earlier in this section, the decision tree is a pictorial representation of the sequence of the decisions. Here, Mr. Singh has to first decide whether he should drill the tubewell or not. This is represented by two branches coming out from the root node 1 (see Fig. 10.3). Similarly, other decisions that lead from these branches are also shown in the decision tree in Fig. 10.3.

Let us now carry out the calculations. We know that in the roll-back technique, we start the calculations from right and move towards the left as explained below:

$$\begin{aligned}\text{Expected monetary value (EMV) at chance node B} &= 0.3 \times (-3000) \\ &\quad + 0.70 \times (-20000) \\ &= -900 - 14000 = -\text{Rs } 14900\end{aligned}$$

But if Mr. Singh decides to drill upto 150 ft, then he has to pay Rs 27000 as drill charges for 90 ft (= 150 – 60). Therefore,

$$\text{total EMV at node B} = \text{Rs}(-14900 - 27000) = -\text{Rs } 41900$$

Since node 2 is a decision node, the EMV at node 2 = $\max \{-41900, -20,000\}$
 $= -\text{Rs } 20000$

Since node A is a chance node, the EMV at node A = $0.4 \times (-3000) + 0.6 \times (-20000)$
 $= -\text{Rs}(1200 + 12000) = -\text{Rs } 13200$

But if Mr. Singh decides to drill up to 60 ft, then he has to pay Rs 18000 as drill charges. Therefore,

total EMV at node A = $\text{Rs}(-13200 - 18000) = -\text{Rs } 31200$

Finally, node 1 is a decision node, therefore EMV at node 1 = $\max \{-31200, -20,000\}$
 $= -\text{Rs } 20000$

The decision tree representing this information is shown in Fig. 10.3.

Now, study node 1 in Fig. 10.3. The EMV at node 1 corresponds to the course of action A_1 .

Hence, the suggestion of the decision maker to Mr. Singh should be:

Mr. Singh should not go for drilling a tubewell at all. He should continue to buy water from his neighbour.

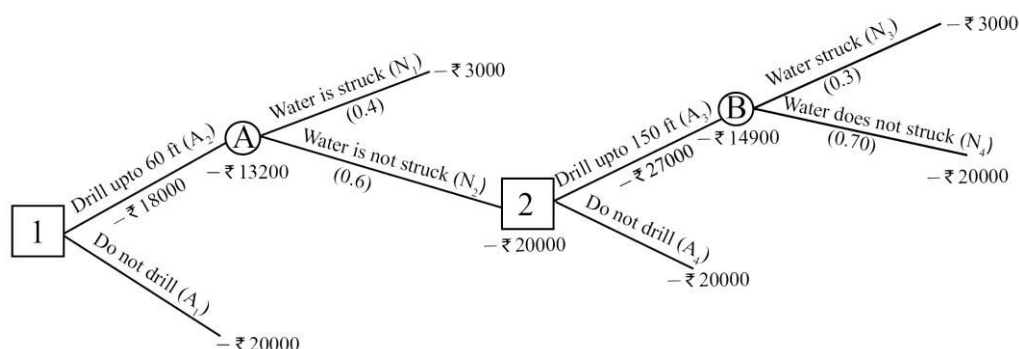


Fig. 10.3: Decision tree for Example 4.

Example 5: An investor has a certain amount of money to invest for five years and only three portfolios X, Y and Z are available to him for investing his money. The estimated profits per rupee on the basis of past experience are shown in Table 10.4. These are subject to the economic condition of the economy in future.

Table 10.9: Estimated Profits per Rupee Subject to the Future Condition of Economy

Economic condition	Probability	Estimated Profit per INR on an Average basis during the Period of Next Five Years		
		In Case of Portfolio X	In Case of Portfolio Y	In Case of Portfolio Z
Economy Grows	0.5	2.5	3	2
Economy Remains Stable	0.2	1.5	2	1.5
Economy Declines	0.3	-0.5	-1.5	-1

Form a decision tree for the given situation and suggest an optimum portfolio to the investor.

Solution: We first define the courses of action and states of nature.

Courses of Action

A_1 : Invest in portfolio X

A_2 : Invest in portfolio Y

A_3 : Invest in portfolio Z

States of Nature

N_1 : Economy grows

N_2 : Economy remains stable

N_3 : Economy declines

Since all amounts represent profit, we shall keep their signs unchanged.

These types of problems (i.e., single stage decision making problems) have already been solved in the previous section without using the decision tree and roll-back technique. But here we want to solve it using the roll-back technique as explained below:

We know that under the roll-back technique, we start the calculations from right and move towards left as follows:

Since the nodes A, B, C are chance nodes, the expected monetary values (EMV) at these nodes are given as:

$$\text{EMV at node A} = 0.5 \times 2.5 + 0.2 \times 1.5 + 0.3 \times (-0.5) = 1.25 + 0.3 - 0.15 = \text{Rs}1.40$$

$$\text{EMV at node B} = 0.5 \times 3.0 + 0.2 \times 2.0 + 0.3 \times (-1.5) = 1.5 + 0.40 - 0.45 = \text{Rs}1.45$$

$$\text{EMV at node C} = 0.5 \times 2 + 0.2 \times 1.5 + 0.3 \times (-1) = 1 + 0.30 - 0.30 = \text{Rs}1.0$$

Since node 1 is the decision node, the EMV at node 1 = $\max \{1.40, 1.45, 1\} = \text{Rs}1.45$

Hence, portfolio B is the optimum portfolio for the investor as per the information given in this case.

The decision tree representing the entire information is shown in Fig 10.4.

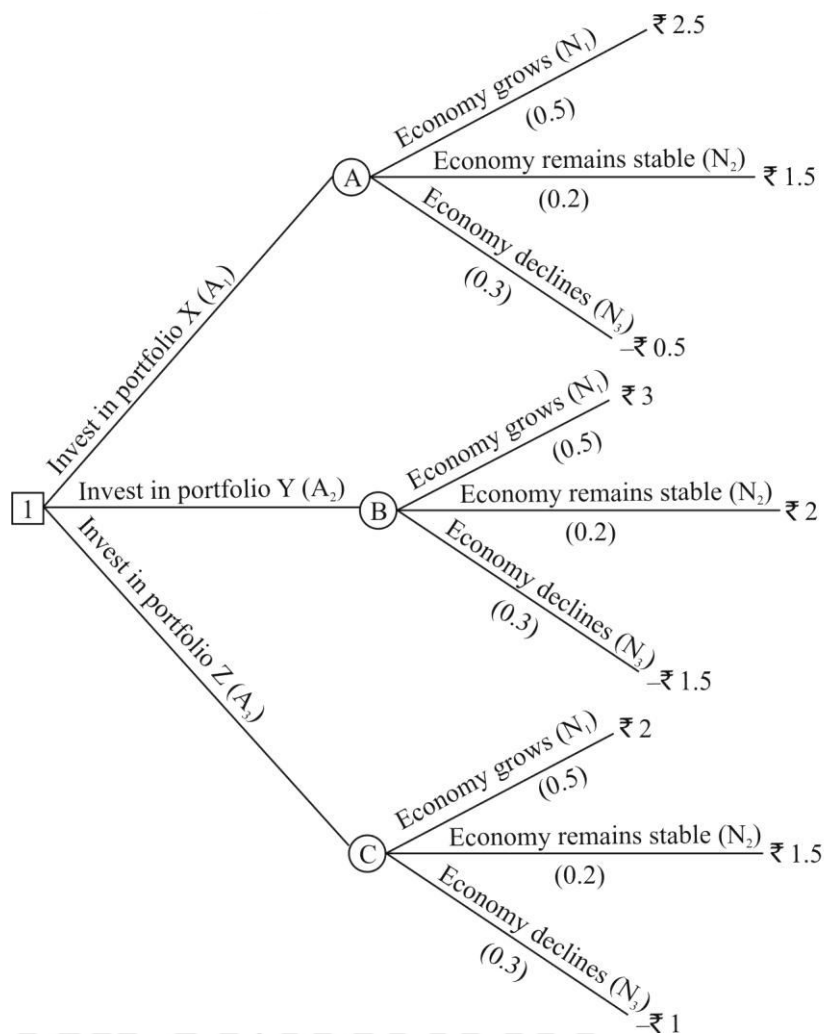


Fig. 10.4: Decision tree for the investor.

You may like to try the following exercises for practice.

- E4)** Company X is planning to launch a new product of direct-to-home (DTH) cables which can be introduced initially either in eastern Madhya Pradesh (MP) or in the entire state of MP. If the product is introduced only in eastern MP, the investment outlay will be Rs 80 lakh for 5 years. After one year, the company can evaluate the project to determine whether it should cover the entire state of MP. For such an expansion, it will have to incur an additional investment of Rs 40 lakh for 4 years. To introduce the product in the entire state of MP right at the beginning would involve an outlay of Rs 140 lakh for five years. The product, in any case, will have a life of 5 years after which it will have zero net value.

If the product is introduced only in eastern MP, demand would be high, moderate or low with the probabilities of 0.5, 0.3 and 0.2, respectively, with annual cash inflow of Rs 30 lakh, Rs 25 lakh and Rs 20 lakh, respectively.

If the product is introduced in the entire state of MP right in the beginning, the demand would be high, moderate or low with the probabilities of 0.4, 0.5 and 0.1, respectively, with annual cash inflows of Rs 50 lakh, 40 lakh and Rs 35 lakh, respectively.

Based on the observed demand in eastern MP, if the product is introduced in the entire state of MP, the following probabilities would exist for high, moderate and low demands:

Table 10.10: Probabilities for High, Moderate and Low Demands if the Product is Introduced in the Entire State of MP as per Observed Demand in Eastern MP

Eastern MP	Entire MP		
	High Demand	Moderate Demand	Low Demand
High Demand	0.7	0.2	0.1
Moderate Demand	0.6	0.2	0.2
Low Demand	0.3	0.2	0.5

On the basis of this information, form a decision tree and using the roll-back technique, obtain the optimal path that should be followed by the company. Assume that all amounts are calculated in terms of the present value.

E5) Suppose one of your friends has a certain amount of money to invest for a period of one year and he/she wants to opt for one of the following three choices available to him/her:

- To lend the money to his neighbour at the interest rate of 12% p.a.
- To invest in real estate.
- To invest in gold.

On the basis of past experience, he/she has the following information about the returns from real estate and gold:

Real Estate: Return may vary in percentage depending on the state of the market as given below:

Table 10.11: Probability of Earning in Percentage for an Investment in Real Estate

Earning in Percentage	Probability
40	0.2
30	0.4
10	0.3
5	0.1

If he/she invests in gold, the chances of earning 15% or 9% return on the investment are 0.6 and 0.4, respectively. On the basis of this information, suggest an optimum option to your friend.

Let us now summarise the main points that we have discussed in this unit.

10.4 SUMMARY

- 1) We find the **expected monetary values** for each course of action by multiplying the payoff value for each course of action with the probability of the corresponding state of nature and taking the sum of these products.
- 2) **Expected opportunity loss** criterion suggests the course of action that minimises our expected opportunity loss.
- 3) If different branches and sub branches of a tree are associated with courses of action, states of nature, probabilities of the states of nature and payoff values of a given decision making problem, then the tree so obtained is known as a **decision tree**.
- 4) **Decision Point:** A point in the pictorial representation of a decision tree having courses of action as immediate sub branches is known as a decision point.

- 5) **Chance Point:** A point in the pictorial representation of decision tree having states of nature as immediate sub branches is known as chance point, chance node or event point.
- 6) **Root Node:** Generally, we keep the direction of moving the decision tree from left to right and so the leftmost node is known as the root node.
- 7) **Decision and Chance Branches:** The branches that represent courses of action are known as **decision branches** and the branches that represent states of nature are known as **chance branches**.
- 8) **End Points:** The points in the pictorial representation of a decision which are neither followed by any decision branch nor by any chance branch are known as end points of the decision tree.

10.5 SOLUTIONS/ANSWERS

E1) The steps involved in the EMV criterion are explained below:

Step 1: We first define the courses of action, states of nature and obtain a payoff matrix. The quantity of flowers the seller will order to the supplier on the previous day for the next day's sale is under the control of the seller. So the flowers he will buy (in dozens) form the courses of action. If A_1, A_2, A_3, A_4 , denote the courses of action, then

$$A_1 = 50, A_2 = 51, A_3 = 52, A_4 = 53$$

But the flowers he will sell (in dozens) on the next day is not under the control of the seller. Hence, the next day's demand form the states of nature. If N_1, N_2, N_3, N_4 denote the states of nature, then

$$N_1 = 50, N_2 = 51, N_3 = 52, N_4 = 53$$

The probabilities for each state of nature are obtained by dividing the respective frequencies by the sum of all frequencies (using the relative frequency approach). Hence, we obtain 0.10, 0.40, 0.40, and 0.10, respectively, as probabilities for these states of nature.

Cost price of 1 dozen flowers = Rs 8

Selling price of 1 dozen flowers = Rs 15

Profit on selling 1 dozen flowers = Rs $(15 - 8) = \text{Rs } 7$

Loss per dozen on unsold flowers = Rs 8

The payoff values can be obtained as

$$\begin{aligned} \text{Payoff value} &= 7 \times (\text{Number of flowers sold in dozens}) \\ &\quad - 8 \times (\text{Number of unsold flowers in dozens}) \end{aligned}$$

Calculation of payoff values for different combinations of courses of action and states of nature are shown in Table 10.12.

Table 10.12: Payoff Table for the Flower Seller

States of Nature	Prob.	Courses of Action			
		A ₁ (50)	A ₂ (51)	A ₃ (52)	A ₄ (53)
N ₁ (50)	0.10	$7 \times 50 - 8 \times 0 = 350$	$7 \times 50 - 8 \times 1 = 342$	$7 \times 50 - 8 \times 2 = 334$	$7 \times 50 - 8 \times 3 = 326$
N ₂ (51)	0.40	$7 \times 50 - 8 \times 0 = 350$	$7 \times 51 - 8 \times 0 = 357$	$7 \times 51 - 8 \times 1 = 349$	$7 \times 51 - 8 \times 2 = 341$
N ₃ (52)	0.40	$7 \times 50 - 8 \times 0 = 350$	$7 \times 51 - 8 \times 0 = 357$	$7 \times 52 - 8 \times 0 = 364$	$7 \times 52 - 8 \times 1 = 356$
N ₄ (53)	0.10	$7 \times 50 - 8 \times 0 = 350$	$7 \times 51 - 8 \times 0 = 357$	$7 \times 52 - 8 \times 0 = 364$	$7 \times 53 - 8 \times 0 = 371$

Step 2: We now obtain the expected monetary value (EMV) for each course of action as follows:

$$\begin{aligned} \text{EMV for } A_1 &= 0.10 \times 350 + 0.40 \times 350 + 0.40 \times 350 + 0.10 \times 350 \\ &= (0.10 + 0.40 + 0.40 + 0.10) \times 350 = 1 \times 350 = 350 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_2 &= 0.10 \times 342 + 0.40 \times 357 + 0.40 \times 357 + 0.15 \times 357 \\ &= 34.2 + (0.40 + 0.40 + 0.10) \times 357 = 34.2 + 0.9 \times 357 \\ &= 34.2 + 321.3 = 355.5 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_3 &= 0.10 \times 334 + 0.40 \times 349 + 0.40 \times 364 + 0.10 \times 364 \\ &= 33.4 + 139.6 + 145.6 + 36.4 = 355 \end{aligned}$$

$$\begin{aligned} \text{EMV for } A_4 &= 0.10 \times 326 + 0.40 \times 341 + 0.40 \times 356 + 0.10 \times 371 \\ &= 32.6 + 136.4 + 142.4 + 37.1 = 348.5 \end{aligned}$$

Step 3: $\text{Max}\{350, 355.5, 355, 348.5\} = 355.5$

Step 4: The maximum EMV corresponds to the course of action A_2 .

Hence, under this criterion, A_2 is the optimum course of action.

E 2) To identify an optimum course of action using the expected opportunity loss (EOL) criterion, we follow the steps given below:

Step 1: We first define the courses of action, states of nature and obtain the payoff table with one additional column of probabilities. But we have already done these calculations while solving E1.

Step 2: We now obtain the opportunity loss matrix. The payoff values in the payoff table obtained in Step 1 represent profit of the flower seller. So to obtain the opportunity loss table, we first calculate the maximum payoff value (Max PV) for each state of nature as follows:

$$\text{Max PV for state of nature } N_1 = \max\{350, 342, 334, 326\} = 350$$

$$\text{Max PV for state of nature } N_2 = \max\{350, 357, 349, 341\} = 357$$

$$\text{Max PV for state of nature } N_3 = \max\{350, 357, 364, 356\} = 364$$

$$\text{Max PV for state of nature } N_4 = \max\{350, 357, 364, 371\} = 371$$

Then we subtract all payoff values corresponding to different courses of action for the state of nature N_1 from 350. We do the same calculation for the states of nature N_2 , N_3 and N_4 by subtracting the payoffs from 357, 364 and 371, respectively, as shown in Table 10.13.

Table 10.13: Opportunity Loss Table for the Flower Seller

States of Nature	Prob.	Courses of Action			
		$A_1 (50)$	$A_2 (51)$	$A_3 (52)$	$A_4 (53)$
$N_1 (50)$	0.10	$350 - 350 = 0$	$350 - 342 = 8$	$350 - 334 = 16$	$350 - 326 = 24$
$N_2 (51)$	0.40	$357 - 350 = 7$	$357 - 357 = 0$	$357 - 349 = 8$	$357 - 341 = 16$
$N_3 (52)$	0.40	$364 - 350 = 14$	$364 - 357 = 7$	$364 - 364 = 0$	$364 - 356 = 8$
$N_4 (53)$	0.10	$371 - 350 = 21$	$371 - 357 = 14$	$371 - 364 = 7$	$371 - 371 = 0$

Step 3: We now obtain the expected opportunity loss (EOL) values for each course of action as follows:

$$\begin{aligned} \text{EOL for } A_1 &= 0.10 \times 0 + 0.40 \times 7 + 0.40 \times 14 + 0.10 \times 21 \\ &= 0 + 2.8 + 5.6 + 2.1 = 10.5 \end{aligned}$$

$$\begin{aligned} \text{EOL for } A_2 &= 0.10 \times 8 + 0.40 \times 0 + 0.40 \times 7 + 0.10 \times 14 \\ &= 0.8 + 0 + 2.8 + 1.4 = 5 \end{aligned}$$

$$\begin{aligned} \text{EOL for } A_3 &= 0.10 \times 16 + 0.40 \times 8 + 0.40 \times 0 + 0.10 \times 7 \\ &= 1.6 + 3.2 + 0 + 0.7 = 5.5 \end{aligned}$$

$$\begin{aligned} \text{EOL for } A_4 &= 0.10 \times 24 + 0.40 \times 16 + 0.40 \times 8 + 0.10 \times 0 \\ &= 2.4 + 6.4 + 3.2 + 0 = 12 \end{aligned}$$

Step 4: We select the minimum from among the EOL values obtained in Step 3:

$$\text{Min} \{10.5, 5, 5.5, 12\} = 5$$

This corresponds to the course of action A_2 . Hence, the optimum course of action under the EOL criterion is A_2 .

E3) First of all, we have to calculate expected profit with perfect information (EPPI) and by definition it is given as

$$\text{EPPI} = \sum_i P(N_i) \times \left(\text{payoff value of optimum course of action for the state of nature } N_i \right)$$

The calculations are shown in Table 10.14.

Table 10.14: Calculation for Expected Profit with Perfect Information (EPPI)

States of Nature	Prob.	Optimum Course of Action	Payoff Value for Optimum Course of Action	Weighted Opportunity Loss
(1)	(2)	(3)	(4)	(2) $\times$ (4)
N_1	0.10	A_1	350	$0.10 \times 350 = 35$
N_2	0.40	A_2	357	$0.40 \times 357 = 142.8$
N_3	0.40	A_3	364	$0.40 \times 364 = 145.6$
N_4	0.10	A_4	371	$0.10 \times 371 = 37.1$

Thus, the expected profit with perfect information (EPPI) is given by

$$\text{EPPI} = 35 + 142.8 + 145.6 + 37.1 = 360.5$$

The expected monetary value (EMV*) under the environment of risk has already been calculated in the solution of E1, i.e.,

$$EMV^* = 348.5.$$

Thus, the expected value of perfect information (EVPI) is given by

$$EVPI = EPPI - EMV^* = 360.5 - 348.5 = \text{Rs } 12$$

E4) We first define the courses of action and states of nature:

Courses of Action

A_1 : Launch the new product in eastern MP

A_2 : Launch the new product in entire MP

A_3 : Expand the product in entire MP after launching it in eastern MP

A_4 : Do not expand the product in entire MP after launching it in eastern MP

States of Nature

N_1 : Demand is high

N_2 : Demand is moderate

N_3 : Demand is low

We now use the roll-back technique to obtain an optimum sequence of courses of action.

We know that under the roll-back technique, we start the calculations from right and move towards left as explained below:

Since node C is a chance node, the expected monetary value (EMV) per annum at node C

$$= \text{Rs} [0.7 \times 50 + 0.2 \times 40 + 0.1 \times 35] \text{ lakh} = \text{Rs } 46.5 \text{ lakh}$$

$$\text{EMV at chance node C for 4 years} = \text{Rs} (4 \times 46.5 - 40) \text{ lakh} = 146 \text{ lakh}$$

Since node 2 is a decision node, EMV at node 2

$$= \text{Rs} [\max \{146, 0\} + 30 \text{ lakh cash inflow}] \text{ lakh} = \text{Rs } 176 \text{ lakh}$$

$$\text{EMV per annum at chance node D} = \text{Rs} [0.6 \times 50 + 0.2 \times 40 + 0.2 \times 35]$$

$$= \text{Rs } 45 \text{ lakh}$$

$$\therefore \text{EMV at chance node D for 4 years} = \text{Rs} [4 \times 45 - 40] \text{ lakh} = 140 \text{ lakh}$$

$$\text{EMV at decision node 3} = \text{Rs} [\max \{140, 0\} + 25 \text{ lakhs cash inflow}]$$

$$= \text{Rs} [140 + 25] \text{ lakh} = \text{Rs } 165 \text{ lakh}$$

$$\text{EMV per annum at chance node E} = \text{Rs} [0.3 \times 50 + 0.2 \times 40 + 0.5 \times 35] \text{ lakh}$$

$$= \text{Rs } 40.5 \text{ lakh}$$

$$\text{EMV at chance node E for 4 years} = \text{Rs} [4 \times 40.5 - 40] \text{ lakh} = \text{Rs } 122 \text{ lakh}$$

$$\text{EMV at decision node 4} = \text{Rs} [\max \{122, 0\} + 20 \text{ lakh cash inflow}]$$

$$= \text{Rs} [122 + 20] \text{ lakh} = 142 \text{ lakh}$$

Since node A is a chance node, EMV

$$\text{at node A} = \text{Rs} [0.5 \times 176 + 0.3 \times 165 + 0.2 \times 142 - 80] \text{ lakh}$$

= Rs 85.9 lakh

Since node B is a chance node, EMV per annum

at node B = Rs $[0.4 \times 50 + 0.5 \times 40 + 0.1 \times 35]$ lakh = Rs 43.5 lakh

EMV at chance node B for 5 years = Rs $[5 \times 43.5 - 140]$ lakh = Rs 77.5 lakh

Finally, since node 1 is a decision node, EMV at node 1 = $\max \{85.9, 77.5\}$
= 85.9 lakh

Hence, the optimum path for the company X is as follows:

- The company should launch the product initially in eastern MP.
- If the demand of the product is high, the company should launch the product in entire MP.
- In the case of moderate or low demand, the company should not launch the product in entire MP.

All the information given above is shown in the decision tree diagram (Fig. 10. 5).

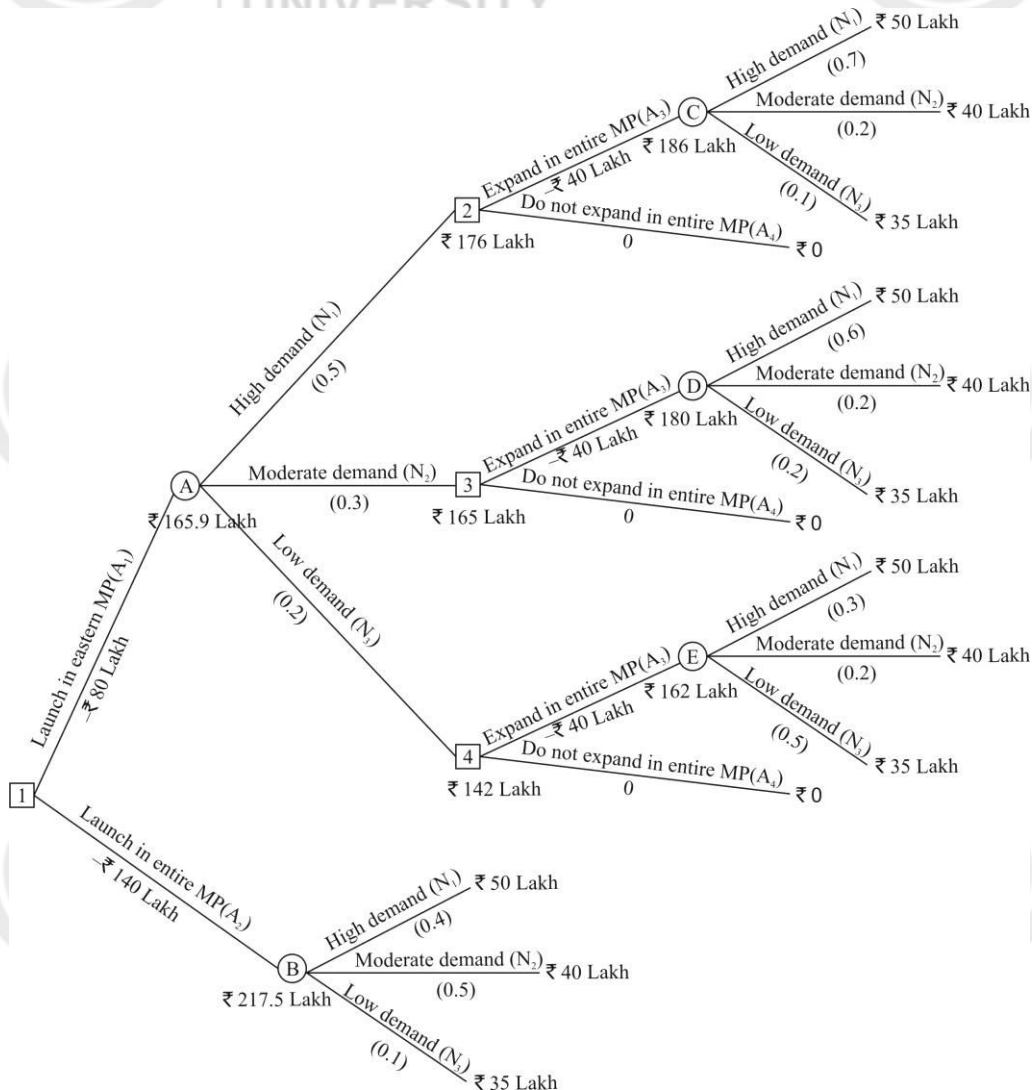


Fig. 10.5: Decision tree for company X.

Note: The above calculations can also be expressed in the form of a table, as shown in Table 10.15.

Node	Course of Action	States of Nature	Probabilities	Conditional Expected Monetary Value (in lakhs)	Expected Monetary Value (EMV) (in lakhs)
2	Expand in Entire MP	High	0.7	50	$0.7 \times 50 = 35$ (per annum)
		Moderate	0.2	40	$0.2 \times 40 = 8$ (per annum)
		Low	0.1	35	$0.1 \times 35 = 3.5$ (per annum)
					Total = 46.5 (per annum) For 4 years = $4 \times 46.5 = 186$ Cost = - 40 EMV = $186 - 40 = 146$
	Do not Expand in Entire MP			0	0
3	Expand in Entire MP	High	0.6	50	$0.6 \times 50 = 30$ (per annum)
		Moderate	0.2	40	$0.2 \times 40 = 8$ (per annum)
		Low	0.2	35	$0.2 \times 35 = 7$ (per annum)
					Total = 45 (per annum) For 4 years = $4 \times 45 = 180$ Cost = - 40 EMV = $180 - 40 = 140$
	Do not Expand in Entire MP			0	0
4	Expand in Entire MP	High	0.3	50	$0.3 \times 50 = 15$ (per annum)
		Moderate	0.2	40	$0.2 \times 40 = 8$ (per annum)
		Low	0.5	35	$0.5 \times 35 = 17.5$ (per annum)
					Total = 40.5 (per annum) For 4 years = $4 \times 40.5 = 162$ Cost = - 40 EMV = $162 - 40 = 122$
	Do not Expand in Entire MP			0	0
1	Launch in Eastern M.P.	High	0.5	$30 + 146 = 176$	$0.5 \times 176 = 88$ (for 5 years)
		Moderate	0.3	$25 + 140 = 165$	$0.3 \times 165 = 49.5$ (for 5 years)
		Low	0.2	$20 + 122 = 142$	$0.2 \times 142 = 28.4$ (for 5 years)
					Total = 165.9 (for 5 years) Cost = - 80 EMV = $165.9 - 80 = 85.9$
	Launch in Entire MP	High	0.4	50	$0.4 \times 50 = 20$ (per annum)
		Moderate	0.5	40	$0.5 \times 40 = 20$ (per annum)
		Low	0.1	35	$0.1 \times 35 = 3.5$ (per annum)
					Total = 43.5 (per annum) For 5 years = $5 \times 43.5 = 217.5$ Cost = - 140 EMV = $217.5 - 140 = 77.5$

Since node 1 is a decision node, the EMV at node 1 = $\max \{85.9, 77.5\} = 85.9$.

This is the same as we calculated without using the table. So the optimum path for the company is the same as suggested earlier.

Now, if you feel comfortable in doing the calculations without using a table, you are free to do so. However, if you feel more comfortable in doing the calculations in the format of a table as shown above, you may adopt the same method.

E 5) First of all, we define the courses of action and states of nature for this situation.

Courses of Action:

A_1 : Lend the money to his/her neighbour

A_2 : Invest in real estate

A_3 : Invest in gold

States of Nature

In case money is loaned to his/her neighbour, there is only one state of nature: He/she will earn 12% in terms of interest for a period of one year.

In the case of investment in real estate there are four states of nature:

N_1 : Property value increases by 40% after one year of investment

N_2 : Property value increases by 30% after one year of investment

N_3 : Property value increases by 10% after one year of investment

N_4 : Property value increases by 5% after one year of investment

In the case of investment in gold there are two states of nature

N_5 : Rate of gold increases by 15% after one year of investment

N_6 : Rate of gold increases by 9% after one year of investment

Before solving this problem using the roll-back technique, let us assume that the money invested is Rs 100. The outcomes of different states of nature on the basis of an investment of Rs 100 are shown against each terminal branch in Fig. 10.6.

Roll-back Technique: We know that under the roll-back technique, we start the calculations from right and move towards left as explained below:

Node A is a chance node, so EMV at node A = $Rs\ 1 \times 112 = Rs\ 112$

Similarly,

$$\begin{aligned}\text{EMV at chance node B} &= Rs(0.2 \times 140 + 0.4 \times 130 + 0.3 \times 110 + 0.1 \times 105) \\ &= Rs(28 + 52 + 33 + 10.5) = Rs\ 123.5\end{aligned}$$

$$\begin{aligned}\text{EMV at chance node C} &= Rs(0.6 \times 115 + 0.4 \times 109) \\ &= Rs(69 + 43.6) = Rs\ 112.6\end{aligned}$$

Now, node 1 is a decision node. So the optimum decision will correspond to the maximum EMV among the EMVs at chance nodes A, B, C.

$$\text{So } \max\{112, 123.5, 112.6\} = 123.5.$$

The decision tree representing this information is shown in Fig. 10.6.

Hence, as a decision maker, you should suggest that your friend should invest in real estate.

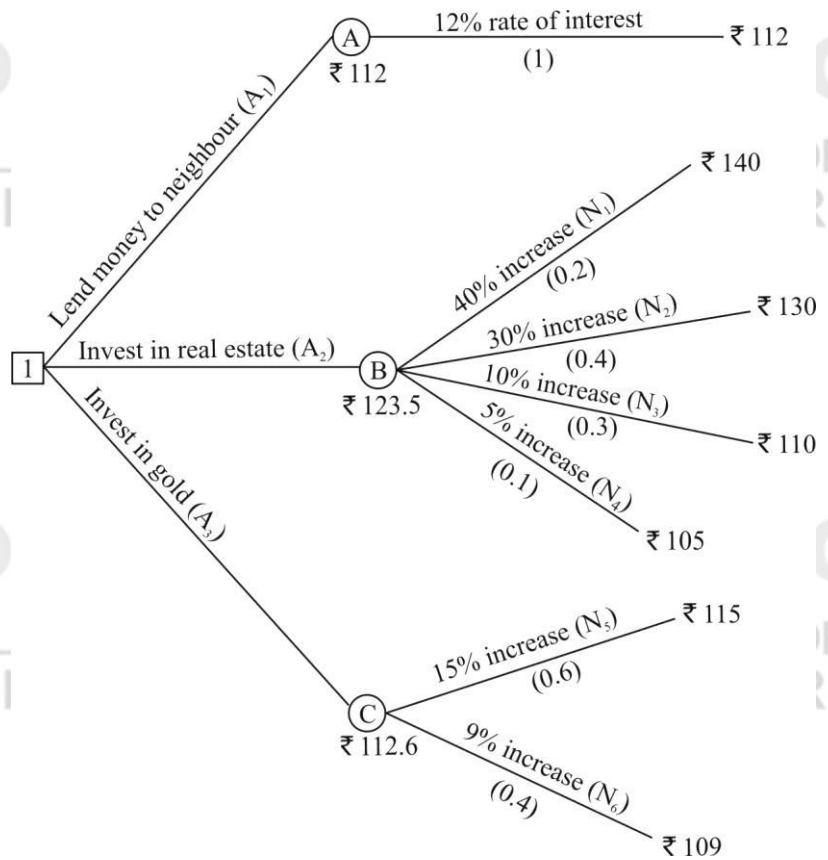


Fig. 10.6: Decision tree for the investment of your friend.

UNIT 11 TWO-PERSON ZERO-SUM GAMES WITH SADDLE POINT

Structure

- 11.1 Introduction
 - Objectives
- 11.2 Key Terms Used in Game Theory
- 11.3 The Maximin-Minimax Principle
- 11.4 Summary
- 11.5 Solutions/Answers

11.1 INTRODUCTION

In “Game Theory”, the word game is not used in the way it is commonly used in different types of sports such as hockey, cricket, football, etc. Also, it does not refer to computer games. In the usual sports or games, the main objective of the opponents is to win the game. But in the games discussed under game theory between two opposing parties with conflicting interests, winning means selecting an optimal strategy, e.g., selecting an optimal course of action as we have discussed in Units 9 and 10. Game theory deals with decision making processes of players in conflicting and competitive situations where the strategy (or action or move) of a player depends upon the move of the opponent.

Recall that in Sec. 9.5 of Unit 9, we have discussed four types of environments under which a decision maker may have to make decisions, namely:

- Decision making under certainty;
- Decision making under uncertainty;
- Decision making under risk; and
- Decision making under conflict.

We have discussed the first three types of environments in detail in Units 9 and 10. We now discuss the fourth type of environment which is present in games between two or more players. In Sec. 11.2, we introduce the key terms used in game theory. Saddle point (explained in Sec. 11.3) is one of the key concepts in game theory. On the basis of whether a saddle point exists in the game or not, games can be further classified as:

- Games with saddle point, and
- Games without saddle point.

The games with saddle point are discussed in the present unit and the games without saddle point are discussed in Unit 12. In Sec. 11.3, we discuss Maximin-minimax principle for solving two-person zero-sum games with saddle point.

In the next unit, we shall discuss the games without saddle point.

Objectives

After studying this unit, you should be able to:

- define the key terms involved in game theory;
- solve the prisoner’s dilemma game; and
- solve two-person zero-sum games using the maximin-minimax principle.

The notion of game theory existed even before John Von Neumann (1903-1957). But it is John Von Neumann who is known as the father of game theory. He first published a paper on a mathematical treatment of game theory in 1928. In 1944 he published a book in collaboration with Oskar Morgenstern on game theory entitled “Theory of Games and Economic Behaviour”.

11.2 KEY TERMS USED IN GAME THEORY

In this section, we introduce the key terms and terminology used commonly in game theory. Then we explain what is meant by a game in game theory. But before doing so, let us consider the following situation:

Suppose two children X and Y agree that:

- Each one of them will simultaneously place a coin on the table.
- Each one of them will show the outcome (head or tail).
- If the faces of both coins match (i.e., either both coins show head or both show tail), child X wins and gets Rs 1 from child Y.
- If the coins do not match, child Y wins and gets Rs 1 from child X.

We can present this information in the form of a matrix as shown below. The first numeral in the four cells having entries $(1, -1)$, $(-1, 1)$, $(-1, 1)$, $(1, -1)$, respectively, represents the amount won by child X and the second numeral represents the amount won by child Y. When child X wins Rs 1 from child Y, the winning amount for X is 1 and the winning amount for Y is -1 (i.e., loss of 1).

		Child Y (Player II)	
		Head	Tail
Child X (Player I)	Head	1, -1	-1, 1
	Tail	-1, 1	1, -1

In game theory, the situation discussed above is known as **Coin Matching Game** and its solution is provided in E2 in Sec. 12.8 of Unit 12.

Let us consider another situation.

Two persons, say X and Y, are arrested by the police with enough evidence for a minor crime. The police suspect that they are responsible for a murder, but do not have enough evidence. Both persons are put in separate cells so that they have no way of communicating with each other. The police starts interrogating them in separate rooms (as shown in Fig.11.1). Each person either confesses or does not confess. Also, each one knows the consequences of confession, which are given below:

- If both of them confess, both go to jail for 5 years.
- If one of them confesses and the other does not, then the one who confessed turns government's witness while the other who did not confess goes to jail for 20 years.
- If both do not confess, both go to jail for one year.

Assume that each one of them has to protect his self-interest, which means that each person tries to act in such a way that he would have to go to jail for a shorter period of time, regardless of the way the other acts. Also assume that they have no way of communicating with each other. What should X and Y do?

We can present this information in matrix form as given below. The first numeral in the four cells having entries $(-5, -5)$, $(0, -20)$, $(-20, 0)$, $(-1, -1)$ represents the time to be spent in prison by person X and the second numeral represents the time to be spent in prison by person Y. A negative sign is attached to the numerals because the time spent in prison is similar to a loss.

		Person Y (Player II)	
		Confesses	Does not Confess
Person X (Player I)	Confesses	-5, -5	0, -20
	Does not Confess	-20, 0	-1, -1

In game theory, the situation discussed above is known as **the Prisoner's Dilemma**. Its solution is provided in Example 4 in Sec. 11.3.



Fig. 11.1: Persons X and Y facing questions of policemen in separate rooms.

We now define some key terms used in game theory.

Player: A participant or competitor who makes decisions is known as a player. A player may be an individual or a group of individuals. For example, in coin matching game, child X and child Y are players, and in the Prisoner's Dilemma game, person X and person Y are players.

Action: The options available to the players are known as actions or courses of action or **moves**.

For example, in coin matching game, the actions are "head" and "tail"; and in the prisoner's dilemma game, "confesses" and "does not confess" are actions.

Play: A play is said to occur when each player selects a course of action.

Strategy: A predetermined rule by which a player decides his/her course(s) of action among the actions available to him/her is known as strategy for the player. A strategy may be of two types:

- (i) **Pure Strategy:** A strategy is said to be a **pure strategy** if the player selects a particular course of action each time, i.e., if a player selects, say, the i^{th} course of action (A_i) each time from among n courses of actions, $A_1, A_2, \dots, A_n$, available to him or her. This means that he/she assigns probability 1 to the i^{th} course of action and zero probability to each of the other courses of action $A_1, A_2, \dots, A_{i-1}, A_{i+1}, \dots, A_n$. So the pure strategy is denoted by $(0, 0, \dots, 0, 1, 0, \dots, 0)$ having values 0 at $(n-1)$ places except the i^{th} position having value 1. For example, in the prisoner's dilemma game, person X will use pure strategy $(1, 0)$ as you will see in the solution of this game, which is provided in Example 4 in Sec. 11.3.
- (ii) **Mixed Strategy:** A strategy for a player is said to be a **mixed strategy** if the player selects a combination of more than one courses of action by assigning a fixed probability to each course of action. That is, if there are n courses of action $A_i, (1 \leq i \leq n)$ available to the player, he/she assigns probabilities $p_1, p_2, \dots, p_n$ to the courses of action $A_1, A_2, \dots, A_n$, respectively, such that $p_1 + p_2 + \dots + p_n = 1$ and $p_i \geq 0$ for all $i, 1 \leq i \leq n$. For example, in the coin matching game, child X will use the mixed strategy $(1/2, 1/2)$ as you will see in the solution of E2, which is provided in Sec. 12.8 of Unit 12.

Note 1: Pure strategy is a particular case of mixed strategy because in the case of pure strategy

$$p_i = 1 \text{ for some } i, \text{ and } p_j = 0 \text{ for all } j \neq i$$

Payoff Values and Payoff Matrix: You have learnt about payoff values and payoff matrices in Unit 9. Here the payoff value means money or anything else

that motivates players and when these payoff values are represented in matrix form, the resulting matrix is known as payoff matrix. For example, in the coin matching game, payoff values are the money that child X gets from child Y or vice versa, while in prisoner's dilemma game, the time (in years) to be spent in prison represents payoff values.

Optimal Strategy: The strategy for a player, which optimises his/her payoff irrespective of the strategy of his /her competitor is known as an optimal strategy.

For example, in the coin matching game, the optimal strategy for child X is $(1/2, 1/2)$, which is explained in the solution of E2 in Sec. 12.8 of Unit 12.

Now, we are in a position to define what we mean by a game in game theory.

Definition of Game: A competitive situation is called a game if

- (i) The number of competitors, called players, is finite.
- (ii) The number of possible courses of action for each player is finite. However, the courses of action need not be the same for each player.
- (iii) Each player selects a course of action simultaneously from among the courses of action available to him/her without directly communicating with the other player.
- (iv) Every combination of courses of action results in an outcome known as payoff value, which motivates the players. Payoff value may represent loss or gain or any other thing of interest. The payoff values may be positive, negative or zero.

n-Person Game: If the number of players involved in the game is $n (> 2)$, it is known as n-person game.

Two-Person Game: If the number of players involved in the game is 2, it is known as two-person game.

Zero-Sum Game: If the algebraic sum of the payoff values of all players after each play is zero, the game is known as a zero-sum game. Mathematically, for zero-sum game,

if a_i , $(1 \leq i \leq n)$ represents the payoff value of the i^{th} player, then $\sum_{i=1}^n a_i = 0$

Non-Zero-Sum Game: A game is said to be non-zero-sum game if there exists at least one play such that the algebraic sum of all the payoff values is not equal to zero. For example, the prisoner's dilemma game is a non-zero-sum game.

Two-Person Zero-Sum Game: If in a game, each payoff value of one player is negative of the payoff value of the other player, it is known as two-person zero-sum game. For example, coin matching game is a two-person zero-sum game.

Note 2: In two-person zero-sum game, if we call the two players as player I and player II, we see that:

each payoff value of player II is **equal in magnitude** to the payoff value of player I but **opposite in sign**.

Hence, if the payoff value of player I is known, the payoff value of player II is automatically known. So, for a two-person zero-sum game, generally, we write the payoff values of only player I instead of writing payoff values of both players. For example, payoff matrix in the case of coin matching game can simply be written as follows:

		Child Y (Player II)	
		Head	Tail
Child X (Player I)	Head	1	-1
	Tail	-1	1

Note 3: In this unit and in Unit 12 we shall limit our discussion to two-person zero-sum games. Only in Example 4, we provide the solution of prisoner's dilemma game, which is a non-zero sum game. So henceforth, in the payoff matrix we shall write only payoff values of player I as explained in Note 2.

11.3 THE MAXIMIN-MINIMAX PRINCIPLE

In Sec. 9.5 of Unit 9, you have studied the maximin-minimax criterion for a decision making situation. The basic idea of the criterion remains the same except that here we are working under the environment of conflict whereas in Unit 9 we solved problems of decision making under uncertainty. The second distinguishing feature is that here two players are making decisions simultaneously, while in Unit 9, there was only one decision maker.

With this clarification, let us define the principle:

Under this principle, first of all, a player lists the worst possible payoff values of all strategies available to him/her. Then he/she selects the strategy corresponding to the optimum payoff value from among the worst possible payoff values.

Let us consider a two-person zero-sum game to explain this principle. The payoff matrix for the game is given below:

		Player B	
		I	II
Player A	I	1	4
	II	2	3
	III	-3	5

If a_{ij} represents the payoff value when player A chooses his/her i^{th} , ($i = \text{I, II, III}$) strategy and player B chooses his/her j^{th} , ($j = \text{I, II}$) strategy, then

$a_{11} = 1 \Rightarrow$ Player A will get Rs 1 from player B

$a_{12} = 4 \Rightarrow$ Player A will get Rs 4 from player B

$a_{21} = 2 \Rightarrow$ Player A will get Rs 2 from player B

$a_{22} = -3 \Rightarrow$ Player A will get Rs 3 from player B

$a_{31} = -3 \Rightarrow$ Player A will pay Rs 3 to player B

$a_{32} = 5 \Rightarrow$ Player A will get Rs 5 from player B

Let us first analyse this game from the point of view of player A: Here payoff values in the payoff matrix represent gains of the player A. So when player A chooses a particular strategy, player B would move in such a way that payoff to player A is minimum for that particular strategy because the interests of player A and player B are conflicting. Thus, if player A employs strategy I, he/she may gain Rs 1 or Rs 4 depending upon the strategy adopted by player B. Now, whatever strategy is adopted by player B, player A will gain at least $\min\{1, 4\} = \text{Rs } 1$. Similarly, if player A employs strategy II, he/she will gain at least $\min\{2, 3\} = \text{Rs } 2$, and if player A employs strategy III, then he/she will gain at least $\min\{-3, 5\} = -\text{Rs } 3$. Obviously, player A would like to opt for the strategy, which maximises his/her minimum gains. Since $\max\{1, 2, -3\} = 2$, player A should adopt strategy II. You may think that if player A adopts strategy III

and player B adopts strategy II, the gain of player A will be Rs 5, which is greater than Rs 2. You are right but there is no guarantee that B will adopt strategy II. He/she may employ strategy I, which will result in a loss of Rs 3 to player A. So, player A should go for strategy II because, in game theory, we assume that both players are equally intelligent.

Now, we analyse this game from the point of view of player B: If player B employs strategy I, then he/she may face a loss of Rs 1, a loss of Rs 2 or a gain of Rs 3 depending upon the strategy adopted by player A. Now, whatever strategy is adopted by player A, player B cannot incur a loss of more than $\max\{1, 2, -3\} = \text{Rs } 2$. Similarly, if player B employs strategy II, he/she cannot incur a loss of more than $\max\{4, 3, 5\} = \text{Rs } 5$. Obviously, player B would like to opt for the strategy, which minimises his/her maximum losses. Since $\min\{2, 5\} = \text{Rs } 2$, player B should adopt strategy I.

From the above discussion, we note that player A should opt for the strategy which corresponds to the maximum payoff value among the row minimum values, i.e., maximum among the minima. Hence, it is known as **maximin value** and is denoted by $\max_i \min_j \{a_{ij}\}$. Player B should opt for the strategy, which corresponds to the minimum payoff value from among the column maximum values, i.e., minimum among the maxima. Hence, it is known as **minimax value** and is denoted by $\min_j \max_i \{a_{ij}\}$.

If $\text{maximin value} = \text{minimax value} = v$ (say),

then v is known as the **value of the game** and the corresponding strategies of players A and B are known as their **optimal strategies**. Also, the position of the element corresponding to the optimal strategies of the two players is known as **saddle point**. In the above example, the value of the game is Rs 2 and strategy II of player A is the optimal strategy for player A while strategy I of player B is the optimal strategy for player B. Also the position of the saddle point is (2,1), i.e., corresponding to second row and first column.

You should follow the steps given below for applying maximin-minimax principle to numerical problems:

Step 1: Identify the minimum element in each row of the payoff matrix and select the largest element among these row minima. This is the maximin value.

Step 2: Identify the maximum element in each column of the payoff matrix and select the minimum among these column maxima. This is the minimax value.

Step 3: If $\text{maximin value} = \text{minimax value} = v$ (say) i.e., $\max_i \min_j \{a_{ij}\} = \min_j \max_i \{a_{ij}\} = v$ (say) and lies at the intersection of the row of maximin value and the column of the minimax value, we say that the game is solved by **maximin-minimax principle**. The maximin (minimax) value v is called the **value of the game**. The strategies corresponding to the row of maximin value and the column of minimax value are termed the **pure optimal strategies** for player A and player B, respectively. Also, the position of the element where the row of maximin value and the column of the minimax value intersect is known as the **saddle point**.

Note 4: If $\text{maximin value} \neq \text{minimax value}$, we say that the game does not have a saddle point. So we cannot obtain the solution of the game in

terms of pure strategies by applying the maximin-minimax principle. Games without a saddle point have mixed strategies as their solutions, which are discussed in Unit 12.

Note 5: By solution of a game, we mean an optimal strategy for each player and the value of the game.

We now define a few more terms before taking up examples.

Lower and Upper Values of the Game: If a_{ij} denotes the payoff value when player I chooses his/her i^{th} , ($i=1, 2, \dots, m$) strategy and player II chooses his/her j^{th} , ($j=1, 2, \dots, n$) strategy, then the maximin value = $\max_i \min_j \{a_{ij}\}$ is

known as the **lower value** of the game and is generally denoted by $\underline{v}$. The minimax value = $\min_j \max_i \{a_{ij}\}$ is known as the **upper value** of the game and is generally denoted by $\bar{v}$.

Strictly Determinable Game: A game is said to be **strictly determinable** if $\underline{v} = v = \bar{v}$, where $\underline{v} = \max_i \min_j \{a_{ij}\}$, $\bar{v} = \min_j \max_i \{a_{ij}\}$, v = value the game

Fair Game: A game is said to be **fair** if $\underline{v} = v = \bar{v} = 0$.

Let us now apply the maximin-minimax principle to a few games.

Example 1: You are given a game having the following payoff matrix:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	5	7	4
	A ₂	4	2	0
	A ₃	6	1	3

Obtain the

- Optimal strategy for player A,
- Optimal strategy for player B,
- Value of the game, and
- Saddle point.

Also answer the questions: Is the game strictly determinable? Is the game fair?

Solution: We apply the maximin-minimax principle to the game. The following three steps are involved in applying this principle:

Step 1: We identify the minimum element of each row and then select the maximum among these minimum elements. This is called the maximin value as indicated in Table 11.1.

Step 2: Then we identify the maximum element of each column and select the minimum among these maximum elements. It is called the minimax value as indicated in Table 11.1.

Table 11.1: Calculation of Maximin-Minimax Principle

		Player B			Step 1	
		B ₁	B ₂	B ₃	Row Minima	Maximin Value
Player A	A ₁	5	7	4	4	$\max \{4, 0, 1\} = 4$
	A ₂	4	2	0	0	
	A ₃	6	1	3	1	
Step 2	Column Maxima	6	7	4		
	Minimax Value	$\min \{6, 7, 4\} = 4$				

Step 3: We check whether the maximin and minimax values are equal or not. In this case, maximin value = minimax value = 4. So, the game is solved using the maximin-minimax principle and we obtain the following results:

- i) Optimal strategy for player A is A_1 as it corresponds to the maximin value,
- ii) Optimal strategy for player B is B_3 as it corresponds to the minimax value,
- iii) The value of the game is 4 [$\because$ maximin value = minimax value = 4]

Now,

$\underline{v}$ = the lower value of the game = maximin value = 4

$\bar{v}$ = the upper value of the game = minimax value = 4

v = the value of the game = 4

$\because \underline{v} = v (= 4) = \bar{v}$, the game is strictly determinable.

But $v = 4 \neq 0$. So the game is not fair.

Note 6: In the above example, player A always has to employ strategy A_1 to maximise his/her minimum gains. So he should associate probability 1 to A_1 and a zero probability to each of A_2 and A_3 .

Hence, the optimal strategy of player A can also be written as a pure strategy (1, 0, 0). Similarly, optimal strategy of player B can also be written as pure strategy (0, 0, 1).

Example 2: Find the maximin and minimax values for the game having the payoff matrix given below. Does the game have a saddle point? If the game has a saddle point, solve it.

		Player B		
		B_1	B_2	B_3
Player A	A_1	2	4	5
	A_2	6	3	7
	A_3	4	1	-1

Solution: Calculations for the maximin and the minimax values are shown in Table 11.2 given below:

Table 11.2: Calculation of Maximin-Minimax Values

		Player B			Step 1	
		B_1	B_2	B_3	Row Minima	Maximin Value
Player A	A_1	2	4	5	2	$\max\{2, 3, -1\} = 3$
	A_2	6	3	7	3	
	A_3	4	1	-1	-1	
Step 2	Column Maxima	6	4	7		
	Minimax Value	$\min\{6, 4, 7\} = 4$				

In the above table, we see that

Maximin value = 3 and minimax value = 4.

Since maximin value $\neq$ minimax value, the game has no saddle point.

Hence, the solution of the game cannot be obtained by using the maximin-minimax principle.

Note 7: Solutions of the games without saddle points are discussed in Unit 12.

Example 3: Determine the range of values of λ and μ that will make the position (2, 2) a saddle point for the game having the payoff matrix given below:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	1	3	5
	A ₂	8	4	λ
	A ₃	2	μ	9

Solution: Since it is given that the position (2, 2) is the saddle point, the maximin value = minimax value = 4.

Now, maximin value = 4 $\Rightarrow$ 4 is the minimum element of the second row.

$$\therefore \lambda \geq 4$$

Also minimax value = 4 $\Rightarrow$ 4 is the maximum element of the second column.

$$\therefore \mu \leq 4$$

Hence, the range of λ is $\lambda \geq 4$ and the range of μ is $\mu \leq 4$.

Example 4: What are the optimal strategies for person X and person Y in the Prisoner's Dilemma game?

Solution: We have described the Prisoner's Dilemma game in Sec. 11.2 and obtained the payoff matrix for it. Let us rewrite the payoff matrix given there.

		Person Y (player II)	
		Confesses	Does not Confess
Person X (Player I)	Confesses	-5, -5	0, -20
	Does not Confess	-20, 0	-1, -1

It is not a zero sum game. Obviously, if players have the facility to communicate with each other, then not to confess is the optimal strategy for both players because in this case both will get prison for only one year. But it is given that they cannot communicate with each other and both persons have to protect their self-interest. What is the optimal strategy for both players? Let us first find the optimal strategy for player X. If Y confesses, then X will go to jail for 5 years if he confesses, and 20 years if he does not confess. So, if Y confesses, it is better for X to "confess" rather than to "not confess". If Y does not confess, then X will be free (0 year in jail) if he confesses, and get 1 year jail term if he does not confess. Also if Y does not confess, it is better for X to "confess" rather than to "not confess".

Thus, for X it is better to confess rather than to not confess, irrespective of whether Y confesses or not. Therefore, the optimal strategy for X is to confess.

Let us now find the optimal strategy for Y. If X confesses, then Y will go to jail for 5 years if he confesses, and 20 years if he does not confess. So, if X confesses, then it is better for Y to "confess" rather than to "not confess". If X does not confess, then Y will be free (0 year jail) if he confesses, and get 1 year jail term if he does not confess. Also if X does not confess, it is better for Y to "confess" rather than to "not confess".

Hence, under the given situation the optimal strategy for both X and Y is to confess. So, the optimal pure strategy for both persons is (1, 0).

You may like to try the following exercises to check your understanding of the key terms and concepts explained in Secs. 11.2 and 11.3.

In the exercises E1) to E5), choose the correct option.

E1) Game theory is the study of:

- (A) Computer games
- (B) Usual sports
- (C) Identifying optimal strategies
- (D) Two-person zero-sum games

E2) If $\underline{v}$ and $\bar{v}$ denote the lower and upper values of the game, then the game is said to be fair if

- (A) $\underline{v} < \bar{v}$ (B) $\underline{v} = \bar{v} \neq 0$ (C) $\underline{v} = \bar{v} = 0$ (D) $\underline{v} < 0 < \bar{v}$

E3) A saddle point exists in a game when

- (A) Maximax Value = Minimin Value
(B) Maximin value = Minimax Value
(C) Maximax value = Maximin value
(D) Maximin value = Minimin value

E4) Three strategies are available to a player in a game between two players. There is a pure optimal strategy in the game for him/her. Which of the following cannot be his/her pure strategy?

- (A) $(1/3, 1/3, 1/3)$ (B) $(1, 0, 0)$ (C) $(0, 1, 0)$ (D) $(0, 0, 1)$

E5) If a player A associates probabilities $p_1, p_2, \dots, p_m$ with m strategies available to him, which one of the following is the case of pure strategy?

- (A) $p_1 = p_2 = \dots = p_m = \frac{1}{m}$
(B) $p_1 = \frac{1}{2}, p_2 = \frac{1}{2}, p_3 = p_4 = \dots = p_m = 0$
(C) $p_1 = p_2 = \dots = p_{m-2} = 0, p_{m-1} = \frac{1}{2}, p_m = \frac{1}{2}$
(D) $p_i = 1$ for some i and $p_j = 0$ for all $j \neq i$

E6) Solve the game for which the payoff matrix is given by

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	2	4	3
	A ₂	1	-2	-3
	A ₃	0	6	1

Let us now summarise the main points that we have covered in this unit.

11.4 SUMMARY

- Player:** A participant or competitor who makes decisions is known as a player. A player may be an individual or a group of individuals. For example, in coin matching game, child X and child Y are players, and in the Prisoner's Dilemma game, person X and person Y are players.
- Action:** The options available to the players are known as actions or courses of action or **moves**.
- Play:** A play is said to occur when each player selects a course of action.
- Strategy:** A predetermined rule by which a player decides his/her course(s) of action among the actions available to him/her is known as strategy for the player. A strategy may be of two types:
 - Pure Strategy:** A strategy is said to be a **pure strategy** if the player selects a particular course of action each time.
 - Mixed Strategy:** A strategy for a player is said to be a **mixed strategy** if the player selects a combination of more than one courses of action by assigning a fixed probability to each course of action.

- 5) **Optimal Strategy:** The strategy for a player which optimises his/her payoff irrespective of the strategy of his /her competitor is known as an optimal strategy.
- 6) **Definition of Game:** A competitive situation is called a game if
 - (i) The number of competitors, called players, is finite.
 - (ii) The number of possible courses of action for each player is finite. However, the courses of action need not be the same for each player.
 - (iii) Each player selects a course of action simultaneously from among the courses of action available to him/her without directly communicating with the other player.
 - (iv) Every combination of courses of action results in an outcome known as payoff value, which motivates the players. Payoff value may represent loss, gain or any other thing of interest. The payoff values may be positive, negative or zero.
- 7) **n-Person Game:** If the number of players involved in the game is $n (> 2)$, it is known as n-person game.
- 8) **Two-Person Game:** If the number of players involved in the game is 2, it is known as two-person game.
- 9) **Zero-Sum Game:** If the algebraic sum of the payoff values of all players after each play is zero, the game is known as a zero-sum game.
- 10) **Non-Zero-Sum Game:** A game is said to be non-zero-sum game if there exists at least one play such that the algebraic sum of all the payoff values is not equal to zero. The prisoner's dilemma game is a non-zero-sum game.
- 11) **Two-Person Zero-Sum Game:** If in a game, each payoff value of one player is negative of the payoff value of the other player, it is known as two-person zero-sum game. The coin matching game is a two-person zero-sum game.
- 12) **Lower and Upper Values of the Game:** If a_{ij} denotes the payoff value when player I chooses his/her i^{th} ($i = 1, 2, \dots, m$) strategy and player II chooses his/her j^{th} ($j = 1, 2, \dots, n$) strategy, then the maximin value $= \max_i \min_j \{a_{ij}\}$ is known as the **lower value** of the game and is generally denoted by $\underline{v}$. The minimax value $= \min_j \max_i \{a_{ij}\}$ is known as the **upper value** of the game and is generally denoted by $\bar{v}$.
- 13) If $\text{maximin value} = \text{minimax value} = v$ (say), then v is known as the **value of the game** and the corresponding strategies of players A and B are known as their **optimal strategies**. Also, the position of the element corresponding to the optimal strategies of the two players is known as **saddle point**.
- 14) **Strictly Determinable Game:** A game is said to be **strictly determinable** if $\underline{v} = v = \bar{v}$.
- 15) **Fair Game:** A game is said to be fair if $\underline{v} = v = \bar{v} = 0$.

11.5 SOLUTIONS/ANSWERS

E1) In game theory, we try to identify the optimal strategy for each player. So, part (C) is the correct option.

E2) We know that a game is said to be fair if both lower value and upper value of the game are equal to zero. So, part (C) is the correct option.

E3) We know that saddle point is the point of intersection of the row of maximin value and column of minimax value. Hence the saddle point exists in a game if maximin value = minimax value. So, part (B) is the correct option.

E4) We know that among m strategies $A_1, A_2, \dots, A_m$ available to a player A , a strategy say A_i , ($1 \leq i \leq m$) is said to be a pure optimal strategy if player A has to use A_i if he/she wants to be in a comfortable position irrespective of the strategy adopted by his/her opponent B . That is, for A_i to be a pure optimal strategy, we should have

$$p_i = 1 \text{ and } p_j = 0 \text{ for all } j \neq i$$

where $p_1, p_2, \dots, p_m$ are the probabilities associated with the strategies $A_1, A_2, \dots, A_m$, respectively.

Hence, except for the part (A), all others are cases of pure strategies. So, the correct option is (A).

E5) As explained in the solution of E4, the correct option is (D).

E6) We solve the game by using the maximin-minimax principle. The following three steps are involved in applying this principle:

Step 1: We identify the minimum element of each row and then select the maximum of these minimum elements. It is the maximin value as indicated in Table 11.3.

Step 2: We identify the maximum element of each column and then select the minimum of these maximum elements. It is the minimax value as indicated in Table 11.3.

Table 11.3: Calculation of Maximin-Minimax Values

		Player B			Step 1	
		B_1	B_2	B_3	Row Minima	Maximin Value
Player A	A_1	2	4	3	2	$\max\{2, -3, 0\} = 2$
	A_2	1	-2	-3	-3	
	A_3	0	6	1	0	
Step 2	Column Maxima	2	6	3		$\min\{2, 6, 3\} = 2$
	Minimax Value					

Step 3: We check whether the maximin and minimax values are equal or not. Here maximin value = minimax value = 2. So, the game is solved using the maximin-minimax principle, and the solution is:

Optimal strategy for player A is A_1 or (1, 0, 0).

Optimal strategy for player B is B_1 or (0, 0, 1).

The value of the game is 2.

UNIT 12 TWO-PERSON ZERO-SUM GAMES WITHOUT SADDLE POINT

Structure

- 12.1 Introduction
 - Objectives
- 12.2. Solution of 2×2 Two-Person Zero-Sum Games without Saddle Point
- 12.3 Dominance Rules for Reducing the Size of the Game
- 12.4 Graphical Method for Reducing the $2 \times n$ and $m \times 2$ Games to 2×2 Games
 - Graphical Method for Reducing $2 \times n$ Games to 2×2 Games
 - Graphical Method for Reducing $m \times 2$ Games to 2×2 Games
- 12.5 Solution of 3×3 Two-Person Zero-Sum Game
- 12.6 Flow Chart
- 12.7 Summary
- 12.8 Solutions/Answers

12.1 INTRODUCTION

In Unit 11, you have learnt the definitions of key terms used in game theory. We have also discussed the solution of two-person zero-sum games with saddle point using the maximin-minimax principle. In this unit, we discuss some methods of solving two-person zero-sum games without saddle point. In Sec. 12.2, we describe the algebraic method for solving 2×2 two-person zero-sum games without a saddle point. However, there are rectangular games of sizes other than 2×2 and to obtain the solution for such games, we need to know the dominance rules. These rules help in reducing the size of the games and are described in Sec. 12.3. However, using the dominance rules, we cannot reduce all rectangular games into games of sizes 2×2 . So, we need methods to solve games of sizes other than 2×2 . The graphical method discussed in Sec. 12.4 is one such method, which reduces the games of size $2 \times n$ and $m \times 2$ to size 2×2 . Once the game is reduced to size 2×2 , we can solve it by using the algebraic method discussed in Sec. 12.2.

In Sec. 12.5, we solve a popular 3×3 two-person zero-sum game known as Stone-Paper-Scissors game by extending the technique of solving 2×2 two-person zero-sum games discussed in Sec. 12.2. A flow chart showing the approach followed in this unit to solve the two-person zero-sum games is given in Sec. 12.6.

Objectives

After studying this unit, you should be able to:

- solve 2×2 two-person zero-sum games by applying the algebraic method;
- apply dominance rules to reduce the size of a game;
- solve $2 \times n$ and $m \times 2$ two-person zero-sum games by applying the graphical method; and
- solve 3×3 two-person zero-sum games.

12.2 SOLUTION OF 2×2 TWO-PERSON ZERO-SUM GAMES WITHOUT SADDLE POINT

In Unit 11, you have learnt how to solve two-person zero-sum games with saddle point using the maximin-minimax principle. You have also learnt that the existence of a saddle point depends on the equality of maximin value and minimax value, where

$$\text{Maximin value} = \text{maximum among row minima} = \max_i \min_j \{a_{ij}\} \quad \dots (1)$$

$$\text{Minimax value} = \text{minimum among column maxima} = \min_j \max_i \{a_{ij}\} \quad \dots (2)$$

and a_{ij} is the $(i, j)^{\text{th}}$ entry of the payoff matrix of the row player.

The operations for obtaining the maximin and minimax values from equations (1) and (2) are different. So, the values obtained from equations (1) and (2) may or may not be equal. If they are equal, the game has a saddle point and can be solved using the maximin-minimax principle, which gives a pure strategy as a solution. But if they are not equal, the game does not have a saddle point. So we need methods to solve the games without saddle point. One such method for solving 2×2 two-person zero-sum games without saddle point is discussed in this section. But before describing the method, we state a theorem required for solving such games.

Fundamental Theorem of Rectangular Games or the Minimax Theorem

It states that for every finite two-person zero-sum game:

- (1) There exists a number v , called the value of the game,
- (2) There exists at least one mixed strategy for player I such that the gain of player I $\geq v$, irrespective of the strategy adopted by player II, and
- (3) There exists at least one mixed strategy for player II such that the loss of player II $\leq v$, irrespective of the strategy adopted by player I.

Note 1: Finite two-person zero-sum game means that the courses of action available to both players should be finite in number.

This theorem simply states that there always exists a solution for every finite two-person zero-sum game in terms of mixed strategies.

The solution of 2×2 two-person zero-sum games in terms of mixed strategies is provided by the **algebraic method**. We first give a statement of the method and then its proof.

Statement: Every 2×2 two-person zero-sum game without a saddle point having the following payoff matrix for player A,

		Player B	
		B ₁	B ₂
Player A	A ₁	a	b
	A ₂	c	d

has a solution in terms of mixed strategies (p_1, p_2) , (q_1, q_2) for player A and player B, respectively. The value of the game is v , and p_1 , p_2 , q_1 , q_2 and v are given by

$$p_1 = \frac{d - c}{(a - b) + (d - c)} \quad \dots (3a)$$

$$p_2 = \frac{a - b}{(a - b) + (d - c)} \quad \dots (3b)$$

$$q_1 = \frac{d-b}{(a-b)+(d-c)}$$

...(3c)

$$q_2 = \frac{a-c}{(a-b)+(d-c)}$$

...(3d)

$$v = \frac{ad-bc}{(a-b)+(d-c)}$$

...(3e)

such that

$$p_1, p_2, q_1, q_2 \geq 0, p_1 + p_2 = 1 \text{ and } q_1 + q_2 = 1$$

...(3f)

Proof: We are given that the payoff matrix for player A has no saddle point. So there is no solution in terms of pure optimal strategies. But from the fundamental theorem of game theory, there exists a solution in terms of mixed strategies. Let (p_1, p_2) and (q_1, q_2) be the optimal mixed strategies for player A and player B, respectively, and v be the value of the game. We now rewrite the payoff matrix indicating the above probabilities as follows:

			Player B	
			q_1	q_2
			B_1	B_2
Player A	p_1	A_1	a	b
	p_2	A_2	c	d

The expected gain of player A is

$$ap_1 + cp_2 \quad \text{when player B employs strategy } B_1$$

$$bp_1 + dp_2 \quad \text{when player B employs strategy } B_2$$

Similarly, the expected loss of player B is

$$aq_1 + bq_2 \quad \text{when player A employs strategy } A_1$$

$$cq_1 + dq_2 \quad \text{when player A employs strategy } A_2$$

Since v is the value of the game and (p_1, p_2) a mixed optimal strategy for player A, we have

$$\underset{\text{I}}{ap_1} + \underset{\text{II}}{cp_2} = v = \underset{\text{III}}{bp_1} + dp_2$$

$$\Rightarrow ap_1 + cp_2 = bp_1 + dp_2$$

$$\Rightarrow ap_1 + c(1-p_1) = bp_1 + d(1-p_1) \quad [\because p_1 + p_2 = 1]$$

$$\Rightarrow (a-c-b+d)p_1 = d-c \Rightarrow p_1 = \frac{d-c}{(a-b)+(d-c)}$$

$$\therefore p_2 = 1-p_1 = 1 - \frac{d-c}{(a-b)+(d-c)} = \frac{(a-b)+(d-c)-d+c}{(a-b)+(d-c)} = \frac{a-b}{(a-b)+(d-c)}$$

$$\begin{aligned} \text{The expected gain of player A} = v = ap_1 + cp_2 &= \frac{a(d-c)}{(a-b)+(d-c)} + \frac{c(a-b)}{(a-b)+(d-c)} \\ &= \frac{ad-ac+ca-bc}{(a-b)+(d-c)} = \frac{ad-bc}{(a-b)+(d-c)} \end{aligned}$$

Similarly, since v is the value of the game and (q_1, q_2) is a mixed optimal strategy for player B, we have

$$\begin{array}{lll} aq_1 + bq_2 = v = cq_1 + dq_2 \\ \text{I} \quad \quad \quad \text{II} \quad \quad \quad \text{III} \end{array}$$

$$\Rightarrow aq_1 + bq_2 = cq_1 + dq_2 \Rightarrow aq_1 + b(1 - q_1) = cq_1 + d(1 - q_1)$$

$$\Rightarrow (a - b - c + d)q_1 = d - b \Rightarrow q_1 = \frac{d - b}{(a - b) + (d - c)}$$

$$\therefore q_2 = 1 - q_1 = 1 - \frac{d - b}{(a - b) + (d - c)} = \frac{(a - b) + (d - c) - d + b}{(a - b) + (d - c)} = \frac{a - c}{(a - b) + (d - c)}$$

$$\begin{aligned} \text{The expected gain of player B} = v = aq_1 + bq_2 &= \frac{a(d - b)}{(a - b) + (d - c)} + \frac{b(a - c)}{(a - b) + (d - c)} \\ &= \frac{ad - ab + ab - bc}{(a - b) + (d - c)} = \frac{ad - bc}{(a - b) + (d - c)} \end{aligned}$$

Thus, we have proved that

$$\begin{aligned} p_1 &= \frac{d - c}{(a - b) + (d - c)}, \quad p_2 = \frac{a - b}{(a - b) + (d - c)} \\ q_1 &= \frac{d - b}{(a - b) + (d - c)}, \quad q_2 = \frac{a - c}{(a - b) + (d - c)} \quad \text{and} \quad v = \frac{ad - bc}{(a - b) + (d - c)} \end{aligned}$$

Now, it only remains to prove that $p_1, p_2, q_1, q_2 \geq 0$.

Remember that we are given that the payoff matrix of player A has no saddle point.

Therefore, either $a \geq b$ or $a \leq b$.

First, suppose that $a \geq b$. This implies that b is the minimum element of the first row

$$\Rightarrow b < d.$$

Otherwise, b will also become the maximum element of the second column and so a saddle point.

Now, $b < d \Rightarrow d$ is the maximum element of the second column.

$$\Rightarrow d > c.$$

Otherwise, d will also become the minimum element of the second row and so a saddle point.

Again, $d > c \Rightarrow c$ is the minimum element of the second row.

$$\Rightarrow c < a.$$

Otherwise, c will also become the maximum element of the first column and so a saddle point.

Finally, $c < a \Rightarrow a$ is the maximum element of the first column.

$$\Rightarrow a > b.$$

Otherwise, a will also become the minimum element of the first row and so a saddle point.

Thus, we obtain the result that if $a \geq b$ then $b < d, d > c, c < a$ and $a > b$.

Due to symmetry, if $a \leq b$ then $b > d, d < c, c > a$, and $a < b$.

Thus, in both cases, either both $(a - b)$ and $(d - c)$ are simultaneously positive or both are simultaneously negative. Therefore, $p_1 \geq 0, p_2 \geq 0$.

Similarly, we can show that $q_1 \geq 0$, $q_2 \geq 0$.

Hence proved.

Note 2: For numerical problems you have to substitute the values of a , b , c , d directly in equations (3a to e) for p_1, p_2, q_1, q_2 and v to obtain the solution.

Let us solve the following example using the algebraic method.

Example 1: Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B	
		B ₁	B ₂
Player A	A ₁	3	2
	A ₂	1	5

Solution: We first check whether a saddle point exists or not.

		Player B		Row Minima	Maximin Value
		B ₁	B ₂		
Player A	A ₁	3	2	2	max{2, 1} = 2
	A ₂	1	5	1	
Column Maxima		3	5		
Minimax Value		min{3, 5} = 3			

Since the maximin value (=2) $\neq$ minimax value (=3), there is no saddle point.

Hence, we have to solve the game in terms of mixed strategies. Let (p_1, p_2) and (q_1, q_2) be the optimal mixed strategies for player A and player B, respectively and v be the value of the game. From equations (3a to e), we have

$$p_1 = \frac{d-c}{(a-b)+(d-c)}, p_2 = \frac{a-b}{(a-b)+(d-c)}, q_1 = \frac{d-b}{(a-b)+(d-c)}$$

$$q_2 = \frac{a-c}{(a-b)+(d-c)} \text{ and } v = \frac{ad-bc}{(a-b)+(d-c)}$$

In this game, $a = 3$, $b = 2$, $c = 1$ and $d = 5$.

$$\therefore p_1 = \frac{5-1}{(3-2)+(5-1)} = \frac{4}{5}, p_2 = \frac{3-2}{(3-2)+(5-1)} = \frac{1}{5}, q_1 = \frac{5-2}{(3-2)+(5-1)} = \frac{3}{5}$$

$$q_2 = \frac{3-1}{(3-2)+(5-1)} = \frac{2}{5} \text{ and } v = \frac{3 \times 5 - 2 \times 1}{(3-2)+(5-1)} = \frac{13}{5}$$

Hence, the solution of the game is as follows:

The optimal strategy for player A is $(4/5, 1/5)$, the optimal strategy for player B is $(3/5, 2/5)$ and the value of the game is $13/5$.

Now, you can try the following exercises for solving a couple of games.

E1) Solve the two-person zero-sum game having the following payoff matrix for player A:

		Players B	
		B ₁	B ₂
Player A	A ₁	-2	-1
	A ₂	4	-3

E2) Solve the coin matching game explained in Sec. 11.2 of Unit 11.

12.3 DOMINANCE RULES FOR REDUCING THE SIZE OF THE GAME

In Sec. 12.2, we have derived and applied the algebraic method to solve 2×2 two-person zero-sum games without saddle point. So this method cannot be applied directly to $m \times n$ two-person zero-sum games without a saddle point if either m or n or both are greater than 2. But some of these games can be reduced to 2×2 games by applying the dominance rules, if applicable. In this section, we state these rules and explain how to apply them. But before you learn the dominance rules, you should understand inferior and superior strategies, which we explain below.

Inferior and Superior Strategies: Let $a = (a_1, a_2, \dots, a_n)$ and $b = (b_1, b_2, \dots, b_n)$ be two n -tuples. We say that n -tuple b **dominates** n -tuple a if

$$a_i \leq b_i \text{ for all } i, 1 \leq i \leq n$$

If $a_i < b_i$ for all $i, 1 \leq i \leq n$ then we say that n -tuple b **strictly dominates** n -tuple a . Suppose that a_{ij} , ($i = 1, 2, \dots, m; j = 1, 2, \dots, n$) represents the payoff value of payoff matrix of player A corresponding to the i^{th} strategy of row player A and j^{th} strategy of column player B. Then we say that the r^{th} strategy of row player A is **inferior** to his/her s^{th} strategy (or s^{th} strategy is **superior** to r^{th} strategy for row player A) if

$$a_{rj} \leq a_{sj} \text{ for all } j, 1 \leq j \leq n$$

that is, all the elements of the r^{th} row are less than or equal to the corresponding elements of the s^{th} row.

Remember that in the payoff matrix of row player A, payoff values represent gains of player A while for column player B, they represent losses. So, from the point of view of player B, we say that the r^{th} strategy of column player B is **inferior** to his/her s^{th} strategy (or s^{th} strategy is **superior** to r^{th} strategy for column player B) if

$$a_{ir} \geq a_{is} \text{ for all } i, \text{ or } 1 \leq i \leq m \quad \left[\begin{array}{l} \because \text{payoff values represent} \\ \text{losses for player B} \end{array} \right]$$

that is, if all the elements of the r^{th} column are greater than or equal to the corresponding elements of the s^{th} column.

So, in general, we define the dominant strategy as follows:

Dominant Strategy: A strategy is said to be the dominant strategy for a player if it is better than another strategy available to that player irrespective of the strategy adopted by the other player.

We now state the dominance rules to be used for reducing the size of the payoff matrix.

Rule 1

If all elements in a row (say, the i^{th} row) of a payoff matrix for row player are **less than or equal to** the corresponding elements of the other row (say j^{th} row), then the row player A will never employ his/her i^{th} strategy because it is dominated by his/her j^{th} strategy. So we can delete the i^{th} row from the payoff matrix. Remember that payoff values are gains for row player.

Rule 2

If all the elements in a column (say, the r^{th} column) of a payoff matrix for row player are **greater than or equal to** the corresponding elements of the other column (say, the s^{th} column), then the column player B will never employ his/her r^{th} strategy because it is dominated by his/her s^{th} strategy. So we can delete the r^{th} column from the payoff matrix. Remember that payoff values are losses for column player.

Note 3: It is important to note that in the case of row player A, the inferior row is deleted but in the case of column player B, the superior column is deleted. This is due to the fact that payoff values are gains for row player A but losses for column player B.

Rule 3

A strategy can also be deleted if it is dominated by average (or any convex combination) of two or more other pure strategies.

Note 4: A combination $\alpha_1 x_1 + \alpha_2 x_2 + \dots + \alpha_n x_n$ of $x_1, x_2, \dots, x_n$ is said to be a convex combination if $\alpha_i \geq 0$ for all i , $1 \leq i \leq n$ and $\alpha_1 + \alpha_2 + \dots + \alpha_n = 1$.

If you want to know more about convex combination, refer to Unit 1 of the Course MSTE-002.

Let us apply these rules to a couple of examples.

Example 2: Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	-2	-4	6
	A ₂	5	4	1
	A ₃	3	2	8

Solution: We first check whether a saddle point exists or not.

		Player B			Row Minima	Maximin Value
		B ₁	B ₂	B ₃		
Player A	A ₁	-2	-4	6	-4	$\max\{-4, 1, 2\} = 2$
	A ₂	5	4	1	1	
	A ₃	3	2	8	2	
Column Maxima		5	4	8		
Minimax Value		$\min\{5, 4, 8\} = 4$				

Since maximin value $\neq$ minimax value, there is no saddle point. Hence, we shall obtain the solution in terms of mixed strategies. Let us first reduce the payoff matrix by applying the dominance rules.

Step 1: We note that each element of the first row is less than the corresponding element of the third row. This implies that the first row is dominated by the third row. So we can eliminate the first row from the given payoff matrix. Thus, we write the reduced payoff matrix as:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₂	5	4	1
	A ₃	3	2	8

Step 2: Each element of the first column is greater than the corresponding element of the second column. This implies that the first column is dominated by the second column. So we can eliminate the first column and write the reduced payoff matrix as:

		Player B	
		B ₂	B ₃
Player A	A ₂	4	1
	A ₃	2	8

In the reduced payoff matrix given above neither any row dominates any other row nor any column dominates any other column. So this matrix cannot be reduced any further using dominance rules. Since the reduced payoff matrix is of order 2×2 , we can apply the algebraic method for 2×2 two-person zero-sum game.

Let (p_1, p_2, p_3) and (q_1, q_2, q_3) be the mixed strategies for players A and B, respectively, for the original payoff matrix. Therefore, from the reduced payoff matrix, (p_2, p_3) and (q_2, q_3) are the mixed strategies for the players A and B, respectively. If v is the value of the game, then from equations (3 a to e), we have

$$p_2 = \frac{d-c}{(a-b)+(d-c)}, p_3 = \frac{a-b}{(a-b)+(d-c)}, q_2 = \frac{d-b}{(a-b)+(d-c)}$$

$$q_3 = \frac{a-c}{(a-b)+(d-c)} \text{ and } v = \frac{ad-bc}{(a-b)+(d-c)}$$

In this case $a = 4, b = 1, c = 2, d = 8$.

$$\therefore p_2 = \frac{8-2}{(4-1)+(8-2)} = \frac{6}{9} = \frac{2}{3}, p_3 = \frac{4-1}{(4-1)+(8-2)} = \frac{3}{9} = \frac{1}{3}$$

$$q_2 = \frac{8-1}{(4-1)+(8-2)} = \frac{7}{9}, q_3 = \frac{4-2}{(4-1)+(8-2)} = \frac{2}{9}$$

$$v = \frac{4 \times 8 - 1 \times 2}{(4-1)+(8-2)} = \frac{30}{9} = \frac{10}{3}$$

Hence, the solution of the game is:

The optimal strategy for player A is $(0, 2/3, 1/3)$, the optimal strategy for player B is $(0, 7/9, 2/9)$ and the value of the game is $10/3$.

Example 3: Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B				
		B ₁	B ₂	B ₃	B ₄	B ₅
Player A	A ₁	6	1	3	2	3
	A ₂	5	7	-5	1	4
	A ₃	8	4	-1	2	2
	A ₄	3	3	-2	2	5

Solution: We first check whether a saddle point exists or not.

		Player B					Row Minima	Maximin Value
		B ₁	B ₂	B ₃	B ₄	B ₅		
Player A	A ₁	6	1	3	2	3	1	$\max\{1, -5, -1, -2\} = 1$
	A ₂	5	7	-5	1	4	-5	
	A ₃	8	4	-1	2	2	-1	
	A ₄	3	3	-2	2	5	-2	
Column Maxima		8	7	3	2	5		
Minimax Value		$\min\{8, 7, 3, 2, 5\} = 2$						

Since maximin value $\neq$ minimax value, the saddle point does not exist. Hence, we have to obtain the solution in terms of mixed strategies. Let us first reduce the payoff matrix by applying the dominance rules.

Step 1: We note that each element of the first and fifth columns is greater than or equal to the corresponding elements of the fourth column. This implies that the fourth column dominates both first and fifth columns. So we can eliminate the first and fifth columns from the given payoff matrix and write the reduced payoff matrix as:

		Player B		
		B ₂	B ₃	B ₄
Player A	A ₁	1	3	2
	A ₂	7	-5	1
	A ₃	4	-1	2
	A ₄	3	-2	2

Step 2: Each element of the fourth row is less than or equal to the corresponding element of the third row. This implies that the fourth row is dominated by the third row. So we can eliminate the fourth row from the reduced payoff matrix. The reduced payoff matrix after this step is:

		Player B		
		B ₂	B ₃	B ₄
Player A	A ₁	1	3	2
	A ₂	7	-5	1
	A ₃	4	-1	2

Step 3: In this reduced payoff matrix, neither any row is directly dominated by any other row nor any column is directly dominated by any other column. But we observe that the average of the first and second columns of the payoff matrix dominates the third column. So, we can eliminate the third column from this reduced payoff matrix and write the reduced payoff matrix as:

		Player B	
		B ₂	B ₃
Player A	A ₁	1	3
	A ₂	7	-5
	A ₃	4	-1

Step 4: Again, in this reduced payoff matrix neither any row is directly dominated by any other row nor any column is directly dominated by any other column. But we observe that the average of the first and second rows of this reduced payoff matrix dominates the third row. So we can eliminate the third row from this reduced payoff matrix. The reduced payoff matrix after this step is:

		Player B	
		B ₂	B ₃
Player A	A ₁	1	3
	A ₂	7	-5

In this reduced payoff matrix, neither any row dominates any other row nor any column dominates any other column. So it cannot be reduced further using dominance rules. Since the reduced payoff

matrix is of order 2×2 , we can apply the algebraic method for 2×2 two-person zero-sum game. Let (p_1, p_2, p_3, p_4) and $(q_1, q_2, q_3, q_4, q_5)$ be the mixed strategies for players A and B, respectively, for the original payoff matrix. Thus, (p_1, p_2) and (q_2, q_3) are the mixed strategies from the reduced payoff matrix for players A and B, respectively. If v is the value of the game, then from equations (3 a to e) we have

$$p_1 = \frac{d - c}{(a - b) + (d - c)}, p_2 = \frac{a - b}{(a - b) + (d - c)}, q_2 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_3 = \frac{a - c}{(a - b) + (d - c)} \text{ and } v = \frac{ad - bc}{(a - b) + (d - c)}$$

In this case $a = 1, b = 3, c = 7, d = -5$.

$$\therefore p_1 = \frac{-5 - 7}{(1 - 3) + (-5 - 7)} = \frac{-12}{-14} = \frac{6}{7}, p_2 = \frac{1 - 3}{(1 - 3) + (-5 - 7)} = \frac{-2}{-14} = \frac{1}{7}$$

$$q_2 = \frac{-5 - 3}{(1 - 3) + (-5 - 7)} = \frac{-8}{-14} = \frac{4}{7}, q_3 = \frac{1 - 7}{(1 - 3) + (-5 - 7)} = \frac{-6}{-14} = \frac{3}{7}$$

$$v = \frac{1 \times (-5) - 3 \times 7}{(1 - 3) + (-5 - 7)} = \frac{-26}{-14} = \frac{13}{7}$$

Hence, the solution of the game is:

The optimal strategy for player A is $(6/7, 1/7, 0, 0)$, the optimal strategy for player B is $(0, 4/7, 3/7, 0, 0)$ and the value of the game is $13/7$.

You may like to try the following exercise for practice.

E3) Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B			
		B ₁	B ₂	B ₃	B ₄
Player A	A ₁	4	3	2	1
	A ₂	6	4	5	0
	A ₃	1	2	0	3

Note 5: You have seen that dominance rules (if applicable) reduce the size of the game. You can apply these rules to any game whether it has a saddle point or not. So we can also apply these rules to the games discussed in Unit 11. If we do so, the size of the game reduces to 1×1 . And if the size of the game reduces to 1×1 , it implies that the game has been solved. For example, let us apply these rules to Example 1 of Unit 11.

Step 1: We note that each element of the first row is greater than or equal to the corresponding element of the second row. This implies that the first row dominates the second row. So we can eliminate the second row from the given payoff matrix and write the reduced payoff matrix as:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	5	7	4
	A ₃	6	1	3

Step 2: Each element of the first column is greater than or equal to the corresponding element of the third column. This implies that the third column dominates the first column. So we can eliminate the first column from the reduced payoff matrix. The reduced payoff matrix after this step is:

		Player B	
		B ₂	B ₃
Player A	A ₁	7	4
	A ₃	1	3

Step 3: Each element of the first row is greater than or equal to the corresponding element of the second row. This implies that the first row dominates the second row. So we can eliminate the second row from the reduced payoff matrix. The reduced payoff matrix after this step is:

		Player B	
		B ₂	B ₃
Player A	A ₁	7	4

Step 4: The single element 7 of the first column is greater than the corresponding single element of the second column. This implies that the second column dominates the first column. So we can eliminate the first column from the reduced payoff matrix. The reduced payoff matrix after this step is:

		Player B
		B ₃
Player A	A ₁	4

Thus payoff matrix reduces to order 1×1 . Hence the solution of the game is:

The optimal strategy for player A is (1, 0, 0), the optimal strategy for player B is (0, 0, 1) and the value of the game is 4. You can match the solution obtained here with the one obtained in Example 1 of Unit 11.

12.4 GRAPHICAL METHOD FOR REDUCING THE $2 \times n$ AND $m \times 2$ GAMES TO 2×2 GAMES

In Sec. 12.3, you have learnt the dominance rules for reducing the size of the game if the payoff matrix has some inferior strategies. But many-a-times we have to deal either directly with $2 \times n$ and $m \times 2$ games or they reduce to one of these forms after applying dominance rules. Therefore, it becomes important to learn how to solve $2 \times n$ and $m \times 2$ games. In this section, we discuss one such method known as the **graphical method**, which reduces the $2 \times n$ and $m \times 2$ games into 2×2 games. Once a game is reduced to size 2×2 , we can solve it using the algebraic method discussed in Sec. 12.2. So, let us first discuss the graphical method for $2 \times n$ games.

12.4.1 Graphical Method for Reducing $2 \times n$ Games to 2×2 Games

In $2 \times n$ games, the row player A has two moves and the column player B may have any finite number of moves (say, n). Let the payoff matrix for player A for a $2 \times n$ game be:

		Player B					
		B_1	B_2	...	B_j	...	B_n
Player A	A_1	a_{11}	a_{12}	...	a_{1j}	...	a_{1n}
	A_2	a_{21}	a_{22}	...	a_{2j}	...	a_{2n}

If the game has a saddle point, it can be solved using the maximin-minimax principle discussed in Unit 11, which gives pure strategies as a solution. If the game does not have a saddle point, its solution will be in terms of mixed strategies. Let (p_1, p_2) and $(q_1, q_2, \dots, q_j, \dots, q_n)$ be the optimal mixed strategies for player A and player B, respectively. Let us rewrite the payoff matrix with these probabilities.

			Player B					
			q_1	q_2	...	q_j	...	q_n
			B_1	B_2	...	B_j	...	B_n
Player A	p_1	A_1	a_{11}	a_{12}	...	a_{1j}	...	a_{1n}
	p_2	A_2	a_{21}	a_{22}	...	a_{2j}	...	a_{2n}

Now, if $e_1, e_2, \dots, e_n$ be the expected payoffs for player A corresponding to different strategies of player B, then these are given by:

$$\begin{aligned} e_1 &= a_{11}p_1 + a_{21}p_2 && \text{if player B employs strategy } B_1 \\ e_2 &= a_{12}p_1 + a_{22}p_2 && \text{if player B employs strategy } B_2 \\ &\vdots && \vdots \\ e_n &= a_{1n}p_1 + a_{2n}p_2 && \text{if player B employs strategy } B_n \end{aligned}$$

But $p_1 + p_2 = 1 \Rightarrow p_2 = 1 - p_1$. So writing the above expected payoffs for player A in terms of p_1 , we get

$$\begin{aligned} e_1 &= (a_{11} - a_{21})p_1 + a_{21} && \text{if player B employs strategy } B_1 \\ e_2 &= (a_{12} - a_{22})p_1 + a_{22} && \text{if player B employs strategy } B_2 \\ &\vdots && \vdots \\ e_n &= (a_{1n} - a_{2n})p_1 + a_{2n} && \text{if player B employs strategy } B_n \end{aligned}$$

The above expected payoffs for player A are linear equations in p_1 . We know from Sec. 2.5 of Unit 2 of MST-001 that the graph of a linear function given by

$$y = ax + b$$

is a straight line (here we have the expected payoff in place of y and p_1 in place of x). Hence, the n expected payoffs of row player A corresponding to different strategies employed by column player B represent n straight lines. Let us now explain how to plot these lines on a graph paper. First of all, we take values of p_1 along the horizontal axis. However, the values of p_1 vary from 0 to 1

($0 \leq p_1 \leq 1$) because p_1 represents probability. Therefore, we restrict the horizontal axis at 1 unit, starting from $p_1 = 0$ and ending at $p_1 = 1$ (see Fig. 12.1a).

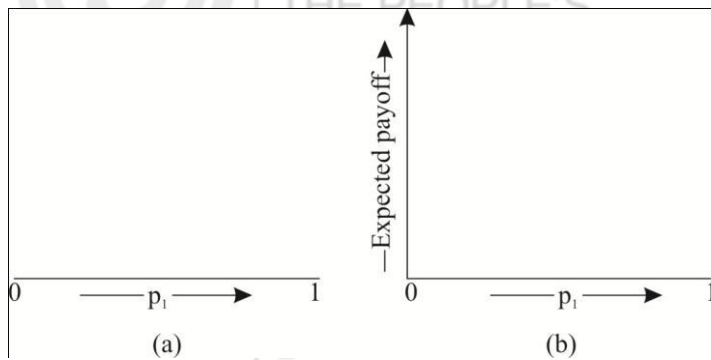


Fig. 12.1: Horizontal and vertical axes for the graph of player A.

We take the expected payoff along the vertical axis (see Fig. 12.1b).

Now, consider the first line:

$$e_1 = (a_{11} - a_{21})p_1 + a_{21}$$

Actually, two points are sufficient to draw the graph of a straight line (read the margin remark). So we obtain two points corresponding to $p_1 = 0$ and $p_1 = 1$, respectively.

At $p_1 = 0$, $e_1 = a_{21}$

At $p_1 = 1$, $e_1 = (a_{11} - a_{21})(1) + a_{21} = a_{11}$

Thus, the two points on the first line are $(0, a_{21})$ and $(1, a_{11})$.

Similarly, two points on the second line are $(0, a_{22})$, $(1, a_{12})$.

⋮
⋮

Two points on the n^{th} line are $(0, a_{2n})$, $(1, a_{1n})$.

Let us plot these lines: we first draw two vertical lines: the first along $p_1 = 0$ and the second along $p_1 = 1$ (see Fig. 12.2a).

We observe that the second coordinates of all points having first coordinate 0 are the same as the payoff values in the second row. Further, the second coordinate of the points having first coordinate 1 are the same as payoff values in the first row. Thus, for numerical problems we need not obtain two points for each line. We can simply take payoff values of the second row on the vertical line along $p_1 = 0$ and payoff values of the first row on the vertical line along $p_1 = 1$ (see Fig. 12.2a). Here we have made one assumption that

$a_{21} < a_{22} < \dots < a_{2n}$ and $a_{12} < a_{11} < a_{13} < a_{14} < \dots < a_{1n}$. The lines can be plotted by simply joining the points a_{21} to a_{11} , a_{22} to a_{12} , ..., a_{2n} to a_{1n} (see Fig. 12.2b).

Recall from Sec. 2.5 of Unit 2 of MST-001, how to draw the graph of a straight line. We simply obtain any three points from the equation of the straight line, plot them on the graph paper and finally join the points. In that unit, we took three points just to keep an automatic check on our calculations. Because if the three points do not fall on the line, it means that we have made some mistake in obtaining them. We do not get this advantage if we work with two points.

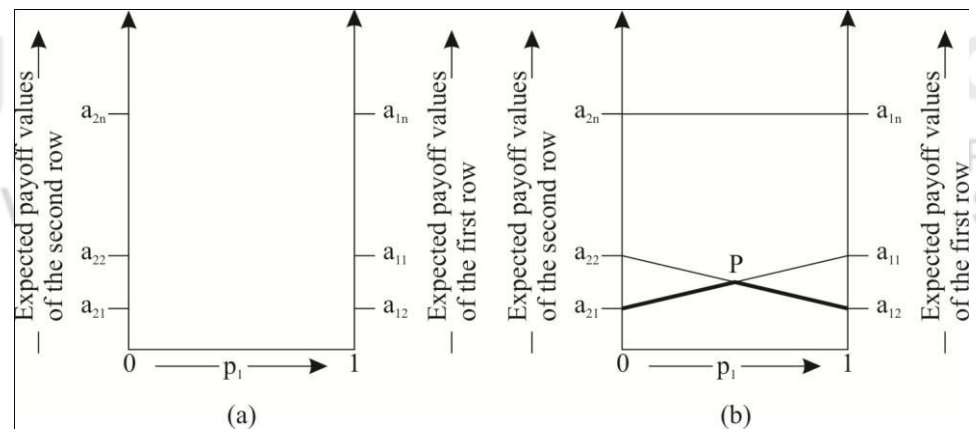


Fig. 12.2: Representation of expected payoff values and expected payoff line segments.

Note that these lines represent expected payoffs of row player A. So the lowest boundary of these lines will give the minimum expected payoff as a function of p_1 . Pay attention to the lowest boundary formed by the lower line segments of expected payoff lines in Fig. 12.2b. It is known as the **lower envelope** and indicated by **bold line segments**. But, obviously player A would like to maximise his/her expected payoffs (gains). This can be achieved by selecting the highest point on the lower envelope in Fig. 12.2b, it is indicated by the point P. If there are only two straight lines passing through the highest point, then we identify the two strategies corresponding to these lines and eliminate all other strategies of player B from the payoff matrix. This reduces the game to a 2×2 two-person zero-sum game and can be solved using the algebraic method discussed in Sec 12.2.

But if there are more than two straight lines passing through the highest point, then there will be alternative solutions. To obtain one solution, we select any two straight lines from among the lines passing through the highest point and identify the strategies corresponding to these two lines. This will again reduce the game into 2×2 two-person zero-sum game and can be solved using the algebraic method discussed in Sec. 12.2. In brief, we can summarise the above procedure in the following steps:

- Step 1:** Draw a horizontal line of length one unit which extends from $p_1 = 0$ to $p_1 = 1$.
- Step 2:** Draw two vertical lines passing through the two points $p_1 = 0$ and $p_1 = 1$ on the horizontal axis.
- Step 3:** Represent the payoff values of the second row on the vertical line $p_1 = 0$ and payoff values of the first row on the vertical line $p_1 = 1$.
- Step 4:** Join a_{21} to a_{11} , a_{22} to a_{12} , ..., a_{2n} to a_{1n} . The n straight lines thus plotted will represent the expected payoffs of row player A.
- Step 5:** Identify the lowest boundary of the expected payoff lines as obtained in Step 4, called the lower envelope. Draw bold lines along the line segments of the boundary.
- Step 6:** Identify the highest point on the lower envelope.
- Step 7:** Identify the strategies corresponding to the expected payoff lines, which pass through the highest point. If there are only two such strategies, then eliminate all other strategies. This reduces the game to 2×2 two-person zero-sum game. Solve it using the algebraic method. If there are more than two such strategies, select any two among

them and eliminate all other strategies. This will again reduce the game into 2×2 two-person zero-sum game and can be solved using the algebraic method.

Let us consider an example to explain the graphical method for solving $2 \times n$ games.

Example 4: Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B		
		B ₁	B ₂	B ₃
Player A	A ₁	1	2	7
	A ₂	6	4	2

Solution: We first check if a saddle point exists or not.

		Player B			Row Minima	Maximin Value
		B ₁	B ₂	B ₃		
Player A	A ₁	1	2	7	1	$\max \{1, 2\} = 2$
	A ₂	6	4	2	2	
Column Maxima		6	4	7		
Minimax Value		$\min \{6, 4, 7\} = 4$				

Since maximin value $\neq$ minimax value, there is no saddle point. Hence, we have to obtain the solution in terms of mixed strategies. Moreover, this game cannot be reduced by using dominance rules, because it has neither any inferior row nor any inferior column. But this game is of the type $2 \times n$ where $n = 3$. Hence, we can apply the graphical method for $2 \times n$ games which will reduce the game into a 2×2 two-person zero-sum game. Let (p_1, p_2) and (q_1, q_2, q_3) be the optimal mixed strategies for players A and B, respectively. Then the expected payoffs of player A corresponding to the moves B₁, B₂, B₃ of the player B are given by

$$e_1 = 1p_1 + 6p_2 = p_1 + 6(1 - p_1) = -5p_1 + 6 \quad \dots (i)$$

$$e_2 = 2p_1 + 4p_2 = 2p_1 + 4(1 - p_1) = -2p_1 + 4 \quad \dots (ii)$$

$$e_3 = 7p_1 + 2p_2 = 7p_1 + 2(1 - p_1) = 5p_1 + 2 \quad \dots (iii)$$

We now follow the steps given below to apply the graphical method for this 2×3 game. Refer to Fig. 12.3.

Step 1: We draw a horizontal line of length 1 unit extending from the point R($p_1 = 0$) to point S($p_1 = 1$).

Step 2: We draw two vertical lines $p_1 = 0$ and $p_1 = 1$ passing through the points R and S, respectively, on the horizontal axis.

Step 3: We represent the payoff values of the second row (6, 4, 2) along the vertical line $p_1 = 0$ and payoff values of the first row (1, 2, 7) along the vertical line $p_1 = 1$.

Step 4: We plot the straight lines given by (i), (ii) and (iii) by joining the payoff values of two rows with their corresponding values. Here we plot the lines by joining 6 to 1, 4 to 2 and 2 to 7 as shown in Fig. 12.3.

Step 5: We draw bold lines along the line segments, which form the lowest boundary of expected payoff lines given by (i), (ii) and (iii) to obtain the lower envelope (see Fig. 12.3).

Step 6: We identify the highest point on the lower envelope. In Fig. 12.3, the highest point is P, which is the point of intersection of expected payoff lines given by (ii) and (iii). Thus, the given game reduces to:

		Player B	
		q_2	q_3
		B_2	B_3
Player A	p_1	A_1	2
	p_2	A_2	4
			7
			2

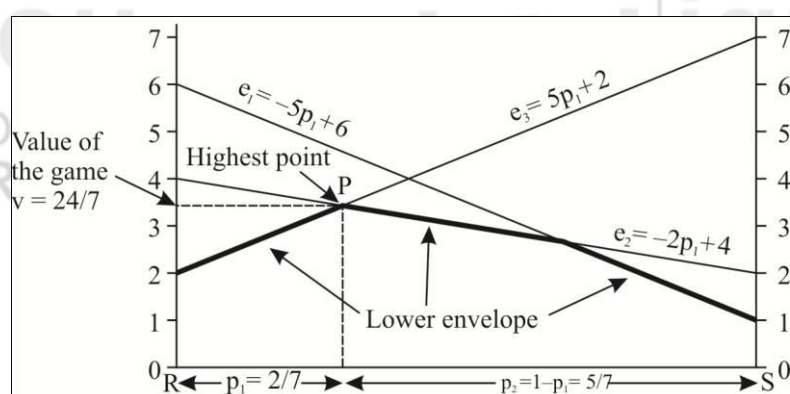


Fig. 12.3: Graph for player A.

We know that for 2×2 two-person zero-sum games p_1, p_2, q_2, q_3 and v are given from equations (3 a to e) as:

$$p_1 = \frac{d-c}{(a-b)+(d-c)}, p_2 = \frac{a-b}{(a-b)+(d-c)}, q_2 = \frac{d-b}{(a-b)+(d-c)}$$

$$q_3 = \frac{a-c}{(a-b)+(d-c)} \text{ and } v = \frac{ad-bc}{(a-b)+(d-c)}$$

In this case, $a = 2, b = 7, c = 4, d = 2$.

$$\therefore p_1 = \frac{2-4}{(2-7)+(2-4)} = \frac{-2}{-7} = \frac{2}{7}, p_2 = \frac{2-7}{(2-7)+(2-4)} = \frac{-5}{-7} = \frac{5}{7}$$

$$q_2 = \frac{2-7}{(2-7)+(2-4)} = \frac{-5}{-7} = \frac{5}{7}, q_3 = \frac{2-4}{(2-7)+(2-4)} = \frac{-2}{-7} = \frac{2}{7}$$

$$v = \frac{2 \times 2 - 7 \times 4}{(2-7)+(2-4)} = \frac{-24}{-7} = \frac{24}{7}$$

Hence, the optimal strategies for original game for players A and B are $(2/7, 5/7)$ and $(0, 5/7, 2/7)$, respectively, and the value of the game is $24/7$.

We now discuss the graphical method for $m \times 2$ games.

12.4.2 Graphical Method for Reducing $m \times 2$ Games to 2×2 Games

In an $m \times 2$ game, the column player B has two moves and the row player A may have any finite number of moves (say, m). Let the payoff matrix for player A for an $m \times 2$ game be:

		Player B	
		B ₁	B ₂
Player A	A ₁	a ₁₁	a ₁₂
	A ₂	a ₂₁	a ₂₂
	⋮	⋮	⋮
	A _i	a _{i1}	a _{i2}
	⋮	⋮	⋮
	A _m	a _{m1}	a _{m2}

If the game has a saddle point, it can be solved using the maximin-minimax principle discussed in Unit 11, which gives pure strategies as a solution. Suppose the game does not have a saddle point, which means that the solution of the game will be in terms of mixed strategies. Let $(p_1, p_2, \dots, p_m)$ and (q_1, q_2) be the optimal mixed strategies for player A and player B, respectively. Let us rewrite the payoff matrix with these probabilities as follows:

			Player B	
			q_1	q_2
			B_1	B_2
Player A	p_1	A_1	a_{11}	a_{12}
	p_2	A_2	a_{21}	a_{22}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	p_i	A_i	a_{i1}	a_{i2}
	$\vdots$	$\vdots$	$\vdots$	$\vdots$
	p_m	A_m	a_{m1}	a_{m2}

If $e_1, e_2, \dots, e_m$ are the expected payoffs for player B corresponding to different strategies of player A, then these are given by:

$$\begin{aligned}
 e_1 &= a_{11}q_1 + a_{12}q_2 && \text{if player A employs strategy } A_1 \\
 e_2 &= a_{21}q_1 + a_{22}q_2 && \text{if player A employs strategy } A_2 \\
 &\vdots && \vdots \\
 e_m &= a_{m1}q_1 + a_{m2}q_2 && \text{if player A employs strategy } A_m
 \end{aligned}$$

But $q_1 + q_2 = 1 \Rightarrow q_2 = 1 - q_1$.

So we can write the expected payoffs for player B in terms of q_1 as follows:

$$\begin{aligned}
 e_1 &= (a_{11} - a_{12})q_1 + a_{12} && \text{if player A employs strategy } A_1 \\
 e_2 &= (a_{21} - a_{22})q_1 + a_{22} && \text{if player A employs strategy } A_2 \\
 &\vdots && \vdots \\
 e_m &= (a_{m1} - a_{m2})q_1 + a_{m2} && \text{if player A employs strategy } A_m
 \end{aligned}$$

The expected payoffs for player B given above are linear equations in q_1 and hence represent straight lines. We can plot these lines on a graph paper as explained in Sec. 12.4.1. We take the values of q_1 along the horizontal axis from $q_1 = 0$ to $q_1 = 1$ (i.e., $0 \leq q_1 \leq 1$). So the horizontal axis is restricted at 1 unit starting from $q_1 = 0$ and ending at $q_1 = 1$ (see Fig. 12.4). We take expected payoffs along the vertical axis.

Let us consider the first line:

$$e_1 = (a_{11} - a_{12})q_1 + a_{12}$$

We know that a straight line can be plotted easily by joining two points on it.

So, we take the points $q_1 = 0$ and $q_1 = 1$ on the line:

$$\text{At } q_1 = 0, e_1 = a_{12}$$

$$\text{At } q_1 = 1, e_1 = (a_{11} - a_{12})(1) + a_{12} = a_{11}$$

Thus, two points on the first line are $(0, a_{12})$ and $(1, a_{11})$.

Similarly, two points on the second line are $(0, a_{22})$ and $(1, a_{21})$.

⋮

Two points on the m^{th} line are $(0, a_{m2})$ and $(1, a_{m1})$.

Let us plot these lines. For convenience, we draw two vertical lines: the first along $q_1 = 0$ and the second along $q_1 = 1$ as shown in Fig 12.4. We observe that the second coordinates of all points having first coordinate 0 are the same as the payoff values in second column. Moreover, the second coordinates of the points having first coordinate 1 are the same as the payoff values in the first column. Thus, for numerical problems we need not obtain two points for each line. We can simply take payoff values of the second column along the vertical line $q_1 = 0$ and payoff values of the first column along the vertical line $q_1 = 1$.

The lines can be plotted by simply joining a_{11} to a_{12} , a_{21} to a_{22} , ..., a_{m1} to a_{m2} .

Note that these lines represent expected payoffs (losses) of column player B. So the uppermost boundary of these lines will give the maximum expected payoffs (losses) as a function of q_1 . The uppermost boundary formed by the uppermost line segments of expected payoff lines is known as the **upper envelope** and indicated by bold line segments.

Now, player A would like to minimise his/her expected payoffs (losses). This can be achieved by selecting the lowest point on the upper envelope. If only two straight lines pass through the lowest point, then we identify the strategies corresponding to these lines and eliminate all other strategies of player A from the payoff matrix. This reduces the game to a 2×2 two-person zero-sum game. So it can be solved using the algebraic method.

However, if more than two straight lines pass through the lowest point, then there will be alternative solutions. To obtain one solution, we select any two straight lines from among the lines passing through the lowest point and identify the strategies corresponding to these two lines. This will again reduce the game to a 2×2 two-person zero-sum game, which can be solved using the algebraic method. In brief, we can summarise the above procedure in the following steps.

Step 1: Draw a horizontal line of length one unit, which extends from the point $R(q_1 = 0)$ to the point $S(q_1 = 1)$.

Step 2: Draw two vertical lines passing through the points R and S, respectively, on the horizontal axis.

Step 3: Represent the payoff values of the second column on the vertical line along $q_1 = 0$ and payoff values of the first column on the vertical line along $q_1 = 1$.

Step 4: Join a_{11} to a_{12} , a_{21} to a_{22} , ..., a_{m1} to a_{m2} . The m straight lines thus plotted will represent the expected payoffs (losses) of column player B.

Step 5: Identify the uppermost boundary of expected payoff lines obtained in Step 4, called the upper envelope. Draw bold lines along the line segments of the boundary.

Step 6: Identify the lowest point on the upper envelope.

Step 7: Identify the strategies corresponding to expected payoff lines, which pass through the lowest point. If there are only two such strategies, eliminate all other strategies. This will reduce the $m \times 2$ game to a 2×2 two-person zero-sum game. We can solve it using the algebraic method. But if there are more than two such strategies, select any two from them and eliminate all other strategies. This will again reduce the game to a 2×2 two-person zero-sum game, which can be solved using the algebraic method.

Let us consider an example to explain the graphical method for $m \times 2$ games.

Example 5: Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B	
		B ₁	B ₂
Player A	A ₁	5	-2
	A ₂	-3	6
	A ₃	-2	3

Solution: We first check if a saddle point exists or not.

		Player B		Row Minima	Maximin Value
		B ₁	B ₂		
Player A	A ₁	5	-2	-2	$\max\{-2, -3, -2\} = -2$
	A ₂	-3	6	-3	
	A ₃	-2	3	-2	
Column Maxima		5	6		
Minimax Value		$\min\{5, 6\} = 5$			

Since maximin value $\neq$ minimax value, there is no saddle point. Hence, we have to obtain the solution in terms of mixed strategies. Also, we cannot reduce this game using dominance rules, because there is neither any inferior row nor any inferior column. But this game is of the type $m \times 2$ where $m = 3$. Hence, we can apply the graphical method for $m \times 2$ games, which will reduce the game to a 2×2 two-person zero-sum game. Let (p_1, p_2, p_3) and (q_1, q_2) be the optimal mixed strategies for players A and B, respectively. The expected payoffs of player B corresponding to the moves A_1, A_2, A_3 of the player A are given by

$$e_1 = 5q_1 - 2q_2 = 5q_1 - 2(1 - q_1) = 7q_1 - 2 \quad \dots (i)$$

$$e_2 = -3q_1 + 6q_2 = -3q_1 + 6(1 - q_1) = -9q_1 + 6 \quad \dots (ii)$$

$$e_3 = -2q_1 + 3q_2 = -2q_1 + 3(1 - q_1) = -5q_1 + 3 \quad \dots (iii)$$

We now follow the steps given below to apply the graphical method for $m \times 2$ games.

Step 1: We draw a horizontal line, 1 unit in length, extending from the point $R(q_1 = 0)$ to the point $S(q_1 = 1)$. (see Fig. 12.4).

Step 2: We draw two vertical lines, $q_1 = 0$ and $q_1 = 1$, passing through the two points R and S on the horizontal axis.

Step 3: We represent the payoff values of the second column (i.e., $-2, 6, 3$) along the vertical line $q_1 = 0$ and payoff values of the first column (i.e., $5, -3, -2$) along the vertical line $q_1 = 1$ (see Fig. 12.4).

Step 4: We plot the straight lines given by (i), (ii) and (iii) by joining the payoff values of the two columns with their corresponding values. Here we plot the line by joining 5 to -2 , -3 to 6 and -2 to 3 as shown in Fig. 12.4.

Step 5: We draw bold lines along the line segments, which are the uppermost boundary of expected payoff lines given by (i), (ii) and (iii), to obtain the upper envelope.

Step 6: We identify the lowest point on the upper envelope. Here the lowest point is P, which is the point of intersection of expected payoff lines given by (i) and (ii). Thus, the given game reduces to:

			Player B	
			q ₁	q ₂
			B ₁	B ₂
Player A	p ₁	A ₁	5	−2
	p ₂	A ₂	−3	6

We know that for 2×2 two-person zero-sum games, p_1, p_2, q_1, q_2 and the value of the game v are given from equations (3a to e) as:

$$p_1 = \frac{d - c}{(a - b) + (d - c)}, p_2 = \frac{a - b}{(a - b) + (d - c)}, q_1 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_2 = \frac{a - c}{(a - b) + (d - c)} \text{ and } v = \frac{ad - bc}{(a - b) + (d - c)}$$

In this case $a = 5, b = -2, c = -3, d = 6$.

$$\therefore p_1 = \frac{6 - (-3)}{(5 - (-2)) + (6 - (-3))} = \frac{9}{16}, p_2 = \frac{5 - (-2)}{(5 - (-2)) + (6 - (-3))} = \frac{7}{16}$$

$$q_1 = \frac{6 - (-2)}{(5 - (-2)) + (6 - (-3))} = \frac{8}{16} = \frac{1}{2}, q_2 = \frac{5 - (-3)}{(5 - (-2)) + (6 - (-3))} = \frac{8}{16} = \frac{1}{2}$$

$$\text{and } v = \frac{5 \times 6 - (-2)(-3)}{(5 - (-2)) + (6 - (-3))} = \frac{24}{16} = \frac{3}{2}$$

Hence, the optimal strategies for the original game for players A and B are $\left(\frac{9}{16}, \frac{7}{16}, 0\right)$ and $\left(\frac{1}{2}, \frac{1}{2}\right)$, respectively, and the value of the game is $\frac{3}{2}$.

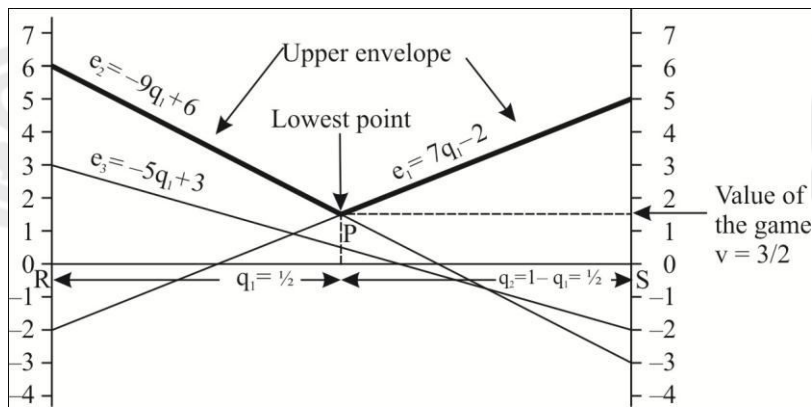


Fig. 12.4: Graph for player B.

You may try the following exercises for applying the graphical methods to solve $2 \times n$ and $m \times 2$ games.

E4) Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B				
		B ₁	B ₂	B ₃	B ₄	B ₅
Player A	A ₁	3	4	5	-2	2
	A ₂	1	6	-3	3	7

E5) Solve the two-person zero-sum game having the following payoff matrix for player A:

		Player B	
		B ₁	B ₂
Player A	A ₁	8	-2
	A ₂	5	1
	A ₃	2	3
	A ₄	4	7

12.5 SOLUTION OF 3×3 TWO-PERSON ZERO-SUM GAME

In Sec. 12.2, we have derived a general result for solving 2×2 two-person zero-sum games. In this section, we are not going to derive such a result for 3×3 two-person zero-sum games. We shall simply solve a 3×3 two-person zero-sum game to give you an idea how such games can be solved. Let us explain this idea with the help of a very popular game known as Stone-Paper-Scissors game.

Stone-Paper-Scissors Game

Let us call the two players playing the game as player A and player B. The rules of the game are as follows:

- In each turn, both players will simultaneously have to produce by a gesture of their hands, either a stone, a paper or a pair of scissors (see Fig. 12.5).
- Stone beats scissors (since a stone can crush scissors), scissors beat paper (since scissors can cut paper) and paper beats stone (since a stone can be covered by paper).

- If both players produce the same gesture of their hands, it is a tie.
- In each turn, the loser has to pay Rs 1 to the winner, and in case of a tie both get rupee zero as payoff from each other.

What are the optimal strategies for players A and B?

Solution: The above information can be represented by a payoff matrix as given below. Note that the first entry of each of the nine cells represents the payoff value of row player A and the second entry represents the payoff value of column player B. You should also study Fig. 12.5 to understand the game better.

		Player B		
		Stone(B_1)	Paper(B_2)	Scissors(B_3)
Player A	Stone(A_1)	0, 0	-1, 1	1, -1
	Paper(A_2)	1, -1	0, 0	-1, 1
	Scissors(A_3)	-1, 1	1, -1	0, 0

Since it is a two-person zero-sum game, the payoff values of row player are equal in magnitude but opposite in sign to the payoff values of the column player in each play. So we can simply represent the game by the following payoff matrix for the row player (refer to Note 2 in Unit 11 of this course):

		Player B		
		Stone(B_1)	Paper(B_2)	Scissors(B_3)
Player A	Stone(A_1)	0	-1	1
	Paper(A_2)	1	0	-1
	Scissors(A_3)	-1	1	0

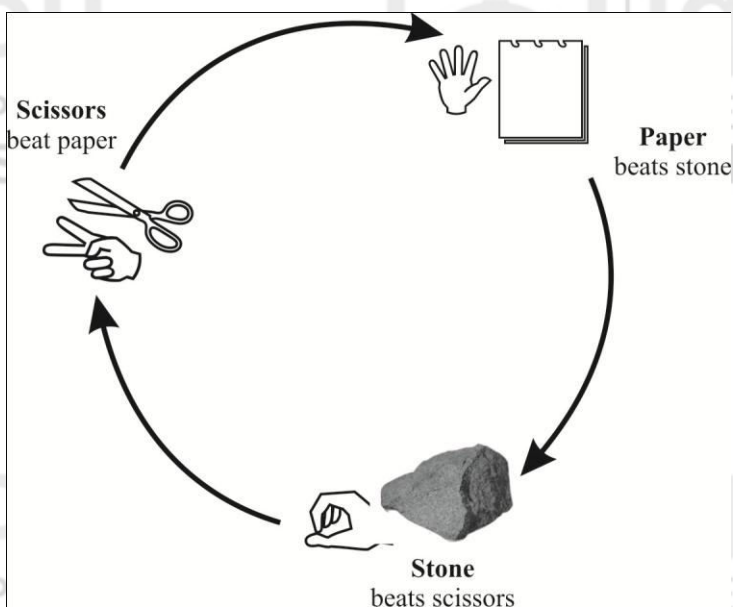


Fig. 12.5: Gestures for the stone-paper-scissors game.

We first check for the saddle point as follows:

		Player B			Row Minima	Maximin Value
		Stone(B_1)	Paper(B_2)	Scissors(B_3)		
Player A	Stone(A_1)	0	-1	1	-1	$\max\{-1, -1, -1\} = -1$
	Paper(A_2)	1	0	-1	-1	
	Scissors(A_3)	-1	1	0	-1	
Column Maxima		1	1	1		
Minimax Value		$\min\{1, 1, 1\} = 1$				

Since the maximin value $\neq$ minimax value, there is no saddle point. Hence, we have to obtain the solution in terms of mixed strategies. Also, we cannot reduce this game by using dominance rules because there is neither any inferior row nor any inferior column. Further, it is neither a $2 \times n$ nor an $m \times 2$ game. So it cannot be reduced to a 2×2 two-person zero-sum game using the graphical method. Hence, we cannot obtain its solution using any of the methods discussed so far. What do we do now? Let us find out.

We first apply the fundamental theorem of game theory (stated in Sec. 12.2) to this game. According to the theorem, since this is a finite two-person zero-sum game, it has a solution in terms of mixed strategies. Let

(p_1, p_2, p_3) and (q_1, q_2, q_3) be the optimal mixed strategies for players A and B, respectively, and v be the value of the game. Note that there is one commonality between 2×2 games (discussed in Sec. 12.2) and the 3×3 game. Both are square games. So let us proceed as in Sec. 12.2 and rewrite the payoff matrix with their probabilities:

			Player B		
			q_1	q_2	q_3
			Stone(B_1)	Paper(B_2)	Scissors(B_3)
Player A	p_1	Stone(A_1)	0	-1	1
	p_2	Paper(A_2)	1	0	-1
	p_3	Scissors(A_3)	-1	1	0

Now, the expected gain of player A is

$$0p_1 + 1p_2 + (-1)p_3 = p_2 - p_3 \quad \text{when player B employs strategy } B_1$$

$$(-1)p_1 + 0p_2 + 1p_3 = -p_1 + p_3 \quad \text{when player B employs strategy } B_2$$

$$1p_1 + (-1)p_2 + 0p_3 = p_1 - p_2 \quad \text{when player B employs strategy } B_3$$

Similarly, the expected loss of player B is

$$0q_1 + (-1)q_2 + 1q_3 = -q_2 + q_3 \quad \text{when player A employs strategy } A_1$$

$$1q_1 + 0q_2 + (-1)q_3 = q_1 - q_3 \quad \text{when player A employs strategy } A_2$$

$$(-1)q_1 + 1q_2 + 0q_3 = -q_1 + q_2 \quad \text{when player A employs strategy } A_3$$

Since v is the value of the game and (p_1, p_2, p_3) is the optimal mixed strategy for player A, we have

$$\begin{matrix} p_2 - p_3 = & -p_1 + p_3 = & p_1 - p_2 = & v \\ \text{I} & \text{II} & \text{III} & \text{IV} \end{matrix}$$

Now,

$$\text{I \& II} \Rightarrow p_1 + p_2 - 2p_3 = 0 \quad \dots \text{(i)}$$

$$\text{and II \& III} \Rightarrow -2p_1 + p_2 + p_3 = 0 \quad \dots \text{(ii)}$$

$$\text{Also } p_1 + p_2 + p_3 = 1 \quad \dots \text{(iii)}$$

Subtracting (i) from (iii) gives

$$3p_3 = 1 \Rightarrow p_3 = \frac{1}{3}$$

Subtracting (ii) from (iii) gives

$$3p_1 = 1 \Rightarrow p_1 = \frac{1}{3}$$

Putting the values of p_1 and p_3 in (iii), we get

$$\frac{1}{3} + p_2 + \frac{1}{3} = 1 \Rightarrow p_2 = \frac{1}{3}$$

$$\text{Again I \& IV} \Rightarrow v = p_2 - p_3 \Rightarrow v = \frac{1}{3} - \frac{1}{3} \Rightarrow v = 0$$

Also, v is value of the game and (q_1, q_2, q_3) is the optimal mixed strategy for the player B.

$$\therefore \underset{\text{I}}{-q_2} + \underset{\text{II}}{q_3} = \underset{\text{III}}{q_1} - \underset{\text{IV}}{q_3} = -q_1 + q_2 = v$$

$$\text{I \& II} \Rightarrow -q_1 - q_2 + 2q_3 = 0 \quad \dots \text{(iv)}$$

$$\text{II \& III} \Rightarrow 2q_1 - q_2 - q_3 = 0 \quad \dots \text{(v)}$$

$$\text{Also } q_1 + q_2 + q_3 = 1 \quad \dots \text{(vi)}$$

Adding (iv) and (vi) gives

$$3q_3 = 1 \Rightarrow q_3 = \frac{1}{3}$$

Adding (v) and (vi) gives

$$3q_1 = 1 \Rightarrow q_1 = \frac{1}{3}$$

Putting the values of q_1 and q_3 in (vi), we get

$$\frac{1}{3} + q_2 + \frac{1}{3} = 1 \Rightarrow q_2 = \frac{1}{3}$$

$$\text{I \& IV} \Rightarrow v = -q_2 + q_3 \Rightarrow v = -\frac{1}{3} + \frac{1}{3} \Rightarrow v = 0$$

Hence, the optimal strategy for both players A and B is $(1/3, 1/3, 1/3)$ and the value of the game is 0.

12.6 FLOWCHART

In Unit 11, we have introduced some key terms involved in the game theory. You have studied the maximin-minimax principle to solve two-person zero-sum games with saddle point. In Sec. 12.2, we have derived and applied the algebraic method to solve 2×2 two-person zero-sum games. In Sec. 12.3, you have learnt the dominance rules and applied them to reduce the size of the payoff matrix by eliminating inferior row(s) or column(s) or both. In Sec. 12.4, we have discussed the graphical method, which reduces $2 \times n$ and $m \times 2$ games into 2×2 games. We have also solved some games to explain the steps involved in this method. Finally, in Sec. 12.5, you have seen how 3×3 two-person zero-sum games can be solved by extending the technique for 2×2 two-person zero-sum games discussed in Sec 12.2. We have also solved a commonly played game known as Stone-Paper-Scissors game. At this stage, you may wish to know: How do we solve the $m \times n$ games, which are not

reducible to any of the games discussed in this unit? The answer is that $m \times n$ two-person zero-sum games can be solved by converting them into linear programming problems. But this discussion is beyond the scope of this course. However, if you are keen to learn this technique, you should first learn how to solve linear programming problems. This is discussed in detail in the course MSTE-002. Then you may refer to any book recommended at the end of this block or any other book available to you. We end the discussion by giving a flow chart in Fig. 12.6, which shows all the information that we have explained in this unit sequentially.

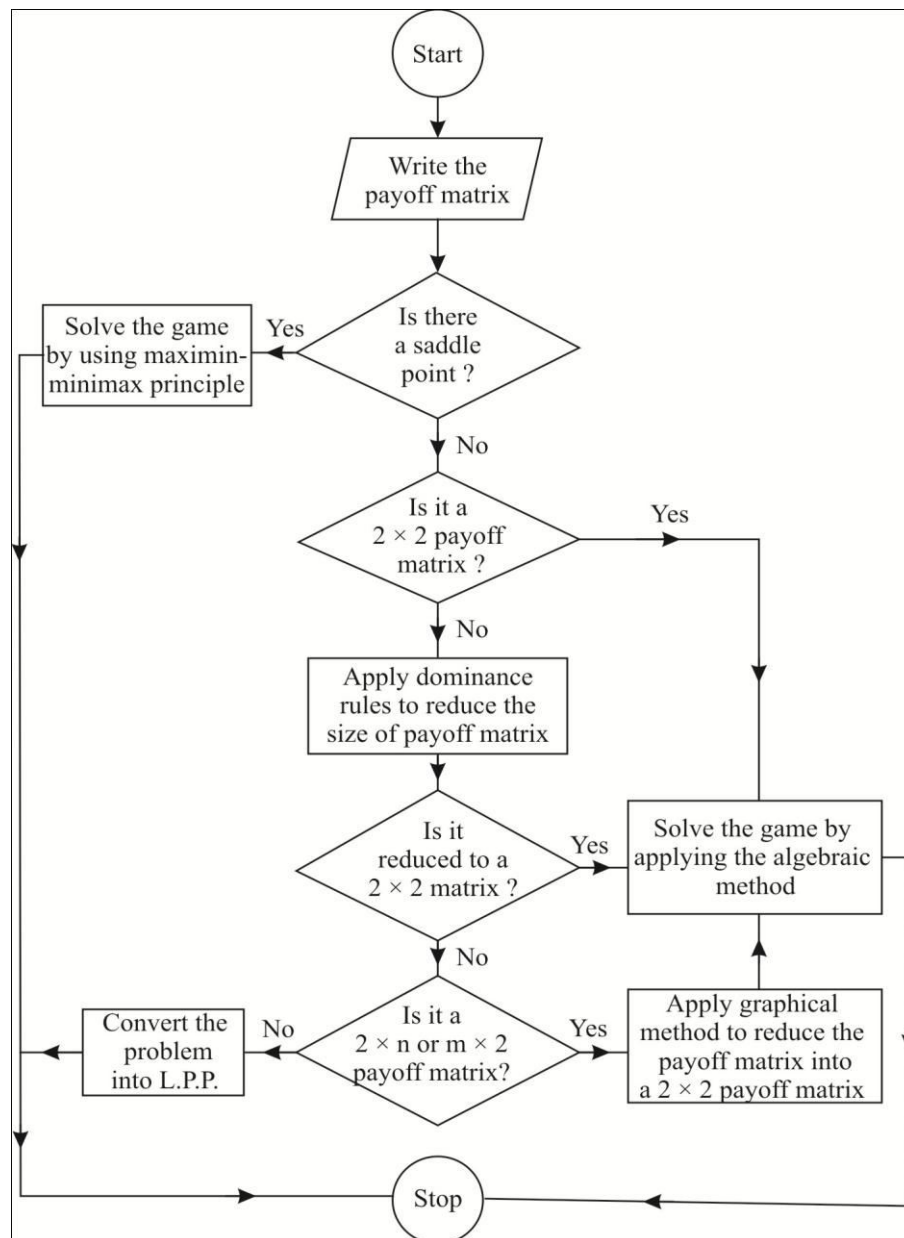


Fig. 12.6: Flowchart of game theory approach.

Let us now present a summary of the main points discussed in this unit.

12.7 SUMMARY

- 1) The solution of a 2×2 two-person zero-sum games in terms of mixed strategies is provided by the **algebraic method**:

$$p_1 = \frac{d - c}{(a - b) + (d - c)}$$

$$p_2 = \frac{a - b}{(a - b) + (d - c)}$$

$$q_1 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_2 = \frac{a - c}{(a - b) + (d - c)}$$

$$v = \frac{ad - bc}{(a - b) + (d - c)}$$

- 2) **Dominant Strategy:** A strategy is said to be the dominant strategy for a player if it is better than another strategy available to that player irrespective of the strategy adopted by the other player.

- 3) The dominance rules for reducing the size of the payoff matrix are:

Rule 1

If all the elements in a row (say, the i^{th} row) of a payoff matrix for row player are **less than or equal to** the corresponding elements of the other row (say, the j^{th} row), then the row player A will never employ his/her i^{th} strategy because it is dominated by his/her j^{th} strategy. So we can delete the i^{th} row from the payoff matrix.

Rule 2

If all the elements in a column (say, the r^{th} column) of a payoff matrix for column player are **greater than or equal to** the corresponding elements of the other column (say, the s^{th} column), then the column player B will never employ his/her r^{th} strategy because it is dominated by his/her s^{th} strategy. So we can delete the r^{th} column from the payoff matrix.

Rule 3

A strategy can also be deleted if it is dominated by average (or any convex combination) of two or more other pure strategies.

- 4) $2 \times n$ and $m \times 2$ games are solved by using the graphical method. The **graphical method** reduces $2 \times n$ and $m \times 2$ games to 2×2 games. Once a game is reduced to size 2×2 it can be solved by the algebraic method.

12.8 SOLUTIONS/ANSWERS

E1) We first check for the saddle point:

		Player B		Row Minima	Maximin Value
		B ₁	B ₂		
Player A	A ₁	-2	-1	-2	$\max\{-2, -3\} = -2$
	A ₂	4	-3	-3	
Column Maxima		4	-1		
Minimax Value		$\min\{4, -1\} = -1$			

Since the maximin value $(-2) \neq$ minimax value (-1) , there is no saddle point. Hence, we have to obtain the solution in terms of mixed strategies. Let (p_1, p_2) and (q_1, q_2) be the optimal mixed strategies for player A and player B, respectively, and v be the value of the game. Then from equations (3a to e), we have

$$p_1 = \frac{d-c}{(a-b)+(d-c)}, p_2 = \frac{a-b}{(a-b)+(d-c)}, q_1 = \frac{d-b}{(a-b)+(d-c)}$$

$$q_2 = \frac{a-c}{(a-b)+(d-c)} \text{ and } v = \frac{ad-bc}{(a-b)+(d-c)}$$

In this case, $a = -2, b = -1, c = 4, d = -3$.

$$\therefore p_1 = \frac{-3-4}{-2-(-1)+(-3)-4} = \frac{-7}{-8} = \frac{7}{8}, p_2 = \frac{-2-(-1)}{-2-(-1)+(-3)-4} = \frac{-1}{-8} = \frac{1}{8}$$

$$q_1 = \frac{-3-(-1)}{-2-(-1)+(-3)-4} = \frac{-2}{-8} = \frac{1}{4}, q_2 = \frac{-2-4}{-2-(-1)+(-3)-4} = \frac{-6}{-8} = \frac{3}{4}$$

$$v = \frac{(-2)(-3)-(-1) \times 4}{-2-(-1)+(-3)-4} = \frac{10}{-8} = -\frac{5}{4}$$

Hence, the solution of the game is:

The optimal strategy for player A is $(7/8, 1/8)$, the optimal strategy for player B is $(1/4, 3/4)$ and the value of the game is $-5/4$.

- E2)** In Note 2 of Unit 11, we have explained that the coin matching game is a two-person zero-sum game between child X (player I) and child Y (player II). The payoff matrix for player I is given below:

		Child Y (player II)	
		Head	Tail
Child X (Player I)	Head	1	-1
	Tail	-1	1

We first check for the saddle point.

		Player II		Row Minima	Maximin Value
		B ₁	B ₂		
Player I	A ₁	1	-1	-1	$\max\{-1, -1\} = -1$
	A ₂	-1	1	-1	
Column Maxima		1	1		
Minimax Value		$\min\{1, 1\} = 1$			

Since maximin value $(-1) \neq$ minimax value (1) , there is no saddle point. Hence, we obtain the solution in terms of mixed strategies.

Let (p_1, p_2) and (q_1, q_2) be the optimal mixed strategies for player A and player B, respectively, and v be the value of the game.

From equations (3a to e),

$$p_1 = \frac{d-c}{(a-b)+(d-c)}, p_2 = \frac{a-b}{(a-b)+(d-c)}, q_1 = \frac{d-b}{(a-b)+(d-c)}$$

$$q_2 = \frac{a-c}{(a-b)+(d-c)}, \text{ and } v = \frac{ad-bc}{(a-b)+(d-c)}$$

In this case, $a = 1, b = -1, c = -1, d = 1$

$$\therefore p_1 = \frac{1 - (-1)}{1 - (-1) + 1 - (-1)} = \frac{2}{4} = \frac{1}{2}, p_2 = \frac{1 - (-1)}{1 - (-1) + 1 - (-1)} = \frac{2}{4} = \frac{1}{2}$$

$$q_1 = \frac{1 - (-1)}{1 - (-1) + 1 - (-1)} = \frac{2}{4} = \frac{1}{2}, q_2 = \frac{1 - (-1)}{1 - (-1) + 1 - (-1)} = \frac{2}{4} = \frac{1}{2}$$

$$v = \frac{1 \times 1 - (-1) \times (-1)}{1 - (-1) + 1 - (-1)} = \frac{0}{4} = 0$$

Hence, the solution of the game is:

The optimal strategy for both child X (player I) and child Y (player II) is $(1/2, 1/2)$ and the value of the game is 0.

E3) We first check for the saddle point.

		Player B				Row Minima	Maximin Value
		B ₁	B ₂	B ₃	B ₄		
Player A	A ₁	4	3	2	1	1	max{1, 0, 0} = 1
	A ₂	6	4	5	0	0	
	A ₃	1	2	0	3	0	
Column Maxima		6	4	5	3		
Minimax Value		min{6, 4, 5, 3} = 3					

Since the maximin value $\neq$ minimax value, there is no saddle point. Hence, we obtain the solution in terms of mixed strategies. Let us first reduce the payoff matrix by applying the dominance rules.

Step 1: We note that each element of the first column is greater than or equal to the corresponding element of the third column. This implies that the third column dominates the first column. So, we eliminate the first column from the given payoff matrix. Thus, the reduced payoff matrix can be written as follows:

		Player B		
		B ₂	B ₃	B ₄
Player A	A ₁	3	2	1
	A ₂	4	5	0
	A ₃	2	0	3

Step 2: In the reduced payoff matrix given above, neither any row is directly dominated by any other row nor any column is directly dominated by any other column. But we observe that the average of the second and the third columns of the reduced payoff matrix dominates the first column. So, we can eliminate the first column from the reduced payoff matrix and obtain a further reduced payoff matrix as:

		Player B	
		B ₃	B ₄
Player A	A ₁	2	1
	A ₂	5	0
	A ₃	0	3

Step 3: In the reduced payoff matrix given above, neither any row is directly dominated by any other row nor any column is directly dominated by any other column. But we observe that the average of the second and third rows of the reduced payoff

matrix dominates the first row. So, we can eliminate the first row from the reduced payoff matrix and obtain a further reduced payoff matrix as:

	Player B		
		B ₃	B ₄
	Player A	A ₂	A ₃
		5	0
		0	3

Now, neither any row dominates any other row nor any column dominates any other column. So, this reduced payoff matrix cannot be reduced further by using dominance rules. But this reduced payoff matrix is of the order 2×2 . So, we can apply the algebraic method for 2×2 two-person zero-sum games. Let $(p_1, p_2, p_3), (q_1, q_2, q_3, q_4)$ be the mixed strategies for players A and B, respectively, for the original payoff matrix. The (2×2) reduced payoff matrix tells us that (p_2, p_3) and (q_3, q_4) are the mixed strategies for players A and B, respectively. If v is the value of the game, then from equations (3a to e), we have

$$p_2 = \frac{d - c}{(a - b) + (d - c)}, p_3 = \frac{a - b}{(a - b) + (d - c)}, q_3 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_4 = \frac{a - c}{(a - b) + (d - c)} \text{ and } v = \frac{ad - bc}{(a - b) + (d - c)}$$

In this case, $a = 5, b = 0, c = 0, d = 3$.

$$\therefore p_2 = \frac{3 - 0}{(5 - 0) + (3 - 0)} = \frac{3}{8}, p_3 = \frac{5 - 0}{(5 - 0) + (3 - 0)} = \frac{5}{8}$$

$$q_3 = \frac{3 - 0}{(5 - 0) + (3 - 0)} = \frac{3}{8}, q_4 = \frac{5 - 0}{(5 - 0) + (3 - 0)} = \frac{5}{8}$$

$$v = \frac{5 \times 3 - 0 \times 0}{(5 - 0) + (3 - 0)} = \frac{15}{8}$$

Hence, the solution of the game is:

The optimal strategy for player A is $(0, 3/8, 5/8)$, the optimal strategy for player B is $(0, 0, 3/8, 5/8)$ and the value of the game is $15/8$.

E4) We first check for the saddle point.

		Player B					Row Minima	Maximin Value
		B ₁	B ₂	B ₃	B ₄	B ₅		
	Player A	A ₁	A ₂	A ₃	A ₄	A ₅		
		3	4	5	-2	2	-2	$\max \{-2, -3\} = -2$
		1	6	-3	3	7	-3	
Column Maxima		3	6	5	3	7		
Minimax Value		$\min \{3, 6, 5, 3, 7\} = 3$						

Since maximin value $\neq$ minimax value, there is no saddle point.

Hence, we have to obtain the solution in terms of mixed strategies. But column four dominates the second and fifth columns. So, by dominance rules, both second and fifth columns can be eliminated from this payoff matrix and the reduced payoff matrix can be written as:

		Player B		
		B ₁	B ₃	B ₄
Player A	A ₁	3	5	-2
	A ₂	1	-3	3

This payoff matrix cannot be reduced further by using dominance rules. But this game is of the type $2 \times n$ where $n = 3$. Hence, we can apply the graphical method for $2 \times n$ games, which will reduce the game to a 2×2 two-person zero-sum game. Let (p_1, p_2) and

$(q_1, q_2, q_3, q_4, q_5)$ be the optimal mixed strategies for players A and B, respectively, for the original payoff matrix. Then the expected payoffs of player A corresponding to the moves B₁, B₃, B₄ of player B are given by:

$$e_1 = 3p_1 + p_2 = 3p_1 + (1 - p_1) = 2p_1 + 1 \quad \dots (i)$$

$$e_3 = 5p_1 - 3p_2 = 5p_1 - 3(1 - p_1) = 8p_1 - 3 \quad \dots (ii)$$

$$e_4 = -2p_1 + 3p_2 = -2p_1 + 3(1 - p_1) = -5p_1 + 3 \quad \dots (iii)$$

We now follow the steps given below to apply the graphical method for $2 \times n$ games as shown in Fig. 12.7:

Step 1: We draw a horizontal line, 1 unit in length, extending from the point R($p_1 = 0$) to point S($p_1 = 1$).

Step 2: We draw two vertical lines $p_1 = 0$ and $p_1 = 1$ passing through the two points R and S on the horizontal axis.

Step 3: We represent the payoff values of the second row (i.e., 1, -3, 3) along the vertical line $p_1 = 0$ and payoff values of the first row (i.e., 3, 5, -2) along the vertical line $p_1 = 1$.

Step 4: We plot the straight lines given by (i), (ii) and (iii) by joining the payoff values of two rows with their corresponding values. Here we plot the lines by joining 3 to 1, 5 to -3 and -2 to 3 as shown in Fig. 12.7.

Step 5: We draw bold lines along the line segments which form the lowest boundary of expected payoff lines given by (i), (ii), (iii). It is the lower envelope.

Step 6: We identify the highest point P on the lower envelope. Here it is the point of intersection of expected payoff lines given by (ii) and (iii). Thus, the given game reduces to:

			Player B	
			q ₃	q ₄
			B ₃	B ₄
Player A	p ₁	A ₁	5	−2
	p ₂	A ₂	−3	3

From equations (3a to e), we have

$$p_1 = \frac{d - c}{(a - b) + (d - c)}, p_2 = \frac{a - b}{(a - b) + (d - c)}, q_2 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_3 = \frac{a - c}{(a - b) + (d - c)} \text{ and } v = \frac{ad - bc}{(a - b) + (d - c)}$$

Here $a = 5$, $b = -2$, $c = -3$, $d = 3$.

$$\therefore p_1 = \frac{3 - (-3)}{(5 - (-2)) + (3 - (-3))} = \frac{6}{13}, p_2 = \frac{5 - (-2)}{(5 - (-2)) + (3 - (-3))} = \frac{7}{13}$$

$$q_2 = \frac{3 - (-2)}{(5 - (-2)) + (3 - (-3))} = \frac{5}{13}, q_3 = \frac{5 - (-3)}{(5 - (-2)) + (3 - (-3))} = \frac{8}{13}$$

$$v = \frac{5 \times 3 - (-2) \times (-3)}{(5 - (-2)) + (3 - (-3))} = \frac{9}{13}$$

Hence, the optimal strategies for original game for the players A and B

are $\left(\frac{6}{13}, \frac{7}{13}\right)$ and $\left(0, 0, \frac{5}{13}, \frac{8}{13}, 0\right)$, respectively, and the value of the

game is $\frac{9}{13}$.

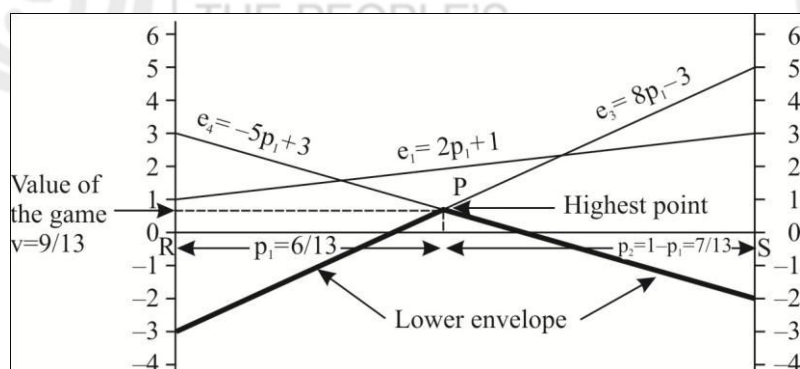


Fig. 12.7: Graph for player A in E4.

E 5) We first check for the saddle point.

		Player B		Row Minima	Maximin Value
		B ₁	B ₂		
Player A	A ₁	8	−2	−2	max { −2, 1, 2, 4 } = 4
	A ₂	5	1	1	
	A ₃	2	3	2	
	A ₄	4	7	4	
Column Maxima		8	7		
Minimax Value		min { 8, 7 } = 7			

Since maximin value $\neq$ minimax value, there is no saddle point. Hence, we have to obtain the solution in terms of mixed strategies. Since the third row is dominated by the fourth row, using dominance rules, we can eliminate the third row from this payoff matrix. The reduced payoff matrix can be written as:

		Player B	
		B ₁	B ₂
Player A	A ₁	8	-2
	A ₂	5	1
	A ₄	4	7

This payoff matrix cannot be reduced further by using dominance rules.

But this game is of the type $m \times 2$, where $m = 3$. Hence, we can apply the graphical method for $m \times 2$ games, which will reduce the game to a 2×2

two-person zero-sum game. Let (p_1, p_2, p_3, p_4) and (q_1, q_2) be the optimal mixed strategies for players A and B, respectively, for the original payoff matrix. The expected payoffs of player B corresponding to the moves A_1, A_2, A_4 of player A are given by

$$e_1 = 8q_1 - 2q_2 = 8q_1 - 2(1 - q_1) = 10q_1 - 2 \quad \dots (i)$$

$$e_2 = 5q_1 + 1q_2 = 5q_1 + 1(1 - q_1) = 4q_1 + 1 \quad \dots (ii)$$

$$e_4 = 4q_1 + 7q_2 = 4q_1 + 7(1 - q_1) = -3q_1 + 7 \quad \dots (iii)$$

We follow the steps given below to apply the graphical method for solving $m \times 2$ games as shown in Fig.12.8:

Step 1: We draw a horizontal line, 1 unit in length, extending from the point $R(q_1 = 0)$ to point $S(q_1 = 1)$ (see Fig. 12.8).

Step 2: We draw two vertical lines $q_1 = 0$ and $q_1 = 1$ passing through the two points R and S on the horizontal axis.

Step 3: We represent the payoff values of the second column (i.e., -2, 1, 7) along the vertical line $q_1 = 0$ and the payoff values of the first column (i.e., 8, 5, 4) along the vertical line $q_1 = 1$ as shown in Fig. 12.8.

Step 4: We plot the straight lines given by (i), (ii) and (iii) by joining the payoff values of two columns with their corresponding values. Here we plot the line by joining 8 to -2, 5 to 1 and 4 to 7 as shown in Fig. 12.8.

Step 5: We draw bold lines along the line segments, which form the uppermost boundary of the expected payoff lines given by (i), (ii), (iii). This is the upper envelope.

Step 6: We identify the lowest point P on the upper envelope. Here it is the point of intersection of expected payoff lines given by (i) and (iii). Thus, the given game reduces to:

			Player B	
			q ₁	q ₂
			B ₁	B ₂
Player A	p ₁	A ₁	8	−2
	p ₄	A ₄	4	7

From equations (3a to e), we have

$$p_1 = \frac{d - c}{(a - b) + (d - c)}, p_4 = \frac{a - b}{(a - b) + (d - c)}, q_1 = \frac{d - b}{(a - b) + (d - c)}$$

$$q_2 = \frac{a - c}{(a - b) + (d - c)} \text{ and } v = \frac{ad - bc}{(a - b) + (d - c)}$$

In this case, $a = 8, b = -2, c = 4, d = 7$.

$$\therefore p_1 = \frac{7 - 4}{(8 - (-2)) + (7 - 4)} = \frac{3}{13}, p_4 = \frac{8 - (-2)}{(8 - (-2)) + (7 - 4)} = \frac{10}{13}$$

$$q_1 = \frac{7 - (-2)}{(8 - (-2)) + (7 - 4)} = \frac{9}{13}, q_2 = \frac{8 - 4}{(8 - (-2)) + (7 - 4)} = \frac{4}{13}$$

$$v = \frac{8 \times 7 - (-2) \times 4}{(8 - (-2)) + (7 - 4)} = \frac{64}{13}$$

Hence, the optimal strategies for original game for players A and B are

$\left(\frac{3}{13}, 0, 0, \frac{10}{13}\right)$ and $\left(\frac{9}{13}, \frac{4}{13}\right)$, respectively, and the value of the game is

$$\frac{64}{13}.$$

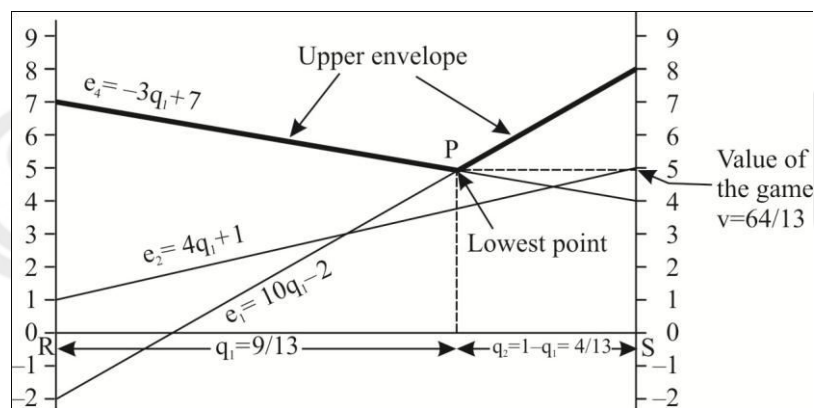


Fig. 12.8: Graph for player B.

FURTHER READING

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- 3) Mathematics Programming Techniques by N. S. Kambo; Affiliated East-West Press Pvt. Ltd., New Delhi (1984) [Chapter 16].

Block

4**RELIABILITY THEORY**

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BLOCK 4 RELIABILITY

In Block 3, you have learnt how statistical tools help in decision making in different situations. In the same block you have also learnt how to solve games between two opponents.

This block discusses another important issue related to each component/product/device/unit/system: that of reliability. By reliability of a component or system, we mean how long it performs its function (intended function, in technical terms) successfully which it is expected to perform under stated conditions. Our discussion is mostly limited to identical components. This block is divided into four units.

In **Unit 13** entitled **Introduction to Reliability** we define reliability and explain some basic functions of reliability. We also discuss how to estimate the basic functions from test generated complete data.

Unit 14 entitled **Reliability Evaluation of Simple Systems** discusses the concepts of series, parallel and mixed configurations. We also explain how to evaluate reliabilities of the systems having components in either of these configurations.

In **Unit 15** entitled **Reliability Evaluation of k-out-of-n and Standby Systems** we explain how to evaluate the reliability of two other commonly used configurations known as k-out-of-n and standby systems. However, reliability of standby systems is discussed only under certain assumptions.

Unit 16 entitled **Reliability Evaluation of Complex Systems** discusses three general techniques to evaluate the reliability of complex systems. These techniques can also be used to evaluate the reliability of simple systems discussed in Unit 14.

Studying this block requires the knowledge of the first 8 units of the course MST-001 and Units 1 to 3, 5, 8, 9, 15 and 16 of MST-003. So you should keep these courses at hand and should refresh the concepts discussed therein.

Notations and Symbols

Sec.	:	Section
Secs.	:	Sections
Fig.	:	Figure
$N_s(t)$	:	Number of components that are operating at time t
$N_f(t)$	:	Number of components that have failed at time t
$R(t)$	:	Value of reliability function at time t
$F(t)$	:	Value of cumulative failure distribution function at time t
$f(t)$	:	Failure density function
$\lambda(t)$	:	Hazard rate function
MTTF	:	Mean time to failure

UNIT 13 INTRODUCTION TO RELIABILITY

Structure

- 13.1 Introduction
 - Objectives
- 13.2 Definition of Reliability
- 13.3 Basic Functions in Reliability and their Relationship
 - Reliability Function
 - Cumulative Failure Distribution Function
 - Failure Density Function
 - Hazard Rate
 - Relationship between the Functions $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$
 - Mean Time to Failure and Median of the Random Variable T
- 13.4 Bathtub Curve
- 13.5 Estimation of Reliability Functions from Failure Data
- 13.6 Summary
- 13.7 Solutions/Answers

13.1 INTRODUCTION

Due to the rapid advancement of science and technology and to meet the ever-increasing needs of the society, industries are increasingly introducing automation for producing goods ranging from the simplest to highly complex systems. However, all processes and products are prone to failure. For example, a television remote control may stop functioning or a malfunction may occur in any other household appliance; an automobile engine starter may fail, an aeroplane may crash due to the failure of some of its component(s), credit card transactions may fail, and so on. You can think of many more similar examples from your daily life. So we would like to understand failures, present them in a different manner and see how we can try to prevent the interruption due to failure! Reliability is nothing but a science of better presentation of failure. For example, the statement that 20% students of the class failed the test can be better presented as 80% students of the class passed the test!

The mathematical foundations of reliability were laid down when queueing theory and renewal theory were developed in the early 1930s and 1940s. You will study some basic concepts of queueing theory in Unit 6 of MSTE-002. However, reliability as a subject gained importance only after World War II because a high failure rate of electronic equipment was observed during the war. The first journal on the subject, Institute of Electrical and Electronics Engineers (IEEE) – Transactions on Reliability, came out in 1963. A number of textbooks on the subject were published in the 1960s.

We give a formal definition of reliability in Sec. 13.2. Four basic functions in reliability, namely, the reliability function, cumulative failure distribution function, failure density function and the hazard rate and their relations are described in Sec. 13.3. We also explain the terms mean time to failure and median of the random variable, time to failure in Sec. 13.3. When a population of identical components is tested, the curve of the relative failure rate of the entire population of the components without replacement is generally of the shape of a bathtub. This curve, called the bathtub curve, is discussed in

It is obvious that without the possibility of one type of failure or another insurance industry would not exist. The science which deals with the concepts used in the insurance industry is known as actuarial science. That is why many concepts of reliability and actuarial science are the same except the terminology used there.

Sec. 13.4. We explain how the basic reliability functions can be estimated from failure data in Sec. 13.5.

In the next unit, we shall discuss reliability evaluation of simple systems.

Objectives

After studying this unit, you should be able to:

- define reliability;
- explain the basic functions in reliability, namely, reliability function, cumulative failure distribution function, failure density function and hazard rate;
- explain three phases of the bathtub curve; and
- estimate basic reliability functions from complete failure data.

13.2 DEFINITION OF RELIABILITY

In articles on reliability, many authors suggest that the word ‘reliability’ was first coined by the poet Samuel Taylor Coleridge when he wrote a piece in praise of his friend, the poet Robert Southey in 1816. However, reliability as a subject of study was taken up only after World War II when high failure rate of electronic equipment used in the war was observed. Today it has taken a new shape by blending itself in all phases of the product life cycle from proposal and design to manufacturing. So in the literature on reliability we find that the definitions of reliability vary from one practitioner or researcher to another. Still a comprehensive and widely accepted definition of reliability may be stated as follows:

Definition of Reliability: Reliability of a component/device/equipment/unit/system is the **probability** that it performs its **intended function adequately** for a specified period of **time** under the **given operating conditions**.

To understand what this definition actually means, we have to clearly understand the meaning of the above-mentioned bold words, which are the key words in this definition. Let us explain what we mean by these key words:

Probability: When a number of identical components operate under similar conditions, the failure times of components generally vary from component to component and we cannot predict their failure times exactly in advance. But we can describe the phenomenon of failure in probabilistic terms. For example, suppose we put N_0 identical components in operation at time $t = 0$ under identical conditions. Suppose after t units of time, we have

$N_s(t)$ = number of components that are operating at time t

$N_f(t)$ = number of components that have failed at time t

Then reliability $R(t)$ at time t is defined as

$$R(t) = \frac{N_s(t)}{N_0} \quad \dots (1)$$

Thus, reliability of a component/system at time t is the probability that it performs its function without failure. Being a probability, its value lies between 0 and 1. From equation (1), we can see that

if $N_s(t) = 0$, $R(t) = 0$ and

if $N_s(t) = N_0, R(t) = 1$.

By definition $N_s(t)$ satisfies the following relation:

$$0 \leq N_s(t) \leq N_0 \quad \text{for any time interval } [0, t]$$

$$\Rightarrow \frac{0}{N_0} \leq \frac{N_s(t)}{N_0} \leq \frac{N_0}{N_0} \quad [\text{on dividing by } N_0]$$

$$\Rightarrow 0 \leq R(t) \leq 1$$

What does this mean? Suppose that the reliability of the refrigerator of a particular company at 10 years is $R(t) = 0.9452$. Then this can be interpreted as follows: Out of 10000 refrigerators of this particular company, approximately 9452 refrigerators worked without failure for 10 years.

Intended Function: The intended function of a product must be described in unambiguous terms early on in the design process so that the expected requirements of the customers can be ensured/addressed. Thus, **intended function** of a product is the function/work/job which it is expected to perform when we put it in operation under stated conditions. For example, if a pump is designed to deliver at least 300 gallons of water per minute, the intended function of the pump is to deliver 300 gallons or more water per minute.

Adequately: A product is said to perform **adequately** if it is performing its intended function as expected. For example, if the intended function of a water pump is to deliver 300 gallons or more water and it is delivering 300 gallons or more water per minute, it means that it is performing adequately. But if the pump delivers less than 300 gallons of water per minute we say that it has failed or is not performing adequately.

Time: You should always keep in mind that reliability can be meaningful only if it is related to a time interval or period of time. For example, we can say that the reliability of a component is 0.98 for a mission time of 100 hours. But the statement 'reliability of a component is 0.95' is meaningless since the time interval is unknown. From equation (1), reliability at time 't' can be defined as follows:

$$R(t) = \frac{\text{Number of components performing intended function at time 't'}}{\text{Number of component at start (i.e., when 't' = 0)}} \quad \dots (2)$$

In the above definition, time 't' means the interval $[0, t]$.

In equation (2), $R(0) = 1$ since no component fails at $t = 0$, and $R(t \rightarrow \infty) = 0$ since no component can survive forever. This point will become clearer to you in the next section when we discuss the wear out period of a component.

Given Operating Condition: By given operating condition, we mean that the performance of the product should be observed under normal stated conditions in which it is expected to perform. The environmental conditions (such as temperature, humidity, shock, vibration, altitude, etc.), design loads (such as voltages, pressure, etc.) and operating conditions (such as maintenance, storage, etc.) affect the reliability of a product. So a product will perform better in the field if its design takes into account and is representative of how it is actually used by the customers. For example, a car which is designed for smooth roads will not perform well if a customer uses it on rough roads.

You may like to try the following exercises to check your understanding of the definition of reliability.

- E1)** Which of the following is not a part of the definition of reliability?
- A) Probability B) Intended function
C) Time D) Certainty
- E2)** The maximum possible value of reliability for a given period of time is
- A) 1 B) 0 C) 0.1 D) 100

13.3 BASIC FUNCTIONS IN RELIABILITY AND THEIR RELATIONSHIP

The goal of this section is to define the following basic functions and explain the relationship between them:

- Reliability function,
- Cumulative failure distribution function,
- Failure density function, and
- Hazard rate function.

In this section, we also obtain expressions for the mean and median of the random variable called the **time to failure** or failure time.

Let us explain these functions one at a time.

13.3.1 Reliability Function

In Sec. 13.2, you have learnt that reliability is a probability. If reliability is a probability, then a random variable has to be associated with it because probability is always associated with a random variable. You are familiar with the concept of a random variable as it has been discussed in detail in the course MST-003. In MST-003, you have also studied some discrete as well as continuous probability distributions. Now, can you guess what the random variable in the case of reliability is? Think for a while.

We hope that you have given this a thought and arrived at the following idea: Generally, the random variable in the case of reliability is generally time – the time in which a component/system fails. Let this random variable, called **time to failure** or **failure time** of the component/system, be denoted by T. Then, by definition, reliability [R(t)] at time t, of the component/system is given by

$$R(t) = \text{Probability that the time to failure of the component/system is } > t$$

$$= P[T > t] \geq 0 \quad \left[\because \text{probability of any event A} \right]$$

$$\quad \quad \quad \text{always satisfies } 0 \leq P(A) \leq 1 \quad \dots (3)$$

The reliability of a component/system when expressed in terms of time t is known as **reliability function** and is generally denoted by $R(t)$.

Let us consider two particular cases of the reliability function:
when $t = 0$ and when $t \rightarrow \infty$.

$$R(0) = P[T > 0] = 1 \quad \left[\because \text{we assume that at } t = 0, \text{ the component / system is performing its intended function.} \right]$$

$$R(t \rightarrow \infty) = P[T > t(t \rightarrow \infty)] = 0 \left[\begin{array}{l} \because \text{no component / system can perform} \\ \text{its intended function forever.} \end{array} \right]$$

The probability of survival of a component decreases as the life time of the component increases. Ultimately this probability will approach zero, since no

component can perform its intended function forever. A typical shape of the reliability function is shown in Fig. 13.1.

From Fig. 13.1, you can see that the reliability function is a non-increasing function in time (t).

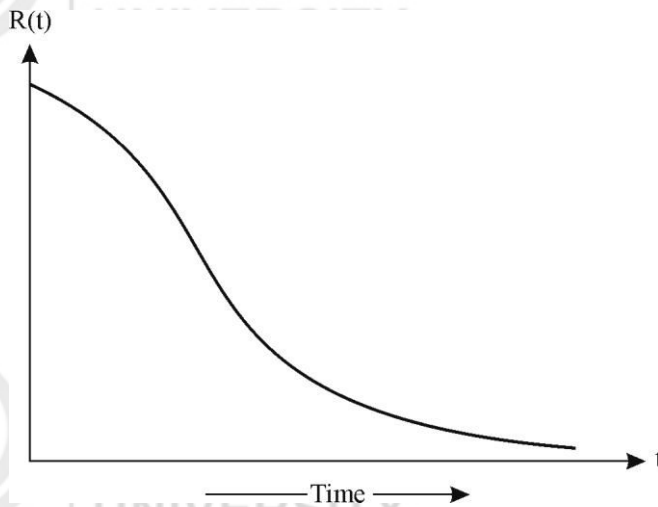


Fig. 13.1: Typical shape of reliability function $R(t)$.

13.3.2 Cumulative Failure Distribution Function

Let us define the function $F(t)$ by

$$F(t) = 1 - R(t) \quad \dots (4)$$

$$\therefore F(0) = 1 - R(0) = 1 - 1 = 0$$

$$\text{and } F(t)_{t \rightarrow \infty} = 1 - R(t)_{t \rightarrow \infty} = 1 - 0 = 1$$

Using equation (3), we can write equation (4) as

$$\begin{aligned} F(t) &= 1 - R(t) = 1 - P[X > t] \\ &= P[X \leq t] \end{aligned} \quad \left[\begin{array}{l} \because \text{from Unit 5 of MST-003, you know that} \\ \text{events } P[X > t] \text{ and } P[X \leq t] \text{ are complementary} \\ \text{to each other. Hence, } P[X > t] + P[X \leq t] = 1. \end{array} \right]$$

$$\text{or } F(t) = P[T \leq t] \quad \dots (5)$$

$$\text{Also, as shown above } F(0) = 0, F(\infty) = 1 \quad \dots (6)$$

Study equations (5) and (6). Can you identify a probability distribution function given in Unit 5 of MST-003, which has similar characteristics? Recall that the cumulative distribution function has these characteristics. But in reliability terminology, this distribution function is known as **cumulative failure distribution function** and gives **unreliability** of the component up to time t . So if we want to calculate the probability of failure of a component at time t (known as **unreliability** of the component), then we have to simply obtain the value of the function $F(t)$. Therefore, if we denote reliability and unreliability of a component by R and Q , respectively, then they satisfy the relation:

$$R + Q = 1 \quad \dots (7)$$

Note that the probability of failure of a component increases after its useful life period is over. Ultimately this probability will approach 1. A typical shape of cumulative failure distribution is shown in Fig. 13.2.

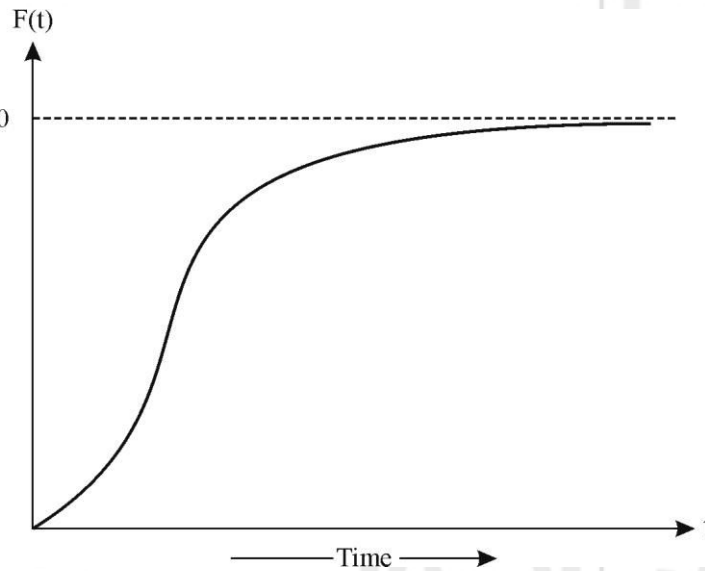


Fig. 13.2: Typical shape of cumulative failure distribution.

13.3.3 Failure Density Function

You have learnt in Unit 5 of MST-003 that the derivative of cumulative distribution function of a random variable is known as the probability density function (pdf) of the random variable. But in the terminology of reliability, the probability density function is known as **failure density function** (fdf). As explained in Sec. 13.3.2, the cumulative distribution function in reliability terminology is the cumulative failure distribution function. So if $f(t)$ denotes failure density function (life time density function), then

$$f(t) = \frac{d}{dt}(F(t)) \quad \dots (8a)$$

From Unit 5 of MST-003, recall that the following relationship exists between $f(t)$ and $F(t)$:

$$F(t) = \int_{-\infty}^t f(t)dt \quad \dots (8b)$$

Since t represents time and so $t \geq 0$, we can write equation (8b) as

$$F(t) = \int_0^t f(t)dt \quad \dots (9)$$

From equation (4), we have

$$R(t) = 1 - F(t) = 1 - \int_0^t f(t)dt \quad \dots (10a)$$

Since $f(t)$ is the pdf of the random variable T , $\int_0^{\infty} f(t)dt = 1$, i.e., the sum of all probabilities of a random variable is always equal to 1. So we can replace 1 in equation (10a) by the integral $\int_0^{\infty} f(t)dt$ and write

$$R(t) = \int_0^{\infty} f(t)dt - \int_0^t f(t)dt \quad \dots (10b)$$

$$R(t) = \left[\int_0^t f(t)dt + \int_t^\infty f(t)dt \right] - \int_0^t f(t)dt \left[\because \text{from Unit 8 of MST-001, we know that} \right. \\ \left. \int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx, a < c < b \right]$$

or $R(t) = \int_t^\infty f(t)dt$... (10c)

Thus, we have

$$F(t) = \int_0^t f(t)dt \quad \dots (11a)$$

and $R(t) = \int_t^\infty f(t)dt$... (11b)

If use of t within the integral sign as well as in the limits of the integral creates confusion for you, then the variable t inside the integral sign can be changed to any other dummy variable, say u . Then we can write $F(t)$ and $R(t)$ as:

$$F(t) = \int_0^t f(u)du \quad \dots (11c)$$

and $R(t) = \int_t^\infty f(u)du$... (11d)

These functions may be interpreted geometrically as areas under the curve of $f(t)$ as shown in Fig 13.3.

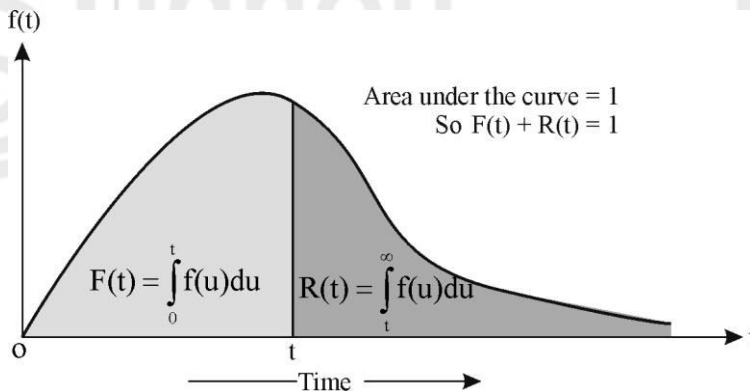


Fig. 13.3: Typical shape of failure density function and geometrical interpretation of $F(t)$ and $R(t)$.

13.3.4 Hazard Rate

The instantaneous rate of failure at time t is known as the **hazard rate** and is generally denoted by $\lambda(t)$.

To define it mathematically, let T denote the random variable, time to failure of the component/system. Let us consider the time interval $[t, t + \Delta t]$. Then, we have

$$P[t \leq T \leq t + \Delta t] = F(t + \Delta t) - F(t) \left[\because P[a \leq X \leq b] = F(b) - F(a) \text{ as you} \right. \\ \left. \text{have learnt in Unit 5 of MST-003} \right] \\ = [1 - R(t + \Delta t)] - [1 - R(t)] \quad [\because R(t) + F(t) = 1] \\ = R(t) - R(t + \Delta t) \quad \dots (12)$$

Conditional probability of failure in the time interval $[t, t + \Delta t]$ given that the component/system has performed its intended function till time t is given by

$$P[t \leq T \leq t + \Delta t | T > t] = \frac{P[t \leq T \leq t + \Delta t]}{P[T > t]}$$

$\therefore P(A|B) = \frac{P(A \cap B)}{P(B)}$. If you want to refresh your memory, you may refer to Unit 3 of MST-003.

Using equation (12) and $R(t) = P[T > t]$, we get

$$P[t \leq T \leq t + \Delta t | T > t] = \frac{R(t) - R(t + \Delta t)}{R(t)}$$

Then the conditional probability of failure per unit time is given by

$$\frac{R(t) - R(t + \Delta t)}{R(t)\Delta t},$$

and instantaneous rate of failure at time t (i.e., hazard rate $\lambda(t)$) is given by

$$\lambda(t) = \lim_{\Delta t \rightarrow 0} \frac{R(t) - R(t + \Delta t)}{R(t)\Delta t} = -\frac{1}{R(t)} \lim_{\Delta t \rightarrow 0} \frac{R(t + \Delta t) - R(t)}{\Delta t}$$

$$= -\frac{1}{R(t)} \frac{d}{dt}(R(t)) \quad \left[\text{on using the definition of derivative; refer to Unit 6 of MST-001.} \right]$$

$$\text{Since } f(t) = \frac{d}{dt}(F(t)) = \frac{d}{dt}(1 - R(t)) = -\frac{d}{dt}(R(t)),$$

$$\lambda(t) = \frac{-\frac{d}{dt}(R(t))}{R(t)} = \frac{f(t)}{R(t)} \quad \dots (13)$$

13.3.5 Relationship between the Functions $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$

So far we have defined the functions $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$. In this section, you will learn how these functions can be obtained from the failure data collected by testing a fixed number of identical components. And finally, we form a table showing the relationship between them, which will be very useful when we solve numerical problems.

Suppose that initially a fixed number (N_0) of identical components are put to test. Let

$N_s(t)$ = number of components that are performing the intended function adequately at time t

$N_f(t)$ = number of components that have failed at time t

$$\text{where } N_s(t) + N_f(t) = N_0 \quad \dots (14a)$$

Now, from equation (1), we know that reliability at time t is given by

$$R(t) = \frac{N_s(t)}{N_0} = \frac{N_0 - N_f(t)}{N_0} \quad [\because N_s(t) + N_f(t) = N_0]$$

$$\text{or } R(t) = 1 - \frac{N_f(t)}{N_0} \quad \dots (14b)$$

Differentiating $R(t)$ with respect to t , we get

$$\frac{d}{dt}(R(t)) = -\frac{1}{N_0} \frac{d}{dt}(N_f(t)) \quad \dots (15)$$

Also, from equations (8a) and (4), we know that

$$\begin{aligned} f(t) &= \frac{d}{dt}(F(t)) = \frac{d}{dt}(1-R(t)) \\ &= -\frac{d}{dt}(R(t)) \quad \dots (16a) \end{aligned}$$

$$= -\left(-\frac{1}{N_0} \frac{d}{dt}(N_f(t))\right) \quad [\text{Using equation (15)}]$$

$$\text{or } f(t) = \frac{1}{N_0} \frac{d}{dt}(N_f(t)) \quad \dots (16b)$$

By definition, hazard rate is given by

$$\lambda(t) = \frac{\text{number of failures per unit time at time } t}{\text{number of components which are exposed to failure at time } t}$$

Since derivative $\frac{dy}{dx}$ = rate of change of y with respect to x, therefore, the numerator in terms of derivative can be written as follows :

$$\lambda(t) = \frac{\frac{d}{dt}(N_f(t))}{N_s(t)} \quad \dots (17)$$

$$\begin{aligned} \text{or } \lambda(t) &= \frac{N_0}{N_s(t)} \frac{1}{N_0} \frac{d}{dt}(N_f(t)) \quad [\text{on multiplying and dividing by } N_0] \\ &= \frac{1}{R(t)} f(t) \quad \left[\begin{array}{l} \because R(t) = \frac{N_s(t)}{N_0} \text{ from equation} \\ (1) \text{ and using equation (16b)} \end{array} \right] \end{aligned}$$

$$\text{or } \lambda(t) = \frac{f(t)}{R(t)} \quad \dots (18)$$

$$= -\frac{1}{R(t)} \frac{d}{dt}(R(t)) \quad [\text{using equation (16a)}]$$

$$\therefore \lambda(t) = -\frac{1}{R(t)} \frac{d}{dt}(R(t)) \quad \dots (19)$$

Integrating $\lambda(t)$ with respect to t on both sides within the limits t = 0 to t = t, we get

$$\begin{aligned} \int_0^t \lambda(t) dt &= -\int_0^t \frac{1}{R(t)} \frac{d}{dt}(R(t)) dt = -\int_0^t \frac{\frac{d}{dt}(R(t))}{R(t)} dt \\ &= -[\ln|R(t)|]_0^t \quad \left[\because \int_a^b \frac{\frac{d}{dx}(f(x))}{f(x)} dx = [\ln|f(x)|]_a^b \right] \\ &= -(\ln|R(t)| - \ln|R(0)|) \\ &= -\ln|R(t)| - \ln 1 \quad [\because R(0) = 1] \end{aligned}$$

$$\begin{aligned}
 &= -\ln(R(t)) - 0 \\
 &= -\ln R(t) \\
 &\Rightarrow -\int_0^t \lambda(t) dt = \ln R(t) \quad \dots (20a)
 \end{aligned}$$

$$\Rightarrow R(t) = e^{-\int_0^t \lambda(t) dt} \quad \left[\because \ln a = m \Rightarrow a = e^m \right]$$

$$\text{or } R(t) = \exp\left(-\int_0^t \lambda(t) dt\right) \quad \dots (20b)$$

Note that if $\lambda(t)$ is constant and $\lambda(t) = \lambda$, then equation (20b) gives

$$R(t) = e^{-\lambda t} \quad \dots (20c)$$

We now put together the relations between $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$ in Table 13.1 as follows:

Table 13.1: Relationship between the Functions $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$

	$R(t)$	$F(t)$	$f(t)$	$\lambda(t)$
$R(t)$	—	$1 - F(t)$	$\int_t^\infty f(t) dt$	$\exp\left(-\int_0^t \lambda(t) dt\right)$
$F(t)$	$1 - R(t)$	—	$\int_0^t f(t) dt$	$1 - \exp\left(-\int_0^t \lambda(t) dt\right)$
$f(t)$	$-\frac{d}{dt}(R(t))$	$\frac{d}{dt}(F(t))$	—	$\lambda(t) \exp\left(-\int_0^t \lambda(t) dt\right)$
$\lambda(t)$	$-\frac{d}{dt}(\ln R(t))$	$\frac{\frac{d}{dt}(F(t))}{1 - F(t)}$	$\frac{f(t)}{\int_0^t f(t) dt}$	—

13.3.6 Mean Time to Failure and Median of the Random Variable T

In reliability terminology, the mean of the random variable T in the absence of repair and replacement is known as the **mean time to failure** (MTTF). Usually the users of a product/component are interested in knowing the average life of the product/component rather than the complete failure details. MTTF is a measure which simply gives a number (in units of time in which life of the product/component is measured) that tells us on an average how long the product/component performs its intended function successfully. It is calculated by taking the mean of the lifetimes obtained on the basis of results of a sample of such identical products/components tested under stated conditions. For example, suppose we put 10 identical components to test under the stated conditions. Let t_i , ($i = 1, 2, \dots, 10$) denote the time for which i^{th} component performs its intended function successfully. The results of the sample are shown in Fig. 13.4. Thus, mean time to failure (MTTF) of such components on the basis of the results of this sample is given by

$$\text{MTTF} = \frac{1}{10} \sum_{i=1}^{10} t_i = \frac{1}{10} (5 + 2 + 11 + 6 + 9 + 1 + 5 + 4 + 6 + 1) = \frac{50}{10} = 5 \text{ hr}$$

In general, if n components are put to test and $t_i, (i = 1, 2, 3, \dots, n)$ denotes the time for which the i^{th} component performs its intended function, then mean time to failure (MTTF) of such components is given by

$$\text{MTTF} = \frac{1}{n} \sum_{i=1}^n t_i$$

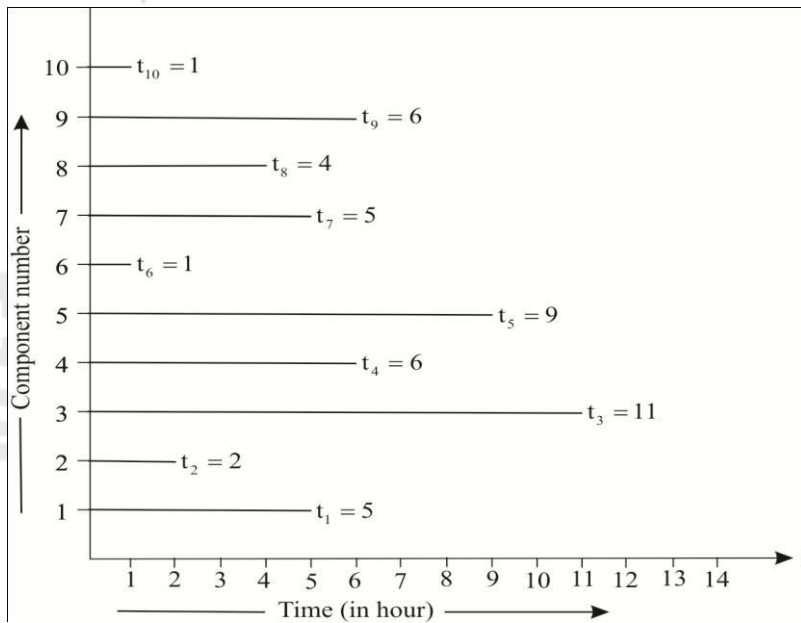


Fig. 13.4: Times for which each of the 10 components operates successfully.

However, if we are given the basic functions of a component then by definition, MTTF is given by

$$\begin{aligned} \text{MTTF} &= E(T) = \int_0^{\infty} t f(t) dt \quad [\text{refer to Unit 8 of MST-003}] \quad \dots (21) \\ &= \int_0^{\infty} t \left(-\frac{d}{dt} R(t) \right) dt \quad \left[\because f(t) = -\frac{d}{dt} R(t) \text{ from equation (16a)} \right] \end{aligned}$$

I II

Integrating by parts, we get

$$\text{MTTF} = t(-R(t)) \Big|_0^{\infty} - \int_0^{\infty} (1)(-R(t)) dt = tR(t) \Big|_{t \rightarrow \infty} - 0 + \int_0^{\infty} R(t) dt$$

Since $R(t) = e^{-\int_0^t \lambda(t) dt}$ and $\lim_{x \rightarrow \infty} x^n e^{-x} = 0$ for $n > -1$,

$$\text{MTTF} = \int_0^{\infty} R(t) dt \quad \dots (22)$$

To calculate MTTF, you should use equation (22) instead of equation (21), because, in general, dealing with equation (22) is easier.

Note 1: If failure rate is constant, i.e., $\lambda(t) = \lambda$, we have

$$R(t) = e^{-\int_0^t \lambda dt} = e^{-\lambda t}$$

$$\therefore \text{MTTF} = \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-\lambda t} dt = \left[\frac{e^{-\lambda t}}{-\lambda} \right]_0^{\infty} = -\frac{1}{\lambda} (0 - 1) = \frac{1}{\lambda}$$

$$\text{i.e. } \boxed{\text{MTTF} = \frac{1}{\text{constant failure rate}}} \text{ or } \boxed{\text{Constant failure rate} = \frac{1}{\text{MTTF}}} \dots (23)$$

Now, to obtain an expression for median of the random variable T , we have to recall the definition of median from MST-002 or MST-003. Median is that value of the variate which divides the distribution of the variate into two equal parts. Therefore, if t_{md} denotes the median of the random variable T , then

$$R(t_{md}) = P[T > t_{md}] = 0.5 \quad \dots (24)$$

Let us now consider a few examples to apply the concepts discussed in this section to numerical problems.

Example 1: The failure density function of the random variable T is given by

$$f(t) = \begin{cases} 0.012 e^{-0.012t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

Calculate

- Reliability of the component.
- Reliability of the component for a 100 hour mission time.
- Mean time to failure (MTTF).
- Median of the random variable T .
- What is the life of the component if a reliability of 0.96 is desired?

Solution: Failure density function of the random variable T is given as

$$f(t) = \begin{cases} 0.012 e^{-0.012t}, & t \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

- From equation (11b), reliability is given by

$$\begin{aligned} R(t) &= \int_t^{\infty} f(t) dt = \int_t^{\infty} 0.012 e^{-0.012t} dt \\ &= \left[\frac{0.012 e^{-0.012t}}{-0.012} \right]_t^{\infty} \quad \left[\because \int_a^b e^{kt} dt = \left[\frac{e^{kt}}{k} \right]_a^b \right] \\ &= - \left[0 - e^{-0.012t} \right] \quad \left[\because e^{-0.012t} \rightarrow 0 \text{ as } t \rightarrow \infty \right] \\ &= e^{-0.012t} \quad \dots (i) \end{aligned}$$

- The reliability of the component for a 100 hour mission time is obtained by replacing t by 100 in (i):

$$R(100) = e^{-0.012 \times 100} = e^{-1.2} \approx 0.3012$$

- We know from equation (22) that

$$\begin{aligned} \text{MTTF} &= \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-0.012t} dt = \left[\frac{e^{-0.012t}}{-0.012} \right]_0^{\infty} \\ &= - \frac{1}{0.012} [0 - 1] = \frac{1}{0.012} \approx 83.3333 \end{aligned}$$

- iv) Let t_{md} denote the median time to failure. Then from equation (24), we have

$$R(t_{md}) = P[T > t_{md}] = 0.5$$

$$\Rightarrow \int_{t_{md}}^{\infty} f(t) dt = 0.5$$

$$\Rightarrow e^{-0.012t_{md}} = 0.5 \quad [\text{replacing } t \text{ by } t_{md} \text{ in (i)}]$$

Taking natural logarithm on both sides

$$-0.012t_{md} = \ln(0.5) \Rightarrow -0.012t_{md} \approx -0.6931$$

$$\Rightarrow t_{md} \approx \frac{0.6931}{0.012} \approx 57.7583 \text{ hr}$$

- v) The life of the component for a reliability of 0.96 can be obtained by solving (i) for t and replacing $R(t)$ by 0.96 as follows:

$$0.96 = e^{-0.012t}$$

Taking natural logarithm on both sides, we get

$$\ln(0.96) = -0.012t \Rightarrow -0.0408 \approx -0.012t \Rightarrow t \approx \frac{0.0408}{0.012} = 3.4 \text{ hr}$$

Example 2: The hazard rate of a component is given by $\lambda(t) = 0.6 \times 10^{-6} t$, where t is in years.

Calculate the reliability of the component for the first 2 years.

Solution: It is given that the hazard rate of a component is $\lambda(t) = 0.6 \times 10^{-6} t$, where t is in years.

We know from equation (20b) that

$$\begin{aligned} R(t) &= \exp\left(-\int_0^t \lambda(t) dt\right) = \exp\left(-\int_0^t 0.6 \times 10^{-6} t dt\right) = \exp\left(\left[\frac{-0.6 \times 10^{-6} t^2}{2}\right]_0^t\right) \\ &= \exp(-0.3 \times 10^{-6} t^2 - 0) = \exp(-0.3 \times 10^{-6} t^2) \end{aligned}$$

Now reliability of the component for the first two years is given by just replacing t by 2. Therefore,

$$R(2) = e^{-0.3 \times 10^{-6} \times 2^2} = e^{-1.2 \times 10^{-6}}$$

Example 3: Show that if the hazard rate of a component is constant, say λ , then the failure distribution of the component follows exponential distribution. Also show that $MTTF = 1/\lambda$.

Solution: We are given that the hazard rate of a component = constant = λ . From equation (20b), we have

$$\begin{aligned} R(t) &= e^{-\int_0^t \lambda(t) dt} = e^{-\int_0^t \lambda dt} \quad [\because \lambda(t) = \lambda = \text{constant}] \\ &= e^{-[\lambda t]_0^t} = e^{-\lambda(t-0)} = e^{-\lambda t} \end{aligned}$$

Using equation (4), we have

$$F(t) = 1 - R(t) = 1 - e^{-\lambda t}$$

which is nothing but the cumulative distribution function of an exponential distribution. Hence, failure distribution of the component follows exponential distribution.

Now we calculate mean time to failure (MTTF) of the component. From equation (22),

$$\begin{aligned} \text{MTTF} &= \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-\lambda t} dt = \left[\frac{e^{-\lambda t}}{-\lambda} \right]_0^{\infty} \left[\because \int_a^b e^{kt} dt = \left[\frac{e^{kt}}{k} \right]_a^b \right] \\ &= -\frac{1}{\lambda} [0 - 1] \quad \left[\because \text{as } t \rightarrow \infty, e^{-\lambda t} \rightarrow 0 \right] \\ &= \frac{1}{\lambda} \end{aligned}$$

$$\text{Hence MTTF} = \frac{1}{\lambda} = \frac{1}{\text{constant hazard rate}}$$

Note 2: Thus we see that if hazard rate = constant = λ = then

$$\text{MTTF} = \frac{1}{\lambda} \text{ or } \lambda = \frac{1}{\text{MTTF}}.$$

Now, you may like to solve the following numerical problems to obtain MTTF.

E3) Calculate MTTF using the information given in Example 2.

E4) If hazard rate of a component is 0.2×10^{-4} , then obtain MTTF.

13.4 BATHTUB CURVE

In Sec. 13.3.4, we have discussed the hazard rate of a component. Consider a population of new identical components and suppose that all of them enter the field at some point in time, say, $t = 0$. In general, the curve of relative failure rate of the entire population of the components without replacement has the typical shape of a bathtub. Therefore, it is known as the **bathtub curve** as shown in Fig. 13.5. The bathtub curve is nothing but a graphical representation of the failure rate of a population of identical components versus time.

The shape of this curve has three different phases, which are explained as follows:

Early Failure or Infant Mortality

When a population of new identical components starts to perform its intended function, a high failure rate is observed in initial stages due to many factors such as manufacturing errors, poor quality or substandard items, incorrect adjustment or positioning, bad assembly, human factors, improper design, etc. This phase of the bathtub curve is known as the period of **early failure** or **infant mortality**. It is also known as the burn-in or decreasing failure rate, or debugging period. Most failures in this period are due to manufacturing errors, design problems or poor quality for which manufacturer may be responsible. So this period is generally covered by a warranty period by the manufacturer. The duration of this period varies from component to component. However, it is typically for the period of time till failure rate decreases.

The hazard rate in this phase can be reduced by increasing quality control at the production level (you have studied about quality control in detail in the first eight units of this course). However, even increased quality control at the production stage cannot completely eliminate infant mortality. Therefore, components should be tested at the factory before delivering them to the customers. That is why, good companies generally test the components before supplying them to users.

Normal Life, Useful Life or Chance Failure Period

Most components having manufacturing error, improper design, bad assembly, etc. have been failed in the early failure period. So for a long period of time (length of the period may vary from component to component), fewer failures are reported and failure rate remains almost at a constant level. This phase of the life of the component is known as **normal life, useful life or chance failure period** as shown in Fig. 13.5.

Usually failures in this period are due to stress-related causes, random fluctuations, etc. But these failures cannot be predicted exactly and happen randomly. That is why this phase is known as chance failure period. The appropriate distribution used in this phase is the exponential distribution as shown in Example 3. We know that in the case of constant failure rate, the failure distribution of the component follows an exponential distribution.

Wear out Failures

Since no component is perfect, it cannot last forever. So after the span of useful life, the failure rate of components starts increasing due to ageing of the components. This phase of the bathtub curve when the components begin to deteriorate is known as the period of **wear out failures** as shown in Fig. 13.5. In the wear out phase, failure rate increases with time. So if we wish to minimise this interruption in the smooth working of a real life system, one of the strategies which can be helpful is replacement of the component when it reaches mean time to failure of its life. The probability distributions generally used to study the failure characteristics in this phase are normal (due the shape of the failure curve in this phase shown in Fig. 13.8), Weibull and Gamma (due to the presence of shape parameter in these distributions).

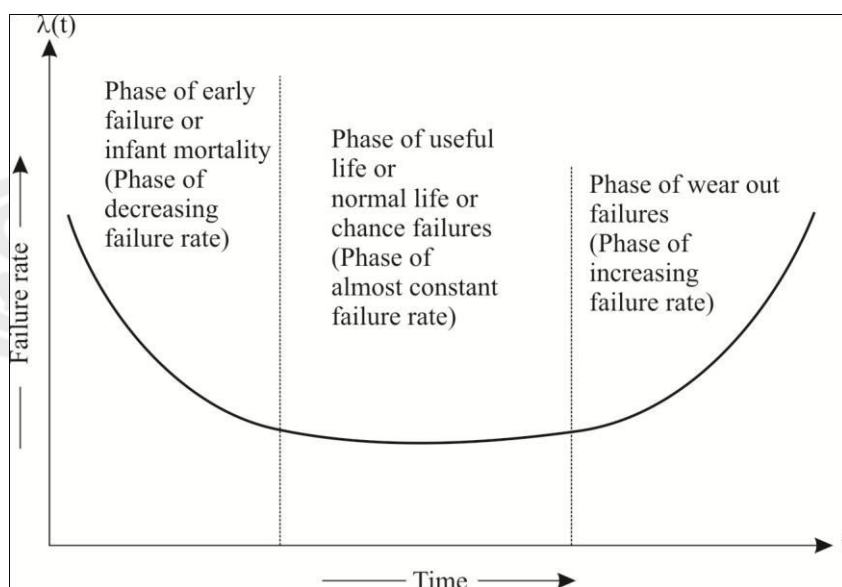


Fig. 13.5: Bathtub curve.

Now, you can try the following exercises based on the concept of the bathtub curve.

- E5)** In different phases of the bathtub curve, which one is the phase of increasing failure rate? Choose the most appropriate alternative.
 A) Infant mortality B) Useful life C) Wear out D) Early failures
- E6)** In the phase of useful life of a component, which distribution is the best suited to study the failure phenomenon? Choose the most appropriate alternative.
 A) Exponential B) Normal C) Weibull D) Gamma

13.5 ESTIMATION OF RELIABILITY FUNCTIONS FROM FAILURE DATA

In Sec. 13.3, we have discussed the basic reliability functions, namely, reliability, cumulative failure distribution, failure density and hazard rate function. It is a real challenge for the people working in the field of reliability to estimate the basic reliability functions from the failure data of a sample of identical components obtained either from test-generated failures or collected from actual field of operation of these components.

The following two approaches are generally used for this purpose:

- (i) Fitting a theoretical distribution, which is appropriate to the failure data.
- (ii) Deriving a function for hazard rate directly from the failure data and consequently other reliability functions using the relations mentioned in Table 13.1.

Here we follow the second approach as the first approach is beyond the scope of the course. Further, the choice of the method in the second approach also varies depending upon whether we have complete failure data or censored (incomplete) failure data. Let us first explain what we mean by complete and censored failure data.

Complete and Censored Failure Data: Suppose a sample of n identical components numbered from 1 to n is put to test. We record the lifetime of the components that fail. If the test is allowed to run until the failure of all n components, we shall have a record of the lifetimes of all n components. The data thus obtained is known as **complete failure data**. But if the test is not allowed to run until the failure of all n components due to some reason, we shall not have a record of the lifetimes of all n components. The data thus obtained is known as **censored failure data**.

Here we discuss only the case of complete failure data as censored failure data is beyond the scope of the course. Let us explain the procedure of estimating the basic reliability functions with the help of the following example. The data considered in this example is purely hypothetical.

Example 4: The failure data for 1000 electronic components is shown in Table 13.2.

Table 13.2: Failure Data of 1000 Electronic Components

Operating Time (in hours)	0-10	10-20	20-30	30-40	40-50	50-60	60-70	70-80
Number of Failures	167	116	98	76	65	56	44	36
Operating Time (in hours)	80-90	90-100	100-110	110-120	120-130	130-140	140-150	
Number of Failures	48	64	77	61	46	27	19	

Estimate and plot the reliability, cumulative failure distribution, failure density and failure rate functions.

Solution: In this example, the number of components in the sample is large. So instead of recording the time to failure of each component individually, it is a common practice to record the number of failures in some suitable intervals of time. In this example, the interval of time is taken as 10 hours. The estimate of reliability, cumulative failure distribution, failure density and failure/hazard rate functions are given in columns 5, 6, 7 and 8 of Table 13.3. The calculations for different columns are explained below.

Calculations of Columns 1 and 2

In column 1, we simply enter time in hours starting from the lower bound of first class interval up to the upper bound of last class interval for the data given in Table 13.2. Entries in the second column represent number of failures observed during test in each interval. The number of failures (frequencies) listed against each class interval of time in Table 13.2 is the sum of the numbers of all components that fail during that time interval. In the second column of Table 13.3, this frequency is entered against the upper bound of the class interval during which it is observed.

Calculation of Column 3

Entries in this column simply represent the cumulative frequencies of the frequencies written in column 2. You are familiar with the concept of cumulative frequencies as it has been explained and used by you a number of times in other courses of this programme. If $N_f(t)$ denotes the number of components that have failed by time t , then the first three entries of the column are calculated as follows:

First entry $N_f(0) = 0$ = same as the first entry of column 2

Second entry $N_f(10) = 0 + 167 = 167$ = sum of first two entries of column 2

Third entry $N_f(20) = 167 + 116 = 283$ = sum of first three entries of column 2

You can calculate the remaining cumulative frequencies as explained above.

Calculation of Column 4

Entries of column 4 represent the number of components that are performing their intended function adequately at time t . Therefore if $N_s(t)$ denotes the number of components that are performing their intended function at time t , then

$$N_s(0) = 1000 \quad \left[\begin{array}{l} \because \text{initially we started with 1000} \\ \text{components and it is assumed that} \\ \text{no component fails at time } t = 0 \end{array} \right]$$

$$N_s(10) = 1000 - 167 = 833 \quad \left[\begin{array}{l} \because 167 \text{ components have failed within the} \\ \text{first 10 hours of their operational life.} \end{array} \right]$$

$$N_s(20) = 833 - 116 = 717 \quad \left[\begin{array}{l} \because \text{out of the remaining 833 components,} \\ 116 \text{ have failed within the next} \\ 10 \text{ hours of their operational life} \end{array} \right]$$

The other entries of this column are calculated in the same way.

Calculations of Column 5

Entries in column 5 represent values of reliabilities at different points in time.

If $R(t)$ denotes the reliability at time t , then by definition [equation (1)]

$$R(t) = \frac{N_s(t)}{N_0} \quad \left[\begin{array}{l} \text{where } N_0 = \text{the size of the sample that} \\ \text{we start with and } N_s(t) \text{ is the same as} \\ \text{explained in calculations of column 4} \end{array} \right]$$

The first three entries of this column are calculated as follows:

$$R(0) = \frac{N_s(0)}{N_0} = \frac{1000}{1000} = 1, \quad R(10) = \frac{N_s(10)}{N_0} = \frac{833}{1000} = 0.833$$

$$R(20) = \frac{N_s(20)}{N_0} = \frac{717}{1000} = 0.717$$

In fact, entries of this column can simply be calculated by dividing the corresponding entries of column 4 by 1000 (the total number of components in the beginning).

Calculations of Column 6

Entries in column 6 represent values of cumulative failure distribution at different points in time. Therefore, if $F(t)$ denotes cumulative failure distribution at time t , then

$$F(t) = \frac{N_f(t)}{N_0}, \quad \left[\begin{array}{l} \text{where } N_f(t) \text{ denotes the number of} \\ \text{components that have failed by time } t \end{array} \right]$$

The first three entries of this column are calculated as follows:

$$F(0) = \frac{N_f(0)}{N_0} = \frac{0}{1000} = 0, \quad F(10) = \frac{N_f(10)}{N_0} = \frac{167}{1000} = 0.167$$

$$F(20) = \frac{N_f(20)}{N_0} = \frac{283}{1000} = 0.283$$

In fact, entries of this column can simply be calculated by dividing the corresponding entries of column 3 by 1000 (the total number of components in the beginning).

Calculations of Column 7

Entries of this column represent values of failure density function at different points of time. If $f(t)$ denotes the value of failure density function at time t , then from equation (16b), we have

$$\begin{aligned} f(t) &= \frac{\frac{d}{dt}(N_f(t))}{N_0} \\ &= \lim_{\Delta t \rightarrow 0} \frac{N_f(t + \Delta t) - N_f(t)}{N_0 \Delta t} \quad [\text{by definition of derivative}] \end{aligned}$$

Thus, approximate values of $f(t)$ can be obtained from the expression

$$f(t) = \frac{N_f(t + \Delta t) - N_f(t)}{N_0 \Delta t}, \quad \text{where } \Delta t = 10 \text{ in this example}$$

$$= \frac{N_f(t + 10) - N_f(t)}{10 \times 1000}, \quad \text{where } t = 0, 10, 20, 30, \dots, \text{ and } N_0 = 1000$$

The first three entries of this column are calculated as follows:

$$f(0) = \frac{N_f(10) - N_f(0)}{10000} = \frac{167 - 0}{10000} = 0.0167$$

$$f(10) = \frac{N_f(20) - N_f(10)}{10000} = \frac{283 - 167}{10000} = \frac{116}{10000} = 0.0116$$

$$f(20) = \frac{N_f(30) - N_f(20)}{10000} = \frac{98}{10000} = 0.0098$$

In fact, entries of this column can simply be calculated by dividing the entries of column 2 by 10000.

Calculations of Column 8

Entries of this column represent values of failure rate or hazard rate at different points in time. If $\lambda(t)$ denotes the value of failure rate at time t , then from equation (17), we have

$$\lambda(t) = \frac{\frac{d}{dt}(N_f(t))}{N_s(t)}$$

$$= \lim_{\Delta t \rightarrow 0} \frac{N_f(t + \Delta t) - N_f(t)}{N_s(t) \Delta t} \quad [\text{by definition of derivative}]$$

Thus, approximate values of $\lambda(t)$ can be obtained from the expression

$$\lambda(t) = \frac{N_f(t + \Delta t) - N_f(t)}{N_s(t) \Delta t}, \quad \text{where } \Delta t = 10 \text{ in this example}$$

$$= \frac{N_f(t + 10) - N_f(t)}{10 N_s(t)}, \quad \text{where } t = 0, 10, 20, 30, \dots,$$

The first three entries of this column are calculated as follows:

$$\lambda(0) = \frac{N_f(10) - N_f(0)}{10 N_s(0)} = \frac{167 - 0}{10 \times 1000} = \frac{167}{10000} = 0.0167$$

$$\lambda(10) = \frac{N_f(20) - N_f(10)}{10 N_s(10)} = \frac{283 - 167}{10 \times 833} = \frac{116}{8330} \approx 0.0139$$

$$\lambda(20) = \frac{N_f(30) - N_f(20)}{10 N_s(20)} = \frac{98}{10 \times 717} = \frac{98}{7170} \approx 0.0137$$

In fact, entries of this column can be calculated by dividing the entries of column 2 by the preceding entries of column 4 and further dividing the resulting value by 10.

Table 13.3: Estimation of $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$

Time	Number of Failures	Cumulative Failures	Number of Survivors	Reliability	Cumulative Failure Distribution	Failure Density Function	Hazard Rate
t	f	$N_f(t)$	$N_s(t)$	$R(t)$	$F(t)$	$f(t)$	$\lambda(t)$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0	0	0	1000	1	0	0.0167	0.0167
10	167	167	833	0.833	0.167	0.0116	0.0139
20	116	283	717	0.717	0.283	0.0098	0.0137
30	98	381	619	0.619	0.381	0.0076	0.0123
40	76	457	543	0.543	0.457	0.0065	0.0120
50	65	522	478	0.478	0.522	0.0056	0.0117
60	56	578	422	0.422	0.578	0.0044	0.0104
70	44	622	378	0.378	0.622	0.0036	0.0095
80	36	658	342	0.342	0.658	0.0048	0.0140
90	48	706	294	0.294	0.706	0.0064	0.0218
100	64	770	230	0.230	0.770	0.0077	0.0335
110	77	847	153	0.153	0.847	0.0061	0.0399
120	61	908	92	0.092	0.908	0.0046	0.0500
130	46	954	46	0.046	0.954	0.0027	0.0587
140	27	981	19	0.019	0.981	0.0019	0.1000
150	19	1000	0	0	1	—	—

The plot of the reliability, cumulative failure distribution, failure density and hazard rate functions are shown in Figs. 13.6 to 13.9, respectively.

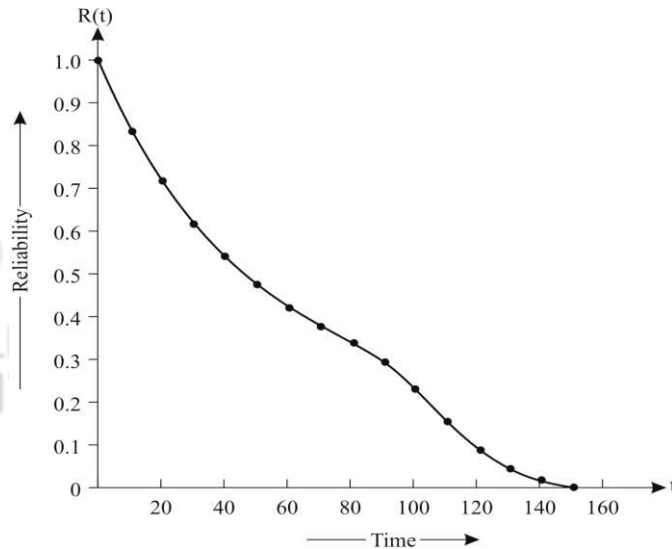


Fig. 13.6: Reliability function.

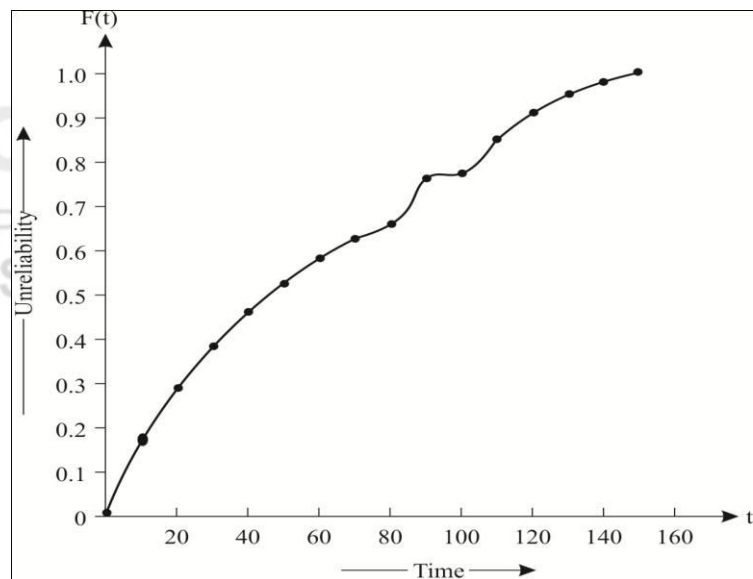


Fig. 13.7: Cumulative failure distribution function.

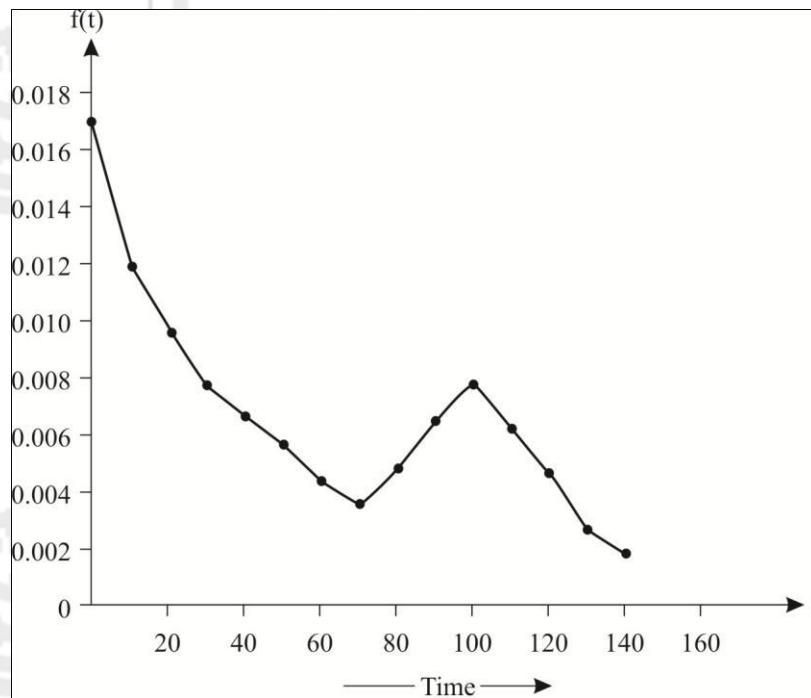


Fig.13.8: Failure density function

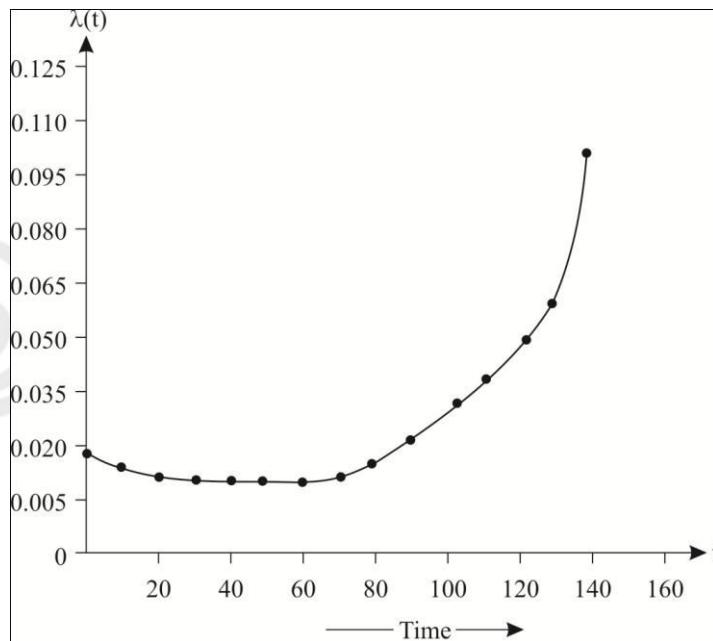


Fig.13.9: Failure rate.

Now, you can try the following exercises to obtain all these functions.

- E7)** The failure data of 10 electronic components are shown in Table 13.4 given below. Estimate the reliability, cumulative failure distribution, failure density and failure rate functions.

Table 13.4: Failure Data of 10 Components

Failure Number	1	2	3	4	5	6	7	8	9	10
Operating Time (in hours)	2	6	31	51	76	116	140	182	250	302

- E8)** Check whether the following statement is true or false. Give reason in support of your answer.

‘A sample of n identical components numbered from 1 to n is put to test. The test is allowed to run until all n components. The data of records of the lifetimes of all n components thus obtained is an example of censored failure data’.

13.6 SUMMARY

1. **Definition of Reliability:** Reliability of a component/device/equipment/unit/system is the **probability** that it performs its **intended function adequately** for a specified period of **time** under the given operating **conditions**.
2. Reliability of a component/system when expressed in terms of time t is known as **reliability function** and is generally denoted by $R(t)$.
3. In reliability terminology, distribution function of the random variable, time to failure (T), is known as **cumulative failure distribution function**.
4. In reliability terminology, probability density function of the random variable, time to failure (T), is known as **failure density function**.
5. Instantaneous rate of failure at time t is known as the **hazard rate function** at time t and is generally denoted by $\lambda(t)$.
6. Some commonly used relations between $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$ are given as follows:

$$R(t) = 1 - F(t) = \int_t^{\infty} f(t) dt = \exp\left(-\int_0^t \lambda(t) dt\right)$$

$$F(t) = 1 - R(t) = \int_0^t f(t) dt = 1 - \exp\left(-\int_0^t \lambda(t) dt\right)$$

$$f(t) = -\frac{d}{dt}(R(t)) = \frac{d}{dt}(F(t)) = \lambda(t) \times \exp\left(-\int_0^t \lambda(t) dt\right)$$

$$\lambda(t) = -\frac{d}{dt}(\ln R(t)) = \frac{\frac{d}{dt}(F(t))}{1 - F(t)} = \frac{f(t)}{\int_0^t f(t) dt}$$

7. The bathtub curve has three phases known as **Early Failure**, **Normal Life** and **Wear out Failure**.

13.7 SOLUTIONS/ANSWERS

- E1)** The definition of reliability is:

Reliability of a component/device/equipment/unit/system/ is the probability that it performs its intended function adequately for a specified period of time under the given operating conditions.

From the above definition, it is clear that the correct option is D.

- E2)** Reliability of a component/device/equipment/unit/system is the probability that it performs its intended function. Being a probability, its value will vary from 0 to 1. Hence, A is the correct option.

- E3)** In Example 2, we have calculated

$$R(t) = e^{-0.3 \times 10^{-6} t^2}$$

$$\therefore \text{MTTF} = \int_0^{\infty} R(t) dt = \int_0^{\infty} e^{-0.3 \times 10^{-6} t^2} dt$$

Putting $0.3 \times 10^{-6} t^2 = x$, we get

$$t = \frac{\sqrt{x}}{\sqrt{0.3 \times 10^{-3}}}$$

$$\text{or } dt = \frac{10^3}{\sqrt{0.3}} \frac{1}{2\sqrt{x}} dx$$

Also, when $t = 0, x = 0$

and when $t \rightarrow \infty, x \rightarrow \infty$

$$\therefore \text{MTTF} = \int_0^{\infty} e^{-x} \frac{10^3}{2\sqrt{0.3}} x^{-1/2} dx = \frac{10^3}{2\sqrt{0.3}} \int_0^{\infty} x^{1/2-1} e^{-x} dx$$

$$\begin{aligned} &= \frac{10^3}{1.0954} \left[\frac{1}{2} \right] \left[\because \int_0^{\infty} x^{n-1} e^{-x} dx, n > 0 \right] \\ &= \frac{10^7}{10954} \sqrt{\pi} \left[\because \left[\frac{1}{2} \right] = \sqrt{\pi} \right] \end{aligned}$$

E4) We know that if hazard rate of a component = constant = λ , then

$$\text{MTTF} = \frac{1}{\lambda}$$

Here $\lambda = 0.2 \times 10^{-4}$, which is a constant.

$$\therefore \text{MTTF} = \frac{1}{0.2 \times 10^{-4}} = \frac{10^4}{0.2} = \frac{10^5}{2} = 50000$$

E5) When a population of identical components is put to test, the failure rate of the components starts increasing due to ageing of the components. In the bathtub curve, this phase is known as wear out. Hence the correct option is (C).

E6) We know that in the phase of the useful life of a population of identical components, hazard rate = constant and constant hazard rate implies that the distribution of failures follows the exponential distribution. Hence, (A) is the correct alternative.

E7) Here the number of components under test is small in size. So we can record the failure time of each component easily, which is given in Table 13.5. Calculations of the entries of different columns have already been explained in Example 4 with necessary clarifications. You can follow similar steps and calculate the entries of different columns of Table 13.5. It is a small exercise given to you for practice.

Table 13.5: Estimation of $R(t)$, $F(t)$, $f(t)$ and $\lambda(t)$

Time	Number of Failures	Cumulative Failures	Number of Survivors	Reliability	Cumulative Failure Distribution	Failure Density Function	Hazard Rate
t		$N_f(t)$	$N_s(t)$	$R(t)$	$F(t)$	$f(t)$	$\lambda(t)$
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
0	0	0	10	$\frac{10}{10} = 1$	0	$\frac{1}{10 \times (2 - 0)} = 0.05$	$\frac{1}{10(2 - 0)} = 0.05$
2	1	1	9	$\frac{9}{10} = 0.9$	$\frac{1}{10} = 0.1$	$\frac{1}{10(6 - 2)} = 0.025$	$\frac{1}{9(6 - 2)} \square 0.0278$
6	1	2	8	$\frac{8}{10} = 0.8$	$\frac{2}{10} = 0.2$	$\frac{1}{10(31 - 6)} = 0.004$	$\frac{1}{8(31 - 6)} = 0.005$
31	1	3	7	$\frac{7}{10} = 0.7$	$\frac{3}{10} = 0.3$	$\frac{1}{10(51 - 31)} = 0.005$	$\frac{1}{7(51 - 31)} \square 0.007$
51	1	4	6	$\frac{6}{10} = 0.6$	$\frac{4}{10} = 0.4$	$\frac{1}{10(76 - 51)} = 0.004$	$\frac{1}{6(76 - 51)} \square 0.0067$
76	1	5	5	$\frac{5}{10} = 0.5$	$\frac{5}{10} = 0.5$	$\frac{1}{10(116 - 76)} = 0.0025$	$\frac{1}{5(116 - 76)} = 0.005$
116	1	6	4	$\frac{4}{10} = 0.4$	$\frac{6}{10} = 0.6$	$\frac{1}{10(140 - 116)} \square 0.0042$	$\frac{1}{4(140 - 116)} \square 0.0104$
140	1	7	3	$\frac{3}{10} = 0.3$	$\frac{7}{10} = 0.7$	$\frac{1}{10(182 - 140)} \square 0.0024$	$\frac{1}{3(182 - 140)} \square 0.0079$
182	1	8	2	$\frac{2}{10} = 0.2$	$\frac{8}{10} = 0.8$	$\frac{1}{10(250 - 182)} \square 0.0015$	$\frac{1}{2(250 - 182)} \square 0.0074$
250	1	9	1	$\frac{1}{10} = 0.1$	$\frac{9}{10} = 0.9$	$\frac{1}{10(302 - 250)} \square 0.0019$	$\frac{1}{1 \times (302 - 250)} \square 0.0192$
302	1	10		$\frac{0}{10} = 0$	$\frac{10}{10} = 1$	—	—

E8) This statement is false because we know that if we have failure data of each component of the sample under test then data thus obtained are known as complete failure data, while failure data set is said to be censored if the test is not allowed to run until all n components fail due to one reason or another.

UNIT 14 RELIABILITY EVALUATION OF SIMPLE SYSTEMS

Structure

- 14.1 Introduction
 - Objectives
- 14.2 Definition of a Simple System
- 14.3 Reliability Evaluation of a System
- 14.4 Reliability of a Series System
- 14.5 Reliability of a Parallel System
- 14.6 Reliability of a Mixed System
- 14.7 Summary
- 14.8 Solutions/Answers

14.1 INTRODUCTION

In Unit 13, we have defined reliability and basic reliability functions, namely, reliability function, cumulative failure distribution function, failure density function and hazard rate function. We have also discussed how to estimate these functions for a population of identical components through test-generated data.

One of the main objectives of getting estimates of these functions is to get an estimate of the reliability of the component, of which a sample is tested.

We now consider how to estimate the reliability of a system. By a system, we mean assemblages of, say, n identifiable components that perform some function. The main objective of this unit is to discuss some techniques of predicting/estimating the reliability of systems, from the reliabilities of the components of the system. We restrict our discussion to simple systems, which are defined in Sec. 14.2. In Sec. 14.3, we explain what is meant by the reliability evaluation of a system.

Techniques of reliability evaluation of a system depend upon the configuration of its components. In practice, several types of configurations are used. But we shall discuss the reliability evaluation techniques only for the following configurations of components of a system:

- Series configuration
- Parallel configuration
- Mixed configuration
- k -out-of- n configuration
- Standby configuration
- Complex configuration

Reliability evaluation of series and parallel systems is discussed in Secs. 14.4 and 14.5, respectively. In Sec. 14.6, we discuss the reliability of systems with mixed configuration of components. In Unit 15, we discuss reliability evaluation of k -out-of- n and standby configurations. Reliability evaluation of complex configurations is discussed in Unit 16.

Objectives

After studying this unit, you should be able to:

- define a system, a simple system and reliability of the system; and
- evaluate the reliability of a system when its components are in series, in parallel and in mixed configurations.

14.2 DEFINITION OF A SIMPLE SYSTEM

The definition of a simple system involves the concept of series and parallel configurations. So, to understand it better, you should first learn the definitions of series and parallel configurations. These are given below.

Series Configuration

Suppose a system has n components. From a reliability point of view, the components of a system are said to be in **series configuration** if they are connected in such a way that

- for the successful operation of the system all n components must perform their intended function, and
- only one needs to fail for system failure.

The string of lights (bulbs or LEDs) used in festivals and weddings is a good example of a series system. The system performs successfully only when all bulbs/LEDs are working and only one of them needs to fail for the system to fail.

The reliability block diagram for logical connectivity of a series configuration generally looks as shown in Fig. 14.1, where each of the n components is illustrated by a block.

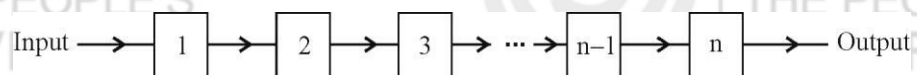


Fig. 14.1: Series configuration for n components.

In reliability literature, logical connectivity of n components, namely, 1, 2, ..., n , in series configuration is also illustrated by a reliability graph as shown in Fig. 14.2.

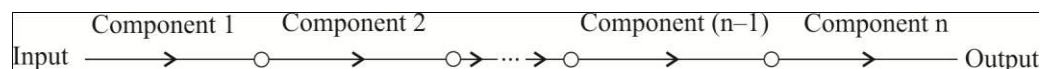


Fig. 14.2: Series configuration for n components.

Parallel Configuration

Let a system have n components. From a reliability point of view, the components of a system are said to be in parallel configuration if they are connected in such a way that

- only one component needs to work for the successful operation of the system, and
- all components must fail for system failure.

The phone line is a good example of a parallel system.

The reliability block diagram for logical connectivity of the parallel configuration generally looks as shown in Fig. 14.3.

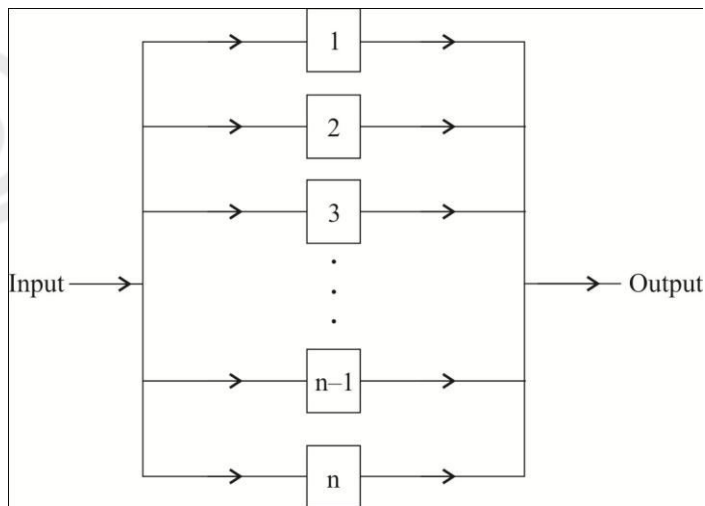


Fig. 14.3: Parallel configuration for n components.

In reliability literature, logical connectivity of n components, namely, 1, 2, ..., n in parallel configuration is also illustrated by a reliability graph as shown in Fig. 14.4.

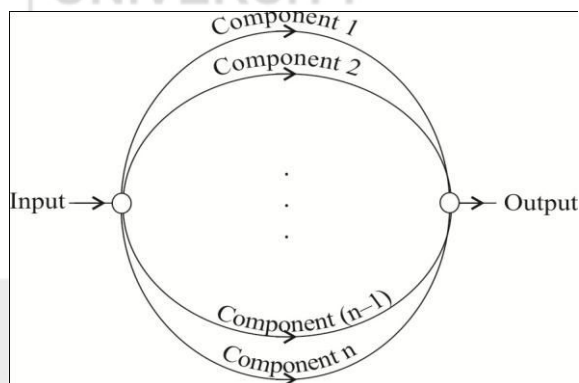


Fig. 14.4: Parallel configuration for n components.

In this unit, we shall use reliability block diagrams for series and parallel configurations.

Let us now define a simple system.

Simple System

A system is said to be simple if either its components are connected in parallel, in series or in a combination of both. In other words, a system is said to be simple if its reliability block diagram can be reduced into subsystems having independent components connected either in parallel or in series.

14.3 RELIABILITY EVALUATION OF A SYSTEM

In reliability theory,

A system is an assemblage of, say, n identifiable components that perform some function.

The n individual components are known as **elementary components**. The reliability of a system depends on the reliability of elementary components. To evaluate the reliability of a system, we have to apply the rules of probability theory such as the addition rule, multiplication rule, conditional probability, independence of events or their combination. You are familiar with these rules

as all these have been discussed in detail in Units 1 to 3 of MST-003. The choice of rules from among these for a given situation depends on the logical connectivity of the elementary components or subsystems made up of elementary components in the system. In general, we follow the steps given below to evaluate the reliability of a system:

- Step 1:** We identify the elementary components or subsystems, which constitute the given system such that either their individual reliabilities are given or can be estimated. Let us call them 'units' comprising the system.
- Step 2:** We evaluate the reliabilities of those units whose reliabilities are not directly given to us.
- Step 3:** We draw a reliability block diagram of the system to represent the logical connectivity of the units (i.e., elementary components and subsystems).
- Step 4:** We determine the constraints that should be fulfilled for the successful operation of the system.
For example, do we need successful operation of all units for the successful operation of the system? Is the successful operation of only one of them enough or some other combination of components needs to operate successfully?
- Step 5:** We apply the rules of probability theory such as addition rule, multiplication rule, conditional probability, independence of events or their combination to evaluate the reliability of the system.

Let us explain the 5 steps given above with the help of a reliability block diagram shown in Fig. 14.5. The number written inside each block indicates the numbering of the component and the number written at the top of each block represents reliability of the corresponding component.

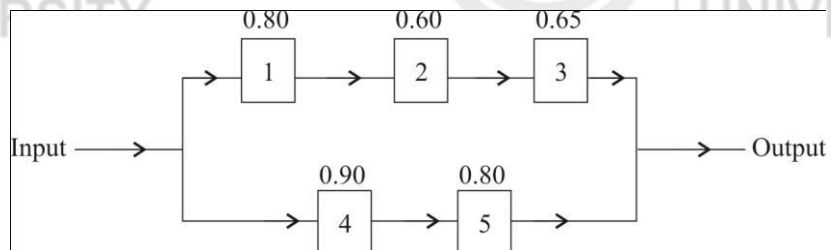


Fig. 14.5: A reliability block diagram.

- Step 1:** There are 5 elementary components in the system shown in Fig. 14.5, and these are numbered as 1, 2, 3, 4, 5.

Two sub-systems constitute the system, as explained below:

- (i) Elementary components 1, 2 and 3 are in series configuration and these form a subsystem. Let us call it unit 1 and denote it by U_1 .
- (ii) Elementary components 4 and 5 are also in series configuration and form another subsystem. Let us call it unit 2 and denote it by U_2 .

- Step 2:** To evaluate the reliability of the system, we first evaluate the reliabilities of the units U_1 and U_2 . We discuss the methods of evaluating reliabilities in Secs. 14.4 to 14.6.

Step 3: The reliability block diagram of the system is shown in Fig. 14.5.

Step 4: The constraints that should be fulfilled for the successful operation of the system are listed below:

- (i) For the successful operation of unit U_1 , all components (1, 2 and 3) must work successfully as they are in series configuration.
- (ii) For the successful operation of unit U_2 , both components 4 and 5 must work successfully as they are also in series configuration.
- (iii) For successful operation of the system, either unit U_1 or unit U_2 or both should work successfully as these units are arranged in parallel configuration.

Step 5: The reliability of the system is evaluated by calculating the probability of the event $(U_1 \cup U_2)$ using the addition law of probability theory.

You have learnt these calculations in Unit 3 of MST-003.

In Secs. 14.4 to 14.6 and in Units 15 and 16, we discuss the reliability evaluation of systems with different configurations of elementary components. We begin by explaining reliability evaluation of a series system.

14.4 RELIABILITY OF A SERIES SYSTEM

In Sec. 14.2, we have defined series configuration of a number of components. A series system is nothing but a system, all of whose components are in series configuration. A series system made up of n components is represented as shown in Fig. 14.6 for the purpose of reliability evaluation.

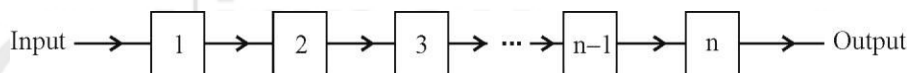


Fig. 14.6: Series system.

Note that Fig. 14.6 simply shows the logical connectivity of the system. Actual physical connectivity of the components in the system may vary. But always remember the following two characteristics of the series system (which have already been listed in Sec. 14.2):

- (1) For the successful operation of the series system, it is necessary that all n components of the system perform their intended function successfully, and
- (2) Only one component needs to fail for system failure.

Let us now obtain an expression for the reliability of the series system.

Let E_i be the event that component i performs its intended function successfully, where $i = 1, 2, 3, \dots, n$.

Let R_i denote the reliability of the component i , where $i = 1, 2, 3, \dots, n$ for a mission of t units of time.

If R denotes the reliability of the series system, then, from the definitions of reliability and series system, we have

$$R = P(E_1 \cap E_2 \cap E_3 \cap \dots \cap E_n) \quad \dots (1)$$

In equation (1), we take the intersection of events because reliability is a probability and the definition of series system implies that for the successful operation of the system, all components must work.

Reliability Theory

If events $E_i, (i=1,2,\dots,n)$ are not independent then we apply conditional probability and get

$$R = P(E_1)P(E_2|E_1)P(E_3|E_1E_2)\dots P(E_n|E_1E_2\dots E_{n-1}), \quad \dots (2)$$

where $E_1E_2 = E_1 \cap E_2, E_1E_2E_3 = E_1 \cap E_2 \cap E_3$, etc.

But if events $E_i, (i=1,2,\dots,n)$ are independent, then we have

$$R = P(E_1)P(E_2)P(E_3)\dots P(E_n) \quad \left[\because P[E_i|E_j] = P[E_i] \text{ if events } E_i \text{ and } E_j \text{ are independent, etc.} \right] \quad \dots (3)$$

$$\text{or } R = \prod_{i=1}^n R_i, \quad \text{for a mission of } t \text{ units of time,} \quad \dots (4a)$$

$$\text{where } R_i = P(E_i) \quad \dots (4b)$$

that is, the reliability of the series system is equal to the product of the individual reliabilities of the components of the system.

In particular, if a system has two components, then

$$R = R_1R_2 \quad \text{for a mission of } t \text{ units of time} \quad \dots (5a)$$

and if the system has three components, then

$$R = R_1R_2R_3 \quad \text{for a mission of } t \text{ units of time} \quad \dots (5b)$$

Let us consider some examples for evaluating the reliability of a series system.

Example 1: Evaluate the reliability of the series system having the reliability block diagram shown in Fig. 14.7 for a mission of 500 hours, where the given reliabilities of the components are for a mission of 500 hours. Assume that the failure of any component does not affect the functioning of the other components.



Fig. 14.7: Reliability block diagram for Example 1.

Solution: We know that the reliability of a series system is equal to the product of the individual reliabilities of the components. Therefore, the required reliability of the series system is given from equation (4a) as

$$\begin{aligned} R &= 0.60 \times 0.75 \times 0.95 \times 0.80 && \text{for a mission of 500 hours} \\ &= 0.342 && \text{for a mission of 500 hours} \end{aligned}$$

Note 1: Note the answer of this example. Does it make you wonder why the reliability of the system is less than that of the worst component? Do not worry. The answer is correct. This problem arises due to limitations of the series system. One of the main limitations of the series system is that reliability of the series system can never be better than the reliability of the worst component in the system. The second major limitation of the series system is that reliability of the system decreases as the number of components in the system increases. These two limitations are illustrated in Examples 2 and 3.

Example 2: A system has three components connected in series having reliabilities 0.40, 0.70, 0.80, respectively, for a mission of 400 hours. What is the percentage increase in the reliability of the system in each of the following cases?

- Reliability of the first component is increased by 0.1 and that of the second and third components remains the same.
- Reliability of the second component is increased by 0.1 and that of the first and third components remains the same.
- Reliability of the third component is increased by 0.1 and that of the first and second components remains the same.

Solution: First of all, we calculate the reliability of the system as per the given reliabilities of the components. From equation (4a or 5b),

$$R = 0.4 \times 0.7 \times 0.8 \quad \text{for a mission of 400 hours}$$

$$= 0.224 \quad \text{for a mission of 400 hours}$$

Now, reliabilities of the system and percentage increase in the reliability of the system in the cases (i), (ii) and (iii) are shown in columns 5 and 6, respectively, of Table 14.1.

Table 14.1: Computation of System Reliability and Percentage Increase in the Reliability of the System for cases (i), (ii) and (iii) of Example 2

	Reliability of the First Component	Reliability of the Second Component	Reliability of the Third Component	Reliability of the System	Percentage Increase in the Reliability of the System
(1)	(2)	(3)	(4)	(5)	(6)
	0.4	0.7	0.8	0.224	–
i)	0.5	0.7	0.8	0.280	$\frac{0.280 - 0.224}{0.224} \times 100 = 25\%$
ii)	0.4	0.8	0.8	0.256	$\frac{0.256 - 0.224}{0.224} \times 100 = 14.29\%$
iii)	0.4	0.7	0.9	0.252	$\frac{0.252 - 0.224}{0.224} \times 100 = 12.5\%$

Note 2: From column 6 of Table 14.1, note that the improvement in the reliability of the system (in percentage) is higher when reliability of the weakest component is increased by 0.1 in comparison with the cases when reliabilities of the other two components are increased one at a time by the same amount (0.1). This suggests that if a system engineer wants to improve the reliability of the series system, he/she should **concentrate** on the **improvement** of the reliability of the **poorest component**. You may have heard a saying that a chain is only as strong as its weakest link. This saying best fits the series system. So we can say that the unit (component/subsystem), which is the poorest in the system from a reliability point of view, dictates the reliability of a series system.

Example 3: Consider a component having reliability 0.95 for a mission of 100 hours. Evaluate the reliability of a series system in the cases having 2, 3, 4, ..., 10 identical components having reliability 0.95 for the same mission of 100 hours.

Solution: If the single component of a system has reliability 0.95 for a mission of 100 hours, then the system will also have the same reliability for a mission of 100 hours. But the reliability of the series system for a mission of 100 hours will vary if it has 2, 3, 4, ..., 10 such identical components connected in series

from a reliability point of view. In fact, the reliability of the system decreases with an increase in the number of its components. The calculations are shown in Table 14.2.

Table 14.2: Calculation of Reliability as the Number of Components varies from 1 to 10

Number of Components	Reliability of Series System	% Decrease in the Reliability of the System Compared to the System having Single Component
1	0.95	—
2	$(0.95)^2 = 0.9025$	$\frac{0.95 - 0.9025}{0.95} \times 100 = 5\%$
3	$(0.95)^3 = 0.857375$	$\frac{0.95 - 0.857375}{0.95} \times 100 = 9.75\%$
4	$(0.95)^4 = 0.814506$	$\frac{0.95 - 0.814506}{0.95} \times 100 = 14.26\%$
5	$(0.95)^5 = 0.773781$	$\frac{0.95 - 0.773781}{0.95} \times 100 = 18.55\%$
6	$(0.95)^6 = 0.735092$	$\frac{0.95 - 0.735092}{0.95} \times 100 = 22.62\%$
7	$(0.95)^7 = 0.698337$	$\frac{0.95 - 0.698337}{0.95} \times 100 = 26.49\%$
8	$(0.95)^8 = 0.663420$	$\frac{0.95 - 0.663420}{0.95} \times 100 = 30.17\%$
9	$(0.95)^9 = 0.630249$	$\frac{0.95 - 0.630249}{0.95} \times 100 = 33.66\%$
10	$(0.95)^{10} = 0.598737$	$\frac{0.95 - 0.598737}{0.95} \times 100 = 36.98\%$

Note 3: From the third column of Table 14.2, note that the reliability of a series system decreases as we increase the number of components in it.

Before we end this section, we also define the unreliability (Q) of a system. It is simply given by

$$Q = 1 - R \quad \dots (6)$$

You may like to solve the following exercises to evaluate the reliability of a series system.

-
- E1)** Three components of a system are connected in series configuration from a reliability point of view. The reliabilities of these components for a mission of 200 days are 0.80, 0.95 and 0.96, respectively. Evaluate the reliability of the system for a mission of 200 days. Assume that the components are independent.
- E2)** Evaluate the unreliability of the system given in E1.
- E3)** The design of a system is such that it requires 100 identical components in series. Further it is desired that the reliability of the system must be 0.95 for a mission of 800 hours. Determine the minimum reliability of each component.
-

14.5 RELIABILITY OF A PARALLEL SYSTEM

In Sec. 14.4, you have learnt that the reliability of the series system is worse than the reliability of its poorest component. It means that we cannot improve

the reliability of a series system with less reliable components. But this can be done in the case of parallel system, which is discussed in this section.

We have defined parallel configuration in Sec. 14.2. A parallel system is nothing but a system all of whose components are in parallel configuration. A parallel system made up of n components is represented as shown in Fig. 14.8 for the purpose of reliability evaluation.

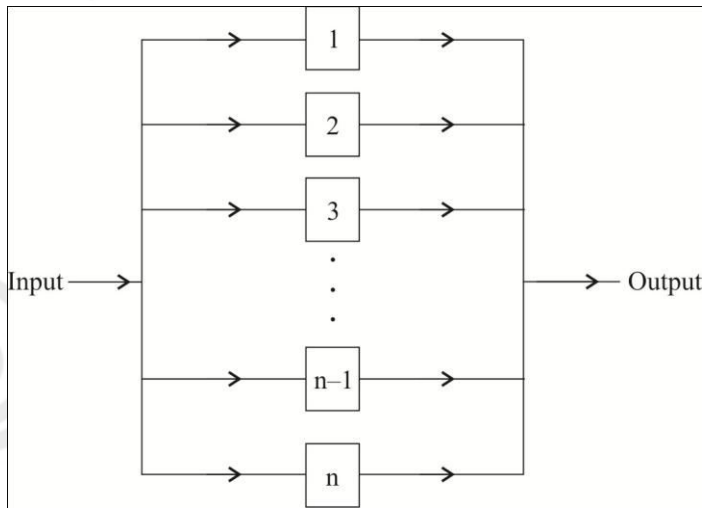


Fig. 14.8: Parallel system.

Note that Fig. 14.8 simply shows logical connectivity of the system. The actual physical connectivity of the components in the system may vary. But always remember the following two characteristics of a parallel system:

- (1) Only one component needs to work successfully for the successful operation of the system, and
- (2) All the components of the system must fail for system failure.

Let us now obtain an expression for the reliability evaluation of the parallel system.

Let E_i be the event that component i performs its intended function successfully, where $i = 1, 2, 3, \dots, n$. Let $\bar{E}_i$ ($i = 1, 2, 3, \dots, n$) denote the event complementary to the event E_i .

Let R_i denote the reliability of the component i , where $i = 1, 2, 3, \dots, n$ for a mission of t units of time.

If R denotes the reliability of the parallel system for the mission of t units of time, then by definition of reliability and parallel system, we have

$$R = P(E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n) \quad \dots (7a)$$

We take the union of events in equation (7a) because the definition of reliability implies that reliability is a probability and the definition of parallel system implies that for the successful operation of the parallel system only one component needs to work.

Further, from Unit 1 of MST-003 we know that for any event, say, A :

$P(A) + P(\bar{A}) = 1$. Here $A = E_1 \cup E_2 \cup E_3 \cup \dots \cup E_n$. Thus (7a) may be written as:

$$R = 1 - P(\bar{E}_1 \cup \bar{E}_2 \cup \bar{E}_3 \cup \dots \cup \bar{E}_n) \quad \dots (7b)$$

Using De Morgan's law (which you have studied in Unit 1 of MST-001), we have

$$R = 1 - P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3 \cap \dots \cap \bar{E}_n) \quad \dots (7c)$$

If events $E_i, (i = 1, 2, \dots, n)$ are not independent, then on applying conditional probability, we get

$$R = 1 - P(\bar{E}_1)P(\bar{E}_2|\bar{E}_1\bar{E}_2)\dots P(\bar{E}_n|\bar{E}_1\bar{E}_2\dots\bar{E}_{n-1}) \quad \dots (8a)$$

But if $E_i, (i = 1, 2, \dots, n)$ are independent then $\bar{E}_i, (i = 1, 2, \dots, n)$ are independent [this result has been proved in Unit 3 of MST-003]. Then, we get

$$\begin{aligned} R &= 1 - P(\bar{E}_1)P(\bar{E}_2)P(\bar{E}_3)\dots P(\bar{E}_n) \\ &= 1 - (1 - P(E_1))(1 - P(E_2))(1 - P(E_3))\dots(1 - P(E_n)) \end{aligned}$$

Since $R_i = P(E_i)$, therefore

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3)\dots(1 - R_n) \quad \dots(8b)$$

$$\text{or } R = 1 - \prod_{i=1}^n (1 - R_i) \quad \text{for a mission of } t \text{ units of time} \quad \dots (8c)$$

Thus, the reliability of the parallel system is equal to one minus the product of unreliabilities of the elementary components of the system.

In particular, if the parallel system has only two components, then

$$R = 1 - (1 - R_1)(1 - R_2) \quad \text{for a mission of } t \text{ units of time} \quad \dots (9)$$

and if the parallel system has three components, then

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3) \quad \text{for a mission of } t \text{ units of time} \quad \dots (10)$$

Let us consider some examples for evaluating the reliability of parallel systems.

Example 4: Evaluate the reliability of the parallel system, which has the reliability block diagram shown in Fig. 14.9 for a mission of 90 days. The reliabilities of the components given in Fig. 14.9 are for a mission of 90 days. Assume that the failure of any component does not affect the functioning of the other components.

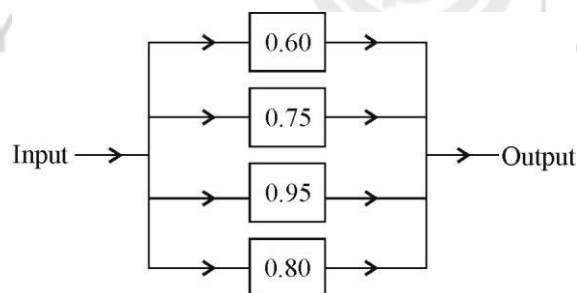


Fig. 14.9: Reliability block diagram for Example 4.

From equation (8b), we know that the reliability of a parallel system (R) having n components with reliabilities R_i , $i = 1, 2, 3, \dots, n$, is given by

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3) \dots (1 - R_n)$$

In this case, $R_1 = 0.60, R_2 = 0.75, R_3 = 0.95$ and $R_4 = 0.80$.

$$\therefore R = 1 - (1 - 0.60)(1 - 0.75)(1 - 0.95)(1 - 0.80)$$

$$= 1 - 0.40 \times 0.25 \times 0.05 \times 0.20 = 1 - 0.001$$

$$= 0.999$$

for a mission of 90 days

Note 4: Note the answer of this example. It tells us that the reliability of the system is very good compared to the series system of Example 1 having components of the same reliability. Here the reliability of the system is even better than the reliability of the best component of the system. But you should keep in mind that for the successful operation of the parallel system only one component needs to work. So this improvement in the system reliability comes at the cost of an initial investment of three additional components. Thus, we have to balance the cost and reliability of a system. Whether to give importance to the cost or to reliability depends on the severity of consequence of system failure. For example, there should be at least one additional parallel engine in the aeroplane, whatever be the initial cost that the airline has to pay. So in this case reliability is more important than cost. But in the case of a component of, say TV, importance may be given to cost instead of reliability because failure of a TV has little consequence compared to the crash of an aeroplane, which may result in loss of lives of hundreds of people.

Let us see the effect of a component's reliability on the reliability of the parallel system with the help of the following example.

Example 5: A system has three components connected in parallel from a reliability point of view having reliabilities 0.20, 0.40, 0.50, respectively, for a mission of 400 hours. What is the percentage increase in the reliability of the system in each of the following cases?

- (i) Reliability of the first component is increased by 0.1 and that of the second and third components remains the same.
- (ii) Reliability of the second component is increased by 0.1 and that of the first and third components remains the same.
- (iii) Reliability of the third component is increased by 0.1 and that of the first and second components remains the same.

Solution: We first calculate the reliability of the system as per the given reliabilities of the components using equation (8b):

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3), \text{ where } R_1 = 0.2, R_2 = 0.4, R_3 = 0.5$$

$$= 1 - (1 - 0.20)(1 - 0.4)(1 - 0.5) = 1 - 0.8 \times 0.6 \times 0.5 = 1 - 0.240$$

$$= 0.760$$

for a mission of 400 hours

Now, reliabilities of the system and percentage increase in the reliability of the system for the cases (i), (ii) and (iii) are shown in columns 5 and 6, respectively, in Table 14.3.

Table 14.3: Computation of System Reliability and Percentage Increase in the Reliability of the System for the Cases (i) , (ii) and (iii)

	Reliability of the I Component	Reliability of the II Component	Reliability of the III Component	Reliability of the System	Percentage Increase in the Reliability of the System
(1)	(2)	(3)	(4)	(5)	(6)
	0.2	0.4	0.5	0.760	—
i)	0.3	0.4	0.5	0.790	$\frac{0.790 - 0.760}{0.760} \times 100 \square 3.95\%$
ii)	0.2	0.5	0.5	0.800	$\frac{0.800 - 0.760}{0.760} \times 100 \square 5.26\%$
iii)	0.2	0.4	0.6	0.808	$\frac{0.808 - 0.760}{0.760} \times 100 \square 6.32\%$

Note 5: From column 6 of Table 14.3, you can see that the improvement in the reliability of the system (in percentage) is higher when reliability of the best component is increased by 0.1 as compared to the cases when reliabilities of the other two components are increased one at a time by the same amount (0.1). This suggests that if a system engineer wants to improve the reliability of a parallel system, he/she should **concentrate** on the **improvement** of the reliability of the **best component**. Thus, the best component dictates the reliability of the parallel system. Recall that the poorest component dictates the reliability of the series system.

Let us see the effect of the number of components on the reliability of a parallel system with the help of Example 6:

Example 6: Consider a component having reliability 0.75 for a mission of 100 hours. Evaluate the reliability of a parallel system when the system has 2, 3, 4, ..., 10 identical components for the same mission of 100 hours.

Solution: If a system has a single component of reliability 0.75 for a mission of 100 hours, it will also have the same reliability for a mission of 100 hours. But the reliability of the parallel system for a mission of 100 hours will vary in the cases having 2, 3, 4, ..., 10 identical components connected in parallel from reliability point of view. In fact, it increases with the number of components as the calculations given in Table. 14.4 below show.

Table 14.4: Calculation of Reliability as Number of Components varies from 1 to 10

Number of Components	Reliability of Parallel System	% Increase in the Reliability of the System Compared to the System having Single Component
1	0.75	—
2	$1 - (1 - 0.75)^2 = 0.9375$	$\frac{0.9375 - 0.75}{0.75} \times 100 = 25\%$
3	$1 - (1 - 0.75)^3 = 0.984375$	$\frac{0.984375 - 0.75}{0.75} \times 100 = 31.25\%$
4	$1 - (1 - 0.75)^4 \square 0.996094$	$\frac{0.996094 - 0.75}{0.75} \times 100 \square 32.81\%$
5	$1 - (1 - 0.75)^5 \square 0.999023$	$\frac{0.999023 - 0.75}{0.75} \times 100 \square 33.20\%$
6	$1 - (1 - 0.75)^6 \square 0.999756$	$\frac{0.999756 - 0.75}{0.75} \times 100 \square 33.30\%$
7	$1 - (1 - 0.75)^7 \square 0.999939$	$\frac{0.999939 - 0.75}{0.75} \times 100 \square 33.33\%$
8	$1 - (1 - 0.75)^8 \square 0.999985$	$\frac{0.999985 - 0.75}{0.75} \times 100 \square 33.33\%$
9	$1 - (1 - 0.75)^9 \square 0.999996$	$\frac{0.999996 - 0.75}{0.75} \times 100 \square 33.33\%$
10	$1 - (1 - 0.75)^{10} \square 0.999999$	$\frac{0.999999 - 0.75}{0.75} \times 100 \square 33.33\%$

Note 6: From the third column of Table 14.4, you can see that the reliability of a parallel system increases by a great extent on the addition of the first redundant component to the one-component system as compared to the addition of second redundant component, third redundant component, and so on. The same column further suggests that to provide more than two redundant components for a one-component system from a reliability point of view is not of much benefit. That is why an aeroplane which requires two engines for successful operation is generally provided two redundant engines to improve the reliability for successful operation.

Example 7: Suppose you are an industrial statistician in a company. Suppose a system engineer visits you and explains his/her problem to you as follows:

‘I want to design a parallel system with an overall reliability of 0.98 by using identical components, each having individual reliability of 0.25. What is the minimum number of components that I should connect?’

What will your answer be?

Solution: Let R_s be the overall reliability of the system and R , the reliability of each component. If the required number of components is n , then from equation (8b), we have

$$R_s = 1 - (1 - R)^n$$

$$\Rightarrow 0.98 = 1 - (1 - 0.25)^n \Rightarrow (1 - 0.25)^n = 1 - 0.98 \Rightarrow (0.75)^n = 0.02$$

Taking natural logarithm on both sides, we get

$$n \ln (0.75) = \ln (0.02) \quad \left[\because \ln(m^n) = n(\ln m) \right]$$

$$\Rightarrow -0.2877n = -3.9120 \quad [\text{Using scientific calculator}]$$

$$\Rightarrow n = \frac{3.9120}{0.2877} = 13.5975$$

Since the number of components cannot be in fraction, he/she should connect 14 components in parallel to achieve a reliability of 0.98.

You can now try the following exercises to evaluate the reliability of a parallel system.

-
- E4)** A system has three components, which are connected in parallel configuration from a reliability point of view. The reliabilities of these components for a mission of 300 days are 0.60, 0.55 and 0.70, respectively. Evaluate the reliability of the system for a mission of 300 days. Assume that the components are independent.
- E5)** Evaluate the unreliability of the system given in E4.
- E6)** Suppose you are an industrial statistician in a company. Suppose a system engineer visits you and explains his/her simple problem to you as follows:

‘I want to design a parallel system with an overall reliability of the system as 0.95 for a mission of 900 hours. The system will have 10 identical components. How poor can a component in the system be?’

Assuming that the components are independent, what will your answer be?

In Secs. 14.4 and 14.5, you have learnt how to evaluate the reliability of series and parallel systems, respectively. But in practice, there are many real systems having components connected in series as well as in parallel. Such systems are known as mixed systems and we discuss them in the next section.

14.6 RELIABILITY OF A MIXED SYSTEM

A system is said to be a mixed system if the components of the system are connected both in series and in parallel configurations. The reliability block diagram shown in Fig. 14.5 corresponds to a mixed system where components 1, 2, 3 and 4, 5 are separately connected in series while the two subsystems – one consisting the components 1, 2, 3 and another consisting the components 4, 5 themselves – are in parallel.

To evaluate the reliability of a mixed system, we first break the reliability block diagram into series or parallel subsystems. Then we evaluate the reliability of each subsystem. Finally, we evaluate the reliability of the given mixed system by combining the reliabilities of the subsystems and applying appropriate probability law(s). The procedure is explained below with the help of an example.

Example 8: Evaluate the reliability of the system for which the reliability block diagram is shown in Fig. 14.10, for a mission of 100 hours. Assume that all components are independent and the reliability of each component is given for a mission of 100 hours as follows:

$$R_1 = 0.80, R_2 = 0.75, R_3 = 0.50, R_4 = 0.65, R_5 = 0.76, R_6 = 0.60, R_7 = 0.95, R_8 = 0.90,$$

where R_i denotes the reliability of the component i , ($i = 1, 2, 3, \dots, 8$).

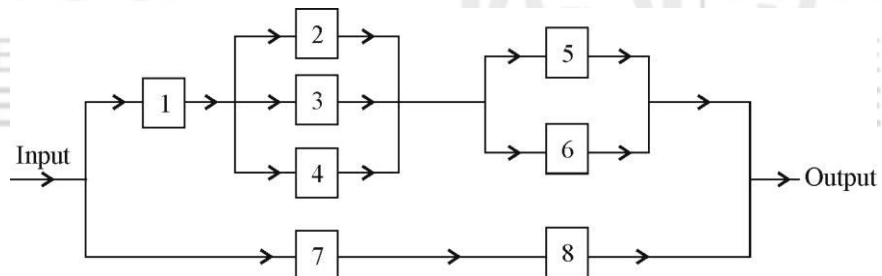


Fig. 14.10: Reliability block diagram for Example 8.

Solution: The components of the given system are connected in both series and parallel configuration. So it is a mixed system. To evaluate the reliability of this mixed system, we have to break the system into subsystems such that all components of a subsystem are either in series or in parallel. This can be done as follows:

- **Reduction I:** Combine the components 2, 3, 4 in parallel configuration to form an equivalent component 9 (see reduction I in Fig. 14.11).
- **Reduction II:** Combine the components 5, 6 in parallel configuration to form an equivalent component 10 (see reduction II in Fig. 14.11).
- **Reduction III:** Combine the components 1, 9, 10 in series configuration to form an equivalent component 11 (see reduction III in Fig. 14.11).

- **Reduction IV:** Combine the components 7, 8 in series configuration to form an equivalent component 12 (see reduction IV in Fig. 14.11).
- **Reduction V:** Finally, combine the components 11 and 12 in parallel configuration to form an equivalent component 13 (see reduction V in Fig. 14.11). The component 13 represents the complete system.

The step by step reduction process is shown in Fig. 14.11.

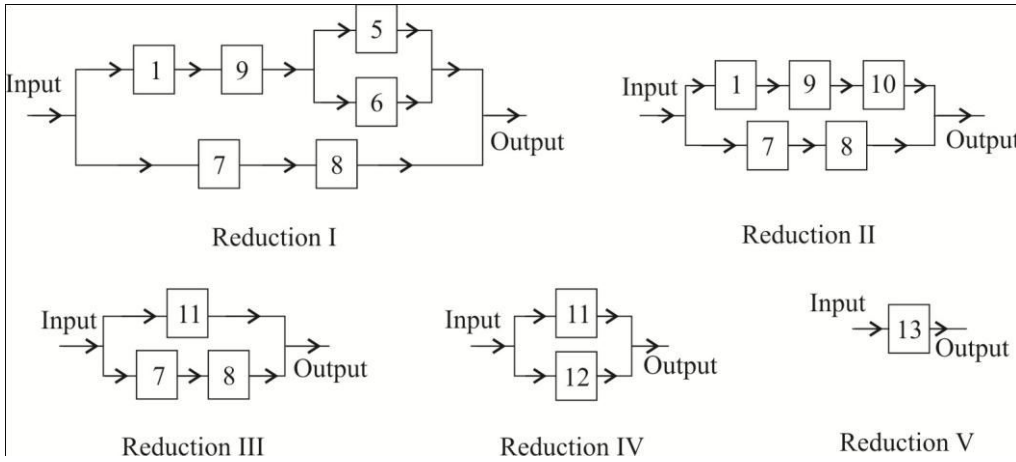


Fig. 14.11: Reductions I to V of Fig. 14.10.

We are given that $R_1 = 0.80$, $R_2 = 0.75$, $R_3 = 0.50$, $R_4 = 0.65$, $R_5 = 0.76$, $R_6 = 0.60$, $R_7 = 0.95$, $R_8 = 0.90$ where R_i denotes the reliability of the given component i , ($i = 1, 2, 3, \dots, 8$). Similarly, if R_9 to R_{13} denote the reliabilities of the equivalent components 9 to 13, then using equations (8b and 4a), we have

$$\begin{aligned}
 R_9 &= 1 - (1 - R_2)(1 - R_3)(1 - R_4) && \left[\because \text{components 2, 3, 4 are} \right. \\
 &= 1 - (1 - 0.75)(1 - 0.50)(1 - 0.65) && \left. \text{in parallel configuration} \right] \\
 &= 1 - 0.25 \times 0.50 \times 0.35 \quad \square \quad 1 - 0.0438 = 0.9562
 \end{aligned}$$

$$\begin{aligned}
 R_{10} &= 1 - (1 - R_5)(1 - R_6) && \left[\because \text{components 5, 6 are} \right. \\
 &= 1 - (1 - 0.76)(1 - 0.60) && \left. \text{in parallel configuration} \right] \\
 &= 1 - 0.24 \times 0.40 = 1 - 0.096 = 0.904
 \end{aligned}$$

$$\begin{aligned}
 R_{11} &= R_1 R_9 R_{10} && \left[\because \text{components 1, 9, 10 are} \right. \\
 &\quad \square \quad 0.80 \times 0.9562 \times 0.904 \quad \square \quad 0.6915 && \left. \text{in series configuration} \right]
 \end{aligned}$$

$$\begin{aligned}
 R_{12} &= R_7 R_8 && \left[\because \text{components 7, 8 are} \right. \\
 &= 0.95 \times 0.90 = 0.855 && \left. \text{in series configuration} \right]
 \end{aligned}$$

$$\begin{aligned}
 R_{13} &= 1 - (1 - R_{11})(1 - R_{12}) && \left[\because \text{components 11, 12 are} \right. \\
 &\quad \square \quad 1 - (1 - 0.6915)(1 - 0.855) = 1 - 0.3085 \times 0.145 \quad \square \quad 1 - 0.0447 = 0.9553 && \left. \text{in parallel configuration} \right]
 \end{aligned}$$

Hence, the reliability of the given mixed system is 0.9553.

You can now try the following exercise to evaluate the reliability of a mixed system.

E7) Evaluate the reliability of the system for which the reliability block diagram is shown in Fig. 14.12, for a mission of 500 hours. Assume that all components are independent. The reliability of each component is given below for a mission of 500 hours:

$$R_1 = 0.40, R_2 = 0.30, R_3 = 0.60, R_4 = 0.80, R_5 = 0.85, R_6 = 0.60,$$

$$R_7 = 0.70, R_8 = 0.95, R_9 = 0.96.$$

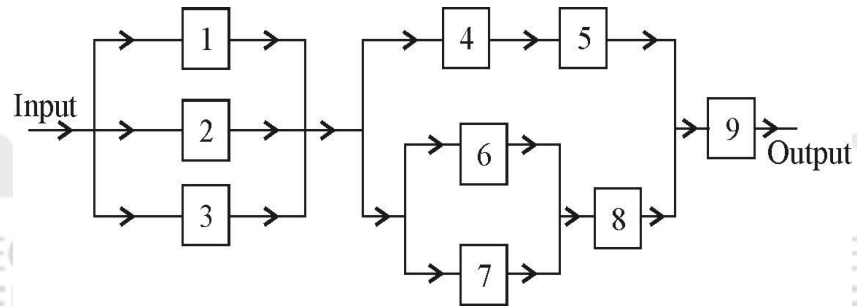


Fig. 14.12: Reliability block diagram for E7.

Let us summarise the main points that we have covered in this unit.

14.7 SUMMARY

1. An assemblage of, say n , identifiable components that perform some function is known as a **system** in reliability.
2. A system is said to have its components in **series configuration** from a reliability point of view if they are connected in such a way that:
 - for the successful operation of the system, it is necessary that all n components perform their intended function successfully, and
 - only one component needs to fail for the system failure.
3. A system is said to have its components in **parallel configuration** from a reliability point of view if they are connected in such a way that:
 - only one component needs to work for the successful operation of the system, and
 - all components must fail for system failure.
4. A system is said to be **simple** if either its components are connected in parallel, in series or in combinations of both. In other words, a system is said to be simple if its reliability block diagram can be reduced into subsystems having independent components either in parallel or in series.
5. In a **series system**, all components of the system are in series configuration.
6. In a **parallel system**, all components of the system are in parallel configuration.
7. In a **mixed system**, the components of the system are connected both in series and in parallel.

14.8 SOLUTIONS/ANSWERS

- E1)** The reliability block diagram of the given series system is shown in Fig. 14.13.

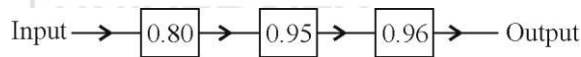


Fig.14.13: Reliability block diagram for E1.

We know that reliability of a series system is equal to the product of the individual reliabilities of the components [see equation (4a)].

Therefore, the required reliability of the series system is given by

$$\begin{aligned}
 R &= 0.80 \times 0.95 \times 0.96 && \text{for a mission of 200 days} \\
 &= 0.7296 && \text{for a mission of 200 days}
 \end{aligned}$$

- E2)** From equation (6), we know that unreliability (Q) of a system is given by.

$$\begin{aligned}
 Q &= 1 - R \\
 \Rightarrow Q &= 1 - 0.7296 && [\text{using the result of E1}] \\
 \Rightarrow Q &= 0.2704
 \end{aligned}$$

Therefore, the probability that the system will fail before a mission of 200 days is 0.2704.

- E3)** Since all 100 components are identical, each component has the same reliability. Let it be denoted by R. If R_s denotes the reliability of the series system, then from equation (4a), we have

$$\begin{aligned}
 R_s &= R^{100} \\
 \Rightarrow 0.95 &= R^{100} \\
 \Rightarrow R &= (0.95)^{1/100} \\
 &\approx 0.9995 && [\text{using scientific calculator}]
 \end{aligned}$$

- E4)** The reliability block diagram of the given parallel system is shown in Fig. 14.14.

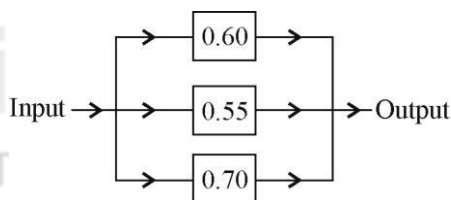


Fig. 14.14: Reliability block diagram for E4.

We know that the reliability (R) of a parallel system having n components with reliabilities R_i , $i = 1, 2, 3, \dots, n$ is given by equation (8b) as

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3) \dots (1 - R_n)$$

In this case, $R_1 = 0.60$, $R_2 = 0.55$, $R_3 = 0.70$ and $n = 3$.

$$\begin{aligned}
 \therefore R &= (1 - 0.60)(1 - 0.55)(1 - 0.70) = 1 - 0.40 \times 0.45 \times 0.30 = 1 - 0.054 \\
 &= 0.946 && \text{for a mission of 300 days}
 \end{aligned}$$

E5) We know that the unreliability (Q) of a system is given by

$$Q = 1 - R$$

$$\text{or } Q = 1 - 0.946 \quad [\text{using result of E4}]$$

$$= 0.054$$

Therefore, the probability that the system will fail before a mission of 300 days is 0.054.

E6) Let the reliability of each component be denoted by R and that of the system by R_s . We know that the reliability (R) of a parallel system having n components with reliabilities R_i , $i = 1, 2, 3, \dots, n$ is given by equation (8b) as

$$R = 1 - (1 - R_1)(1 - R_2)(1 - R_3) \dots (1 - R_n)$$

In this case $R = R_s$, $R_1 = R_2 = R_3 = \dots = R_n = R$ and $n = 10$

$$\therefore R_s = 1 - (1 - R)^{10}$$

$$\Rightarrow 0.95 = 1 - (1 - R)^{10}$$

$$\Rightarrow (1 - R)^{10} = 1 - 0.95 \quad \text{as } R_s = 0.95 (\text{given})$$

$$\Rightarrow (1 - R)^{10} = 0.05$$

$$\Rightarrow 1 - R = (0.05)^{1/10}$$

$$\Rightarrow 1 - R \approx 0.7411 \quad [\text{Using scientific calculator}]$$

$$\Rightarrow R \approx 1 - 0.7411 = 0.2589$$

Hence, your answer to the system engineer should be that each component cannot be poorer than a component having reliability 0.2589.

E7) The components of the given system are connected both in series and in parallel. So it is a mixed system. To evaluate the reliability of this mixed system, we have to break the system into subsystems such that either all the components of a subsystem are in series or in parallel. This can be done as follows.

Reduction I: Combine the components 1, 2, 3 in parallel configuration to form an equivalent component 10 (see reduction I in Fig. 14.15).

Reduction II: Combine the components 4, 5 in series configuration to form an equivalent component 11 (see reduction II in Fig. 14.15).

Reduction III: Combine the components 6, 7 in parallel configuration to form an equivalent component 12 (see reduction III in Fig. 14.15).

Reduction IV: Combine the components 12, 8 in series configuration to form an equivalent component 13 (see reduction IV in Fig. 14.15).

Reduction V: Combine the components 11, 13 in parallel configuration to form an equivalent component 14 (see reduction V in Fig. 14.15).

Reduction VI: Combine the components 10, 14, 9 in series configuration to form an equivalent component 15 (see reduction VI in Fig. 14.15). The component 15 represents the complete system.

This step by step reduction process is shown in Fig. 14.15.

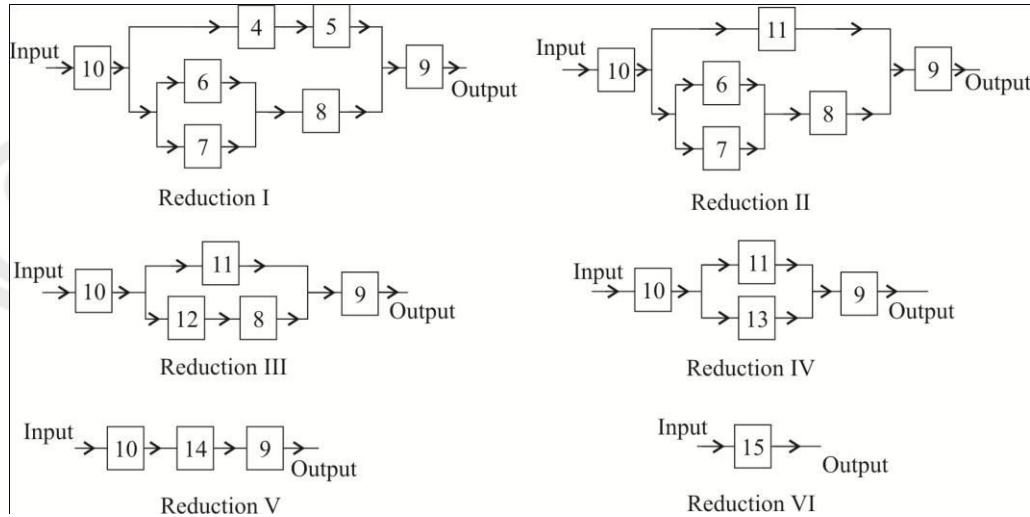


Fig. 14.15: Reliability block diagram for E7.

We are given that $R_1 = 0.40$, $R_2 = 0.30$, $R_3 = 0.60$, $R_4 = 0.80$, $R_5 = 0.85$, $R_6 = 0.60$, $R_7 = 0.70$, $R_8 = 0.95$, $R_9 = 0.96$.

Similarly, if R_{10} to R_{15} denote the reliabilities of the equivalent components 10 to 15, then from equations (4a) and (8b), we have

$$\begin{aligned} R_{10} &= 1 - (1 - R_1)(1 - R_2)(1 - R_3) && \left[\because \text{components 1, 2, 3 are} \right. \\ &= 1 - (1 - 0.40)(1 - 0.30)(1 - 0.60) && \left. \text{in parallel configuration} \right] \\ &= 1 - 0.60 \times 0.70 \times 0.40 = 1 - 0.168 = 0.832 \end{aligned}$$

$$\begin{aligned} R_{11} &= R_4 R_5 && \left[\because \text{components 4, 5 are} \right. \\ &= 0.80 \times 0.85 = 0.68 && \left. \text{in series configuration} \right] \end{aligned}$$

$$\begin{aligned} R_{12} &= 1 - (1 - R_6)(1 - R_7) && \left[\because \text{components 6, 7 are} \right. \\ &= 1 - (1 - 0.60)(1 - 0.70) && \left. \text{in parallel configuration} \right] \\ &= 1 - 0.40 \times 0.30 = 1 - 0.12 = 0.88 \end{aligned}$$

$$\begin{aligned} R_{13} &= R_{12} R_8 && \left[\because \text{components 12, 8 are} \right. \\ &= 0.88 \times 0.95 = 0.836 && \left. \text{in series configuration} \right] \end{aligned}$$

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$$\begin{aligned} R_{14} &= 1 - (1 - R_{11})(1 - R_{13}) \\ &= 1 - (1 - 0.68)(1 - 0.836) \\ &= 1 - 0.32 \times 0.164 = 1 - 0.05248 = 0.94752 \end{aligned}$$

$\left[\because \text{components 11,13 are} \right]$
 $\left[\text{in parallel configuration} \right]$

$$R_{15} = R_{10} R_{14} R_9$$

$\left[\because \text{components 10,14,9 are} \right]$
 $\left[\text{in series configuration} \right]$

$$= 0.832 \times 0.94752 \times 0.96 = 0.7568032$$

Hence, reliability of the given system is 0.7568032 for a mission of 500 hours.

UNIT 15 RELIABILITY EVALUATION OF k-OUT-OF-n AND STANDBY SYSTEMS

Structure

- 15.1 Introduction
 - Objectives
 - 15.2 Redundancy
 - 15.3 Reliability of k-out-of-n System
 - 15.4 Reliability of Standby System
 - 15.5 Summary
 - 15.6 Solutions/Answers
-

15.1 INTRODUCTION

In Unit 14, we have discussed reliability evaluation of series, parallel and mixed systems. You have learnt that for the successful operation of a system having n components:

- it is necessary that all n components must perform their intended function successfully in the case of series system, and
- only one needs to work successfully in the case of parallel system.

However, in practice, there exist systems, which require the successful operation of at least k , ($1 \leq k \leq n$), components out of n identical components. Such systems are known as **k-out-of-n systems** or **partially redundant** systems. Further, in parallel systems discussed in Unit 14, all n components work simultaneously, while for successful operation of a parallel system only one needs to work successfully. This is known as **active redundancy**. We explain what is meant by redundancy in Sec. 15.2. In Sec. 15.3, we discuss how to evaluate the reliability of k-out-of-n system of identical components.

The redundancy complementary to active redundancy is known as the **standby (passive) redundancy**, which is discussed in Sec. 15.4.

In Unit 16, we shall discuss three techniques to evaluate the reliability of complex systems.

Objectives

After studying this unit, you should be able to:

- define redundancy, active redundancy and standby (passive) redundancy;
 - evaluate the reliability of a k-out-of-n system; and
 - evaluate the reliability of a standby system under certain assumptions.
-

15.2 REDUNDANCY

Redundancy in a system means that there exists an alternative parallel path for the successful operation of the system. For example, suppose a system has n components connected in parallel and at least k , ($1 \leq k \leq n$), components are needed for the successful operation of the system. In such a system, we say that

the number of **basic components** is k and the remaining $(n - k)$ components are known as **redundant** components. In fact, this system is known as a k -out-of- n system and is discussed in the next section of this unit. The basic purpose of introducing redundant components in a system is to improve the reliability of the system. As we have seen in Example 6 in Unit 14, the reliability of a system increases as we introduce redundant components in a system. Redundancy can be classified as shown in Fig. 15.1.

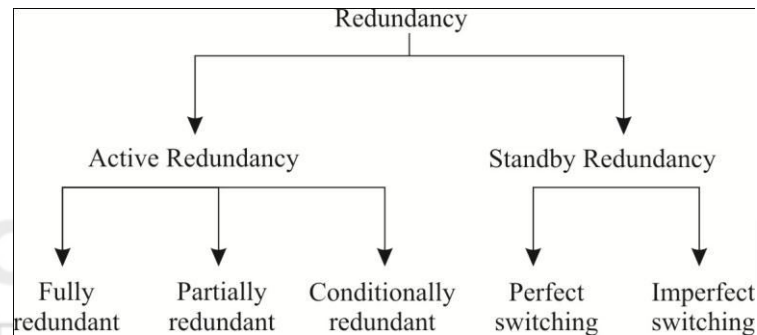


Fig. 15.1: Classification of redundancy.

Let us explain the terms used in Fig. 15.1.

Fully Redundant System: A k -out-of- n system is known as a **fully redundant** system if $k = 1$. But for $k = 1$, the k -out-of- n system is the same as the parallel system, which we have discussed in Sec. 14.5 of Unit 14. Thus, the parallel system is nothing but a 1-out-of- n system and is also known as a **fully redundant** system.

Partially Redundant System: Before defining a partially redundant system, let us first define a non-redundant system. A k -out-of- n system is said to be a **non-redundant system** if $k = n$. But for $k = n$, the k -out-of- n system is simply the series system, which we have discussed in Sec. 14. 4 of Unit 14. Thus, a series system is nothing but an n -out-of- n system and is also known as a **non-redundant** system. Now, we define a partially redundant system. A k -out-of- n system is said to be a **partially redundant system** if $1 < k < n$.

As far as conditional redundancy is concerned, it is beyond the scope of the course. On the other hand, what we mean by standby redundancy is discussed in Sec. 15.4 of this unit. Let us now determine the reliability of the k -out-of- n system.

15.3 RELIABILITY OF k -OUT-OF- n SYSTEM

A system having n components is said to be k -out-of- n system if and only if at least k components out of n are required to function successfully for the successful operation of the system.

In k -out-of- n system, although all n components are in operation, the k components that are essential for the system to operate are called the **basic components**. The $(n - k)$ components that are added with the objective to improve the reliability of the system are known as **redundant components**. Thus, a series system is an n -out-of- n system and a parallel system is a 1-out-of- n system. That is, the series and parallel systems are particular cases of k -out-of- n system when $k = n$ and $k = 1$, respectively. The possible paths for a 2-out-of-3 system having components 1, 2 and 3 are shown in Fig. 15.2.

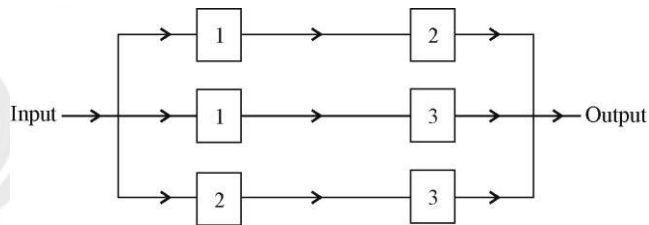


Fig. 15.2: Possible paths for 2-out-of-3 system.

So, for reliability evaluation of k-out-of-n system, we have to find a technique, which serves our purpose for all values of k varying from 1 to n. For simplicity, suppose all the n components of the system are identical and independent. In fact, we impose these two restrictions on the components of the system intentionally so that we can apply binomial distribution for evaluating the reliability of the system. Recall from Unit 9 of MST-003 that the probability mass function of the binomial distribution is given by

$$p(x) = P[X = x] = {}^nC_x p^x q^{n-x}, \quad \dots (1)$$

where

n = number of trials in the random experiment,

x = number of successes out of n trials,

p = probability of success in each trial,

q = 1 – p = probability of failure in each trial.

nC_x = binomial coefficient which represents number of ways of occurrence of x successes out of n trials.

You may like to refer to Unit 4 of MST-001 if needed.

We now consider the main objective of this section: To obtain an expression for the reliability of k-out-of-n system. Let us first see whether we can apply binomial distribution or not. Let us understand the above listed meanings of n, x, p, q in terms of our k-out-of-n system. So we first define success and probability of success in our case as follows.

Successful operation of a component in k-out-of-n system represents a success. Since there are n components, the number of successes for a given time may be 0, 1, 2, ..., n. As far as the probability of success is concerned, reliability of each component represents the probability of success. But for applying the binomial distribution, the probability of success should be the same for each trial. However, we have already assumed that all the n components are identical. Therefore, the reliability of each component is assumed to be the same. Hence, the requirement of constant probability in each trial is satisfied.

Further, to apply binomial distribution, the probability of success in each trial should be independent. But we have already assumed that n identical components in the system are independent. Thus, all the requirements for applying binomial distribution are satisfied by the k-out-of-n system of identical and independent components.

Therefore, if R_s denotes the reliability of k-out-of-n system for a given time, then

R_s = Probability of success of at least k components

$$\text{or } R_s = \sum_{x=k}^n {}^nC_x R^x Q^{n-x}, \quad \dots (2)$$

Reliability Theory

where

n = number of components in the system,

k = minimum number of components required for the success of the system,

R = reliability of each component,

$Q = 1 - R$ = unreliability of each component. ... (3)

Let us explain the procedure of evaluating R_s with the help of an example.

Example 1: Consider a piping system having 5 pipes connected in parallel as shown in Fig. 15.3. Assume that all pipes are identical and independent. If reliability of smooth flow of the liquid from each pipeline is 0.80 for a mission of 1 year, evaluate the reliability of the system working successfully. The system is said to work successfully if at least 3 pipelines perform their intended function successfully.

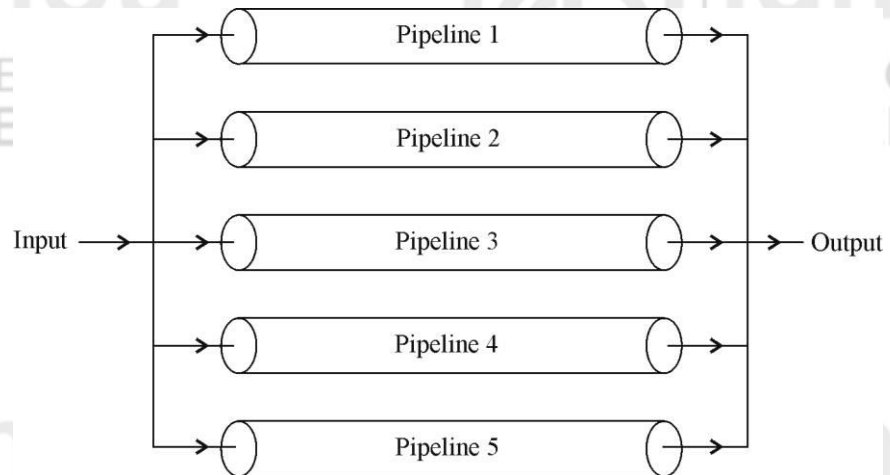


Fig. 15.3: System of 5 pipelines connected in parallel.

Solution: Since it is given that for successful operation of the system, at least 3 pipelines should work successfully, it is a k -out-of- n system, where $k = 3$ and $n = 5$.

If R denotes the reliability of each pipeline and R_s that of the 3-out-of-5 system, then applying the binomial distribution given in equation (2), we get

$$\begin{aligned}
 R_s &= \sum_{k=3}^5 {}^5C_k R^k (1-R)^{5-k} \\
 &= {}^5C_3 R^3 (1-R)^2 + {}^5C_4 R^4 (1-R) + {}^5C_5 R^5 (1-R)^0 \\
 &= 10(0.8)^3 (1-0.8)^2 + 5(0.8)^4 (1-0.8) + (0.8)^5 \\
 &= 10 \times 0.512 \times 0.04 + 5 \times 0.4096 \times 0.2 + 0.32768 \\
 &= 0.2048 + 0.4096 + 0.32768 \\
 &= 0.94208
 \end{aligned}$$

for a mission of 1 year

Let us explain the procedure of handling the case of non-identical components in k -out-of- n system with the help of an example.

Example 2: For the system given in Example 1, assume that instead of 5 pipelines, we have 4 pipelines, which are independent but not identical. It is given that the reliabilities of 4 pipelines for a mission of 1000 hours are $R_1 = 0.60$, $R_2 = 0.70$, $R_3 = 0.80$ and $R_4 = 0.75$, respectively. Evaluate the reliability of 2-out-of-4 system.

Solution: Here we are given that $k = 2$ and $n = 4$. Since the components (pipelines) are non-identical, we cannot apply the binomial distribution in this case. This is because one of the requirements for applying the binomial distribution is that the probability of success in each trial should be the same. This requirement is not met in this case due to the presence of non-identical components. The case of non-identical components is handled by considering all possible mutually exclusive and exhaustive cases for each possible value of k . Here the values of k may be 2, 3 or 4. Let us find the mutually exclusive and exhaustive cases for each value of k one at a time. This is explained below:

For $k = 2$,

the number of mutually exclusive cases = the number of ways of exactly
2 successes out of 4

$$= {}^4C_2 = \frac{|4|}{|2|4-2} = \frac{4 \times 3 \times |2|}{2 \times |2|} = 6$$

These six mutually exclusive cases are $R_1 R_2 \bar{R}_3 \bar{R}_4$, $R_1 \bar{R}_2 R_3 \bar{R}_4$, $R_1 \bar{R}_2 \bar{R}_3 R_4$, $\bar{R}_1 R_2 R_3 \bar{R}_4$, $\bar{R}_1 R_2 \bar{R}_3 R_4$, $\bar{R}_1 \bar{R}_2 R_3 R_4$, where $\bar{R}_i$ denotes the unreliability of the component i , $i = 1, 2, 3, 4$. These mutually exclusive cases are shown in Fig. 15.4a.

For $k = 3$,

the number of mutually exclusive cases = the number of ways of exactly
3 successes out of 4

$$= {}^4C_3 = \frac{|4|}{|3|4-3} = \frac{4 \times |3|}{|3| \times 1} = 4$$

These four mutually exclusive cases are $R_1 R_2 R_3 \bar{R}_4$, $R_1 R_2 \bar{R}_3 R_4$, $R_1 \bar{R}_2 R_3 R_4$, $\bar{R}_1 R_2 R_3 R_4$ as shown in Fig 15.4b.

For $k = 4$,

the number of mutually exclusive cases = the number of ways of exactly
4 successes out of 4

$$= {}^4C_4 = \frac{|4|}{|4|4-4} = \frac{|4|}{|4| \times |0|} = 1 \text{ as } |0| = 1$$

There is only one possibility of 4 successes out of 4, which is $R_1 R_2 R_3 R_4$. This case of 4 successes is shown in Fig. 15.4c.

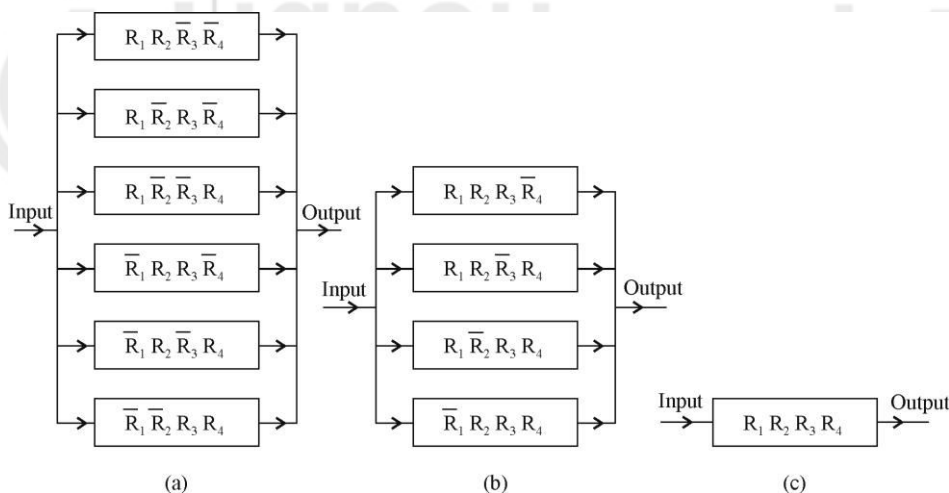


Fig. 15.4: Mutually exclusive paths for (a) $k = 2$; (b) $k = 3$ and (c) $k = 4$.

Let R_a , R_b and R_c denote the reliabilities for the cases when $k = 2$, $k = 3$ and $k = 4$, respectively.

Now, to evaluate reliability for $k = 2$, we simply have to add the reliabilities of the six mutually exclusive cases as shown in Fig. 15.4a (by addition law of probability for mutually exclusive events).

$$\begin{aligned}
 \therefore R_a &= R_1 R_2 \bar{R}_3 \bar{R}_4 + R_1 \bar{R}_2 R_3 \bar{R}_4 + R_1 \bar{R}_2 \bar{R}_3 R_4 + \bar{R}_1 R_2 R_3 \bar{R}_4 \\
 &\quad + \bar{R}_1 R_2 \bar{R}_3 R_4 + \bar{R}_1 \bar{R}_2 R_3 R_4 \\
 &= 0.6 \times 0.7 \times (1-0.8)(1-0.75) + 0.6 \times (1-0.7)0.8(1-0.75) \\
 &\quad + 0.6 \times (1-0.7)(1-0.8) \times 0.75 + (1-0.6) \times 0.7 \times 0.8(1-0.75) \\
 &\quad + (1-0.6) \times 0.7(1-0.8)0.75 + (1-0.6)(1-0.7)0.8 \times 0.75 \\
 &= 0.6 \times 0.7 \times 0.2 \times 0.25 + 0.6 \times 0.3 \times 0.8 \times 0.25 + 0.6 \times 0.3 \times 0.2 \times 0.75 \\
 &\quad + 0.4 \times 0.7 \times 0.8 \times 0.25 + 0.4 \times 0.7 \times 0.2 \times 0.75 + 0.4 \times 0.3 \times 0.8 \times 0.75 \\
 &= 0.021 + 0.036 + 0.027 + 0.056 + 0.042 + 0.072 \\
 &= 0.254
 \end{aligned}$$

$$\begin{aligned}
 R_b &= R_1 R_2 R_3 \bar{R}_4 + R_1 R_2 \bar{R}_3 R_4 + R_1 \bar{R}_2 R_3 R_4 + \bar{R}_1 R_2 R_3 R_4 \\
 &= 0.6 \times 0.7 \times 0.8(1-0.75) + 0.6 \times 0.7(1-0.8)0.75 + 0.6(1-0.7)0.8 \times 0.75 \\
 &\quad + (1-0.6)0.7 \times 0.8 \times 0.75 \\
 &= 0.084 + 0.063 + 0.108 + 0.168 \\
 &= 0.423
 \end{aligned}$$

$$R_c = R_1 R_2 R_3 R_4 = 0.6 \times 0.7 \times 0.8 \times 0.75 = 0.252$$

The reliability of the whole system is given by

$$R = R_a + R_b + R_c = 0.254 + 0.423 + 0.252 = 0.929$$

You may like to try the following exercise to evaluate the reliability of a k-out-of-n system.

E1) For the system given in Example 1, evaluate the reliability of the system for $k = 1, 2, 3, 4, 5$ and comment on the results.

15.4 RELIABILITY OF STANDBY SYSTEM

Refer to Fig. 15.1, where we have classified redundancy broadly into two categories, namely, active redundancy and standby redundancy. By active redundancy, we mean that all the components connected in parallel are turned on at the beginning of operation of the system, and continue to perform until they fail. Thus, in active redundancy, all components are simultaneously in the operating mode. We have discussed two such systems, namely, the parallel system and k-out-of-n system, where all the components of the system are turned on at the beginning of the operation of the system. So all the components are simultaneously in the operating mode.

On the other hand, in standby redundancy, the components are connected in parallel but do not start operating simultaneously from the beginning of the operation of the system. In **standby redundancy**, the component(s) in operating mode, is/are known as **normally operating component(s)**. The

component(s) kept in standby or reserve mode is/are known as **standby component(s)**. Other than this, there is a changeover device. The function of the changeover device is to sense the failure of normally operating component and in case of a failure, to bring a standby component into the normal operating mode. To explain the concept of standby system, let us first consider the simplest case of the two components (say, A and B) standby system.

A switch in the standby systems can put any component into operation. In the 2-component standby system, initially this switch is connected to component A and turns it on, as shown in Fig. 15.5. In Fig. 15.5, the switch is represented by S. In this setup, component B will remain in standby (reserve or inoperative) mode till such time as component A performs its function successfully. As soon as component A fails, the switch senses the failure and puts the component B into operation.

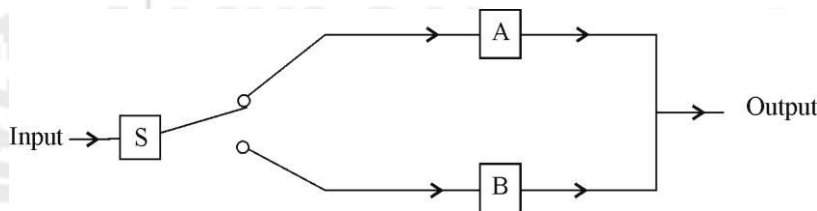


Fig. 15.5: A typical two components standby system.

In general, if we have a standby system having n components, namely, 1, 2, 3, ..., n , then there exists a switch, say S, which can put any one of the n components into operation. The system works in the same way as the two-component standby system.

Initially the switch is connected to component 1 and turns it on (see Fig. 15.6). Here we are assuming that only one component is in the normally operating mode. Until the component 1 performs its intended function successfully, the remaining $(n-1)$ components, namely 2 to n , remain in the standby (reserve or inoperative) mode. As soon as component 1 fails, the switch senses the failure and turns component 2 on. Till such time as the component 2 performs its intended function successfully, the remaining $(n-2)$ components, namely 3 to n , remain in the standby (reserve or inoperative) mode. As soon as component 2 fails, the switch senses the failure and turns the component 3 on. This process continues until the failure of the n^{th} component. When the n^{th} component fails, the system will go into failure mode. If there are more than one components in the normally operating mode, the process will work in the same way as it works for one normally operating component.

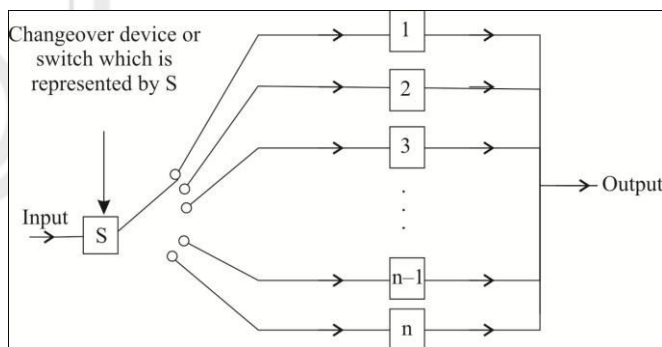


Fig. 15.6: A typical n components standby system for the case of only one component in normally operating mode.

In the discussion so far, we have not considered a key point of the standby system: that the switch itself may fail during the operation of a component or during the time of changeover of a normally operating component to a standby

component. Due to this feature of the standby system, we divide the discussion of this section under the following two heads:

- Standby system with perfect switching, and
- Standby system with imperfect switching.

Let us discuss these one at a time, in brief.

Standby System with Perfect Switching

By perfect switching, we mean that the changeover device/switch will neither fail during the operation of a component nor during the changeover of a failed component to the next available standby component. In reliability terms, we are assuming that the changeover device is 100% reliable. To evaluate the reliability of the standby system with perfect switching, we assume that the standby component(s) will not fail in its/their standby position. Now, let E_i be the event that component i performs its intended function successfully, where $i = 1, 2, 3, \dots, n$. Let Q_i denote the unreliability of the component i , given that components 1 to $(i-1)$ have failed. Further, if R and Q denote the reliability and unreliability of the standby system, then

$$Q = P(\bar{E}_1 \cap \bar{E}_2 \cap \bar{E}_3 \cap \dots \cap \bar{E}_n), \text{ where } \bar{E}_i \text{ denotes complement of event } E_i$$

$$= P(\bar{E}_1)P(\bar{E}_2 | \bar{E}_1)P(\bar{E}_3 | \bar{E}_1 \bar{E}_2) \dots P(\bar{E}_n | \bar{E}_1 \bar{E}_2 \dots \bar{E}_{n-1}) \quad \dots (4a)$$

[using the concept of conditional probability]

$$= Q_1 Q_2 Q_3 \dots Q_n \quad \dots (4b)$$

$$\therefore \text{reliability (R) of the standby system is given by } R = 1 - Q \quad \dots (5)$$

In particular, in the case of one normally operating component and one standby component, the unreliability of the system is given by

$$Q = Q_1 Q_2 \quad \dots (6)$$

and reliability (R) of the standby system is given by equation (5).

Note1: Here the unreliability Q_2 of the component 2 is not the same as it is in the case of parallel system because in a standby system, component 2 is used for a shorter duration as compared to the case of parallel system.

Let us consider an example.

Example 3: A standby system has three components 1, 2, 3, where component 1 is normally operating and components 2, 3 are standby components. The reliability of component 1 is 0.95. The reliability of component 2 given that component 1 has failed is 0.96 and that of component 3 given that components 1 and 2 have failed is 0.98. Evaluate the reliability of the system under the assumption that the switch is perfect.

Solution: We know that for an n -component standby system (with one normally operating component), if Q_i is the unreliability of component i given that components 1 to $(i-1)$ have failed, then reliability (R) of the system is given by

$$R = 1 - Q_1 Q_2 Q_3 \dots Q_n$$

In this case,

$$Q_1 = 1 - 0.95 = 0.05, \quad Q_2 = 1 - 0.96 = 0.04, \quad Q_3 = 1 - 0.98 = 0.02$$

$$\therefore R = 1 - Q_1 Q_2 Q_3$$

$$= 1 - (0.05)(0.04)(0.02) = 1 - 0.00004 = 0.99996$$

We now discuss the reliability evaluation of the standby system in the case of imperfect switching.

Standby System with Imperfect Switching

By imperfect switching, we mean that the changeover device/switch may fail either during the operation of a component or during the time of bringing a standby component in place of a normally operating failed component. To explain the concept, let us consider a simple two-component standby system with one normally operating component (say, A) and one standby component (say, B). To evaluate the reliability of the system, let us assume that the switch fails only at the time of changeover. Therefore, the unreliability of the system is given by

$$Q = P(\text{successful changeover}) \times P(\text{system failure given successful changeover}) \\ + P(\text{unsuccessful changeover}) \times P(\text{system failure given unsuccessful changeover})$$

$$\Rightarrow Q = P_s Q_A Q_B + \bar{P}_s Q_A, \quad \left[\begin{array}{l} \because \text{in case of successful changeover, system} \\ \text{failure} \Rightarrow \text{A fails and then component B fails.} \\ \text{In case of unsuccessful changeover, system} \\ \text{failure} \Rightarrow \text{failure of component A.} \end{array} \right] \dots(7)$$

where,

P_s = probability of successful changeover

$\bar{P}_s = 1 - P_s$ = probability of unsuccessful changeover

Q_A = unreliability of component A

Q_B = unreliability of component B given that component A has failed

Let us consider an example.

Example 4: Consider a two-component standby system with A as normally operating component and B as standby component. The reliability of component A is 0.90 while the reliability of component B given that A has failed is 0.95. Assume that the switch can fail only at the time of changeover with a probability of failure 0.03. Evaluate the reliability of the system.

Solution: We know that if

P_s = probability of successful changeover

$\bar{P}_s = 1 - P_s$ = probability of unsuccessful changeover

Q_A = unreliability of component A

Q_B = unreliability of component B given that component A has failed

then from equation (7), the unreliability (Q) of the two-component standby system under the assumption that the switch can only fail during the time of changeover is given by

$$Q = P_s Q_A Q_B + \bar{P}_s Q_A \\ = (1 - 0.03) \times (1 - 0.90)(1 - 0.95) + 0.03 \times (1 - 0.90) \quad [\text{using given reliabilities}] \\ = 0.97 \times 0.10 \times 0.05 + 0.03 \times 0.10 = 0.00485 + 0.003 = 0.00785$$

Hence, the reliability of the system = $1 - 0.00785 = 0.99215$

You can try the following exercise to evaluate the reliability of a standby system.

E2) If in Example 2, the probability of unsuccessful changeover is 0.25 instead of 0.03, evaluate the reliability of the system.

Let us summarise the main points that we have covered in this unit.

15.5 SUMMARY

1. **Redundancy** in a system means that there exists an alternative parallel path for the successful operation of the system. If a system has n components connected in parallel and at least k , ($1 \leq k \leq n$) components are needed for the successful operation of the system, we say that the number of **basic components** is k and the remaining $(n - k)$ components are known as **redundant components**.
2. **Fully Redundant System:** A k -out-of- n system is known as a **fully redundant** system if $k = 1$.
3. A k -out-of- n system is said to be a **non-redundant** system if $k = n$.
4. A k -out-of- n system is said to be a **partially redundant** system if $1 < k < n$.
5. **Reliability of k -out-of- n system** is obtained by using **binomial distribution** in the case of **identical components**. In the case of **non-identical components**, it is calculated by first listing out all **mutually exclusive** and **exhaustive** cases for different values of k and then by applying the addition law of probability.
6. **Reliability of standby system** having n components, namely 1, 2, ..., n , in the case of **perfect switching** is given by

$$Q = Q_1 Q_2 Q_3 \dots Q_n,$$
 where Q_i denotes the unreliability of the component i , given that components 1 to $(i - 1)$ have failed.
7. **Reliability of standby system** having n components, namely 1, 2, ..., n , in the case of **imperfect switching** under the assumption that the changeover device (switch) fails only at the time of changeover (in case of two components) is given by

$$R = 1 - Q$$
 where $Q = P_s Q_A Q_B + \bar{P}_s Q_A$, and
 P_s = probability of successful changeover
 $\bar{P}_s = 1 - P_s$ = probability of unsuccessful changeover
 Q_A = unreliability of component A
 Q_B = unreliability of component B given that component A has failed

15.6 SOLUTIONS/ANSWERS

E1) If R_s denotes the reliability of the system and R that of each component, where $R = 0.80$, then for different values of k , R_s is calculated below:

For $k = 1$,

$$\begin{aligned}
 R_s &= \sum_{k=1}^5 {}^5C_k R^k (1-R)^{5-k} \\
 &= {}^5C_1 R (1-R)^4 + {}^5C_2 R^2 (1-R)^3 + {}^5C_3 R^3 (1-R)^2 \\
 &\quad + {}^5C_4 R^4 (1-R) + {}^5C_5 R^5 \\
 &= 5 \times 0.8 (1-0.8)^4 + 10 (0.8)^2 (1-0.8)^3 + 10 (0.8)^3 (1-0.8)^2 \\
 &\quad + 5 (0.8)^4 (1-0.8) + (0.8)^5 \\
 &= 0.0064 + 0.0512 + 0.2048 + 0.4096 + 0.32768 \\
 &= 0.99968
 \end{aligned}$$

For $k = 2$,

$$\begin{aligned}
 R_s &= \sum_{k=2}^5 {}^5C_k R^k (1-R)^{5-k} \\
 &= {}^5C_2 R^2 (1-R)^3 + {}^5C_3 R^3 (1-R)^2 + {}^5C_4 R^4 (1-R) + {}^5C_5 R^5 (1-R)^0 \\
 &= 0.0512 + 0.2048 + 0.4096 + 0.32768 \\
 &\quad \text{[using already calculated values]} \\
 &= 0.99328
 \end{aligned}$$

For $k = 3$,

$$\begin{aligned}
 R_s &= \sum_{k=3}^5 {}^5C_k R^k (1-R)^{5-k} \\
 &= {}^5C_3 R^3 (1-R)^2 + {}^5C_4 R^4 (1-R)^1 + {}^5C_5 R^5 (1-R)^0 \\
 &= 0.2048 + 0.4096 + 0.32768 \text{ [Using already calculated values]} \\
 &= 0.94208
 \end{aligned}$$

For $k = 4$,

$$\begin{aligned}
 R_s &= \sum_{k=4}^5 {}^5C_k R^k (1-R)^{5-k} = {}^5C_4 R^4 (1-R) + {}^5C_5 R^5 (1-R)^0 \\
 &= 0.4096 + 0.32768 \quad \text{[using already calculated values]} \\
 &= 0.73728
 \end{aligned}$$

For $k = 5$,

$$R_s = \sum_{k=5}^5 {}^5C_k R^k (1-R)^{5-k} = {}^5C_5 R^5 (1-R)^0 = 0.32768$$

Let us put all the calculated values in Table 15.1 as follows.

Table 15.1: Reliability of the System for $k = 1, 2, 3, 4, 5$

Value of k	Reliability of the System for a Mission of 1 year	% Decrease in Reliability of the System Compared to the Value for $k = 1$
1	0.99968	—
2	0.99328	$\frac{0.99968 - 0.99328}{0.99968} \times 100 \square 0.64\%$
3	0.94208	$\frac{0.99968 - 0.94208}{0.99968} \times 100 \square 5076\%$
4	0.73728	$\frac{0.99968 - 0.73728}{0.99968} \times 100 \square 26.25\%$
5	0.32768	$\frac{0.99968 - 0.32768}{0.99968} \times 100 \square 67.22\%$

Comment: From Table 15.1, we observe that the reliability of the system decreases as k increases. The maximum decrease occurs when $k = 5 = n$, i.e., in the case when k -out-of- n system coincides with the series system.

E2) We know that if

P_s = probability of successful changeover

$\bar{P}_s = 1 - P_s$ = probability of unsuccessful changeover

Q_A = unreliability of component A

Q_B = unreliability of component B given that component A has failed

then unreliability (Q) of the two-component standby system under the assumption that the switch can only fail during the time of changeover is given by

$$Q = P_s Q_A Q_B + \bar{P}_s Q_A$$

$$= (1 - 0.25) \times (1 - 0.90)(1 - 0.95) + 0.25 \times (1 - 0.90)$$

[using given reliabilities]

$$= 0.75 \times 0.10 \times 0.05 + 0.25 \times 0.10 = 0.00375 + 0.025 = 0.02875$$

Hence, the reliability of system = $1 - 0.02875 = 0.97125$

UNIT 16 RELIABILITY EVALUATION OF COMPLEX SYSTEMS

Structure

- 16.1 Introduction
 - Objectives
- 16.2 Definition of Complex System
- 16.3 Decomposition Method or Conditional Probability Approach
- 16.4 Minimal Path Set and Cut Set
- 16.5 Cut Set Method
- 16.6 Tie Set Method
- 16.7 Summary
- 16.8 Solutions/Answers

16.1 INTRODUCTION

In Units 14 and 15, you have learnt how to evaluate the reliability of series, parallel and mixed systems, and k-out-of-n systems. We have also introduced the concept of standby redundancy in Unit 15.

However, there exist systems, which cannot be reduced to parallel, series or mixed systems with independent components. Such systems are known as **complex systems**. In this unit, we discuss some techniques of evaluating the reliability of complex systems. We first define the complex system in Sec.16.2. In Sec. 16.3, we discuss the decomposition method or the conditional probability approach for evaluating the reliability of a complex system. We discuss two more techniques of evaluating the reliability of a complex system, which use the concepts of minimal path set and minimal cut set. Therefore, in Sec. 16.4 we explain these concepts. Then we discuss the cut set method in Sec. 16.5 and the tie set method in Sec. 16.6 for evaluation of reliability.

Objectives

After studying this unit, you should be able to:

- define a complex system;
- explain the concept of minimal path set and minimal cut set; and
- evaluate the reliability of complex systems using the decomposition method or conditional probability approach, cut set method and tie set method.

16.2 DEFINITION OF COMPLEX SYSTEM

Recall the definition of a simple system from Sec. 14.2 of Unit 14:

A system is said to be **simple** if either its components are connected in parallel, in series or in combinations of both. The reliability block diagram of a simple system consists of series or parallel configurations of independent components. Else, it can be reduced into subsystems having independent components connected either in parallel or in series.

We now define a complex system. A system is said to be **complex** if the reliability block diagram of the system cannot be reduced into subsystems having independent components connected either in parallel or in series.

A commonly used example of a complex system known as the bridge configuration is shown in Fig. 16.1.

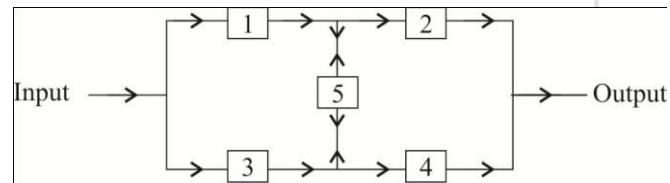


Fig. 16.1: Bridge configuration.

You can see that the reliability block diagram of the bridge configuration shown in Fig. 16.1 cannot be reduced into subsystems having independent components connected either in parallel or in series.

To evaluate the reliability of a complex system, we reduce the given system into subsystems and obtain the reliability of the subsystems using the concepts of series or parallel systems. Then we combine these reliabilities to obtain the reliability of the complex system. There are different techniques of reducing a complex system into series or parallel subsystems. We discuss three such techniques in the next three sections of this unit. We explain the conditional probability approach or decomposition method in Sec. 16.3.

16.3 DECOMPOSITION METHOD OR CONDITIONAL PROBABILITY APPROACH

To evaluate the reliability of a complex system using the decomposition method or conditional probability approach, we follow the steps explained below:

Step 1: We first choose a keystone component, say, K. In general, we select that component of a complex system as a keystone component, which appears to bind together its reliability block diagram. In fact, we can choose any component in the system as the keystone component and we shall get the same answer. But a poor choice may increase the number of steps in the calculations. Choosing a keystone component should not be difficult for you, once you study Example 1.

Step 2: We decompose the original system first by considering the keystone component K to be good, which means that it is 100% reliable. Secondly, we consider it as bad (which means that it is not reliable at all or has failed). Then the reliability (R) of the system is given by

$$R_s = P(\text{system success} | \text{component K is good})P(\text{component K is good}) + P(\text{system success} | \text{component K is bad})P(\text{component K is bad}) \dots (1)$$

If the component K is good, we replace it by a line in the reliability block diagram of the system. This means that the information can flow in either direction without any interruption. If the component K is bad, we remove it from the reliability block diagram of the system. This is because if component K is bad, it means that the path(s) of information which goes/go through component K is/are interrupted. Hence, information cannot pass through component K.

The unreliability (Q) of the original system is given by

$$Q = P(\text{system failure} | \text{component K is good})P(\text{component K is good}) + P(\text{system failure} | \text{component K is bad})P(\text{component K is bad}) \dots (2)$$

Step 3: If the reduced subsystems are in series or in parallel configuration, then we first evaluate the reliability of each subsystem by applying the procedure discussed in Unit 14. Finally, we combine the two reliabilities or unreliabilities using equation (1) or (2). But, if both or one of the subsystems are/is not in series or parallel configuration, then we select a keystone component again and apply the same method. This process continues until the most recently reduced subsystems are in series or in parallel configurations.

Let us explain this procedure with the help of an example. Note that in this example and in the calculations in this unit, we do not follow our standard practice of restricting calculations up to four decimal places (except at some places). Instead, we shall take the calculated values as such without rounding them off because we shall be comparing the results obtained using different techniques for the same problem.

Example 1: Using the decomposition method or conditional probability approach, evaluate the reliability of the system having reliability block diagram shown in Fig. 16.1 for a mission of 1000 hours. Assume that each component has a reliability of 0.95 for a mission of 1000 hours. Also assume that the components are independent.

Solution: The calculations involved in evaluating the reliability of the complex system for this example using the decomposition method are explained as follows:

Step 1: We first choose a keystone component. Here we choose component 5 as the keystone component because this choice will reduce the reliability block diagram to a series or parallel configuration or a combination of both in a single step. What if we were to choose any component other than component 5? Note that we would have to further choose a keystone component to obtain a series or parallel or a combination of the two configurations. So, here component 5 is the key component K.

Step 2: Now, component 5 can be good or bad. The reduced subsystems corresponding to these cases are shown in Fig. 16.2.

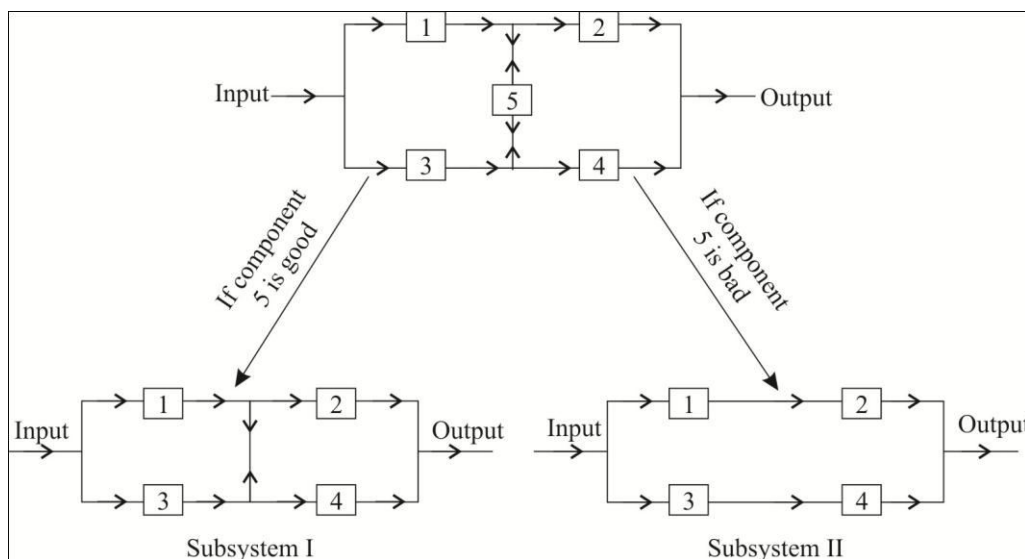


Fig. 16.2: Reduction of given system as per the good and bad status of the component 5.

If R_s denotes the reliability of the original system, then

$$\begin{aligned}
 R_s &= P(\text{system success} | \text{component 5 is good}) P(\text{component 5 is good}) \\
 &\quad + P(\text{system success} | \text{component 5 is bad}) P(\text{component 5 is bad}) \\
 &= 0.95 \times P(\text{system success} | \text{component 5 is good}) \\
 &\quad + (1 - 0.95) \times P(\text{system success} | \text{component 5 is bad}) \quad \dots (i)
 \end{aligned}$$

$$\left[\begin{array}{l} \because \text{reliability of component 5 is } 0.95, \\ \therefore \text{unreliability} = 1 - 0.95 \end{array} \right]$$

Step 3: Let R_i denote the reliability of the component i , ($i = 1, 2, 3, 4, 5$). In this step, we first evaluate the reliabilities of each reduced subsystem. Note that in subsystem I, components 1 and 3 are in parallel, components 2 and 4 are in parallel and the two parallel configurations are in series (see Fig. 16.2).

$$\begin{aligned}
 \therefore P(\text{system success} | \text{component 5 is good}) &= [1 - (1 - R_1)(1 - R_3)] \times \\
 &\quad [1 - (1 - R_2)(1 - R_4)] \\
 &= [1 - (1 - R_3 - R_1 + R_1 R_3)] [1 - (1 - R_4 - R_2 + R_2 R_4)] \\
 &= (R_1 + R_3 - R_1 R_3)(R_2 + R_4 - R_2 R_4) \quad \dots (ii)
 \end{aligned}$$

In subsystem II, components 1 and 2 are in series, components 3 and 4 are in series and the two series configurations are in parallel (see Fig. 16.2).

$$\begin{aligned}
 \therefore P(\text{system success} | \text{component 5 is bad}) &= 1 - (1 - R_1 R_2)(1 - R_3 R_4) \\
 &= 1 - (1 - R_3 R_4 - R_1 R_2 + R_1 R_2 R_3 R_4) \\
 &= R_1 R_2 + R_3 R_4 - R_1 R_2 R_3 R_4 \quad \dots (iii)
 \end{aligned}$$

Using equations (ii) and (iii) in equation (i), we get

$$\begin{aligned}
 R_s &= 0.95(R_1 + R_3 - R_1 R_3)(R_2 + R_4 - R_2 R_4) \\
 &\quad + 0.05(R_1 R_2 + R_3 R_4 - R_1 R_2 R_3 R_4)
 \end{aligned}$$

Since we are given that $R_1 = R_2 = R_3 = R_4 = R_5 = 0.95$, therefore

$$\begin{aligned}
 R_s &= 0.95(0.95 + 0.95 - (0.95)^2)(0.95 + 0.95 - (0.95)^2) \\
 &\quad + 0.05[(0.95)^2 + (0.95)^2 - (0.95)^4] \\
 &= 0.95[1.90 - 0.9025][1.90 - 0.9025] \\
 &\quad + 0.05[0.9025 + 0.9025 - 0.81450625] \\
 &= 0.95(0.9975)(0.9975) + 0.05(0.99049375) \\
 &= 0.9452559375 + 0.0495246875 \\
 &= 0.994780625
 \end{aligned}$$

You may like to try the following exercise to apply the decomposition method.

- E1)** The reliability block diagram of a system is shown in Fig. 16.3. Evaluate the reliability of the system using the decomposition method for a mission of 1 year. Assume that the reliability of each component is 0.90 for a mission of 1 year and the components are independent.

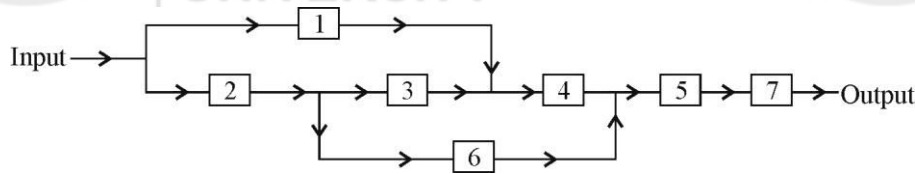


Fig. 16.3: Reliability block diagram for E1.

16.4 MINIMAL PATH SET AND CUT SET

In Sec. 16.3, we have discussed the conditional probability approach or decomposition method for evaluating the reliability of complex systems. We now discuss the concept of minimal path set and cut set, which form the basis of the other two techniques used for reliability evaluation of complex systems.

Minimal Path Set

Let a system consist of n components, namely, 1, 2, 3, ..., n . Let C denote the set of all components of the system. Then

$$C = \{1, 2, 3, \dots, n\}$$

A **path set** P is a subset of C such that if all the components of P are successful, then there exists a path between the input and output of the system. A path set P is said to be the **minimal path set** if there exists no proper subset of P as a path set in the system. In other words, a path set P is said to be minimal path set if the failure of even a single component of P results in the failure of the path between the input and output.

For example, the path sets for the system having the reliability block diagram shown in Fig. 16.4 are as follows:

$$P_1 = \{1, 3\}, P_2 = \{2, 3\}, P_3 = \{1, 2, 3\}$$

Of these paths, P_1 and P_2 are minimal paths because there is no proper subset of P_1 and P_2 , which is also a path. But the path P_3 is not a minimal path because there exist paths P_1 and P_2 , which are proper subsets of P_3 .

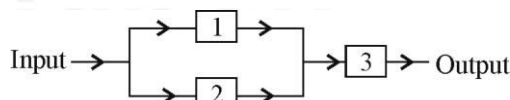


Fig. 16.4: Reliability block diagram of a system having components 1 and 2 in parallel. The parallel configuration of 1 and 2 is in series with component 3.

Minimal Cut Set

Let a system consist of n components, namely, 1, 2, 3, ..., n . Let C denote the set of all components of the system. Then

$$C = \{1, 2, 3, \dots, n\}$$

A **cut set** C_1 is a subset of C such that a failure of all components of C_1 results in the failure of the system.

A cut set C_1 is said to be the **minimal cut set** if there exists no proper subset of C_1 as a cut set in the system.

For example, the cut sets for the system having the reliability block diagram shown in Fig 16.4 are:

$$C_1 = \{3\}, C_2 = \{1, 3\}, C_3 = \{2, 3\}, C_4 = \{1, 2\}, C_5 = \{1, 2, 3\}$$

Of these cut sets, only C_1 and C_4 are minimal cut sets, because there exist no proper subsets of C_1 and C_4 , which are cut sets. For all other cut sets mentioned above, there exists at least one proper subset, which is a cut set. So these subsets are not minimal cut sets.

Try to solve the following exercise to obtain minimal path sets and cut sets.

E2) For the system having the reliability block diagram shown in Fig. 16.1, obtain the minimal path sets and minimal cut sets.

So far, you have learnt the concepts of minimal path sets and minimal cut sets. We now discuss the other two techniques of evaluating the reliability of a complex system based on these concepts.

16.5 CUT SET METHOD

In this method, we first find minimal cut sets for the given system. Let $C_1, C_2, C_3, \dots, C_k$ be the minimal cut sets for the system consisting of n components, namely, $1, 2, 3, \dots, n$. Let m_i denote the number of components in the cut set C_i , $i = 1, 2, 3, \dots, k$. By definition, all components of a minimal cut set must fail for the system to fail. This implies that the components of a cut set are connected in parallel from a reliability point of view. Further, the occurrence of any one cut set among $C_1, C_2, C_3, \dots, C_k$ results in the failure of the system. This implies that $C_1, C_2, C_3, \dots, C_k$ are connected in series from a reliability point of view. Thus, in terms of $C_1, C_2, C_3, \dots, C_k$, the reliability block diagram of the system would be as shown in Fig. 16.5,

where the m_i components of cut set C_i are denoted by $C_{i1}, C_{i2}, \dots, C_{im_i}$, $1 \leq i \leq k$.

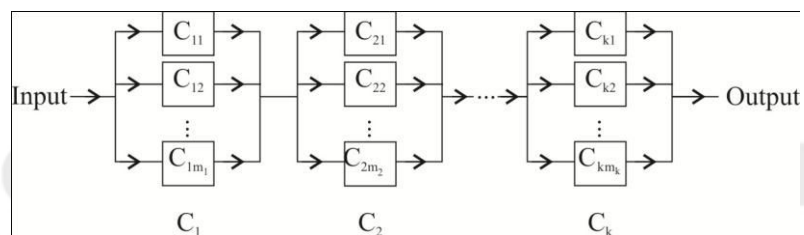


Fig. 16.5: A typical reliability block diagram in terms of k cut sets.

Refer to Fig. 16.5. Note that the cut sets $C_1, C_2, \dots, C_k$ are connected in series. If all the components of C_i and C_j , ($1 \leq i, j \leq k$, but $i \neq j$) are distinct, you can evaluate the reliability of the system using the procedure explained in Sec. 14.6. However, in general, for complex systems, one or more component(s) of one cut set appears/appear in other cut sets (e.g., in Fig. 16.6, component 1 appears in C_1 and C_3 , etc.). Due to this reason, we cannot apply the procedure explained in Sec. 14.6. Always remember this important point before you apply the cut set method. Then you can give a thought as to how we can evaluate the reliability of such a system. Let us give you some hints:

- This method evaluates the probability of failure of the system (i.e., unreliability) and reliability is obtained by subtracting unreliability from 1.
- The system will fail if at least one of the cut sets $C_1, C_2, \dots, C_k$ occurs (recall the definition of a cut set).

Now, you have to simply identify a law of probability which gives you the probability of failure. With these hints, have you been able to arrive at the answer? Recall the concepts discussed in Unit 3 of MST-003. It is the addition law of probability, which gives us the probability of occurrence of at least one event from among several events. Thus, if Q_s denotes the unreliability of the system, then

$$Q_s = P(C_1 \cup C_2 \cup C_3 \cup \dots \cup C_k) \quad \dots(3)$$

[$\because$ occurrence of either of $C_i, 1 \leq i \leq k$,
will interrupt all the paths from input to output.]

Let us explain the procedure with the help of an example.

Example 2: Using the cut set method, evaluate the reliability of the system, which has the reliability block diagram shown in Fig. 16.1 for a mission of 1000 hours. It is given that each component has reliability of 0.95 for a mission of 1000 hours. Assume that the components are independent.

Solution: For applying the cut set method, we first find all minimal cut sets for the system having the reliability block diagram shown in Fig. 16.1. The minimal cut sets for this system have already been listed in E2. So, we can write

$$C_1 = \{1, 3\}, C_2 = \{2, 4\}, C_3 = \{1, 5, 4\}, C_4 = \{3, 5, 2\}.$$

From a reliability point of view, these cut sets can be shown as in Fig. 16.6.

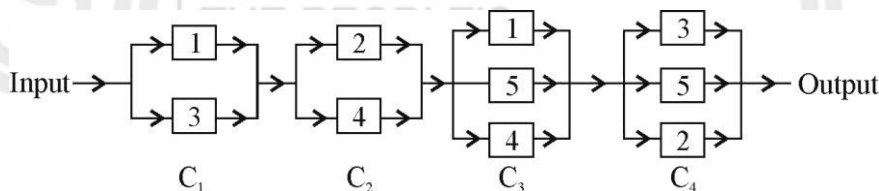


Fig. 16.6: Cut sets for the reliability block diagram of Fig. 16.1.

As mentioned earlier, you should always remember that although C_1, C_2, C_3, C_4 seem to be connected in series in Fig. 16.6, we cannot use the concept of series system because these are not made up of independent components. Note that component 4 appears in both C_2 and C_3 , component 2 appears in both C_2 and C_4 , etc.

Therefore, if Q_i denotes the unreliability of the component i , ($i = 1, 2, 3, 4, 5$) and Q_s that of the system, then by cut set method, we get

$$Q_s = P(C_1 \cup C_2 \cup C_3 \cup C_4)$$

Applying the addition law of probability (inclusion-exclusion rule), we get

$$\left. \begin{aligned} Q_s = & P(C_1) + P(C_2) + P(C_3) + P(C_4) - P(C_1 \cap C_2) - P(C_1 \cap C_3) \\ & - P(C_1 \cap C_4) - P(C_2 \cap C_3) - P(C_2 \cap C_4) - P(C_3 \cap C_4) \\ & + P(C_1 \cap C_2 \cap C_3) + P(C_1 \cap C_2 \cap C_4) + P(C_1 \cap C_3 \cap C_4) \\ & + P(C_2 \cap C_3 \cap C_4) - P(C_1 \cap C_2 \cap C_3 \cap C_4) \end{aligned} \right\} \dots (i)$$

Reliability Theory

$$\left[\begin{array}{l} \because \text{in Unit 3 of the course MST-003, you have studied that} \\ P(A \cup B) = P(A) + P(B) - P(A \cap B) \\ P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) \\ \quad + P(A \cap B \cap C), \text{ etc.} \end{array} \right]$$

Let us first evaluate each term of R.H.S. of (i) separately.

$$P(C_1) = Q_1 Q_3 \quad \left[\because \text{occurrence of cut set } C_1 \text{ means that} \right. \\ \left. \text{all components of } C_1 \text{ must fail.} \right]$$

Similarly,

$$P(C_2) = Q_2 Q_4$$

$$P(C_3) = Q_1 Q_5 Q_4$$

$$P(C_4) = Q_3 Q_5 Q_2$$

$$P(C_1 \cap C_2) = P(C_1)P(C_2) = Q_1 Q_2 Q_3 Q_4$$

$$P(C_1 \cap C_3) = P(C_1)P(C_3) = Q_1 Q_3 Q_4 Q_5 \quad [\because \text{we know that } P(A \cap A) = P(A)]$$

$$P(C_1 \cap C_4) = P(C_1)P(C_4) = Q_1 Q_2 Q_3 Q_5$$

$$P(C_2 \cap C_3) = P(C_2)P(C_3) = Q_1 Q_2 Q_4 Q_5$$

$$P(C_2 \cap C_4) = P(C_2)P(C_4) = Q_2 Q_3 Q_4 Q_5$$

$$P(C_3 \cap C_4) = P(C_3)P(C_4) = Q_1 Q_2 Q_3 Q_4 Q_5$$

$$P(C_1 \cap C_2 \cap C_3) = P(C_1 \cap C_2 \cap C_4) = P(C_1 \cap C_3 \cap C_4) = P(C_2 \cap C_3 \cap C_4) \\ = Q_1 Q_2 Q_3 Q_4 Q_5$$

$$P(C_1 \cap C_2 \cap C_3 \cap C_4) = P(C_1)P(C_2)P(C_3)P(C_4) = Q_1 Q_2 Q_3 Q_4 Q_5$$

We are given that all components (1, 2, 3, 4, 5) have the same reliability 0.95 for a mission of 1000 hours. Hence, their unreliability will also be the same.

Let $Q_1 = Q_2 = Q_3 = Q_4 = Q_5 = Q$ (say). Then

$$P(C_1) = Q^2$$

$$P(C_2) = Q^2$$

$$P(C_3) = Q^3$$

$$P(C_4) = Q^3$$

$$P(C_1 \cap C_2) = P(C_1 \cap C_3) = P(C_1 \cap C_4) = P(C_2 \cap C_3) = P(C_2 \cap C_4) = Q^4$$

$$P(C_3 \cap C_4) = Q^5$$

$$P(C_1 \cap C_2 \cap C_3) = P(C_1 \cap C_2 \cap C_4) = P(C_1 \cap C_3 \cap C_4) = P(C_2 \cap C_3 \cap C_4) = Q^5$$

$$P(C_1 \cap C_2 \cap C_3 \cap C_4) = Q^5$$

Putting these values in (i), we get

$$Q_s = (Q^2 + Q^2 + Q^3 + Q^3) - (5Q^4 + Q^5) + (4Q^5) - Q^5 \\ = 2Q^2 + 2Q^3 - 5Q^4 + 2Q^5$$

Since the reliability of each component for a mission of 1000 hours = 0.95,
the unreliability of each component up to a mission
of 1000 hours = $Q = 1 - 0.95 = 0.05$

Hence, unreliability of the system (Q_s) is given by

$$\begin{aligned} Q_s &= 2(0.05)^2 + 2(0.05)^3 - 5(0.05)^4 + 2(0.05)^5 \\ &= 0.005 + 0.00025 - 0.00003125 + 0.000000625 \\ &= 0.005219375 \end{aligned} \quad \dots (ii)$$

Therefore, the reliability of the system (R_s) is given by

$$R_s = 1 - Q_s = 1 - 0.005219375 = 0.994780625$$

Note 1: Suppose we call $P(C_1), P(C_2), P(C_3), P(C_4)$ as first order terms;

$P(C_1 \cap C_2), P(C_1 \cap C_3)$, etc. as second order terms;

$P(C_1 \cap C_2 \cap C_3), P(C_1 \cap C_2 \cap C_4)$, etc. as third order terms and so on. Then for components having high reliability (that is, having low unreliability), the second and higher order terms can be ignored without affecting the value of reliability much. For example, in Example 2, if we neglect second order and higher order terms, then

$$Q_s = (2Q^2 + 3Q^3) = 0.00525 \text{ and } R_s = 1 - 0.00525 = 0.99475$$

The percentage error in the value of R_s is

$$\frac{0.994780625 - 0.99475}{0.994780625} \times 100\% \approx 0.0030786\% = 3.0786 \times 10^{-3}\%$$

i.e., 0.0030% (approx) which can be tolerated. This approximation saves a lot of calculations! This is one of the main advantages of the cut set method. It will be clearer to you when you do the exercises E3 and E4. The second main advantage of this method is that the calculations can be done by programming on a digital computer.

Try the following exercises to apply the cut set method.

-
- E3)** Evaluate the reliability of the system for which the reliability block diagram is shown in Fig. 16.3 for a mission of 1 year, using the cut set method. It is given that each component has reliability of 0.9 for a mission of 1 year. Assume that the components are independent. Use only first order terms in your calculations.
- E4)** You may be interested in knowing the amount of calculations involved in E3 if the calculations were not restricted to first order terms. Repeat the calculations keeping terms of all orders. Note that such exercises will not form a part of the examination due to time constraints.
-

We now discuss the third technique for evaluating the reliability of a complex system based on the concept of minimal path sets.

16.6 TIE SET METHOD

The tie set method is based on the concept of minimal path sets and in this method we evaluate the reliability of the system. Note that this method is different from the cut set method in which we evaluate the unreliability of the system. In the tie set method, we first find minimal path sets for the system. Let

$P_1, P_2, \dots, P_k$ be the minimal path sets for the system consisting of n components, namely, $1, 2, \dots, n$. Let m_i denote the number of components in the minimal path set P_i , ($i = 1, 2, \dots, k$).

By definition, all components of a minimal path set must work successfully for the successful operation of the system along this path. Note that this implies that the components of a minimal path set must be connected in series from a reliability point of view. Further, the existence of any one path from among $P_1, P_2, \dots, P_k$, from input to output ensures the successful operation of the system. This implies that the minimal path sets are connected in parallel from a reliability point of view. Thus, the reliability block diagram of such a system can be shown in terms of $P_1, P_2, \dots, P_k$ as in Fig. 16.7. But in Fig. 16.7, the minimal path sets $P_1, P_2, \dots, P_k$ are represented by minimal tie sets $T_1, T_2, \dots, T_k$. The name tie is given to the minimal paths because the nodes of the reliability graph (see Fig. 14.2) tie the different branches together to form a path between input and output.

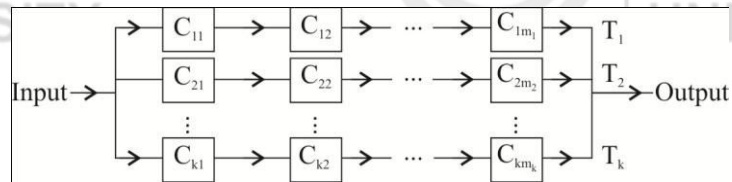


Fig. 16.7: A typical reliability block diagram in terms of k minimal path sets.

As explained for cut set method, the reliability of the system (R_s) can be obtained by expanding the expression given below using the addition law of probability:

$$R_s = P(T_1 \cup T_2 \cup T_3 \cup \dots \cup P_k) \quad \dots (4)$$

Let us explain the procedure with the help of an example.

Example 3: Evaluate the reliability of the system for which the reliability block diagram is shown in Fig. 16.1 for a mission of 500 hours by using the tie set method. It is given that each component has reliability of 0.95 for a mission of 500 hours. Assume that all components are independent.

Solution: For applying the tie set method, we first find the minimal path sets for the system having the reliability block diagram shown in Fig. 16.1. The minimal path sets for this system have already been listed in E2. So we repeat them as follows:

$$P_1 = \{1, 2\}, P_2 = \{3, 4\}, P_3 = \{1, 5, 4\}, P_4 = \{3, 5, 2\}$$

Let us denote these minimal path sets by minimal tie sets T_1, T_2, T_3 and T_4 , respectively. The reliability block diagram in terms of tie sets is shown in Fig. 16.8.

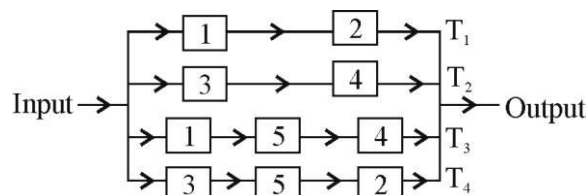


Fig. 16.8: Minimal path sets for Example 3.

You should remember one important point here: Although T_1, T_2, T_3, T_4 seem to be connected in parallel, you should not use the concept of parallel system because they are not made up of independent components. Note that component 1 appears in both T_1 and T_3 , component 2 appears in T_1 and T_4 , etc.

Therefore, if R_i denotes the reliability of the component i , ($i = 1, 2, 3, 4, 5$), and R_s that of the system, then by tie set method, we have

$$R_s = P(T_1 \cup T_2 \cup T_3 \cup T_4)$$

Applying the addition law of probability (which you have studied in Unit 3 of MST-003), we get

$$R_s = P(T_1) + P(T_2) + P(T_3) + P(T_4) - P(T_1 \cap T_2) - P(T_1 \cap T_3) - P(T_1 \cap T_4) \\ - P(T_2 \cap T_3) - P(T_2 \cap T_4) - P(T_3 \cap T_4) + P(T_1 \cap T_2 \cap T_3) \\ + P(T_1 \cap T_2 \cap T_4) + P(T_1 \cap T_3 \cap T_4) + P(T_2 \cap T_3 \cap T_4) \\ - P(T_1 \cap T_2 \cap T_3 \cap T_4) \quad \dots(i)$$

Let us first evaluate each term of R.H.S. of (i) separately.

$$P(T_1) = R_1 R_2 \quad \left[\begin{array}{l} \because \text{occurrence of tie set } T_1 \text{ means that there} \\ \text{exists a path between input and output} \\ \text{through } T_1. \text{ This implies that both components} \\ \text{1 and 2 must work successfully.} \end{array} \right]$$

Similarly,

$$P(T_2) = R_3 R_4$$

$$P(T_3) = R_1 R_5 R_4$$

$$P(T_4) = R_3 R_5 R_2$$

$$P(T_1 \cap T_2) = P(T_1)P(T_2) = R_1 R_2 R_3 R_4$$

$$P(T_1 \cap T_3) = P(T_1)P(T_3) = R_1 R_2 R_4 R_5$$

$$\left[\begin{array}{l} \because \text{occurrence of } T_1 \cap T_3 \text{ means that there exists a path between input and output} \\ \text{through the paths } P_1 \text{ and } P_3. \text{ This requires that components 1, 2, 4, 5 must work} \\ \text{successfully. But component 1 appears in both paths and hence its probability is} \\ \text{taken only once.} \end{array} \right]$$

In the same way, we can write

$$P(T_1 \cap T_4) = P(T_1)P(T_4) = R_1 R_2 R_3 R_5$$

$$P(T_2 \cap T_3) = P(T_2)P(T_3) = R_1 R_3 R_4 R_5$$

$$P(T_2 \cap T_4) = P(T_2)P(T_4) = R_2 R_3 R_4 R_5$$

$$P(T_3 \cap T_4) = P(T_3)P(T_4) = R_1 R_2 R_3 R_4 R_5$$

$$P(T_1 \cap T_2 \cap T_3) = P(T_1)P(T_2)P(T_3) = R_1 R_2 R_3 R_4 R_5$$

$$P(T_1 \cap T_2 \cap T_4) = P(T_1)P(T_2)P(T_4) = R_1 R_2 R_3 R_4 R_5$$

$$P(T_1 \cap T_3 \cap T_4) = P(T_1)P(T_3)P(T_4) = R_1 R_2 R_3 R_4 R_5$$

$$P(T_2 \cap T_3 \cap T_4) = P(T_2)P(T_3)P(T_4) = R_1 R_2 R_3 R_4 R_5$$

$$P(T_1 \cap T_2 \cap T_3 \cap T_4) = P(T_1)P(T_2)P(T_3)P(T_4) = R_1 R_2 R_3 R_4 R_5$$

Since all components 1, 2, 3, 4, 5 have the same reliability $0.95 = (R, \text{ say})$ for a mission of 500 hours, we can replace R_1, R_2, R_3, R_4, R_5 by R in all expressions obtained above. So we get

$$P(T_1) = R^2, P(T_2) = R^2, P(T_3) = R^3, P(T_4) = R^3$$

$$P(T_1 \cap T_2) = P(T_1 \cap T_3) = P(T_1 \cap T_4) = P(T_2 \cap T_3) = P(T_2 \cap T_4) = R^4$$

$$P(T_3 \cap T_4) = R^5$$

$$P(T_1 \cap T_2 \cap T_3) = P(T_1 \cap T_2 \cap T_4) = P(T_1 \cap T_3 \cap T_4) = P(T_2 \cap T_3 \cap T_4) = R^5$$

$$P(T_1 \cap T_2 \cap T_3 \cap T_4) = R^5$$

Putting these values in (i), we get

$$R_s = (R^2 + R^2 + R^3 + R^3) - (5R^4 + R^5) + (4R^5) - R^5$$

$$= 2R^2 + 2R^3 - 5R^4 + 2R^5$$

$$= 2(0.95)^2 + 2(0.95)^3 - 5(0.95)^4 + 2(0.95)^5 \quad [\because R = 0.95 \text{ given}]$$

$$= 1.805 + 1.71475 - 4.07253125 + 1.547561875$$

$$= 0.994780625$$

which is the same as obtained by the cut set method in Example 2.

Now, you can try the following exercise to apply the tie set method.

E5) Evaluate the reliability of the system for which the reliability block diagram is shown in Fig. 16.3 for a mission of 1 year, using the tie set method. It is given that each component has reliability of 0.9 for a mission of 1 year. Assume that all components are independent.

Before we end the discussion, we would like to tell you that the techniques discussed in this unit are general. You can also apply these techniques for simple systems discussed in Unit 14.

Let us now summarise the main points that we have covered in this unit.

16.7 SUMMARY

1. A system is said to be **complex** if the reliability block diagram of the system cannot be reduced into subsystems having independent components connected either in parallel or in series.

2. To evaluate the reliability of a complex system using the **decomposition method** or **conditional probability approach**, we first choose a keystone component, say K. In general, that component is selected as a keystone component which appears to bind together the reliability block diagram of the system. The reliability of the system is given by

$$R_s = P(\text{system success} | \text{component K is good}) P(\text{component K is good}) \\ + P(\text{system success} | \text{component K is bad}) P(\text{component K is bad})$$

3. For a system consisting of n components, namely, 1, 2, 3, ..., n with C being the set of all components of the system:

$$C = \{1, 2, 3, \dots, n\}$$

a **path set** P is a subset of C such that if all the components of P are successful, then there exists a path between the input and output of the system. A path set P is said to be **minimal path set** if there exists no proper subset of P as a path in the system. In other words, a path set P is said to be minimal path set if the failure of even a single component of P results in the failure of the path between input and outputs.

A **cut set** C_1 is a subset of C such that the failure of all components of C_1 results in the failure of the system.

A cut set C_1 is said to be **minimal cut set** if there exists no proper subset of C_1 which forms a cut set for the failure of the system.

4. From the cut set method, unreliability of the system is given by

$$Q_s = P(C_1 \cup C_2 \cup C_3 \cup \dots \cup C_k)$$

5. From the tie set method, the reliability of the system is given by

$$R_s = P(T_1 \cup T_2 \cup T_3 \cup \dots \cup T_k)$$

16.8 SOLUTIONS/ANSWERS

- E1) The calculations involved in evaluating the reliability of the complex system having reliability block diagram shown in Fig.16.3 using decomposition method are explained as follows:

Step 1: We choose component 3 as the keystone component. The reliability block diagram shown in Fig.16.3 reduces to subsystems I and II as shown in Fig.16.9 depending on whether component 3 is good or bad. The reliability of subsystem II can be directly obtained using the concepts of series and parallel configurations. But subsystem I needs further decomposition. For further decomposition, consider component 4 as the keystone component. Further reduction of the subsystem I into subsystems III and IV is also shown in Fig. 16.9. The reliabilities of subsystems III and IV can be evaluated using the concepts of series and parallel configurations.

Step 2: If R_s denotes the reliability of the original subsystem, then

$$R_s = P(\text{system success} | \text{component 3 is good}) P(\text{component 3 is good}) \\ + P(\text{system success} | \text{component 3 is bad}) P(\text{component 3 is bad})$$

If R_I and R_{II} denote the reliabilities of the subsystems I and II, respectively, and R_i denotes the reliability of the component i , ($i = 1, 2, 3, \dots, 7$), then

$$R_s = R_I R_3 + R_{II} (1 - R_3) \quad \dots (i)$$

The reliability of the subsystem I can be expressed as

$$R_I = P(\text{system success} | \text{component 4 is good}) P(\text{component 4 is good}) \\ + P(\text{system success} | \text{component 4 is bad}) P(\text{component 4 is bad})$$

If R_{III} and R_{IV} denote reliabilities of the subsystems III and IV, respectively, then

$$R_I = R_{III} R_4 + R_{IV} (1 - R_4) \quad \dots (ii)$$

Step 3: We evaluate the reliabilites of each of the reduced subsystems.

Using the concepts of series and parallel systems, the reliabilities of subsystems II, III and IV are given by

$$R_{II} = [1 - (1 - R_1 R_4)(1 - R_2 R_6)] R_5 R_7 \\ = [1 - (1 - R_2 R_6 - R_1 R_4 + R_1 R_2 R_4 R_6)] R_5 R_7 \\ = (R_1 R_4 + R_2 R_6 - R_1 R_2 R_4 R_6) R_5 R_7$$

We are given that $R_i = 0.90, \forall i, 1 \leq i \leq 7$ for a mission of 1 year.

$$\therefore R_{II} = [(0.9)^2 + (0.9)^2 - (0.9)^4] (0.9)^2 \\ = (0.81 + 0.81 - 0.6561) \times 0.81 \\ = 0.780759 \quad \dots (iii)$$

$$R_{III} = [1 - (1 - R_1)(1 - R_2)] R_5 R_7 = (R_1 + R_2 - R_1 R_2) R_5 R_7 \\ = (0.9 + 0.9 - 0.9 \times 0.9) (0.9)^2 = (1.8 - 0.81) (0.81) \\ = 0.8019 \quad \dots (iv)$$

$$R_{IV} = R_2 R_6 R_5 R_7 = (0.9)^4 = 0.6561 \quad \dots (v)$$

Using (iv) and (v) in (ii), we get

$$R_I = 0.8019 \times 0.9 + 0.6561 \times (1 - 0.9) = 0.72171 + 0.06561 \\ = 0.78732 \quad \dots (vi)$$

Using (iii) and (vi) in (i), we get

$$R_s = 0.78732 \times 0.9 + 0.780759 \times (1 - 0.9) \\ = 0.708588 + 0.0780759 \\ = 0.7866639$$

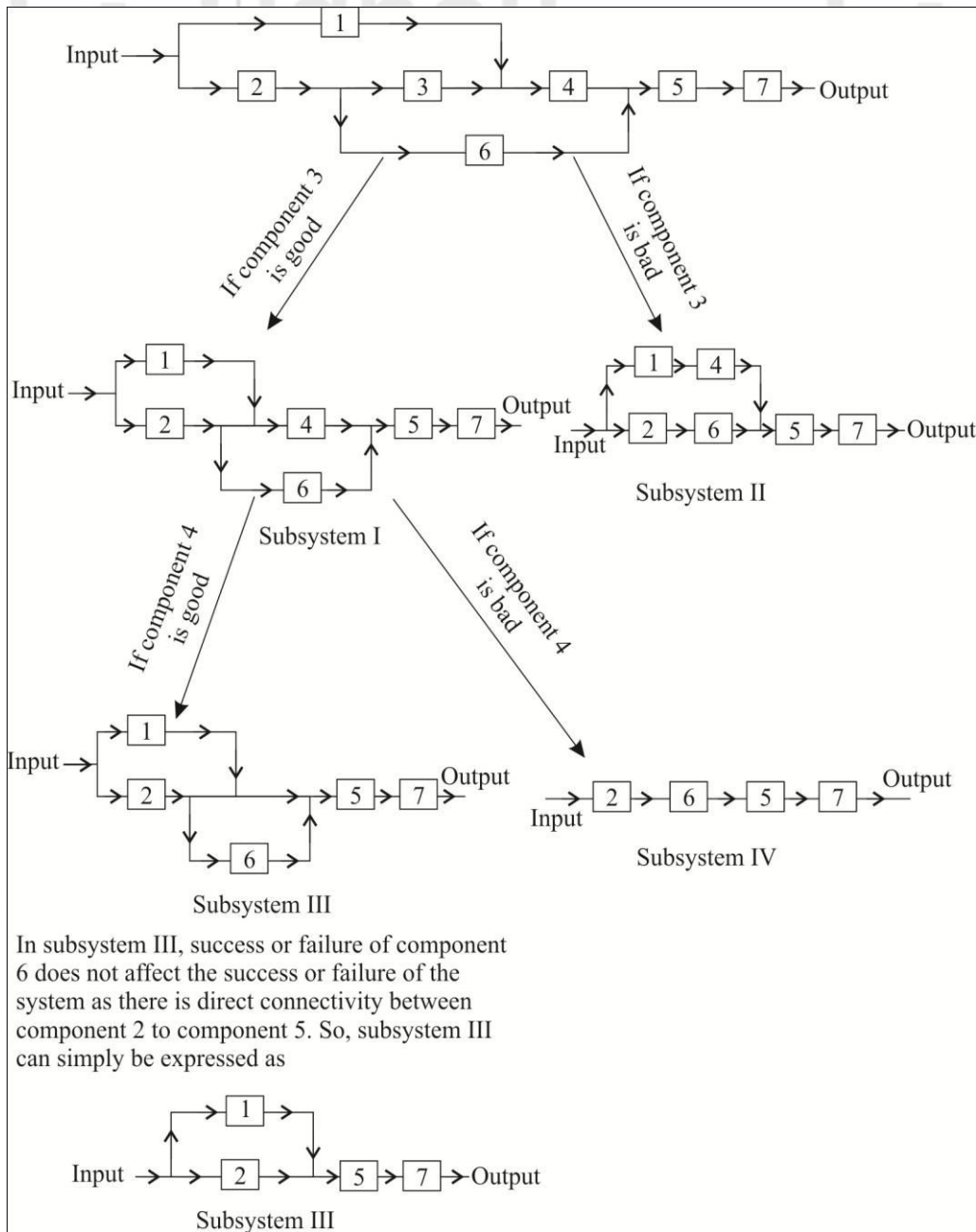


Fig. 16.9: Reduction of the given system into subsystems I, II, III, IV.

E2) Minimal path sets of the system having the reliability block diagram shown in Fig. 16.1 are given by

$$P_1 = \{1, 2\}, P_2 = \{3, 4\}, P_3 = \{1, 5, 4\}, P_4 = \{3, 5, 2\}.$$

The minimal cut sets for the same system are given by

$$C_1 = \{1, 3\}, C_2 = \{2, 4\}, C_3 = \{1, 5, 4\}, C_4 = \{3, 5, 2\}.$$

E3) For applying the cut set method, we have to find the minimal cut sets for the system having the reliability block diagram shown in Fig. 16.3. Minimal cut sets for Fig. 16.3 are given by

$$C_1 = \{1, 2\}, C_2 = \{2, 4\}, C_3 = \{4, 6\}, C_4 = \{1, 3, 6\}, C_5 = \{5\}, C_6 = \{7\}$$

These cut sets can be expressed from a reliability point of view as shown in Fig. 16.10.

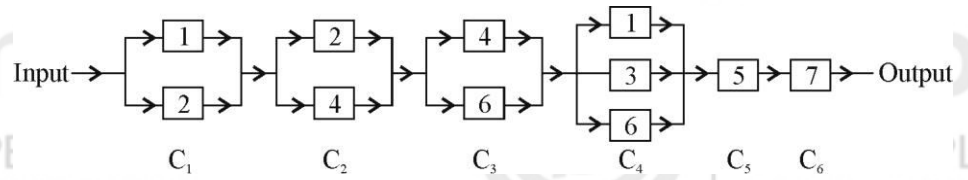


Fig. 16.10: Cuts sets for Fig. 16.3.

If Q denotes the unreliability of the system, then applying the cut set method, we get

$$Q = P(C_1 \cup C_2 \cup C_3 \cup C_4 \cup C_5 \cup C_6) \quad \dots (i)$$

If we apply the addition rule of probability in R.H.S. of (i), we shall get $6(= {}^6C_1)$ first order terms, $15(= {}^6C_2)$ second order terms, $20(= {}^6C_3)$ third order terms, $15(= {}^6C_4)$ fourth order terms, $6(= {}^6C_5)$ fifth order terms and $1(= {}^6C_6)$ sixth order term. But here it is given that we have to evaluate the reliability of the given system using only the first order terms. Therefore, unreliability (and hence reliability) using only first order terms is given by

$$\begin{aligned} Q &= P(C_1) + P(C_2) + P(C_3) + P(C_4) + P(C_5) + P(C_6) \\ &= Q_1 + Q_2 + Q_4 + Q_6 + Q_1 + Q_3 + Q_6 + Q_5 + Q_7 \quad \dots (ii) \end{aligned}$$

where Q_i denotes the unreliability of component i , ($1 \leq i \leq 7$).

It is given that all the components 1 to 7 have the same reliability 0.9 for a mission of 1 year. Hence, their unreliability will also be the same. So, let

$$Q_1 = Q_2 = \dots = Q_7 = Q \text{ (say)}$$

where $Q = 1 - 0.9 = 0.1$ for a mission of 1 year.

Therefore, equation (ii) gives

$$\begin{aligned} Q &= (0.1)^2 + (0.1)^2 + (0.1)^2 + (0.1)^3 + 0.1 + 0.1 \\ &= 3(0.1)^2 + (0.1)^3 + 2(0.1) = 0.03 + 0.001 + 0.2 = 0.231 \end{aligned}$$

Hence, reliability (R) of the system only using first order terms is given by

$$\begin{aligned} R &= 1 - Q = 1 - 0.231 \\ &= 0.769 \text{ for a mission of 1 year.} \end{aligned}$$

Note 2: Let us measure the error that occurs because of restricting the calculations to first order terms. In E1, the reliability of the same system has been obtained without any approximation. So on comparing the two results obtained in E1 and E3, the percentage error in calculation done only up to first order terms is given by

$$\text{error in \%} = \frac{0.7866639 - 0.769}{0.7866639} \times 100\% \approx 2.25\%$$

This error is not very high compared to the saving in calculations. Refer to solution of E4 for more detail.

Further, this error reduces very fast as the string of 9s (e.g., 0.99, 0.999, 0.9999, ...) increases in the reliability of the components of the system.

- E4)** If we want to evaluate unreliability (and hence reliability) of the system by considering terms of all orders then expanding the R.H.S. of (i) by applying addition law of probability, we get (to save space here we will use the convention $C_i C_j$ in place of $C_i \cap C_j$, etc.).

$$\begin{aligned}
 Q = & P(C_1) + P(C_2) + P(C_3) + P(C_4) + P(C_5) + P(C_6) \\
 & - P(C_1 C_2) - P(C_1 C_3) - P(C_1 C_4) - P(C_1 C_5) - P(C_1 C_6) \\
 & - P(C_2 C_3) - P(C_2 C_4) - P(C_2 C_5) - P(C_2 C_6) - P(C_3 C_4) \\
 & - P(C_3 C_5) - P(C_3 C_6) - P(C_4 C_5) - P(C_4 C_6) - P(C_5 C_6) \\
 & + P(C_1 C_2 C_3) + P(C_1 C_2 C_4) + P(C_1 C_2 C_5) + P(C_1 C_2 C_6) \\
 & + P(C_1 C_3 C_4) + P(C_1 C_3 C_5) + P(C_1 C_3 C_6) + P(C_1 C_4 C_5) \\
 & + P(C_1 C_4 C_6) + P(C_1 C_5 C_6) + P(C_2 C_3 C_4) + P(C_2 C_3 C_5) \\
 & + P(C_2 C_3 C_6) + P(C_2 C_4 C_5) + P(C_2 C_4 C_6) + P(C_2 C_5 C_6) \\
 & + P(C_3 C_4 C_5) + P(C_3 C_4 C_6) + P(C_3 C_5 C_6) + P(C_4 C_5 C_6) \\
 & - P(C_1 C_2 C_3 C_4) - P(C_1 C_2 C_3 C_5) - P(C_1 C_2 C_3 C_6) - P(C_1 C_2 C_4 C_5) \\
 & - P(C_1 C_2 C_4 C_6) - P(C_1 C_2 C_5 C_6) - P(C_1 C_3 C_4 C_5) - P(C_1 C_3 C_4 C_6) \\
 & - P(C_1 C_3 C_5 C_6) - P(C_1 C_4 C_5 C_6) - P(C_2 C_3 C_4 C_5) - P(C_2 C_3 C_4 C_6) \\
 & - P(C_2 C_3 C_5 C_6) - P(C_2 C_4 C_5 C_6) - P(C_3 C_4 C_5 C_6) \\
 & + P(C_1 C_2 C_3 C_4 C_5) + P(C_1 C_2 C_3 C_4 C_6) + P(C_1 C_2 C_3 C_5 C_6) \\
 & + P(C_1 C_2 C_4 C_5 C_6) + P(C_1 C_3 C_4 C_5 C_6) + P(C_2 C_3 C_4 C_5 C_6) \\
 & - P(C_1 C_2 C_3 C_4 C_5 C_6) \quad \dots (i)
 \end{aligned}$$

It is given that all components (1 to 7) have the same reliability 0.9 for a mission of 1 year. Hence, their unreliability will also be the same.

Let $Q_1 = Q_2 = \dots = Q_7 = Q$ (say) where $Q = 1 - 0.9 = 0.1$ for a mission of 1 year.

Let us first evaluate each term of R.H.S. of (i) separately in terms of Q .

$$P(C_1) = Q_1 Q_2 = Q^2$$

$$P(C_2) = Q_2 Q_4 = Q^2$$

$$P(C_3) = Q_4 Q_6 = Q^2$$

$$P(C_4) = Q_1 Q_3 Q_6 = Q^3$$

$$P(C_5) = Q_5 = Q$$

$$P(C_6) = Q_7 = Q$$

$$P(C_1C_2) = Q_1Q_2Q_4 = Q^3$$

[$\because$ occurrence of C_1C_2 or $C_1 \cap C_2$ means that all components of C_1 and C_2 have failed. But component 2 appears in both C_1 and C_2 and so its unreliability is taken only once.]

Similarly,

$$P(C_1C_3) = Q_1Q_2Q_4Q_6 = Q^4$$

$$P(C_1C_4) = Q_1Q_2Q_3Q_6 = Q^4$$

$$P(C_1C_5) = Q_1Q_2Q_5 = Q^3$$

$$P(C_1C_6) = Q_1Q_2Q_7 = Q^3$$

$$P(C_2C_3) = Q_2Q_4Q_6 = Q^3$$

$$P(C_2C_4) = Q_1Q_2Q_3Q_4Q_6 = Q^5$$

$$P(C_2C_5) = Q_2Q_4Q_5 = Q^3$$

$$P(C_2C_6) = Q_2Q_4Q_7 = Q^3$$

$$P(C_3C_4) = Q_1Q_3Q_4Q_6 = Q^4$$

$$P(C_3C_5) = Q_4Q_5Q_6 = Q^3$$

$$P(C_3C_6) = Q_4Q_6Q_7 = Q^3$$

$$P(C_4C_5) = Q_1Q_3Q_5Q_6 = Q^4$$

$$P(C_4C_6) = Q_1Q_3Q_6Q_7 = Q^4$$

$$P(C_5C_6) = Q_5Q_7 = Q^2$$

$$P(C_1C_2C_3) = Q_1Q_2Q_4Q_6 = Q^4$$

$$P(C_1C_2C_4) = Q_1Q_2Q_3Q_4Q_6 = Q^5$$

$$P(C_1C_2C_5) = Q_1Q_2Q_4Q_5 = Q^4$$

$$P(C_1C_2C_6) = Q_1Q_2Q_4Q_7 = Q^4$$

$$P(C_1C_3C_4) = Q_1Q_2Q_3Q_4Q_6 = Q^5$$

$$P(C_1C_3C_5) = Q_1Q_2Q_4Q_5Q_6 = Q^5$$

$$P(C_1C_3C_6) = Q_1Q_2Q_4Q_6Q_7 = Q^5$$

$$P(C_1C_4C_5) = Q_1Q_2Q_3Q_5Q_6 = Q^5$$

$$P(C_1C_4C_6) = Q_1Q_2Q_3Q_6Q_7 = Q^5$$

$$P(C_1C_5C_6) = Q_1Q_2Q_5Q_7 = Q^4$$

$$P(C_2C_3C_4) = Q_1Q_2Q_3Q_4Q_6 = Q^5$$

$$P(C_2C_3C_5) = Q_2Q_4Q_5Q_6 = Q^4$$

$$P(C_2C_3C_6) = Q_2Q_4Q_6Q_7 = Q^4$$

$$P(C_2C_4C_5) = Q_1Q_2Q_3Q_4Q_5Q_6 = Q^6$$

$$P(C_2C_4C_6) = Q_1Q_2Q_3Q_4Q_6Q_7 = Q^6$$

$$P(C_2C_5C_6) = Q_2Q_4Q_5Q_7 = Q^4$$

$$P(C_3C_4C_5) = Q_1Q_3Q_4Q_5Q_6 = Q^5$$

$$P(C_3C_4C_6) = Q_1Q_3Q_4Q_6Q_7 = Q^5$$

$$P(C_3C_5C_6) = Q_4Q_5Q_6Q_7 = Q^4$$

$$P(C_4C_5C_6) = Q_1Q_3Q_5Q_6Q_7 = Q^5$$

$$P(C_1C_2C_3C_4) = Q_1Q_2Q_3Q_4Q_6 = Q^5$$

$$P(C_1C_2C_3C_5) = Q_1Q_2Q_4Q_5Q_6 = Q^5$$

$$P(C_1C_2C_3C_6) = Q_1Q_2Q_4Q_6Q_7 = Q^5$$

$$P(C_1C_2C_4C_5) = Q_1Q_2Q_3Q_4Q_5Q_6 = Q^6$$

$$P(C_1C_2C_4C_6) = Q_1Q_2Q_3Q_4Q_6Q_7 = Q^6$$

$$P(C_1C_2C_5C_6) = Q_1Q_2Q_4Q_5Q_7 = Q^5$$

$$P(C_1C_3C_4C_5) = Q_1Q_2Q_3Q_4Q_5Q_6 = Q^6$$

$$P(C_1C_3C_4C_6) = Q_1Q_2Q_3Q_4Q_6Q_7 = Q^6$$

$$P(C_1C_3C_5C_6) = Q_1Q_2Q_4Q_5Q_6Q_7 = Q^6$$

$$P(C_1C_4C_5C_6) = Q_1Q_2Q_3Q_5Q_6Q_7 = Q^6$$

$$P(C_2C_3C_4C_5) = Q_1Q_2Q_3Q_4Q_5Q_6 = Q^6$$

$$P(C_2C_3C_4C_6) = Q_1Q_2Q_3Q_4Q_6Q_7 = Q^6$$

$$P(C_2C_3C_5C_6) = Q_2Q_4Q_5Q_6Q_7 = Q^5$$

$$P(C_2C_4C_5C_6) = Q_1Q_2Q_3Q_4Q_5Q_6Q_7 = Q^7$$

$$P(C_3C_4C_5C_6) = Q_1Q_3Q_4Q_5Q_6Q_7 = Q^6$$

$$P(C_1C_2C_3C_4C_5) = Q_1Q_2Q_3Q_4Q_5Q_6 = Q^6$$

$$P(C_1C_2C_3C_4C_6) = Q_1Q_2Q_3Q_4Q_6Q_7 = Q^6$$

$$P(C_1C_2C_3C_5C_6) = Q_1Q_2Q_4Q_5Q_6Q_7 = Q^6$$

$$P(C_1C_2C_4C_5C_6) = Q_1Q_2Q_3Q_4Q_5Q_6Q_7 = Q^7$$

$$P(C_1C_3C_4C_5C_6) = Q_1Q_2Q_3Q_4Q_5Q_6Q_7 = Q^7$$

$$P(C_2C_3C_4C_5C_6) = Q_1Q_2Q_3Q_4Q_5Q_6Q_7 = Q^7$$

$$P(C_1C_2C_3C_4C_5C_6) = Q_1Q_2Q_3Q_4Q_5Q_6Q_7 = Q^7$$

Putting these values in (i), we get

$$\begin{aligned}
 Q &= (2Q + 3Q^2 + Q^3) - (Q^2 + 8Q^3 + 5Q^4 + Q^5) + (8Q^4 + 10Q^5 + 2Q^6) \\
 &\quad - (5Q^5 + 9Q^6 + Q^7) + (3Q^6 + 3Q^7) - Q^7 \\
 &= (0.2 + 0.03 + 0.001) - (0.01 + 0.008 + 0.0005 + 0.00001) \\
 &\quad + (0.0008 + 0.0001 + 0.000002) \\
 &\quad - (0.00005 + 0.000009 + 0.0000001) \\
 &\quad + (0.000003 + 0.0000003) - (0.0000001) \\
 &= 0.231 - 0.01851 + 0.000902 - 0.0000591 + 0.0000033 - 0.0000001 \\
 &= 0.2133361
 \end{aligned}$$

Hence, the reliability of the system is given by

$$R = 1 - Q = 1 - 0.2133361 = 0.7866639$$

which is the same as obtained in E1 using the decomposition method.

E5) For applying the tie set method, we first find the minimal path sets for the system having the reliability block diagram shown in Fig. 16.3. The minimal path sets for this system are given by

$$P_1 = \{1, 4, 5, 7\}, P_2 = \{2, 3, 4, 5, 7\}, P_3 = \{2, 6, 5, 7\}$$

A minimal path set is also known as tie set. If we denote path sets

P_1, P_2, P_3 by tie sets T_1, T_2, T_3 , respectively, then from a reliability point of view these tie sets can be shown as in Fig. 16.11.

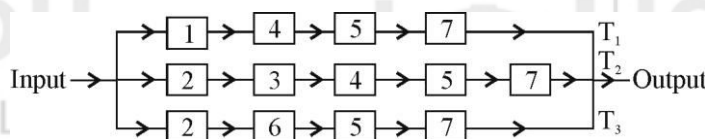


Fig. 16.11: Minimal tie sets for E5.

Here tie sets T_1, T_2 and T_3 are in parallel but we cannot apply the procedure of pure parallel configuration because they are not made up of independent components. Hence, if R_i denotes the reliability of the component i , $i = 1, 2, \dots, 7$, and R_s that of the system, then applying the tie set method, we get

$$\begin{aligned}
 R_s &= P(T_1 \cup T_2 \cup T_3) \\
 &= P(T_1) + P(T_2) + P(T_3) - P(T_1 \cap T_2) - P(T_1 \cap T_3) \\
 &\quad - P(T_2 \cap T_3) + P(T_1 \cap T_2 \cap T_3) \quad \dots (i)
 \end{aligned}$$

Let us first evaluate each term of R.H.S. of (i) separately.

$$\begin{aligned}
 P(T_1) &= R_1 R_4 R_5 R_7 \left[\begin{array}{l} \because \text{occurrence of tie set } T_1 \text{ means that there} \\ \text{exists a path between input and output} \\ \text{through } T_1. \text{ This implies that the components} \\ 1, 4, 5 \text{ and } 7 \text{ must work successfully.} \end{array} \right] \\
 &= (0.9)^4
 \end{aligned}$$

Similarly,

$$P(T_2) = R_2 R_3 R_4 R_5 R_7 = (0.9)^5$$

$$P(T_3) = R_2 R_6 R_5 R_7 = (0.9)^4$$

$$P(T_1 \cap T_2) = R_1 R_2 R_3 R_4 R_5 R_7 = (0.9)^6$$

$$P(T_1 \cap T_3) = R_1 R_2 R_4 R_5 R_6 R_7 = (0.9)^6$$

$$P(T_2 \cap T_3) = R_2 R_3 R_4 R_5 R_6 R_7 = (0.9)^6$$

$$P(T_1 \cap T_2 \cap T_3) = R_1 R_2 R_3 R_4 R_5 R_6 R_7 = (0.9)^7$$

Putting all these values in (i), we get

$$\begin{aligned} R_s &= (0.9)^4 + (0.9)^5 + (0.9)^4 - (0.9)^6 - (0.9)^6 - (0.9)^6 + (0.9)^7 \\ &= 2(0.9)^4 + (0.9)^5 - 3(0.9)^6 + (0.9)^7 \\ &= 1.3122 + 0.59049 - 1.594323 + 0.4782969 \\ &= 0.7866639 \quad \text{for a mission of 1 year} \end{aligned}$$

which is the same as obtained in E1 and E4.

Note 3: At this stage you may be thinking that the tie set method is easier as compared to the cut set method. It is true for this system. But there exist other systems whose reliability block diagrams have lesser number of cut sets than tie sets. In that case, the cut set method will involve less calculation. The moral of the story is that if you are solving problems manually and wish to know which method (cut set method or tie set method) to apply, then do so using the following information:

- Apply the cut set method using only first order terms if you can tolerate marginal error and components have high reliability;
- However, if you want to calculate the reliability using all order terms then apply the tie set method if the number of tie sets is less as compared to the number of cut sets and cut set method if the number of cut sets is less as compared to the number of tie sets.

FURTHER READING

1. Reliability Evaluation of Engineering Systems: Concepts and Techniques by Roy Billinton and Ronald N Allan; Plenum Press New York and London (1983) [Chapters 4, 5 and 6].
2. An introduction to Reliability and Maintainability engineering by Charles E. Ebeling; Tata McGraw Hill Education Private Limited (2000) [Chapters 1, 2, 3 and 5].
3. Reliability Engineering by E Balagurusamy; Tata McGraw Hill Education Private Limited (1984) [Chapters 1 to 6].
4. Concepts in reliability with an introduction to Maintainability and Availability by L.S. Srinath; Affiliated East-West Press PVT. LTD. (1975) [Chapters 2, 3, 4 and 6].